Alessio Del Bue · Cristian Canton ·
Jordi Pont-Tuset · Tatiana Tommasi (Eds.)

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Computer Vision – ECCV 2024 Workshops

Milan, Italy, September 29–October 4, 2024
Proceedings, Part II

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Jordi Pont-Tuset · Tatiana Tommasi
Editors

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Foreword

Welcome to the proceedings of the European Conference on Computer Vision (ECCV) 2024, the 18th edition of the conference, which has been held biennially since its founding in France in 1990. We are holding the event in Italy for the third time, in a convention centre where we can accommodate nearly 7000 in-person attendees.

The journey to ECCV 2024 started at CVPR 2019, where the general chairs met at a sketchy bar in Long Beach to discuss ideas on how to organize an amazing future ECCV. After that, Covid came and changed conferences probably forever, but we maintained our excitement to organize an awesome in-person conference in 2024, particularly having seen the wonderful event that was ECCV 2022 in Tel Aviv. And here we are, 5 years later, delighted that we finally get to welcome all of you to Milan!

First and foremost, ECCV is about the exchange of scientific ideas, and hence the most important organizational task is the selection of the programme. To our Program Chairs, Aleš Leonardis, Elisa Ricci, Gül Varol, Olga Russakovsky, Stefan Roth, and Torsten Sattler, our entire community owe a huge debt of gratitude. As you will read in their Preface to the proceedings, they dealt with over 8,500 submissions, and their diligence, thoroughness, and sheer effort has been inspirational. From the early stages of designing the call for papers, through recruiting and selecting area chairs and reviewers, to ensuring every paper has a fair and thorough assessment, their professionalism, commitment, and attention to detail has been exemplary. From all the choices that we made to organize ECCV, choosing this amazing team of Program Chairs has been, without a doubt, our very best decision.

The Program Chairs were advised and supported by an Ethics Review Committee, Chloé Bakalar, Kate Saenko, Remi Denton, and Yisong Yue, who provided invaluable input on papers where reviewers or Area Chairs had raised ethical concerns.

The Publication Chairs, Jovita Lukasik, Michael Möller, François Brémond, and Mahmoud Ali, did greatwork in assembling the camera-ready papers into these Springer volumes, dealing with numerous issues that arose promptly and professionally.

With the ever-expanding importance of workshops, tutorials, and demos at our major conferences comes a corresponding increase in the challenge of organizing over 70 workshops and 9 tutorials attached to the main conference. The Workshop and Tutorial Chairs, Alessio Del Bue, Jordi Pont-Tuset, Cristian Canton, and Tatiana Tommasi, and Demo Chairs, Hyung Chang and Marco Cristani, worked tirelessly to coordinate the very varied demands and structures of these different events, yielding an extremely rich auxiliary program which will enhance the conference experience for everyone who attends.

The Industry Chairs, Shaogang Gong and Cees Snoek, working with Limor Urfaly and Lior Gelfand, managed to attract numerous companies and organizations from all around the world to the industrial Expo, making ECCV 2024 an international event that not only showcases excellent foundational research, but also presents how such research

can be transformed in real products. It is notable that together with the big companies, several start-ups chose ECCV to present their business ideas and activities.

As usual in the recent main conference, we also organized Speed Mentoring and Doctoral Consortium events. While the latter is an institutional occasion allowing senior PhDs to show their work, the former is especially important to answer to the many doubts and uncertainties young scholars have while undertaking a career in Computer Vision and Artificial Intelligence. We warmly thank our Social Activities Chairs, Raffaella Lanzarotti, Simone Bianco and Giovanni Farinella, and the Doctoral Consortium Chairs, Cigdem Beyan and Or Litany, for their dedication to organize such events, so important for our young colleagues, as well as all the mentors who were available to share their experience.

As the conference becomes larger and larger, the overall budget increases to a level where uncertainties in estimates of attendance can result in significant losses, perhaps enough to significantly damage the prospects of running the conference in the future. This risk imposed the need to require authors to commit to full conference registrations, with the undesirable consequent possibility to cause hardship to authors, particularly student authors, from organizations where funding attendance is difficult. To mitigate this effect, we allocated travel grants to students who might otherwise have had difficulty in attending. Our Diversity Chairs, David Fouhey and Rita Cucchiara, put tremendous effort into a wonderfully careful and thoughtful programme, balancing diversity in all its forms with a changing budget landscape, allowing us to support the participation of many people who might not otherwise have been able to attend.

This year's conference also marks the move to a unified, multi-year conference website, based on that used for other leading computer vision and AI conferences. This was an effort that we decided to take on to make the ECCV website more familiar to the community and support the organization of the future ECCV editions. Our thanks go to Lee Campbell of Eventhosts who managed the entire design and development of the new website and provided responsive support and updates throughout the conference organization timeline.

Our Finance Chairs, Gérard Medioni and Nicole Finn, provided invaluable advice and assistance on all aspects of the financial planning of the conference, in conjunction with the professional conference organizers AIM Group, where Lavinia Ricci led a fantastic team that managed thousands of details from space planning to food to registrations. In particular, Elder Bromley and Mara Carletti were a tremendous support to the entire conference organization. The conference centre, Mico DMC, also provided an excellent service, showing considerable flexibility in adapting the venue and processes to the particular demands of an academic-focused conference.

Finally, we cannot forget to thank our Social Media Chair, Kosta Derpanis, supported by Abby Stylianou and Jia-Bin Huang, who, with their presence on social media, spread ECCV news and responded to small and big questions in a timely manner, helping the community to promptly receive information and fix their problems.

October 2024

Andrew Fitzgibbon
Laura Leal-Taixé
Vittorio Murino

Preface

Welcome to the workshop proceedings of the 18th European Conference on Computer Vision (ECCV 2024).

This year, the main ECCV event was accompanied by 73 workshops, scheduled on September 29 and 30, 2024. We received 131 workshop proposals covering a wide range of computer vision topics but, due to space constraints, we were unable to accept several high-quality submissions. To optimize space allocation, all workshops were scheduled in a half-day format.

In the selection process, we sought to achieve a thoughtful balance among the topics by including a range of long-standing as well as newly introduced series.

The following 73 workshops took place at ECCV 2024:

3D Vision

W01 – Recovering 6D Object Pose
W02 – Half-century of Structure-from-Motion (50SfM)
W03 – Dense Neural SLAM (NeuSLAM)
W04 – Geometry in the Large Model Era
W05 – Spatial AI
W06 – Transparent and Reflective Objects in the Wild Challenges (TRICKY)
W07 – Wild3D – 3D Modeling, Reconstruction, and Generation in the Wild
W08 – AI3DCC – AI for 3D Content Creation
W09 – 3D Vision and Modeling Challenges in eCommerce

Applications

W10 – FashionAI: Exploring the Intersection of Fashion and Artificial Intelligence for Reshaping the Industry
W11 – Computer Vision for Ecology (CV4E)
W12 – Computer Vision in Plant Phenotyping and Agriculture (CVPPA)
W13 – Computer Vision for Metaverse (CV4Metaverse)
W14 – Computer Vision for Videogames (CV2)
W15 – Vision-Based Industrial Inspection (VISION)

Art

W16 – AI for Visual Arts Workshop and Challenges (AI4VA)
W17 – Vision for Art (VISART)
W18 – AI4DH: Artificial Intelligence for Digital Humanities

Autonomous Driving and Robotics

W19 – ROAD Workshop and Challenge: Event Detection for Situation Awareness in Autonomous Driving
W20 – Vision-Centric Autonomous Driving (VCAD)
W21 – Robust, Out-of-Distribution and Multi-modal Models for Autonomous Driving (ROAM)
W22 – Autonomous Vehicles Meet Multimodal Foundation Models
W23 – Multimodal Perception and Comprehension of Corner Cases in Autonomous Driving: Towards Next-Generation Solutions
W24 – Multi-agent Autonomous Systems Meet Foundation Models: Challenges and Futures

Detection, Recognition, and Low-Level Vision

W25 – Visual Object Tracking and Segmentation Challenge (VOTS)
W26 – Advances in Image Manipulation (AIM) Workshop and Challenges
W27 – Instance-Level Recognition
W28 – Large-Scale Video Object Segmentation (LSVOS)

Efficiency

W29 – Efficient Deep Learning for Foundation Models
W30 – Computational Aspects of Deep Learning

Human

W31 – Assistive Computer Vision and Robotics (ACVR)
W32 – Foundation Models for 3D Humans
W33 – Artificial Social Intelligence
W34 – Towards a Complete Analysis of People (T-CAP): Fine-Grained Understanding for Real-World Applications
W35 – Observing and Understanding Hands in Action
W36 – Workshop and Competition on Affective Behavior Analysis-in-the-Wild
W37 – Expressive Encounters: Co-speech Gestures Across Cultures in the Wild

Medical and Bio-inspired Vision

W38 – BioImage Computing (BIC)
W39 – Human-Inspired Computer Vision (HCV)

Machine Learning

W40 – Knowledge in Generative Models
W41 – Self-supervised Learning: What Is Next?
W42 – Traditional Computer Vision in the Age of Deep Learning (TradiCV)
W43 – Uncertainty Quantification for Computer Vision
W44 – Emergent Visual Abilities and Limits of Foundation Models (EVAL-FoMo)
W45 – Beyond Euclidean: Hyperbolic and Hyperspherical Learning for Computer Vision
W46 – Unlearning and Model Editing (U&ME)
W47 – Out-of-Distribution Generalization in Computer Vision Foundation Models
W48 – Visual Concepts
W49 – Sometimes Less Is More: Dataset Distillation Challenge
W50 – Quantum Computer Vision and Machine Learning (QCVML)
W51 – More Exploration, Less Exploitation (MELEX)
W52 – Synthetic Data for Computer Vision

Multimodal

W54 – Audio-Visual Generation and Learning (AVGenL)
W56 – Multimodal Agents
W57 – Enabling Complex Perception Through Vision and Language Foundational Models (OmniLabel)
W58 – Perception Test Challenge

Responsible AI

W59 – The Dark Side of Generative AIs and Beyond
W60 – Foundation Models Creators Meet Users (FOCUS)
W61 – Fairness and Ethics Towards Transparent AI: Facing the Challenge Through Model Debiasing (FAILED)
W62 – Explainable AI for Computer Vision: Where Are We and Where Are We Going?
W63 – Trust What You Learn (TWYN): Trustworthiness in Computer Vision
W64 – Women in Computer Vision (WICV)
W65 – Privacy-Preserving Computer Vision
W66 – Critical Evaluation of Generative Models and Their Impact on Society (CEGIS)
W67 – Explainable and Interpretable Artificial Intelligence for Biometrics (xAI4Biometrics)
W68 – Green Foundation Models

Scene Understanding

W69 – Scalable 3D Scene Generation and 3D Geometric Scene Understanding
W70 – OpenSUN3D: Open-Vocabulary 3D Scene Understanding
W71 – Map-Free Visual Relocalization: Metric Pose Relative to a Single Image
W72 – Neuromorphic Vision (NeVi): Advantages and Applications of Event Cameras
W73 – Neural Fields Beyond Conventional Cameras
W74 – GigaVision: When Gigapixel Videography Meets Computer Vision
W75 – Eyes of the Future: Integrating Computer Vision in Smart Eyewear
These LNCS volumes collect 574 accepted papers from 53 of the 73 workshops, distributed as follows:

Part I, LNCS 15623: W01, W02, W06, W10
Part II, LNCS 15624: W11, W13, W14
Part III, LNCS 15625: W12
Part IV, LNCS 15626: W15
Part V, LNCS 15627: W16
Part VI, LNCS 15628: W17, W18
Part VII, LNCS 15629: W19, W21, W23, W25
Part VIII, LNCS 15630: W24
Part IX, LNCS 15631: W26 (part)
Part X, LNCS 15632: W26 (part), W27, W28
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Part XII, LNCS 15634: W31, W32
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Part XIX, LNCS 15641: W49, W51, W54, W56
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Part XXI, LNCS 15643: W61, W62, W63
Part XXII, LNCS 15644: W64, W66, W67, W68
Part XXIII, LNCS 15645: W69, W70, W71, W75
Part XXIV, LNCS 15646: W72

We sincerely thank the ECCV general chairs for trusting us with the responsibility for the workshops, the workshop organizers for their hard work in putting together exciting programs, and the workshop presenters and authors for contributing to the conference.

January 2025

Alessio Del Bue
Cristian Canton
Jordi Pont-Tuset
Tatiana Tommasi

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W02 – Half-century of Structure-from-Motion (50SfM)

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W06 – Transparent and Reflective Objects in the Wild Challenges (TRICKY)

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W11 – Computer Vision for Ecology (CV4E)

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W12 – Computer Vision in Plant Phenotyping and Agriculture (CVPPA)

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W16 – AI for Visual Arts Workshop and Challenges (AI4VA)

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W17 – Vision for Art (VISART)

Organizers

Amanda Wasielewski	Uppsala Universitet, Sweden
Nanne van Noord	University of Amsterdam, Netherlands
Noa Garcia	Osaka University, Japan
Peter Bell	Universität Marburg, Germany
Stuart James	Durham University

W18 – AI4DH: Artificial Intelligence for Digital Humanities

Organizers

Lorenzo Baraldi	University of Modena and Reggio Emilia, Italy
Silvia Cascianelli	University of Modena and Reggio Emilia, Italy
Marcella Cornia	University of Modena and Reggio Emilia, Italy
Francesca Matrone	Politecnico di Torino, Italy
Guido Borghi	University of Modena and Reggio Emilia, Italy
Marina Paolanti	University of Macerata, Italy
Roberto Pierdicca	Università Politecnica delle Marche, Italy

W19 – ROAD Workshop and Challenge: Event Detection for Situation Awareness in Autonomous Driving

Organizers

Fabio Cuzzolin	Oxford Brookes University, UK
Salman Khan	Oxford Brookes University, UK

Reza Javanmard Alitappeh	University of Science and Technology of Mazandaran, Iran
Eleonora Giunchiglia	Imperial College London, UK
Izzeddin Teeti	Oxford Brookes University, UK
Mihaela Catalina Stoian	University of Oxford, UK
Andrew Bradley	Oxford Brookes University, UK
Gurkirt Singh	ETH Zürich, Switzerland
Jorge Dias	Khalifa University, UAE
Nadya Abdel Madjid	Khalifa University, UAE
Majid Khonji	Khalifa University, UAE
Bilal Hassan	Khalifa University, UAE
Chi-Hsi Kung	National Yang Ming Chiao Tung University, Taiwan
Yi-Hsuan Tsai	Google, USA
Yi-Ting Chen	National Yang Ming Chiao Tung University, Taiwan

W20 – Vision-Centric Autonomous Driving (VCAD)

Organizers

Yiming Li	NYU and NVIDIA Research, USA
Yue Wang	University of Southern California and NVIDIA Research, USA
Hang Zhao	Tsinghua University, China
Vitor Guizilini	Toyota Research Institute, USA
Yiyi Liao	Zhejiang University, China
Xinshuo Weng	NVIDIA Research, Canada
Chen Feng	NYU, USA
Marco Pavone	Stanford University and NVIDIA Research, USA

W21 – Robust, Out-of-Distribution and Multi-modal Models for Autonomous Driving (ROAM)

Organizers

Marcelo Contreras	University of Alberta, Canada and Universidad de Ingeniería y Tecnología (UTEC), Peru
Dunant Cusipuma	Artificio, Peru

David Ortega — Artificio, Peru
Victor Flores-Benites — Artificio and Universidad de Ingeniería y Tecnología (UTEC), Peru
Arturo Deza — Artificio and Universidad de Ingeniería y Tecnología (UTEC), Peru

W22 – Autonomous Vehicles Meet Multimodal Foundation Models

Organizers

Yan Wang — NVIDIA, USA
Yurong You — NVIDIA, USA
Kashyap Chitta — University of Tübingen, Germany
Yiming Li — NYU, USA
Yue Wang — University of Southern California and NVIDIA Research, USA
Yiyi Liao — Zhejiang University, China
Li Erran Li — AWS AI, Amazon and Columbia University, USA
Deva Ramanan — CMU, USA
Kilian Q. Weinberger — Cornell University, USA
Laura Leal-Taixe — NVIDIA and Technical University of Munich, Germany

W23 – Multimodal Perception and Comprehension of Corner Cases in Autonomous Driving: Towards Next-Generation Solutions

Organizers

Kai Chen — Hong Kong University of Science and Technology, China
Ruiyuan Gao — Chinese University of Hong Kong, China
Lanqing Hong — Huawei, China
Hang Xu — Huawei, China
Xu Jia — Dalian University of Technology, China
Holger Caesar — TU Delft, Netherlands
Dengxin Dai — Huawei, China
Bingbing Liu — Huawei, China
Dzmitry Tsishkou — Huawei, China
Songcen Xu — Huawei, China

Chunjing Xu	Huawei, China
Qiang Xu	Chinese University of Hong Kong, China
Huchuan Lu	Dalian University of Technology, China
Dit-Yan Yeung	Hong Kong University of Science and Technology, China

W24 – Multi-agent Autonomous Systems Meet Foundation Models: Challenges and Futures

Organizers

Haibao Yu	University of Hong Kong and Tsinghua University, China
Jianing Qiu	Chinese University of Hong Kong, China
Yao Mu	University of Hong Kong, China
Jiankai Sun	Stanford University, USA
Li Chen	Shanghai AI Lab, China
Walter Zimmer	Technical University of Munich, Germany
Jiaru Zhong	Beijing Institute of Technology, China
Dandan Zhang	Imperial College London, UK
Fei Gao	Zhejiang University, China
Shanghang Zhang	Peking University, China
Mac Schwager	NVIDIA and Stanford University, USA
Ping Luo	University of Hong Kong, China
Zaiqing Nie	Tsinghua University, China

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Je-Seok Ham	Electronics and Telecommunications Research Institute
Hao Wei	Chinese University of Hong Kong, China
Carlos Plou	University of Zaragoza, Spain

W25 – Visual Object Tracking and Segmentation Challenge (VOTS)

Organizers

Matej Kristan	University of Ljubljana, Slovenia
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Pavel Tokmakov	Toyota Research Institute, USA
Michael Felsberg	Linköping University, Sweden
Luka Čehovin Zajc	University of Ljubljana, Slovenia
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Johanna Björklund	Umeå University, Sweden

W26 – Advances in Image Manipulation (AIM) Workshop and Challenges

Organizers

Radu Timofte	University of Würzburg, Germany
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Marcos V. Conde	University of Würzburg, Germany
Dmitriy Vatolin	Moscow State University, Russia
Eduardo Pérez-Pellitero	Noah's Ark Lab, UK

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Cheng-Ming Chiang	MediaTek, Taiwan
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Shuhang Gu	University of Electronic Science and Technology of China, China
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Kyoung Mu Lee	Seoul National University, South Korea
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Yawei Li	ETH Zürich, Switzerland
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Wenjun Zeng	Microsoft Research, China
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Yulun Zhang	Shanghai Jiao Tong University, China
Jun-Yan Zhu	CMU, USA
Wangmeng Zuo	Harbin Institute of Technology, China

W27 – Instance-Level Recognition

Organizers

Andre Araujo	Google DeepMind (Primary Contact)
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Noa Garcia	Osaka University, Japan
Bohyung Han	Seoul National University, South Korea
Guangxing Han	Columbia University, USA
Giorgos Tolias	Czech Technical University in Prague, Czech Republic
Hao Yang	Amazon, USA
Nikolaos-Antonios Ypsilantis	Czech Technical University in Prague, Czech Republic
Xu Zhang	Amazon, USA

W28 – Large-Scale Video Object Segmentation (LSVOS)

Organizers

Lingyi Hong	Fudan University, China
Henghui Ding	Fudan University, China
Chang Liu	Nanyang Technological University, Singapore

Ning Xu	Apple, USA
Linjie Yang	ByteDance, USA
Yuchen Fan	Meta Reality Labs, USA

W29 – Efficient Deep Learning for Foundation Models

Organizers

Hongxu (Danny) Yin	NVIDIA, USA
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Ji Lin	OpenAI, USA
Maying Shen	NVIDIA, USA
Jason Clemons	NVIDIA, USA
Xin Wang	Microsoft Research, USA
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Xueyan Zou	University of California, San Diego, USA
Xiaolong Wang	University of California, San Diego, USA
Song Han	MIT, USA
Jan Kautz	NVIDIA, USA

W30 – Computational Aspects of Deep Learning

Organizers

Giuseppe Fiameni	NVIDIA, Italy
Iuri Frosio	NVIDIA, Italy
Claudio Baecchi	Small Pixels, Italy
Laura Morselli	CINECA, Italy
Lorenzo Baraldi	University of Modena and Reggio Emilia, Italy
Dario Garcia Gasulla	Barcelona Supercomputing Center, Spain

W31 – Assistive Computer Vision and Robotics (ACVR)

Organizers

Giovanni Maria Farinella	University of Catania, Italy
Antonino Furnari	University of Catania, Italy
Marco Leo	CNR, Italy
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Francesco Ragusa	University of Catania, Italy
Mohan Trivedi	University of California, USA

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Jim Regh	Georgia Tech, USA
Sara Colantonio	CNR, Italy
Lorenzo Mur-Labadia	University of Zaragoza, Spain
Andrea Loddo	University of Cagliari, Italy

W32 – Foundation Models for 3D Humans

Organizers

Yao Feng	Meshcapade and MPI-IS, Germany and ETH Zürich, Switzerland
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Naureen Mahmood	Meshcapade, Germany
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Weiyang Liu	MPI-IS, Germany and University of Cambridge, UK
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Jing Lin	Tsinghua University, China
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Siyuan Bian	Shanghai Jiaotong University, China
Tingting Liao	Mohamed bin Zayed University of Artificial Intelligence, UAE
Zeyu Cai	Hong Kong University of Science and Technology, China

W33 – Artificial Social Intelligence

Organizers

Leena Mathur	CMU, USA
Evonne Ng	Meta, USA
Fiona Ryan	Georgia Tech, USA
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Paul Liang	MIT, USA
Gül Varol	École des Ponts, ParisTech, France

Shiry Ginosar	University of California, Berkeley and Google, USA
Hanbyul Joo	Seoul National University, South Korea
Justine Cassell	Inria, France
James Rehg	University of Illinois, Urbana-Champaign, USA
Louis-Philippe Morency	CMU and Meta, USA

W34 – Towards a Complete Analysis of People (T-CAP): Fine-Grained Understanding for Real-World Applications

Organizers

Mohamed Daoudi	IMT Lille Douai, France
Xavier Alameda-Pineda	Inria, France
Federico Becattini	University of Siena, Italy
Guido Borghi	University of Modena and Reggio Emilia, Italy
Marcella Cornia	University of Modena and Reggio Emilia, Italy
Claudio Ferrari	University of Parma, Italy
Tomaso Fontanini	University of Parma, Italy
Andrea Pilzer	NVIDIA, Italy

W35 – Observing and Understanding Hands in Action

Organizers

Hyung Jin Chang	University of Birmingham, UK
Rongyu Chen	National University of Singapore, Singapore
Zicong Fan	ETH Zürich, Switzerland
Otmar Hilliges	ETH Zürich, Switzerland
Kun He	Meta Reality Labs, USA
Take Ohkawa	University of Tokyo, Japan
Yoichi Sato	University of Tokyo, Japan
Elden Tse	National University of Singapore, Singapore
Linlin Yang	Communication University of China, China
Lixin Yang	Shanghai Jiao Tong University, China
Angela Yao	National University of Singapore, Singapore
Linguang Zhang	Facebook Reality Labs (Oculus), USA

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Takuma Yagi	AIST, Japan
Zerui Chen	Inria, France
Zhiying Leng	Beihang University, China
Zhongqun Zhang	University of Birmingham, UK

W36 – Workshop and Competition on Affective Behavior Analysis-in-the-Wild

Organizers

Dimitrios Kollias	Queen Mary University of London, UK
Stefanos Zafeiriou	Imperial College London, UK
Irene Kotsia	Cogitat, UK
Abhinav Dhall	Flinders University, Australia
Shreya Ghosh	Curtin University, Australia

Data Chairs

Chunchang Shao	Queen Mary University of London, UK
Guanyu Hu	Queen Mary University of London, UK and Xi'an Jiaotong University, China

W37 – Expressive Encounters: Co-speech Gestures Across Cultures in the Wild

Organizers

Viktor Schmuck	King's College London, UK
Ariel Gjaci	Università degli Studi di Genova, Italy
Chaitanya Ahuja	Meta AI, USA
Gustav Eje Henter	KTH Royal Institute of Technology, Sweden
Rajmund Nagy	KTH Royal Institute of Technology, Sweden
Youngwoo Yoon	Electronics and Telecommunications Research Institute, South Korea
Oya Celiktutan	King's College London, UK

W38 – BioImage Computing (BIC)

Organizers

Alexander Krull	University of Birmingham, UK
Peter Bajcsy	NIST, USA
Jan Funke	HHMI Janelia, USA
Dagmar Kainmueller	BIH/MDC, Germany
Khaled Khairy	St. Jude Children's Research Hospital, USA
Qingjie Meng	University of Birmingham, UK
Virginie Uhlmann	EMBL-EBI, UK
Martin Weigert	EPFL, Switzerland

W39 – Human-Inspired Computer Vision (HCV)

Organizers

Lucia Schiatti	Istituto Italiano di Tecnologia, Italy
Mengmi Zhang	Nanyang Technological University and A*STAR, Singapore
Yen-Ling Kuo	University of Virginia, USA
Vittorio Cuculo	University of Modena and Reggio Emilia, Italy
Martin Schrimpf	EPFL, Switzerland
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Tongxuan Tian	University of Virginia, USA
Francesca Pia Villani	Università Politecnica delle Marche, Italy
Christopher Wang	MIT, USA
Dongyang Xu	Tsinghua University, China
Ruizhe Zheng	Fudan University, China

W40 – Knowledge in Generative Models

Organizers

Anand Bhattad	Toyota Technological Institute at Chicago, USA
Xiaodan Du	Toyota Technological Institute at Chicago, USA
Nick Kolkin	Adobe, USA
Grace Luo	University of California, Berkeley, USA
Shuang Li	University of Toronto and Vector Institute, Canada
Greg Shakhnarovich	Toyota Technological Institute at Chicago, USA

W41 – Self-supervised Learning: What Is Next?

Organizers

Michael Dorkenwald	University of Amsterdam, Netherlands
Mert Bulent Sariyildiz	Naver Labs Europe, France
Christian Rupprecht	University of Oxford, UK
Hilde Kuehne	University of Bonn, Germany
Serge Belongie	University of Copenhagen, Denmark

W42 – Traditional Computer Vision in the Age of Deep Learning (TradiCV)

Organizers

Federica Arrigoni	Politecnico di Milano, Italy
Adrien Bartoli	Université Clermont Auvergne, France
Andrea Fusiello	University of Udine, Italy
Vladislav Golyanik	MPI Informatik, Germany
Zuzana Kukelova	Czech Technical University in Prague, Czech Republic
Luca Magri	Politecnico di Milano, Italy
Tomas Pajdla	Czech Technical University in Prague, Czech Republic
Shaifali Parashar	LIRIS, INSA-Lyon, France
Matteo Poggi	University of Bologna, Italy
Fabio Tosi	University of Bologna, Italy

W43 – Uncertainty Quantification for Computer Vision

Organizers

Andrea Pilzer	NVIDIA, Italy
Gianni Franchi	ENSTA Paris, France
Andrei Bursuc	valeo.ai, France
Arno Solin	Aalto University, Finland
Martin Trapp	Aalto University, Finland
Rui Li	Aalto University, Finland

Marcus Klasson	Aalto University, Finland
Gaëtan Brison	Hi!Paris, France
Angela Yao	National University of Singapore, Singapore
Ivor Simpson	University of Sussex, UK
Neill D. F. Campbell	University of Bath, UK

W44 – Emergent Visual Abilities and Limits of Foundation Models (EVAL-FoMo)

Organizers

Ashkan Khakzar	University of Oxford, UK
Boyi Li	University of California, Berkeley, USA
Francesco Pinto	University of Oxford, UK
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Ameya Prabhu	University of Tübingen, Germany
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Neil Chowdhury	OpenAI and MIT, USA
Aida Nematzadeh	Google DeepMind, UK

Senior Advisors

Trevor Darrell	University of California, Berkeley, USA
Bo Li	University of Chicago, USA
Philip Torr	University of Oxford, UK

W45 – Beyond Euclidean: Hyperbolic and Hyperspherical Learning for Computer Vision

Organizers

Aiden Durrant	University of Aberdeen, UK
Fabio Galasso	Sapienza University of Rome, Italy
Michael Kampffmeyer	Arctic University of Norway (UiT), Norway
Georgios Leontidis	University of Aberdeen, UK
Pascal Mettes	University of Amsterdam, Netherlands
Leyla Mirvakhabova	Qualcomm AI Research, USA
Adín Ramírez Rivera	University of Oslo, Norway

Indro Spinelli	Sapienza University of Rome, Italy
Stella Yu	University of Michigan, USA

W46 – Unlearning and Model Editing (U&ME)

Organizers

Diego Garcia-Olano	Meta AI, USA
Tal Hassner	Weir, USA
Iacopo Masi	Sapienza, University of Rome, Italy
Mayank Vatsa	Indian Institute of Technology, Jodhpur, India

W47 – Out-of-Distribution Generalization in Computer Vision Foundation Models

Organizers

Bingchen Zhao	University of Edinburgh, UK
Wufei Ma	Johns Hopkins University, USA
Artur Jesslen	University of Freiburg, Germany
Jiahao Wang	Johns Hopkins University, USA
Siwei Yang	University of California, Santa Cruz, USA
Sagar Vaze	University of Oxford, UK
Haoqin Tu	University of California, Santa Cruz, USA
Zijun Wang	University of California, Santa Cruz, USA
Xinyu Yang	CMU, USA
Huaxiu Yao	University of North Carolina, Chapel Hill, USA
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W48 – Visual Concepts

Organizers

Joy Hsu	Stanford University, USA
Yunzhi Zhang	Stanford University, USA
Jiayuan Mao	MIT, USA
Elliott Wu	Stanford University, USA
Daniel Cohen-Or	Tel-Aviv University, Israel
Jiajun Wu	Stanford University, USA

W49 – Sometimes Less Is More: Dataset Distillation Challenge

Chairs

Kai Wang	National University of Singapore, Singapore
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George Cazenavette	MIT, USA
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Samir Khaki	University of Toronto, Canada
Yuchen Zhang	National University of Singapore, Singapore
Tongzhou Wang	MIT, USA
Bo Zhao	Shanghai Jiao Tong University, China
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Yang You	National University of Singapore, Singapore

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Zhiwei Deng	Google Research, USA
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Yanqing Liu	National University of Singapore, Singapore
Yukun Zhou	National University of Singapore, Singapore
Jiawei Liu	National University of Singapore, Singapore
Junhao Zhang	National University of Singapore, Singapore
Songhua Liu	National University of Singapore, Singapore
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Angela Yao	National University of Singapore, Singapore
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Joey Tianyi Zhou	Centre for Frontier AI Research (CFAR), Singapore

W50 – Quantum Computer Vision and Machine Learning (QCVML)

Organizers

Jan-Nico Zaech	INSAIT, Switzerland
Tolga Birdal	Imperial College London, UK

Vladislav Golyanik	MPI, Germany
Michele Sasdelli	University of Adelaide, Australia

W51 – More Exploration, Less Exploitation (MELEX)

Organizers

Vasileios Belagiannis	Friedrich-Alexander-Universität Erlangen-Nürnberg, Germany
Azade Farshad	TU München, Germany
Angela Yao	National University of Singapore, Singapore
Gustavo Carneiro	University of Surrey, UK

W52 – Synthetic Data for Computer Vision

Organizers

Lucia Cascone	University of Salerno, Italy
Zilong Huang	TikTok, Singapore
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Generating Binary Species Range Maps

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Abstract. Accurately predicting the geographic ranges of species is crucial for assisting conservation efforts. Traditionally, range maps were manually created by experts. However, species distribution models (SDMs) and, more recently, deep learning-based variants offer a potential automated alternative. Deep learning-based SDMs generate a continuous probability representing the predicted presence of a species at a given location, which must be binarized by setting per-species thresholds to obtain binary range maps. However, selecting appropriate per-species thresholds to binarize these predictions is non-trivial as different species can require distinct thresholds. In this work, we evaluate different approaches for automatically identifying the best thresholds for binarizing range maps using presence-only data. This includes approaches that require the generation of additional pseudo-absence data, along with ones that only require presence data. We also propose an extension of an existing presence-only technique that is more robust to outliers. We perform a detailed evaluation of different thresholding techniques on the tasks of binary range estimation and large-scale fine-grained visual classification, and we demonstrate improved performance over existing pseudo-absence free approaches using our method.

Keywords: Species Range Maps · Thresholding · SINR

1 Introduction

Accurate range maps, depicting where species are present and absent, are indispensable tools for conservation. Statistics derived from their ranges are some of the key information used to estimate the threatened status of different species [39]. However, traditional expert-derived range maps are costly and time consuming to create. To alleviate this, advancements in statistics and machine learning have enabled the development of species distribution models (SDMs) that can estimate the probability of a species being present, among other quantities, at different locations [2]. Recently, deep learning-based multi-species SDMs,

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trained on crowd-sourced observation data, have been demonstrated to be capable of predicting the presence of thousands of different species simultaneously for an input location of interest [4,11,12,24,30,41]. However, to obtain a binary range map from these continuous probabilities, they need to be binarized using a threshold. Binary range maps are used in visualization in addition to being necessary for computing spatial metrics for determining the threatened status of a species.

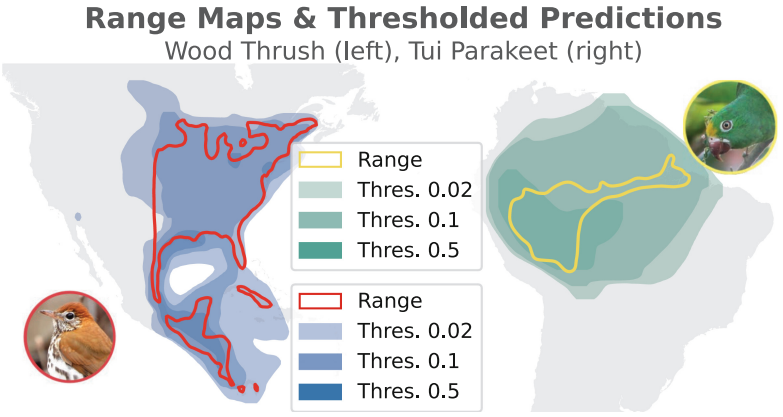

Fig. 1. Binary range maps for two different species. These ranges are generated by the SINR [11] species distribution model (SDM) for Wood Thrush (left) and Tui Parakeet (right), where the expert range map is denoted via solid outline. Converting the continuous SDM outputs to binary range maps requires setting thresholds (*e.g.* 0.02, 0.1, or 0.5) which result in very different range maps depending on the values chosen. More importantly, here the same threshold value is not the best for both species.

Finding the most effective threshold is a challenging problem, as the one that leads to the most accurate range maps will not necessarily be the same across different species (see Fig. 1). This is because the outputs of multi-species deep SDMs can be highly miscalibrated, which is likely the result of a number of factors such as imbalances in the species present in the training data, spatial biases, and issues stemming from species co-occurrences. The threshold selection task has been previously explored in the ecology literature under the setting where calibration data, which denotes a subset of locations where the species of interest is present or absent, is available [27]. However, the more challenging setting, and the one explored in this work, is where we do not have any information related to confirmed absences [29]. This presence-only data paradigm is most relevant to the opportunistic data available from large citizen science platforms such as iNaturalist [20], which contains over 200 million species observations to date.

A number of different thresholding approaches have been proposed for the case when only species presence observations are available [29]. However, many of

these approaches require the generation of 'pseudo-absences', *i.e.* hallucinated data that simulates locations where a species is said to not be found. As a result, the precise design choices made when generating these pseudo-absences can have a large impact on the accuracy of the resulting binary range maps. In this work, we compare these approaches to alternative thresholding techniques that do not require absence data of any form. Furthermore, we outline a simple extension of these absence-free techniques that obtain superior performance. Our resulting approach is conceptually simple and more efficient than the more computationally expensive pseudo-absence-based alternatives.

We make the following contributions: (i) We perform a detailed evaluation of multiple different thresholding techniques for generating binary species range maps. Our experiments are conducted using recent multi-species SDMs over a large set of expert-derived evaluation range data. (ii) We show that binary range maps can also be used as priors to improve large-scale fine-grained image classification and that they are superior to existing continuous priors. (iii) Finally, we introduce an extension of an existing thresholding approach that does not require any pseudo-absence data yet still outperforms existing presence-only methods on both range estimation and image classification tasks. Code to reproduce our experiments is available here: https://github.com/filipgdorm/binary_range_maps

2 Related Work

Species Distribution Models (SDMs). There is a large body of work on developing models to estimate the spatial and temporal distributions of species from raw observation data [2,13]. These models are typically trained using one of two different types of data: presence-only (PO) or presence-absence data. PO data consists of a set of observations (*i.e.* locations in space and time where a species has been observed), without any confirmed absences. PO data is significantly easier to collect compared to presence-absence data, requiring only the recording of locations where a species has been observed, without the need for extensive, resource-intensive surveys to confirm absences. As a result, PO data is abundant and widely available through online citizen science initiatives such as iNaturalist [20].

However, a challenge in training SDMs with PO data is the need for negative signal (*i.e.* species' absences) during training. Without having absences in the training data, SDMs can naively learn to simply predict presence, for every species, at every location. One common approach to overcome this issue is to create *pseudo*-absences (PAs), *i.e.* artificially generated data points representing species' absences [1]. Various approaches for generating PAs have been proposed, with popular techniques including selecting locations uniformly at random (*i.e.* 'random background points') or locations where other species have been observed (*i.e.* 'target background points') [5,37,38]. The effectiveness of different methods for generating PAs for training multi-species deep SDMs has recently been explored in [43].

Recently, [11] introduced a deep learning-based approach for modeling species' distributions, termed Spatial Implicit Neural Representations (SINRs). SINRs are parameterized as fully connected neural networks and can be trained on noisy crowd-sourced PO data. The authors demonstrated the ability to jointly estimate the ranges of thousands of species within a single model using PO training data. They introduced new benchmarks for evaluating the accuracy of predicted range maps by comparing model outputs to expert-derived ranges. They also quantified the improvement in fine-grained visual classification for different species when using SDMs as priors to modify image classifier predictions based on the location where the image was captured [30]. However, one of the limitations of [11]'s range estimation evaluation is that it uses a continuous, *i.e.* area under the curve, metric. As a result, the task of *binary* range estimation is not directly evaluated as the evaluation protocol is agnostic to the choice of threshold used to produce a binary range map. We use the models and data from [11] for our evaluation and demonstrate that the threshold required can vary significantly across species and how to select the best one is not obvious.

Binarizing Range Maps. The output of SDMs such as [11] are a set of continuous predictions representing species' presences at the evaluated locations. Binarizing these predictions to create binary range maps requires setting thresholds, which can result in very different maps depending on the thresholds selected. Furthermore, as seen in Fig. 1, the choice of best threshold can be species dependent. Without true absence data for calibration [15,27], the problem of converting continuous outputs to binary ranges, given only PO data, is challenging.

The problem of selecting appropriate thresholds for SDMs has been previously explored in the ecology literature [27–29]. However, these existing works tend to focus on an older generation of single species machine learning methods such as Maxent [36], Boosted Regression Trees [14], and Random Forests [6]. Additionally, these experiments are often limited to a small number of species, sometimes using only simulated data [29], and constrained to smaller geographic regions. Previous work has shown that fixed constant thresholds (*e.g.* 0.5) do not work well when the training data is imbalanced [17,22]. In the context of PO data, the impact of different thresholding techniques has been evaluated in [28,29]. Existing techniques can be broadly categorized based on the data they require, *e.g.* no data, presences, or presences and PAs. When no data is available, approaches such as choosing a fixed threshold or simply setting the threshold to be the average of the predictions can be used. Alternatively, if presence data is available, one approach is to choose the lowest/minimum presence threshold based on the SDM predictions associated with the presence observations in the training data [34,36]. In essence, this approach attempts to maximize recall by minimizing the number of false negatives. Another approach is to generate continuous predictions for all test locations (*i.e.* not just training presences) and choose the threshold that results in a fixed percentage of the lowest predictions being set as absences [34]. However, the selected percentage is heavily influenced by the ratio between presence and absence data. If one generates PAs, then it is possible to select the threshold that directly maximizes

the downstream evaluation metric of interest (*e.g.* F1 score) [29] or to achieve a desired target value (*e.g.* 95% sensitivity) [33]. However, these approaches are sensitive to mismatches between the distribution of true absences and generated PAs.

We build on the above existing work by extending the evaluation of thresholding techniques beyond simulation data, and instead use a recent dataset of globally distributed expert-derived range maps of 3,000 different species from [11]. Furthermore, our evaluation is based on recent multi-species neural-network-based SDMs that are jointly trained on PO data from thousands of species.

SDMs as Geo Priors for Vision. In addition to directly generating range maps, SDMs have also been shown to be helpful for assisting image classification. Several works have demonstrated that combining the probabilistic outputs of image classifiers with an estimate of the spatial distribution, *i.e.* as a 'geo prior', of the categories of interest can improve classification accuracy. Additional image metadata, which encodes the geographic location where each image was captured, can be used to estimate the spatial distributions of different classes. Once trained, the geo prior downweights the classifier predictions for categories that are unlikely to be present, according to the prior, at the location where the test image was taken. Different approaches have been explored in the literature, ranging from methods that spatially bin observations [3], ones that jointly train classifiers with additional location inputs [9,40], or methods that combine classifier predictions with separate spatial distribution models [30]. In our later experiments we show that binary range maps provide a small, but non-trivial, improvement in classification accuracy compared to existing approaches.

3 Method

Our goal is to generate binary species range maps. We begin with an overview of how SDMs can be trained to generate continuous range maps, and then introduce different approaches for converting these continuous predictions to binary ones.

3.1 Predicting Species' Presence

Recently [11] introduced a neural network-based approach, SINR, for predicting continuous range maps. Their model is trained simultaneously on thousands of different species from crowd-sourced presence-only data. At test time, the input to the model is a longitude-latitude tuple denoting a location of interest, and the output is a multi-label classification indicating the probability that each of several thousand different species is present there. Below we provide an explanation of how their model is parameterized and subsequently trained.

SINR denotes an input location by its longitude and latitude as $\mathbf{x}_i = [lon, lat]$. As in [30], these inputs are wrapped using a sinusoidal encoding to limit the impact of boundary effects. Other more complex input transformations are possible, but this type of wrapped input is still competitive on related spatial representation learning tasks [42]. While it is possible to use other input

features such as ones representing local environmental conditions, such as temperature and rainfall, in this work we mainly focus on location only models. The true absence (0) or presence (1) of S different species at $\mathbf{x}_i$ is denoted by the multi-label vector $\mathbf{y}_i \in \{0,1\}^S$. In practice, in the presence-only setting we do not have any training data that indicates if a species is absent. Instead, similar to [10], the observed training data is represented as $\mathbf{z}_i \in \{1, \emptyset\}^S$, where $z_{ij} = 1$ if a species is present and $z_{ij} = \emptyset$ if the presence or absence of the species is unknown. SINR models estimate $\mathbf{y}_i$ at any location $\mathbf{x}_i$ within a spatial domain $\mathcal{X}$, given observational training data $\{(\mathbf{x}_i, \mathbf{z}_i)\}_{i=1}^N$.

The range estimation model we wish to learn is denoted as $\hat{\mathbf{y}}_i = h_\phi(f_\theta(\mathbf{x}_i))$, where $f_\theta : \mathcal{X} \to \mathbb{R}^k$ is a location encoder parameterized by θ, and $h_\phi : \mathbb{R}^k \to [0,1]^S$ is a multi-label classifier parameterized by ϕ. The location encoder f_θ is a fully connected neural network with residual connections and the species classifiers in h_ϕ are simple per-species linear projections followed by sigmoid non-linearities. The prediction $\hat{\mathbf{y}}_i \in [0,1]^S$ represents a prediction of the probability of each species being present at location $\mathbf{x}_i$. The parameters of both components of the model can be estimated using stochastic gradient descent:

$$\theta^*, \phi^* = \underset{\theta,\phi}{\operatorname{argmin}} \frac{1}{N} \sum_{i=1}^{N} \mathcal{L}(\hat{\mathbf{y}}_i, \mathbf{z}_i), \tag{1}$$

where $\mathcal{L}$ is a suitable loss function and $\hat{\mathbf{y}}_i = h_\phi(f_\theta(\mathbf{x}_i))$ is the model's prediction. The output of this model is thus a probability distribution indicating species' presence, or absence, over a location of interest.

Training Loss. [11] evaluate different loss functions $\mathcal{L}$ for training SINR models. At their core, these losses amount to different ways of addressing the lack of true absence data. To address this, different mechanisms for generating pseudo-absences (PAs) are explored. For example, taking inspiration from the task of single positive multi-label learning [10], the heuristic that all unobserved labels are negative is used. This assumption also holds for species range estimation because, given an arbitrary location, most species will not actually be present there. Here we focus on their best performing approach, 'Full Assume Negative' ($\mathcal{L}_{\text{AN-full}}$), which is inspired by losses introduced in [30]. We describe the other losses in Appendix C.

For the $\mathcal{L}_{\text{AN-full}}$ loss function, PAs are selected in two different ways. First, each species presence is paired with a PA at a location $\mathbf{r}$ that is sampled uniformly from the surface of the earth, *i.e.* $\mathbf{r} \sim \text{Uniform}(\mathcal{X})$. Second, each species observation is also paired with a PA at the same location for a *different* species, which generates PAs that align with the distribution of the data. The final loss is thus represented as:

$$\mathcal{L}_{\text{AN-full}}(\hat{\mathbf{y}}_i, \mathbf{z}_i) = -\frac{1}{S} \sum_{j=1}^{S} \left[\mathbb{1}_{[z_{ij}=1]} \lambda \log(\hat{y}_{ij}) + \mathbb{1}_{[z_{ij}\neq 1]} \log(1 - \hat{y}_{ij}) + \log(1 - \hat{y}'_j) \right], \tag{2}$$

where $\hat{y}_{ij}$ is the predicted probability of species j being present, z_{ij} indicates the presence or absence of species j, $\lambda > 0$ is a hyperparameter used to prevent

PAs from dominating the loss, and $\hat{\mathbf{y}}' = h_\phi(f_\theta(\mathbf{r}))$ are the predicted probabilities of the species of interest being present at a randomly chosen location $\mathbf{r}$. This loss is efficient in that it provides a training signal for all entries in $\hat{\mathbf{y}}$ for each training example, where a positive signal is provided for the observed present species and a negative signal from PAs is provided for all other species.

3.2 Generating Binary Range Maps via Thresholding

Thresholding techniques aim to estimate species-specific thresholds that lead to the most accurate binary range maps. Thresholds are applied after the SDM has been trained and after the predictions for each location have been generated. Once a threshold is identified, all predictions above or equal to it are marked as presences, and all below as absences. Next we outline different threshold generation techniques.

Single Fixed Threshold. This uses a single constant threshold (*e.g.* 0.5) across all species [26,31,32]. If the model is well calibrated, this should be sufficient.

Single Best Threshold. This method also uses a single threshold, but unlike the previous approach here we use the presence and absence data from the expert ranges in the evaluation data to select the threshold that leads to the highest mean F1 score. This threshold is selected by generating 20 linearly spaced candidate thresholds in the range 0–1 and choosing the best one.

Random Sampling. When true absences are not available, one common choice is to generate pseudo-absences (PAs) [11,43]. This method takes presence observations from the *training* data and generates randomly distributed PAs for each species where it has not been observed and selects per-species thresholds that maximize the mean F1 score on this data. This is achieved by binning the observations into H3 [19] cells of resolution four. The cells chosen as presences are those where the species has been observed, the absences are randomly sampled from the rest of the cells. In our later evaluation, we explore the impact of varying the ratio between the true presence and generated PA data.

Target Sampling. To match the distribution of the training data, this approach generates PAs in locations where other species have been observed, but the species of interest is absent [11,43]. We use H3 [19] cells of resolution four to bin the presence data, where cells with at least one observation of a species are counted as presences. All cells where there are at least N_t observations across all species, but the species of interest is not present, are counted as PAs. The value of N_t is calculated by first identifying all the cells where the species of interest has been observed at least once. These cells are ranked based on the ratio of the total number of observations in each cell to the number of observations specifically for the species of interest within that cell. N_t represents the 95th percentile value of this ratio, indicating the minimum number of total observations in a cell needed to be reasonably confident that the species of interest is not present in that cell. Given the resulting presence and PA data, we can select per-species thresholds that maximize the F1 score.

Threshold Classifier. For this method we introduce a novel supervised machine learning approach to predict the best threshold per-species. For the input features for each species we use the final learned species embedding vector from [11], where the assumption is that species with similar embeddings will require similar thresholds. For the output of the learning algorithm, we bin the thresholds into 20 possible values, evenly distributed between 0-1, and treat it as a classification problem. 75% of the 2,418 IUCN species are used to train the classifier and the remaining 25% are used for evaluation. As a result, the final scores for this approach are not directly comparable to the other techniques. We explore two different variants of this approach: a Random Forest (RF) classifier which uses the default values from scikit-learn [35] or a Multilayer Perceptron (MLP) with two hidden layers, containing 100 neurons each.

Mean Predicted Threshold. Following the approach outlined in [29], we select the threshold based on the average of the SDM predictions. Specifically, for each species, we generate predictions across all locations. Subsequently, we determine the threshold by calculating the mean predicted value from a randomly selected subset of these locations.

Lowest Presence Threshold (LPT). This approach extracts the continuous predictions for a species of interest associated with each presence observation from the training data and selects the threshold as the lowest value from this set [34]. This then ensures that the resulting binary range is guaranteed to encompass each of the presence observations.

Lowest Presence Threshold - Robust (LPT-R). This method builds on the previous `LPT` approach [34], but we extend it to make it more robust to outliers. The motivation is that `LPT` is highly susceptible to outliers as inaccurate observations or 'vagrants' (individuals lost outside their ranges [25]) can massively bias the threshold selection. So instead of choosing a threshold that includes all training presences, we instead select one that encompasses the vast majority of them. Specifically, we perform the same steps as `LPT` but set the threshold by sorting all the predicted continuous scores and choosing it to be equal to the 5th percentile. Figure A5 in the appendix illustrates a visual example of this method.

4 Implementation Details

4.1 Datasets

Presence-Only Training Data. Our training data is publicly available crowdsourced occurrence data from iNaturalist [20]. The original raw data contains annotated images of species with geographic locations and timestamps. We use the same subset of 35.5 million total observations containing 47,375 species from [11]. Here, observations are of 'research grade' status, meaning that there is a higher confidence that the classifications are correct. While iNaturalist offers

numerous advantages, it also has limitations. For instance, the data contained within is biased towards the Global North.

Presence-Absence Evaluation Data. For evaluation, we require data that contains both confirmed presence *and* absence observations distributed across the globe. The S&T and IUCN datasets from [11] satisfy these requirements. The *eBird Status and Trends* (S&T) dataset covers 535 bird species biased to North America [16]. This dataset contains estimated relative abundance maps which have been converted into binary range maps. The second dataset consists of ranges of 2,418 species from the *International Union for Conservation of Nature* (IUCN) [21]. In contrast to the S&T dataset, this dataset is more taxonomically diverse and more globally distributed. The two evaluation datasets are represented as H3 [19] cells at resolution five, which corresponds to 2,016,842 evaluation locations distributed across the entire globe.

Image Classification Evaluation Data. In addition to range estimation, we also evaluate different models as geographic priors on the task of assisting image classification predictions as in [3,11,30]. We again use the same dataset from [11], which contains 282,974 images from iNaturalist, which represents a subset of 39,444 species from our training set. The evaluation metric for this task is the top-1 image classification accuracy resulting from combining, in our case binary, range predictions with the probabilistic classification outputs of an Xception classifier [8] trained on images from iNaturalist.

4.2 Evaluation Metrics

Continuous range predictions can be evaluated using area under the curve-based metrics such as average precision. These scores can then be averaged across different species to come up with a single final score as in [11]. However, our goal is to evaluate binary range maps, not continuous predictions. Thus after thresholding a continuous probability of presence output from a model, we consider all locations above the threshold as representing predicted presences and all below as predicted absences. Evaluation is performed using the F1 score, which combines precision and recall into a single value and is particularly useful in binary classification problems where there is an imbalance between the classes, which is the case here as the presences are typically outnumbered by absences. The F1 score is computed per-species after binarization:

$$F_1 = 2 \cdot \frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}} = \frac{\text{TP}}{\text{TP} + \frac{1}{2}(\text{FP} + \text{FN})}, \tag{3}$$

where TP denotes True Positives, FP False Positives, and FN False Negatives. The final reported mean F1 score is the average per-species F1 score across the species in each of the respective evaluation sets. Unless stated otherwise, the held-out evaluation datasets are completely detached from any threshold identification methods to ensure that there is no data leakage in the process.

5 Experiments

In this section we evaluate different techniques for generating binary range maps. Unless specified otherwise, experiments are carried out with the $\mathcal{L}_{\text{AN-full}}$ SDM from [11] capped at 1000 training samples per-species without environmental input features. The model's predictions are binarized using different thresholding techniques and then evaluated by computing the mean F1 score against the held-out test set. This score is presented together with information about how far the performance is from the upper bound. We also provide additional quantitative results, including different input features, and analysis in Appendix A.

5.1 Binarizing Range Maps

Evaluating Different Thresholding Techniques. Table 1 presents the mean F1 score for a wide set of binarization techniques when evaluated on the IUCN dataset from [11]. The robust Lowest Presence Threshold (`LPT-R`) method is the highest-performing technique, closely followed by the `Target Sampling` approach. Both the `LPT` and `LPT-R` methods have the advantage that they do not require any pseudo-absences to be defined. Neither `Threshold Classifier` techniques work particularly well here when predicting thresholds, indicating that

Table 1. Binary range estimation performance of different thresholding techniques. Here we report the average mean F1 score for five different random initializations of the $\mathcal{L}_{\text{AN-full}}$ SDM on the IUCN evaluation set, where the upper bound is 67.2%. (†) denotes threshold classifier scores which are computed on a 25% subset of the full evaluation set, as they are trained on the rest. (✓) indicates whether a thresholding technique uses true absences (TA), pseudo-absences (PA), or one single overall threshold (ST). Bold entries indicate best methods, and underline are second best.

Thresholding Method	ST	PA	TA	↑ Mean F1	↓ Upper Bound Δ
Threshold Classifier - RF †			✓	55.4	−11.8
Threshold Classifier - MLP †			✓	56.3	−10.9
Single Fixed Threshold - 0.5	✓			40.2	−27.0
Single Best Threshold - 0.1	✓		✓	55.2	−12.0
Random Sampling - #Absences=#Presences		✓		57.6	−9.6
Random Sampling - #Absences=5#Presences		✓		58.1	−9.1
Random Sampling - #Absences=10#Presences		✓		56.9	−10.3
Random Sampling - 100 Absences		✓		57.3	−9.9
Random Sampling - 1000 Absences		✓		57.6	−9.6
Random Sampling - 10000 Absences		✓		43.9	−23.3
Target Sampling		✓		<u>59.7</u>	<u>−7.5</u>
Mean Predicted Threshold				37.7	−29.5
Lowest Presence Threshold (LPT)				54.3	−12.9
Lowest Presence Threshold - Robust (LPT-R)				**60.8**	**−6.4**

the relationship between SDM species' weights and final thresholds is not very effective for this type of model. Still, these methods, along with the vast majority of techniques, work better than using one `Single Fixed Threshold` of 0.5 for binarization. The main exception to this is the `Mean Predicted Threshold` technique which performs worse than this baseline. Unlike [29], which highlights this approach as viable in the PO setting, our evaluation on a larger set of globally distributed species indicates that it is not very effective. Generally the SDM outputs are very small because most species are absent from most places. Thus, selecting the threshold by averaging predictions results in a small threshold.

Qualitative examples of the resulting binary ranges for several species are depicted in Fig. 2. This figure displays a representative set of species and highlights how the two best-performing methods compare to expert-derived range maps. In many cases, such as the *Wood Thrush* and *Tui Parakeet*, the methods perform similarly. The former case is an example of a North American bird, a species that has abundant PO training data. The latter is a South American species with much less training data, but where the two methods yield almost identical threshold and range maps. The *Tolai Hare* and *Eastern Green Snake* are both examples where the `LPT-R` approach generates much more accurate ranges. Both of these species are in regions with less training data and have large ranges, as seen in expert ranges on the left. In these examples, the `Target Sampling` greatly underestimates the ranges. This seems to often be the case for species with large ranges and few training samples. However, for species like the *Yosemite Toad* with few training samples and a small range, the `Target Sampling` approach's more conservative range estimation appears to work better.

Evaluating Different SDMs. Next, in Table 2 we evaluate the best-performing techniques more using different input SDMs and additional datasets to quantify their robustness. Here, the different SDMs correspond to training SINR models with one of three different loss functions. We include results on the smaller S&T dataset from [11], which is biased toward birds from North America with larger ranges. As can be seen in the results, for the more challenging IUCN dataset, the `LPT-R` approach outperforms the other methods for all three input SDMs. The `Target Sampling` approach is the second best-performing method for $\mathcal{L}_{\text{AN-full}}$ and $\mathcal{L}_{\text{AN-SSDL}}$, while being the second worst method for $\mathcal{L}_{\text{AN-SLDS}}$. The performance gap is at most a few percentage points, but considering that the `LPT-R` approach foregoes several computation steps associated with PA generation used by the other methods, this highlights its utility. As `LPT-R` performs well on the IUCN dataset, it perhaps implies that it may work better for a wider set of species, even those that are less prevalent on iNaturalist. This is important, given that these types of species are often the ones that require monitoring.

For the S&T evaluation dataset, the `Target Sampling` approach generally performs best. However, for the $\mathcal{L}_{\text{AN-SLDS}}$ SDM, the performance of the different top techniques is almost identical. As noted earlier, the S&T dataset has a North American bias, which is also where most of the iNaturalist training data is from. Similar to the findings in [11], the $\mathcal{L}_{\text{AN-SLDS}}$ SDM performs well on the S&T

dataset, which is likely due to the fact that this loss function uses a version of target background sampling when generating its pseudo-absences which will also result in a bias towards North America in the context of this dataset.

Table 2. Binary range estimation across different input SDMs. Here we report the mean F1 score for three different input SDMs, where each is trained with a different loss function. The reported scores are the result of five repeated runs, with different initializations, for each SDM. We also report results for an ensemble of the five different $\mathcal{L}_{\text{AN-full}}$ SDMs, where we average the model predictions before thresholding.

		S&T		IUCN	
Model	Method	↑ Mean F1	↓ Upper Bound Δ	↑ Mean F1	↓ Upper Bound Δ
$\mathcal{L}_{\text{AN-full}}$	Random Sampling	67.4	−8.7	57.6	−9.6
	Target Sampling	**70.5**	**−5.6**	59.7	−7.5
	LPT	60.0	−16.1	54.3	−12.9
	LPT-R	69.1	−7.0	**60.8**	**−6.4**
$\mathcal{L}_{\text{AN-SSDL}}$	Random Sampling	57.8	−11.9	50.5	−9.6
	Target Sampling	**64.1**	**−5.6**	53.1	−7.0
	LPT	58.2	−11.5	49.9	−10.2
	LPT-R	61.3	−8.4	**53.5**	**−6.6**
$\mathcal{L}_{\text{AN-SLDS}}$	Random Sampling	**71.1**	**−5.1**	38.9	−8.5
	Target Sampling	71.0	−5.2	37.4	−10.0
	LPT	59.5	−16.7	30.0	−17.4
	LPT-R	**71.1**	**−5.1**	**39.4**	**−8.0**
Ensemble $\mathcal{L}_{\text{AN-full}}$	Random Sampling	67.5	−9.4	58.3	−10.0
	Target Sampling	**71.0**	**−5.9**	60.5	−7.8
	LPT	62.8	−14.1	56.2	−12.1
	LPT-R	70.0	−6.9	**61.7**	**−6.6**

Ensembling SDMs. Finally, in the bottom rows of Table 2 we study the impact of ensembling five different random initializations of the $\mathcal{L}_{\text{AN-full}}$ SDM. This ensemble was used for both threshold identification and, subsequently, evaluation on both S&T and IUCN datasets. The ensemble results in more accurate range maps compared to a single model, as illustrated by the higher mean F1 score for both datasets. We also observe a reduction in the difference between `LPT-R` and `Target Sampling` on S&T as a result of ensembling.

5.2 Geo Priors for Image Classification

SDMs can also be used as 'geo priors' to assist image classifiers by combining their outputs with a classifier's probabilistic predictions. Here, we conduct an evaluation similar to the one carried out in [11], with the main difference

being that with binary range maps, locations outside a species range would be downweighted to zero, *i.e.* the SDM would prevent the classifier from predicting that species. We use the image classification dataset from [11], which consists of 282,974 images of different species collected from iNaturalist, covering 39,444 species from the PO training set. Each image is accompanied by the latitude and longitude at which it was taken. In Table 3, the results of using different binary range maps are presented. Using a geo prior to weight predictions increases the Top-1 accuracy in all experiments compared to the vision-only prediction model (75.4%), except when binarizing with a constant threshold of 0.5 for all species. As previously established, different species require different thresholds, and a threshold of 0.5 is generally too high. This results in most species being incorrectly marked as absent from nearly all locations, which degrades model performance. However, setting a lower threshold of 0.1 for binarizing all species significantly improves accuracy, and outperforms the vision-only model. This shows that the vision model benefits from a geo prior, even if the range maps are not very accurate when making predictions.

Switching to more refined methods of setting species-specific thresholds improves the performance even further. Out of all methods, the `Target Sampling` and `LPT-R` approaches work best. However, Table 3 shows that this performance is lower than when using the raw prediction scores without binarizing them, *i.e.* the original continuous outputs of the SDM (81.6%). We hypothesize that the continuous scores allow the model to express uncertainty about a species being at a given location rather than choosing between the two extremes of classifying as either present or absent. To address this, we add a small constant $+\delta$ to the predicted absences after thresholding when using them as priors, *e.g.* 0.01. In all cases, adding this constant improves the performance of all binarization techniques as it prevents them from saying that a species is never present at

Table 3. Image classification with geo priors using different thresholding techniques. Here we use a $\mathcal{L}_{\text{AN-full}}$ SDM as a geo prior to assist image classification. The results are the average of five different random initializations of the SDM. '$+\ \delta$' indicates that a small constant is added after thresholding to ensure that it is not possible for a thresholding technique to predict that a species is never present.

Thresholding Method	Top-1 Acc.	
		$+\delta$
No Prior (*i.e.* vision-only)	75.4	-
Continuous - No Threshold	81.6	81.4
Single Fixed Threshold - 0.5	52.7	75.7
Single Fixed Threshold - 0.1	77.6	81.2
Single Fixed Threshold - 0.01	80.9	81.3
Random Sampling	78.5	81.5
Target Sampling	80.1	<u>81.7</u>
LPT	80.4	80.8
LPT-R	80.0	**81.8**

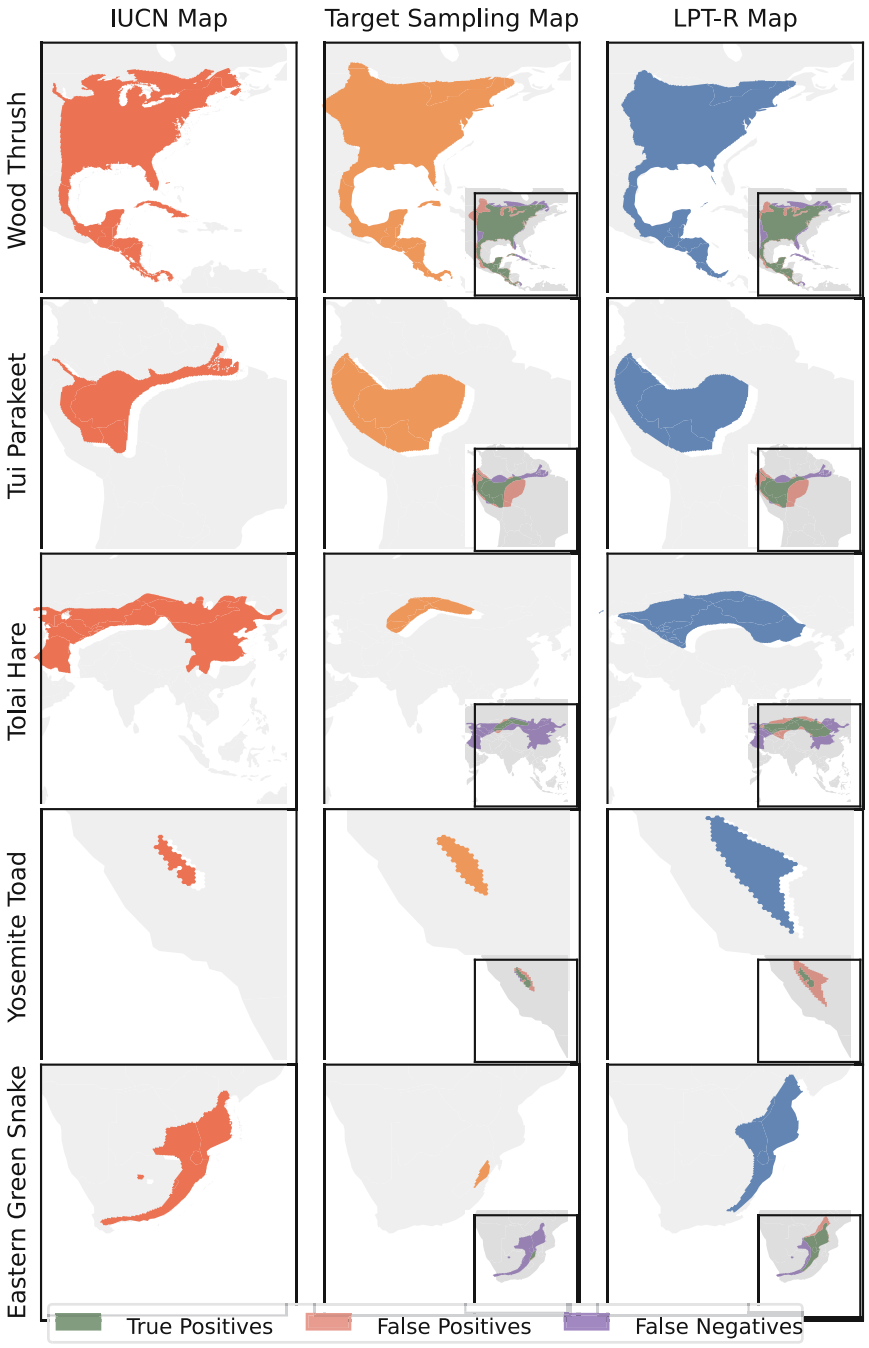

Fig. 2. Qualitative examples of estimated binary ranges. Each row depicts a different species, and the columns show the expert-derived ranges and the outputs from the `Target Sampling` and `LPT-R` approaches, respectively. Inset, we also display the different types of errors. We use an ocean mask for visualization purposes.

a location. As a result, the binarized range maps generated through both `Target Sampling` and `LPT-R` outperform the continuous baseline.

5.3 Limitations

Despite the promise of our work, there are still some important limitations to note. First, the training data contains biases, both spatial biases at the national (*e.g.* Europe and North America are disproportionately represented) and local scale (*e.g.* it is biased towards locations that are easier for people to get to), in addition to some species being much more prevalent than others. Spatial biases in citizen science data have been well characterized in the literature [18] and various approaches have been proposed to mitigate it [7,23]. We leave the investigation of such methods for future work. Another data related issue is that the expert-derived range maps in the test set can contain errors or simply be outdated [11]. However, to the best of our knowledge the test set used represents the best source of large-scale global evaluation data available. The main results in this work use species distribution models that only use coordinate features as input. Results can change if additional environmental covariates are used as input, see Appendix A. We also do not perform any regularization on the binary maps to make them more spatially contiguous. This could be investigated in future work, although we note that our predictions are typically already contiguous. Finally, caution should be exercised when using the resulting binary range maps in conservation or assessment applications as the results demonstrate that performance is still likely insufficient for many species.

6 Conclusion

We explored the problem of automatically converting continuous species distribution model outputs into binary range maps via thresholding. Through detailed evaluation, we compared the performance of multiple different thresholding techniques, using different underlying models, across a range of different species. We also proposed an extension of an existing method, which we call `LPT-R`, and demonstrate that leads to more accurate binary range maps in a number of cases. Our approach circumvents the need to create pseudo-absences, making it computationally cheaper while also yielding more accurate ranges. Additionally, we showed that binary range maps can also be used as geographic priors to improve fine-grained image classification accuracy. A possible future extension to our work would be to leverage temporal data to generate time-conditioned range maps, which could significantly enhance our understanding of distribution changes over time.

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WildFusion: Individual Animal Identification with Calibrated Similarity Fusion

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Abstract. We propose a new method – WildFusion – for individual identification of a broad range of animal species. The method fuses deep scores (e.g., MegaDescriptor or DINOv2) and local matching similarity (e.g., LoFTR and LightGlue) to identify individual animals. The global and local information fusion is facilitated by similarity score calibration. In a zero-shot setting, relying on local similarity score only, WildFusion achieved mean accuracy, measured on 17 datasets, of 76.2%. This is better than the state-of-the-art model, MegaDescriptor-L, whose training set included 15 of the 17 datasets. If a dataset-specific calibration is applied, mean accuracy increases by 2.3% points. WildFusion, with both local and global similarity scores, outperforms the state-of-the-art significantly – mean accuracy reached 84.0%, an increase of 8.5% points; the mean relative error drops by 35%. We make the code and pre-trained models publicly available, enabling immediate use in ecology and conservation (https://github.com/WildlifeDatasets/wildlife-tools).

1 Introduction

Identifying individual animals is essential in various domains of wildlife research. It help us understand the complexities of species dynamics [46,62], which is necessary for developing efficient conservation strategies. Besides, it can improve the accuracy of population density estimation, which is important in problems like disease monitoring and control [45], the role of the animal in the ecosystem [51], monitoring invasive species [11] and measuring the involvement of humans in the animal's habitat and ecological restoration [8]. Accurate identification requires domain knowledge and is extremely time-consuming due to the need for manual data processing. Therefore, considerable progress has been made in the development of methods for automating this process. Even though identifying animal individuals from images is challenging, machine learning and computer

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vision methods applied to species with unique patterns already enhance ecological research [67]. The automation of animal re-identification is typically based on (i) *deep learning*, (ii) *local feature matching*, or (iii) *species-specific methods*.

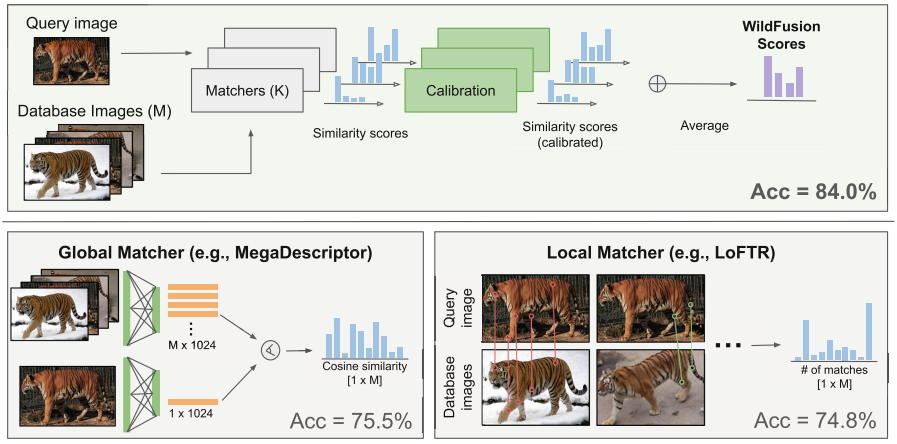

Fig. 1. Calibrated similarity fusion. Fusing local (in the $[0, \mathcal{R}]$ range) and global matching scores (e.g., cosine similarity) is not possible without calibration. By calibrating the outputs of any local and global matcher, we can easily fuse them and achieve better performance. In terms of accuracy and evaluated on 17 datasets, we increased the performance by 8.5% on average and reduced relative error by 35%.

The *deep learning*-based approaches [10,16,23,34,40,61] use either a standard classification-like approach or metric learning. Even though those approaches perform well, the models need relatively large annotated datasets, and fine-tuning requires considerable computational resources. On the other hand, methods based on *local descriptors* (e.g., SIFT [39], SuperPoint [18]) can be used without fine-tuning. [6,22,47,50]. Indeed, the overall accuracy of matching local descriptors does not achieve a performance of deep learning methods [13], but those approaches are still very popular due to the existence of open-source tools (e.g., HotSpoter, WildMe) that are based on local descriptors. Additionally, matching requires a pairwise comparison between all query and database samples. As the identity database grows in size, the computational time quickly becomes unfeasible; therefore, local feature matching remains a viable option only for moderately sized datasets. The *species-specific* methods are usually tailored to suit species without any visual characteristics [5,7,28,31,65]. Existing methods focus, for example, on the shape of an elephant's ear, the facial characteristics of primates, or the fluke shape of whales. However, due to their idiosyncratic nature, these methods are difficult to transfer to other species.

In light of that, we propose WildFusion, a new state-of-the-art approach to zero-shot animal re-identification. It fuses calibrated deep similarity functions (i.e., MegaDescriptor and/or DINOv2 feature similarity) and local matching

scores (i.e., number of matches from descriptors such as LoFTR and LightGlue) to select an identity from a database. For reference, see the illustration in Fig. 1. With this straightforward approach, WildFusion significantly outperforms the current state-of-the-art without domain adaptation or fine-tuning.

The Main Contributions of This Paper are:

- A new ensembling framework (WildFusion) that allows a combination of deep- and local-feature matching scores.
- A state-of-the-art performance on a set of animal identification problems, outperforming current methods by 8.5% on average; measured on 17 datasets.
- Comprehensive evaluation of selected state-of-the-art deep-learning and local feature-matching methods for image-matching and animal re-identification.
- Showing that WildFusion works without the need for fine-tuning and provides state-of-the-art performance out of the box, even in a zero-shot setting.

2 Related Work

Local Feature Matching Methods: Early work on animal re-identification used hand-crafted features such as cheetah's spots [31] or zebra's stripes [33]. Since these approaches suffer from poor performance and are non-transferable to other species, methods extracting local patterns such as SIFT [39] or ORB [52] were soon widely used. They extract descriptors from a database of images and match the descriptors from image pairs. Popular SW, e.g., WildID [9], HotSpotter [15] and I^3S are using such approach for years. Recently, the focus moved to local features extracted by deep networks such as ALIKED [71], DISK [60] or SuperPoint [18]. The classical matching of local descriptors could be simply replaced by deep methods such as LightGlue [36], Superglue [53], and LoFTR [57] that allow both extracting and matching of the local features. These matching methods return potential matches and their confidence scores. They require manual thresholding to determine which features are matched. In animal re-identification, deep local features are slowly coming into focus; for example, [48] used a combination of the SuperPoint features with the SuperGlue matching.

Deep Embedding Methods: The applications of deep methods in animal re-identification are relatively new [12]. The simplest use case consists of extracting embeddings from a neural network and feeding them to an SVM classifier [14,32,41]. This approach has low computational demands, but the network cannot be fine-tuned. Another simple approach involves fine-tuning a pre-trained neural network [23,54]. These approaches usually require a fixed number of classes (individuals), which is not realistic. For this reason, metric learning methods (e.g., ArcFace [13], Siamese networks [30], and Triplet loss [19,40]) became popular. Instead of classifying images into a pre-determined set of classes, they measure differences between images and are, therefore, able to generalize into new individuals. Another approach is to use publicly available large-scale, foundational models pre-trained on large datasets such as BioCLIP

[56], DINOv2 [44], and MegaDescriptor [13]. Since these models are primarily designed for general computer vision tasks, they are not adapted for the nuances of wildlife re-identification, which heavily relies on fine-grained patterns. This was addressed by MegaDescriptor [13], the Swin-based foundational model for animal re-identification that was trained on over 30 datasets (collected using WildlifeDatasets) using ArcFace loss [17].

Species-specific Methods are usually tailored to a particular species or closely related species and involve pre-processing steps such as extracting patches from regions of interest or accurately aligning images. Besides, they are not transferable to other species. Examples of these methods are Amphident [21] which find matching pixels within newt patterns or CurvRank v2 [42] and finFindR [66], which match the fin curvatures to identify mantas, dolphins, or whales.

3 Methodology

In this section, we describe the similarity scores based on deep embeddings and local feature matching and introduce the proposed WildFusion, a method for finding an image from a database of images $x_1, \ldots, x_D$ closest to query image x_q.

Note: For wildlife re-identification, we employ a standard setting inspired by practical applications in animal ecology, widely used in automated animal re-id studies [4,13]. This setting corresponds to the image retrieval problem, where the goal is to find the most visually similar images (whose identity is used as prediction) in the database for a given query image based on a similarity metric.

3.1 Global Similarity Score

Given an image x, we use a neural network $f(x)$ to extract a fixed-length embedding. The network $f(x)$ is a complex function that maps images into embedding space where the representations of images depicting the same animal are closer together, while those of different individual animals are distinctively separated. Common architectures of neural networks include convolutional [38,68] or transformer-based [20,37] architectures and are often trained with metric learning, e.g., ArcFace [17] and Triplet loss [55], to promote separability in the embedding space. The similarity between images is calculated as the similarity between their representation in the embedding space. Formally, we define the *global similarity* between two images x_0 and x_1 as the cosine similarity between their corresponding deep embeddings extracted by neural network f:

$$s_G(x_0, x_1) = \frac{f(x_0) \cdot f(x_1)}{\|f(x_0)\| \|f(x_1)|}. \tag{1}$$

3.2 Matching Based Similarity Score

We derive a similarity metric based on local feature matching as the number of found significant matches. The feature matching methods return a list of

potential matches and their confidence score. We declare a match to be significant if its confidence score is above some threshold μ. Formally, given two images x_0 and x_1 with the number of matches $M(x_0, x_1)$ with confidence scores $c_m(x_0, x_1)$, we define the *local similarity* metric as

$$s_L(x_0, x_1) = \sum_{m=1}^{M(x_0,x_1)} I\big(c_m(x_0, x_1) > \mu\big), \tag{2}$$

where I is the counting (0/1) function. This approach requires the tuning of the thresholding hyperparameter μ, where large values of μ allow for a small number of high-quality matches, while low values of μ result in a larger amount of matches with potentially lower quality. As we empirically show in Sect. 6.1, all considered feature matching methods are robust to μ selections with $\mu = 0.5$ being reasonable choice in most scenarios.

3.3 Score Calibration

Calibration refers to rescaling model outputs so that they could be interpreted probabilistically. In our case, it is required to normalize the outputs of multiple models to the common range $[0, 1]$. Predictions of well-calibrated model reflect confidence in the given class predictions [27].

We apply calibration to the predicted similarity scores. The similarity scores are used for comparison and ranking in image retrieval, with the magnitude of the scores having no direct interpretation. We use calibration to ensure that the predicted similarity score corresponds to the probability that the images in a pair have the same identity. We construct calibrated scores from either global or matching-based scores using a calibration function $f_{\text{cal}} : \mathbb{R} \to [0, 1]$.

$$\hat{s}(x_0, x_1) = f_{cal}(s(x_0, x_1)). \tag{3}$$

A common approach for constructing f_{cal} is Platt scaling [49], which involves fitting a single-variable logistic regression with uncalibrated scores as inputs. Another widely used method is isotonic regression [70], a variant of binning regression with a monotonicity constraint. Given uncalibrated scores, it learns a non-decreasing piecewise constant function. However, for our application in ranking and image retrieval, we require a strictly increasing function to handle ties in scores. To achieve this, we first apply the isotonic regression and second interpolate the bin centers using a Piecewise Cubic Hermite Interpolating Polynomial (PCHIP) [25], which performs cubic interpolation while preserving monotonicity. This procedure results in the required strictly increasing calibration function.

3.4 WildFusion - Calibrated Score Ensembling

To construct an ensemble, we consider K models with similarity scores $s_i(x_0, x_1)$ for a pair of images (x_0, x_1). The calibrated scores $\hat{s}_i(x_0, x_1)$ are interpreted as

estimates of same true probability $P(\mathrm{id}(x_0) = \mathrm{id}(x_1) \mid x_0, x_1)$. To denoise the prediction, we assume the probabilities to be independent observations with additive noise. The WildFusion score is thus a weighted average of n calibrated scores:

$$s_F(x_0, x_1) = \sum_{i=1}^{K} w_i \hat{s}_i(x_0, x_1), \tag{4}$$

where the weights should reflect the variance of the additive noise in the score. If we assume that all scores have similar variance, equal weights $w_i = \frac{1}{n}$ are selected, and the weighted average reduces to the simple average.

Fig. 2. Distinct animal features for re-identification. Based on the natural visual appearance, the most distinguishable features for animals are spots, stripes, facial landmarks, and the shape of body parts (e.g., ears for elephants and fin for whales).

4 Datasets

We evaluate WildFusion on 17 datasets[1] that include diverse species of animals. The datasets were acquired with the help of the recently developed library Wildlife Datasets [13], which allows easy access to the datasets and provides unified dataset splits. We selected subsets of the datasets that are saturated in performance or have a large number of images. Sample images from selected datasets are shown in Fig. 2. For basic statistics of the datasets, see Table 1.

To construct the appropriate training (*database*) and test (*query*) datasets, we followed the methodology proposed in [13]. However, while analyzing this procedure, we discovered inconsistencies caused by the incorrect loading of images for ATRW and NDD20 datasets due to multiple identities in one image. For these datasets, we fixed the loading by applying the appropriate bounding box or segmentation mask. Therefore, as loaded images are not exactly the same, the achieved results for these two datasets are higher than in the original work.

[1] For zero-shot, we use only two that were not used in the MegaDescriptor training.

Table 1. Characteristics of selected datasets. †Used in zero-shot scenario.

	category	# of images	# of individuals
ATRW [34]	tigers	5,415	182
CowDataset† [26]	cows	1485	13
GiraffeZebraID [47]	giraffes, zebras	6,925	2,056
Giraffes [40]	giraffes	1,393	178
HyenaID2022 [59]	Hyenas	3,129	256
LeopardID2022 [59]	leopards	6,806	430
NyalaData [19]	nyalas	1,942	237
SealID [43]	seals	2,080	57
SeaStarReID2023† [63]	starfish	2187	95
SeaTurtleID [4]	sea turtles	7,774	400
WhaleSharkID [29]	whale sharks	7,693	543
ZindiTurtleRecall [3]	sea turtles	12,803	2,265
BelugaID [2]	belugas	5,902	788
CTai [24]	chimpanzees	4,662	71
IPanda50 [64]	pandas	6,874	50
NDD20 [58]	dolphins	2,657	82
NOAARightWhale [1]	whales	4,544	447

5 Experiments

Evaluation Protocol. We consider the same closed-set splits as in [13], meaning that all individual animals are both in the database (training) and query (test) sets. We approach the problem as image retrieval: for each image in the query set, we find an image in the database and make the query prediction have the same identity as the image from the database. Performance in all experiments is measured as top-1 accuracy.

WildFusion relies on finding hyper-parameters and fitting calibration models. The standard approach involves splitting the training set into development and validation parts, which are used for the selection of the best hyper-parameters and fitting calibration models. However, this approach is not applicable in our case because the MegaDescriptor was trained on the whole training set and cannot be used for validation. We addressed this issue by splitting the original test set into a validation set and a new, smaller test set using a 0.5 ratio. We estimated both μ and the calibration function on the validation set and utilized them for the final prediction on the test set. Due to this change in test set, our results are not directly comparable to the results reported by [13].

Technical Details. To construct the global scores, we use embeddings extracted by MegaDescriptor [13] and DINOv2 [44]. For local matching scores,

we use LightGlue [36] feature matching with local descriptors ALIKED [71], DISK [60], and SuperPoint [18]. We use at most 512 keypoints and their appropriate descriptors, extracted from images resized to 512×512. For matching with LoFTR [57], we use the outdoor variant trained on the MegaDepth [35] dataset. On input, we use image pairs with both images resized to 512×512. A total of four local feature matching methods were considered to construct matching-based scores. All these methods were taken off-the-shelf, and none were fine-tuned or retrained. We perform the experiments on the datasets described in Sect. 4. In the baseline WildFusion, we search for optimal hyperparameter μ from Eq. 2 separately for each dataset. The calibrated scores are given equal weights w_i. The summary of settings is in Table 2.

Table 2. WilfFusion settings overview. We test a variety of state-of-the-art local and global methods for animal re-identification and image retrieval. The calibration is done using Logistic or Isotonic regression.

Components:	Local matching methods:	Global similarity methods:
	– LoFTR – LightGlue + SuperPoint – LightGlue + Disk – LightGlue + Aliked	– MegaDescriptor-L-384 – DINOv2-512
Calibration:	Isotonic regression with PCHIP interpolation Logistic regression	
Fusion:	Average with equal weights w_i	

5.1 Baseline Performance

WildFusion clearly outperforms MegaDescriptor in most scenarios. When using all available scores, WildFusion shows superior performance on 16 out of 17 datasets, with only one dataset, ZindiTurtleRecall, showing a slight decrease in accuracy. The average accuracy improvement is substantial, with WildFusion (all) achieving 84.0% compared to MegaDescriptor's 75.5%, representing a notable average gain of 8.5% points. The most significant improvements are seen in datasets like NDD20, WhaleSharkID, SeaStarReID2023, and SealID, where WildFusion shows accuracy gains of over 14% points.

Interestingly, even when using only local matching scores, WildFusion maintains competitive performance. It outperforms MegaDescriptor on 11 out of 17 datasets with average accuracy (78.5%), which is better than MegaDescriptor by 3.0% points. This suggests that the local matching scores are quite powerful on their own, without any need for fine-tuning on animal datasets. More details about the results, including per dataset performance, are in Table 3. Besides, we provide a qualitative evaluation in Fig. 3

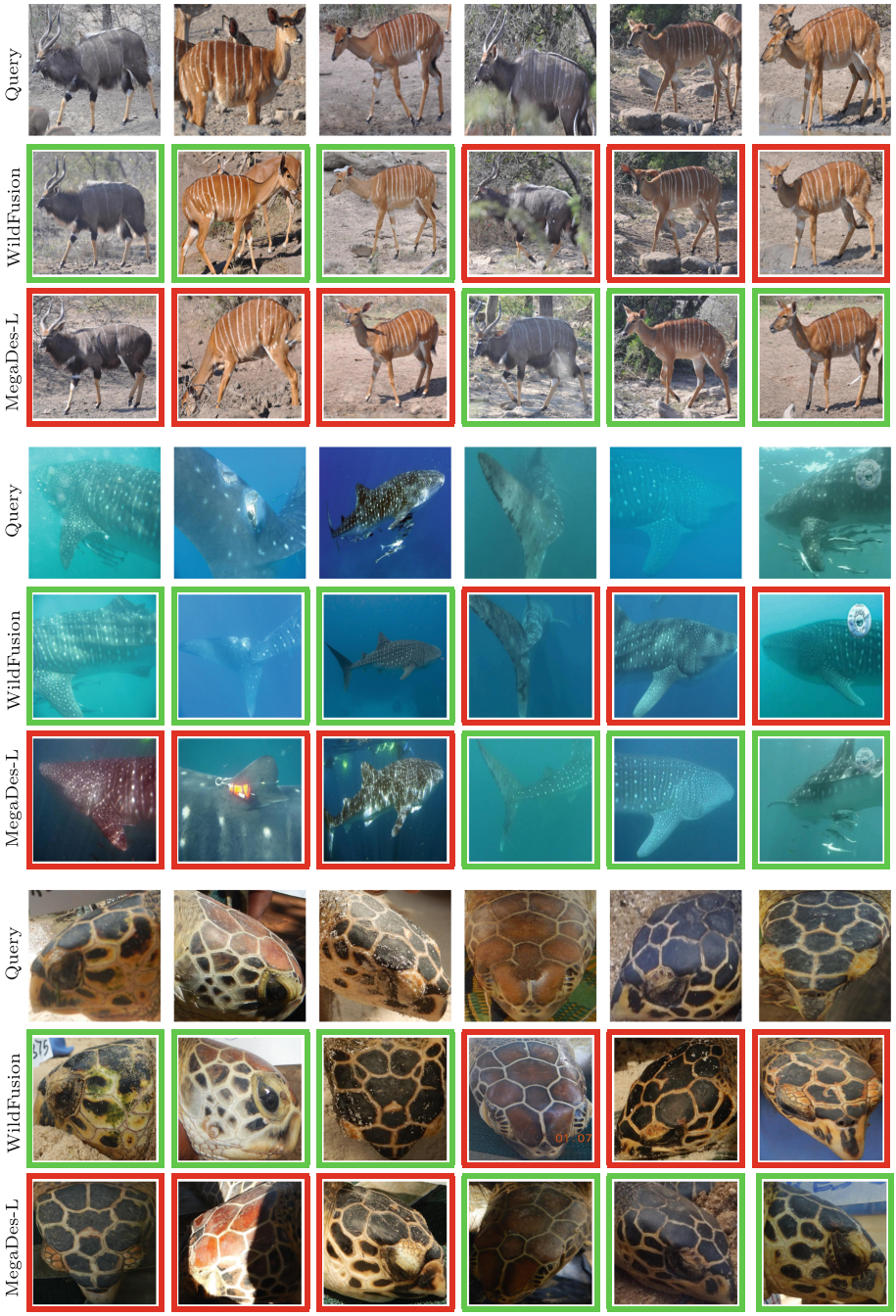

Fig. 3. Qualitative performance. Selected examples where WildFusion changed the decision of the MegaDescriptor-L on NyalaData, WhaleSharkID, and ZindiTurtle; three correct and false samples. We suspect that some wrong matches are mislabeled data.

Table 3. WildFusion's performance in comparison with MegaDescriptor. On average, WildFusion, outperforms MegaDescriptor, even with just *local* descriptors. WildFusion with *all* local and deep descriptors ranks the best on all but two datasets.

	MegaDescriptor	(*all*)		(*local*)	
	Large-384	WildFusion	Δ	WildFusion	Δ
ZindiTurtleRecall	74.24	71.90	-2.34	45.62	-28.62
CTai	91.86	92.08	+0.21	81.80	-10.06
ATRW	97.96	98.51	+0.56	98.33	+0.37
CowDataset	98.66	100.00	+1.34	100.00	+1.34
SeaTurtleIDHeads	91.18	95.00	+3.82	93.82	+2.63
IPanda50	85.76	89.68	+3.92	81.40	-4.36
NyalaData	41.59	46.26	+4.67	25.23	-16.36
BelugaID	67.61	72.46	+4.85	63.07	-4.54
NOAARightWhale	43.25	49.25	+6.00	42.18	-1.07
Giraffes	91.04	99.25	+8.21	98.51	+7.46
HyenaID2022	78.41	90.48	+12.06	88.25	+9.84
GiraffeZebraID	82.98	95.74	+12.77	94.81	+11.84
LeopardID2022	77.82	90.93	+13.11	89.40	+11.58
SealID	78.47	92.82	+14.35	90.91	+12.44
SeaStarReID2023	82.24	99.53	+17.29	100.00	+17.76
WhaleSharkID	62.04	80.33	+18.28	77.68	+15.64
NDD20	38.35	63.53	+25.19	63.16	+24.81
Average	*75.50*	*83.99*	*+8.49*	*78.48*	*+2.98*

6 Ablation Studies

This section presents a set of ablation studies to empirically validate the design choices behind the WildFusion.

6.1 Effect of Local Matching Score Threshold

The hyperparameter μ controls the trade-off between low-quality matches and fewer high-quality matches. When μ is low, the score is influenced by many low-quality matches, often presented in the background. When μ is very high, it filters out most of the matches, leading to a loss of information and resulting in a zero score for nearly all pairs.

Comparing performance scores with constant μ and μ selected based on the validation set suggests that local methods are robust to μ selection, and selecting any μ values between [0.4, 0.6] is a good choice. Interestingly, local methods perform better with μ=0.45 fixed for all datasets than searching for optimal μ on the validation set. When local matching scores are combined with global scores, the range of suitable μ values is wider and extends from 0.4 to 0.8. This shows that adding global scores to the ensemble reduces the downside of having zeros in the score for large μ values (see Fig. 4).

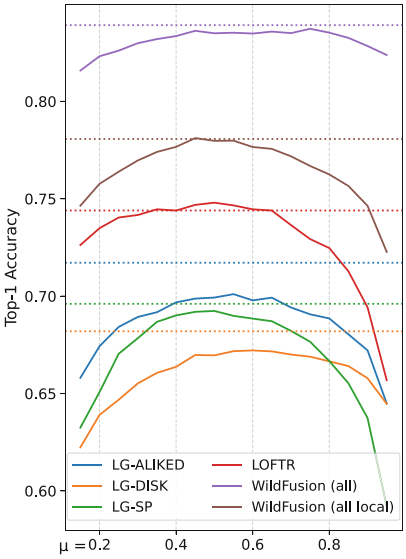

Fig. 4. Effect of μ on performance. Full lines represents constant μ, and dotted lines optimal μ found on validations set for each dataset. Fixing $\mu = 0.5$ provides comparable results to the best μ based on validation set.

6.2 Effect of Score Selection

WildFusion's versatility allows it to fuse any score. As mentioned, using WildFusion only with all local matching scores outperforms the MegaDescriptor-L global score. When we included the global score from the general-purpose feature extractor DinoV2, performance improved only marginally, highlighting the importance of fine-tuning the deep embedding model.

Using the MegaDescriptor-L global score with at least one local matching score significantly outperforms using MegaDescriptor-L alone. Combining it with LG-ALIKED achieves the highest accuracy of 83.0%, followed by LoFTR at 81.4%. LG-SuperPoint and LG-DISK also show comparable performance with accuracies of 80.6% and 81.1%, respectively. Combining the global score with all local matching scores further improves performance, suggesting that the local matching scores are mostly uncorrelated and perform well in the ensemble. More details can be found in Table 4.

Table 4. Ablation on local and global score fusion. We report WildFusion's performance using various local and global methods. Combining local methods with fine-tuned global scores of MegaDetector-L achieves the best results.

Global \ Local	*None*	LG-DISK	LG-SP	LG-ALIKED	LoFTR	all
None	-	68.9	70.1	72.2	74.8	78.5
DINOv2	47.5	70.4	71.6	73.7	74.8	78.8
MegaDescriptor-L	75.5	81.1	80.6	83.0	81.4	84.0

6.3 Effect of Calibration

Using isotonic regression for calibration yields marginally better results on average compared to logistic regression (83.9% accuracy). However, there is a discrepancy in performance between the datasets. For example, using logistic regression was better on NDD20 (+3.0%) and ZindiTurtleRecall (+ 2.7%), but it significantly underperformed on NyalaData (-6.5%) compared to the isotonic regression. This suggests that the poor performance of WildFusion on ZindiTurtleRecall can be related to incorrect calibration.

How Much Data Do We Need for Calibration? To test this, we create variously sized subsets of labeled images from database and validation sets, such that at least 2 positive and 2 negative pairs can be created. Pairs created from this subset are used for calibration and finding μ. We perform additional experiments with μ fixed to 0.5 to isolate the effect of low data calibration from finding μ. As visualized in Fig. 5, isotonic regression performs better than logistic regression in low data scenarios, both with optimized and fixes μ. Calibration with fixed μ is significantly better for a smaller number of samples but yields marginally worse results when a lot of labeled data is available. In general, calibration is very data efficient. For example, fixing μ to 0.5 and calibrating each dataset using only 10 labeled images still gives a reasonable 79.5% accuracy. Adding more data to the calibration further increases the performance of up to 200 samples, where additional data only gives marginal improvements. Our results suggest that WildFusion is viable even with very few labeled samples.

6.4 Constraining Number of Comparisons

Given a database with M samples and a query with N samples, methods based on local features often need to perform pairwise comparisons, needing $M \times N$ comparisons. Since many modern matching algorithms are based on neural networks $M \times N$, neural network inferences are required. With the increasing database size, the computational time quickly becomes unfeasible; therefore, the calculation of all local scores remains a viable option only for moderately sized datasets.

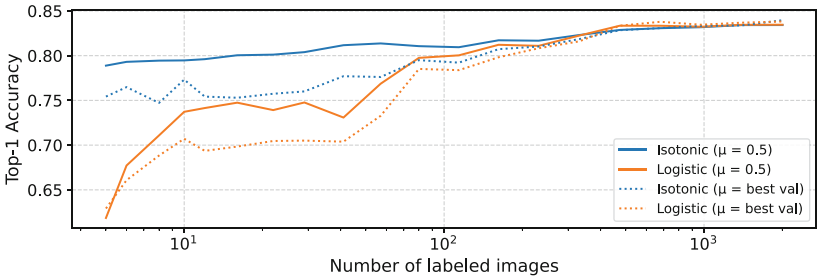

Fig. 5. Ablation on optimal number of images for model calibration. Isotonic regression calibration with fixed $\mu = 0.5$ for all datasets outperforms other approaches in low data scenarios.

Shortlist Strategy. We consider a scenario where we have two types of scores, one cheap to calculate, such as global score s_G from MegaDescriptor or DinoV2, and one expensive, such as WildFusion with local matching scores s_W. We follow the shortlist strategy [69] and use cheap global scores to filter candidate samples. The expensive scores are calculated for a restricted size shortlist to validate and re-rank the top matches. The running time is controlled by a computational budget B in terms of the number of expensive score evaluations per query image.

Results. Using the shortlist strategy, we are efficiently able to utilize the WildFusion scores s_W, which are costly to calculate. On average, budget $B = 300$ is enough to reach accuracy comparable to calculating all scores. For example, on SeaTurtleIdHeads, WildFusion needs only about 200 s_W calculations to reach its peak performance. With a database size of 6063, this results in more than a 30-fold increase in inference speed. Interestingly, performance at $B = 10^3$ is slightly better than using all comparisons. It indicates that local matching scores in WildFusion are prone to some degree of false positive matches when applied

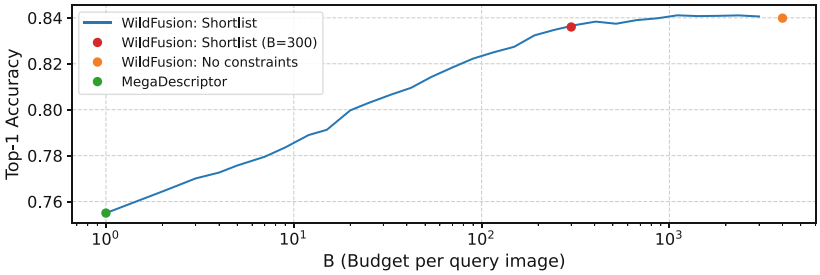

Fig. 6. Rate of performance improvement with increasing budget. The shortlist strategy allows adding more computational resources to improve performance, up to a budget of $B = 200$.

to all images in the database. A more detailed visualization of the speed-up is in Fig. 6.

7 Zero Shot Performance

Encouraged by the fact that calibration works well even with a low number of data points to achieve reasonable performance, we conducted an experiment in a zero-shot setting, meaning no data is needed prior to inference. We split datasets into disjoint subsets. For each score, we trained a single calibration model on one subset and evaluated it on a different subset, with $\mu = 0.5$ fixed for all local matching scores. This differs from the default setting, where subsets of the same dataset were used for calibration and evaluation.

For the zero-shot experiment, we evaluated WildFusion using only local matching scores without incorporating MegaDescriptor's global scores, as the latter had already been trained on the data. Our zero-shot WildFusion approach achieved an average accuracy of 76.2%, which is 2.3% points lower than the accuracy obtained with dataset-specific calibration. Notably, this performance is also 0.7% points higher than the state-of-the-art fine-tuned model MegaDescriptor-L-384, demonstrating the effectiveness of our method in a zero-shot setting without fine-tuning or dataset-specific calibration. For more detail about performance, see Table 5.

For both novel datasets not included in the MegaDescriptor training set, WildFusion with local scores achieved a perfect 100% accuracy. In contrast, for the CowsDataset, DINOv2 reached an accuracy of 96.0%, while MegaDescriptor achieved 98.7%. Similarly, for the SeaStarReID2023 dataset, DINOv2 obtained an accuracy of 82.2%, whereas MegaDescriptor reached 88.8%.

Table 5. WildFusion performance in zero-shot setting. No data from the evaluated dataset was used prior to test time (except Wahltinez et al. [63], which used standard classification setting).

	MegaDescriptor-L	DINOv2	(*local*) WildFusion	[63] Wahltinez et al.
CowsDataset	98.7	88.8	100.0	–
SeaStarReID2023	82.2	96.0	100.0	99.9
17 datasets	–	47.5	76.2	–

8 Conclusion

In this paper, we presented WildFusion, a novel approach to individual animal identification that leverages a calibrated similarity fusion of deep and local

matching scores. By combining deep features extracted from MegaDescriptor or DINOv2 with local matching descriptors (e.g., LoFTR and SuperPoint), WildFusion achieves state-of-the-art performance across a wide range of datasets. Our method is easy to use in real applications as it does not require training and is usable out of the box with any pre-trained deep embedding models and local feature-matching methods. Besides, the code was made public.

Even though the best results were obtained with dataset-specific calibration, we have empirically shown that using WildFusion of only local similarity score and with generic calibration still gives good performance, with mean accuracy dropping only by 2.3% and still reduced the relative error of MegaDescriptor by 44% points. WildFusion's flexibility was also further proven by its strong performance in zero-shot settings tested on species "never seen before."

The scalability and generalization potential of WildFusion makes it suitable for application across different species and environments, contributing significantly to the field of animal re-identification.

Limitations: WildFusion leverages off-the-shelf local matching methods like LoFTR and LightGlue, which were originally trained on datasets featuring static objects. Therefore it is not optimized for matching animals, where the same animal can be observed in various poses, lighting conditions and with occlusions. Therefore, our work can be extended by adapting or retraining these local feature-matching models specifically for animal identification tasks, potentially improving the accuracy and robustness of the WildFusion approach. The same applies to deep descriptors such as MegaDescriptor and DINOv2, which, if not trained on that species, will most likely underperform.

WildFusion is best suited for offline analysis of existing databases. While we introduced a method to address scalability, it remains insufficient for real-time identification. Future research could explore the development of more efficient algorithms to enable real-time processing and online identification.

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Towards Zero-Shot Camera Trap Image Categorization

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Abstract. This paper describes the search for an alternative approach to the automatic categorization of camera trap images. First, we benchmark state-of-the-art classifiers using a single model for all images. Next, we evaluate methods combining MegaDetector with one or more classifiers and Segment Anything to assess their impact on reducing location-specific overfitting. Last, we propose and test two approaches using large language and foundational models, such as DINOv2, BioCLIP, BLIP, and ChatGPT, in a zero-shot scenario. Evaluation carried out on two publicly available datasets (WCT from New Zealand, CCT20 from the Southwestern US) and a private dataset (CEF from Central Europe) revealed that combining MegaDetector with two separate classifiers achieves the highest accuracy. This approach reduced the relative error of a single BEiTV2 classifier by approximately 42% on CCT20, 48% on CEF, and 75% on WCT. Besides, as the background is removed, the error in terms of accuracy in new locations is reduced to half. The proposed zero-shot pipeline based on DINOv2 and FAISS achieved competitive results (1.0% and 4.7% smaller on CCT20, and CEF, respectively), which highlights the potential of zero-shot approaches for camera trap image categorization.

Keywords: Camera Traps · Classification · Retrieval · BLIP · DINOv2 · Zero-shot · Vision and Language · ChatGPT · SAM · MegaDetector

1 Introduction

Camera traps are valuable assets in ecological research. They are commonly used to estimate wildlife populations, species distribution, and their interactions [4,24,31,38]. In many cases, the data (i.e., images) are still processed manually, which is extremely time-consuming, given the relatively high number of operated camera traps and their continuous data flow. Therefore, a concerted effort is being made to automate this process using machine learning and computer vision [22,40]. While some studies have achieved human-level performance [37,40], some challenges persist. For instance, models trained on a set of locations perform poorly in new ones, and models trained in the closed set

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setting must be re-trained. The same applies to new and unseen species, where adaptability is crucial, as it ensures that if a new, rare animal species appears in the monitored area, the classifier can recognize it.

In this paper, we test how existing foundational models perform in automatic camera trap image categorization and if they allow to overcome the above-mentioned problems. First, we evaluate state-of-the-art CNN- and Transformer-based classification architectures on three datasets (i.e., WCT [1] from New Zealand, CCT20 [3] from California, and CEF from Central Europe[1]) from different continents and with different species. Second, we use MegaDetector (MD) [21] and Segment Anything (SAM) [15] models for zero-shot detection and segmentation and test how these models improve classification performance and how they can help mitigate the issue of overfitting to the location/background. At last, we propose and test two approaches using large language and foundational models, such as DINOv2 [8,25], BioCLIP [35], BLIP [16], and ChatGPT [32], in a zero-shot scenario that perform similarly good in *seen* and *new locations* as the best approach based on supervised learning.

2 Related Work

In the past, camera trap image categorization was often done manually, which was neither effective nor fast [11,36]. In response, ecologists and the machine learning community naturally shifted their interest in developing novel methods based on machine learning and computer vision.

Pioneer studies in automatic camera trap image categorization [12,42] were based on local features (e.g., SIFT, SURF) or early adoptions of neural network classifiers. With the fast progress in machine learning, especially CNNs, many studies [20,23,40] focused on fine-tuning standardized architectures for image categorization such as ResNet and VGG. Recent work [7,13,22,33,34] employs a two-step process involving object detection[2] followed by classification. This approach primarily mitigates information loss caused by resizing images to fit the expected classifier input and consequentially reduces overfitting to training locations. Additionally, detecting the animals has clear advantages, as it enables handling multi-species presence and counting the animals present, in addition to categorization.

Even though there is a long track record of methods for camera trap image categorization, most existing approaches share the following drawbacks: (i) Methods that do not use object detection to crop the animal before classification tend to overfit to the background, resulting in poor performance on new and *unseen* locations. Additionally, even with the cropped animal, some background pixels are present and, therefore, can result in overfitting. (ii) All available pre-trained models, trained on a closed set of categories (i.e., species), cannot be effectively deployed in different climates or continents where species not present

[1] Due to animal safety concerns this dataset is available only after signing a Non-Disclosure Agreement (NDA).

[2] Usually using large pre-trained models, e.g., MegaDetector [21].

in the training data are naturally present. (iii) Existing datasets lack standardization, and available categories are usually on a different taxonomical level, i.e., species, genus, family.

3 Datasets

While selecting the datasets for our experiments, we considered the different geographical locations and species. Therefore, we evaluated our experiments on three datasets that originate from North America (Caltech Camera Traps-20), Oceania (Wellington Camera Traps), and Europe (Central Europe Fauna-22).

We use CCT20 as the primary subject in our ablations due to its smaller scale (i.e., faster training) and wide usage. To verify the outcomes in different settings and geographic locations, we use the WCT and CEF22 datasets. Below, we briefly describe each dataset, with statistics provided in Table 1.

Note: *Typically, there is only one animal per image, but in rare cases where multiple species appear, we remove those images from the evaluation. Besides and similarly as in the original work* [3]*, we also remove images without animals.*

Table 1. Statistics for selected camera trapping datasets. †Denotes custom split.

Dataset	Boxes	Categories	Training	Validation	Test
Caltech Camera Traps [3]	✓	16	13,553	5,209	39,102
(our) Central Europe Fauna	–	32	92,603	26,479	–
Wellington Camera Traps† [1]	–	17	140,035	35,009	95,406

The **Caltech Camera Traps-20 (CCT20)** [3] dataset consists of images captured from 20 locations in the Southwestern United States. This dataset includes approximately 58,000 images, covering 15 animal categories as well as one "empty" image category. The data are neither balanced nor filtered, reflecting a more natural distribution of species across both time and location. To allow evaluation of location/environment regularization, the validation set is divided into two subsets: *cis* and *trans*. The *cis* subset includes only locations included in the training set, while the *trans* subset includes just new locations.

The **Wellington Camera Traps (WCT)** [1] dataset originates from 187 locations in New Zealand and contains around 270,000 images across 16 categories, plus one category for empty and unclassifiable images. Since the dataset is not pre-split into train, validation, and test subsets, we allocated 1/3 of the locations for testing and the remaining 2/3 for development, which includes both training and validation. Within the development set, the images are randomly divided in an 80/20 ratio for training and validation, respectively. This approach ensures a balanced distribution for model development and evaluation.

Unlike the previous two datasets, the **Central Europe Fauna-22 (CEF22)** is a private dataset originating from three different trapping projects and multiple distinct regions within Central Europe. The dataset consists of approximately 120,000 images and includes 32 species categories[3]. Notably, all empty

[3] Except two taxon labels (e.g., Aves and Martes) data are labeled with species labels.

images were prefiltered and are not included in the dataset. This curated approach focuses exclusively on images containing identifiable species, enhancing the dataset's utility for training and evaluation purposes.

4 Baseline Performance

To the best of our knowledge, there is no standardized benchmark for camera trap categorization (including the three selected datasets); therefore, we provide a few baseline experiments that evaluate state-of-the-art CNN- and Transformer-based architectures for image classification. This section presents multiple experiments and ablations that focus on benchmarking performance for CNN- and transformer-based classifiers. In the following section, these models are further enhanced with existing foundational models in a zero-shot detection and segmentation context. In Fig. 1, we illustrate four designed approaches.

Evaluation Protocol: In all the experiments and ablations presented in this section, we use the Top1 and Top3 accuracy and macro averaged F1 score ($F1_m$)

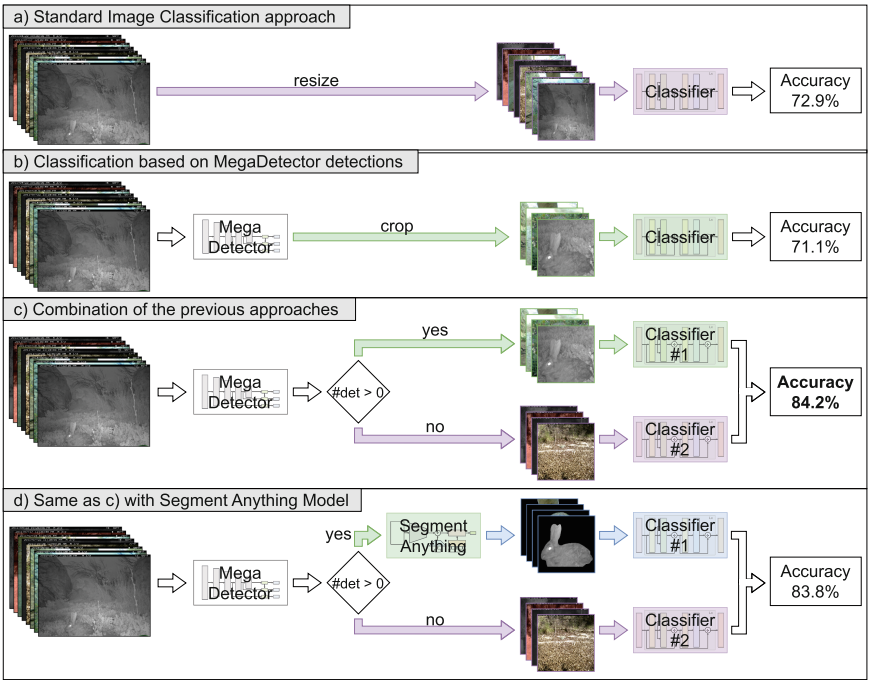

Fig. 1. Four baseline approaches illustration. (a) Standard image classification approach where images are resized and fed to a trained image classifier. (b) A more recent two-stage approach with object detection before the classification. This approach can suffer from missing reject options for images with no detection. (c) With two image classifiers, the problem of (b) can be easily mitigated. (d) Excluding background pixels using SAM could help to prevent overfitting to the location.

as the evaluation metrics. The evaluation is primarily carried out on CCT20 validation subsets (i.e., cis_val + trans_val) and further tested on CEF22 and WCT datasets to verify the results transferability.

Benchmarking Image Classifiers: Based on the results reported in general image classification [5,27] and species classification [28,29], we primarily focus on transformers, but we add ConvNext and ResNext into the mix to compare it for camera trap image analysis.

Experiment Settings: We use pre-trained models from the timm python library [39], version 0.9.12. We train all models using SGD optimizer and the momentum of 0.9 [30] for 40 epochs with a batch size of 64, and the cosine scheduler, which decreases the learning rate from $1e^{-3}$ to $2.54e^{-6}$, and the following augmentations during the training: *RandomResizeCrop* with a scale from 0.8 to 1.0, and RandAugment [6] with a magnitude of 0.2.[4]

Results: Achieved results are reported in Tables 2 and 3. As expected, transformers and CNNs performed competitively, but transformers achieved better performance in new locations. All models underperformed significantly on the unseen *trans locations*, while the CNNs exhibit a higher score difference between

Table 2. Ablation on model architecture. We benchmark selected CNN- and transformer-based architectures on the CCT20 val. sets without empty images. The ranking in camera trap image categorization more-less follows the ranking on ImageNet.

Architecture	Pre-train checkpoint	Input size	*Cis location* Top1	*Cis location* $F1_m$	*Trans location* Top1	*Trans location* $F1_m$	*Both* Top1
ConvNeXt-Base [19]	IN22k	224^2	84.9	68.2	36.2	20.5	65.7
ResNeXt-50 [41]	IN1k	224^2	**86.8**	**69.7**	35.8	16.4	66.7
EfficientViT-B3 [5]	IN1k	224^2	78.9	61.2	40.9	19.4	63.9
ViT-Base/p16 [9]	IN1k	224^2	84.6	68.0	49.4	20.9	70.7
SwinV2-Base/w16 [17]	IN1k	256^2	85.4	66.9	49.1	21.9	71.1
Swin-Base/p4-w7 [18]	IN22k	224^2	83.6	67.0	52.0	24.3	71.1
BEiT-Base/p16 [2]	IN22k	224^2	84.4	67.2	54.4	21.1	72.6
BEiTV2-Base/p16 [27]	IN22k	224^2	85.9	68.6	53.0	21.7	72.9
EfficientViT-L3 [5]	IN1k	224^2	83.5	67.0	**60.4**	**28.6**	**74.4**

Table 3. Transformers performance. All models achieved a competitive accuracy on all three datasets. However, the BEiTv2 model performed much better on the tail species. *fps was measured on NVIDIA A40 (GPU) and AMD EPYC 7543 (CPU).

Architecture	CCT20 dataset Top1	CCT20 dataset Top3	CCT20 dataset $F1_m$	CEF22 dataset Top1	CEF22 dataset Top3	CEF22 dataset $F1_m$	WCT dataset Top1	WCT dataset Top3	WCT dataset $F1_m$	fps*
BEiT-Base/p16	72.6	89.7	60.1	85.3	94.5	68.4	**86.5**	98.0	72.7	331
BEiTV2-Base/p16	72.9	89.6	61.9	**87.5**	**95.5**	**72.2**	86.0	**98.8**	**75.5**	**333**
EfficientViT-L3	**74.4**	**90.8**	**62.7**	85.7	94.9	62.7	85.1	98.7	67.1	295

[4] Selected as sub-optimal for all selected models based on our preliminary experiments.

the *trans* and *cis*, indicating that they overfit more on the locations themselves. For the following experiments, we will use just the top 3 architectures, i.e., BEiT, BEiTV2, and EfficientViT.

5 Zero-Shot Detection and Segmentation

This section analyzes and provides a qualitative and quantitative evaluation of the potential of two pre-trained foundational models, the MegaDetector (MD) [21] and the Segment Anything Model (SAM) [15], in processing camera trap images. Furthermore, we explore decision-making strategies when MD detects no object, including a) considering an image to be empty and b) classifying the whole image. At last, leveraging the insights gained from our ablations, we test three state-of-the-art transformer-based architectures in combination with MD and SAM trained on three distinct datasets with different species and origins.

Image Data Processing: We run MD to obtain detections of objects for all images. Then crop the detections from the image so that the resulting image has the same width and height to avoid breaking the aspect ratio. Furthermore, we use the SAM (initialized with the MD's detection) to remove background pixels and move the object to the center. See Fig. 2 for examples of the detected and segmented objects in the CCT20 dataset.

Classification Models: Since neither MD nor SAM provides class categories, we will use the top three architectures (in terms of accuracy) from the previous ablation. We have four different types of classifiers, each differing in the expected input: (a) full-size images, (b) cropped detections, (c) cropped detections with segmented objects, and (d) both full-size and cropped images. We investigate combinations of these in our experiments and ablation studies.

Building a Classifier for Inference: Once the models are trained, we create a predictor containing two classifiers. First of all, we apply MD on the image. If

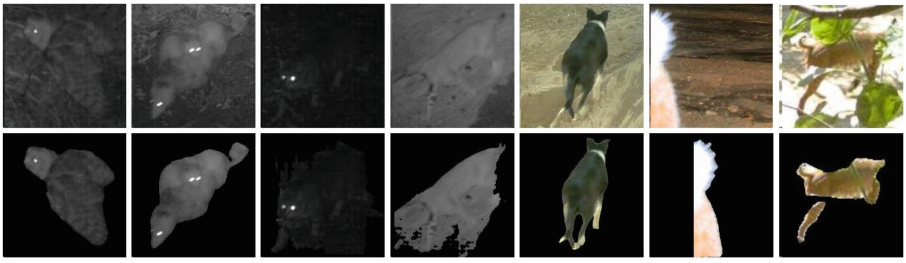

Fig. 2. Zero-shot segmentation with Segment Anything. Random samples from CCT20 datasets are processed by MegaDetector, and resulting detections are fed into SAM. Even with the poor quality of the data, the zero-shot segmentation performs relatively well across a wide range of species. However, with the infra-red images and the small size of an object, the SAM starts to fail (3^{rd} and 4^{th} column from the left).

MD finds an object, we use classifier #1, which is trained to recognize objects from cropped or even segmented images. If no object is found, we use classifier #2 on the whole image. Besides, we perform an experiment in which we use only one classifier on cropped and full-size images.

5.1 Performance Evaluation

We compare three types of classifiers that differ in their expected input data to explore the benefits of using MD and/or SAM on camera trap images. Each classifier expects one of the following inputs: (a) resized images, (b) images cropped based on detection from MegaDetector, and (c) cropped detections with removed background (we use Segment Anything Model, which is fed with the MegaDetector detections). The comparison is made on a filtered validation set of the CCT20 dataset that contains only images with objects. By cropping detections from images, we have improved the Top1 accuracy on average by 10.3%, 5.2%, and 10% on the CCT20, CEF22, and WCT datasets, respectively. The improvement is lower on the dataset with European fauna as it was most likely not used in the training of the MD. Using the segmented animals, the performance decreased by an average of 5.1% on CCT20, 2.8% on CEF22, and 0.8% on WCT. See Table 4 for more comprehensive evaluation.

Location Overfitting: We use the CCT20 *cis* and *trans* subsets to test the hypothesis that image classifiers tend to overfit to background pixels and, therefore, perform poorly in new locations. Following the results listed in Table 4, it is evident that reducing the number of background pixels available for training with both cropping our segmenting the object reduces the location overfitting. Besides training the model on cropped objects, increases the overall classification performance as more detail is provided. Using MD and "feeding" a classifier

Table 4. Ablations on location overfitting. We compare the performance in terms of Top1 Accuracy of three models trained on (a) resized images, (b) cropped animals, and (c) cropped + segmented animals on three datasets. While (a) shows huge overfitting to the location, the (b) and (c) approaches show much better generalization and robustness to changes in testing location. In general, (b) achieves the best performance.

						CCT	
Architecture	MD	SAM	CCT	CEF	WCT	*cis*	*trans*
BEiT-Base/p16	–	–	72.6	85.3	86.5	84.4	54.4
BEiT-Base/p16	✓	–	**83.9**	**91.1**	**96.0**	**89.6**	**75.2**
BEiT-Base/p16	✓	✓	80.8	88.8	95.0	87.3	70.8
BEiTV2-Base/p16	–	–	72.9	87.5	86.0	85.9	53.0
BEiTV2-Base/p16	✓	–	**84.2**	**92.2**	**96.6**	**90.1**	75.0
BEiTV2-Base/p16	✓	✓	83.8	89.2	95.8	89.2	**75.5**
EfficientViT-L3	–	–	74.4	85.7	85.1	83.5	60.4
EfficientViT-L3	✓	–	**83.1**	**90.8**	**95.0**	**87.2**	**76.9**
EfficientViT-L3	✓	✓	81.1	87.6	94.4	85.5	74.2

with cropped detections still results in around 15% lower performance in the new locations. However, it is a significant improvement as without the cropping, the resulting performance in *trans* (*unseen*) locations dropped by roughly 30% in terms of Top1 accuracy, and the error almost doubled.

5.2 Additional Ablations

Dealing with images without detections: Although MD is a powerful foundational model, it has its limitations, such as generating false positives or failing to detect objects altogether. Therefore, we provide an additional ablation that compares two classification methodologies when MD does not detect any objects: (a) using a single classifier for both cropped and full-size images and (b) using separate classifiers for each.

Using a single classifier (a) over the two separate classifiers (b) reduces computational complexity, but it reduces the Top1 accuracy by around 1% for all three architectures. For example, EfficientViT improves accuracy by 1.4% with two classifiers, as shown in the Table 5.

What to do with "empty" images? A common issue in camera trapping is false sensor activation, leading to saving "empty" images with no animals. In this ablation, we test two approaches to distinguish empty images from those with animals. We use the CCT20 dataset, which includes an "empty" category in the validation and test sets, though not in the training set.

We test the following two approaches:

(a) Treat images as "empty" when MD detects no object.
(b) Generate or add object-free data to the training set and train a second classifier on full-size images. We generate average images based on location, date, and time.

Results in Table 6 show that assuming an image is empty when MD does not detect anything significantly reduces performance by up to 18.5%, highlighting the need for better strategies in handling empty images.

Table 5. Classification of images with none MegaDetector detections. Performance of (a) one classifier for cropped images from detections and full-size images and (b) two separated classifiers, one trained on cropped images and the second one on full-size images.

Architecture	Classifier	Top1
BEiT-Base/p16	(a)	66.5
BEiT-Base/p16	(b)	67.5
BEiTV2-Base/p16	(a)	67.4
BEiTV2-Base/p16	(b)	**68.2**
EfficientViT-L3	(a)	65.8
EfficientViT-L3	(b)	67.2

Table 6. Dealing with empty images. We compare a method that trusts the MD to declare the image empty if no object is found with another that uses two classifiers: 1st on a cropped image if an object is detected and 2nd on the full image if not.

If no detection, then	Top1
consider it as *empty image.*	49.1
use 2nd classifier.	67.5
consider it as *empty image.*	49.7
use 2nd classifier.	**68.2**
consider it as *empty image.*	50.1
use 2nd classifier.	67.2

6 Zero-Shot Classification

In addition to the standard classification approach with trained closed-set model for camera trap image categorization, and following the success of LLMs, we further explore the capabilities of existing foundational models for multi-modal processing (e.g., BLIP [16] and ChatGPT [32]) and image retrieval using BioCLIP [35] and DINOv2 [25]. For "classification" based on image retrieval, we use the FAISS library [10,14] developed for efficient similarity search and clustering of deep embeddings. Both evaluated pipelines are illustrated in Fig. 3.

Note: *We are aware of the limited reproducibility when using ChatGPT, but one of the goals of this work is to fully explore what are the current LLMs capabilities in scenarios related to ecology compared to standardly used methods.*

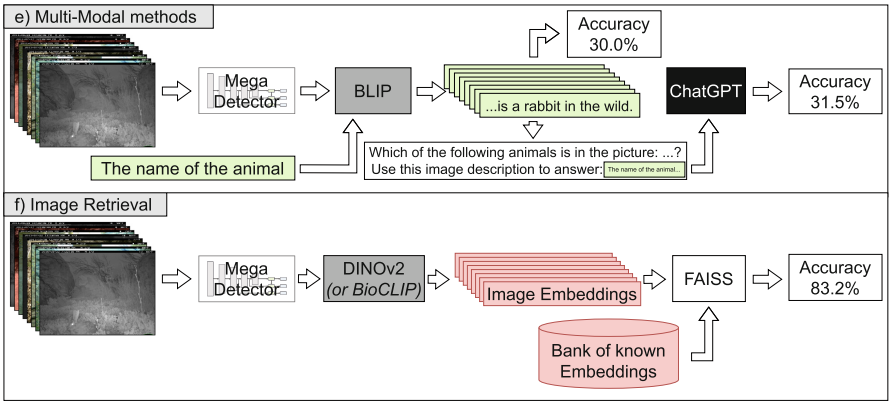

Fig. 3. Zero-Shot approaches. (e) Multi-modal methods extract image info. based on given textual and image prompts. (f) Embeddings are generated from the training set, and during inference, selective search finds similar images from the database.

6.1 Large Language and Multi-modal Models

We use an image captioning model, BLIP, that provides a textual output to a prompt consisting of text and image. To enhance BLIP's focus on animals, we condition various image captions by providing text that includes keywords such as "what," "species," "animal," etc. Furthermore, we use ChatGPT (based on GPT-4) as a filter to identify animals from the given options and image description. We consider the image to be empty if none or more than one of the possible categories is included in the response.

BLIP [16]: We test how the pre-trained multi-modal image captioning model's (i.e., BLIP) performance depends on different conditional captioning inputs (e.g., *"The species of the animal is..."*, or *"The name of the animal..."*). Overall the performance fluctuated inconsistently across the datasets and "prompts," and conditioning the output with no captions can yield relatively good scores. Using *"A running..."* as captioning yielded the best results on the CEF22 datasets; however, slightly underperformed on the CCT20 and WCT. Still, this captioning seems to be a cross-dataset sub-optimum. Interestingly, when the model fails to recognize an animal, it often defaults to "a bear in the wild," even if no bear is present in the dataset, or to phrases like "is not visible"/"is on the camera screen." For detailed performance, see Table 7. Compared to baseline performance with approach (a) and Top1 accuracy of BEiTV2-Base/p16 (i.e., 84.2, 92.2, and 96.6 on CCT20, CEF22, and WCT respectively), BLIP underperformed badly, achieving roughly half of the accuracy.

Table 7. Ablation on "captioning prompting" of BLIP

Conditional image captioning	CCT20	CEF22	WCT
–	21.7	32.6	30.1
The picture shows a cute ...	14.7	35.6	21.8
I see cute ...	19.0	36.6	**30.2**
The species of the animal is ...	20.0	2.5	18.0
The animal in the picture is ...	20.1	1.9	16.8
A running ...	21.5	**40.9**	27.8
A peeking ...	22.5	38.1	29.2
This animal is called ...	24.4	28.3	24.9
The name of the animal ...	**24.9**	37.3	25.7

Fine-tuning BLIP Captions with ChatGPT: ChatGPT has gained popularity due to its suitability across diverse topics and its ability to engage in human-like conversations. However, its capabilities in camera trap image categorization have not been explored yet. Since the ChatGPT API only allows textual inputs, we use it to further process the BLIP's outputs given the image and sub-optimal conditional captioning and put it as an input into the GPT-4 based model [32]. The proposed textual input for the ChatGPT is as follows:

Write a one-word answer to this question: "Which of the following animals is in the picture: <list of categories>?" Consider this image description in the answer: <BLIP generated caption>.

Where *<list of categories>* are all targets without "empty" category, separated with commas and *<BLIP generated caption>* is BLIP's output.

Based on our experiment, ChatGPT can slightly improve the performance of BLIP, with an improvement of approximately 1.5% on the CCT20 dataset. Our observations indicate that, in some cases, ChatGPT adjusts the final prediction by selecting a class from the options provided. However, more frequently, ChatGPT ignores the options and instead propagates the species identified in the caption generated by BLIP, even if it is not among the provided options (e.g., "bear"). Furthermore, ChatGPT tends to list the options from which it claims to have selected its answer, even when the chosen answer is absent from the list.

6.2 Retrieval-Based Classification

In this section, we test the suitability of existing pre-trained foundational models (e.g., BioCLIP [35] and DINOv2 [8,25]) for zero-shot camera trap image categorization in a retrieval-like setting. Zero-shot classification of camera trap images can solve some of the major drawbacks of the existing approaches as it allows the classification of images into categories not seen during training by retrieving and comparing them against a set of predefined class examples. This means that the database can be extended by new species at any time without re-training the model. Furthermore, this approach allows simple updates of the class reference samples, so weekly labeled data and data with no additional value might be

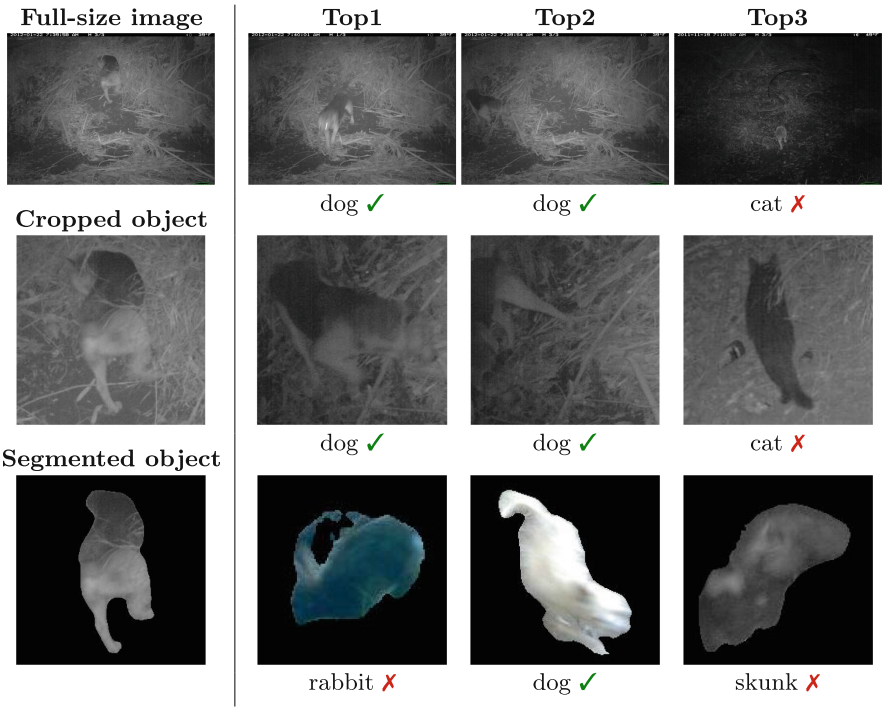

Fig. 4. Top3 closest images to given inputs using DINOv2$_G$ embeddings.

easily filtered out or replaced. On the other hand, after the inference, one more step, i.e., similarity search, is needed. For example predictions, see Fig. 4.

Method: First, we feed forward all the training images into the image encoders and store image embeddings into the *database*. Second, we generate embeddings for all *query* data the same way. At last, we use FAISS library [10,14] to perform an efficient similarity search. In both cases, the images are resized to 256×256 and center-cropped to 224×224. As a baseline, we use the standard L2 distance without feature vector normalization and k-NN classifier with k = 1 to select the species prediction.

Results: Overall, the best-performing model was the DINOv2$_G$, which achieved an outstanding performance comparable to models trained with full supervision. More precisely, the DINOv2$_G$ achieved a Top1 accuracy of 83.2% and 87.5% on CCT and CEF datasets, respectively. Interestingly, the newer and "*supposed-to-be better*" DINOv2$_B$+reg [8] slightly underperformed the vanilla DINOv2$_B$ and the BioCLIP$_B$ heavily underperformed both DINO models.

As in previous experiments, cropping detected animals was also beneficial for image retrieval, and the segmentation seems to be unnecessary. In the scenario with just a single classifier, the DINOv2$_G$ performed similarly on the CEF dataset and slightly better on the CCT dataset. A more comprehensive qualitative and quantitative evaluation is listed in Table 8, and Fig. 4.

Table 8. Camera trap image categorization in retrieval-like setting. We compare BioCLIP, DINOv2 [25], and DINOv2+reg [8], on the CCT20 and CEF20 datasets in terms of Top1 accuracy. To allow direct comparison with the *standard* approaches (i.e., (a), (b), and (c) in Fig. 1), we add scores achieved with BEiTv2 classifiers explicitly trained to classify camera trap images on those datasets.

	MD	SAM	BioCLIP$_B$	DINOv2$_B$+reg	DINOv2$_B$	DINOv2$_G$	*BEiTv2*$_B$
CCT	–	–	60.5	65.5	67.8	74.4	72.9
	✓	–	65.4	77.5	77.8	<u>83.2</u>	**84.2**
	✓	✓	62.7	71.8	70.2	75.6	83.8
CEF	–	–	70.1	80.9	81.5	87.1	87.5
	✓	–	69.9	82.5	82.6	<u>87.5</u>	**92.2**
	✓	✓	61.6	74.2	75.3	81.1	<u>89.2</u>

Ablation on Location Generalization. Given that image retrieval-based classifiers demonstrated comparable results to fine-tuned models, we performed an additional ablation on three African datasets[5] from Snapshot Safari [26]: Enonkishu (ENO), Kgalagadi (KGA), and Kruger (KRU). As these datasets are not split into development and testing subsets, we use images from the first x locations[6] to build the database and the remaining locations for testing. In other

[5] We remove empty images and compare models without using MD or SAM.

[6] We use the first 8, 15, and 26 locations for the ENO, KGA, and KRU datasets, resulting in 20%, 31%, and 29% of the total images being used for testing, respectively.

Table 9. Results of image retrieval-based classifier on additional datasets.

Model	CCT	CEF	WCT	ENO	KGA	KRU
BioCLIP$_B$	60.5	70.1	80.3	44.3	72.8	55.7
DINOv2$_G$	**74.4**	87.1	83.6	**65.9**	79.9	**79.6**
BEiTv2$_B$	72.9	**87.5**	**86.0**	53.5	**83.0**	41.1

words, the results reported in Table 9 for ENO, KGA, and KRU are based solely on images from unseen locations.

Retrieval-based methods achieve results comparable to, and in some cases exceeding, the BEiTv2 fine-tuned models. This is particularly evident on the African datasets, which contain relatively limited training data for classification (3,929, 662, and 1,312 images for ENO, KGA, and KRU, respectively), where model training completely failed for KRU. Considering that the performance of retrieval-based methods can be easily improved by simply adding new data to the database, whereas standard image classifiers require model fine-tuning, retrieval-based approaches prove promising results, especially for small datasets.

Ablation on Matching Strategy. In this experiment, we compare two similarity measures (L_2 norm and cosine similarity) and various k values for the k-NN classifier over DINOv2$_G$ features on five distinct datasets. The results are inconclusive as the best-performing approach is different for each dataset. Therefore, we recommend testing the best approach before using it on your dataset. For further details, refer to Table 10.

Table 10. Matching strategy settings ablation. We compare how different settings affect DINOv2$_G$ retrieval performance using L_2 norm and cosine similarity based on class centroids and various k values in the k-NN algorithm.

	Cosine similarity		L_2 norm				
Dataset	1-NN	3-NN	1-NN	3-NN	5-NN	10-NN	centr.
CCT	**74.4**	72.7	**74.4**	74.1	72.3	71.8	54.8
CEF	87.0	87.7	87.1	**87.8**	**87.8**	87.7	74.2
ENO	66.6	**68.6**	65.9	67.9	68.4	65.9	54.6
KGA	80.6	78.5	79.9	79.5	79.2	83.4	**86.9**
KRU	80.0	79.8	79.6	76.1	77.2	**80.7**	74.6

7 Conclusion and Perspectives

This paper presents a comprehensive comparative analysis of various approaches to automating camera trap image categorization. Our findings highlight the usefulness of integrating two independent classifiers-one specialized for cropped animals and the other for full images-with MegaDetector. This approach, where the

appropriate classifier is selected based on whether the MegaDetector detects an animal, resulted in great improvements and allowed very limited overfitting to the location. More precisely, the combination of the MegaDetector with two fine-tuned BEiTV2 models reduced the relative error by around 42%, 48%, and 75% on the CCT20, CEF22, and WCT test sets, respectively.

In addition, we proposed two alternative approaches for zero-shot classification based on (i) multi-modal models (e.g., BLIP and ChatGPT) and (ii) image retrieval based on deep features from DINOv2 and BioCLIP. For both, we used MegaDetector to detect and crop the animals.

Interestingly, DINOv2$_G$ achieved Top1 accuracy of 83.2% and 87.5% on CCT and CEF, respectively, trailing behind the best approach by 1.0% and 4.7% percentage points. This approach, leveraging robust visual representations and efficient similarity search, offers interesting properties such as (i) no requirement for fine-tuning while introducing new species and (ii) robustness toward location.

The provided analysis underscores the value of using specialized classifiers tailored to different aspects of the image. When initial object detection fails, integrating a secondary classifier for full images substantially enhances the classification of images that MegaDetector deems empty, thus reducing overall classification errors. Besides, our results achieved in a zero-shot scenario showed a promising direction for zero-shot camera trap image classification methods. Future work should focus on developing those methods, as they allow for straightforward adaptability for new species and locations.

The main outcomes derived from this work include the following:

MegaDetector Rules. The high accuracy and robustness in detecting animals, people, and vehicles in camera trap images make MegaDetector an excellent asset for preprocessing raw camera trapping data. Since CNN- and Transformer-based classifiers require resizing the original images to an expected input size, the detection and cropping of animals from the original images reduce the data loss related to resizing and subsequentially results in a considerable increase in performance and also a better regularization towards location changes.

Large Language and Multi-modal Models Do Not Perform Well. While powerful in natural language processing and image captioning, ChatGPT and BLIP perform poorly on camera trap images. This is most likely due to their utility being confined to text- or text+image-based applications, making them unsuitable for the visual analysis of camera trap data.

A Great Potential of Retrieval-like Classification. DINOv2 features used in image retrieval settings for the classification of camera trap data are surprisingly robust, especially when using larger architectures. Comparing it with traditional classifier-based methods offers a promising alternative with similar performance and multiple benefits, e.g., new species or locations can be quickly and easily introduced just by adding a few samples into the *database*, and no fine-tuning is ever needed, which saves a considerable amount of CO_2.

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Larval Hostplant Prediction from *Luehdorfia japonica* Image Using Multi-label ABN

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Abstract. Butterflies are easily recognizable due to their showy coloration, and are a familiar taxon to many enthusiasts. Because of the abundance of specimens collected and the ease of comparison among various species, regional variation in butterfly spots can be observed, which is related to multiple factors such as geological history, topography, climate, and hostplants. In this study, we focus on the relationship between regional variation in butterfly spots and the distribution of larval hostplants and aim to clarify the relationship by classifying the larval hostplants based on images of butterfly spots. Specifically, we focus on the *Luehdorfia japonica*, a species of butterfly with known geographic variation in butterfly spots and a highly understood distribution. We create *Luehdorfia japonica* image dataset based on digital specimens and the metadata about the collection site. We show that the multi-label attention branch network can be trained on the dataset to accurately classify the larval hostplant from the specimen images and that the analysis of the attention map provides the same basis for decision making as the expert knowledge.

Keywords: Attention branch network · Multi-label image classification · Visual explanation · *Luehdorfia japonica* · Regional variation

1 Introduction

Butterflies are highly visible and one of the most familiar species. Because of the abundance of specimens collected, the distribution of their habitat is highly well understood [11]. And, because of the ease of comparison among specimens, many species are known to show regional variation (geographic variation) in their

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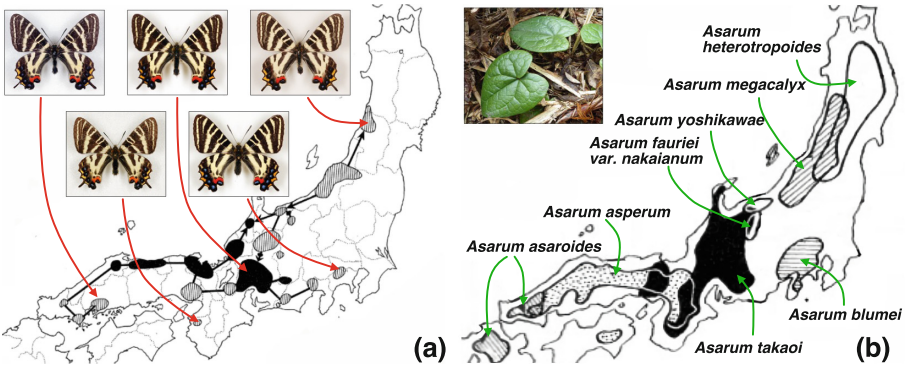


Fig. 1. Overview of the relationship between wing spots and the distribution of the *Asarum*. (a) Schematic illustration of populations based on regional variation in wing shape and spots. The blacker the population, the more developed the black bands of the spots. (b) Conceptual map of the distribution of the *Asarum* used by the *Luehdorfia japonica*. Modified from reference [33], respectively.

spots. Moreover, due to the high degree of distribution elucidation, it is easy to identify regional extinctions and the timing of extinctions and declines. Since various factors such as geographical history, topography, environment, and larval hostplants, plants eaten during the larval stage, are considered to be related to regional variation, we expect to elucidate various problems and factors related to butterflies by correlating butterfly specimen data with various data.

Among various species of butterflies, in this study, we focus on *Luehdorfia japonica*, whose distribution is highly known due to its popularity among butterfly lovers. The wing spots of *Luehdorfia japonica* are well known to vary from region to region. As shown in Fig. 1, *Luehdorfia japonica* is thought to correspond to its regional variation in the distribution of *Asarum* [11,24], which is a larval hostplant and has significant speciation.

In this paper, we aim to clarify the relationship between the regional variation of the butterfly spots and their larval hostplants, based on images of *Luehdorfia japonica* specimens and information on the distribution of *Asarum* in each habitat area. Specifically, a *Luehdorfia japonica* image dataset was created from the Tomoo Fujioka Butterfly Collection (Fujioka Collection) and information on the distribution of larval hostplants. We will then analyze the regional variation in the spotted pattern through visualization of the larval hostplant classification and its basis for judgment from the specimen images[1]. To create the dataset, digital specimen images are preprocessed and image samples are collected. At the same time, each image sample is annotated labels for the hostplants used during the larval stage based on the collection site information. In some cases, the

[1] The technical report versions of this paper are presented in [35,36]. In this paper, we construct a larger *Luehdorfia japonica* dataset and report the results of more detailed experiments and their analysis results.

hostplant label is not a single label, but multiple labels are assigned depending on the habitat. The dataset is trained and classified using the attention branch network (ABN) [7] to improve the accuracy of visual description and image classification tasks.

The number of samples in the created dataset is highly dependent on the type of specimens present and the location of the collection. The number of *Luehdorfia japonica* in each collection site and the number of individuals collected varies widely, resulting in a dataset with potential class imbalance. It is known that the recognition accuracy of a dataset with class imbalance decreases in a class with a small number of samples. In this study, we introduce error functions and data sampling techniques that take class imbalance into account to reduce class imbalance. This will not only improve classification accuracy but also confirm changes in the visual explanations acquired, aiming to acquire appropriate gazing regions that support the relationship between mottled patterns and larval hostplants.

2 Related Work

2.1 Application of Computer Vision to Biology

Several biological analyses have been performed by applying computer vision techniques [3,6,20,21,34,38]. Cuthill et al. [3] quantitatively analyzed Müllerian mimicry in butterflies by utilizing the distance between samples in the embedding space obtained by triplet network [28] and categorical cross-entropy loss.

Fan et al. [6] constructed and classified a dataset of 80 swallowtail butterfly species from Yunnan, China, and Lin et al. [20,21] proposed an image classification model based on deep learning to identify butterfly subfamilies, genus, and subspecies. Both studies identified the biological classification of butterflies as fine-grained image classification tasks. On the other hand, this study deals with a more detailed image identification task because it deals with the regional variation of the spots in a single species of butterflies, *Luehdorfia japonica*. In addition, we will not only classify but also clarify the relationship between larval hostplants and spots using visual explanations (attention maps) obtained in the inference process of the classification model.

2.2 Multi-label Image Classification

This study deals with multi-label image classification tasks. For multi-label image classification, learning methods that take class imbalance into account have been proposed [9,10,32]. Major approaches include devising loss functions [14,25,27] and sampling methods during training [1,2,22,23]. Since the datasets in this study potentially have a class imbalance problem, we train them considering the class imbalance to improve the classification accuracy.

A typical multi-label classification task in the computer vision field is pedestrian attribute recognition, which estimates attribute information such as gender and clothing from pedestrian images [12,13,26,27,31,37]. For the pedestrian

attribute recognition task, approaches that consider class imbalance in multi-label estimation [13,27] and inference models and learning methods that consider the consistency of the gazing region based on the spatial characteristics of the attributes to be estimated [12,26,31,37] have been proposed. Jia et al. [12] propose a regularization such that the features are spatially and semantically consistent. Specifically, they propose a loss function to make the attention map of samples in a mini-batch whose attributes are positive consistent, and an error to learn the consistency of feature maps (vectors) compressed by weighted global average pooling.

As described above, many pedestrian attribute recognition methods improve recognition accuracy by incorporating semantic information of attributes to be classified, which is prior knowledge possessed by humans, and location information where features of each attribute are likely to appear. On the other hand, we do not utilize prior knowledge possessed by humans or experts, but extract and analyze characteristic micro-regions and patterns from label information that are useful for classification. The objective is to elucidate the causes of regional variation in geographic regions. Furthermore, we will not only develop a model for classification by focusing on the same areas as the experts' knowledge but also reveal new characteristic patterns of spots that have not been revealed before.

2.3 Visual Explanations

Visual explanation is a method for presenting the basis for judgments about the inference results of machine learning models. Major approaches include post-hoc methods such as GradCAM [29,30] and models that embed a mechanism to generate an attention map inside the model [7,27,39]. In this study, we utilize the attention branch network (ABN) [7], which incorporates visual explanations into inference results through an attention mechanism, to classify and analyze each class label (larval hostplants) while indicating the areas that were gazed at during classification.

3 Dataset

3.1 Tomoo Fujioka Butterfly Collection

Tomoo Fujioka Butterfly Collection (*Fujioka Collection*) is one of the most valuable collections of butterfly specimens in the world, with approximately 290,000 specimens of all species from Japan and related neighboring countries, and approximately 1,750 specimen boxes. To make effective use of this collection, a database and virtual museum are being developed [15,16]. From 2019, a reorganization of the butterfly collection record information in this database in a format compliant with GBIF (Global Biodiversity Information Facility) and S-Net (Science Museum Net) is underway [17]. Thus, work is underway to revise the data of the above collections into internationally accessible natural history specimen data.

The use of this collection database and related data such as vegetation and flora will make it possible to elucidate the regional variation of butterfly spots. In this study, we utilize the Fujioka Collection to perform quantitative larval host-plant classification based on images of the *Luehdorfia japonica*. We will focus on the Japanese endemic species of the butterfly, *Luehdorfia japonica*, because of its huge mass and remarkable geographical variation of wing patches in the collection. First, we will examine the labels necessary for image classification and build a dataset by utilizing the image data and label data from the Fujioka Collection database. Hereafter, we describe the details of the dataset construction.

Table 1. Distribution of labels in *Luehdorfia japonica* dataset.

Label (*Asarum*)	Use in exp.	# of samples	
		Male	Female
A. curvistigma	✓	235	45
A. nipponicum	✓	124	29
A. yoshikawae	✓	55	10
A. megacalyx	✓	341	99
A. nipponicum var. *saninense*	✓	54	13
A. rigescens var. *brachypodion*	✓	91	21
A. asaroides	✓	92	24
A. tohokuense	✓	42	10
A. takaoi	✓	2,075	517
A. caulescens		1	0
A. fauriei		0	1
A. asperum	✓	340	148
A. fauriei var. *nakaianum*		3	1
A. ikegamii	✓	145	30
A. blumei	✓	359	74

3.2 *Luehdorfia japonica* Dataset

Digital Specimen Imaging and Preprocessing. To capture digital specimen images, we first remove each specimen from the specimen box and photograph it along with its specimen number, collection value label, size scale, and color chart for correction (see the top row in Fig. 2). Multiple images are taken at different depths of field, and then depth composited to produce an image in which all parts of the specimen are in focus. The bottom row in Fig. 2 shows an example of a digital specimen of *Luehdorfia japonica* taken by the photographer. The image size of the digital specimen is approximately 6,500×4,000 pixels.

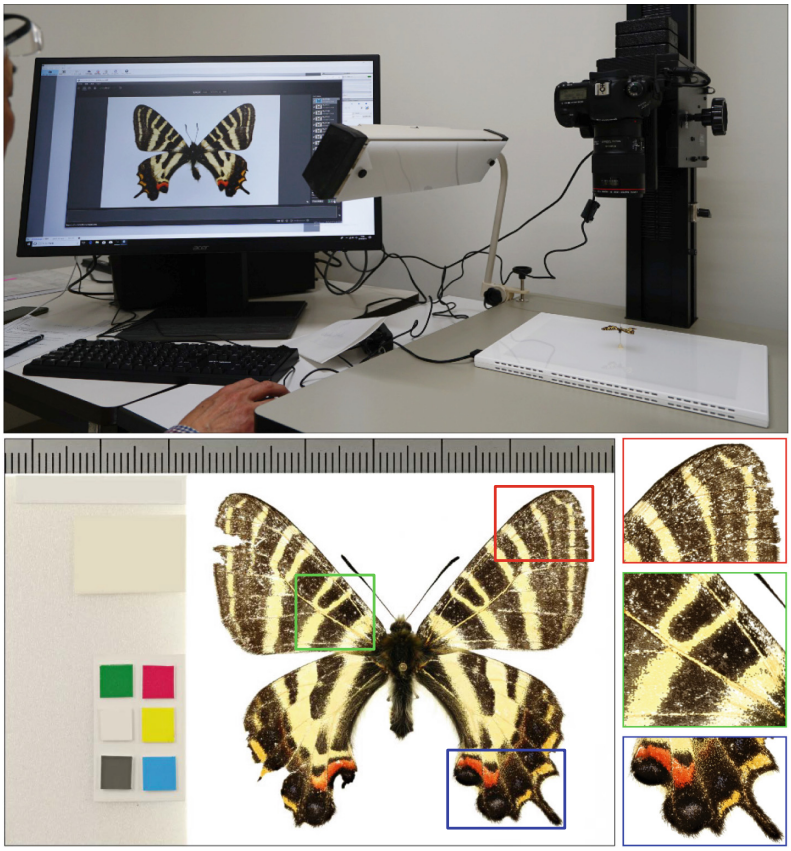


Fig. 2. Photography of the digital specimens of Tomoo Fujioka Butterfly Collection. (Top) Photographing and (Bottom) an example of the digital specimens image data.

As mentioned above, these digital specimens contain color charts and other parts that are unnecessary for image classification. Therefore, we remove them as a preprocessing. Examples of preprocessed images are shown in Fig. 3. All cropped image patches were uniformly sized (see the experimental section for details). The total number of created images is 4,434, of which 3,510 are male and 924 are female.

Assigning Hostplant Label Data. In addition to the image preprocessing, we annotate larval hostplant labels to each of the *Luehdorfia japonica* image samples. Metadata such as place of collection, date of collection, and collector were assigned to each of the *Luehdorfia japonica* samples. Based on the collection location information and the Japanese vegetation information, we assigned larval hostplant labels to each sample [11,24].

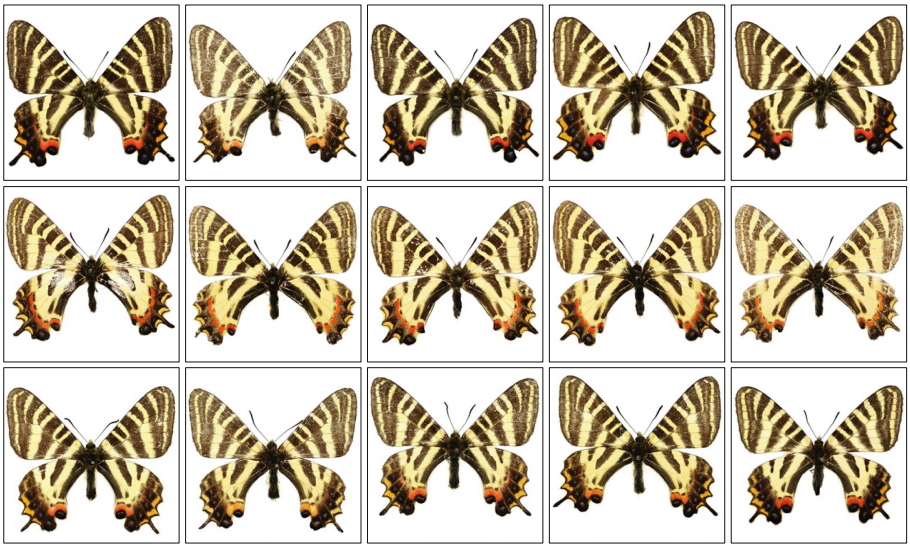


Fig. 3. Examples of pre-processed Luehdorfia japonica images.

Table 1 shows a breakdown of the labels in the *Luehdorfia japonica* dataset. Note that the total number of labels does not equal the number of samples, as a single sample may have multiple labels. From the Table 1, there are several labels for which there are not a sufficient number at this time. Therefore, we will limit the number of labels used in the experiment and conduct the experiments.

4 Method

4.1 Network Model

Figure 4 shows the network structure of the multi-label classification ABN. The feature extractor extracts a feature map common to all labels from an input image, and the extracted feature map is input to the attention branch. The attention branch performs global average pooling (GAP) [18] according to each label and obtains the classification result of each label. At the same time, the attention map, which is a feature map obtained from the convolution layer before GAP, is input to the perception branch. After applying weighting (attention mechanism) to the feature maps output from the feature extractor, the final classification result is output. This makes it possible to visualize the regions gazed at during classification, and to perform classification using features that emphasize the features of those regions.

4.2 Model Training Considering Class Imbalance

As described in the previous section, the dataset used in this study has a class imbalance problem. Therefore, if we train a model naively, learning proceeds

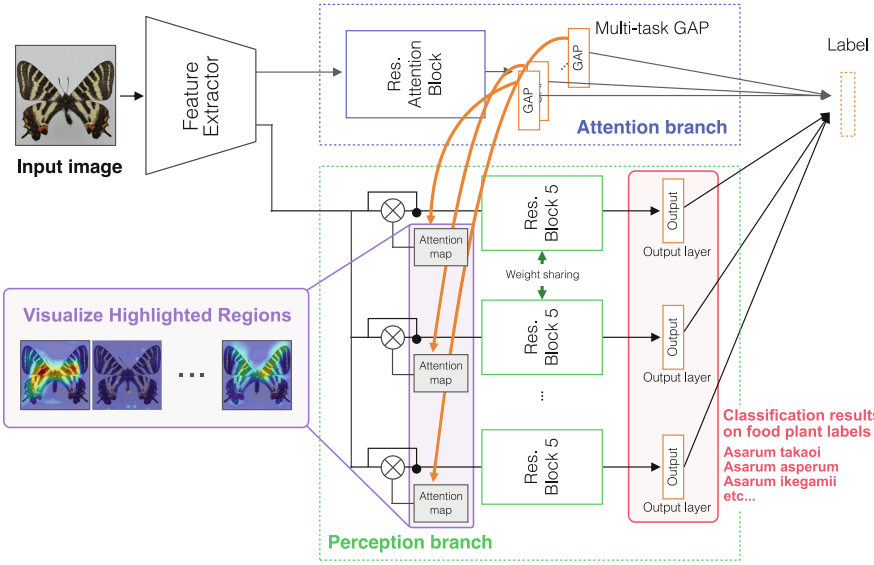


Fig. 4. Network structure of Multi-label classification ABN. The model first extract feature map from an input image via the feature extractor (e.g., ResNet). Then, the extracted feature map is fed into the attention branch, where we further extract feature map and predict classification score passing thrhough global average pooling (GAP). Meanwhile, by using the feature map obtained from the feature extractor and the attention maps by GAP, we compute final prediction results at the perception branch. In addition to the prediction, we can obtain attention map as visual explanation.

mainly on labels with a large number of samples, which may bias the classification accuracy. Furthermore, since the objective of this study is not only to achieve high classification accuracy but also to analyze butterfly spots through visualization of the regions gazed at by the model during classification, it is necessary to obtain appropriate classification and reasonable gazing regions. In this study, we experimentally examine changes in accuracy and acquired attention maps by introducing a learning method that takes class imbalance into account. Specifically, we introduce (i) an error function and (ii) data sampling that accounts for class imbalance.

Loss Functions. The ABN error function L_{abn} is defined as

$$L_{abn} = L_{att} + L_{per}. \tag{1}$$

In other words, in multi-label classification ABN, the errors (L_{att}, L_{per}) between the output of the attention and the perception branches and the correct label is calculated. Conventional multi-label classification ABN uses binary cross-entropy loss (BCE) for each of (L_{att}, L_{per}), but it does not learn enough class

labels with insufficient number of samples, which may cause recognition accuracy degradation. In this study, we use the following error functions that take class imbalance into account.

One is a Weighted Focal Loss (WFL) [27]. WFL is a weighted error function based on the prior distribution of labels in the focal loss [19]. Let x be the logit for a class c and y^c be the positive solution label. Also, when defining the probability $p = \sigma(x)$ using the sigmoid function $\sigma(\cdot)$ and x, WFL is defined as follows:

$$L_{wfl} = -\sum_{c}^{C} w_c \left(y^c (1-p)^\gamma \log(p) + (1-y^c) p^\gamma \log(1-p) \right), \tag{2}$$

where $w_c = e^{-a_c}$ is the weight for class c, calculated from the prior distribution a_c of class c.

The other is an Asymmetric Loss (ASL) [25]. ASL is a loss function that takes into account the balance between positive and negative classes and is defined as follows based on focal loss.

$$L_{asl} = -\sum_{c} \left(y^c (1-p)^{\gamma_+} \log(p) + (1-y^c) p_m^{\gamma_-} \log(1-p_m) \right) \tag{3}$$

where $p_m = \max(p - m, 0)$ is the shifted probability to ignore errors in the easy negative class of classification and m is a parameter representing the margin of probability to ignore errors. By adjusting the balance of the parameter (γ_+, γ_-), the error calculation takes into account the balance between positive and negative classes.

Data Sampling. Approaches that consider balance when learning and classifying with unbalanced data include "oversampling," which samples more samples from rare classes, and "undersampling," which samples fewer samples from major classes. In this study, in addition to the aforementioned error functions, we consider differences in discrimination accuracy and obtain attention maps by adjusting the data sampling method. Specifically, by referring to the number of male labels shown in Table 1, we undersampled the sample of *Asarum takaoi*, which has a significantly large number of samples, and oversampled the data with a small number of samples such as *Asarum ikegamii*.

Thus, we can take the class balance into account in the training process.

5 Experiments

5.1 Experimental Settings

Dataset. As a dataset, we use the *Luehdorfia japonica* dataset described in Sect. 3.2. The image samples are resized to 448×448 pixels and input to the network. Of this dataset, we use only 3,510 male images. Among them, we use 3,503 images with 12 hostplant labels for the experiment (see the "Use in exp."

column in Table 1), as they provide a sufficient number of samples for training and evaluation. The 3,503 samples are divided randomly, and 2,799 are used for training and 704 are used for evaluation. In addition, the following two types of training sample sets are used in the experiment.

Unbalanced A training set that uses the above 2,799 randomly selected samples as they are. Experiments are performed on the dataset with class imbalance. The training dataset is affected by the class balance of the collected dataset, resulting in a large bias in the ratio of the number of labels per attribute.

Balanced A training set in which the data sampling was performed so that the proportion of the number of attribute labels in the training samples is as equal as possible according to the number of attribute labels, as described in Sect. 4.2.

The number of each training sample for each class label are shown in Table 3.

Model. As a network model, we use a multi-label classification ABN with a backbone of ResNet-18 [8] pre-trained on ImageNet [4]. The mini-batch size is 32 and the number of training cycles is 1,000 epochs. During training, we apply weak data enhancements such as horizontal flip, random crop, and contrast transformation. Momentum SGD (learning rate= 0.01, momentum= 0.9) is used as the optimization method for all training, and the learning rate is divided by 10 at the 500 and 750 epochs.

Evaluation Metrics. F1-score and mean average precision (mAP) will be employed as evaluation metrics. Moreover, by visualizing the acquired attention map, a qualitative evaluation will be conducted, and the relationship between the gazing area and the spots will be discussed from a butterfly expert's point of view.

5.2 Accuracy Comparison

Table 2. Classification accuracy [%].

	Training set: unbalanced		Training set: balanced	
	F1-score [%]	mAP [%]	F1-score [%]	mAP [%]
BCE	74.49	92.26	88.68	92.81
WFL [27]	74.61	93.78	88.79	92.74
ASL [25]	81.74	89.23	80.14	92.91

Table 2 shows the F1-score and mAP of averages over every larval host-plant label. Comparing the unbalanced dataset (Unbalanced) with the balanced

Table 3. F1-score over each class [%]. The column of "# of samples" indicates the number of labeled samples on training dataset.

Training set	Label (*Asarum*)	# of samples	BCE	WFL	ASL
Unbalanced	*Asarum curvistigma*	188	100.00	100.00	98.94
	A. nipponicum	99	91.30	95.83	93.61
	A. yoshikawae	44	53.33	30.76	77.77
	A. megacalyx	272	66.66	75.96	67.82
	A. nipponicum var. *saninense*	43	30.76	30.76	77.77
	A. rigescens var. *brachypodion*	72	41.66	53.84	57.14
	A. asaroides	73	97.29	94.44	97.29
	A. tohokuense	33	87.50	87.50	94.11
	A. takaoi	1,659	85.30	80.00	74.28
	A. asperum	272	88.37	89.70	85.00
	A. ikegami	115	52.38	57.14	58.53
	A. blumei	287	99.30	99.30	98.59
	mean	–	74.49	74.60	81.74
Balanced	*A. curvistigma*	282	100.00	100.00	100.00
	A. nipponicum	294	97.95	100.00	100.00
	A. yoshikawae	418	73.68	77.77	72.72
	A. megacalyx	408	80.85	76.11	71.42
	A. nipponicum var. *saninense*	408	95.23	73.68	80.00
	A. rigescens var. *brachypodion*	223	63.41	80.00	68.57
	A. asaroides	365	97.43	100.00	88.37
	A. tohokuense	330	100.00	100.00	56.25
	A. takaoi	610	96.17	96.60	93.57
	A. asperum	408	84.76	88.23	71.35
	A. ikegami	436	74.57	73.07	67.69
	A. blumei	576	100.00	100.00	91.71
	mean	–	88.67	88.79	80.14

dataset (Balanced), the F1-score improves when using Balanced. In particular, the F1-score is significantly improved when BCE and WFL are used, indicating that sample balance adjustment contributes significantly to the accuracy of the loss function without considering the balance between positive and negative classes.

In addition, focusing on the difference in loss functions, ASL obtains a higher F1-score than the other error functions when using Unbalanced. On the other hand, the F1-score is lower for the Balanced dataset than for BCE and WFL. This confirms that ASL is effective when training on datasets with significantly unbalanced samples.

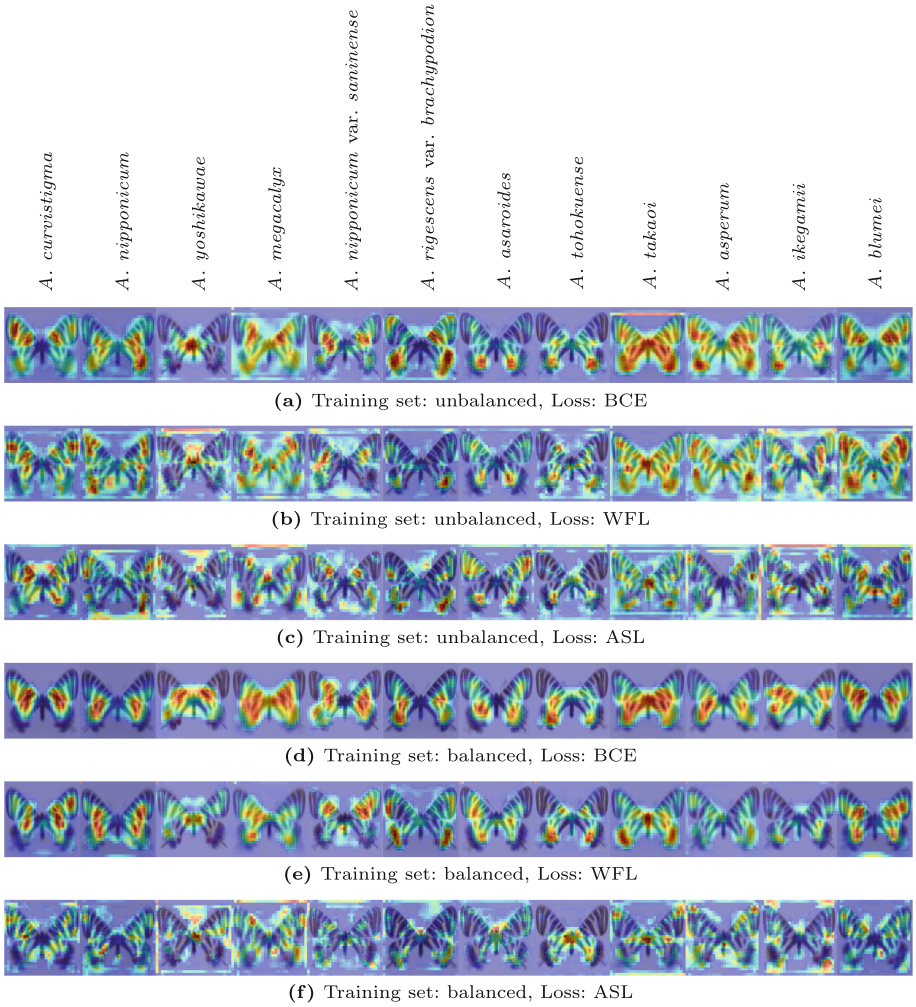
Next, the F1-score for each larval hostplant label is shown in Table 3. The table also shows the number of training samples for each larval hostplant label. The F1-score of the unbalanced case shows that ASL improves the F1-score of the hostplants with a small number of samples, i.e., *Asarum yoshikawae*, *Heterotropa nipponica* var. *saninense*, and *Asarum tohokuense*. ASL greatly improves the accuracy of attribute classes with a small number of samples, i.e., a large number of negative samples.

On the other hand, the results using Balanced showed no significant change in ASL accuracy, but a large improvement in BCE and WFL accuracy. The reason for this improvement in accuracy can be attributed to the strong co-occurrence of the hostplant labels. Specifically, from Fig. 1(b), there is no possibility that both *Asarum blumei* and *Asarum asaroides* are positive, and if *Asarum fauriei* var. *nakaianum* is positive, then *Asarum takaoi* is also positive, and so on. The co-occurrence of positive labels for this larval hostplant classification is considered to be very strong compared to attribute estimation for pedestrian and facial images, which are common computer vision tasks. These results suggest that the distribution of labels is relatively simple and that a simple adjustment of the number of samples greatly improves accuracy. When more complex combinations of positive labels are present, improvements can be expected from loss functions such as ASL, but a detailed analysis of the co-occurrence and distribution of hostplant labels is one of our future works.

5.3 Evaluation of Attention Maps

Figure 5 shows the average images of attention maps for each hostplant label. In the attention maps shown in Fig. 5(a-c) when using the Unbalanced training set, there are cases where the entire wing is gazed at, and cases where strong attention occurs in the background, which is unnecessary for classification. On the other hand, in the result after the sample balance was adjusted (Fig. 5(d-f)), the area of attention to the background was suppressed, and the area of attention existed in the region of the geese. This indicates that the sample balance adjustment contributes to the acquisition of stable attention, and that stable attention maps can be obtained by data sampling.

Next, we compare the differences when the error function is changed in the attention map using the Balance as shown in Fig. 5(d-f). BCE and WFL with high F1-score show almost no strong attention to the background area and focus mainly on the body and wings. On the other hand, ASL showed reduced noise compared to the unbalanced case but gazed at areas that should not be recognized compared to BCE and WFL. Consequently, it is clear that sample balance adjustment plays a significant role in obtaining highly explanatory attention maps, and that changing the error function tends to improve the accuracy but not the explanatory power of the attention maps.



(a) Training set: unbalanced, Loss: BCE

(b) Training set: unbalanced, Loss: WFL

(c) Training set: unbalanced, Loss: ASL

(d) Training set: balanced, Loss: BCE

(e) Training set: balanced, Loss: WFL

(f) Training set: balanced, Loss: ASL

Fig. 5. The attention maps taking average over each hostplant label. The top part of the figure shows the name of the hostplant label to which each column of the attention map corresponds.

5.4 Discussion Based on Expert Findings

Herein, we discuss the relationship between the acquired gazing areas and the spots. Specifically, we present to the butterfly experts the attention map (Fig. 5(e)) obtained by using the sample-adjusted WFL with the highest accuracy from the above results and with a stable gazing on the wings of the butterflies.

As a consideration for the overall trend, the attentions are often hit near the upper margin of the forewing (the middle of the costa to the middle of

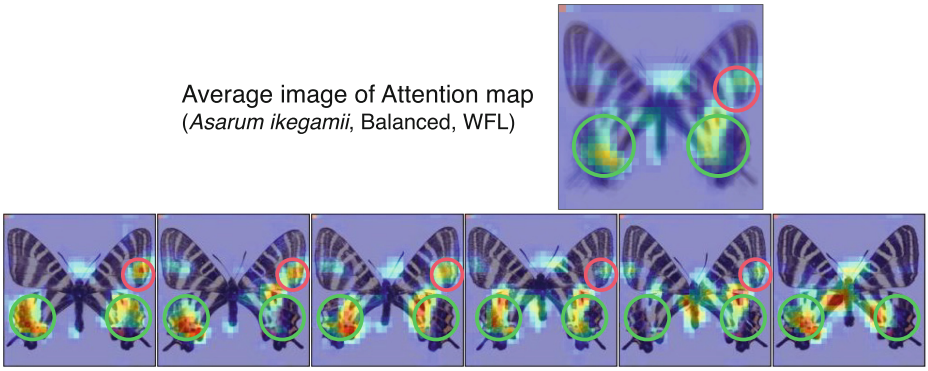


Fig. 6. Attention maps for *Asarum ikegamii*. Top row: the average image of attention maps for *Asarum ikegamii* label. Bottom row: individual attention maps for *Asarum ikegamii* label. Every attention map images are obtained from a model trained with balanced training set and WFL.

the discoidal cell on the forewing). This is known to be a representative area for experts to check when estimating the collection locality [5]. Therefore, we can say that the gazing area acquired by the multi-label classification ABN is consistent with the general findings of experts and that the explanatory nature presented by the attention map is valid.

In addition to the overall evaluation, we focus on the attention map for the *Asarum ikegamii*. Figure 6 shows the average image of attention maps of *Asarum ikegamii* and several individual attention maps. First, the attention of *Asarum ikegamii* is concentrated around the rearwing (green circle in Fig. 6), which corresponds to the conventional expert's finding. Meanwhile, the other attention is concentrated near the lower margin of the forewing (red circle in Fig. 6). This location differs from the expert's findings and is one of the future issues to be addressed, as a detailed analysis in the future may reveal new features of the spots that have not been recognized before.

6 Conclusion

In this paper, we recognize the larval hostplant classes from *Luehdorfia japonica* images and visualize the basis of decision making on the classification results as an attention map using the multi-label classification ABN. We achieved high classification accuracy by training with a loss function that takes into account class imbalance and data sampling. The analysis of larval hostplant classification and butterfly spots using the attention map revealed that the learned attention map is focused on the areas that match the expert's knowledge.

Our future work includes more detailed validation and the identification of new trends in regional variation beyond the experts' findings. In addition, we will gradually add more *Luehdorfia japonica* samples to the dataset and make the database publicly available.

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Depth Any Canopy: Leveraging Depth Foundation Models for Canopy Height Estimation

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Abstract. Estimating global tree canopy height is crucial for forest conservation and climate change applications. However, capturing high-resolution ground truth canopy height using LiDAR is expensive and not available globally. An efficient alternative is to train a canopy height estimator to operate on single-view remotely sensed imagery. The primary obstacle to this approach is that these methods require significant training data to generalize well globally and across uncommon edge cases. Recent monocular depth estimation foundation models have shown strong zero-shot performance even for complex scenes. In this paper, we leverage the representations learned by these models to transfer to the remote sensing domain for measuring canopy height. Our findings suggest that our proposed Depth Any Canopy, the result of fine-tuning the Depth Anything v2 model for canopy height estimation, provides a performant and efficient solution, surpassing the current state-of-the-art with superior or comparable performance using only a fraction of the computational resources and parameters. Furthermore, our approach requires less than $1.30 in compute and results in an estimated carbon footprint of 0.14 kgCO_2. Code, experimental results, and model checkpoints are openly available at github.com/DarthReca/depth-any-canopy.

Keywords: Canopy Height Maps · Remote Sensing · Monocular Depth Estimation

1 Introduction

Measuring tree canopy extent and height accurately is crucial to tracking the health of our world's ecosystem [36]. However, manual in-situ measurements or airborne laser scans (ALS) using LiDAR [1,9] are expensive and slow, making them unscalable for acquiring global measurements [33]. Furthermore, each acquisition method comes with their own sensitivities and errors [29]. Consequently, there is a growing need for more accessible and cost-effective approaches to canopy height estimation (Fig. 1).

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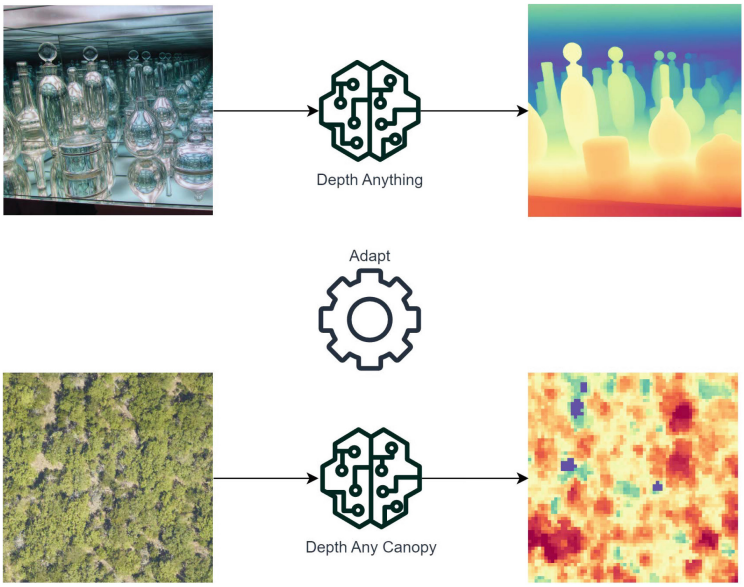


Fig. 1. From Depth Anything [39] to Depth Any Canopy. Depth Anything is a monocular depth estimation foundation model trained on natural imagery. We fine-tune and adapt Depth Anything v2 for the task of estimating tree canopy height in remote sensing imagery, resulting in Depth Any Canopy (DAC).

Remotely sensed satellite data is currently in abundance and an efficient alternative; however, satellite sensors built for measuring canopy height, such as GEDI [6], provide measurements which lack high spatial resolution. With this said, the most reasonable automated solution is to utilize machine learning to extract canopy height learned from satellite imagery geospatially aligned with ALS-derived canopy height maps [8].

One promising alternative is to develop a canopy height estimator that uses single-view remotely sensed image. However, this approach faces the substantial challenge of requiring extensive training data to achieve generalization across diverse global environments. Foundation models have displayed the importance of large-scale pretraining, which allows for the high-quality pretrained weights to be fine-tuned in a low-cost manner for various downstream tasks [4,42].

The task of canopy height estimation from remote sensing imagery can be posed similarly to the depth estimation task, where the camera view is always fixed overhead and we seek to measure distance above ground, with ground represented as zero. Monocular depth estimation for natural imagery has recently experienced significant zero-shot performance improvements by pretraining on complex synthetic data and then pseudo-labeling a large-scale dataset for further semi-supervised training [38]. These models have shown the capability of being adapted for various applications, suggesting the opportunity for cross-domain transfer learning.

In this work, we explore the adaption of depth foundation models to remote sensing domain, specifically for tree canopy height estimation, without requiring significant amount of pretraining on remote sensing imagery. We propose leveraging Depth Anything v2 [39], a state-of-the-art monocular depth estimation model, to enhance the accuracy and efficiency of canopy height estimation from aerial imagery.

Our findings indicate that a proper finetuning of the model not only surpasses the current state-of-the-art performance but does so with fewer computational resources. Using this new model, Depth Any Canopy (DAC), we aim to provide a scalable and efficient solution for global canopy height estimation, making a step forward in providing high-quality canopy height data more accessible and contributing to better forest management and climate change mitigation efforts.

Our contributions can be summarized as follows:

1. We adapt monocular depth estimation foundation models, Depth Anything v2, fine-tuning them for canopy height estimation derived from remote sensing imagery, resulting in Depth Any Canopy (DAC).
2. We achieve superior or comparable results to a larger baseline with significant advantages of being pretrained on millions of in-domain satellite imagery.
3. Our resulting fine-tuning process and models are efficient and low-cost. They can be reproducible using minimal consumer GPU hardware.

The rest of the article is structured as follows. Section 2 discusses the related works in canopy height estimation and foundational models in computer vision and remote sensing. Section 3 describe the datasets employed in the study. Section 4 presents the models, the metrics, and adopted preprocessing. Section 5 presents the experimental results from both quantitative and qualitative perspectives. Section 6 draws the conclusion and presents future works.

2 Related Works

In this section, we discuss the related works in canopy height estimation and monocular depth estimation.

2.1 Canopy Height Estimation

Various techniques have been employed to estimate canopy height, ranging from traditional field measurements [40] to advanced remote sensing technologies [31]. One common tool is Light Detection and Ranging (LiDAR) due to its high spatial resolution and accuracy, allowing for detailed 3D representations of forest canopies [31]. The integration of LiDAR data with optical imagery and the application of machine learning algorithms have further enhanced canopy height estimation, thanks to the ability to deal with complex and heterogeneous environments [35]. The usage of LiDAR technologies is more expensive than RGB cameras, and generating canopy height maps (CHM) using deep learning models trained on remotely sensed satellite or aerial imagery has become popular

as methods have been able to provide accurate and higher-resolution CHM estimates. Lang et al. [16] trained a model to produce a low-resolution 10 m global canopy height map by fusing GEDI and Sentinel-2 satellite imagery. Becker et al. [2] similarly created a low-resolution map by using Bayesian deep learning to estimate canopy height in fused Sentinel-1 SAR and Sentinel-2 optical imagery. Pauls et al. [23] improved upon these works by creating a novel shift resilient loss to adjust their ground truth for the intricacies of the GEDI global height product. Fogal et al. [7] created a 1.5 m resolution multitemporal dataset using aligned SPOT satellite imagery and ALS acquisitions in France. Tolan et al. [32] took a self-supervised learning approach by pretraining a Vision Transformer (ViT) [5] using the DINOv2 [22] method on 18 million crops of globally sampled high-resolution WorldView satellite imagery. The authors then fine-tune this model to estimate canopy height using a dataset of U.S. based imagery and 1 m resolution ALS-derived CHMs from the NEON catalog. While this approach works in practice, a significant amount of compute and efforts to collect these datasets are required.

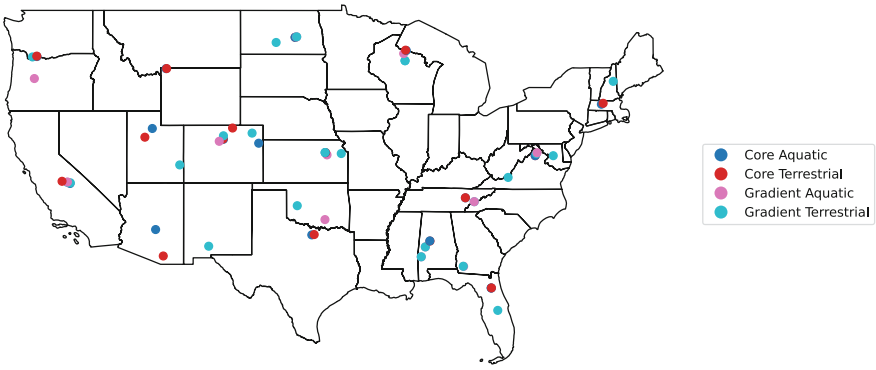


Fig. 2. National Ecological Observatory Network sites across the US. Aquatic sites collect information about aquatic ecosystems, while terrestrial ones collect data about terrestrial ecosystems. Core sites provide long-term support, while Gradient sites are temporary sites to study the ecological response to specific changes.

2.2 Foundation Models in Computer Vision and Remote Sensing

Foundation models are becoming one of the prevalent solutions in many fields when there is no sufficient training time or large data availability [42], which is particularly relevant for transformers. In the computer vision field, ViT [5], and subsequent CLIP [24] and DINOv2 [22] have become well-established and robust encoders employed in many architectures and downstream tasks. Models like Segment Anything [13] and OWL-ViT [21] make use of these advancements

for general-purpose usage. In the remote sensing field, following these advancements, many solutions were proposed providing generative models [12], multimodal models [14], or task-specific solutions [19,41]. With this said, foundation models remain unexplored for the task of canopy height estimation.

2.3 Monocular Depth Estimation

Monocular depth estimation for natural imagery, being a more mainstream area of computer vision research, experiences faster improvements than remote sensing height estimation topics. The MiDaS/DPT family of models [3,17,25] has shown that training for relative depth on a mixture of datasets, instead of training on small individual metric depth datasets such as NYU Depth V2 [30], ETH3D [28], and DIODE [34], experiences better transfer learning performance. MegaDepth [18] and Depth Anything [38,39] have found that pretraining for relative depth estimation on higher quality and large-scale synthetic and pseudo-labeled images further improves zero-shot generalization performance.

Furthermore, Marigold [11] displayed the efficiency of requiring a single consumer NVIDIA RTX 4090 GPU for fine-tuning image generation diffusion models pretrained on large-scale datasets for the task of depth estimation. These advancements in natural imagery provide an interesting and promising alternative to the full training of models for canopy height estimation, and thanks to the pre-training, they provide solid baselines for predicting relative distances.

3 Datasets

In this section, we describe the data sources and the datasets employed for the analysis.

3.1 Data Sources

The primary data source for high-resolution CHM is derived from the National Ecological Observatory Network (NEON) catalog [10]. The network is composed of many sites across the U.S., as shown in Fig. 2. Terrestrial sites collect data to monitor changes in climate, surface-atmosphere interactions, biogeochemical processes, organismal populations, and habitat structure over the land, while aquatic sites collect data to monitor changes in freshwater and biogeochemical processes, organismal populations, and habitat structure in water sources. Core sites provide long-term support, while Gradient sites are temporary sites to study the ecological response to specific changes. In our specific case, the data consists of high-resolution, multi-spectral, and hyperspectral aerial imagery captured from aircraft over various ecological sites. Multi-spectral imagery captures data in several specific wavelengths, typically including visible and near-infrared bands. Hyperspectral imagery captures data in hundreds of contiguous spectral bands. This imagery provides detailed spatial data on vegetation, land cover,

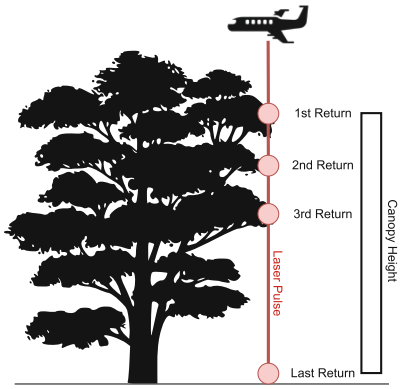


Fig. 3. ALS-derived tree canopy height acquisition process. LiDAR sensors attached to a fixed-wing aircraft, emits a laser pulse that reflects many times until reaching the ground. The canopy height is computed using the time-delta between the first and the last return of the pulse, the pulse obtained when reaching the ground level.

and water bodies, aiding in vegetation health assessment, species distribution, and biomass monitoring.

LiDAR sensors use laser light to measure distances to the Earth's surface and various objects from aircraft or drones. These sensors emit thousands of laser pulses per second toward the ground, capable of penetrating vegetation. These pulses hit various surfaces, such as the canopy, branches, and the ground, reflecting back to the sensor and providing punctual information. Points representing the highest surfaces in the forest canopy are used to create a Digital Surface Model (DSM), including the canopy's top. The Canopy Height Model (CHM) is derived by subtracting the Digital Terrain Model (DTM), which represents the bare earth surface, from the DSM. This provides the height of the canopy above the ground surface [26] as shown in Fig. 3.

3.2 EarthView Dataset

The Satellogic EarthView Dataset [27] is a comprehensive collection of multispectral earth imagery for general-purpose tasks. The dataset is divided into four distinct subsets sourced from Satellogic, Sentinel-1, Sentinel-2 satellites and aerial imagery and ALS-derived CHMs from the NEON catalog. While the dataset provides satellite imagery provided by one commercial satellite (Satellogic) and two publicly available European Space Agency missions (Sentinel-1 and Sentinel-2), we employed only the aerial imagery, which is the only one with canopy height maps by LiDAR sensor, and it is the common imagery type between the two analyzed datasets. The NEON subset is composed of very high-resolution RGB images at 0.1 m paired with 1 m CHMs.

3.3 High Resolution Canopy Height Maps Dataset

The High-Resolution Canopy Height Maps (HRCHM) [32] is a collection of approximately 5,800 NEON CHMs with an area of 1 km × 1 km and a resolution of 1 m per pixel. Samples were selected based on quality, minimal artifacts, and temporal proximity to a collection of Maxar satellite imagery. The CHMs were reprojected and resampled to match the Maxar imagery resolution and paired with a corresponding RGB satellite images to train a canopy height estimation model. The dataset was split into a train and test set, referred to as the *NEON test set*, and preprocessed into 256 × 256 random crops.

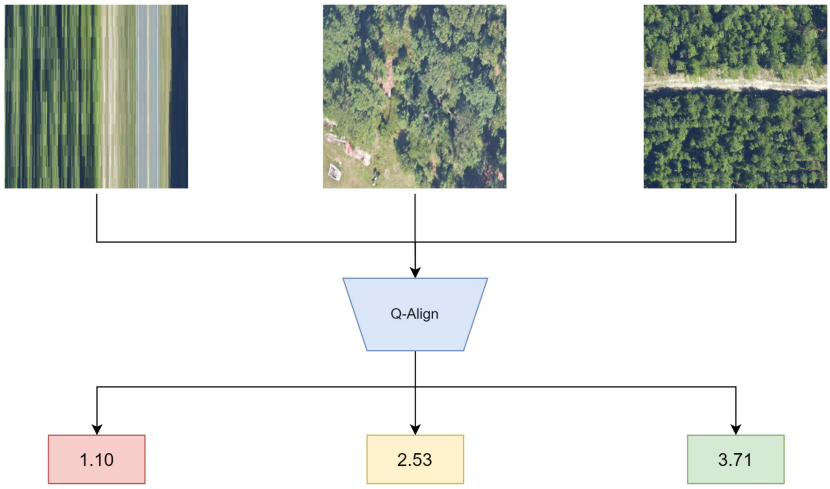


Fig. 4. Example quality scores by Q-Align [37] on NEON RGB images from the EarthView dataset. On the left, a noisy sample scored *1.10*; in the middle, a medium-quality sample scored *2.53*; and on the right, a better-quality sample scored *3.71*. This solution permits the detection and filtering of low-quality samples affected by warping and motion blurs.

4 Methodology

This section outlines the model, preprocessing, and metrics used in the analysis.

4.1 Problem Statement

Let I be an arbitrary RGB image of size $W \times H \times 3$, where W and H are the width and height of the image in pixels, respectively, and the channel depth of the image is 3, corresponding to the red, green, and blue color channels. The objective is to estimate the canopy height map M from this image. M can be represented by a matrix of size $W \times H$, where each element $M_{i,j}$ represents the estimated height of the canopy for the corresponding pixel $I_{i,j}$ in the image I.

4.2 Preprocessing

The NEON imagery in the EarthView dataset contains a significant number of samples with motion blur or warping due to image stitching errors. We find that these samples significantly degrade fine-tuning performance and are likely unuseful for evaluation. Therefore, we seek to filter these samples by employing the Q-Align [37] image quality assessment vision-language model to score the quality of each sample between a range of [0–5] where higher values indicate better quality as shown in Fig. 4. We then filter low-quality samples using a score threshold of $t = 2.5$ (which indicates above-average quality). The training set contains 45781 samples, validation set 5788, and test set 5682. Every sample contains at least one pixel with the canopy height different from zero. According to KS-test the distribution of heights is similar between train-val ($p \approx 0.68$) and train-test ($p \approx 0.75$)

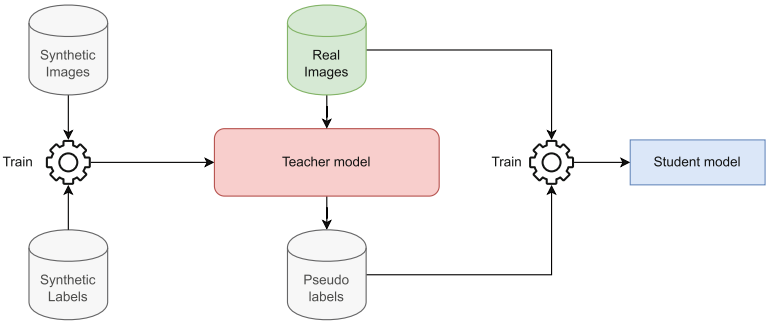


Fig. 5. Depth Anything v2 training procedure. Synthetic images and relative labels are used to train a large teacher model. It is employed to annotate real images to create pseudo-labels. The real images and relative pseudo-labels are used to train a small student model.

4.3 Depth Anything for Canopy Height Estimation

In our study, we used Depth Anything v2 in both zero-shot and with a finetuning on EarthView dataset, called Depth Any Canopy, for addressing the canopy height estimation.

Depth Anything [38] is a state-of-the-art monocular depth estimation model. It features a DINOv2 pretrained ViT encoder [22] and DPT decoder [25]. The model is trained using an online student-teacher distillation with a combination of real depth maps and pseudo-labeled depth maps of 62 million imagery. It excels in zero-shot generalization and improves performance when fine-tuned for metric depth on evaluation benchmark datasets.

The successor, Depth Anything v2 [39], takes a different approach and only pretrains on high-quality synthetic image and depth map pairs due to invalid

pixels and low complexity of scenes in canonical metric depth datasets. It then uses the synthetic pretrained model to generate precise pseudo-labels of 62+ million real images in an offline manner. A student model is then trained on these pseudo-labels as shown in Fig. 5. The model uses scale-and-shift invariant losses to ensure consistency and gradient matching loss for sharper depth predictions [25]. This model not only outperforms previous state-of-the-art models in accuracy, but also speed, and efficiency primarily due to the large-scale and high-quality pretraining dataset.

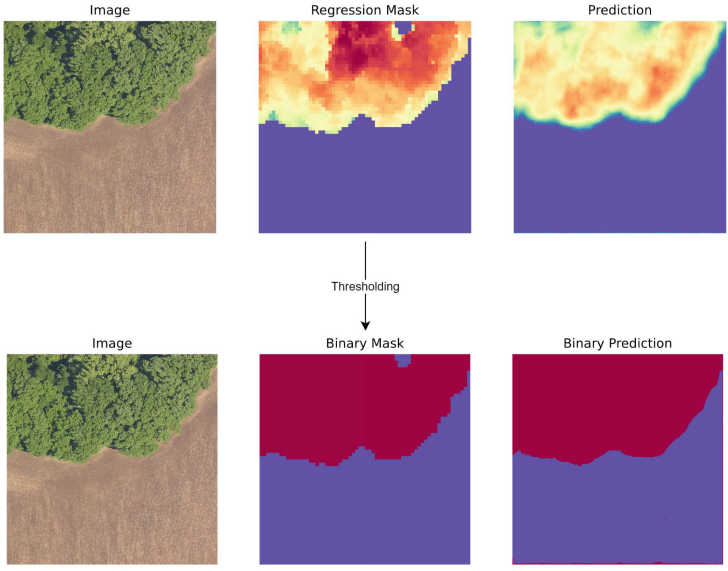


Fig. 6. Tree Canopy Extent evaluation process. We threshold the ground truth and predicted canopy heights to obtain binary masks. We utilize this to evaluate a model's effectiveness in identifying tree extent.

4.4 Baseline

As described in Sect. 2, Tolan et al. [32] proposed to pretrain a model using the DINOv2 on a large-scale dataset of 18 million 256 × 256 RGB crops from global Maxar WorldView satellite imagery. This model was then fine-tuned on the HRCHM dataset described in Sect. 3.3.

4.5 Metrics

In our experimentation, we analyze the predicted performance on the canopy height estimation regression task using the Mean Absolute Error (MAE) (L1

distance). Furthermore, we evaluate the ability of the model to segment the tree canopy from the background by using the same process by Tolan et al. [32] to mask the predicted and ground truth CHM using a threshold of $t = 1E - 4$ (to include also very small trees which were excluded by the previous work) as shown in Fig. 6. We then compute binary segmentation performance using the Intersection-Over-Union (IoU) metric to understand the quality of the predictions of areas considered as trees, tree canopy extent. Lastly we compute the Pearson Correlation (PC) of prediction and ground truth in the tree areas to understand whether relative heights are consistent despite the errors. To compare model size and efficiency, we report the floating-point operations per second (FLOPs) in billions (giga) and the number of parameters in millions.

Table 1. Results on EarthView and HRCHM datasets. The best for each metric is bolded, while the second-best is underlined. *FT* indicates if the model is finetuned on EarthView. *DA* and *DAC* refers to Depth Anything v2 and Depth Any Canopy, while *DAC-S* and *DAC-B* refers to the ViT-S and ViT-B variants, respectively.

Model	FT	# Params	GFLOPs	EarthView [27]			HRCHM [32]		
				MAE ↓	IoU ↑	PC ↑	MAE ↓	IoU ↑	PC ↑
SSL-H [32]	✗	677M	414	0.2236	0.4164	0.1544	**0.0306**	0.485	**0.7441**
DA-S [39]	✗	**24.8M**	**115**	0.4116	0.4164	0.2892	0.5960	**0.6474**	0.1791
DA-B [39]	✗	97.5M	381	0.4607	0.4164	**0.361**	0.5972	**0.6474**	0.1692
DAC-S [39]	✓	**24.8M**	**115**	0.1410	0.5323	0.2740	0.1025	0.5672	0.6102
DAC-B [39]	✓	97.5M	381	**0.1304**	**0.5926**	0.3483	0.1203	0.5494	0.6171

5 Experimental Results

In this section, we present the experimental settings and an analysis of the results.

5.1 Experimental Settings

The models were finetuned with the AdamW optimizer [20] for 3 epochs (until plateau) with a batch size of 8 on the EarthView training set using an NVIDIA RTX A6000 48 GB GPU. We employed a learning rate scheduler with a warmup of 5% of the training steps with linear decay. The maximum learning rate is set to $\alpha = 5E - 6$. We utilize the Mean Squared Error (L2 distance) between the ground truth and predicted canopy height as a loss function. The canopy height maps are min-max normalized (at dataset level) to relative height since EarthView does not provide height in meters, only a relative distance.

We evaluate and fine-tune the Depth Anything v2 ViT-S and ViT-B checkpoints, referred to as DA-S and DA-B, respectively into Depth Any Canopy

(DAC-S) and (DAC-B). We do not utilize the best performing ViT-G (Giant) weights because they have not been released. We baseline against the ViT-H (Huge) checkpoint from Tolan et al. [32], referred to as SSL-H, trained using the process described in Sect. 4.4. We utilize this checkpoint because no other smaller ViT variants of this model were made available.

We note that the SSL-H baseline is comparatively larger than the Depth Anything v2 checkpoints we initialize from. Furthermore, SSL-H is already pretrained in-domain on 18 million remote sensing images and fine-tuned on a large set of NEON aerial RGB imagery for canopy height estimation. Therefore, the Depth Anything v2 weights should be at a significant disadvantage when comparing to this baseline.

5.2 Discussion

Overall Results. We tested the SSL-Huge (SSL-H) [32] on both datasets (EarthView and HRCHM) to understand the performance and generalization capabilities. Depth-Anything (DA) was tested on both datasets in zero-shot and after finetuning on EarthView. This approach provides a comparison of both in-domain and out-of-domain data to understand the capability of generalization.

In Table 1, we reported the results obtained by each tested model. The fine-tuning of Depth Anything v2 on EarthView provides good comparative results and consistent performance under both MAE and IoU metrics, while the PC is generally higher on HRCHM for SSL-H.

SSL-H proves to be best in terms of MAE and PC on HRCHM, while the IoU is lower than the one achieved by DA. The performance is worse on EarthView, particularly on MAE, which decreases by 10 times with respect to the performance on HRCHM. Additionally, we can reach good performance with smaller models.

Zero-Shot Comparison. When comparing Depth Anything v2 in zero-shot to Depth Any Canopy, we can conclude it is necessary to adapt the model to this new unseen task to be an effective solution. For example, shows a reduction in MAE from 0.4116 to 0.1410 on EarthView and from 0.5960 to 0.1025 on HRCHM after fine-tuning DA-S to DAC-S. However, the main advantages in terms of resource efficiency are maintained by providing good results in fewer GFLOPs. In many cases, DAC-S provides the best or second-best performance in around 1/4 of the GFLOPs and 1/27 of the parameters of SSL-H [32]. DAC-S achieves results comparable to DAC-B, indicating it is a preferable solution for the task due to its low resource demands.

Carbon Footprint Analysis. As mentioned in Sect. 5.1, to achieve similar or comparable results to SSL-H by Tolan et al. [32], our DAC fine-tuning requires simply 3 epochs on the EarthView dataset using an NVIDIA RTX A6000 GPU. This requires 1.5 and 2.61 h at an estimated cost of $1.24 and $2.09 using the Lambda Labs hourly pricing for the DAC-S and DAC-B variants, respectively. We calculate that our fine-tuning results in an estimated 0.14 and 0.24 kg of

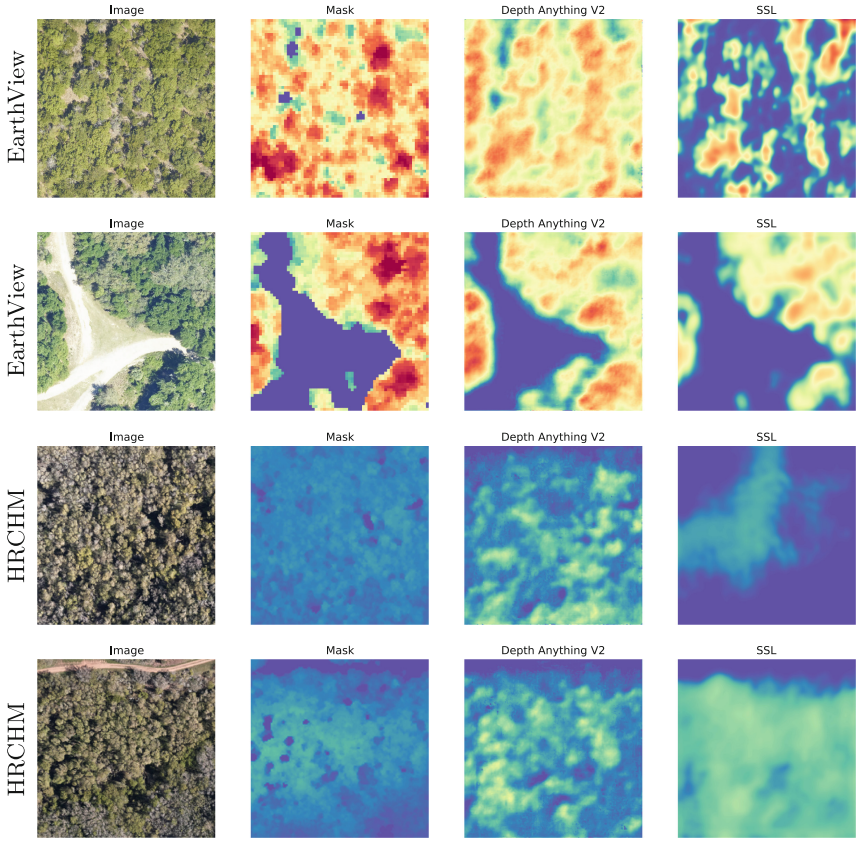


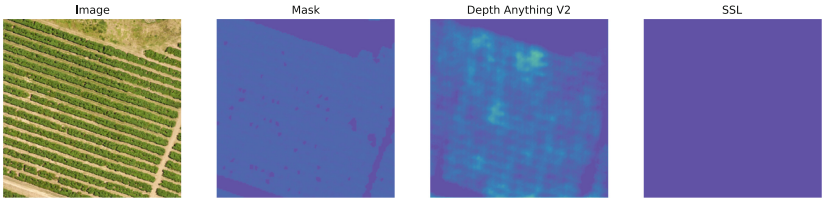
Fig. 7. Example predictions of Depth Any Canopy (DAC-S) and SSL-Huge model from Tolan et al. [32] on NEON imagery from the EarthView and HRCHM datasets. Left to right: NEON RGB Image, Ground Truth Canopy Height Map, DAC-S predicted CHM and SSL-H predicted CHM.

CO_2 emissions of using the ML CO_2 Impact calculator [15]. We believe this to be preferable to the 8kg of CO_2 emissions reported by our SSL-H baseline for fine-tuning, not considering the 1.8T of CO_2 emissions for pretraining.

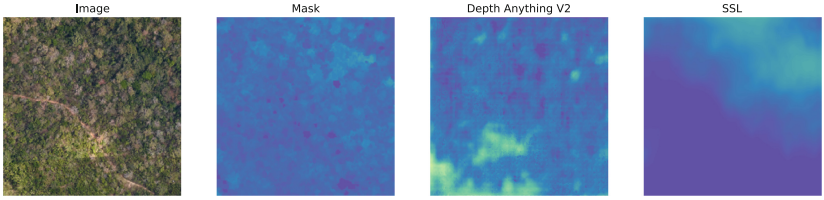
Qualitative Results. Figure 7 provides a qualitative example of predictions by DAC-S and SSL-H on both datasets. On EarthView, SSL-H does not recognize many trees as expected, while DAC-S provides better performance. On HRCHM, we can note that SSL-H underestimates or overestimates some areas, although providing good IoU performance. The variations captured by DAC are generally superior to the ones captured by SSL-H.

Figure 8 provides an overview of other cases. SSL-H avoids prediction on low trees, as in this qualitative example, while DAC-S tries to provide a height map of the area with a good level of detail. This shows the robustness achieved by

Depth Any Canopy. The model can deal with complex scenarios, providing a high level of detail for complex scenes.



(a) Example 1



(b) Example 2

Fig. 8. Examples of limitations of SSL-H addressed by Depth Any Canopy. Through our analysis, we find that the SSL-H model by Tolan et al. [32] tends to generate overly smooth predictions or predicts zero height for smaller vegetation. Because Depth Any Canopy is fine-tuned from the Depth Anything v2 weights it is able to recover vegetation heights for complex scenes and edge cases.

6 Conclusion

In this work, we have presented a novel approach to canopy height estimation by leveraging Depth Anything v2, a state-of-the-art monocular depth estimation foundation model. Our proposed model, Depth Any Canopy, demonstrates superior or comparable performance to current state-of-the-art methods with fewer computational resources. This is crucial for a scalable and cost-effective global canopy height estimation solution. Our findings show the potential of depth estimation foundation models pre-trained on large-scale natural imagery. They can be adapted for specific tasks in the remote sensing domain with minimal additional training. By fine-tuning Depth Anything v2, we have shown that it is possible to achieve high-quality canopy height maps from single-view images, overcoming the limitations of expensive pre-training on large-scale satellite imagery datasets to achieve comparable performance. In future works, we plan to expand the evaluation to a wider variety of forest biomes and geographical regions to account for more diverse environments and to investigate the usage of hyperspectral and radiometric imagery, which could enhance the understanding of the area with complex features.

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Underwater Uncertainty: A Multi-annotator Image Dataset for Benthic Habitat Classification

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Abstract. Continuous inspection and mapping of the seabed allows for monitoring the impact of anthropogenic activities on benthic ecosystems. Compared to traditional manual assessment methods which are impractical at scale, computer vision holds great potential for widespread and long-term monitoring. We deploy an underwater remotely operated vehicle (ROV) in Jammer Bay, a heavily fished area in the Greater North Sea, and capture videos of the seabed for habitat classification. The collected JAMBO dataset is inherently ambiguous: water in the bay is typically turbid which degrades visibility and makes habitats more difficult to identify. To capture the uncertainties involved in manual visual inspection, we employ multiple annotators to classify the same set of images and analyze time spent per annotation, the extent to which annotators agree, and more. We then evaluate the potential of vision foundation models (DINO, OpenCLIP, BioCLIP) for automating image-based benthic habitat classification. We find that despite ambiguity in the dataset, a well chosen pre-trained feature extractor with linear probing can match the performance of manual annotators when evaluated in known locations. However, generalization across time and place is an important challenge.

Keywords: Seabed monitoring · Benthic habitats · Label uncertainty

1 Introduction

Seabed integrity is challenged by anthropogenic activities causing habitat degradation such as fishing, mining and quarrying, and sustainable energy construction [19,28,38]. In the European Union, multiple directives are implemented

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to protect seabed integrity. The Marine Strategy Framework Directive requires member states to achieve Good Environmental Status (GES) in marine habitats [13] and the Habitat Directive and Nature 2000 focus on preserving species and habitats in Marine Protected Areas [10]. We specifically focus on Jammer Bay in the Greater North Sea (Skagerrak ICES area), a region heavily impacted by human activities and thus crucial for management. Despite being included in the Marine Strategy Framework Directive and encompassing several Marine Protected Areas, monitoring efforts in this region are limited.

Monitoring the seabed requires comprehensive, systematic and reproducible data collection methods. Compared to sediment sampling which is invasive, costly and highly localized, recording and analysing video footage is a promising technique for marine biologists [1,6,7,25,34,43,45–47]. When combined with computer vision methods, imaging technology provides unprecedented possibilities to enhance and automate underwater monitoring [4,5,32].

Fig. 1. Satellite view of the 16 data collection locations, color-coded by acquisition date and with image examples on the right.

Fig. 2. Illustration of the recording procedure from deploying the ROV, diving to the seabed, capturing video data of the benthic habitats, and, lastly, returning to the surface. Only the data captured at the seabed is used in this project.

Developing underwater image classifiers requires labelled data for training and evaluation [39]. While there is a growing number of underwater datasets for species identification [29,35,48], publicly available and annotated datasets

of seabed habitats are much scarcer. The recently released BenthicNet [31] is a large-scale compilation of seabed imagery from different sources worldwide. However, the majority of labelled images are from Australia and Tasmania, with coral reefs and clear waters [15,26]; the only labelled images from European waters are in Spain. It is important to establish reference datasets in other environments targeted by anthropogenic activities. In Denmark, the main source of visual data for benthic habitats is the marine raw material database (MARTA) [21] - however the images are mostly provided uncurated, unlabelled, or in poor quality.

To address this gap, we present the JAMBO dataset - the first publicly available multi-annotator benthic habitat classification image dataset from temperate waters. We deploy an underwater remotely operated vehicle (ROV) equipped with lights and cameras to capture footage of the seabed in Jammer Bay. Each curated image is labelled by six annotators with different backgrounds and expertise. Their task is to distinguish between sandy substrates and stone reef, or to mark the image as "bad". Analysing their disagreement allows us to evaluate the difficulty and ambiguity of classifying benthic habitat types from ROV images in this area. We then evaluate several baseline models, leveraging pre-trained vision transformers for feature extraction. To the best of our knowledge, this constitutes the first study of annotator disagreement and automatic benthic habitat classification in turbid environments resembling Jammer Bay.

2 The JAMBO Dataset

This section provides details on the data collection and annotation procedures. The JAMBO dataset can be found at https://vap.aau.dk/jambo.

2.1 Data Acquisition

The study took place in Jammer Bay, a part of the Greater North Sea off the Northwestern coast of Denmark, see Fig. 1. The water depths ranges from 14 to 34 m at the study sites. The environmental conditions in the area is affected by the North Atlantic drift and currents are generally moving from South to North along the Danish West coast. These conditions, combined with weather and resuspended particles, produce high degrees of variance in underwater turbidity and visibility. Relevant dive sites were determined by selecting three regions of management importance (commonly frequented by beam trawl, otter trawl, and Danish seine) and with limited habitat knowlegde, coupled with the utilization of a side-scan sonar to pinpoint areas featuring a combination of sandy substrates and stone reefs.

Data was collected with a Chasing M2 Pro Max underwater ROV featuring a 4K resolution front-facing camera solely used for navigation. A GoPro Hero Black 11 camera was mounted beneath the drone at a 45-degree angle to continuously record the seabed. Two 4000 lumen lights were placed in a downwards-facing position, spaced far apart, and angled in a way that ensured the seabed to be

illuminated in the field of view of the GoPro camera while minimizing backscatter. Camera settings and a picture of the ROV can be found in Appendix A.1.

Recording was initiated prior to submerging the ROV into the water and continued until it resurfaced at the boat as illustrated in Fig. 2. Therefore, the videos included several stages, from deployment to recovery of the ROV, leading to a significant amount of data not suitable for benthic habitat classification. Each dive lasted 20–30 minutes and approximately seven hours of video data was recorded in total. We conducted manual curation to identify the relevant segments where the ROV was in proximity of the seabed. Subsequently, we carefully reviewed the remaining footage to locate and remove erroneous segments, e.g., if the seabed was not in the camera's field of view due to navigation. A frame was extracted for every two seconds from 12 video segments, resulting in a total of 3290 frames to be annotated. Frames were center-cropped at a resolution of 2048×1024 to exclude lens distortion effects from the image.

2.2 Annotation Scheme

The material that makes up the sediment and substrate of a given habitat plays a vital role in the classification procedure [11] (referred to as level 2 in the European nature information system (EUNIS) hierarchy). The noise and lighting from the ROV may affect the behavior of organisms, potentially leading to escape responses, which complicates classification procedures that consider flora and fauna (level 3 and above in the EUNIS hierarchy). Thus, the aim is to build a pipeline for automatically classifying the primary material of the seabed. However, determining the true habitat type from visual inspection of images from Jammer Bay is not trivial due to the turbid and low light conditions in the area. Therefore, we have defined two general habitat types covering Jammer Bay, based on the EUNIS key navigation [18] and GEUS habitats definitions [30], resulting in the following class labels:

- **Sand** habitats are characterized as primarily sand or muddy sand with less than 5% clay and less than 30% cover of stones/boulders, vegetation, and mussel beds. The habitat is stable but under the influence of tidal streams, waves, and turbidity. This habitat classification is modified from the EUNIS classification *MC52 Atlantic circalittoral sand* [17] and corresponds to the 'sand dunes' habitat as defined by GEUS [30].
- **Stone** reef habitats are characterized by having more than 30% seabed cover of stones and boulders and are to some extent affected by tidal streams, waves, and turbidity. This classification is modified from the EUNIS classification *MC12 Atlantic circalittoral rock* [16] and corresponds to GEUS' 'reef' classification [30].
- **Bad** is a class used to label images that cannot be confidently annotated as containing one of the aforementioned habitat types by the annotator due to poor image quality, turbidity, or similar.

To investigate the uncertainty involved in labeling images of benthic habitats in turbid waters, we asked six annotators to annotate the same set of images. The

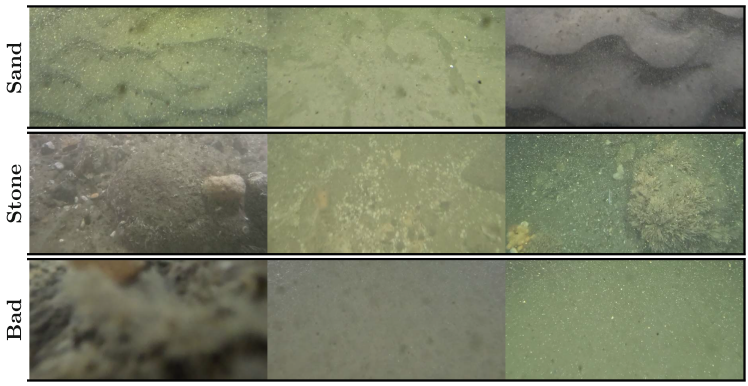


Fig. 3. Dataset samples illustrating variations in turbidity, lighting, and texture of the seabed. These images were unanimously labelled as sand (top), stone (middle) or bad (bottom) by all annotators.

group of annotators consisted of three marine biologists (*Bio1, Bio2, Bio3*) and three computer vision researchers (*CV1, CV2, CV3*) who were all familiar with the framework for classifying habitats, while the level of background knowledge regarding habitat classification varied among the annotators. One participant, *Bio1*, was a field expert with knowledge about the habitats in the Jammer Bay area and was also acquainted with the videos. The other two marine biologists, *Bio2* and *Bio3*, were not acquainted with the videos but had prior knowledge about marine habitats in general. The three computer vision researchers, *CV1, CV2*, and *CV3*, had viewed and processed the videos during the initial sorting phase but had no experience with marine habitat classification.

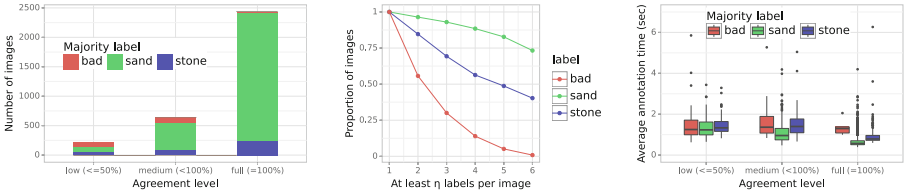
Images were displayed to the annotators at a fixed size of 1920×960, with different monitors used across annotators. For each image, we stored the class label assigned by each of the annotators, as well as the time spent assigning the label. Image examples from the three categories are presented in Fig. 3 showcasing some of the variations encountered within and between the classes.

2.3 Dataset Analysis

To gain insights into the characteristics and quality of the collected data, we first perform an analysis of the image content and manual annotations. We investigate the level of agreement between annotators as well as discrepancies between them, shedding light on the ambiguity associated with image-based habitat classification. We quantify the prevalence of sand vs. stone reefs, the proportion of *bad* images, and the time spent on manual annotation.

Annotator Agreement - Each image in our dataset is associated with six labels (one from each annotator). Its majority label is taken as the most frequent label; in cases where 2 or 3 classes appear with the same frequency across annotators, the majority label is chosen at random. We quantify the level

of annotator agreement for every image by taking the ratio of annotators who chose the majority label to the total number of annotators involved (six). For example, if three annotators labeled an image as *sand* while the remaining three selected different labels, the agreement level would be 50%. As illustrated in Fig. 4a, complete label agreement was observed for 2431 images (73.89% of the dataset), while high disagreement was noted for 209 images (6.35%). We see a notable prevalence of *sand* habitats, accounting for 82.67% (2720 images) of the data. Conversely, *stone* habitats are less frequently encountered, comprising only 11.88% of the dataset (391 images), resulting in a stone-to-sand ratio of 14.38% (almost 7 sand images for every stone image). A subset of 179 images (5.44%) has been categorized as *bad* by the majority of annotators, but only a small number (6) of images were unanimously labeled as *bad*. The graph presented in Fig. 4b elaborates this point by providing an indication of the uncertainty level for each of the classes. The vertical axis represents the proportion of images with respect to the number of images having at least one label of the given class, while the horizontal axis denotes the minimum number of labels present. Notably, approximately 75% of the *sand* images received unanimous labeling by all annotators, whereas for the *stone* images, this proportion is less than 50%. Additionally, the *bad* images show minimal unanimity among annotators.



(a) Number of images and label distribution for three agreement levels.

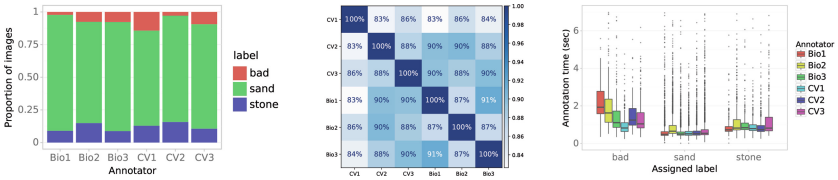
(b) Proportion of images with at least η label(s) for a given class.

(c) Annotation time for different agreement levels. Outliers beyond 7 seconds are not shown for readability.

Fig. 4. Quantitative analysis of inter-annotator agreement.

On average, an image takes less than a second to annotate and *sand* images tend to be the most quickly identified. More time is spent labeling *stone* images, while ambiguous or low-quality images subject to the *bad* class, take the longest time to annotate. Looking at the time spent per label across three agreement levels, we see that images unanimously labeled by all annotators require the least processing time, as depicted in Fig. 4c. This trend holds true for all three classes, with noticeably less variation in the time spent annotating images when everyone agrees compared to when there are discrepancies between annotators.

In summary, this suggests that annotating *stone* images poses a greater challenge than *sand* images. This complexity may arise from uncertainties in assessing variables such as the extent of stone coverage on the seafloor or correctly identifying objects within the frame (is it actually a stone?). Conversely, *sand* images seem to present fewer interpretative difficulties, likely attributable to their relatively clean nature.



(a) Proportion of images assigned to each of the three labels across the six annotators.

(b) Percentage of labels matching between pairs of annotators.

(c) Annotation time for different classes. Outliers >7 seconds not shown for readability.

Fig. 5. Quantitative analysis of individual annotation patterns.

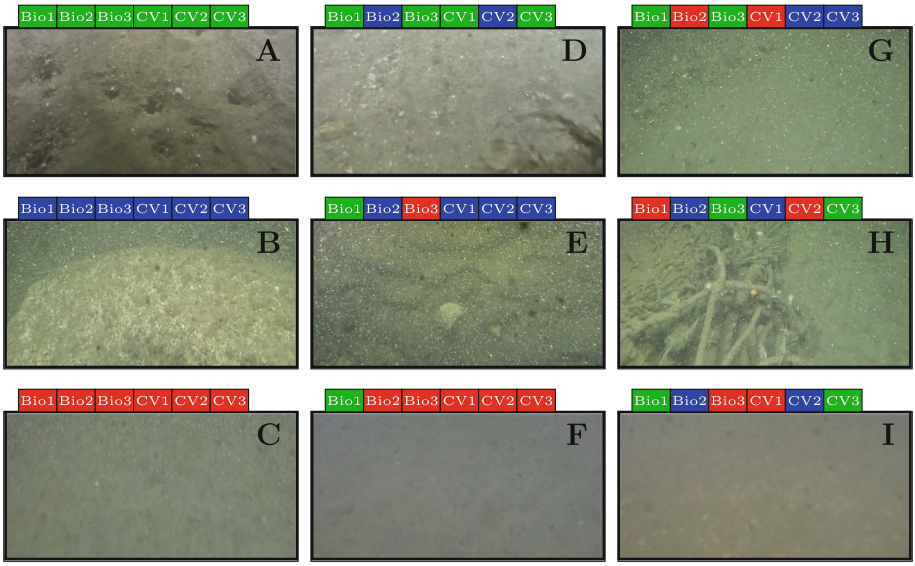
Comparing Individual Annotators - We examine the class distribution at the annotator level to identify potential disparities attributable to educational background. Initially, we look at Fig. 5a, which shows the proportion of images that have been assigned the respective classes by each of the six annotators. We see that *Bio1* and *CV2* were the least likely to label an image as *bad*, but had a significant difference in the number of images labeled as *stone*. The same pattern is seen between *Bio2, Bio3* and *CV3* who agree on the number of *bad* images, while the sand-to-stone ratio varies. Generally, the sand-to-stone ratio varies across annotators from 10.17% (*Bio1*) to 19.31% (*Bio2*) and there is no clear link between the class ratios and the background of the annotator.

Taking a closer look at the pair-wise agreement between the individual annotators, we see the same pattern, which is quantified in Fig. 5b. The general tendency is an agreement level close to 90%, except for *CV1* who is more likely to disagree with the rest, with an average agreement level of 84.26%. However, *CV1* is also the annotator with the highest number of *bad* labels, which could partially explain this pattern. Interestingly, there appears to be no discernible correlation between the background of the annotators and their agreement regarding classifications; annotators with a computer vision (CV) background are not necessarily more likely to agree with each other than with annotators specialized in marine biology (Bio), and vice-versa. Bio1 and Bio3 are the most likely pair of annotators to agree, but no pair exceeds 91% of matching labels.

Besides looking at the distribution of classes, the time spent annotating the images provides further insights regarding the labeling patterns of the individual annotators. In Fig. 5c we can see that the time spent per image varies both depending on the class and the individual annotator. Annotators who are the most conservative with assigning the *bad* label (*Bio1, Bio2* and *CV2*) take the longest time to do so with more than a second on average. On the contrary, *CV1*, who labeled more than 14% of the images as *bad*, typically completed this task in less than one second, suggesting less tolerance for ambiguity. Across all classes and annotators, we see outliers that take significantly longer time to annotate, which suggests a higher label uncertainty for a subset of the images.

Nine concrete examples with varying levels of agreement between the annotators are presented in Fig. 6.The first column contains three images with full agreement. Images Fig. 6(A) and 6(B) contain little motion blur and include

salient features of the seabed, leading to a unanimous labeling of *sand* and *stone*, respectively. In image Fig. 6(A), the seabed is predominantly composed of sand with few small stones, while in image Fig. 6(B), a prominent overgrown stone is visible despite the backscatter caused by resuspended particles. The challenges posed by motion blur, low light conditions, and a scarcity of distinct features render the interpretation of image Fig. 6(C) challenging, resulting in a unanimous *bad* label.



(a) Full agreement (100%). **(b)** Medium agreement (≥ 50%). **(c)** Low agreement (< 50%).

Fig. 6. Image examples for different agreement levels across the six annotators. The slots in the colored bars represent the label assigned by *Bio1, Bio2, Bio3, CV1, CV2,* and *CV3*, respectively. The green, blue, and red colors represent sand, stone, and bad. (Color figure online)

The second column contains images where the majority of the annotators agree on one of the classes. Image Fig. 6(D) contains some motion blur but is otherwise of decent quality with a reasonable amount of light and low turbidity. The image contains a blurred dark object in the lower right corner, smaller stones scattered across the seabed, and no distinct sand ridges. These are all factors that are assumed to contribute to the contradictory labels with four votes on *sand* and two on *stone*. In the second row, image Fig. 6(E) has been labeled as *stone* by the majority, but has also been assigned a *bad* and *sand* label. The label discrepancy is likely due to the image being affected by heavy backscatter, while simultaneously featuring a distinct stone in the center of the image and relatively clear sand ridges. Image Fig. 6(F) is uniformly dark, making it ambiguous whether the background is the seabed or turbid water. Additionally,

there are no distinct habitat markers, except for subtle wave-like patterns in the bottom half (possibly sand ridges) and some salt- and pepper-like spots, which can be interpreted either as particles in the water, or small stones or shells on the seabed.

The last column features images with low annotator agreement. The first image, Fig. 6(G), is labeled as *sand* (2) or *bad* (1) by the marine biologists, whereas the annotators with a background in computer vision label it as *stone* (2) or *bad* (1). Most of the image is uniformly sandy colored and there is a decent amount of backscattering particles. However, in the upper left corner, there is a structure that may be interpreted as a formation of rocks, which is assumed to be the cause of the label discrepancy. We defined no label for abandoned fishing nets and this is likely causing some confusion when labeling image Fig. 6(H). The image is of decent quality with good lighting and only little backscatter, however some observers may view part of the structure as being stones or expect the net to be entangled in stones, while others classify the image based on the sandy right part of the image and ignore the net. Image Fig. 6(I) is blurry and contains few features, except for bright spots, which may be interpreted as either stones or shells.

In summary, after examining the dataset's images, and as exemplified by the images in Fig. 6, we identify three primary motivations for assigning the *bad* label, all contributing to the complexity of image-based habitat classification: (1) issues related to image quality such as darkness, blur, or heavy backscatter (2) objects obscuring the seabed (such as ghosts nets) and (3) a lack of salient features. Disagreement between *sand* and *stone* appears to be influenced by different interpretations of what 30% stone coverage looks like. Distinguishing between small stones, suspended particles, or organisms on the seabed can be difficult, especially when images are blurred and lack distinctive features.

3 Benthic Habitat Classification

3.1 Experimental Set-up

We describe the models used for classifying JAMBO images along with the evaluation procedure.

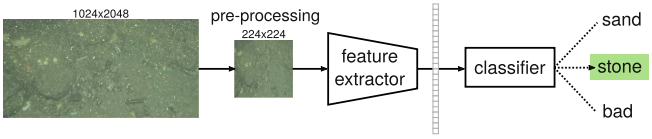


Fig. 7. A pre-trained encoder converts the high-dimensional input image into a 1-dimensional vector capturing its essential characteristics. This feature vector is fed to a linear classifier trained on the JAMBO dataset, which outputs the predicted class.

Models - We apply a three-step pipeline to automatically classify the seabed images, as illustrated in Fig. 7. First, images are resized to 224×224, following

standard practice. Second, considering the limited size of our dataset, we leverage pre-trained vision encoders for feature extraction, which we apply out-of-the-box (frozen weights). Following [44], we compare several models for feature extraction, all using a ViT-B/16 vision transformer [14] as image encoder but differing in their training data and objective (implementation details in Appendix A.2):

- **Sup-IN21K** - a model trained with a standard supervised classification objective on ImageNet-21K [12].
- **DINO** [8] - a general vision-only foundation model, trained in a self-supervised manner on ImageNet-1K [41] using self-distillation.
- **OpenCLIP** [24] - open-source version of CLIP [37] trained on LAION-400M [42] (crawled from random web pages). CLIP is a general multimodal foundation model which learns joint representations of image-text pairs using contrastive learning.
- **BioCLIP** [44] - a multimodal foundational model for the biology domain, trained with a species-classification CLIP objective on TreeOfLife-10M.

Lastly, we train a L2-regularized logistic regression classifier (linear probing) to map the frozen features to class logits. We chose linear probing as it gave consistently better results than a k-Nearest Neighbour (k-NN) classifier on this dataset (see Appendix A.4 for a comparison).

Ground truth - The classifier requires well-defined ground truth labels for training and evaluation. However, the ground truth in our dataset is often ambiguous due to disagreement amongst the six annotators leading to no clear-cut and objective ground truth. Therefore, we compare several supervision schemes:

- **Consensus**: only training and evaluating on images with full agreement between annotators, discarding the rest.
- **Majority**: using the entire dataset, with the majority label as ground truth.
- **Expert**: using the entire dataset, taking the annotations of *Bio1*, who has the highest level of expertise, as ground truth.
- **Noisy**: using all the annotators' labels during training instead of a single label per image. That is, every training image is observed six times by the classifier, each time with a potentially different label. Since there is no single ground truth in this scheme, it is excluded from the quantitative evaluation.

Evaluation & Dataset splits - Considering the class imbalance, we evaluate classification performance on unseen images in terms of class-wise and macro F1 score. We first evaluate stratified cross-validation performance with 20 random folds (90% train, 10% test). We then consider 3 more challenging but realistic dataset splits, where images from a specific date (Sep.26, Nov.7 and Nov.10) are held out as a test set while the other dates are used as training. See Appendix A.3.

3.2 Quantitative Evaluation

We benchmark the classification models against individual human annotators on the JAMBO dataset. The results are summarized in Fig. 8 at the class level

and Table 1 at the test set level. Note that in the consensus scenario, the annotators have perfect performance by definition, as the subset of images has been unanimously labelled by all the annotators. We also note that there is a large variation in classification performance for the *bad* class due to the small number of samples (only six images in the dataset with unanimous *bad* labels).

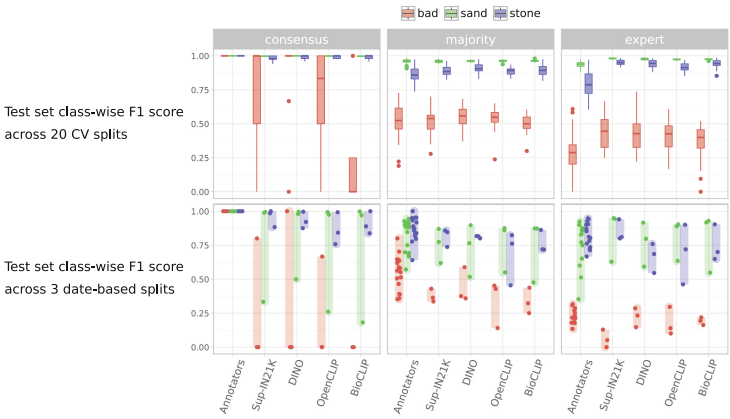


Fig. 8. Class-wise classification performance in the cross-validation setting (top) and on the 3 date-based test sets (bottom). Due to the small sample size, performance on date-based test sets is shown as a scatter plot (one point per test set and model/annotator).

Table 1. 2-class macro F1-score (excluding the *bad* class) in % on different test sets. For the cross-validation (CV) setting, we record the mean score across the 20 test sets. Best score per test set is in bold, and scores greater than the annotators' are underlined.

label scheme	consensus				majority				expert			
test sets	CV	Sep.26	Nov.07	Nov.10	CV	Sep.26	Nov.07	Nov.10	CV	Sep.26	Nov.07	Nov.10
Annotators	*100*	*100*	*100*	*100*	*91.16*	*87.7*	*87.17*	*80.74*	*86.17*	*82.25*	*82.43*	*71.12*
Sup-IN21K	99.03	98.74	**99.75**	60.82	92.25	85.79	75.54	**73.81**	**96.69**	**94.49**	**87.25**	**71.90**
DINO	99.46	99.61	92.75	**71.05**	**93.51**	85.75	78.24	66.63	95.90	74.09	73.08	67.56
OpenCLIP	**99.64**	99.31	90.81	50.92	92.31	81.83	65.70	68.68	94.64	90.18	67.57	67.71
BioCLIP	99.33	**99.90**	92.91	50.98	92.83	**86.75**	**79.66**	59.69	95.73	91.06	81.45	59.85

Looking at cross-validation performance, the classifiers excel at identifying sand and stone images, even surpassing the performance of individual annotators in the majority and expert settings. That is, learning to automate classification from a single annotator can be more accurate than asking a second annotator to manually perform the classification. Models perform the best when following the consensus scheme, aside from the *bad* class (likely due to few *bad* samples). The strong performance was expected, as ambiguous images are excluded from both

training and evaluation. Looking at Fig. 8, we see that class-wise performance echoes the annotator agreement and class distribution analysis: *sand* images have the highest level of annotator agreement and the highest prevalence in the dataset, and are also the easiest to classify. The performance gap between feature extractors widens for the *stone* and *bad* class, which are rarer and more prone to disagreement. When trained and evaluated on images from different dates, classification performance significantly drops compared to the cross-validation setting. The Nov.10 test set is especially challenging due to the dark, cloudy and crowded images, introducing a strong domain shift (cf. examples in Appendix A.3).

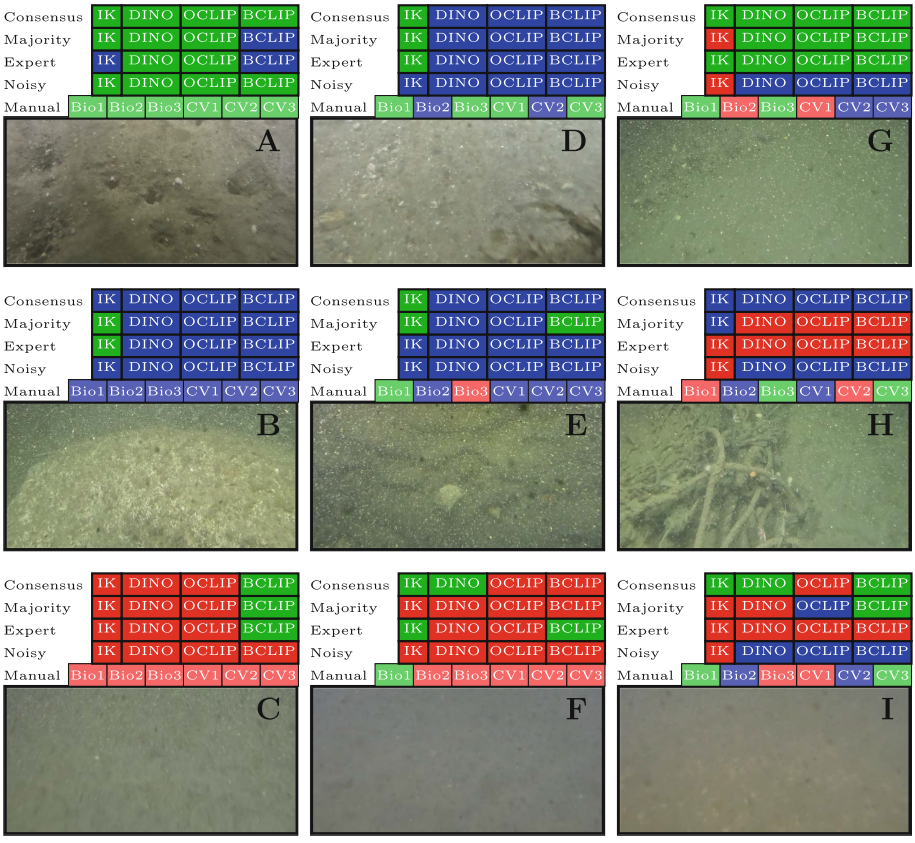
Comparing the four encoders, there is no clear "winner". Although BioCLIP is specifically for the biology domain and includes underwater images, it does not bring consistent performance gains compared to the general-purpose encoders - in fact, it performs the poorest under strong distribution shift (Nov.10 test set), and on the *bad* class (Fig. 8). Sup-IN21K and DINO, both trained on ImageNet, exhibit stronger overall performance on the date-based test sets than CLIP-based models. Sup-IN21K gives the best performance under the expert scheme, while DINO obtains the highest score when averaging across classes, test splits and supervision schemes.

3.3 Qualitative Evaluation

We take a closer look at specific predictions by classifiers trained on the JAMBO dataset. Revisiting the examples from Fig. 6, we hold out these nine images as a test set, and train linear classifiers on extracted features from the rest of the images in the dataset. We obtain sixteen different trained models (four different feature extractors, four supervision schemes) whose predictions on the test set are visualized in Fig. 9. Annotator labels are also shown for reference.

It is interesting to note that although predictions vary across models and supervision schemes, they are rarely unreasonable or unacceptable when compared to the annotators. BioCLIP is the only variant whose predictions does not always match with at least one of the manual annotators: in Fig. 9(C), BioCLIP models predict *sand*, while all annotators agree that the image is *bad* due to poor visibility of the seafloor. As the agreement between annotators decreases, so does the agreement between the sixteen models.

Looking at specific supervision schemes, the noisy label scheme seems to smooth out differences between feature extractors, giving identical predictions across DINO and CLIP-based models. We also observe that under expert supervision, the models' predictions are not necessarily consistent with the manual labels of *Bio1*, despite being trained with the labels of this annotator. This warrants investigating the extent to which features from different pre-trained encoders are able to capture not only the presence of certain objects/cues (e.g. a stone or sand ridge) but also their percentage cover in the image.



(a) Full agreement (100%). **(b)** Medium agreement (≥ 50%). **(c)** Low agreement (< 50%).

Fig. 9. Predictions on images from Fig. 6 by the four models trained on the rest of the dataset with 2 different supervision schemes (noisy labels vs. majority labels). Model names are abbreviated due to space constraints. The green, blue, and red colors represent sand, stone, and bad, respectively. (Color figure online)

4 Discussion

Through our analysis of multi-annotator labels, we identified two key factors contributing to disagreement between annotators: ambiguity in the images and ambiguity in the interpretation of the image content related to the habitat definitions. On one hand, the seabed is not always clearly or fully visible. Currents or movement of the camera may cause motion blur, and turbidity caused by re-suspended particles and organic matter in the water degrades image clarity by scattering the light. Attenuation of the natural light or uneven illumination from artificial lights can obscure details and distort colors. These issues are all amplified as the distance between the camera and seabed increases. On the other hand, even when images are clear, we saw that annotators interpreted the con-

tent differently with respect to the *sand*, *stone*, and *bad* categories, leading to conflicting annotations. This was especially true for images featuring both sand and stones (is it *stone* enough?), or images with poor visibility containing salient features indicating the habitat type (is it *bad* enough?). Special cases such as abandoned fishing gear or muddy habitats were also not explicitly included in our label definition, which may have introduced additional doubt.

While the quality of seabed images captured in turbid waters is difficult to improve, ways to reduce some of the disagreement between annotators could be explored. First, labelling instructions could be defined more comprehensively manner and covers unforeseen scenarios, e.g., by including an *other* class to handle special cases. Second, the labelling interface could provide additional visual cues to annotators, e.g., by displaying the image in multiple resolutions/scales to make high or low frequency features more apparent, or including adjacent frames in the form of a short video clip to provide contextual information. Other and more informative annotation schemes could also be considered. For example, regression-based annotation where the *stone* or *sand* coverage is recorded from 0% to 100%, or a more detailed procedure where objects of interest, such as stones, flora, and fauna, are individually segmented at the instance level. Alternatively, an unstructured labelling scheme could be followed, where annotators provide a free-form text description rather than a single pre-defined category per image. Improved annotation schemes may facilitate more precise evaluations of stone and sand reef coverage and size, which is relevant for assessing the condition of the reefs. For instance, the structural complexity of stone reefs offers insights into their role as nurseries, shelters, and foraging sites, aspects that may be overlooked when conducting simple image classification. However, the manual labelling process becomes costlier with the inclusion of additional information, and the automated analysis pipeline grows more complex.

The models we have evaluated for automating benthic habitat image classification are simple, since they only require training a linear classifier, and achieve promising baseline results. However, the JAMBO dataset opens interesting challenges for future work due to heavy class imbalance, distribution shift, and lack of a single objective ground truth. Class imbalance occurs naturally due to the prevalence of sand habitats in the region and the rarity of bad images in curated video clips. Limited generalization to images from unfamiliar locations is also a well-known problem when deploying computer vision systems in the wild [3,27], which is amplified in the underwater domain due to the cost and scarcity of training data. Observer bias and disagreement, while well-studied in biology [9,22,34,36], is often overlooked in computer vision datasets. An interesting direction is to consider the uncertainty inherent in this dataset as something that is valuable to model [23,33] and evaluate [2]. This could, for example, be achieved by encoding label uncertainty with soft labels [20] or formulating habitat classification as a multiple hypothesis prediction task [40], where the model can produce multiple plausible interpretations for a single image.

Lastly, while this work focuses on habitat classification from seabed images, an equally relevant task to automate is the curation of raw underwater videos,

as only a small fraction of recorded frames actually capture the seabed. Manually sifting through this largely uneventful footage to identify relevant segments is laborious and time-consuming, hindering the adoption of computer vision-based classification. Future research could explore techniques such as keyframe extraction or novelty detection to identify relevant images automatically.

5 Conclusion

We have presented JAMBO, the first publicly available and multi-annotator benthic habitat classification image dataset from the Greater North Sea. This is a valuable resource for researchers and practitioners in the cross-disciplinary fields of seabed mapping and underwater vision. Through detailed analysis, we have not only provided insights into the appearance of benthic habitats in this important area, but also into the uncertainties arising when annotating images of the seabed. Lastly, we have explored the potential of pre-trained vision models for classifying seabed images. These provide a solid baseline without needing to train or fine-tune a feature extractor. However, automatic benthic habitat classification under distribution shift and label ambiguity remains an open problem.

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Deep Learning for Automated Shark Detection and Biometrics Without Keypoints

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Abstract. This study introduces a pipeline designed to aid biologists via automatic detection and biometric analysis of marine animals. Our approach uses detection transformer (DETR) to detect subjects in an image, then generates a segmentation mask over the animal. We also introduce a new method to measure the center line of segmentations, which can be used to assess length during tail movement of animals in images. We test our system on a new dataset of aerial drone imagery of Pacific Nurse Sharks (*Ginglymostoma unami*). The detection model was trained on a dataset of drone-captured images under diverse environmental conditions of varying water clarity and lighting conditions, achieving a recall of 0.96 and precision of 0.80 at an IOU of 0.35. Notably, our method does not require labeled segmentations or keypoints in the dataset, as we find Segment Anything Model (SAM) has strong zero-shot performance. The efficiency of the pipeline was benchmarked against non-expert human annotators, showing a 91% decrease in data analysis time.

Keywords: Object Detection · Biometrics · Ecology

1 Introduction

The physical attributes of individual marine animals, such as body length, age, and body condition are critical indicators of their overall ecology and physiology [4,42]. Studying biometrics at the individual level can be key to estimating population health and how it changes over time [10,35,40]. Long-term monitoring of these metrics is key to understanding how animal populations are responding to potential environmental shifts such as climate change or human disturbance [4,40]. However, measuring biometrics of large marine animals is challenging as

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they can be difficult to approach and physically measure in their natural habitats. In this study, we present a new method for automatically estimating shark length, width, and age from aerial imagery. The pipeline was tested on aerial imagery of Pacific Nurse Sharks *Ginglymostoma unami* in Santa Elena Bay, Costa Rica.

Santa Elena Bay is a marine protected area (MPA) in northwestern Costa Rica (Fig. 1) harboring several endangered marine species including sea turtles, whales, and black coral [11]. The recently described and endangered Pacific Nurse Sharks *Ginglymostoma unami* are also known to aggregate in the bay, and as a newly described species their behavior is particularly poorly understood [13]. Pacific Nurse Sharks are challenging to study because they are highly mobile and elusive. Works such as [17,29] use a variety of remote sensors including underwater cameras and acoustic tags to track and study these cryptic animals throughout their habitats. However, collecting and analyzing this data is laborious and requires significant manual effort. We aim to supplement this research through automatic analysis of manually collected drone imagery.

This research has important ramifications for the MPA and biodiversity in the region. Pacific Nurse Sharks are sometimes found in aggregations of up to 50 individuals, and these aggregations are of particular biological interest since they are potentially relevant for regulation of the social behavior, and courtship of the sharks [29]. This work will help assess factors affecting the unique aggregation behavior, how climate change could potentially affect the species, and the efficacy of the MPA.

In this paper, we present a pipeline for biologists to automatically detect marine animals from aerial drone imagery and compute the length, width, mass, and age of each animal. To do so, we benchmark several SoTA object detectors, which prompt a downstream SAM mask [26]. Then we use a custom heuristic algorithm to calculate shark biometrics. We test our pipeline on a new dataset of Pacific Nurse Shark drone imagery.

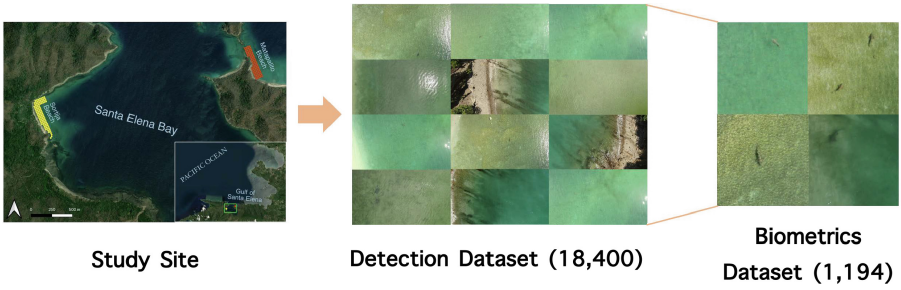

Fig. 1. We curated a dataset for detection and biometric analysis of Pacific Nurse Sharks from aerial imagery. The dataset contains 18,400 images for detection. This includes 1,194 images suitable for biometric analysis where the shark is fully visible. Note: images in the biometrics dataset figure are cropped for clarity.

2 Related Work

2.1 Object Detection

Convolutional Neural Networks (CNNs) have been the foundation of most advances in image recognition and object detection over the past decade [28]. CNNs automatically learn hierarchical feature representations from raw image data, which has proven effective for a wide range of image recognition tasks [1]. The initial layers capture basic image features like edges and textures, while deeper layers learn more complex features specific to the objects within the images.

Two-Stage Object Detectors, built on the architecture of CNNs, have been pivotal in advancing object detection. These models, such as R-CNN and its more advanced successors, Fast R-CNN and Faster R-CNN, prioritize detection accuracy [6,19,32]. The two-stage process involves first generating region proposals where objects might exist and then classifying each proposal into object categories while refining bounding box coordinates. Models like Faster R-CNN integrate these stages by introducing a Region Proposal Network (RPN) that shares full-image convolutional features with the detection network, significantly speeding up the process while maintaining high accuracy. This architecture is particularly beneficial in scenarios requiring precise localization and detailed contextual understanding.

In contrast, One-Stage Object Detectors such as YOLO (You Only Look Once) [25] prioritize inference speed, making them suitable for real-time applications. These detectors simplify the detection pipeline: they do not require a separate region proposal step and instead treat object detection as a simple regression problem, predicting classes and bounding boxes for the entire image in a single pass. While traditionally less accurate than two-stage detectors, improvements in network design, training schemes, and integration of context have narrowed this gap, making them highly competitive.

More recently, the Detection Transformer (DETR) model integrates transformers into object detection [7]. DETR eliminates the need for many hand-engineered components of traditional object detectors by using a transformer to perform direct set prediction of object classes and bounding boxes. The use of global context by the transformer allows DETR to achieve impressive results rivaling non-transformer architectures [7]. This is particularly useful in scenarios like ours where objects are in close proximity in large groups of sharks or partially occluded via camera glare, high turbidity, or other sea life.

2.2 Marine Animal Detection from Aerial Imagery

There has been significant effort towards developing aerial UAV systems for studying wildlife [5,24,36]. Several works seek to plan optimal paths for one or multiple drones to perform aerial surveys in varying environmental conditions [9,36], and others explore the efficacy of various sensor modalities for detecting wildlife [2,34]. Instead, in this work, we focus on developing automated systems for analyzing existing drone imagery.

One of the first studies applying neural network to aerial imagery of marine life trained a vanilla CNN for sea turtle detection [21]. They achieved a precision of 16.3% and a recall of 76.5% using a threshold value of 0.93 equating to an f1-score of 0.134. A reason for its poor performance, as hypothesized by [21], was a lack of a rich internal feature representation caused by the fact that the model was trained from scratch via their dataset instead of using a transfer learning approach. This model had weaker performance than subsequent works, likely due to lack of pretraining and relatively simple model architecture [22,23,38]. In particular, transfer learning - using a pre-trained CNN with some skip connection has shown to improve model performance as in [15] where they trained Faster R-CNN with a ResNet-101 backbone pre-trained on the COCO dataset to achieve an f1-score of 0.91. Another recent work [37] fine-tuned VGG16, pretrained with ImageNet, to draw bounding boxes around a variety of different objects including sharks in the water. Through their approach, they achieved an average precision of 0.904.

In [44], they propose YOLOv7-sea, an object detector, built from YOLOv7, for maritime search and rescue missions using UAVs. YOLOv7-sea adds a prediction head to detect tiny objects and integrates the Simple, Parameter-Free Attention Module (SimAM) to find attention regions. On their dataset, they achieved 59% average precision across all selected thresholds. With this said, their focus seems to be on objects that humans use in the water like boats, jet skis, buoys, and lifesaving equipment.

Pavithra et al. [33] proposes a novel hybrid architecture called SwinConvMixerUNet for underwater image segmentation, combining Swin Transformer and ConvMixer, leveraging the Swin Transformer's ability to capture spatial information and the ConvMixer's channel-mixing capabilities to enhance feature extraction and segmentation accuracy. The SwinConvMixerUNet outperforms all existing models on the SUIM dataset, achieving 84.83% mean intersection of the union (mIOU). However, their task focuses on the segmentation of all objects in the scene from images captured underwater.

To the best of our knowledge, at this time there is no existing work applying recent object detection models to detecting nurse sharks from drone imagery. This presents an opportunity for increased predictive performance as models like DETR have achieved state-of-the-art in object detection and many other downstream applications [16,27,39,45]. However, to our knowledge this model has not yet been applied to aerial shark imagery.

2.3 UAV-Based Animal Biometrics

Collecting marine animal biometrics from drone imagery is of great interest to biologists [31,41]. Open source software packages such as [41] have provided a semi-automated biometrics workflow for whales, but still require human input to select keypoints. Bierlich et al. [3] proposed two deep learning models to automate data collection on whale body size and condition from UAV imagery. The first model, DeteX, uses the YOLOv5 as a detector to identify images containing whales. The second model, XtraX, uses key point detection and the

Segment Anything Model to extract body length and condition measurements from images selected by DeteX. The authors compared the automated measurements to manual measurements of 63 gray whales and found that the automated method was one-ninth as time-consuming and achieved a mean coefficient of variation of 1.46% between automatic and manual length measurements. In our study, we seek to collect biometrics of animals including length, width, mass, and age. Similar to Gray et al. [20], we use segmentation maps and photogrammetry to estimate these parameters. The aforementioned work used PCA to find the major axis of the whale, but this approach would not work for our dataset since while the tails of whales only deform through camber (dorsoventral undulation), sharks tails deform through torsion (lateral undulation). This lateral undulation requires a different approach, hence our design of a custom heuristic described in Sect. 3.3.

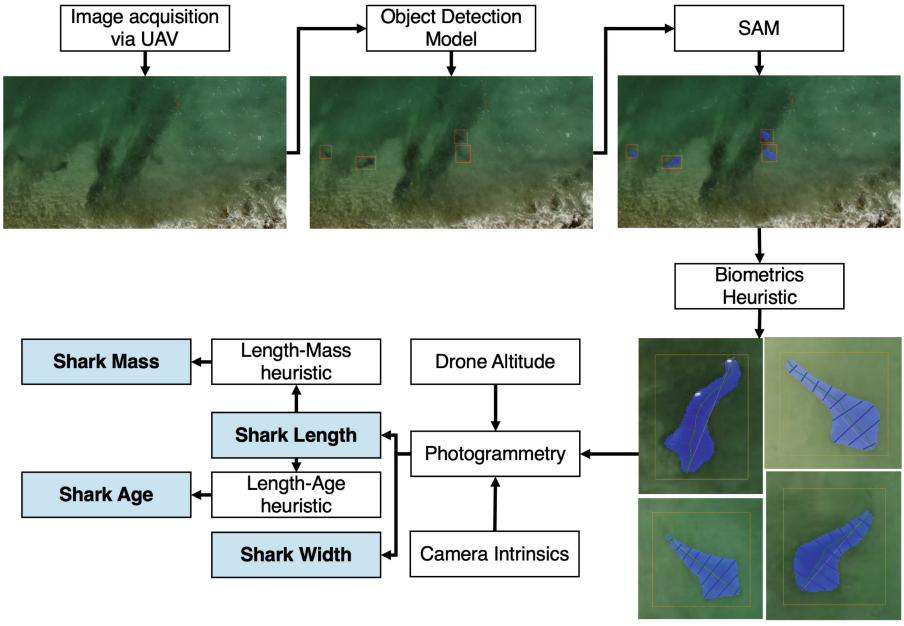

Fig. 2. Biometrics pipeline. Outputs are highlighted in blue. (Color figure online)

3 Methods

3.1 Dataset

The imagery was collected from two field sites (Matapalito Beach and Sortija Beach) in the coastal waters of the Eastern Tropical Pacific Ocean, in Santa Elena Bay (Fig. 1). The dataset was collected from 2022–2024, over a period of 23 months, with varying water visibility (turbidity), illumination, and wind/wave

conditions. Images were collected at each site by flying a DJI Mavic 2 drone on a preprogrammed path, recording a continuous video at 30FPS and 3840×2160 resolution. The drone stopped at predetermined waypoints for 3 s each (Fig. 1, yellow and orange dots).

More than 6 h of video was recorded in total, during 60 drone surveys, resulting in 648,000 total frames captured. For object detection, the dataset was pruned to include a diverse set of 8 videos from the two sites, across varying tidal, turbidity, water surface glare, and wave conditions. Ground truth bounding boxes were added for each Pacific Nurse by two of our team members with the Computer Vision Annotation Tool (CVAT) [12]. The images from 1 video were completely separated from the rest of the dataset, as a test for generalization. The rest of the dataset was time-blocked such that each 45 adjacent frames of a video were placed into a set, and that set was held together when the data was split into training, validation, and test sets. This time-blocking was done to minimize occurrences of consecutive frames being present in the training and test set, which would artificially inflate models' performance metrics. Sharks were present on diverse conditions across videos and time segments, including variable turbidity and substrate. Prior to model training, images were rescaled from 3840×2160 pixels to 1080×1080 pixels. Standard data augmentation techniques, including random rotation, brightness, and hue adjustments, were also applied.

Excluding the images from the video held for generalization testing, our object detection dataset contained 9200 unique positive examples (images containing sharks) in our dataset, and a corresponding 27000 unique negative examples (images containing no sharks). We selected 9200 negative examples to remain in the dataset, such that our dataset contained an equal ratio of images containing sharks and images not containing sharks. The dataset was then randomly split into training, validation, and test sets, in a [80-10-10] ratio, maintaining the images in the 45-frame time blocks. This dataset contains 18400 total images.

We found that while all labeled sharks were suitable for training object detection models, many frames contained partially occluded sharks. These frames were not possible to extract ground truth biometrics from, so we therefore curated a smaller dataset containing 1194 clear shark images suitable for biometrics as seen in Fig. 1.

3.2 Detection Models

In this research, we focused on evaluating a series of models for shark detection, predominantly leveraging the Detectron2 model library [43]. First, we employed several RCNN model architectures, a two-stage detector optimized for accuracy (as opposed to inference speed like many one-stage detectors) [19]. Faster RCNN architectures are composed of a feature extractor backbone (typically a pre-trained CNN) followed by a region proposal network to predict object boundaries and their "objectness" scores, and finally region-of-interest (ROI) pooling to extract uniform-sized features. We trained several RCNN with various backbones

(ResNet and ResNeXt architectures) and feature pyramid networks (FPN) to enhance the effect of objects of various scales in images. All of these models were pre-trained on the Imagenet dataset [14].

We also implemented DETR which uses a transformer architecture to reason about all objects in the scene globally, which to our knowledge is the first work using this model for shark detection from aerial imagery [7]. This setup is expected to be advantageous for detecting sharks in highly dynamic and complex environments. Finally, we trained a single-stage pre-trained YOLOv8 model which is particularly well suited for real-time detection tasks [25]. The pre-trained medium YOLOv8 model was trained for 40 epochs using default hyperparameters (until loss converged). DETR was trained for 120 epochs, and the rest of detectron models were trained for 40 epochs each - all with a maximum of 100 objects detected.

3.3 Biometrics

The image and object detection bounding boxes were used to prompt the Segment Anything Model (SAM) mask predictor [26] to generate a mask containing shark pixels. We designed a custom algorithm to estimate the length of the shark along its centerline, robust to the orientation and deformation of the shark, as follows:

Given a binary mask $\in \{0,1\}^{m\times n}$, compute the largest contour C and its convex hull H. Identify the furthest points $P_1, P_2 \in H$ by calculating the pairwise distances $D_{ij} = \|H_i - H_j\|_2$ and finding the maximum distance. Divide the segment $\overline{P_1P_2}$ into n equal subsegments, and for each dividing point S_i, compute the center of mass of the mask c_i along the line which is perpendicular to $\overline{P_1P_2}$ and intersects S_i. The shark's length L is calculated as the sum of Euclidean distances between P_1, c_i, and P_2:

$$L = \|P_1 - c_0\| + \sum_{i=0}^{n-2} \|c_i - c_{i+1}\| + \|c_{n-1} - P_2\|. \tag{1}$$

To measure the width at each point, we calculated the perpendicular vector to the centerline at each center of mass point c_i. The perpendicular unit vector $\mathbf{u}_i$ is defined as:

$$\mathbf{u}_i = \frac{1}{\|\mathbf{v}_i\|}\begin{pmatrix} -v_{iy} \\ v_{ix} \end{pmatrix} \quad \text{where} \quad \mathbf{v}_i = \frac{(\mathbf{c}_{i+1} - \mathbf{c}_i) + (\mathbf{c}_i - \mathbf{c}_{i-1})}{2} \tag{2}$$

Here, $\mathbf{v}_i$ is the average direction vector at point i, calculated as the average of the direction vectors from c_i to c_{i+1} and from c_{i-1} to c_i. The width at point c_i is then measured by extending a line along $\mathbf{u}_i$ until it intersects the mask boundaries on both sides of the centerline.

The length L and widths W obtained from the mask, measured in pixels, were converted to centimeters using the following function, modified from [41]:

$$L_{\text{cm}} = \left(\frac{S_w}{I_w} \times \frac{A + D}{F} \times 100\right) \times L_{\text{pixels}} \tag{3}$$

where S_w is the sensor width in mm (13.2 mm for a 1" CMOS sensor), I_w is the image width in pixels (3840 pixels for 3840×2160p resolution), A is the altitude of the drone in meters (37 m), D is the depth of the shark in meters, and F is the focal length of the camera in mm (28 mm). We assumed a constant depth D of 1.5 m for this analysis, since nurse sharks were typically observed at approximately 1.5 m depth in snorkeling surveys in Santa Elena Bay. In the future, benthic animal position extracted from camera metadata could be collocated with bathymetry data to improve the accuracy of the depth estimate. The pixel dimension in mm per pixel is calculated as the ratio of the sensor width to the image width. This is converted to meters per pixel, and then the Ground Sampling Distance (GSD) in meters per pixel is calculated. Finally, the GSD is converted from meters per pixel to centimeters per pixel to determine the length and widths in centimeters.

We used length to estimate shark age and mass. To estimate the age, we utilized the von Bertalanffy (vB) growth function with known coefficients for the nurse shark [8,18]. Mass was estimated using a combined mass model for nurse sharks [30]. The full biometrics pipeline is visualized in Fig. 2.

4 Experiments and Results

4.1 Object Detection

We tested our suite of models on an internal test dataset to assess the models' ability to learn representations, as well as a previously-unseen external holdout video taken on a different day to assess the generalizability of the model. Testing the models on both datasets gauges potential overfitting on previously-seen data, while also evaluating for practical usage on drone imagery from new distributions.

Among the models, YOLOv8 and DETR performed the best on the internal holdout test set across all metrics, including mean Average Precision and Average Recall from IOU thresholds between 0.5 to 0.95 and Average Precision and Average Recall at IOU thresholds of 0.35 and 0.5. The exact precision and recall values are shown in Table 1. YOLO had the highest mAP as well as the highest mAR, with DETR performing slightly better at lower IOU thresholds. Both of these performed better than the trained Faster R-CNN models with Feature Pyramid Network and Dilated-C5 backbones. It is important to note that although mAP averaged between 0.5 and 0.95 are traditionally used as performance metrics for object detection models, we found that it is not necessary for predicted bounding boxes to have a high IOU with ground truth boxes in order to obtain accurate results on downstream biometrics methods. Thus, although both YOLO and DETR had a relatively low mAP and mAR of 0.63 and 0.70, performance at a lower IOU threshold of 0.35 was significantly higher-with DETR achieving a near-perfect AP and AR of 0.94 and 0.99, respectively.

As expected, the performance of both YOLOv8 and DETR declined when tested on a completely separate video, with a mAP of 0.22 obtained for YOLOv8 and a mAR of 0.40 obtained for DETR, as seen in Table 2. Similarly, the precision

and recall at lower IOU thresholds were significantly higher and proved to be sufficient for deriving downstream biometrics. The DETR model performed the best at an IOU threshold of 0.35, with an average precision of 0.80 and an average recall of 0.96. A higher recall was heavily prioritized for this shark detection task, with the identification of all sharks crucial given the ease of manual removal of potential false positives.

Table 1. Performance Metrics of Various Object Detection Models on a holdout test set. Best performance for each metric in bold, with larger values indicating higher average precision and recall.

Model	mAP@[0.5:0.95]	AP@0.35	AP@0.50	mAR@[0.5:0.95]	AR@0.35	AR@0.50
YOLOv8	**0.63**	0.93	**0.92**	**0.70**	0.97	**0.97**
Faster R-CNN X101-FPN	0.48	0.63	0.61	0.70	0.99	0.60
Faster R-CNN R101-DC5	0.29	0.43	0.43	0.35	0.43	0.41
DETR	0.43	**0.94**	0.90	0.58	**0.99**	0.97

Table 2. Performance Metrics of YOLOv8 and DETR on holdout video. Best performance for each metric in bold.

Model	mAP@[0.5:0.95]	AP@0.35	AP@0.50	mAR@[0.5:0.95]	AR@0.35	AR@0.50
YOLOv8	**0.22**	0.77	**0.64**	0.34	0.79	0.73
DETR	0.14	**0.80**	0.53	**0.40**	**0.96**	**0.85**

The effect of IOU between the predicted and ground truth boxes versus recall was studied for YOLOv8 on the holdout video, as shown in Fig. 3. We see that the recall is stable through an IOU threshold of 0.4, then rapidly declines as IOU increases. This shows that we detect nearly 80% of the sharks in the holdout video at lower IOU values, thus motivating the non-standard AP and AR calculations at an IOU of 0.35. Precision-recall curves were generated for YOLOv8 and DETR to study the overall performance at this lower IOU threshold on the internal and external test datasets (see Fig. 4). We see that both models performed better on the internal dataset, with high precision and recall achieved at lower confidence thresholds. Similarly, both YOLOv8 and DETR had relatively high performance on the holdout video, excluding low precision at lower confidence thresholds. These results show the large number of low-confidence predictions for both models under a confidence threshold of 0.2, with high precision and recall maintained at higher confidence values.

Although both models had high accuracy at lower IOU thresholds, DETR outperformed YOLOv8 on the holdout video with a nearly perfect recall of 0.96 on completely unseen data. Beyond a confidence threshold of 0.25, precision was maintained; thus all further biometric analysis was performed using YOLOv8 predicted bounding boxes with a confidence threshold of 0.25.

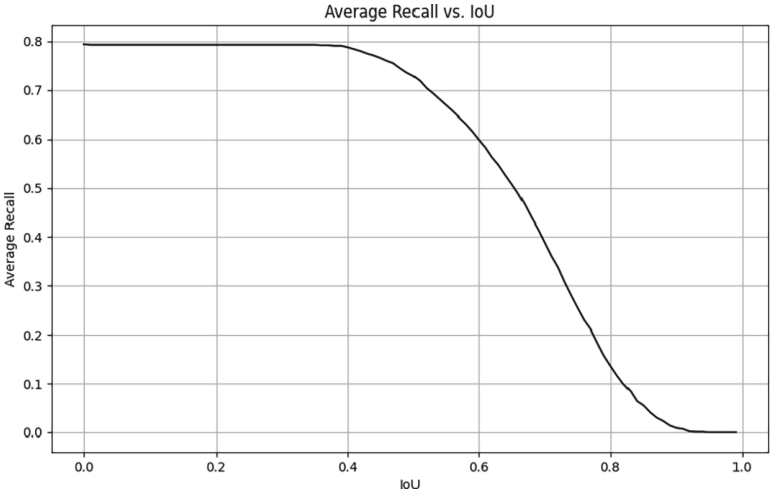

Fig. 3. Recall vs. IOU threshold for YOLOv8 on the holdout video.

4.2 Biometrics

We compared the performance of our approach to manually measured shark lengths from a randomly selected subset of photos (n = 122) containing sharks from the holdout video. We benchmarked our approach based on percent error and coefficient of variation (CV, following Bierlich et al. [3]) between the manually measured and automatically measured shark lengths for each image. In total, we found that SAM, when prompted with YOLOv8 predicted bounding boxes, successfully generated correct segmentations on 71% of the photos (n = 87), while it failed to generate a correct segmentation on 29% of the photos (n = 35). Examples of successful and failed segmentations are found in (Fig. 5 (b) and (c)). The manually-sorted criteria for successful segmentation are that the shark is included in the segmentation mask symmetrically about its true centerline, and that there are no erroneous inclusions (shadows, surface glare) or exclusions (tip of the tail, a single pectoral fin) in the mask. We report results for the automated and manual measurements for the images that SAM generated correct shark segmentations. All length measurements had a CV $< 8\%$, with an average CV between automatic and manual length measurements of 2.71% (n = 87, median = 2.70%, SD = 1.54%, maximum = 5.55%) (Fig. 5 (a)). Likewise, the length measurements had a percent error $< 12\%$, with an average percent error between manual and automatic measurements of 5.54% (n = 87, median = 5.56%, SD = 1.54%, maximum = 11.76%. This represents only a slight decrease in performance compared to ground truth bounding boxes (from which measurements had an average CV of 2.31%), showing that exact bounding box predictions are not necessary for prompting accurate segmentations from SAM. On average, the automatic measurements overestimated length by 2.47 cm (n = 87, median = 2.79 cm, SD = 3.91 cm, minimum = -8.48 cm, maximum =

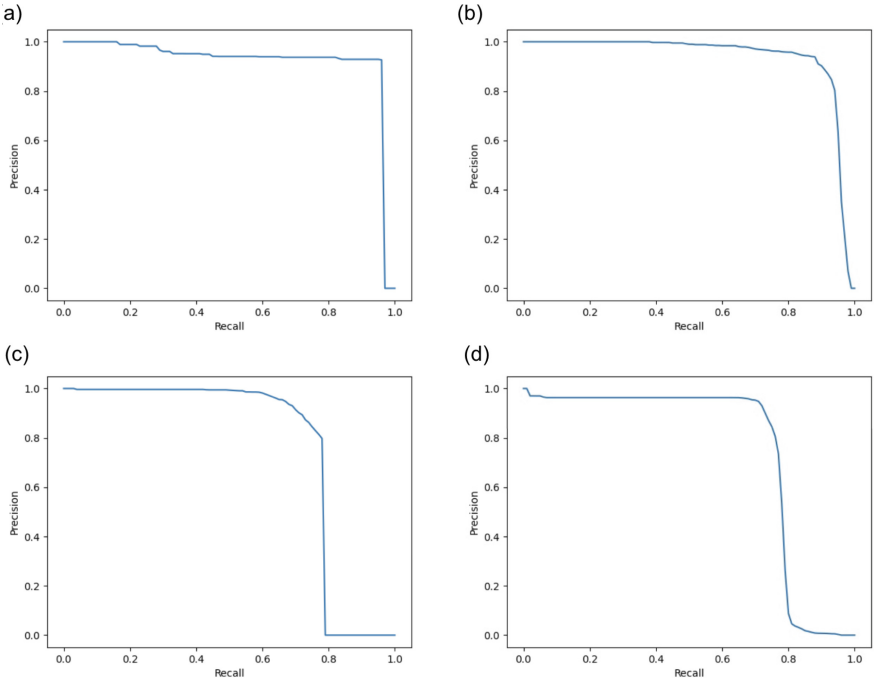

Fig. 4. Precision-Recall curves for models on internal and external test datasets. (a) YOLOv8 on internal test set. (b) DETR on internal test set. (c) YOLOv8 on holdout video. (d) DETR on holdout video.

7.69 cm). The average percent error between manual-length derived mass estimates and automatic-length derived mass estimates was 11.85% (n = 87, median = 11.81%, SD = 6.97%, maximum = 26.09%). The average CV for mass estimates was 5.64% (n = 87, median = 5.63%, SD = 3.20%, maximum = 11.54%). Similarly, the average percent error between manual-length derived age estimates and automatic-length derived age estimates was 9.23% (n = 87, median = 9.45%, SD = 5.36%, maximum = 19.91%). The average CV for age estimates was 4.45% (n = 87, median = 4.51%, SD = 2.53%, maximum = 9.05%).

4.3 User Comparisons

To compare the accuracy and speed of our shark detector to a non-expert human, we randomly drew 100 samples from the holdout video dataset with an even ratio of positive and negative examples. After showing the person examples of negative and positive examples, and basics of CVAT labeling workflow, they labeled the dataset of 100 samples. We found that the human had a classification precision of 0.94 and a recall of 0.83 on the dataset. Our best model had a classification precision of 0.8 and a recall of 0.96 on the same dataset. The human took an average of 10.5 s per frame, while the model only takes an average of

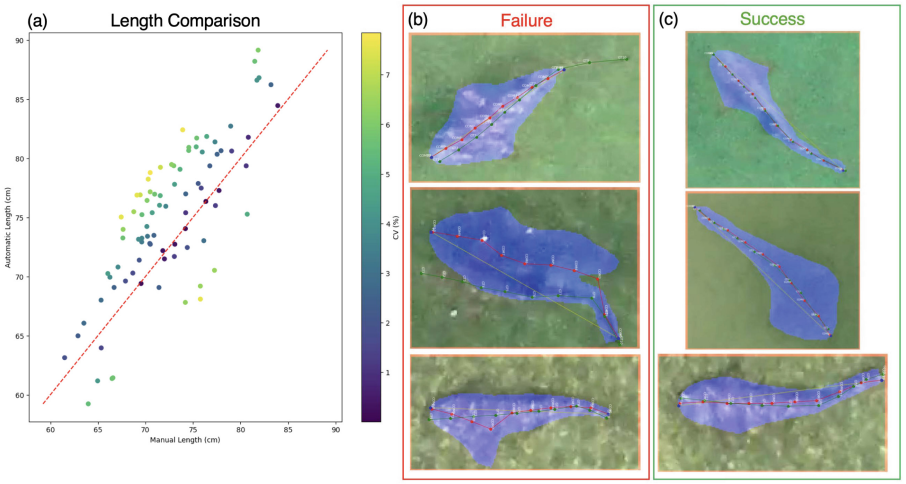

Fig. 5. (a) Manual and automatic length measurement results. The dashed red line represents a 1:1 ratio. Points are colored by coefficient of variation percentage (CV %), where lighter color points indicate a higher CV. (b) and (c) include manual (green) and automatic (red) polylines used to measure the length of each shark. (b) Example failure cases of SAM segmentations, from top to bottom: Missing the end of the tail, segmenting a shadow, and segmenting only one pectoral fin. (c) Examples of successful SAM segmentations and length measurement. (Color figure online)

0.25 s on an NVIDIA A10 GPU. The annotator also measured sharks in 100 images using the CVAT polyline tool in approximately 50 min, so on average, it took a human annotator 40.5 s on average to detect and measure the sharks in each frame. In contrast, the automatic biometrics pipeline takes an average of 3.75 s per image on an NVIDIA A10 GPU. These preliminary results show our model takes 91% less time than a human to perform inference and shark measurement, while achieving human-level performance on classification recall and measurement accuracy.

5 Conclusion

In this study, we develop a new pipeline which significantly improves the efficiency of studying Pacific Nurse Sharks (*Ginglymostoma unami*) through aerial drone imagery. We benchmark several SoTA object detection models such as YOLOv8, Faster R-CNN, and DETR, and conclude that DETR performed the best - enabling 91% faster data processing than manual annotation methods. To our knowledge, this is one of the largest datasets of nearshore shark aerial drone imagery, and the first shark detection work using DETR.

We also demonstrate the efficacy of the pre-trained SAM model on segmenting sharks when prompted by bounding boxes. Using a custom heuristic, we

analyze SAM masks to compute marine animal length, width, mass, and estimated age—which are essential for understanding shark population demographics. We found that SAM generated successful segmentations 77% of the time. On 27 randomly sampled images from our biometrics dataset, our length heuristic had a 2.39% average CV from the manual labels. Our biometrics heuristic, when combined with the shark detection model, provides researchers a way to automatically estimate shark length to centimeter-scale accuracy from drone imagery. However, there are limitations and opportunities for improvement for the biometrics pipeline. The primary limiting factor is the accuracy of the segmentation, as that has downstream effects on the length and width prediction. In turbid water, often the pectoral fins of the shark and the tail will be left out of the segmentation, or the end of the tail will be segmented separately from the rest of the body. Also, when the shark is swimming above the bottom, SAM will sometimes segment the shadow of the shark instead of the shark itself (Fig. 5 (b)). For simplicity, our photogrammetric analysis did not include the variable depth of water the sharks are swimming in, the variance in drone altitude, or corrections due to refraction at the air-water interface, but all of these considerations could feasibly be added if additional measurement accuracy is required.

A key feature of our work is the centerline estimation heuristic, which is novel for marine animals and has broad applicability beyond just measuring nurse shark biometrics. This method is not only valuable for length measurements but also for animal pose estimation, enabling researchers to compute important new metrics such as tailbeat frequency and amplitude.

Future work in object detection will involve testing different data augmentation approaches and exploring different DETR variants. There remains a large amount of unlabeled shark video data in our dataset, so including a wider swath of this data in the training will likely improve detection results. Further development on the biometrics pipeline will incorporate object tracking to maintain the identity of sharks between frames, which would allow biologists to observe a shark's speed, tailbeat frequency, tailbeat amplitude, and other kinematic observations from aerial imagery. These observations are particularly relevant for understanding shark behavior and physiology on a fine scale, and they would allow biologists to quantitatively study the movement patterns of sharks in aggregations. By finetuning SAM on labeled segmentation data, we can extract better masks to feed into the biometrics heuristic, which will improve the quality of results for kinematic observations, which otherwise would require laborious data labeling efforts.

6 Examples

An output video generated from our shark detection and biometrics pipeline can be found in on YouTube.

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Improving in Situ Real-Time Classification of Long-Tail Marine Plankton Images for Ecosystem Studies

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Abstract. The complexity of marine plankton monitoring has highlighted the limitations of conventional machine learning models, particularly when faced with long-tailed data distributions commonly found in natural environments. This paper introduces a comprehensive framework that leverages a novel dataset from the Plankton imager (Pi-10) instrument designed to enhance plankton image monitoring accuracy in a real-time application. We employ cutting-edge image classification architectures, including pre-trained Vision Transformers (ViT) and BERT Pre-training of Image Transformers (BEiT). We integrate Label-Aware Smoothing (LAS) into our training process to address the challenges of long-tailed data distributions. Further, we innovate with dynamic label-aware smoothing, which adjusts smoothing factors based on attention scores from ViTs to tailor model confidence to the significance of different image regions. The results demonstrate improvements in classification performance on the Pi-10 dataset, effectively handling long-tail distribution challenges and setting new benchmarks for real-time image classification in ecological research and biodiversity monitoring. This approach advances biodiversity monitoring and provides a scalable solution adaptable to other domains encountering similar distributional challenges.

Keywords: Transformer · Label-aware smoothing · Long-tail recognition · Plankton image analysis

1 Introduction

Plankton, microscopic organisms in marine and freshwater environments, are key indicators of aquatic ecosystem health. Their sensitivity to environmental

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changes and short life cycles make them crucial for monitoring. By observing plankton, we gain insights into the biogeochemistry and productivity of the oceans, as their presence and behavior reflect the impacts of changing conditions [34,39]. Image processing technologies offer a promising approach to enhancing the accuracy and efficiency of plankton monitoring. These technologies allow for detailed insights into plankton distribution across various scales, reducing the reliance on traditional, labour-intensive methods that involve manual sampling and identification [2,12,19,27].

However, applying machine learning and image processing techniques to plankton data introduces significant challenges, such as data set shifts (DSS), when the characteristics of the plankton images in the training set may not fully represent the diversity and variability present in the open ocean particles that the real-time classifier will encounter during deployment [32]. The DSS complicates the application of conventional machine learning models, as the feature distribution in training sets often fails to mirror real-world conditions [36].

Traditional deep-learning approaches to plankton image classification are particularly challenging due to the high variability of features within classes and the similarity between different classes. These factors complicate practical model training and accurate classification. However, given their significant success in computer vision, ViTs are becoming increasingly popular and widely utilised in visual recognition tasks [15]. In response, a comprehensive study shows that ViTs outperform convolutional neural networks (CNN)-based methods [26]. However, real-world data often exhibits significant class imbalance that can severely skew outcomes in data-driven deep neural networks, making the task of Long-Tailed Recognition (LTR) particularly challenging, where common species (head) are over-represented relative to rarer species (tail). Traditional techniques - such as random downsampling of majority classes or oversampling of minority classes - can be used to address class imbalance, though may lead to the loss of important information or overfitting [4,46]. To overcome these issues, LTR and transfer learning are increasingly utilised to enhance model adaptability and accuracy [16,20,26,28,38].

This paper introduces a new dataset from the Plankton imager (Pi-10) instrument, a state-of-the-art high-speed colour line scan imaging device, alongside a real-time framework designed to elevate image classification efficacy, particularly addressing the complexities introduced by long-tailed data distributions. We have adapted leading-edge, pre-trained image classification architectures, notably the ViT [15] and BEiT [1], which are renowned for their exceptional performance across various tasks. Additionally, We have integrated the advanced Label-Aware Smoothing model (LAS) [51] into our training process to enhance efficacy, boost real-time classification capabilities, and effectively address LTR challenges [21,41]. To further improve classification on highly imbalanced datasets, we propose a novel dynamic label-aware smoothing method that adjusts smoothing factors based on the attention scores computed for each patch by ViTs. This innovation ensures that the label smoothing factors are dynamically tuned, reflecting the varying significance of different image parts as perceived by the transformer architecture. In summary, our contributions are (i) the

adoption of ViT and BEiT models to improve image classification performance, (ii) the integration of dynamic label-aware smoothing to address long-tail data challenges effectively, and (iii) the effective use of transfer learning to enhance model performance on the newly introduced Pi-10 dataset.

2 Related Work

Transformers. The ViT model represents a significant departure from traditional image classification methods, which predominantly utilised Convolutional Neural Networks (CNNs). The ViT model adopts the Transformer architecture for visual data, initially designed for natural language processing tasks [45]. This approach allows the model to effectively handle long-range dependencies between image patches [15], introducing a novel methodology for image analysis. Complementing the ViT, the Bidirectional Encoder Representation from Image Transformers (BEiT) [1] employs a self-supervised learning approach inspired by BERT [14]. This is achieved through a masked image modeling task, enabling the model to learn valuable visual representations without needing labeled data, thus significantly improving its efficacy in downstream tasks.

The BEiT model was pre-trained on extensive datasets [42], and subsequently fine-tuned in a supervised fashion on ImageNet [13]. BEiT improves downstream tasks, potentially surpassing traditional CNNs [6,47]. ViT and BEiT have been used for plankton image analysis and showed promising results in enhancing ecology studies [9,26]. Here, we incorporate ViT and BEiT with LTR strategies, representing significant advancements over traditional CNNs. Our results demonstrate that the ViT model is competitive with the BEiT model when applied to long-tail data, particularly after incorporating dynamic label-aware smoothing.

Transfer Learning in Plankton Image Classification. Previous work on plankton image classification has demonstrated that out-of-domain transfer learning, using models pre-trained on large-scale natural image datasets such as ImageNet1K or ImageNet22K, yields good results [26]. Hence, we use a pre-trained ResNet-18 model with ImageNet as a baseline model. We then employ the ViT model (see 4.2 for more details), which was initially trained on ImageNet-21k and subsequently fine-tuned on ImageNet 2012. Furthermore, we use the BEiT model, pre-trained in a self-supervised manner on the same dataset as ViT (also known as ImageNet-21k [13]). This dataset consists of 14 million images and 21,841 classes, at a resolution of 224×224, and the model is subsequently fine-tuned on the same dataset at the same resolution.

Long-Tail Learning. The challenges posed by imbalanced datasets in machine learning are critical, especially for tasks that rely on accurate representation of minority classes [21,29,41]. Various strategies have been developed to tackle this issue, including Post-hoc correction methods, which adjust a model's outputs after training to better represent minority classes [4,10,18,24,37]. Data modification strategies alter the training dataset by methods like oversampling the

less represented classes or undersampling the overrepresented ones, thereby aiming to balance the training environment [7,8,48]. Additionally, Loss weighting involves adjusting the loss function to impose heavier penalties for misclassifications of minority classes, thus directing the model's focus towards these classes [11,11,17,23,30]. Lastly, Margin modification techniques, including logit adjustment, tailor the decision boundary by modifying class logits to enhance the classifier's fairness towards minority classes [22,25,31,43,49,50].

The challenge posed by the skewed distribution of class frequencies in our plankton dataset was met with a novel integration of LTR strategies during the training phase. We implemented a Label-Aware Smoothing model (LAS) for ViT and compared it with a range of approaches including Class-Balanced (CB), Label-Distribution-Aware Margin (LDAM), Balanced Cross-Entropy (Bal_CE), and an adjusted form of Binary Cross-Entropy (BCE) loss [5,11,31]. The dynamic LAS loss was the most accurate, significantly enhancing model performance across minority classes. Additionally, transfer learning played a pivotal role in refining model accuracy and training efficiency. By incorporating the newly introduced Pi-10 dataset and fine-tuning pre-existing models over a concise series of training epochs, we achieved rapid model convergence and marked improvement in accuracy.

3 Plankton Imager System

The Plankton Imager Pi-10[1] is a high-speed color line scan imaging instrument [12,35,40]. The instrument is connected to the clean sea water underway system (inlet at 4 m depth), which supplies water continuously (flow rate of 34 L min-1). It captures images of passing particles within a size range of 180 μm to 3.5 cm. Images are taken in RGB color using an EPIX E8 frame store, and processed to extract a region of interest, saved in TIFF format with a timestamp and unique identifier. Raw images are stored in 12-bit resolution and then converted to 8-bit for viewing and analysis. The Pi-10 operates continuously during surveys, capturing images of all particles passing through the flow cell, with only mesozooplankton (200 µm – 2 cm) processed and saved due to file-size constraints and operational needs. Due to plankton's skewed distributions towards smaller species, most captured images show small plankton and particles [44]. The annotation process involves a trained taxonomist sorting through the collection of images, categorising them into ecologically relevant groups (supplementary materials). The plankton dataset was gathered during a research cruise on the Research Vessel Cefas Endeavour at a specific site (Fig. 1) in the North Sea, utilising the Plankton Imager (Pi-10 version).

Realtime Access to Plankton Imager Data. We developed a real-time classification pipeline where images are taken and immediately classified. This pipeline bypasses the often long periods between image collection, big data transferring, and image analysis. The real-time data collection pipeline is equipped

[1] More details can be found at www.planktonanalytics.com.

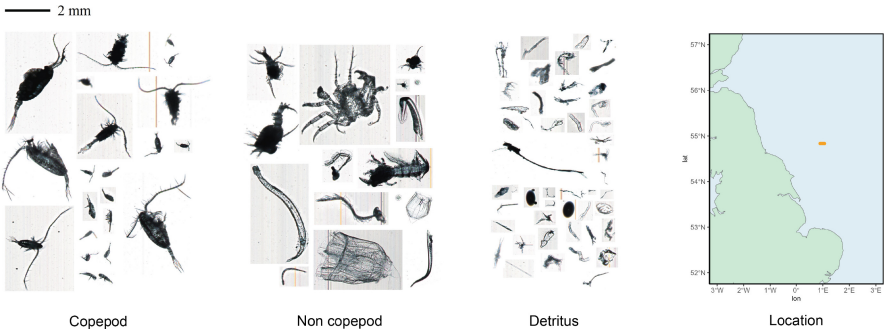

Fig. 1. Sample diversity captured by the Plankton Imager and location of the research vessel: a comparative display of copepods, non-copepods, and detritus.

with a Pi-10 instrument. For in-situ image processing and classification tasks, we utilise the "Edge-AI" The NVIDIA Jetson AGX Orin[2] is a compact and powerful edge device equipped with a GPU explicitly designed for AI and machine learning applications in resource-constrained environments. We chose this device because of its optimal balance of performance and size, which makes it ideal for deploying advanced AI workloads directly at the edge without relying on cloud infrastructure. This system processes the images to categorise them into counts of copepods, non-copepods, and detritus via real-time classification using a transformer model. Summarized data are then transmitted via the ship's broadband satellite communication systems to a cloud-hosted digital dashboard[3], enabling remote access and analysis. The Pi-10 imaging system captures images that are processed in real-time and simultaneously stored on a hard drive for further analysis. In this study, we utilised a comprehensive dataset of 55,570 samples from the Pi-10 dataset gathered during a fisheries survey conducted in the North Sea in August 2023. The data, meticulously labeled by taxonomists, was used to train our proposed model, ensuring accuracy and relevance in the classification process. The evaluation of the final model occurs in two stages: initially, it was assessed using a designated test set; subsequently, it will be integrated into the real-time pipeline for monitoring marine plankton. This dual-phase testing ensures both the efficacy and practical applicability of the model, culminating in its deployment to monitor plankton via a live dashboard actively.

Train/Validation/Test Split. To ensure robust model performance assessment, we employed a data splitting strategy: 70% of the dataset was allocated for training and 10% for validation purposes, while the remaining 20% was reserved for testing. The total number of tests is 11,114, comprising 2,451 for copepod, 8,203 for detritus, and 460 for non-copepod.

[2] www.nvidia.com/en-us/autonomous-machines/embedded-systems/jetson-agx-orin/.
[3] More details can be found at https://planktonapi-dev.cefastest.co.uk.

4 Methodology

4.1 Task Definition

In studying visual recognition with skewed class distributions, we analyse a dataset $D = \{X, Y\}$ consisting of N entries divided among K different classes. Each data point x_i in X corresponds to a label y_i from Y, with y_i ranging from 1 to K. The representation of each class K_i differs markedly, with n_i instances per class, characterizing the dataset's long-tail distribution. We denote the disparity in class representation by the imbalance ratio $\gamma = \frac{n_{\max}}{n_{\min}}$. To address this imbalance, we propose to develop a model $M = \{F_{\theta_f}, W_{\theta_w}\}$, incorporating a feature encoder F_{θ_f} and a classifier W_{θ_w}, tailored to effectively handle these disparities.

4.2 Classification Model

The classification process begins by segmenting each input image into smaller patches. Each patch is linearly transformed and embedded with positional encodings to preserve spatial relationships within the image. These encoded patches are then passed through multiple layers of the ViT encoder, where they undergo multi-headed self-attention and feed-forward processing. The output from the final encoder layer is fed into a classification head that predicts the image class.

The operations within the transformer utilised by the ViT are mathematically described by:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{1}$$

where Q, K, and V represent the query, key, and value matrices derived from the input embeddings, and d_k is the dimensionality of the keys.

4.3 Label-Aware Smoothing

The label-aware smoothing (LAS) [51] is crucial for addressing the overconfidence of predictions typically used with cross-entropy loss and the varying distributions of predicted probabilities across classes.

The LAS loss function is defined as:

$$l(q, p) = -\sum_{i=1}^{K} q_i \log p_i, \tag{2}$$

where q_i is the target distribution after label-aware smoothing:

$$q_i = \begin{cases} 1 - \epsilon_y = 1 - f(N_y), & \text{if } i = y, \\ \frac{\epsilon_y}{K-1} = \frac{f(N_y)}{K-1}, & \text{otherwise.} \end{cases} \tag{3}$$

Here, ϵ is the base smoothing parameter, y is the true class, and N_y represents the number of instances for class y, influencing the function $f(N_y)$. The

label-aware smoothing function $f(N_y)$ is designed to vary the smoothing effect dynamically according to class frequency, making the model more sensitive to less frequent classes. This function is defined in several forms, such as linear, concave, and convex. In this study, we employ the concave form, which has demonstrated promising outcomes in initial experiments:

$$f(N_y) = \epsilon_K + (\epsilon_1 - \epsilon_K) \sin\left(\frac{\pi(N_y - N_K)}{2(N_1 - N_K)}\right) \tag{4}$$

where:

- ϵ_1 and ϵ_K are two hyperparameters.
- N_1 and N_K are the maximum and minimum class frequencies in the dataset, respectively.

The concave form of label smoothing increases regularisation more substantially for classes with fewer instances, effectively strengthening regularisation where it is most needed. This method aims to mitigate bias toward frequently occurring classes by moderating the confidence levels of the predictions, thereby enhancing performance in less common classes. The formula for cross-entropy loss after applying the softmax function is defined as

$$l(y; p) = -\log(p_y) = -w_y^T x + \log\left(\sum_i \exp(w_i^T x)\right), \tag{5}$$

where $y \in \{1, 2, \ldots, K\}$ denotes the label. Here, the feature vector x resides in $\mathbb{R}^M$ and is input into the classifier, and w_i represents the i-th column vector of the weight matrix W. The ideal scenario occurs when $w_y^{*T} x \to \infty$ and $w_i^T x$ for $i \neq y$ remains comparatively low. This optimisation is supported by the adjusted softmax weights:

$$w_i^{*T} x = \begin{cases} \log\left(\frac{(K-1)(1-\epsilon_y)}{\epsilon_y}\right) + c, & \text{if } i = y, \\ c, & \text{otherwise,} \end{cases} \tag{6}$$

where c is a calibration constant.

4.4 Dynamic Label-Aware Smoothing for ViTs

Label-aware smoothing in CNNs adjusts the confidence levels assigned to class labels to mitigate the model's overconfidence. For ViTs, which interpret the global context of an image differently through a self-attention mechanism, we propose a dynamic label-aware smoothing method that adjusts smoothing factors based on the aggregated attention scores computed for each patch. The modified label smoothing formula is given by:

$$q_i = \begin{cases} 1 - \epsilon_y \cdot \bar{\alpha}_i & \text{if } i = \text{true class}, \\ \frac{\epsilon_y \cdot \bar{\alpha}_i}{K-1} & \text{otherwise,} \end{cases} \tag{7}$$

where $\bar{\alpha_i}$ represents the aggregated attention scores for the patches contributing to the prediction of class i, and ϵ_y is the base smoothing factor, chosen based on preliminary experiments to optimise model performance. This adjustment ensures that the smoothing factor is contextually relevant, reflecting the varying significance of different image parts as perceived by the transformer.

4.5 Attention-Weighted Smoothing Factor

In ViTs, the self-attention mechanism assigns different weights to input image patches, reflecting their importance. These weights, $\alpha = \{\alpha_1, \alpha_2, ..., \alpha_N\}$ for N patches, are derived using the softmax function to ensure they are normalised and sum to one, essential for their subsequent use as dynamic factors in label smoothing. The normalised attention weight for each patch i is calculated as:

$$\alpha_j = \frac{\exp(a_j)}{\sum_{z=1}^{N} \exp(a_z)} \tag{8}$$

where a_j is the raw attention score for patch j and $\bar{\alpha}_i = \frac{1}{N} \sum_{j=1}^{N} \alpha_j$. These weights are then used to scale the label smoothing factors dynamically. This dynamic adjustment ensures that patches deemed more significant by the attention mechanism have a greater influence on the smoothed labels, thus aligning the model's learning process more closely with the intrinsic data distribution and enhancing its performance on imbalanced datasets.

5 Experiment Details

5.1 Model Training

For our experiments, we utilised two pre-trained models: Google's ViT [47] and Microsoft's BEiT [13], both adapted to classify the three distinct categories depicted in Fig. 1. The input images underwent preprocessing with transformations, including random resized cropping to (224×224), random horizontal flipping, and rotations of up to 10°C. Additionally, brightness, contrast, saturation, and hue were moderately adjusted to augment the dataset. We applied normalisation using the default ImageNet means and standard deviations. The dataset was split into training (70%) and validation (10%) sets. Class weights were computed and normalised to ensure balanced learning across the categories.

To handle the long-tail distribution, we evaluated several techniques, including LAS loss and Bal_CE loss, with parameters fine-tuned according to the class weights and specific model requirements. Optimisation was conducted using the Adam optimiser with an initial learning rate of 0.00005, with training extending up to 30 epochs or until convergence was observed on the validation set.

The optimal model configurations were determined by achieving the highest validation accuracy. We fine-tuned hyperparameters such as learning rate and batch size in subsequent training runs, with detailed configuration tracking using Weights & Biases [3]. To ensure reproducibility, three random seeds were

used across multiple runs. The results are reported as the best validation accuracy obtained during training, demonstrating that the models-especially ViT—performed well, outperforming traditional CNN-based models and offering competitive results compared to BEiT.

5.2 Hyperparameter Grid Search

To optimise the performance of our models, we conducted a hyperparameter grid search, systematically exploring multiple combinations to identify the best configuration. For the ResNet model, the grid included the number of epochs $\{20, 30, 100\}$ and learning rates $\{0.01, 0.001, 0.0001\}$. The optimal configuration, determined based on validation accuracy, was a learning rate of 0.0001 and 30 epochs. For the fine-tuned ViT and BEiT models, we explored learning rates $\{1 \times 10^{-4}, 5 \times 10^{-5}, 1 \times 10^{-5}\}$, epochs $\{10, 20, 30\}$, a training batch size of 32, an evaluation batch size of 32, and gradient accumulation steps set to 4. The optimiser used was Adam with β values $\{0.9, 0.999\}$ and an epsilon of 1×10^{-8}. We employed a linear learning rate scheduler with a warmup ratio of 0.1 to improve training stability. These hyperparameters were meticulously selected to ensure robust training and high accuracy.

5.3 Metrics

We evaluated each model by calculating the macro-averaged f1-score, which assigns equal importance to all classes, regardless of their frequency. This approach ensures that the f1-score, recall, and precision metrics are calculated in a way that considers each class equally. By using macro-averaging, the method is sensitive to underperformance in less frequent classes, highlighting areas where the model may struggle. Additionally, we normalised the confusion matrix to show percentages, offering a clear visual representation of the model's strengths and weaknesses across different categories (supplementary materials).

6 Results

6.1 Classifier

Our proposed dynamic LAS loss function consistently outperforms alternatives, particularly when integrated with the ViT architecture. The ViT-LAS model achieves the highest macro-average f1-score (0.98 ± 0.2) and non-copepod f1-score (0.95 ± 0.5), setting a new benchmark for balanced multi-class classification in this domain. While models utilising standard cross entropy (ST_CE) and Bal_CE demonstrate competitive performance, they encounter challenges with the non-copepod class, where class imbalance is more severe. BEiT and ViT models trained with BCE and LDAM losses also achieve strong results but exhibit more significant variability across multiple runs. DL-based models and loss functions were compared based on their macro-averaged and minority class (non-copepod) f1-score, shown in Table 1. Our findings highlight the dynamic

Table 1. Comparative performance metrics of deep learning models: this table displays f1-score and standard deviations for various models under different loss functions and training epochs, covering overall macro averages and minority class-specific scores for non-copepods.

Model name	Loss function	Number of epochs	F1-score (macro avg)	F1-score (non-copepod)
ResNet	ST_CE	30	0.93 ± 0.07	0.83 ± 0.2
Beit	Bal_CE	20	0.95 ± 0.05	0.88 ± 0.5
Beit	LDAM	20	0.96 ± 0.2	0.90 ± 0.4
Beit	D-LAS (ours)	20	$\mathbf{0.97 \pm 0.3}$	$\mathbf{0.93 \pm 0.7}$
Beit	BCE	10	0.96 ± 0.1	0.91 ± 0.8
Beit	ST_CE	20	0.94 ± 0.5	0.90 ± 0.3
Beit	CB_CE	20	0.94 ± 0.7	0.86 ± 0.2
ViT	ST_CE	30	0.93 ± 0.4	0.90 ± 0.1
ViT	BCE	20	0.97 ± 0.5	0.93 ± 0.6
ViT	Bal_CE	20	0.97 ± 0.2	0.92 ± 0.6
ViT	CB_CE	10	0.97 ± 0.7	0.92 ± 0.5
ViT	LDAM	10	0.97 ± 0.4	0.94 ± 0.5
ViT	D-LAS (ours)	30	$\mathbf{0.98 \pm 0.2}$	$\mathbf{0.95 \pm 0.5}$

LAS loss function's robustness in addressing class imbalance and improving classification performance, particularly for the underrepresented non-copepod class. The model with dynamic LAS loss function stands out as the top performer, making it the most dependable choice for this classification task.

7 Discussion

The experimental results emphasise the crucial role of Transformers and advanced loss functions in addressing the challenges of plankton image classification. Our approach demonstrated a 0.12% improvement in the f1-score for the minority class—a notable achievement given the severe class imbalance typically found in such datasets. This gain underscores the effectiveness of integrating ViT models with dynamic LAS loss function, highlighting their capability to manage complex, imbalanced data distributions.

Marine particle classification, especially differentiating between the key classes copepod, detritus, and non-copepod presents a significant challenge due to the skewed distribution and high variability within these categories. Our methodology capitalised on cutting-edge techniques such as transfer learning and long-tail learning strategies to tackle this highly imbalanced dataset. Our model achieved substantial performance gains by integrating pre-trained ViT and BEiT architectures with LTR loss functions. Specifically, the proposed dynamic LAS approach achieved an overall f1-score of 98%, with an impressive 95%f1-score for

the tail class (non-copepods), a considerable improvement over the traditional ResNet model. In comparison, while ResNet attained an overall f1-score of 93%, it struggled with the non-copepod category, achieving only 83%.

These results align with existing literature comparing transformer-based models and convolutional neural networks (CNNs) in imbalanced classification tasks [9,26,33]. Research consistently shows that ViT models, leveraging their global attention mechanisms, excel at extracting features from underrepresented classes, leading to superior performance in settings characterised by severe class imbalance. Additionally, the LAS technique complements this by reducing bias towards majority classes and mitigating overfitting, which aligns with our observed improvements.

The integration of ViT with LAS and LTR strategies within an end-to-end trainable framework not only enhances overall classification accuracy but also significantly improves the performance of minority classes. This is particularly valuable in scenarios involving complex visual data with high intra-class variability.

8 Conclusion

Marine scientists studying pelagic ecosystems face significant challenges in managing and analysing vast imagery datasets from diverse habitats and multiple imaging systems. Automated camera-based sensors are extensively utilised in vessel-based research to monitor plankton and marine particles. Despite their utility, annotating data for fully supervised learning in plankton grouping tasks is costly and time-intensive. In response, real-time and open-source software can be adapted to classify images of other marine objects and species, helping transform how scientists study the oceans. This opens the door to a new era of monitoring beyond plankton, where measured variables can be seen in real-time, thus allowing for sampling to be adapted according to visible changes. Studying changes in community structures and biodiversity as they happen will help further our understanding of what drives changes in biodiversity and community structures within the environment, thus increasing our ability to use our seas sustainably. Looking ahead, we plan to enhance our methods for analysing marine plankton in real-time at a higher taxonomic resolution and revising biodiversity from a computer vision perspective to tackle the DSS problem. By utilising foundation models and self-supervised learning methods, we aim to address the limitations of traditional machine learning approaches that require retraining to accommodate new camera systems. These advanced techniques will enable consistent categorisation of particles, even when camera systems change, without the need for extensive retraining of machine learning models.

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Code Availability. The code we used to train and evaluate our models is available at https://github.com/noushineftekhari/ViT-LASNet

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KAN-Mixer: Kolmogorov-Arnold Networks for Gene Expression Prediction in Plant Species

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Abstract. Understanding the intricate relationships between cis-regulatory elements and gene expression is crucial for decoding genetic regulation in ecological systems. In this study, we introduce a novel application of Kolmogorov-Arnold Networks (KANs) for predicting gene expression across diverse plant species, including Arabidopsis thaliana, Solanum lycopersicum, Sorghum bicolor, and Zea mays. Our model, named KAN-Mixer, utilizes k-mers as inductive biases to capture biologically relevant patterns in nucleotide sequences. By employing token embeddings and mixer architectures, KAN-Mixer enhances both the interpretability and usability of KANs. Our results indicate that KAN-Mixer achieves comparable accuracy to ConvNet-based approaches while offering superior interpretability, making it a robust tool for ecological data analysis in a variety of environments.

Keywords: KAN · Genetic Regulation · Ecology

1 Introduction

Gene expression regulation is orchestrated by complex interactions between proteins and nucleic acids, particularly DNA and RNA. A key element in this regulation is cis-regulatory elements (CREs), short DNA sequences near a gene's genomic region recognized by transcription factors. At the transcript level, RNA processing is regulated by nucleotide sequences binding protein factors, forming a sophisticated gene regulatory network. Understanding these regulatory mechanisms is essential for comprehending ecological dynamics across diverse environments.

Current experimental molecular biology techniques often provide a limited understanding of these nucleotide codes due to their reductionistic nature. Consequently, a holistic approach, such as deep learning, can offer substantial insights into the plant gene regulatory code. With the increasing availability of genomic data, deep learning

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methods can more effectively annotate and functionally characterize CREs. Systematic investigations of sequence-to-regulation relationships across various plant species and regulatory domains are essential, as genome-scale identification and annotation of cis-regulatory sequence features remain largely unexplored, especially considering the diverse ecological systems ranging from tropical forests to Arctic tundras, and urban to rural settings.

Deep learning applications in exploring gene regulatory networks, particularly for annotating and functionally characterizing CREs, require a nuanced understanding of neural network architectures, their inductive biases, and interpretability. Multi-layer perceptrons (MLPs) [12], known for their universal approximator capability, excel at learning complex representations but struggle with tasks involving spatial or temporal dependencies, such as identifying CREs in genomic sequences. Their minimal inductive bias requires significant data to uncover relevant features, often leading to overfitting and making the models challenging to interpret.

Inspired by the Kolmogorov-Arnold representation theorem [3,14], Kolmogorov-Arnold Networks (KANs) [19] replace the fixed activation functions of MLPs with learnable activation functions on edges, parameterized as splines. This change allows KANs to outperform MLPs in both accuracy and interpretability. Both theoretically and empirically, KANs exhibit faster neural scaling laws compared to MLPs, suggesting a more efficient learning process. For interpretability, KANs offer intuitive visualization and easy interaction, which is particularly beneficial in scientific research. However, to ensure interpretability, the overly long gene sequences must be flattened and input into the KANs, which introduces a significant parameter burden. This has become a major obstacle hindering the widespread application of KANs in the processing of genetic data.

In this paper, we demonstrate the interpretable use of KANs for gene expression prediction in four plant species *Arabidopsis thaliana*, *Solanum lycopersicum*, *Sorghum bicolor*, and *Zea mays*. We present KAN-Mixer, a straightforward yet interpretable model for predicting gene sequence-expression relationships. Using k-mers as an inductive bias for gene sequence analysis is analogous to applying convolutional filters in convolutional neural networks (ConvNets) for visual recognition. K-mers, which are sequences of fixed length k, encapsulate significant biological signals by assuming that closely positioned nucleotides and recurring patterns hold important biological relevance. Specifically, we propose token embeddings for gene sequences and constrain the modeling of the KANs at the token level, enhancing the usability of KANs while preserving interpretability.

Inspired by the MLP-Mixer architecture [24], we utilize mixer architectures to facilitate communication between different channels and tokens. KAN-Mixer emphasizes specific sequence motifs and their hierarchical organization, offering more insightful and explainable results compared to MLPs and ConvNets. This aligns with the ecological objectives of genomics research, where understanding the underlying mechanisms and functional implications of CREs necessitates models that provide clear and biologically interpretable insights. Our experiments demonstrate that KAN-Mixer is at least as effective as existing ConvNet-based baselines [21], with the added benefit of enhanced interpretability.

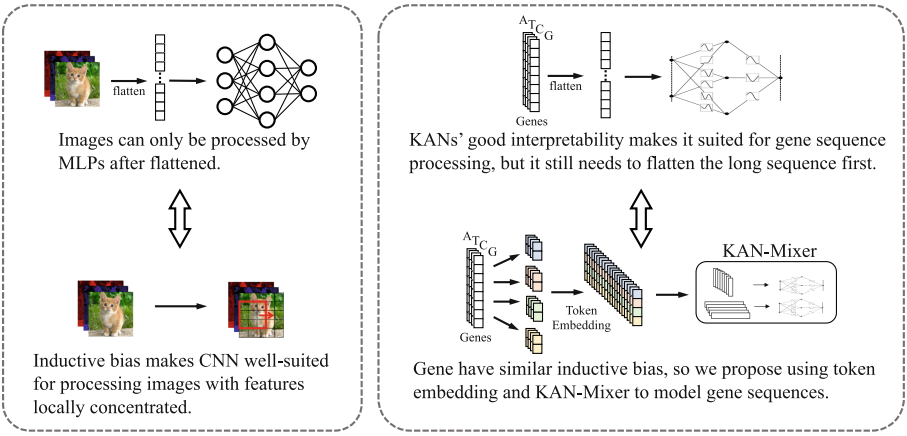

Fig. 1. KANs' notable interpretability makes it suitable for gene sequence processing, yet it struggles with extremely long gene sequences due to significant parameter burdens and performance declines. Inspired by convolutional inductive biases, we introduce KAN-Mixer with Token Embedding to enhance KANs usability in gene sequence analysis without compromising interpretability.

By providing a robust and interpretable model for gene expression prediction, KAN-Mixer contributes to the broader goal of advancing ecological research through the integration of cutting-edge computer vision techniques. Understanding gene regulatory mechanisms across diverse plant species not only enhances our knowledge of fundamental biological processes but also informs conservation strategies, agricultural practices, and ecological management. This work highlights the potential for deep learning models to bridge the gap between complex genetic data and practical ecological applications, fostering a deeper connection between the fields of computer vision and ecology (Fig. 1).

2 Related Work

MLPs and KANs. Multi-layer perceptrons (MLPs) [7,11,12,25] are widely recognized as fundamental components of contemporary deep learning models. Despite their prevalent use, MLPs have significant limitations, particularly in terms of interpretability [6]. Recently, Kolmogorov-Arnold Networks (KANs) [18,19] have been introduced, addressing this issue by employing learnable activation functions directly on the network's edges. These functions are formulated as splines based on the Kolmogorov-Arnold representation theorem [3,14,15,23]. This structural innovation eliminates the need for linear weights and enhances interpretability, allowing scientists to directly identify critical components within the input data. This feature makes KANs particularly valuable for aiding new scientific discoveries. However, in the field of biology, the application of KANs presents unique challenges due to the extremely high dimensionality of data, such as gene sequences. The substantial length of these data imposes a sig-

nificant parameter burden on KANs. Although recent efforts have focused on improving the usability and computational efficiency of KANs, significant challenges remain in their application within biological domains. In this paper, we aim to improve the usability of KANs in modeling gene sequences without compromising their interpretability and performance.

Gene Sequence-Expression Prediction. Although deep learning models often yield accurate results, they frequently lack interpretability, a critical shortcoming in many scientific applications. For instance, in the task of gene sequence-expression prediction, biologists require not only precise predictions but also an understanding of which specific gene segments play key roles, providing insights that lead to new scientific discoveries. In this paper, we specifically focus on the relationship between non-coding regulatory element sequences and gene expression, a fundamental task for understanding gene regulation. Previous work [21] employed ConvNet-based approach to predict gene expression in various plant species. Our approach seeks to enhance predictive effectiveness with deeper biological insights.

3 KAN-Mixer

In this section, we detail our method, KAN-Mixer, to deal with the challenges posed by KANs in processing long gene sequences. As shown in Fig. 3, we introduce a token embedding layer at the beginning of the model to extract features, with interpretability ensured by the k-mers characteristics of gene sequences. Subsequently, we innovatively incorporate KAN-Mixer to simultaneously focus on both channel and token information within the tensor.

3.1 Premilinary

KAN is predicated on the Kolmogorov-Arnold Representation Theorem (Eq. 1) with learnable activation functions on edges and summation operations on nodes. This theorem postulates that any multivariate continuous function on a bounded domain can be expressed as a finite composition of continuous functions of a single variable and the binary operation of addition.

$$f(\mathbf{x}) = \sum_{q=1}^{2n+1} \Phi_q \left(\sum_{p=1}^{n} \phi_{q,p}\left(x_p\right) \right) \tag{1}$$

for a smooth $f : [0,1]^n \to \mathbb{R}$, $\phi_{q,p} : [0,1] \to \mathbb{R}$ and $\Phi_q : \mathbb{R} \to \mathbb{R}$.

KAN uses B-spline basis functions (Eq. 2) to approximate $\phi_{q,p}$ and Φ_q. Although KAN exhibits competitive accuracy and superior interpretability compared to MLP, its computational speed is notably slower. To mitigate this deficiency, we replace the B-spline basis functions (Eq. 2) with *Gaussian Radial Basis Functions* (RBF) (Eq. 4) [2,4,10], following the same rationale as in [18]. These RBFs accurately approximate the B-spline basis (in Fig. 2) and are computationally efficient.

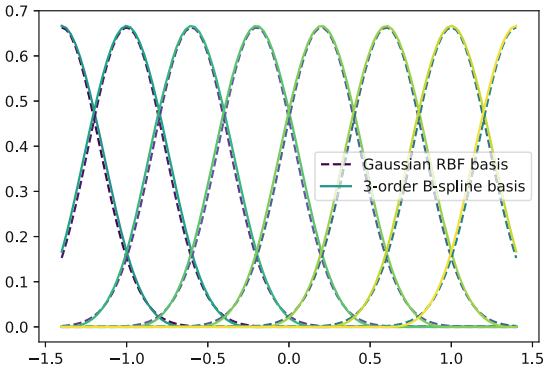

Fig. 2. Gaussian Radial Basis Functions can well approximate 3-order B-Spline basis.

$$B_{i,0}(x) = \begin{cases} 1 & \text{if } x_i \le x < x_{i+1}, \\ 0 & \text{otherwise} \end{cases} \tag{2}$$

$$B_{i,p}(x) = \frac{x - x_i}{x_{i+p} - x_i} B_{i,p-1}(x) + \frac{x_{i+p+1} - x}{x_{i+p+1} - x_{i+1}} B_{i+1,p-1}(x) \tag{3}$$

$$R_i(x) = \exp\left(-\left(\frac{x - x_i}{h}\right)^2\right) \tag{4}$$

Except using RBF instead of B-spline basis function, *LayerNorm* (Eq. 5) [1] is also applied compared to the original KAN. LayerNorm is placed before RBF transform after the input in order to scale inputs to the range of spline grids.

$$\mathrm{L}(x) = \frac{\mathrm{x} - \mathrm{E(x)}}{\sqrt{\mathrm{Var(x)} + \epsilon}} * \gamma + \beta \tag{5}$$

$$\phi_{l,i,j}(\cdot) = R_{l,i,j}(\mathrm{L}(\mathbf{x}_l), g_min, g_max, num_g) \tag{6}$$

where g_min, g_max, num_g determines the grid by changing the range and the smoothness of the RBF basis.

A KAN with layers $[n_0, n_1, \cdots, n_L]$ can be expressed as $\mathrm{KAN}(\mathbf{x}) = (\boldsymbol{\Phi}_L \circ \cdots \boldsymbol{\Phi}_2 \circ \boldsymbol{\Phi}_1)(\mathbf{x})$ with Φ_l shape $[n_{l+1}, n_l]$. The operation from layer n_l to layer n_{l+1} can be represented as

$$\mathbf{x}_{l+1} = \underbrace{\begin{pmatrix} \phi_{l,1,1}(\cdot) & \phi_{l,1,2}(\cdot) & \cdots & \phi_{l,1,n_l}(\cdot) \\ \phi_{l,2,1}(\cdot) & \phi_{l,2,2}(\cdot) & \cdots & \phi_{l,2,n_l}(\cdot) \\ \vdots & \vdots & & \vdots \\ \phi_{l,n_{l+1},1}(\cdot) & \phi_{l,n_{l+1},2}(\cdot) & \cdots & \phi_{l,n_{l+1},n_l}(\cdot) \end{pmatrix}}_{\boldsymbol{\Phi}_l} \mathbf{x}_l, \tag{7}$$

where $\boldsymbol{\Phi}_l$ is the function matrix corresponding to the l^{th} KAN layer, $\mathbf{x}_l$ is the input from the l^{th} KAN layer with shape $[n_l, 1]$ and $\mathbf{x}_{l+1}$ is the output from the $(l+1)^{\text{th}}$ KAN layer with shape $[n_{l+1}, 1]$. The input layer data undergoes layer normalization to transform it into the range of RBF spline grids. Subsequently, the RBF basis is computed based on this normalized data, serving as an operator. These RBF basis-derived operators are then learned through a MLP layer with solely trainable weight parameters. This process constitutes the learning of the $\boldsymbol{\Phi}_l$ component.

3.2 Token Embedding

The utilization of k-mers [13,22] enables the extraction of fixed-length features from gene sequences, analogous to employing fixed-size filters in image processing. This method aligns with token-level analysis, reflecting the principles of k-mers in identifying critical parts of gene sequences.

Convolutional neural networks (ConvNets) [17] possess inherent inductive biases that enhance learning efficiency [5,8,26], particularly through locality and translation invariance. These biases allow ConvNets to exploit spatial hierarchies in data, making them highly effective for image recognition tasks. The architecture of ConvNets enables the learning of local features through convolutional layers, making them particularly suitable for tasks involving spatial hierarchies, such as sequence analysis in genomics. In this context, ConvNets can effectively capture local dependencies within nucleotide sequences, identifying regulatory motifs that are crucial for gene expression regulation. The weight-sharing mechanism in ConvNets reduces the number of parameters, enhancing both generalization and interpretability, as the learned filters often correspond to biologically meaningful motifs.

Our model incorporates a Patch Embedding layer at the outset to ensure input interpretability. This layer transforms one-hot encoded gene sequence data into a format that enhances the model's ability to extract and interpret meaningful features from complex biological sequences. We employ a one-dimensional convolutional layer with both the kernel size and stride set to 8. This configuration allows us to analyze the importance of each input sequence segment in increments of 8 units. By balancing granularity and comprehensibility, we can precisely identify segments that significantly impact the model's predictions. This choice maintains the clarity of the model's decision-making process and aligns with our goal of enhancing the interpretability of deep learning models in genomic research.

3.3 Mixer Architecture

To extract richer features, our token embedding layer transforms the original 4-channel one-hot representation of the gene sequence into 128 channels. This transformation requires subsequent processing to calculate feature relationships not only between different tokens but also across different channels. Inspired by the MLP-Mixer [24], we leverage KANs to both token mixing and channel mixing processes, introducing our method, KAN-Mixer. Each block in KAN-Mixer contains two distinct KANs, dedicated to token mixing and channel mixing, respectively.

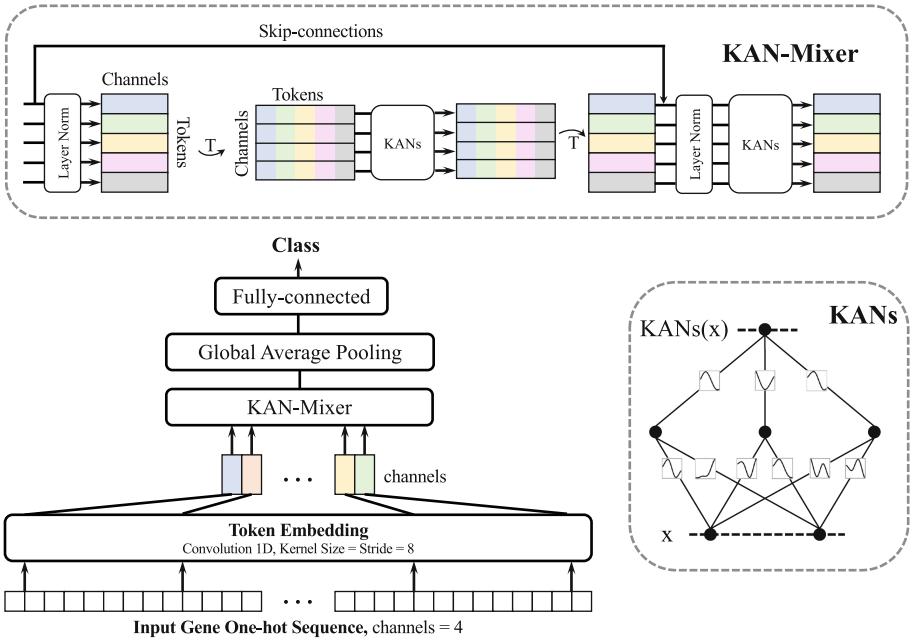

Fig. 3. Structure of our Method. We introduce a token embedding layer at the beginning of our model, setting the smallest modeling unit as a short genetic sequence. To enhance the learning of relationships between adjacent tokens, we innovatively employ the KAN for both token mixing and channel mixing, collectively forming the KAN Mixer.

Token Mixing with KANs. We replace the original fully connected layers with KANs to leverage RBF approximation. This block initially expands the dimensionality of the token representation to a higher-dimensional feature space, providing the necessary operational space for KANs to learn and integrate features across a broader functional landscape.

Channel Mixing with KANs. Similarly, we utilize KANs to process the embedding dimensions across different input channels. This block maintains the original dimensionality but focuses on optimizing inter-channel feature integration directly within the embedding space.

The input first undergoes processing through the Token Mixing block, where features among different tokens (patches) are manipulated and integrated. The output from this block is combined with the input via a residual connection, enhancing information retention and gradient flow across the network. This combined output is then fed into the Channel Mixing block, where features are mixed at the channel level, enabling a comprehensive amalgamation of learned representations across different feature channels. In our experiments, for a vanilla KAN with layer dimensions [3020$\times$4, 128, 2], KAN-Mixer reduces the parameter count from 13.94 million to 1.44 million, demonstrating its improved applicability and lighter parameter load.

3.4 Exploration on CIFAR

We first conducted a comparative study between KAN-Mixer and MLP-Mixer [24] on tasks simpler than gene sequence-expression prediction. Specifically, we performed image classification on the widely used CIFAR-10/100 datasets [16], which consist of 60,000 images divided into 10 and 100 classes, respectively. To ensure a fair comparison, we configured KAN-Mixer with a similar number of parameters as the tiny-sized MLP-Mixer. The experimental results are presented in Table 1.

Table 1. Our method, KAN-Mixer, achieves performance comparable to the MLP-Mixer on CIFAR-10/100 datasets. Accuracy, measured as a percentage, is used to evaluate performance, with higher values indicating better outcomes.

	CIFAR-10	CIFAR-100
MLP-Mixer [24]	89.8	70.1
KAN-Mixer (Ours)	91.2	70.2

In comparison to the state-of-the-art all-MLP architecture, MLP-Mixer, KAN-Mixer demonstrates comparable accuracy in classifying small images. The CIFAR-10/100 experiments were crucial for fine-tuning KAN-Mixer's hyperparameters, *e.g.*, learning rate, normalization, weight decay, stride, layer number, and so on. As CIFAR-10/100 is a standard dataset in deep learning with extensive research, it provides an ideal benchmark for identifying issues in KAN-Mixer. We train our model on the training set and select the suitable hyperparameters based on the test set. Moreover, since MLP-Mixer is a vision model, performing CIFAR-10/100 classification tasks allows for straightforward apples-to-apples comparisons.

During our experiments, we found that optimizing splines (using RBF) is a crucial factor influencing both the performance and stability of KAN-Mixer. Additionally, KAN exhibits a distinct advantage in interpretability, enabling the direct identification of key regions in the input images. Consequently, we extended our study to address the high-dimensional gene sequence expression challenge, thereby highlighting the interpretability of KAN-Mixer.

4 Experiments

We first introduce the experimental setting concerning gene data, models, and metrics for evaluation in Sect. 4.1. Besides, we report an overall comparison between KAN-Mixer and baselines in Sect. 4.2. Finally, we provide the visualization of KAN-Mixer and provide some interpretations in Sect. 4.3.

4.1 Settings

Dataset. Following the same dataset setting[1] of the [21], we utilize the genome assemblies and annotations from the Ensembl Plants database (v52) [9]. These sequences span 500−3000 nucleotides upstream and 100−700 nucleotides downstream of the transcription start site (TSS), and 100−700 nucleotides upstream and 500−3000 nucleotides downstream of the transcription termination site (TTS). We encoded the gene flanking sequences using one-hot encoding, resulting in data shaped (3020, 4).

Labels are derived by estimating the log-transformed transcript per million values (logMaxTPM) and categorizing them as low, medium, or high, based on the lower and upper quartiles of the logMaxTPM distribution. We specifically target genes that fall below the lower quartile or above the upper quartile of this distribution. Consequently, the task is framed as a binary classification problem, aiming to predict genes as either low (below the lower quartile) or high (above the upper quartile).

Model. The model is structured into four key components: a patch embedding layer, a KAN-Mixer layer, an average pooling layer, and a linear head for binary classification. For a fair comparison with the baseline [21], the patch embedding layer extracts an "8-mer" feature by employing a 1D convolution with both a patch size and a stride of 8. This results in 378−dimensional tokens that are subsequently processed by the KAN-Mixer layer, which has a feature dimension of 128, matching that of the final linear head. Notably, the configuration of the KAN-Mixer layer here differs from that used in our CIFAR experiments. Detailed code will be released soon. For optimization, we employ an Adam optimizer with a learning rate of 5e-4 and a weight decay rate of 0.1 for the spline linear layer. Additional details are available in the appendix.

The baseline model, as detailed in the original paper by Peleke et al. [21], comprises three blocks and an MLP. Each block contains two convolutional layers, a max pooling layer, and a dropout layer. Further specifications are provided in the appendix. Notably, the original study used the test set to adjust the learning rate and implement early stopping, and considered the best training accuracy as the final result[2]. We address these methodological flaws by eliminating any test set exposure during training and limiting the training duration to 5 epochs, as both models demonstrate rapid convergence. Additionally, for comparison with KAN-Mixer, our baselines include both traditional KANs and MLPs with layer dimensions of $[3020 \times 4, 128, 2]$.

Metrics. There are two settings in our experiment, *single species* (in-domain), and *cross species* (out-of-domain). We conduct experiments on four plant species, *Arabidopsis thaliana*, *Solanum lycopersicum*, *Sorghum bicolor*, and *Zea mays*.

In the **single species (in-domain)** task, each species had two tissue types examined, root and leaf. To testify to the validity of our models and baselines, we take chromosomal-level cross-validation, in which for each iteration, genes located on one of the chromosomes were used as a validation set and the rest for training. In other

[1] https://github.com/NAMlab/DeepCRE.

[2] Details can be found in the official codebase.

words, for a given chromosome within a specific tissue and species, all other chromosomes from that tissue and species constituted the training set, while that chromosome served as the validation set.

In the **cross species (out-of-domain)** task, for a particular chromosome within a specific tissue and species, all other species with the same tissue type formed the training set, and that chromosome functioned as the validation set.

The metrics evaluated are validation accuracy and the area under the receiver operating characteristic curve (AUC-ROC), which is a performance metric used to evaluate the ability of a binary classification model to distinguish between classes, representing the likelihood that a randomly chosen positive instance is ranked higher than a randomly chosen negative one.

Table 2. Performance comparison on in-domain plant datasets. Our method KAN-Mixer achieves comparable accuracy (Acc) and AUC-ROC scores to the baseline across different plant datasets, with a significant improvement on the Arabidopsis thaliana dataset.

Method	Zea		Sor		Ara		Sol		Average	
	Acc↑	AUC-ROC↑	Acc↑	AUC-ROC↑	Acc ↑	AUC-ROC ↑	Acc↑	AUC-ROC ↑	Acc ↑	AUC-ROC ↑
Vanilla-MLP	75.1	79.3	73.0	75.5	76.8	83.0	78.0	81.6	75.7	79.5
Vanilla-KAN	67.5	66.9	61.9	66.1	67.9	73.0	69.4	71.1	66.7	68.8
KAN-Mixer	76.4	82.5	75.5	81.5	78.9	87.2	79.5	85.5	77.5	83.8
ConvNet [21]	77.7	83.8	76.6	81.9	73.2	80.1	80.2	86.2	77.6	83.6

4.2 Results

In-domain Experiments. As evidenced by the results presented in Table 2, KAN-Mixer achieves highly competitive performance compared to the baseline method across various plant species. Notably, for the Zea mays dataset, KAN-Mixer attains superior accuracy and AUC-ROC score of 78.9% and 87.2%, respectively, outperforming the ConvNet's [21] 73.2% and 80.1%. This remarkable improvement highlights KAN-Mixer's capability to handle complex biological tasks in an internal environment. Moreover, the results demonstrate that KAN-Mixer's performance is on par with the baseline across the remaining plant species, with comparable or marginally lower scores, demonstrating the effectiveness and applicability of KAN-Mixer. A detailed experiment can be found in Fig. 4. More results are shown in the Appendix (Table 4).

Out-domain Experiments. Extending our evaluation to out-domain scenarios, the findings presented in Table 3 underscore KAN-Mixer's remarkable generalization capabilities. While KAN-Mixer's accuracy and AUC-ROC scores exhibit a slight decrease compared to the baseline for certain plant species, such as Arabidopsis thaliana and Solanum lycopersicum, its overall performance remains highly competitive and on par with the baseline method's average scores. Remarkably, KAN-Mixer outperforms the baseline in predicting traits for Zea mays, achieving superior accuracy and AUC-ROC

Table 3. Out-domain performance comparison across plant datasets. While our method KAN-Mixer exhibits slightly lower accuracy and Auc-roc scores compared to the baseline for certain plant species like Arabidopsis thaliana, its overall performance remains highly competitive. Notably, KAN-Mixer outperforms the baseline in predicting traits for Zea mays, achieving higher accuracy of 0.782 and Auc-ROC of 0.867. This showcases KAN-Mixer's ability to effectively transfer knowledge from in-domain tasks to related out-domain contexts, demonstrating its strong generalization capabilities.

Method	Zea		Sor		Ara		Sol		Average	
	Acc ↑	AUC-ROC ↑	Acc ↑	AUC-ROC ↑	Acc ↑	AUC-ROC ↑	Acc ↑	AUC-ROC ↑	Acc ↑	AUC-ROC ↑
KAN-Mixer	78.2	86.7	75.4	83.3	71.0	80.3	70.7	83.9	74.1	84.0
ConvNet [21]	76.8	83.4	74.0	81.9	75.0	83.2	74.0	81.9	75.8	83.4

scores of 78.2% and 86.7%, respectively. This outcome highlights KAN-Mixer's adeptness at transferring knowledge acquired from in-domain tasks to related yet distinct out-domain contexts, thereby enhancing its predictive prowess and generalization capabilities. The consistent and robust performance of KAN-Mixer across both in-domain and out-domain experiments substantiates its efficacy and potential for widespread adoption in diverse biological applications.

4.3 Interpretability and Visualization

KAN-Mixer effectively exchanges the input information in both token and feature dimensions. We plot the different values of the inputs and outputs (left and right) of one KAN-Mixer layer in Fig. 4. The figure is drawn from the root of the Solanum lycopersicum, on the 6th chromosome. The x-axis delineates the token dimensions extracted from 8-mer gene sequences, ranging from 1 to 378 dimensions. The y-axis represents the data dimensions. After training, a total of two hundred test samples contributed to the generation of this graph. Each point (x, y) is colored according to the numerical value of the x-th dimension of the token for the y-th sample, where a greater numerical value corresponds to a bluer color, while a smaller one corresponds to a more yellow hue. Before our KAN-Mixer layer, the data feature has almost no distribution among the token dimensions. However, the distribution among the token dimensions becomes vivid after KAN-Mixer layer, since there is a significant increase in the number of vertical lines in the output graph. Compared to the input, the output distribution exhibits a pattern of interweaving both horizontally and vertically.

KAN-Mixer demonstrates a spontaneous intra-layer pattern in untranslated gene regions and gene expression. Figure 5 adopts the same pruning and visualization of KANs [19], where a scatter figure is plotted to record the pre-layer input and the post-layer output in diverse dimensions. As the token dimension of 128 is too high for visualization, we rank the same importance score and select the top dimension. The red points in the figure represent different data samples. The x-axis plots the value of input value of each sample at the top input dimension. The y-axis plots the value of output value of each sample at the top output dimension. Layer 1 has a rough Gaussian distribution while layer 2 represents a different Sigmoid distribution. Note that we never

Table 4. In-domain performance comparison across plant datasets.

Species	Tissue	Chromosome	Baseline		KAN-Mixer	
			acc ↑	aucroc ↑	acc ↑	aucroc ↑
Arabidopsis thaliana	leaf	1	0.677	0.748	0.757	0.850
Arabidopsis thaliana	leaf	2	0.682	0.753	0.779	0.856
Arabidopsis thaliana	leaf	3	0.799	0.870	0.797	0.890
Arabidopsis thaliana	leaf	4	0.799	0.851	0.776	0.863
Arabidopsis thaliana	leaf	5	0.746	0.803	0.753	0.847
Arabidopsis thaliana	root	1	0.693	0.763	0.768	0.834
Arabidopsis thaliana	root	2	0.741	0.819	0.797	0.893
Arabidopsis thaliana	root	3	0.756	0.839	0.826	0.910
Arabidopsis thaliana	root	4	0.723	0.804	0.824	0.887
Arabidopsis thaliana	root	5	0.702	0.762	0.811	0.887
Solanum lycopersicum	leaf	1	0.748	0.843	0.755	0.845
Solanum lycopersicum	leaf	2	0.749	0.826	0.806	0.843
Solanum lycopersicum	leaf	3	0.754	0.820	0.757	0.796
Solanum lycopersicum	leaf	5	0.853	0.870	0.855	0.884
Solanum lycopersicum	leaf	6	0.790	0.867	0.788	0.860
Solanum lycopersicum	leaf	7	0.807	0.859	0.799	0.851
Solanum lycopersicum	leaf	8	0.800	0.872	0.815	0.863
Solanum lycopersicum	leaf	9	0.801	0.850	0.795	0.848
Solanum lycopersicum	leaf	10	0.828	0.864	0.772	0.852
Solanum lycopersicum	leaf	11	0.856	0.905	0.847	0.900
Solanum lycopersicum	leaf	12	0.793	0.811	0.793	0.823
Solanum lycopersicum	root	1	0.779	0.876	0.771	0.854
Solanum lycopersicum	root	2	0.802	0.854	0.792	0.851
Solanum lycopersicum	root	3	0.784	0.826	0.741	0.812
Solanum lycopersicum	root	4	0.820	0.872	0.811	0.879
Solanum lycopersicum	root	5	0.809	0.854	0.786	0.849
Solanum lycopersicum	root	6	0.849	0.906	0.842	0.897
Solanum lycopersicum	root	7	0.814	0.885	0.791	0.863
Solanum lycopersicum	root	8	0.823	0.896	0.755	0.841
Solanum lycopersicum	root	9	0.788	0.830	0.799	0.864
Solanum lycopersicum	root	10	0.836	0.878	0.788	0.861
Solanum lycopersicum	root	11	0.857	0.906	0.829	0.897
Solanum lycopersicum	root	12	0.759	0.823	0.781	0.799

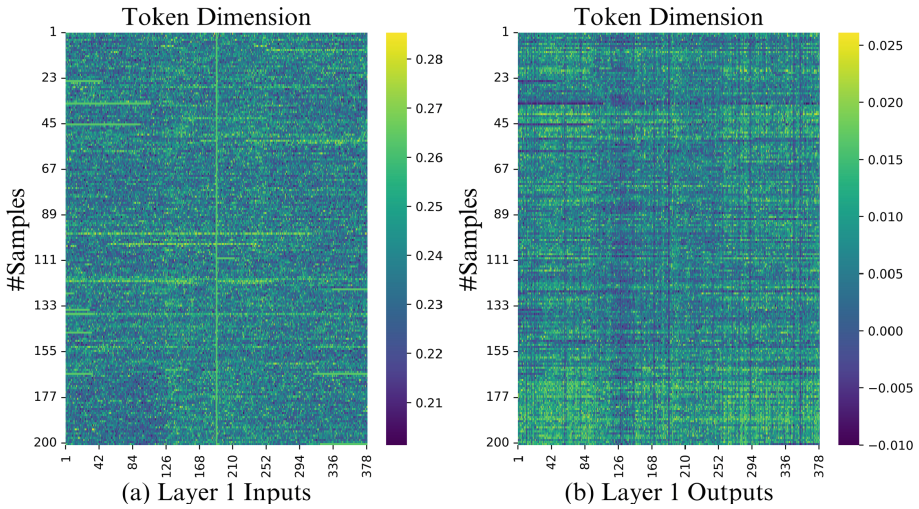

Fig. 4. KAN-Mixer efficiently exchanges input information across both token and feature dimensions. We visually represent the input (a) and output (b) values of a single layer of KAN-Mixer. The x-axis corresponds to the token dimensions derived from 8-mer gene sequences, ranging from 1 to 378 dimensions, while the y-axis represents the data dimensions. This visualization is based on 200 test samples post-training. Each plotted point (x, y) is color-coded based on the numerical value of the x-th dimension of the token for the y-th sample, where higher numerical values are depicted in shades of blue and lower ones in shades of yellow. (Color figure online)

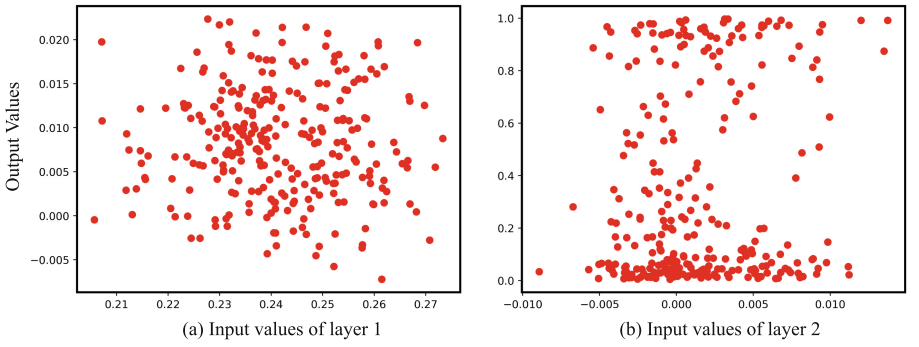

Fig. 5. KAN-Mixer reveals an emergent intra-layer pattern in untranslated gene regions and gene expression. We employ the same pruning and visualization techniques as KAN [19]. Due to the high token dimension of 128, we rank and select the top dimension based on importance scores for visualization. Red points represent different data samples. The x-axis shows the input values of each sample at the top input dimension, while the y-axis shows the output values at the top output dimension. Layer 1 exhibits an approximate Gaussian distribution, whereas layer 2 shows a distinct Sigmoid distribution. (Color figure online)

take both Gaussian and Sigmoid-like functions in our training losses. This spontaneous nature demonstrates the interpretability within different KAN-Mixer layers, which was neither feasible nor practical in previous deep learning models.

5 Discussion

KANs have recently demonstrated their potential in aiding scientists to identify crucial relationships within input dimensions, thereby facilitating new scientific discoveries due to their superior interpretability. In this paper, we explore the applicability of KANs in predicting biological gene sequence expression, specifically within an ecological context. To ensure interpretability, we encountered the challenge of inputting extremely long gene sequences into KAN, resulting in a substantial parameter burden. Drawing inspiration from the concept of k-mer in bioinformatics, we propose introducing a locality-focused weight-sharing inductive bias to KAN, significantly reducing computational costs. Specifically, we employ convolution operators for token embedding to transform one-hot gene features into k-mer tokens. These tokens are then processed by KAN-augmented token mixing and channel mixing blocks, which efficiently exchange information across both token and feature dimensions. Our experiments on gene sequence expression prediction for non-coding regulatory elements indicate that our method, KAN-Mixer, achieves performance comparable to a state-of-the-art ConvNet-based approach. More importantly, this new approach retains KANs' excellent interpretability, accurately identifying key segments within input gene sequences without the need for additional tools. This novel modeling paradigm is poised to advance KANs' role in promoting ecological and biological scientific discoveries in the future. By providing a robust and interpretable framework for gene expression analysis, KAN-Mixer can significantly enhance our understanding of ecological dynamics and biodiversity. This approach has the potential to inform conservation strategies, improve agricultural practices, and contribute to ecological management across various environmental contexts.

Ethics Considerations. In the application of KANs for gene expression prediction, it is essential to consider the ethical implications, particularly in the context of ecological research. Our work emphasizes the importance of transparency and interpretability in AI models, ensuring that the insights derived from these models can be understood and trusted by the broader scientific community. This transparency is crucial for the responsible application of AI in sensitive areas such as conservation and biodiversity, where decisions can have significant environmental impacts. Furthermore, the deployment of such models must respect the ethical guidelines surrounding genetic data usage, including privacy and consent, especially when dealing with indigenous and local communities' genetic resources. Adhering to ethical standards helps in maintaining public trust and fostering collaborations across different sectors. By integrating ethical considerations into our research, we aim to set a precedent for the responsible use of advanced computational techniques in ecological and biological studies, ensuring that scientific progress goes hand in hand with ethical responsibility.

Limitations and Future Work. Our research targets the prediction of gene sequence expression in the biological domain, focusing on identifying crucial segments within well-expressed gene sequences. Despite the promising results, several critical questions remain unanswered, indicating directions for future research. Primarily, KAN-Mixer has concentrated on classifying gene sequence expression in various plant non-coding regulatory elements, offering a more intuitive and interactive framework. Leveraging KANs' interpretability, we aim to deepen our understanding of the underlying mechanisms of gene expression, thereby accelerating scientific discovery. Future research should explore whether a clearer link between input features and outputs can improve our comprehension of gene-phenotype relationships. Moreover, future work should investigate the integration of large-scale pre-trained features from DNA foundation models, such as Evo [20]. Utilizing these pre-trained features could significantly improve the efficiency and effectiveness of our models, leading to more accurate and insightful predictions.

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Multi-scale and Multimodal Species Distribution Modeling

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Abstract. Species distribution models (SDMs) aim to predict the distribution of species by relating occurrence data with environmental variables. Recent applications of deep learning to SDMs have enabled new avenues, specifically the inclusion of spatial data (environmental rasters, satellite images) as model predictors, allowing the model to consider the spatial context around each species' observations. However, the appropriate spatial extent of the images is not straightforward to determine and may affect the performance of the model, as scale is recognized as an important factor in SDMs. We develop a modular structure for SDMs that allows us to test the effect of scale in both single- and multi-scale settings. Furthermore, our model enables different scales to be considered for different modalities, using a late fusion approach. Results on the GeoLifeCLEF 2023 benchmark indicate that considering multimodal data and learning multi-scale representations leads to more accurate models.

1 Introduction

In the face of the current biodiversity crisis, biodiversity models informed by ever-growing data are crucial to support conservation efforts [23]. In particular, information about the suitability of species in areas where few observations are recorded enables robust decision-making [15]. Such information is obtained from species distribution models (SDMs), which relate species occurrence data with environmental variables through statistical methods [1,12,26].

With the increasing availability of species occurrence data, notably through crowd-sourcing [17,28], the use of deep learning (DL) has recently been explored in SDMs [4,25,29]. DL models are of particular interest thanks to their flexibility in terms of architecture and input data types, their scalability, and their ability to model the distributions of many species with a single model [5,7,13,31].

While most SDM approaches use point values of environmental variables as predictors (*i.e.* tabular data), DL facilitates the integration of geospatial information, which allows the model to consider the spatial context surrounding each species observation. Recent works have used convolutional neural

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networks (CNNs) to integrate patches extracted from rasters of environmental variables [4,8], satellite images [10,13], or both [25]. However, the size of the image patches considered by the model is often not justified. Studies on non-DL SDMs have shown that scale affects model performance and that the appropriate scale may depend on the species or the type of environment [14,19,21]. Yet, when handling tabular data, these effects are essentially linked to the resolution of the predictors. To the best of our knowledge, no study has focused on the effect of spatial extent[1] of the image patches on the performance of DL-SDMs.

Furthermore, little work has gone into considering spatial data with different resolutions in multimodal DL-SDMs. One study used the same patch size for all images, treating them as a single stack but resulting in different spatial extents for each data source [8]. Others have aligned the resolutions of the different images in pre-processing, even though this leads to unnecessarily large models for coarser grain images [25]. Finally, another study used satellite images alongside coarser bioclimatic data considered as tabular data, hence not considering the spatial context for the latter modality [10].

In this study, we analyze the effects of scale in a modular structure for SDMs based on CNNs. Inspired by works in multi-scale modeling [6,24] and multimodal modeling with late fusion [10,22], we design a model that can extract features at multiple scales from a single feature map and from multiple modalities with different resolutions. Our architecture enables each modality to be considered at its native resolution and at different scales. Using the GeoLifeCLEF 2023 (GLC23) benchmark [3], we investigate the effect of spatial extent on model performance in both single- and multi-scale settings, as well as uni- and bi-modal models. Our results indicate superior performance of multi-scale, multimodal approaches. Code to replicate our models can be found on GitHub.

2 Methods

Species Data. The GLC23 dataset [3] contains two types of georeferenced plant species observation data. It includes 5 million presence-only (PO) observations across Europe. This type of data consists of opportunistic observations in which the non-observation of a species does not confirm its absence. It is widely available but is subject to many biases [9]. Additionally, presence-absence (PA) data is available for 26k sites in France and Great Britain. PA data is more difficult to obtain as it reflects exhaustive sampling with confirmed species absences. It represents species distribution more accurately but is often not available [11]. To reflect the reality of available data for most species, we train our models on PO data and validate them with PA data. In the GLC23 challenge, all teams used PA data for training [2], thus our results are not comparable to those of the leaderboard. As only part of the PA data is openly available in GLC23, we use 7,438 PA sites for validation. The labels for the remaining 22,404 sites are kept secret for evaluation through the GLC23 Kaggle page. We use this second

[1] We use *scale* or *spatial extent* to refer to the size of image patches considered by the model (i.e. its receptive field), and *resolution* to refer to the size of the pixels.

dataset as a test set, even though it only allows us to report the evaluation metric used by the challenge. For training, we keep the PO occurrences for the 2,173 species in the validation set and merge occurrences recorded at the same location and date, amounting to 2,856,818 training samples.

Model Predictors. We use two modalities among those included in GLC23 dataset [3]: 19 bioclimatic rasters describing temperature and precipitation at a 30-arc seconds resolution ($\approx$ 600 m at 50° N, the median latitude of occurrence records) [18] and images from the Sentinel-2 satellite, providing RGB and a near-infrared (NIR) bands at a 10-meter resolution. All input data is normalized by subtracting the mean and dividing by the standard deviation. From the bioclimatic rasters, patches of various sizes between 1×1 and 25×25 pixels are extracted around species occurrence records, corresponding to a ground footprint from $0.6\text{km} \times 0.6\text{km} = 0.36\text{km}^2$, up to $15\text{km} \times 15\text{km} = 225\text{km}^2$. For the satellite data, the dataset provides patches of 128×128 pixels, centered around the species occurrence records. We extract patches with sizes of 25×25, 59×59, and 115×115 pixels, which correspond to a ground footprint of $0.25\text{km} \times 0.25\text{km} = 0.06\text{km}^2$, $0.59\text{km} \times 0.59\text{km} = 0.35\text{km}^2$, and $1.15\text{km} \times 1.15\text{km} = 1.33\text{km}^2$, respectively. Our models take as input image patches with the size required for the largest spatial extent considered for each modality.

Model. We propose a model structured in three parts: (1) a common encoder for all scales, (2) a spatial module that can have one or multiple branches for single- or multi-scale models, and (3) a linear classification layer with the same number of output neurons as species, 2,174 in our case, that is applied to the concatenated outputs of the previous module (Fig. 1a,b). The sigmoid function is applied to the output to obtain predictions between 0 and 1. When considering multiple modalities, each one is encoded separately and the spatial module is adapted to the resolution and the scales to be considered for each modality. We use late-fusion and concatenate the 1024-dimensional feature vectors output by each branch of the spatial modules before the final classification layer (Fig. 1c). Late-fusion has been shown to work well in single-scale multimodal SDMs [10]. Our approach, where modality-specific streams are only fused at the end, allows us to consider each modality at its native resolution and at different scales.

The encoder for bioclimatic variables is composed of four convolutional layers with a kernel size of 1, keeping the encoder's receptive field at 1 pixel, or $0.6\text{km} \times 0.6\text{km}$. Between each convolution, batch normalization and ReLU are applied. The receptive field of 1 allows the downstream spatial module to consider a small number of pixels as spatial extent, which is pertinent with the coarse resolution of 30-arc seconds. The satellite image encoder consists of nine convolutional layers, corresponding to the first layers up to the second residual block of a ResNet model [16], resulting in a receptive field of 25×25 pixels or $0.25\text{km} \times 0.25\text{km}$.

The spatial module may consist of one or multiple branches, where the number of branches corresponds to the number of scales taken into account. Each branch consists of a series of convolutional and max pooling layers, after which

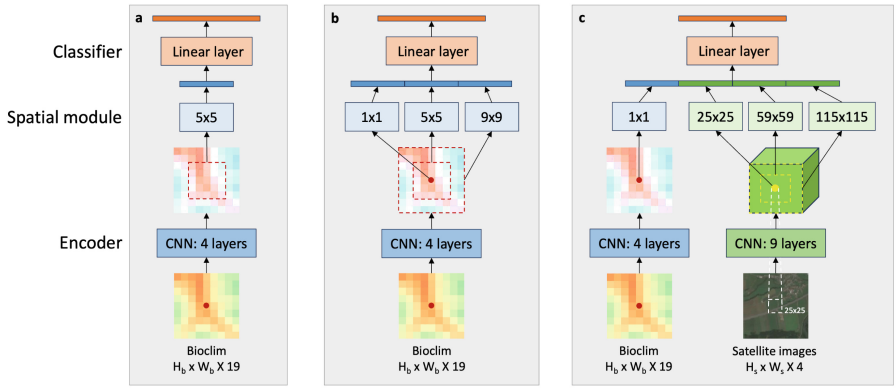

Fig. 1. Example of architectures with our modular structure for SDMs. **a.** Single-scale, unimodal model architecture for bioclimatic variables at scale (5×5). **b.** Multi-scale unimodal model architecture for bioclimatic variables at scales (1×1), (5×5) and (9×9). **c.** Multi-scale multimodal model architecture for bioclimatic variables at scale (1×1), and Sentinel-2 satellite images at scales (25×25), (59×59) and (115×115). The receptive fields after the encoders for bioclimatic variables and satellite image are 1×1 and 25×25 pixels, respectively.

the central pixel of the tensor is extracted to obtain a vector of length 512. The receptive field of the central pixel corresponds to the extent considered by that branch of the spatial module. Finally, a linear layer with $1,024$ outputs and ReLU are applied to each vector.

We use a weighted loss function for multi-label classification for SDMs [29], which was shown to perform well on the GLC23 dataset [30]. We consider the records of other species as negatives and do not sample additional pseudo-absences, to avoid the costly operation of downloading new satellite images on the fly. We use a stochastic gradient descent optimizer with a learning rate of 0.01 and weight decay of 0.0001. All models are trained end-to-end for 30 epochs with a batch size of 256 on NVIDIA A100 with 80GB video memory.

Evaluation. We evaluate our models on the validation set with the area under the receiver operating characteristic curve (AUC). AUC is widely used in SDMs and measures how well the model discriminates presence from absence sites for each species [27]. We consider the median AUC across species. Additionally, we compute the metric used for the GLC23 challenge [2], the micro-F1 score, on the validation and test sets. This metric measures the overlap between the predicted and actual set of species, averaged over the sites. While AUC can be computed directly on the probabilistic output of our models, the F1 requires binary predictions. Although species- or model-specific binarization schemes may be used, we chose a fixed binarization threshold of 0.5 to ensure comparability among models and avoid overfitting on the validation data.

Table 1. Model performance and training time for unimodal models considering various scales in single- and multi-scale settings. The median species AUC is computed on the validation data. The micro-F1 score is computed on the validation and test data. The best and second-best scores per column are in bold and underlined, respectively. Scales are indicated in pixels and performance metrics are in %.

	Validation		Test	
Scales	AUC	F1	F1	Runtime
1	**86.91**	3.05	2.37	1.5 hrs
5	85.69	**3.16**	2.37	2.6 hrs
9	83.63	2.95	2.36	2.0 hrs
17	83.59	2.30	2.08	3.6 hrs
25	83.65	1.98	1.58	9.7 hrs
1, 5	85.73	3.13	**2.39**	2.8 hrs
1, 5, 9	85.37	3.14	2.30	2.9 hrs
1, 5, 9, 17	86.28	3.05	2.35	6.8 hrs
1, 5, 9, 17, 25	85.12	3.13	2.28	17.9 hrs

(a) Bioclimatic variables

	Validation		Test	
Scales	AUC	F1	F1	Runtime
25	80.41	2.76	1.69	3.9 hrs
59	81.38	2.88	1.98	8.0 hrs
115	80.85	2.98	2.00	28.7 hrs
25, 59	80.67	3.19	2.15	9.6 hrs
25, 59, 115	**81.80**	**3.53**	**2.25**	38.6 hrs

(b) Satellite images

3 Results

Unimodal Models. First, we train models with only bioclimatic variables or satellite images as predictors. Table 1 shows the performance and training time of single- and multi-scale models with different spatial extents.

Table 1a shows that, when considering only bioclimatic variables, small spatial extents obtain the best performance. While the performance decreases with increasing spatial extent for single-scale models, the different multi-scale models recover the performance of the best single-scale models, albeit with longer training time. We note that the 1×1 scale obtains the best AUC, and the 5×5 scale yields the best F1 score on the validation set. Interestingly, the combination of these scales slightly outperforms both single-scale models on the test set, indicating the marginal advantage of a multi-scale approach for this modality. The differences in performance among these models are relatively small. This may be explained by the high spatial autocorrelation in bioclimatic variables, leading to limited additional information for medium-sized spatial contexts. However, considering large spatial extents can be disadvantageous, indicating that, beyond a certain extent, spatial context is not informative and may even introduce spurious correlations for modeling the distributions of plant species.

In contrast, the results for models with satellite images in Table 1b indicate that taking multiple scales into account leads to better performance. While the F1 scores generally increase with model complexity, the AUC does not consistently follow this trend, with no clear relationship between spatial extent and performance among the single-scale models. This result suggests a larger

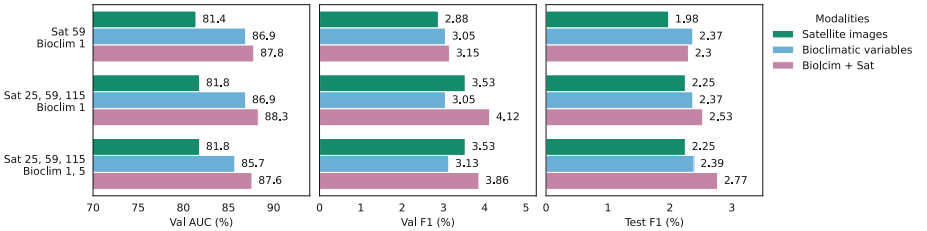

Fig. 2. Performance of bimodal and corresponding unimodal models, quantified by their validation median AUC, and test micro-F1 scores.

variability of which scales are most informative, possibly due to the much higher resolution and semantic content of satellite images compared to bioclimatic variables. We speculate that different scales may be informative for different species or sites and that the multi-scale architecture may learn which spatial extents are relevant, leading to its superior performance.

Multimodal Models. We compare the performance of bimodal models to their unimodal counterparts (Fig. 2). We consider the best scale or combination of scales for each modality. Concerning the AUC, we find that the models with bioclimatic variables outperform those with satellite images, and combining both modalities leads to further improvement. Regarding the site-wise performance, quantified by the F1 scores, we find that the unimodal models with multiple scales for satellite images outperform the models with bioclimatic variables on the validation set, but the opposite is observed on the test set. The bimodal models perform better than both of their unimodal counterparts, in particular with multi-scale feature extraction for the satellite imagery. Furthermore, the inclusion of multiple scales for both bioclimatic variables and satellite images leads to the best F1 score on the test set. These results confirm the advantage of combining modalities describing the environmental conditions with satellite images [10]. Furthermore, they indicate that multi-scale representations for both modalities lead to better species community predictions. However, such complex models require 20-fold longer training times than the simplest unimodal models. One may consider whether the performance increase is worth the carbon emissions associated with training these models, estimated at 6.91 and 0.35 kg, respectively [20].

Species and Site-Level Differences in Performance. While the differences in performance are sometimes relatively small when aggregated, some species or sites have large differences among models. To illustrate this, we compare two models with bioclimatic variables as predictors and two bimodal models (Fig. 3). These pairs of models have differences in median AUC of 1.2% and 0.7%, respectively, and, accordingly, the vast majority of species have a small difference in performance: 90% of species have a difference in AUC of less than 11% between the two pairs of models. The species with larger differences have few records in

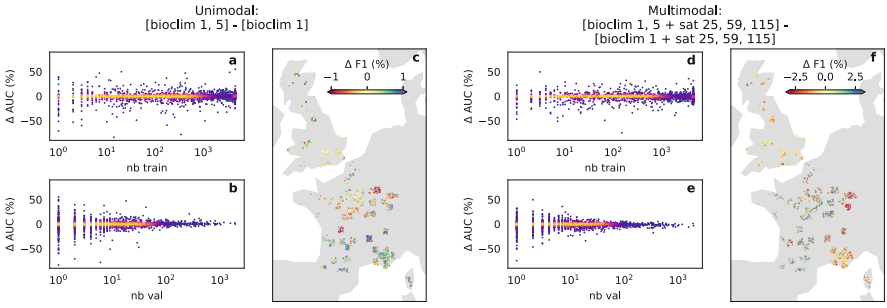

Fig. 3. Difference in median AUC (ΔAUC) and micro-F1 (ΔF1) between two unimodal and two bimodal models. Positive Δ values indicate that models [bioclim 1,5] or [bioclim 1, 5 + sat 25, 59, 115] outperform models [bioclim 1] or [bioclim 1 + sat 25, 59, 115], respectively, and vice-versa for negative values. **a, b, d, e.** ΔAUC values plotted against the number of occurrences in the training data (nb train) and the validation data (nb val) for 2,173 species. Colors indicate point density, with higher densities in yellow. **c, f.** 7,348 validation sites plotted on maps and colored by ΔF1. (Color figure online)

the validation data (Fig. 3a,d), but no clear trend can be found with the number of records in the training data (Fig. 3b,e). While this may be explained by the sensitivity of AUC to small sample sizes [27], our results suggest that rare species may be more sensitive to the scale of the predictors. Mapping the differences in the validation F1 score per site reveals geographical clusters (Fig. 3c,f). These clusters are more locally marked for bioclimatic variables, with a clear preference for a certain model in some regions, despite a median difference in F1 of 0.03%. For the bimodal models, considering a single scale for bioclimatic variables generally leads to better site-wise performance, with a median difference in F1 of −0.23%. However, this difference is less marked in certain regions, such as the north and west of France. These qualitative results indicate that multi-scale models may be more informative for some species or in some regions, characterized by a specific type of ecosystem. We leave to future work the further investigation of these relationships.

4 Conclusion

In this study, we develop a modular structure for SDMs and explore the effect of the scale of spatial predictor variables on the GLC23 dataset for European plant species distributions. We find that small scales are most appropriate when considering bioclimatic variables. When using satellite images, our multi-scale approach showed a clear benefit in performance. Combining the best architectures for each modality with a late fusion scheme leads to further increases in performance, indicating the complementarity of both modalities. Overall, our multi-scale and multimodal model achieved the best performances for both species-wise and site-wise evaluation. Our results suggest that the most informative scales may depend on the species or site. Future work may explore these relationships further, and investigate scale-dependencies in other species groups beyond plants.

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Semantic Segmentation of Benthic Classes in Reef Environments Using a Large Vision Transformer

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Abstract. Coral reefs are crucial for biodiversity and provide vital resources for humankind. But despite such a central role, they are confronted to increasing threats linked to climate change, pollution, and local stressors. To ensure effective conservation, efficient and scalable monitoring is key: this necessitates automated identification of benthic classes and their states on a large scale through semantic segmentation. However, segmentation of underwater videos is challenging, because of visual similarities between benthic classes, underwater distortions and limited available datasets, making it harder to create accurate and robust models. In this paper, we present a method for training a semantic segmentation model on a small dataset of video frames of coral scenes, by fine-tuning a large transformer model. Our approach uses transfer learning on the Segment Anything Model (SAM), incorporating specific training and prediction strategies. We benchmark our model against a CNN for semantic segmentation as a baseline. Our results demonstrate a substantial improvement in model performance, particularly for benthic classes that often appear as small objects and rarer classes, highlighting the potential of our approach in advancing coral reef mapping and monitoring.

1 Introduction

In the face of rising temperatures and various other stressors affecting coral ecosystems, coral monitoring is all the more important to help devise data-driven conservation strategies. Current well-established methods used to monitor corals range from divers photographing the seafloor to advanced multi spectral satellite imaging [15]. The most common approach to monitor benthic cover involves photo quadrats taken by divers, followed by expert analysis to assess coral abundance and bleaching status, which are then extrapolated to larger areas [11]. This process is labour-intensive and challenging to scale due to the need for expert evaluation. Computer vision has been applied to automate quadrat annotation [3,18], but its efficacy hinges greatly on controlled conditions, given the complexities of underwater environments.

However, currently there are no automatic semantic segmentation systems for benthic semantic segmentation from general purpose reef images, as most systems are trained on domain-specific images such as photo quadrats [3,5].

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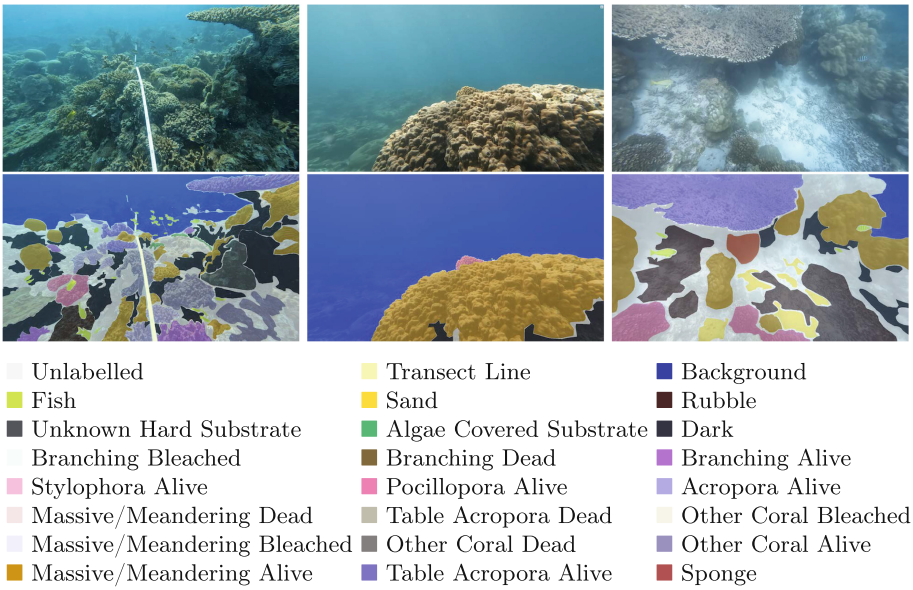


Fig. 1. Example dataset images with their labels. The labels encompass 30 classes with an additional class for unlabeled pixels.

To address these limitations, [12] proposed a coral monitoring approach for 3D mapping, integrating frame-by-frame semantic segmentation and learned Simultaneous Localization and Mapping (SLAM). This technique generates 3D point clouds with RGB data and semantic class labels, offering scalability and efficiency through video-based data acquisition. It streamlines benthic classification, eliminating the requirement for expert image analysis. This method's effectiveness relies on successful SLAM and a robust semantic segmentation algorithm.

In recent literature, most existing models are limited to deep learning approaches primarily relying on convolutional neural networks (CNNs) [8,9]. Following the excellent performance of large scale transformers in other tasks, our paper explores a large vision transformer, a modified version of Segment Anything (SAM [10]) that we name SAMarine, for enhancing the semantic segmentation model. Our model can then serve either as an improved building block for [12], but also as a stand alone tool for semantic segmentation of coral reefs, as it provides insight into the distribution of the main coral types as well as their condition (whether they are bleached, dead, or alive).

A limiting factor in the research is the lack of large and densely annotated datasets. Hence, there is a lack of well-trained transformer models and limited application of transfer learning from large datasets within this domain. Challenges also arise from imaging under water, creating blur, and lack of contrast

due to color attenuation and scattering [1,2,13], which varies with depth, temperature, salinity, the particles suspended in the water, and other factors. Moreover, corals show high morphological plasticity, which can make it extremely challenging to tell some corals apart [16].

We contribute by making a set of changes in the architecture and training procedure deviating from a standard semantic segmentation setup, leading to an mIoU improvement from 0.412 to 0.513 with respect to the original work of [12]. This improvement is seen on a challenging dataset of coral reef images with fine-grained semantic segmentation labels of benthic classes.

2 Dataset

The dataset used in this paper was collected from various coral reef sites in the Red Sea in Israel, Jordan and Djibouti as part of the Transnational Red Sea Center's[1] expeditions. The segmentation dataset is comprised of annotated video frames from a dataset videos gathered by a diver swimming between 1 and 4 m above the reef following a transect line using a GoPro Hero 10 camera at 1080×1920px. The dataset spans a diverse array of underwater environments, ranging from water-dominated scenes to those teeming with diverse coral formations, as illustrated by examples in Fig. 1. Representative video frames of the environment were selected manually and annotated by coral experts with polygons of 30 distinct classes, including one extra class dedicated to unlabeled pixels. The inclusion of this class serves to handle complex scenes where class attribution might be uncertain, while addressing challenging scenarios with difficult to outline class boundaries. These unlabeled pixels fill transitional spaces between class boundaries in such cases. The full dataset consists of 986 training images (61868 polygons) and 99 test images (7247 polygons).

The dataset has significant class imbalance both in the frequency of classes as well as their apparent sizes. This is illustrated in Fig. 2 showing the relation between the median number of pixels per polygon to the number of polygons, where for certain classes the difference in the number of annotated polygons as well as the average polygon size varies with over two orders of magnitude. The rarity of some species, like seagrass, poses challenges for the model to learn effectively, resulting in very poor scores for these classes. Consequently, in our result analysis, we differentiate between the performance of common and rare classes. Furthermore, the imbalance in size among certain classes, exemplified by fish being at least one order of magnitude times smaller than all other classes, impedes learning due to their restricted pixel representation, rendering them more challenging to grasp.

[1] http://trsc.org.

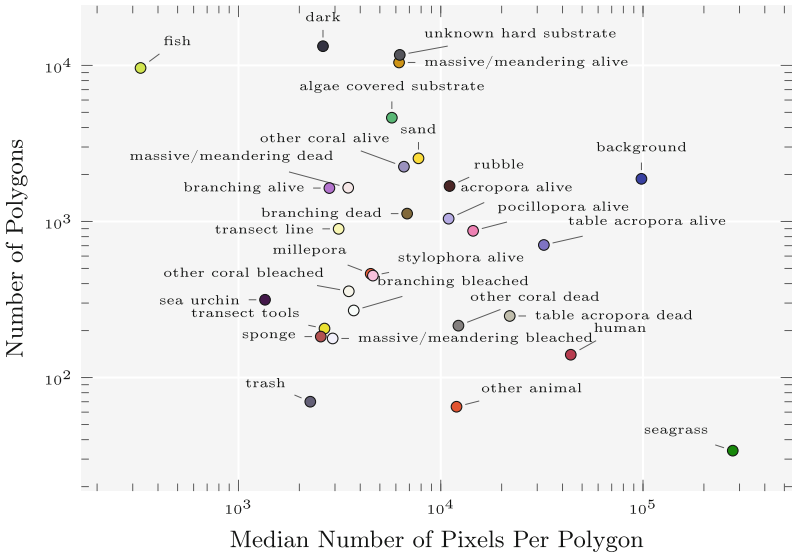


Fig. 2. Class Distribution. Relation of median number of pixels per polygon to the total number of polygons per class.

3 Method

3.1 SAMarine

Model Architecture. Our model, **SAMarine** (Fig. 3), is an adaptation of the pre-trained Segment Anything Model (SAM) [10] to the domain of semantic segmentation of corals. It is based on the architecture proposed by [20] for medical semantic segmentation. For our specific semantic segmentation task, we aim at classifying each pixel into one of the 30 benthic classes, deviating from the binary foreground-background segmentation in SAM. To do so, we design SAMarine to output 30 binary segmentation masks, one per benthic class. In these experiments, we do not actively design prompts, and use the default prompt from the original SAM.

Training. We use a hybrid loss function that combines cross-entropy loss ($Loss_{CE}$) and dice loss ($Loss_{Dice}$) [14], weighted by parameter λ as shown in Eq. 1. This approach allows us to strike a balance between distribution-based and geometry-based optimisation.

$$Loss_{Combined} = \lambda \cdot Loss_{CE} + (1 - \lambda) \cdot Loss_{Dice} \tag{1}$$

Moreover, we perform a number of augmentations (center / random cropping and resizing, rotation of up to 15°, jittering, horizontal flips) and vary their strength. To tackle the size of the pre-trained backbones, we use LoRa [7] and identify the best rank through ablations.

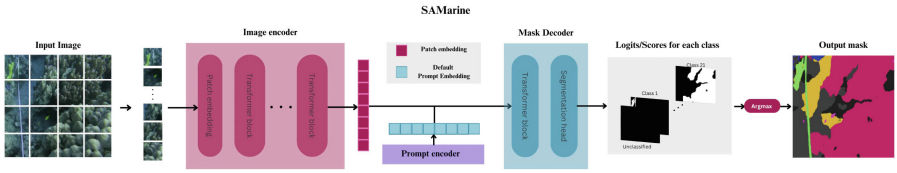


Fig. 3. SAMarine Architecture. Overview of our SAMarine approach. On the left side, the input image undergoes a patch-based embedding process before being forwarded to the transformer encoder. The resulting image embedding is then queried by a default prompt embedding in the mask decoder. The mask decoder generates a class-specific score mask, which leads to the output segmentation mask on the right side of the figure. Illustration inspired by [20].

Inference. To handle the loss of resolution that would be necessary if we fed an entire image (see next section Sect. 3.2), we feed the large input image to SAMarine in smaller patches and then reconstruct the full resolution map, where the softmax probabilities are averaged in the overlapping parts.

3.2 Baseline

The baseline method is a standard semantic segmentation setup. A DeeplabV3+ [4] model with a *resnext50_32x4d* backbone pre-trained on ImageNet [6], is trained using cross-entropy loss, excluding the unlabelled class, with classes weighted inversely to the square root of the pixel count to address class imbalance. We use random resized crops, horizontal flips, rotations and colour jitter as augmentations. For handling large images at inference, we resize the input image from 1080×1920px to 352×608px, and then upsample the prediction to the original size, as is standard practice in many semantic segmentation setups [17,19].

4 Results

4.1 Implementation Details

We assess the baseline method using both the resizing and the patch-based prediction (as in SAMarine) with a patch size of 1024. Subsequently, we evaluate the SAMarine model, with the 'Vit-H' backbone with a LoRA rank of 128. The model is assessed using patch-based prediction with single patches of size 512. Unless specified otherwise, $\lambda = 1$ (a pure cross-entropy loss) is used with medium-strength augmentations.

To assess our models, we use mean Intersection over Union (mIoU) as the evaluation metric, considering the class imbalance. We categorise the results into common and rare classes, focusing on the former as the latter exhibits high volatility, making their outcomes less reliable.

4.2 Results

Results are shown in Table 1, where we observe an overall performance increase for the baseline from the resizing to the patching approach (0.412 to 0.431), mostly related to an improved performance in the *Common* classes (increase from 0.587 to 0.632). This shows that for underwater coral scenes, due to the abundance of details, the loss of resolution due to the resizing approach leads to a loss in performance.

Table 1. Numerical comparison between our SAMarine and the DeepLabV3+ baseline.

Method	mIoU		
	Overall	*Common*	*Rare*
Baseline (Resizing)	0.412	0.587	0.271
Baseline (Patching)	0.431	0.632	0.281
SAMarine (Patching, $\lambda = 1$)	0.492	0.685	0.337
+ Balanced combined loss ($\lambda = 0.5$)	0.494	0.688	0.336
+ Stronger augmentations	**0.513**	**0.693**	**0.369**

Most importantly, there is a significant performance leap observed from the baseline to our proposed SAMarine, visible in both rare and common classes. Our experiments revealed that employing a balanced combined loss with $\lambda = 0.5$, alongside more severe data augmentation strategies, resulted in improved performance, leading to an overall increase from 0.431 (baseline) to 0.492 (for the base SAmarine) and 0.513 (for the final model). The common classes enjoy an improvement of 6%, while the rare classes of 8%. The increased performance for smaller classes is attributed partly to the use of patch-based prediction, which preserves resolution and maintains accuracy in predicting these compared to resized prediction method. Additionally, the overall performance boost across all classes could stem from the efficacy of the transformer architecture employed or the benefits of transfer learning from pre-training on a large segmentation dataset.

Ablation studies about the different model architectures and the hyper-parameter λ are reported in Fig. 4. As illustrated in Fig. 4a, superior performance was attained using the larger 'Vit-H' backbone compared to 'Vit-B', along with a larger rank of 128. Additionally, in Fig. 4b, a $\lambda = 0.5$ yielded the best results, contrasting sharply with the inferior performance observed when using only the dice loss ($\lambda = 0$).

The visual analysis in Fig. 5 reveals insights into model performance on small classes. Compared to the baseline resizing approach, the patch-based method shows a marked improvement in predicting fish instances. Notably, the SAMarine model surpasses others by predicting more small fish instances with greater boundary delineation accuracy. This underscores SAMarine's effectiveness in capturing fine-grained details.

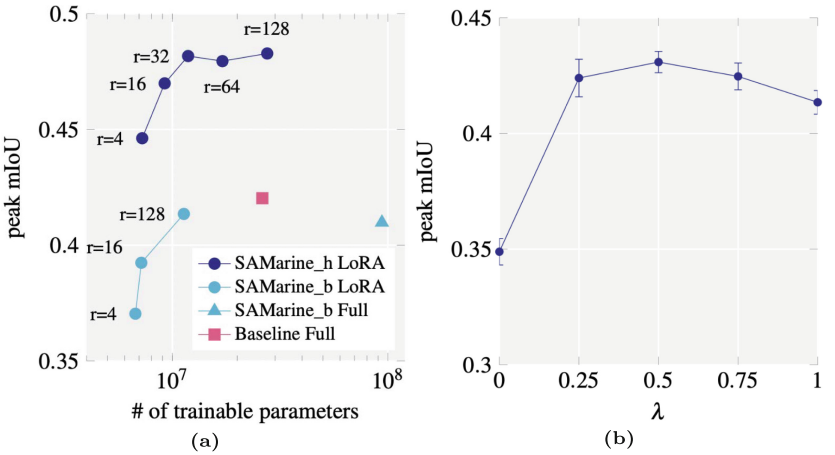
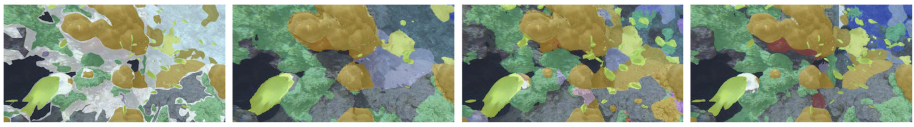


Fig. 4. Ablation Analysis. Figure (a) illustrates the ablations conducted to determine the best backbone architecture for SAMarine and the best rank, found to be using 'Vit-H' and a LoRA rank of 128. Figure (b) shows the best value for the loss function hyper-parameter λ as 0.5.



(a) Label (b) Baseline (Resiz.) (c) Baseline (Patch.) (d) SAMarine

Fig. 5. Qualitative Analysis. Example predictions using various methods. The fish class, highlighted in yellow, demonstrates superior performance and detail capture with the SAMarine approach. (Color figure online)

5 Conclusion

We present a visual transformer-based model for automatic semantic segmentation of benthic classes in reef scenes on a dataset of expert-annotated video frames. This methodology incorporates transfer learning from a large transformer model (Segment Anything), strong augmentations, a combined loss function and patch-based prediction. Our model SAMarine outperforms the baseline, particularly in smaller classes, and is able to accurately predict boundaries around these. This is significant, as there are a lot of small benthic classes, which are challenging to discern in these unprocessed large underwater scenes. These findings also highlight the need for larger datasets of densely labeled images of benthic habitats in coral reef scenes, with a special focus on rare or naturally small organisms. Overall, improved performance in automatic semantic segmentation marks a solid step toward implementing coral monitoring at larger scale.

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POLO – Point-Based, Multi-class Animal Detection

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Abstract. Automated wildlife surveys based on drone imagery and object detection technology are a powerful and increasingly popular tool in conservation biology. Most detectors require training images with annotated bounding boxes, which are tedious, expensive, and not always unambiguous to create. To reduce the annotation load associated with this practice, we develop POLO, a multi-class object detection model that can be trained entirely on point labels. POLO is based on simple, yet effective modifications to the YOLOv8 architecture, including alterations to the prediction process, training losses, and post-processing. We test POLO on drone recordings of waterfowl containing up to multiple thousands of individual birds in one image and compare it to a regular YOLOv8. Our experiments show that at the same annotation cost, POLO achieves improved accuracy in counting animals in aerial imagery.

Keywords: Animal detection · Wildlife censuses · Annotation cost

1 Introduction

Frequent animal censuses are a key requirement for successful conservation management and become particularly important when dealing with endangered species. Wildlife in vast and open landscapes can be surveyed efficiently using aerial imagery recorded either from airplanes or unmanned aerial vehicles (UAVs), with the latter being used more and more thanks to the reduced operating costs and safety risks [9,13]. Given the large amounts of data collected during these flights, machine learning approaches are often employed to count animals in the images, allowing biologists to estimate the development of populations. For this purpose, convolutional neural networks (CNNs) are among the most popular techniques [4–6,10].

While CNNs hold the promise of high detection accuracy, this potential is limited by the volume of labeled data available for model training [1]. Since the creation of labeled data implies manual annotation, often to be provided in the form of bounding boxes, accurate counting of animals from aerial imagery comes

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at a significant annotation cost, which limits the scalability of deep learning-based conservation efforts. To reduce these costs, bounding boxes can be automatically created from point annotations, which can be obtained with higher speed and are thus much cheaper to produce [7,14]. This approach simply consists of generating square boxes around the point annotations and using these pseudo-labels to train conventional object detection architectures. However, animals in aerial images are regularly of very small size (*i.e.*, a few pixels in length), partially occluded, difficult to separate if standing close together, or distorted due to perspective and motion blur. All of these factors contribute to animal boundaries being difficult to demarcate, hence significantly affecting the quality of automatically generated boxes.

In the present study, we avoid these issues without raising the annotation cost by developing an object detection framework that can be trained entirely on point labels. We approach this task by modifying the YOLOv8 algorithm [8], one of the most advanced and widely used bounding box detection models. The proposed architecture is called POLO (**P**oint-Y**OLO**). We compare two different loss functions to train it, and introduce a new metric to perform non-maximum-suppression (NMS) on point predictions.

We test POLO on a challenging dataset, which involves counting five species of waterfowl in drone images of the Izembek lagoon in Alaska. Moreover, we compare POLO's performance to that of a regular YOLOv8 trained on pseudo-labels, and find that while the latter approach yields high counting accuracy, results improve when using POLO.

2 Related Work

2.1 Point-Based Object Detection and Counting

Point labels represent a popular annotation type in the field of crowd counting, where they are used to train density estimation models that take images as input and output a heat map encoding the estimated density of people in every pixel [2,12,19] – an approach motivated by the observation that conventional detection algorithms struggle with scale variation (*i.e.*, people in the background appearing smaller than people close to the camera) and partial occlusion of individuals by other individuals.

In object detection on the other hand, the point format has been used as part of mixed labeled sets, where a small set of bounding box annotations is complemented by a larger set of point labels to train detection models to ouptut bounding box predictions [3,7,18].

Our solution differs from the above methods, since we output point detections (rather than density maps or bounding boxes) and do not use any bounding box annotations for training. This makes for a simpler learning task than generating bounding box outputs from point labels, and suffices for our purpose of counting animals in drone imagery. The present work is most similar to the efforts of Song *et al.* [16], who also develop a detection model that trains on and outputs point labels. However, their architecture cannot distinguish between different classes, making it unsuitable for animal censuses involving multiple species.

2.2 The YOLO Algorithm

To identify objects, YOLOv8 employs a one-stage detection strategy that divides images into a predefined number of grid cells. Each grid cell is then processed on two separate branches, one predicting bounding boxes that enclose objects lying within the cell, and the other predicting the class of said objects. Importantly, YOLOv8 does not directly regress bounding box coordinates, but samples a probability distribution to obtain the most likely number of pixels by which the four bounding box edges are offset from the center of grid cells.

The training objective of YOLOv8 consists of three loss terms: a *binary cross-entropy* loss ($\mathcal{L}_{BCE}$) for the class predictions, an *Intersection-over-Union* loss ($\mathcal{L}_{IoU}$) to learn the geometric prediction of bounding boxes, and a *Distribution Focal Loss* ($\mathcal{L}_{DFL}$) [11] to optimize the probability distribution used for predicting bounding box offsets. The overall loss value is then computed as a weighted sum of these three components, where the class, IoU, and DFL loss are scaled by factors 0.5, 7.5, and 1.5, respectively.

3 Method

To achieve compatibility with point labels, we apply the following modifications to the YOLOv8 architecture:

1. **Output dimensions:** By default, the number of output channels in the final convolutional layer of YOLOv8 is $4 \cdot 16$. These channels encode probabilities of the different bounding box edges being offset by $[0, 1, ..., 15]$ pixels from the center of a grid cell.
   We change the architecture to predict the x- and y- coordinates of the center points of objects that lie within grid cells, using only two channels:

$$\hat{p}_x = \sigma(a_1) \cdot 2 - 0.5 + c_x \tag{1}$$

$$\hat{p}_y = \sigma(a_2) \cdot 2 - 0.5 + c_y \tag{2}$$

   where $\hat{p}_x$ and $\hat{p}_y$ are the predicted coordinates, a_1 and a_2 are the activation values of a grid cell in the first and second output channel, $\sigma(\cdot)$ is the sigmoid function, and c_x and c_y are the coordinates of the grid cell's top left corner. This way, predicted values can range between –0.5 and 1.5, allowing the model to locate objects that do not fit entirely into one cell. We borrow this approach from YOLOv8's predecessor, YOLOv5 [8], where it is used to regress the center point of bounding boxes.
2. **Loss function:** Neither the IoU-, nor the DFL-loss term can be applied to our point model. We consider the following alternatives:
   (a) Average Hausdorff-Distance [15]:

$$\mathcal{L}_{AH}(\hat{P}, P) = \frac{1}{|P|} \sum_{i=1}^{|P|} \mathtt{min}_{\hat{\mathbf{p}} \in \hat{P}} d(\hat{\mathbf{p}}, \mathbf{p}_i) + \frac{1}{|\hat{P}|} \sum_{j=1}^{|\hat{P}|} \mathtt{min}_{\mathbf{p} \in P} d(\hat{\mathbf{p}}_j, \mathbf{p}) \tag{3}$$

   where $\hat{P}$ is the set of predicted points, P is the set of ground truth locations, $|\cdot|$ denotes the cardinality, and $d(\cdot, \cdot)$ is the Euclidean distance.

(b) Mean Squared Error (MSE) [16]:

$$\mathcal{L}_{MSE} = \frac{1}{|P|} \sum_{i=1}^{|P|} ||\mathbf{p}_i - \hat{\mathbf{p}}_i||_2^2 \tag{4}$$

Again, P denotes the set of ground truth locations and $\hat{\mathbf{p}}_i$ is the prediction corresponding to ground truth $\mathbf{p}_i$.

3. **Postprocessing:** To remove redundant detections, we implement a variation of the traditional NMS algorithm. Specifically, we define the *Distance-over-Radius* (DoR) metric, and use it to replace the IoU. Here, the DoR is computed by dividing the distance of a predicted point to its ground truth label by a radius value r_c specified by the user for each object/animal class in the dataset:

$$DoR = \frac{d(\hat{\mathbf{p}}, \mathbf{p})}{r_c} \tag{5}$$

with $d(\hat{\mathbf{p}}, \mathbf{p})$ denoting the Euclidean distance between a predicted point and the ground truth location. During NMS, low-confidence detections are removed if their DoR to higher-confidence detections falls below a specified threshold.

4 Experiments

4.1 Experimental Setup

Dataset. We evaluate and test all models in this study on a publicly available set of drone images [17] from the Izembek lagoon in Alaska hosted on the LILA BC data repository. The dataset includes 9,267 images of waterfowl of size 8688 × 5792 pixels, and a total of 521,270 pseudo-boxes for the classes "Brant goose" (424,790 boxes), "Canada goose" (47,561 boxes), "Gull" (5,631 boxes), "Emperor goose" (2,013 boxes), and "Other" (5,631 boxes). The data was divided into a training (80%), validation (5%), and test set (15%). Images were split into 640 × 640 patches with 10% overlap, to match the input size expected by YOLOv8 and POLO. 95% of patches that did not contain animals were discarded. The remaining 5% were used as negative samples to reduce false positive detections. Bounding boxes that span multiple patches were clipped to the patch limits if at least 15% of the area of the box lied within the patch in question.

Implementation Details. Throughout experiments, we employ a batch size of 32 and set the training duration to 300 epochs, while activating YOLO's early stopping mechanism with a patience value of 50. Counting accuracy was assessed by applying the trained models to our test set. There too, we split images into

640 × 640 patches with 10% overlap, but map the patch-level predictions back to image-level after inference to obtain a global animal count for the entire image. An additional round of NMS is applied after mapping patch predictions to the image-level in order to remove redundancy in the patch overlap regions.

4.2 Loss Functions Comparison

In a first step we compare the counting accuracy achieved with POLO models trained on the Hausdorff and MSE loss, respectively. We set a DoR threshold of 0.3 for post-processing during this initial comparison, and use the YOLOv8 loss balancing scheme mentioned in Sect. 2.2; *i.e.*, we assign the Hausdorff-/MSE-loss a weight of 7.5, and scale the classification loss by 0.5. We use a 40 pixel radius for all classes, except for the Gull category, where we set the radius to 30 pixels. These values were obtained by manually measuring the length of animals in the training images, and adding a buffer of 10–15 pixel to account for variations in ground sampling distance between images, and in the appearance of birds. For example, when in flight, animals will occupy more pixels compared to when they are resting on the water surface due to being closer to the drone and their wings being spread. Table 1 displays the mean absolute error (MAE) per image obtained when training POLO with the Hausdorff/MSE loss function.

Table 1. MAE scores obtained with the MSE/Hausdorff loss function.

	Brant Goose	Other	Gull	Canada Goose	Emperor Goose
POLO MSE	**5.91**	**4.57**	0.93	2.26	0.25
POLO Hausdorff	6.57	4.64	**0.91**	**2.19**	**0.22**

As can be seen, the Hausdorff model beats the MSE model in three out of five categories. However, for the most abundant species of the dataset (Brant goose), the MSE loss reduces the MAE considerably. We hence decided to use the latter for all subsequent experiments.

4.3 Loss Balancing

To optimize the balance between the MSE and classification loss terms, we introduce a hyperparameter $\alpha \in [1, 9]$, based on the value range used for loss scaling in YOLOv8. We then train a total of nine different models using the DoR threshold and radii specified in Sect. 4.2, where the loss is composed as follows:

$$\mathcal{L} = \alpha \cdot \mathcal{L}_{MSE} + (10 - \alpha) \cdot \mathcal{L}_{BCE} \tag{6}$$

Table 2 contains the MAE scores for different values of α. We find the model trained with $\alpha = 1$ to perform consistently well across classes: while it is never the most accurate, it yields the second- or third-lowest MAE score in every category. Consequently, we choose the model trained with $\alpha = 1$ for the remaining experiments.

Table 2. MAE scores for varying loss weights. The **first**, second, and third best scores are colored in green, blue, and orange, respectively.

α	Brant Goose	Other	Gull	Canada Goose	Emperor Goose
1	5.3	4.17	0.73	1.89	0.22
2	5.58	4.11	0.86	1.94	0.22
3	5.48	4.2	0.74	2.03	0.2
4	5.51	4.23	**0.69**	**1.81**	0.2
5	5.34	**4.07**	0.86	1.95	0.22
6	5.63	4.31	0.78	2.02	0.21
7	**5.25**	4.77	0.83	2.05	**0.19**
8	5.31	4.28	0.81	2.02	0.22
9	5.72	4.62	0.79	1.97	0.23

4.4 Radius and DoR Threshold

We finally perform a full grid search over combinations of radius values and DoR thresholds. We probe DoR thresholds between [0.1, 1] and multiply the radii defined in Sect. 4.1 by scaling factors in the range of [0.25, 2]. Figure 1 visualizes the effect different radii and DoR thresholds have on the MAE achieved for the Brant goose class. As can be seen, two strategies maximize counting accuracy: using intermediate values for both, radius and DoR threshold, or pairing opposite extremes; *i.e.*, combining large radii with low DoR tresholds and vice versa. It should be noted though that the MAE is affected less strongly when large radii are used with high DoR thresholds, compared to the combination of small radii and low DoR values. We observe similar behavior in the remaining classes, where a scaling factor of 1.25 and a DoR-threshold of 0.6 achieve top MAE scores across categories.

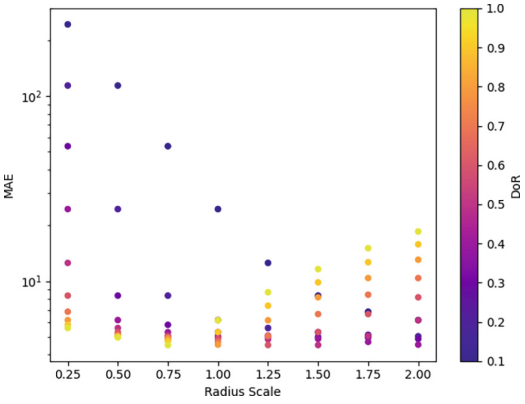

Fig. 1. MAE scores achieved for the Brant goose class depending on radius and DoR value.

5 Results

The animal counts and MAE values obtained with YOLOv8 and POLO are showcased in Table 3. Based on the above results, we use a POLO model trained with the MSE loss function scaled by factor 1 in this experiment. We multiply the radii mentioned in Sect. 4.2 by factor 1.25, and set the DoR threshold to 0.6.

Table 3. Counting accuracy of YOLOv8 and POLO.

	Brant Goose		Other		Gull		Canada Goose		Emperor Goose	
	Count	MAE	Count	MAE	Count	MAE	Count	MAE	Count	MAE
YOLOv8	68,732	5.86	6,256	**3.78**	1,251	1.05	7,113	1.98	213	0.26
POLO	67,501	**4.51**	6,436	4.08	961	**0.69**	6,923	**1.87**	149	**0.22**
Ground Truth	64,764		4,910		584		7,233		225	

Overall, both models yield reasonable counts, but overestimate the abundance of the classes "Brant Goose", "Other", and "Gull" is overestimated, while under-detecting "Canada Goose" and "Emperor Goose". Importantly, POLO achieves a lower MAE in four out of five categories.

Qualitatively, both architectures struggle to separate animals in close proximity of each other, and tend to mistake bright water structures for birds. Differences lie mostly in the classes assigned to these false positive predictions (*cf.* Fig. 2).

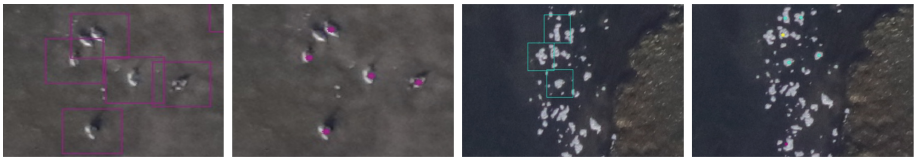

Fig. 2. True- (columns 1 & 2) and false-positive (columns 3 & 4) detections obtained with YOLOv8 and POLO (magenta = Brant Goose, turquoise = Other, yellow = Gull). (Color figure online)

6 Conclusion

Our results show that YOLOv8 achieves surprisingly accurate animal counts when trained on pseudo-labels. However, we manage to improve counting accuracy with POLO, and obtain consistently lower MAE scores across categories. As this leads to more conservative estimates compared to YOLOv8, POLO appears less prone to false positive detections. The modifications proposed in this work

hence represent an improved way to minimize the annotation cost of conventional object detection. Subsequent efforts will be directed towards assessing the difference between POLO and YOLOv8 when hand-crafted bounding boxes are used for training the latter. We further plan to study the behavior of POLO under different data acquisition scenarios, such as flight altitude, camera angle, etc.

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Mining Field Data for Tree Species Recognition at Scale

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Abstract. Individual tree species labels are particularly hard to acquire due to the expert knowledge needed and the limitations of photointerpretation. Here, we present a methodology to automatically mine species labels from public forest inventory data, using available pretrained tree detection models. We identify tree instances in aerial imagery and match them with field data with close to zero human involvement. We conduct a series of experiments on the resulting dataset, and show a beneficial effect when adding noisy or even unlabeled data points, highlighting a strong potential for large-scale individual species mapping.

1 Introduction

Monitoring trees is essential for addressing biodiversity loss and climate change, maintaining ecosystem services, and guiding global conservation and restoration efforts [6,18]. Tree data serve to provide baselines, track changes over time, train and validate ecological models, as well as inform policies and decision making [9]. While forest inventories provide high-quality tree data, such as species and ground measurements, they are labor-intensive, costly, and often limited in spatial and temporal scope [9]. Remote sensing methods through aerial and satellite images offer broader spatiotemporal coverage for tree monitoring, but lack the precision and resolution of ground surveys [6,9]. Recent advances in deep learning have begun to optimize this tradeoff between detail and scale by linking field and image data [14]. This has, for instance, enabled the leap from general tree cover assessments to individual tree counts and traits such as height or biomass using aerial and satellite images [4,24,33]. Yet, our current capabilities fall short in identifying tree species at large scales, hindering new avenues for ecological research and environmental management (Fig. 1).

Bridging the gap between remote sensing and terrestrial data is needed to perform species identification. While high-resolution aerial imagery is now widely accessible in many countries, species data at the individual tree level is often nonexistent or not publicly available [35]. National Forest Inventories (NFIs) are invaluable repositories of tree species data, meticulously collected through extensive efforts, which have evolved from focusing on productive purposes to

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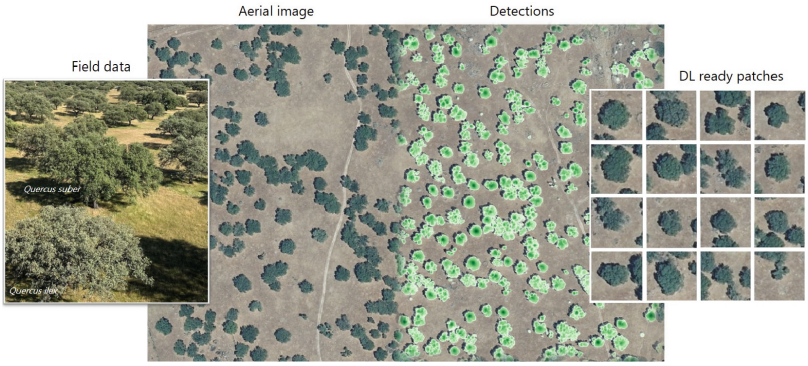

Fig. 1. National forest inventories provide a standardized and extensive source of species information at individual level. We perform individual tree detection on aerial photography to extract individual positions and match them with the field data to extract deep learning-ready patches with species labels.

embracing a broader range of ecological and environmental variables [35]. However, trees outside NFIs plots, as well as trees outside of forests such as those in agricultural and urban landscapes, are left unmonitored [23]. Furthermore, NFIs often do not provide the precise locations of the parcels (and thus individual trees) [27,30], which are key for matching ground-based with remotely sensed information [12].

In this study, we develop a framework for performing tree species recognition at scale using publicly available aerial images (with pixel resolution of 15–25 cm) and publicly available forest inventory data. Our approach involves three main steps: (1) mining individual trees detections in aerial images using pretrained deep models; (2) matching detected trees and NFI data to build a dataset of individual trees; and (3) training and validating a deep learning model for species recognition across forest and non-forest ecosystems.

2 Related Work

Recent years have seen groundbreaking advances in mapping overstory trees in satellite and aerial imagery at scale using deep learning approaches, including tree counting [4], bounding box detection [37], tree segmentation [10,24], and heatmap peak detection [3]. But identifying individual tree species can be challenging due to insufficient image resolution [2], the variability in species traits and the structural complexity of the ecosystem [2,8]. A number of studies have classified species using remote sensing with a certain accuracy [2,8,25,31,40,41], with a notable effort recently on building large-scale datasets of urban tree species [1,5], one species over large areas [16], or monospecific forested areas [11] using public data. However, to our knowledge, there is no existing dataset for individual tree species recognition "in the wild", *i.e.* in mixed forests and trees outside of forests at scale, with no assumption of species distribution.

A key challenge hampering individual species identification is the lack of reliable labels at individual level. While NFIs do produce individual level measurements (mostly in Europe) [9], there is a high uncertainty on individual positions due to the accumulation of noise from GPS measurements, image registration, and temporal mismatches. This particular situation of high volumes of data that also come with a high level of noise is not uncommon in remote sensing, and has been addressed through various works exploiting unreliable labels at large scale for canopy height mapping [20], individual tree detection [37], or building mapping [13]. Parallel to this line of work, recent advances in semi-supervised learning [32,38,39] are a valuable toolbox for building label-efficient methods.

3 Dataset Construction

We collected aerial imagery from publicly available campaigns conducted by the Spanish National Plan for Aerial Orthophotography (PNOA) [26]. The images have a ground sampling distance (GSD) of 15–25 cm, and include four bands as RGB + NIR. We covered over 65,000 km^2 in the provinces of Badajoz, Cáceres, Madrid, and León, chosen for their diverse Mediterranean climates, from xeric to humid and continental, and altitudes ranging from 200 to 2,648 m above sea level. For the field data, we made use of the 4th NFI of Spain [17], which collects extensive data at national-level about forests and their evolution, both from a forest management and ecological perspective. It distributes plots roughly at the intersections of a 1 km^2 cell grid, which are monitored every 10 years. Plots consist on circular parcels, where all trees with a trunk diameter $\geq 7,5$ cm are recorded in a 25 m radius. This includes identifying tree species, recording their position using local coordinates from the center of the plot, measuring functional traits such as diameter and height, and evaluating their health status among other attributes.

3.1 Individual Tree Detection

We use the aerial imagery and available models to perform individual tree detection. In recent years, there has been signficant progress in tree detection, from a fruitful combination of deep vision models and large-scale labeling efforts [3,4,7,10,21,36].

An ensemble of models was employed to predict individual tree positions. The ensemble includes two types of models:

– Models trained for pixel-wise classification (segmentation) on rasterized polygon labels for individual crowns. We used a combination of the Tversky [29] and Focal [22] loss functions. The Tversky loss encourages a high overlap between predictions and labels, while the Focal loss focuses on misclassified pixels, making them complimentary.

$$L_{\text{seg}} = 0.6 * L_{\text{tversky}} + 0.4 * L_{\text{focal}} \tag{1}$$

– Models trained for individual tree detection with Gaussian heatmap modelling [34], on point labels. Each point p at position (x, y) generates a 2D Gaussian kernel

$$\mathbf{h}_{i,j}^{p} = \exp\left(-\frac{(i-x)^2 + (j-y)^2}{2\sigma^2}\right) \tag{2}$$

with a fixed-size standard deviation σ. Individual Gaussian kernels are max-pooled pixel-wise to give a target heatmap, for which we optimize with a simple MSE loss

$$L_{\text{hm}} = \|\hat{\mathbf{h}} - \mathbf{h}\|_2, \quad \mathbf{h} = \max_{p}(\mathbf{h}^p) \tag{3}$$

Segmentation models predict the extent and shape of the canopy cover, but tend to output a continuous mask in dense forests. On the other hand, heatmap-based detection models give an equal weight to each individual tree which encourages an accurate positioning and count of predictions, but can fail to capture the extent of the crown area since they train with a fixed σ. With pixel-wise ensembling, we exploit the synergy between those two approaches and get predictions that are more stable, more accurate, and that better predict crown area and center location. In both cases, we use a simple UNet [28] architecture with a ResNet50 [15] backbone.

We trained models on two datasets. The first dataset contains around 100k individual polygons with aerial imagery in Rwanda [24], RGB 25 cm GSD. The second dataset contains around 40k individual points with aerial imagery in France, RGB 20 cm GSD. The Rwanda dataset was used with segmentation and heatmap detection (after converting the polygons to their centroids), and the France dataset was used with heatmap detection only. Every dataset/model combination was systematically trained on three random 80/20 train/val splits. This gives a total of 9 models.

After predicting with each model separately, the 9 outputs are merged into a final raster with pixel-level arithmetic averaging. Detections are extracted as points with local maxima identification with a kernel size $\mathrm{K} = 2\,\mathrm{m}$ and a confidence threshold $\mathrm{T} = 0.25$. These values were manually chosen to balance over-prediction on closed canopies and under-prediction on sparse areas or smaller trees, they can also be optimized with a sweep on a validation set, if available [3].

3.2 Matching with Field Data

We consider a list of N detections and M field-measured trees on a local patch. First, we conduct 1-to-1 matching by minimizing a patch-level cost matrix, where pairwise assignment cost is the intercenter Euclidean distance:

$$c_{ij} = \begin{cases} \|p_i - \hat{p_j}\|_2 & \text{if } \|p_i - \hat{p_j}\|_2 < 4\text{m} \\ \infty & \text{otherwise,} \end{cases} \tag{4}$$

with a threshold of 4 m to ensure that matches stay realistic. Compared to an exhaustive assignment to the closest candidate, 1-to-1 matching better handles dense areas where the closest candidate might not be the best choice if considering a wider context. We will refer to matched detections as "**verified**", and unmatched detections as "**unverified**".

After matching, we differentiate two situations:

A the parcel is monospecific, *i.e.* only has trees of the same species. In that case, we assign the same species to all detections within the radius. Verified and unverified detections are kept.
B the parcel has multiple species. In that case, we only assign species to verified detections and keep them, and ignore others.

While verified detections can be considered as "ground truth", suitable for training and validating classification models, unverified detections in monospecific plots are not as reliable. We treat these cases as noisy labels, and use them only for training.

3.3 Statistics

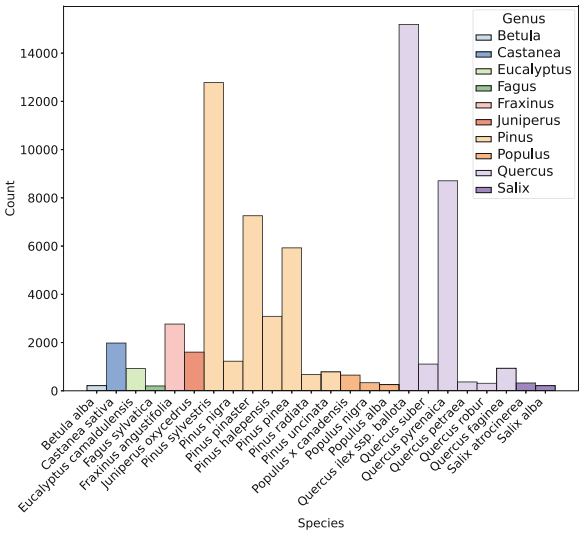

Fig. 2. Our dataset has a variety of species with a typical imbalanced distribution. We only plot species with more than 200 individuals.

The final dataset contains 69180 individual trees, of which 20.4% were matched with a field observation with a maximum distance of 4 m, the remaining being detected in a monospecific parcel and assigned the corresponding label. Of the 1518 parcels, 61.6% (935) are monospecific, the rest having multiple species.

The dataset contains 56 unique species belonging to 28 different genera. We plot the species histogram in Fig. 2.

3.4 Quality Control

We asked an expert in forestry to label a random selection of 48 parcels. We compared the hand-labels with the field-measured tree positions at these parcels, and with the ensemble model predictions, to assess A) if the detection model maintains its accuracy on the new areas we consider here, B) the level of agreement between remote observations and field measures. We conduct one-to-one matching to collect true positive, false positives, false negatives, and report the F1 score, the harmonic mean of precision and recall.

We indicate results in Table 1. We note a high agreement between manual labels and model predictions, and a low agreement of the latter two with field measures. Notably, field data reports a significantly lower count of trees overall. This mismatch highlights the level of uncertainty on tree positions, due to inconsistencies between NFI standards (only measuring trees above a certain radius and within a certain area, instrument and human errors) and photointerpretation from remote sensing data (only using the visible crown to identify trees, data and model errors).

Most interestingly perhaps, we note that the model predictions agree more with the field data than human labels do. In other words, the performance of our ensemble deep model surpasses that of a human annotator, if we consider the field data as close to being "ground truth". This encourages us to consider model predictions as a good source of tree positions for mining species labels.

Table 1. Agreement between detections from aerial photography vs. manual labels from photointerpretation vs field-reported tree positions.

Reference	Matched	Difference	F1	Avg. distance
Field data	Manual labels	+1342	41.4	1.2 m
Field data	Predictions	+1099	47.9	1.1 m
Manual labels	Predictions	−243	76.2	0.6 m

4 Experiments

We trained a simple ResNet34 classifier on our dataset. We conduct experiments to assess the potential of automatically mining tree species from monospecific plots. We report three metrics: overall accuracy (OA), class-averaged intersection over union (mIoU), and class-averaged recall (AR). While OA indicates how well the model predicts species overall, class-averaged metrics remove the bias towards frequent species.

We conduct three experiments (Table 2): training only on verified labels, training on verified and unverified labels, and training on all labels + a volume

Table 2. Species classification experiments. Progressively adding noisy and unlabeled data enhances performance.

	OA	mIoU	AR
Only verified	40.3±2.5	12.9±1.6	24.8±4.1
+ unverified	44.4±1.5	16.8±1.5	31.2±2.9
+ 50k unlabeled	45.5±0.7	23.3±0.2	42.6±0.4
+ 500k unlabeled	54.9±1.2	29.7±1.2	54.8±2.4

of unlabeled patches. For this last setup, we apply a recent semi-supervised approach, deep label propagation [19], which builds a similarity graph between labeled and unlabeled data points to extract pseudo-labels. We experiment with 50k (around 50% labeled data) and 500k (around 10% labeled data) additional unlabeled patches that we randomly extracted from areas that did not contain NFI parcels.

We note a significant and clear positive effect when adding first unverified patches, then unlabeled patches. Class-averaged metrics in particular were increased by a factor of 2 between a model trained only on verified patches and a model trained with only 10% of labeled data, indicating that unlabeled patches contribute to building more balanced models, that do not ignore less-represented species.

5 Conclusion

We introduced a pipeline to automatically match field observations with predictions on aerial photography from an ensemble of pretrained models for individual tree detection and segmentation. Comparing model predictions with hand-labels, we noted that the ensemble surpasses human performance, a strong signal for using predictions at large scale. Importantly, the pretrained models were **not** fine-tuned on our target area, making ensembling a promising strategy for generalizable tree detection.

Our experiments on the automatically generated dataset of tree species show that a classifier is able to learn from this not-so-reliable data, and even benefits from large volumes of unverified or unlabeled examples. While forest inventory data can be unreliable and difficult to work with from a remote sensing perspective, we show here that it is possible to harness the sheer volume of data with modern deep learning methods and learn balanced species classifiers. Note that this strategy can be effortlessly generalized to estimate biomass, height, health, and other attributes commonly collected in forest inventories.

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MaskSDM: Adaptive Species Distribution Modeling Through Data Masking

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Abstract. Species distribution models (SDMs) correlate species occurrences with environmental conditions and underpin much of ecological research. A key consideration in developing SDMs is selecting the optimal set of environmental predictor variables, which vary depending on the specific application and species involved. Existing SDMs approaches are limited to a fixed set of predictors defined a priori. This becomes problematic whenever predictors are suboptimal for a particular species or research question to be answered, or when some predictors are unavailable at a given location. To address this, we introduce MaskSDM, a versatile approach that allows end-users to choose relevant variables and gain insights into their contributions to predictions. Our approach employs masked data modeling to learn robust data representations. This allows MaskSDM to effectively handle missing data during both training and inference, addressing a common challenge in real-world geospatial datasets. Evaluations against alternative methods demonstrate that MaskSDM offers improved predictive performance and facilitates valuable analyses of variable contributions.

Keywords: Species Distribution Modeling · Deep learning · Masked Data Modeling · Ecology

1 Introduction

With the ongoing biodiversity crisis, species distribution models (SDMs) play a key role in monitoring and predicting the distribution of species across various ecosystems. These models, which relate species occurrences to environmental conditions [13], are essential for conservation biology [21]. They are of great interest for understanding current species distributions [15], identifying potential habitats [20], and predicting future changes driven by climate change, habitat destruction, and invasive species [2,31]. Such a wide range of applications,

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however, requires careful determination of problem-specific characteristics, as ecological environments are diverse and modeling challenges manifold. A critical aspect in this regard is the selection of appropriate environmental predictor variables for the model [1,4,14,29]. For instance, the distribution of many species is impacted by the human footprint [16]; however, when aiming to understand species' ecological niches, human influence is typically excluded from the model. Conventional SDMs are generally incapable of adapting to such divergent research questions using a single model. Being able to explicitly select input variables of SDMs for each specific task in a generic modeling pipeline presents several challenges, which are outlined below.

First, existing multi-species distribution models (which learn the distributions of multiple species with a single model [25]) assume the same set of input variables during training and inference. This prevents the selection of input variables based on the differing needs of different species. We argue that end-users of SDMs should ideally decide which input variables are significant for their specific applications and species. *Second*, the SDMs currently used for ecological explanations are often too simplistic [13,30], limiting their ability to capture complex relationships between the input variables. This hinders accurately providing ecological insights by revealing which input variables contribute to the distribution of species and how they do so [6]. *Third*, existing SDMs require the availability of all the considered input variables for the region of interest. However, some input variables may be missing or known to be noisy or uncertain for some locations during training or inference. This calls for multi-modal solutions able to handle situations with variable (numbers of) inputs.

To address these challenges, we introduce MaskSDM, a method inspired by recent advancements in deep learning, which has demonstrated innovative applications in ecology [35] and SDMs [5,7,41]. MaskSDM enables end-users to select relevant variables at inference and provide insights into their contributions to predictions, while effectively handling missing data. To do so, we leverage masked data modeling, which has proven to be effective in learning robust data representations in natural language processing [9], computer vision [22], and various other data types [28]. Masked data modeling involves randomly hiding parts of the input data, forcing the model to learn and infer the missing information from the remaining predictors. This is typically achieved through a reconstruction objective during a pre-training step, where the model predicts the missing parts [9,22,27]. We adopt an approach in which the model directly learns the species distribution in a supervised manner but with missing masked input variables. Our MaskSDM approach offers several key benefits:

1. Flexibility for end-users. MaskSDM allows to select only the variables deemed relevant for specific tasks and species of interest.
2. Analysis of variable contributions. MaskSDM enables a detailed analysis of how different input variables contribute to species predictions.
3. Robustness to missing input variables. MaskSDM handles missing input variables during both training and inference. This ensures that all variables can be included in the model, even if some are unavailable for certain samples.

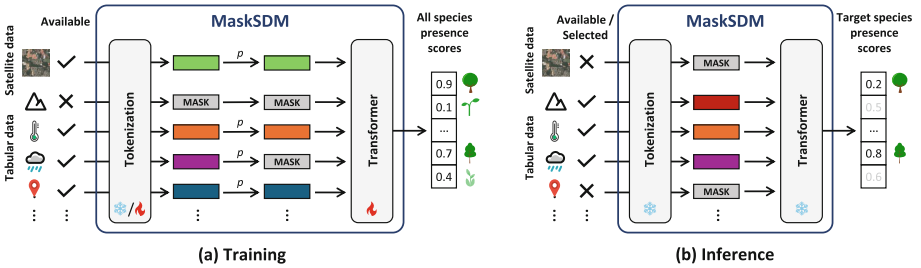


Fig. 1. Overview of MaskSDM. (a) During training, our method employs a mask token to indicate missing input variables to the Transformer model. Additionally, this mask token is used to randomly mask each input variable with a probability p. (b) During inference, MaskSDM can take any subset of variables as input to predict the presence of species of interest.

We evaluate MaskSDM using sPlotOpen [34], a global dataset comprising plant species survey data. Our approach consistently outperforms other SDM methods across various sets of input variables. Furthermore, we demonstrate the effectiveness of MaskSDM through examples that showcase its ability to analyze variable contributions and its adaptability in diverse scenarios, paving the way for the creation of a foundation model for SDMs [3].

2 Method

SDMs typically predict the likelihood of observing a species in a given location based on a predefined set of input variables $\mathcal{E} = \{x^1, x^2, \ldots, x^N\}$ representing environmental conditions. These variables are usually assumed to be consistently available and of the same data type [36]. Our approach, called MaskSDM, is capable of learning from multiple types of data inputs, referred to as modalities, while effectively handling missing modalities. This enables MaskSDM to predict the presence of species of interest by exploiting any subset of variables $\mathcal{E}' \subset \mathcal{E}$. To achieve this, we employ masked data modeling to learn species distributions in a supervised manner. The general MaskSDM approach (Fig. 1) is explained in the following, while the specific implementation details are provided in Sect. 3.

Tokenization. The different modalities are first converted into a standardized format through a process called *tokenization*, which works by projecting inputs to high-dimensional feature vectors (*tokens*) of prescribed size [19,28]. The functions f^j producing these tokens $t_i^j = f^j(x_i^j)$ are the *tokenizers*, with each modality having a dedicated tokenizer tailored to its specific characteristics. Pre-trained tokenizers, trained on larger datasets, are available for some variables [26,28]. They can be utilized to produce more informative and general tokens and save computational time. Otherwise, the tokenizers can be trained alongside the rest of the model parameters. While our experiments focus on tabular data and satellite image features as these are commonly used in SDMs

[10,17], MaskSDM can be easily extended to incorporate other types of data since each modality has its own tokenizer [28]. Given the diverse distributions of variables in tabular data, we tokenize each input variable individually [18,19]. Then, to model complex, non-linear interactions between the tokens of the different input variables, all the tokens are given as input to a transformer encoder model [38].

Masked Data Modeling. During training, MaskSDM employs the masked modeling paradigm to learn robust species distributions. This is achieved by replacing missing input variables with a *mask token* t_{MASK}, signaling to the transformer encoder that a variable is absent. This mask token is learned as part of the model training process, alongside the other parameters of the tokenizers and the transformer encoder. By using mask tokens, we can leverage all available samples during training, even if some values are missing. This is particularly beneficial for real-world datasets, which are often incomplete. For example, we use a dataset containing meta-information (see Sect. 3) that helps explain and disentangle the contributions of different input variables. However, these metadata are not available for all samples. Our approach ensures effective utilization of the dataset despite missing values.

To ensure that the model is robust to any subset of input variables, we randomly mask additional input variables during training, even when they are available [28]. At each iteration, we draw a random number uniformly between 0 and 1, which is the probability of masking each input variable. This stochastic approach forces the model to handle scenarios in which only a limited number of variables are accessible at times, while nearly all variables may be available at others. During inference, MaskSDM offers the capability to replace variables that are missing, or else deemed unsuitable or irrelevant, with the mask token. Moreover, it allows to predict species distributions using different subsets of variables, observing how these choices affect prediction maps and model performance. As a result, users can gain valuable insights into the collective impacts of various input variables on the model's output.

3 Experimental Setup

Dataset. We use the sPlotOpen dataset [34], which comprises 95,104 vegetation plots recorded worldwide. This dataset includes plant species presence and absence data, where the presence of a species is recorded if it is observed in a given plot. To reduce the training time of the model and ensure sufficient data for each species, we focus on predicting the distributions of species with at least 1,000 observations, resulting in a total of 228 species. We use two different methods to divide the data into training, validation, and test sets. The first method involves random splitting of the plots to evaluate the model's *interpolation* capacity. The second method employs spatial blocked cross-validation [32] to assess the model's *extrapolation* capacity (see Appendix A for splits). This approach simulates more realistic scenarios, such as predicting the spread of

invasive species [2], ensuring that the model can generalize to new, unseen areas. We collect predictor variables from WorldClim [24] and SoilGrids [23] for each vegetation plot. WorldClim provides essential climate data on temperature and precipitation, widely used as predictors in SDMs [14]. SoilGrids supplies information on soil properties, such as organic carbon content, pH levels, and texture. Additionally, we incorporate human influence data using human footprint maps, encompassing nine variables representing human pressure [39], such as population density and nightlights. The longitude and latitude coordinates are also provided to the model, as spatial information has been shown to enhance the performance of SDMs in certain contexts where geographic processes are more influential [11,13]. The sPlotOpen dataset also includes supplementary variables for some plots, such as topographic information (elevation, aspect, slope), and metadata like location uncertainty, surface area, and the coverage and heights of various vegetation layers. This results in a total of 61 tabular data variables, all standardized before input into the model. It is crucial to note that while the WorldClim, SoilGrids, and coordinate variables are consistently available for every plot, other variables are frequently absent (see Table 2 of Sabatini et al. [34]). Finally, image features derived from Sentinel-2 satellite images are incorporated using SatCLIP models [26]. These are available for all plots.

Model Architecture and Training. The input variables in the tabular data are tokenized using periodic activation functions [18], followed by a linear layer and ReLU. This approach has demonstrated improved performance in encoding numerical values [18] and is particularly effective for geographic coordinates in capturing features at different scales [33]. For the satellite image features, we utilize the SatCLIP encoder (with 40 Lagrange polynomials) [26], distilled from the ViT16 model [12] trained on satellite images. This encoder serves as the tokenizer, generating tokens that represent characteristics of the satellite images. The transformer encoder follows the architecture used in the FTTransformer [19]. It consists of three identical blocks, each processing tokens of size 192 as both input and output. The number of tokens corresponds to the number of input variables. Each block employs self-attention with 8 heads and a feed-forward network, interleaved with layer normalization and dropout with probability 0.1. Following these blocks, average pooling aggregates the tokens outputted by the final block into a vector of size 192. Subsequently, a linear prediction head with the sigmoid function is applied to generate suitability scores for the 228 species. We optimize the model parameters using stochastic gradient descent with a learning rate of 0.01 and a batch size of 256. To address the class imbalance between presences and absences, we exploit a weighted binary cross-entropy for multi-label classification, with species weights defined according to Zbinden et al. [40]. The model was trained for 300 epochs, with early stopping based on the area under the receiver operating curve (AUC) on the validation set.

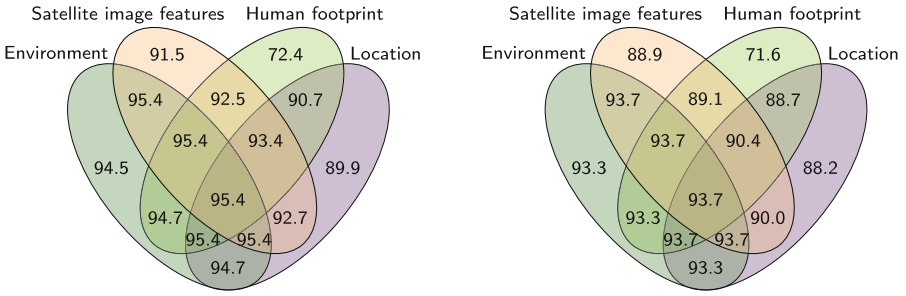
4 Results

We analyze the effectiveness of MaskSDM compared to common baselines for tabular data (MLP, ResNet, and FTTransformer) [19], where missing values are imputed using the mean of the respective variable [37]. Table 1 shows the average AUC across all species on the extrapolation test set using different subsets of input variables. MaskSDM consistently outperforms the baselines, with the performance gap widening as fewer variables are available. Notably, the important difference between MaskSDM and the baseline model with the same architecture (FTTransformer) and training procedure highlights the effectiveness of the masked data modeling approach. This is because the mean imputations in the baselines are treated as genuine data points, while MaskSDM explicitly marks missing variables with a mask token. In addition, the ResNet baseline performs better than the FTTransformer baseline, suggesting that there is potential for improvement in the base implementation of the FTTransformer architecture.

Table 1. AUC performance comparison of MaskSDM to different baselines for various subsets of input variables. The number in parentheses indicates the number of input variables in each subset. Note that average temperature is included as part of the WorldClim data. The other subsets are not overlapping.

Input Variable (#)	Avg. Temperature (1)	✓	✗	✗	✓	✓	✓	✓	✓	✓	✓
	WorldClim (19)	✗	✗	✗	✓	✓	✓	✓	✓	✓	✓
	SoilGrids (8)	✗	✗	✗	✗	✓	✓	✓	✓	✓	✓
	Topographic (3)	✗	✗	✗	✗	✗	✓	✓	✓	✓	✓
	Location (2)	✗	✓	✗	✗	✗	✗	✓	✓	✓	✓
	Human footprint (9)	✗	✗	✗	✗	✗	✗	✗	✓	✓	✓
	Plot metadata (20)	✗	✗	✗	✗	✗	✗	✗	✗	✓	✓
	Satellite image features	✗	✗	✓	✗	✗	✗	✗	✗	✗	✓
Method	MLP [19]	69.9	75.5	N/A	88.1	89.0	89.7	91.1	91.2	91.5	N/A
	ResNet [19]	72.5	80.7	N/A	87.3	90.7	91.5	**93.4**	**93.4**	**94.7**	N/A
	FTTransformer [19]	72.2	75.3	70.2	82.1	86.0	87.3	91.8	91.9	93.7	94.3
	MaskSDM (ours)	**80.3**	**88.2**	**88.9**	**91.6**	**92.6**	**93.3**	93.3	**93.4**	**94.7**	**94.8**

In Fig. 2, we analyze the contributions of different input variables to the performance of the model, comparing the test scenarios of interpolation (Fig. 2a) and extrapolation (Fig. 2b). Unlike traditional ablation studies that require retraining for each combination of input variables, our approach utilizes a single model for all combinations (same for Table 1), showcasing the simplicity and efficiency of MaskSDM in analyzing variable contributions. As anticipated, the AUCs are higher for interpolation splits than those for more challenging extrapolation splits. In both cases, environmental variables alone provide strong performance, with modest further improvements seen when adding satellite image features. However, incorporating human footprint variables does not significantly improve performance when combined with other variables. Likewise, location



(a) Interpolation (b) Extrapolation

Fig. 2. Comparison of the MaskSDM AUC performance between interpolation and extrapolation testing tasks using different subsets of input variables. The "Environment" subset includes variables from WorldClim, SoilGrids, and topographic data.

data yields only minimal performance gains for interpolation and no improvements for extrapolation, which aligns with our expectations.

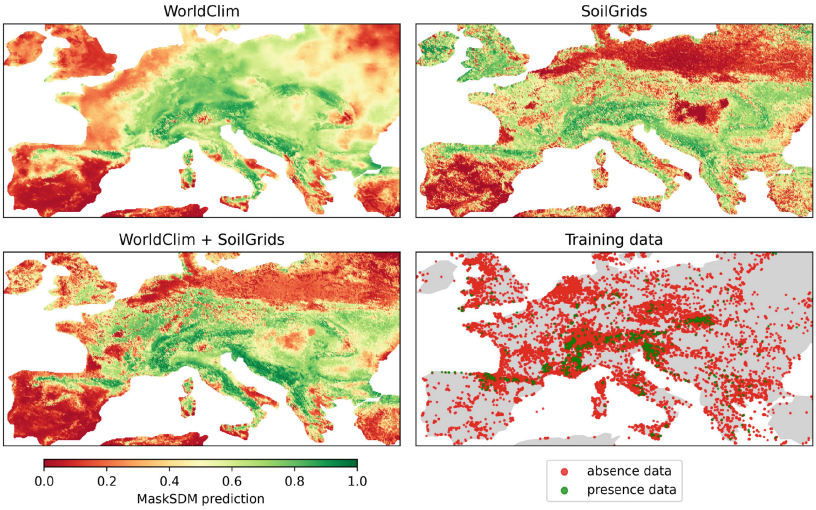


Fig. 3. Distribution of *Anthyllis vulneraria* with different subsets of input variables.

Finally, we illustrate the utility of MaskSDM in visually disentangling the contributions of each variable to the predictions by comparing prediction maps generated using different subsets of input variables, as shown in Fig. 3. We focus on the distribution of *Anthyllis vulneraria*, a medicinal plant native to Europe, whose distribution is significantly influenced by temperature, precipitation, and soil nutrients [8]. The prediction maps confirm the complementary roles of

climatic and soil variables, with climatic variables producing coarser predictions compared to the finer-grained predictions of soil variables.

5 Conclusion

In this work, we introduce MaskSDM, a flexible approach for handling missing input variables in SDMs that can be tailored to the specific needs of end-users, while facilitating the analysis of variable contributions. By leveraging masked data modeling with transformers, MaskSDM effectively integrates various data sources, even when faced with missing or incomplete modalities. We plan to evaluate MaskSDM with additional data modalities and presence-only datasets, which are prevalent in SDMs. We also aim to conduct in-depth analyses of variable contributions to derive ecological insights, ultimately supporting efforts to address the ongoing biodiversity crisis.

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Fine-Tuning for Bird Sound Classification: An Empirical Study

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Abstract. In biodiversity research scientists now routinely acquire audio recordings of vocalizing bird species and are then faced with the task of identifying the species audible in these recordings. Here, we analyze the accuracy (precision, recall and F_1 score) of several deep networks, in conjunction with pre-training and data augmentation techniques, for classifying audio recordings of twelve bird species under multiple data scarcity settings.

Keywords: Bird Sound Recognition · Classification · Fine-tuning

1 Introduction

A well-established indicator of ecosystem health is the variety of bird species [10]. Traditionally, ornithologists identify and count bird species in the study areas. This approach is time-consuming and expensive. Another approach is the combination of passive acoustic monitoring [7,17,27,35] and automated bird sound classification [24] which facilitates the implementation of monitoring systems. These systems can enable long-term measurements of biodiversity and can assist ornithologists. The latter approach is shown to be effective, e.g., in [25,43]. The annual BirdCLEF Challenge [11,19,20,22,23] fosters innovation in model and algorithm development for bird sound classification, the most effective of which are deep networks [11,13,21,23–25,32,34].

Inferring decisions with a deep network from data distributed unlike the data it was trained on can result in lower accuracy. Thus, a fine-tuning of the parameters of the network can become necessary. This requires the annotation of additional data, which can be expensive.

In order to study this data efficiency problem in the context of passive acoustic monitoring of bird species, we make the following contributions: Firstly, we collect and annotate audio recordings of twelve bird species in a rather challenging environment. Secondly, we learn the parameters of several convolutional networks with a varying fraction of available training data, both from a random initialization and from an initialization learned on a large pre-training dataset. Thirdly, we compare the different models and analyze how much data is necessary to obtain satisfactory results. The source code to reproduce the experiments is available from [37]. The collected data and annotations are available upon request.

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2 Related Work

The task of bird sound classification is studied in [11,13,21,23–25,32,34], most notably BirdNET [25]. The authors of [39] evaluate transfer learning of hybrid CNN-LSTMs [1,2], attention-based CNN-LSTMs [47] and multi-head transformers [9,41] for the task of bird sound classification. An interpretable convolutional network for bird sound classification is proposed in [16], which computes prototypes of bird sounds.

Apart from the task of classification, the authors of [6,28,33,38] study the task of clustering bird sounds by k means clustering [33], k nearest neighbor clustering [6], clustering with respect to all elements of three given clusters [28, Section 2.2] and correlation clustering [38].

Self-supervised representation learning for natural images is studied in [3–5,14,36,46]. The authors of [29] benchmark the contrastive self-supervised methods SimCLR [5], Barlow Twins [46] and FroSSL [36] for the task of few-shot bird sound classification, showing competitive results compared to supervised learning approaches. A joint contrastive and generative pretext task for the task of audio representation learning is suggested in [12].

The authors of [42] propose a model for sound separation that separate multiple audio sources audible in the same sound recording. This approach has been shown to increase the accuracy of bird sound classification [8].

3 Models

3.1 Data Representation

We consider a finite, non-empty set S of sound recordings, a feature space $X \subseteq \mathbb{R}^n$ with $n \in \mathbb{N}$, a finite, non-empty set C of classes, and a label space $Y = \{0,1\}^C$, i.e., a multi-label classification problem. Moreover, we consider the features $x\colon S \to X$ and labels $y\colon S \to Y$. In our experiments, each recording $s \in S$ has a duration of 3 s and its features x_s are given by the pixel values of the mel spectrogram computed with a frame width of 1024 samples, an overlap of 768 samples and 128 mel bins, rescaled to 128×384 pixels.

3.2 Architectures

We employ the convolutional networks ResNet-{18, 34, 50} [15], EfficientNet (small) [40] and WideResNet50 [45]. We adjust these architectures by setting the number of input channels for the first convolutional layer to one. Apart from these architectures, we re-implement BirdNet [25]. In addition, we evaluate the BirdNet-Analyzer [18]. Each model defines a function $f_\theta\colon X \to \mathbb{R}^C$ where $\theta \in \Theta = \mathbb{R}^m$ are the $m \in \mathbb{N}$ parameters of the respective architecture.

4 Learning

The parameters of the models are learned by minimization of the logistic loss. In practice, we compute the parameters $\hat{\theta} \in \mathbb{R}^m$ heuristically by stochastic gradient descent with an adaptive learning rate. As software, we employ AdamW [26] with mini-batches $B \subseteq S$. As hardware, we use a single NVIDIA A100 GPU with 16 AMD EPYC CPU 7352 cores, equipped with 32 GB of RAM.

We set the initial learning rate to 10^{-4}, the batch size to 64, and the maximum number of steps to 400,000. We terminate the optimization as soon as the F_1 score on the training set has not improved for 2,000 steps. The class-distribution within each batch is on average uniform, i.e., we use over-sampling to counteract class imbalance.

5 Experiments

5.1 Dataset

In this section, we describe the fine-tuning and pre-training datasets. We optionally apply data augmentation during learning, more specifically, horizontal and vertical roll, SpecAugment [30], and the addition of white noise, pink noise or environmental noise from ESC-50 [31]. We apply each type of augmentation with a probability of 20%. Thus, a sample can be augmented by multiple techniques but at most one type of noise. For further detail, we refer to the source code [37].

Fine-Tuning Dataset. We collect recordings at a sampling rate of 32 kHz with a total of 60 AudioMoths, a low cost recorder [17], deployed at 75 different sites. Using BirdNET [25] we filter 3 s-chunks of audio for the presence of one of 12 bird species. These chunks are validated and, if possible, corrected by ornithologists. This defines a dataset $\mathcal{D}$; see Table 1 and Fig. 1. We partition $\mathcal{D}$ into five stratified folds $\{\mathcal{D}_i\}_{i\in\{0,1,2,3,4\}}$. Each fold defines a training and a test set, $\text{TRAIN}_i = \mathcal{D} \setminus \mathcal{D}_i$

Table 1. Below, we report for the fine-tuning dataset $\mathcal{D}$ and for each species $c \in C$ the number of sites the species has been detected at and the number of samples n_c.

Bird Species	# Sites	n_c
Alauda arvensis	46	1056
Lanius collurio	12	606
Motacilla flava	41	491
Passer montanus	20	268
Saxicola torquatus	13	340
Sylvia curruca	10	636
Anthus pratensis	25	367
Emberiza calandra	21	660
Emberiza citrinella	40	626
Linaria cannabina	20	480
Saxicola rubetra	6	117
Sylvia communis	24	611

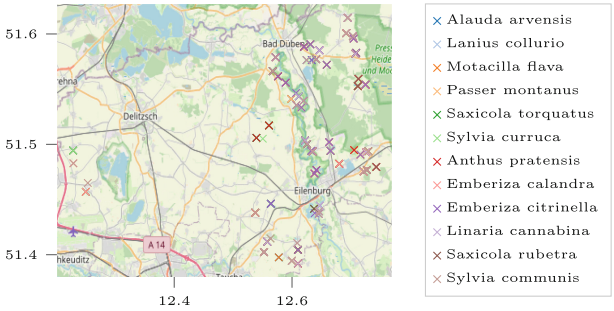

Fig. 1. Above, we show for the fine-tuning dataset $\mathcal{D}$ and for each species $c \in C$ the geographic coordinates of the sites the species has been detected at.

and $\text{TEST}_i = \mathcal{D}_i$. Let n_c be the number of instances in TRAIN_i containing the bird species $c \in C$. With respect to an additional parameter $f \in [0, 1]$, we define the training datasets $\text{TRAIN}_{i,f}$, which is a randomly sampled subset of TRAIN_i and contains $\lfloor f \cdot n_c \rfloor$ samples of bird species c. We sample in such a way that for any $f' \geq f$, $\text{TRAIN}_{i,f} \subseteq \text{TRAIN}_{i,f'}$.

Pre-training Dataset. From Xeno-Canto [44], we collect 317,656 recordings of a total of 8,968 bird species of quality A or B, each recording containing only a single species. We resample the files to 44,100 Hz and split them into 3 s-chunks. We apply the signal detector proposed in [25] to exclude non-salient chunks. This defines the dataset PRE-TRAIN consisting of 1,026,539 audio chunks.

Metrics. In order to measure the accuracy of decisions $\hat{y} \colon S \to \{0, 1\}^C$, we compute the number of true positives (TP), true negatives (TN), false positives (FP) and false negatives (FN) for every class $c \in C$ individually. From these, we compute the macro-averaged precision, recall and F_1 score.

5.2 Experiments

We consider three initializations of the parameters θ of the models defined in Sect. 3.2: (1) randomly distributed parameters, (2) parameters learned from PRE-TRAIN without augmentation, and (3) parameters learned from PRE-TRAIN with augmentation. We learn the initializations (2) and (3) as described in Sect. 4. For all initializations, we learn the parameters on the datasets $\text{TRAIN}_{i,f}$, with and without augmentation, and apply it to the independent dataset TEST_i, for every $i \in \{0, 1, 2, 3, 4\}$ and for every $f \in \{0.1j \mid 1 \leq j \leq 10\}$. After completion of the optimization and for every model, we infer independent decisions $\hat{y}^s \in \{0, 1\}^C$ for every recording $s \in S$ and class $c \in C$ by asking whether $f_\theta(x^s)_c \geq 0$ ($\hat{y}^s_c = 1$) or $f_\theta(x^s)_c < 0$ ($\hat{y}^s_c = 0$).

In Fig. 2 we show the mean, minimum and maximum F_1 score over all folds, $0 \leq i \leq 4$, as a function of the fraction of samples f, and for models initialized

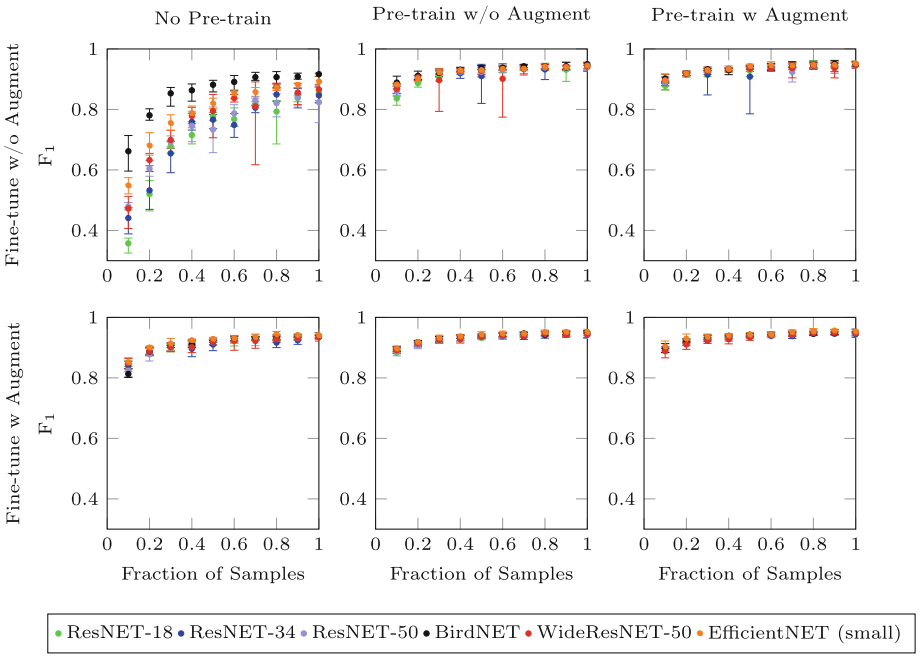

Fig. 2. We depict above the F_1 score of the models ResNET-{18, 34, 50}, BirdNET, WideResNET-50 and EfficientNet (small) as a function of the fraction of samples per species. We show these for models initialized with randomly distributed parameters (left) and from parameters learnt on PRE-TRAIN without (middle) and with (right) augmentations. Also, we report these metrics without (top) and with (bottom) augmentation during fine-tuning. Each datapoint shows the mean, minimum and maximum of the F_1 score across the five test folds.

with random parameters (Column 1), with parameters learned from PRE-TRAIN without augmentation (Column 2), with parameters learned from PRE-TRAIN with augmentation (Column 3); and with parameters fine-tuned on $\text{TRAIN}_{i,f}$, without augmentation (Row 1) and with augmentation (Row 2). In Table 2, we report for each model at $f = 1.0$ the mean of the macro-averages of the precision, the recall and the F_1 score across the five test folds. In Table 3, we report for $f = 1.0$ and the model with the highest mean F_1 score the precision, recall and F_1 score for each species of bird.

From Fig. 2 (Column 1), we observe: On the one hand, learning without augmentations during fine-tuning (Row 1) leads to unstable results for the F_1 score across the five test folds, e.g., the difference between the minimum and maximum F_1 score across the five folds for the BirdNET architecture at $f = 0.1$ can be as high as 10%. On the other hand, applying augmentations during fine-tuning (Row 2) leads to more consistent F_1 scores across the five folds for all considered models and all fractions of samples f. Moreover, when learning without data augmentations (Row 1) no model except the BirdNET architecture

Table 2. Below, we report the median of the macro-averaged precision, recall and F_1 score across the five test folds for $f = 1$. We mark in boldface the best entry for a specific model, and underline the best entry across all models.

	PRE-TRAIN	FT Aug.	Precision	Recall	F_1
BirdNet-Analyzer (conf. = 0.1) [18]					
0.	-	-	88.5	81.0	83.3
ResNet-18					
1.	no	no	93.1	79.4	85.3
2.	no	yes	94.8	92.7	93.7
3.	yes	no	**95.9**	92.6	94.2
4.	yes	yes	95.5	93.7	94.5
5.	yes + Aug.	no	95.8	**93.8**	**94.7**
6.	yes + Aug.	yes	95.5	93.7	94.5
ResNet-34					
1.	no	no	92.2	79.2	84.7
2.	no	yes	95.0	92.7	93.8
3.	yes	no	95.5	93.6	94.5
4.	yes	yes	95.1	93.7	94.3
5.	yes + Aug.	no	**95.8**	**94.3**	**95.0**
6.	yes + Aug.	yes	95.4	93.7	94.5
ResNet-50					
1.	no	no	90.1	78.3	82.4
2.	no	yes	94.8	92.9	93.7
3.	yes	no	**95.5**	92.9	94.1
4.	yes	yes	94.9	94.0	94.4
5.	yes + Aug.	no	**95.5**	93.9	94.7
6.	yes + Aug.	yes	95.4	**94.5**	**94.9**

	PRE-TRAIN	FT Aug.	Precision	Recall	F_1
BirdNet [25]					
1.	-	no	92.5	91.1	91.7
2.	-	yes	94.1	93.4	93.6
3.	yes	no	**94.7**	95.3	95.0
4.	yes	yes	94.3	95.9	95.1
5.	yes + Aug.	no	94.5	95.2	94.8
6.	yes + Aug.	yes	94.5	**96.0**	**95.2**
WideResNet-50					
1.	no	no	92.1	82.6	86.6
2.	no	yes	94.9	92.4	93.5
3.	yes	no	95.5	93.2	94.2
4.	yes	yes	95.2	93.6	94.3
5.	yes + Aug.	no	95.6	93.9	94.6
6.	yes + Aug.	yes	**95.7**	**94.1**	**94.8**
EfficientNet (small)					
1.	no	no	91.4	87.4	89.2
2.	no	yes	94.5	93.7	94.0
3.	yes	no	95.2	93.9	94.4
4.	yes	yes	95.1	95.2	95.1
5.	yes + Aug.	no	**95.5**	94.8	95.1
6.	yes + Aug.	yes	95.4	**95.3**	**95.3**

shows saturation of the F_1 score, even when using all data samples, $f = 1.0$. In contrast, learning with data augmentations (Row 2) leads to almost saturated F_1 scores, even for $f \geq 0.3$. Lastly, we observe an increase in the F_1 score when using augmentations as opposed to not using augmentations, which is higher for smaller datasets used for fine-tuning, i.e., a low f.

In Fig. 2 (Columns 2–3), we observe: Initializing the parameters of the models with parameters learned from PRE-TRAIN, with or without augmentation, leads to the F_1 score being more consistent across the five test folds, irrespective of whether or not augmentations are applied during fine-tuning. For $f \geq 0.3$, we obtain almost saturated F_1 scores with only marginal improvements for increasing the fine-tuning dataset size, i.e., increasing f. Applying augmentations during fine-tuning for both initializations leads to marginal improvements in the F_1 score for both model initializations.

In Table 2, we observe: Both using augmentations during fine-tuning and initializing the models with parameters learned from PRE-TRAIN lead to increasing performance of the model with respect to the F_1 score. With an F_1 score of 95.3%, the EfficientNet architecture initialized with parameters learned from PRE-TRAIN with augmentations and fine-tuned with augmentations (Row 6) is the best performing model in this study. Interestingly, from a comparision of ResNet-18 (Row 3), BirdNet (Row 6) and EfficientNet (Row 6), we observe that the EfficientNet architecture is neither the best performing model with respect to precision (by a margin of 0.5%) or recall (by a margin of 0.7%). Hence, the

Table 3. Below, we report the mean of the precision, recall and F_1 score across the five test folds per species and for the overall best performing model Efficient-Net (small).

Bird Species	Precision	Recall	F_1
Alauda arvensis	90.7	90.4	90.6
Lanius collurio	99.3	98.5	98.9
Motacilla flava	92.6	93.5	93.0
Passer montanus	95.9	95.4	95.6
Saxicola torquatus	95.1	90.7	92.8
Sylvia curruca	99.2	99.5	99.4
Anthus pratensis	96.8	97.6	97.1
Emberiza calandra	92.0	94.5	93.2
Emberiza citrinella	90.9	92.2	91.5
Linaria cannabina	98.9	98.3	98.6
Saxicola rubetra	97.4	95.7	96.5
Sylvia communis	95.7	97.5	96.6

EfficientNet architecture offers the best tradeoff between precision and recall in this study.

In Table 3, we observe: On the one hand, even though Alauda arvensis is the majority species in the dataset (cfg. Table 1), the EfficientNet architecture performs worst for this species with respect to the F_1 score. On the other hand, the model's F_1 score for the minority class Saxicola rubetra is substantially higher than for the species Emberiza citrinella, Emberiza calandra and Motacilla flava for which we have more training data.

6 Conclusion

We examine six neural network architectures for bird sound classification empirically, in different data scarcity settings. For those audio recordings of 12 bird species considered in the experiments, we conclude that both data augmentation and initialization with parameters learned from a pre-training datasets stabilizes the accuracy of the respective models across different test folds and improves accuracy across all data scarcity settings. With all the available training data, the most accurate architecture is EfficientNet. With only 10% of the training data, i.e., between 10–80 recordings per species, we still obtain satisfactory results for the classification of the 12 species considered in this study. Adjusting the parameters to the data distribution increases the accuracy compared to a non-adjusted, pre-trained model by a margin of 12%.

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Multimodal Fusion Strategies for Mapping Biophysical Landscape Features

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Abstract. Multimodal aerial data are used to monitor natural systems, and machine learning can significantly accelerate the classification of landscape features within such imagery to benefit ecology and conservation. It remains under-explored, however, how these multiple modalities ought to be fused in a deep learning model. As a step towards filling this gap, we study three strategies (Early fusion, Late fusion, and Mixture of Experts) for fusing thermal, RGB, and LiDAR imagery using a dataset of spatially-aligned orthomosaics in these three modalities. In particular, we aim to map three ecologically-relevant biophysical landscape features in African savanna ecosystems: rhino middens, termite mounds, and water. The three fusion strategies differ in whether the modalities are fused early or late, and if late, whether the model learns fixed weights per modality for each class or generates weights for each class adaptively, based on the input. Overall, the three methods have similar macro-averaged performance with Late fusion achieving an AUC of 0.698, but their per-class performance varies strongly, with Early fusion achieving the best recall for middens and water and Mixture of Experts achieving the best recall for mounds. The code is available at https://github.com/lgordon99/fusion-strategies-eccv.

Keywords: multimodal · fusion · aerial imagery · ecology · remote sensing · rhino middens · termite mounds · few-shot · class imbalance

1 Introduction

In response to widespread land degradation and biodiversity loss, in its Fifteenth Sustainable Development Goal "Life on Land", the United Nations calls for protecting, restoring, and promoting the sustainable use and management of terrestrial ecosystems [3]. Successfully conserving habitats and species necessitates effective ecosystem and biodiversity monitoring [10]. Towards this goal, ecologists and protected area managers map biophysical landscape features, often through a combination of remote sensing and ground surveys.

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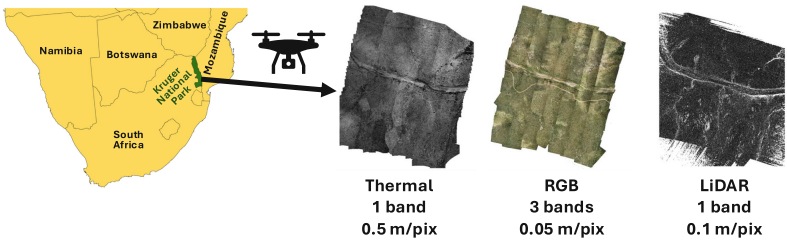


Fig. 1. Multi-band, multi-resolution aerial imagery collected for a site in Kruger National Park, South Africa.

In African savanna ecosystems, three examples of such features of interest are rhino middens, termite mounds, and water. Rhino middens are communal defecation sites used by rhinos for territorial marking and social communication [9]. Termite mounds, composed of soil, termite saliva, and dung, are islands of enhanced nutrient and moisture content in the landscape, facilitating the growth of vegetation, which in turn attracts herbivores [8]. Water, which is here defined as a river, stream, or watering hole, is a key resource on which many animal species are dependent. These landscape features are often distributed throughout vast, inaccessible areas, making manual annotation of remotely sensed imagery tedious and a complete ground survey beyond available time and resources. This challenge motivates the use of deep learning (DL) to automate the detection of biophysical landscape features in UAV imagery.

We develop DL models to map these features in aerial imagery taken of a 284-hectare site in Kruger National Park, South Africa in January 2020 [5]. The UAV was equipped with a thermal camera, an RGB camera, and a LiDAR (light detection and ranging) scanner, which simultaneously collected thermal, RGB, and elevation data at 0.5-m, 0.05-m, and 0.1-m resolution, respectively (see Fig. 1). Using this multimodal data, we study the question of: **How to most effectively fuse multi-resolution thermal, RGB, and LiDAR data in a DL model for multi-class image classification using limited, imbalanced training data?** In particular, is it better to perform fusion earlier or later in the feature extraction process? And how should the modalities be weighted when fused later?

2 Related Work

This work is an extension of Gordon *et al.* [5], which develops an active learning methodology for the detection of rhino middens in this same imagery. While the purpose of that work was to demonstrate the success of a domain-inspired active learning technique, this work explores different fusion methods and architectures and also generalizes to a multi-class setting by considering termite mounds and water in addition to rhino middens. Burke *et al.* [1] summarizes two general fusion strategies. In *early fusion*, multiple input modalities are stacked and fed

into a single neural network as multiple input channels. In contrast, *late fusion* involves using a separate network for each of the modalities to extract modality-specific features. The features extracted by these multiple networks are then concatenated and fed into a prediction layer.

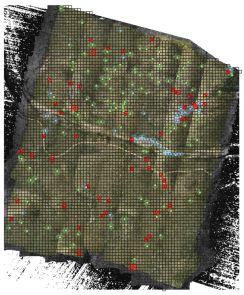
Yeh *et al.* [12] compares the performance of early and late fusion for predicting wealth across Africa from Landsat satellite imagery and nightlight data. In their early fusion method they stacked the Landsat and nightlight bands and passed them through a single network, while in their late fusion method they used two separate ResNet-18 models, one for the Landsat imagery and one for the nightlight data, to extract features from each of the modalities that were then merged and passed through a fully connected layer. They found that late fusion outperformed early fusion. Late fusion can be beneficial when different modalities are more or less useful for classifying images belonging to different classes. For example, Gordon *et al.* [5] found that thermal imagery was most useful for detecting rhino middens due to their strong heat signature. In this dataset, while the height of termite mounds makes them stand out in LiDAR imagery, the same is not true of water, which is at ground level.

A *mixture of experts* in the context of DL [4] contains separate neural networks, or "experts," that specialize in different parts of the input space, which here correspond to modalities. The outputs of the different networks are combined using weights specified by a *gating network*, which maps each input to a distribution over the experts using a multilayer perceptron (MLP) with a final softmax layer to normalize the resultant weight values. The expert outputs weighted by these gating weights then serve as the final predictions. In contrast to late fusion that has fixed modality-specific weights for each class, using a gating network makes it possible to weight the modalities differently for different inputs. This is useful, for example, if some termite mounds are tall and others are short, meaning that LiDAR will be more useful for the former than the latter.

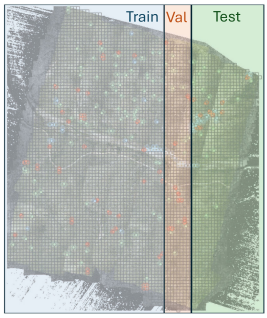
3 Dataset

The dataset consists of spatially aligned thermal, RGB, and LiDAR orthomosaics in GeoTIFF format along with latitude-longitude coordinates for the rhino middens, termite mounds, and water bodies in the site, which we gridify into 20×20-m cells (see Fig. 2a). We use QGIS to assign each grid cell a label corresponding to the aforementioned landscape feature it contains, and it is labeled "empty" if it contains none of them. Because the band number and resolution of the orthomosaics vary across modalities, the same cell is represented by a different-sized tile in each of the modalities: (40,40) for thermal, (3,400,400) for RGB, and (200,200) for LiDAR. As this is a geospatial dataset, we perform a spatially-aware train-validation-test partition [11] (see Fig. 2b), taking the first 50 columns of the grid for training, the next 9 for validation, and the remaining 22 for testing. To address the severe class imbalance (most cells are empty), within the training set we randomly undersample grid cells in the empty class and oversample cells in the midden and water classes so that there are 88 cells

for each class. The validation set contains 704 empty cells, 12 midden cells, 18 mound cells, and 17 water cells. The test set contains 1,036 empty cells, 10 midden cells, 16 mound cells, and 21 water cells. The dataset is proprietary and sensitive due to rhino poaching but could be made available upon request.



(a) Thermal, RGB, and LiDAR gridded orthomosaics with landscape features: rhino middens, termite mounds, and water.



(b) The site is spatially split into training, validation, and testing areas.

Fig. 2. Spatial distribution of landscape features (a) and regional data split (b).

4 Methods

To address the limited training data (88 cells per class) available, we utilize transfer learning. All our models are based on a ResNet-50 [6] pretrained on ImageNet-1k [2], a dataset of ground-level RGB images. ResNet-50 takes in three-channel images and outputs a vector of class predictions. We modify the final fully connected layer in the architecture to yield a four-dimensional output to match the four classes in our dataset. We adapt the architecture further in various ways to implement three distinct fusion methods visualized in Fig. 3.

EARLY FUSION. The EARLY FUSION strategy (see Fig. 3a) first involves upsampling the thermal and LiDAR TIFFs using cubic resampling to match the resolution of the RGB TIFF. Each grid cell then corresponds to three tiles (thermal, RGB, LiDAR) of shapes (400,400), (3,400,400), and (400,400), respectively. We concatenate these three tiles to form a (5,400,400) array for each grid cell. Because the pre-trained ResNet-50 architecture accepts a three-channel input, we modify the first convolutional layer to accept five channels and set the weights of the non-RGB channels to the mean of the weights of the RGB channels and rescale all the weights in that first layer by 3/5, following the procedure in Yeh *et al.* [12].

LATE FUSION. The LATE FUSION strategy (see Fig. 3b) passes the tiles of different modalities through separate feature extractors. Since the thermal and LiDAR tiles have a single band, we modify the first convolutional layer of a

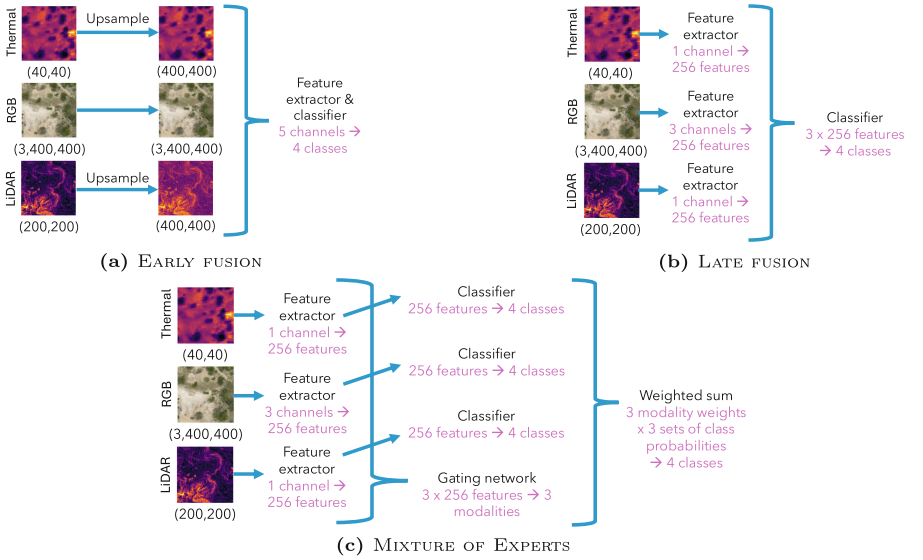


Fig. 3. Diagrams for the three fusion methods studied.

ResNet-50 to accept a single-channel input. We set the weights of that channel to the mean of the weights of the RGB channels and rescale all the weights in that first layer by 3, consistent with the procedure described for Early fusion. The final fully connected layer of all three networks is modified to output 256 features. The three sets of 256 features output by the three feature extractors are then concatenated and fed into a final fully connected layer that outputs class predictions.

Mixture of Experts. The Mixture of Experts (MoE, see Fig. 3c) method first passes the tiles of different modalities through separate feature extractors as in Late fusion. Next, the features extracted for each modality are passed through separate fully connected layers that output class predictions. The three sets of 256 features are also concatenated and passed through a gating network consisting of a fully connected layer outputting 256 features, a ReLU function, a fully connected layer with a three-dimensional output (for the three modalities), and a softmax function. The output of the gating network are then normalized weights on each of the modalities. These gating weights are used to combine the individual expert predictions in a weighted average to yield the final class predictions.

5 Experiments

Setup. We use a batch size of 64 and continue training until the AUC on the validation set has been less than the best AUC for more than 10 consecutive epochs. We normalize all the imagery using the training statistics such that each

band in the training imagery has mean 0 and standard deviation 1. We perform 50 trials for each method, where the random seed for trial i is i. Training for Early fusion and Late fusion utilizes cross entropy loss and Mixture of Experts uses negative log likelihood loss. We use an Adam optimizer with a learning rate of 0.001 for Early fusion and Late fusion, and 0.0001 for Mixture of Experts. We tuned these learning rates by selecting the best AUC on the validation set during the training period when using a learning rate of 0.1, 0.01, 0.001, 0.0001, and 0.00001. All model parameters are fine-tuned during training. All experiments were carried out using one NVIDIA A100 SXM4 40 GB graphics card. Early fusion required 40 GB of RAM and Late fusion and Mixture of Experts required 20 GB.

Results. The overall performance of the three methods is plotted in Fig. 4a, which shows all three methods have very similar precision and AUC on the test set. Comparing Figs. 4b–4e shows that even after balancing classes in the training set, precision was by far highest on the empty class for all methods and rather low for all landscape features, which means that all methods lead to a high number of false positives, which could be attributed to the low number of instances available for training for each class. Recall is highest for termite mounds, with Mixture of Experts achieving the best performance. We hypothesize that this is because different termite mounds can look very different in the various modalities depending on their height and vegetation, so Mixture of Experts, which applies input-specific modality weights, is well-suited. We observe a large performance difference between middens and mounds, and mounds' higher recall could be due to the larger number of distinct images in the training set. The ranking of the three methods on recall is also reversed for middens and mounds, which could be due to the approximately 3x fewer parameters in the Early fusion model being a better fit to the smaller amount of distinct midden instances in the training data. Mixture of Experts does not introduce many more parameters than Late fusion, but the adaptive modality weighting might lead to overfitting for the midden and water classes. Though Late fusion does not perform best for any of the individual classes, it generalizes best across them, leading to its best AUC overall. For all three landscape features we see that recall is much higher than precision. For this application, recall is more important than precision since ecologists would rather have to manually discard some empty images than miss some instances of the landscape features. Nevertheless, future work could explore data augmentation and alternative class-balancing strategies [7] in order to improve the F1 score for all methods.

Mixture of Experts is the most interpretable among these fusion methods, as we can inspect the gating weights assigned to each of the modalities for test images belonging to different classes, visualized in Table 1. We see that for midden images, the LiDAR modality is weighted highest. While middens show slight depressions in the LiDAR imagery due to rhinos' scraping behavior, their warmth gives them the strongest signal in thermal data [5], so MoE's poor midden recall could be explained by it not having learned to weight thermal

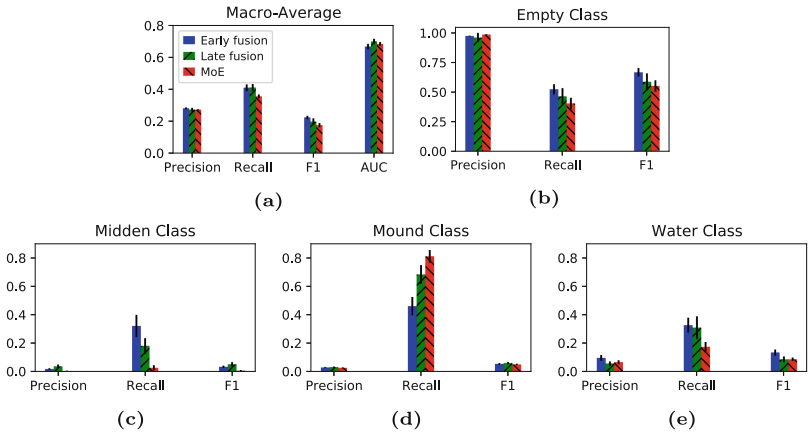


Fig. 4. Performance evaluated for three fusion methods both macro-averaged over all classes (a) and stratified by class (b–e). Mean and two standard errors are displayed.

Table 1. Mixture of Experts per-modality gating weights averaged across test images in each class and pooled across trials. Mean and two standard errors are displayed.

	Rhino midden	Termite mound	Water
Thermal	0.35 ± 0.02	0.30 ± 0.03	$\mathbf{0.40 \pm 0.03}$
RGB	0.25 ± 0.02	0.27 ± 0.03	0.24 ± 0.03
LiDAR	$\mathbf{0.40 \pm 0.02}$	$\mathbf{0.43 \pm 0.04}$	0.36 ± 0.04

highest for this class. For mound images, the LiDAR modality is weighted highest, even more strongly than for middens. This is expected because mounds tend to be even taller than middens but often resemble other vegetation in thermal and RGB. MoE achieves high recall for mounds, so its intuitive modality weighting is consistent with strong performance. For water images, the thermal modality is weighted highest. While water's cooler temperature gives it some signal in thermal data, we expect its blue color visible in the RGB imagery to be an even stronger signal. Incorporating new data from other sites or using data augmentation techniques could help MoE learn to more appropriately weight the modalities for the midden and water classes.

6 Conclusion

We studied three methods for fusing thermal, RGB, and LiDAR imagery for mapping three biophysical landscape features: rhino middens, termite mounds, and water. Early fusion concatenates multimodal tiles of the same resolution and passes them through a 5-channel ResNet. Late fusion passes tiles of different modalities through separate ResNet feature extractors before fusing

them in a fully connected layer. Mixture of Experts uses a gating network to weight the predictions from each modality as a function of the input. Overall, the results show that while the three methods have similar macro-averaged performance with Late fusion achieving an AUC of 0.698, their per-class performance varies strongly, with Early fusion achieving the best recall for middens and water and Mixture of Experts achieving the best recall for mounds.

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I-Design: Personalized LLM Interior Designer

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Abstract. Interior design allows us to express our identities and live according to our preferences, with each design reflecting our unique personalities. However, it is not easy for non-professionals to realize these visions, as it requires aligning functional and aesthetic expectations with the constraints of physical and digital spaces. This complexity often renders interior design a luxury. To make interior design more accessible, we present **I-Design**, a personalized interior design tool that enables users to generate and visualize their design goals through natural language communication. I-Design utilizes a team of large language model agents that engage in dialogue and logical reasoning to transform textual user input into feasible scene graph designs with relative object relationships. Following this, an effective placement algorithm determines the optimal location for each object within the scene. The final design is then constructed in 3D by retrieving and integrating assets from an existing object database. Additionally, we propose a new evaluation protocol that employs a vision-language model and complements the design pipeline. Extensive experiments demonstrate that I-Design outperforms existing methods in delivering high-quality 3D design solutions and aligning with abstract concepts that match user input, showcasing its advantages in both detailed 3D arrangement and conceptual fidelity.

Keywords: LLMs · Text-to-3D · Scene graphs · Retrieval · Interior design · 3D indoor scene synthesis

1 Introduction

Our lives are intimately connected to the spaces we inhabit, as they become the backdrop for our memories, hobbies, and time with loved ones. However, finding or creating the perfect space to match our lifestyle, aspirations, and needs is not always straightforward, and professional assistance can be considered a

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luxury. Even with expert help, the gap between what inhabitants truly desire and their ability to convey it to professionals often leads to unsatisfactory results. This discrepancy can leave people with ill-fitting living spaces, impacting their physical and mental well-being.

Advancements in Metaverse and supporting technologies now offer people the opportunity to create their own digital spaces to experience with AR in their physical space or to visit in the digital realm. Interior design in the Metaverse is as complex as it is in the physical world, with many professionals offering services as designers or architects of the Metaverse. Designing interior spaces that better suit individual needs, whether in the physical or digital realm, should be accessible to everyone. To this end, we tackle the challenging task of 3D Indoor Scene Synthesis (**3DISS**). Given a user's unstructured textual description of preferences, our goal is to deliver 3D design solutions that align with their vision.

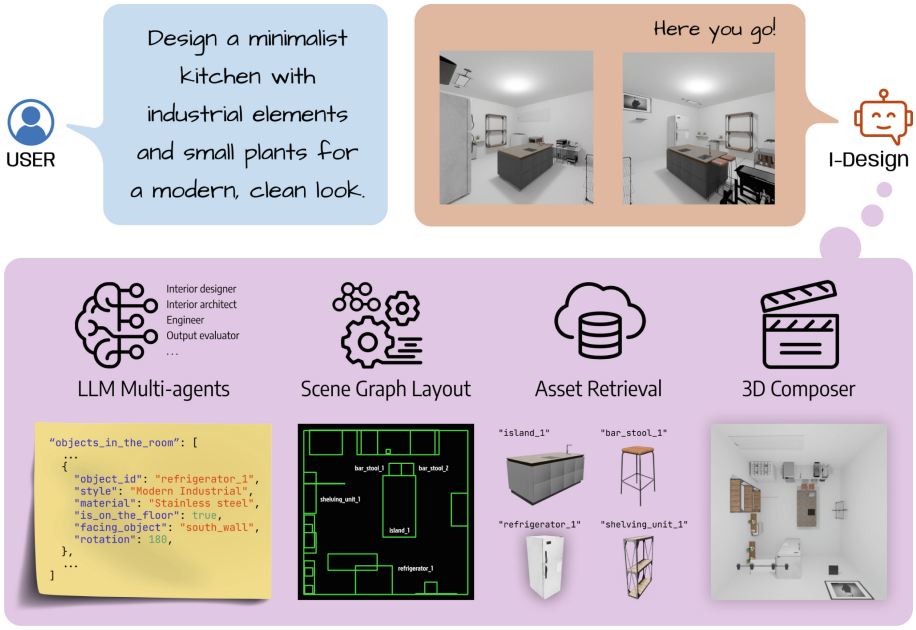

Fig. 1. Overview of I-Design. Starting from the user specification of design preferences in plain text, we query LLM agents to come up with room items, their properties, and their relative relationships in the form of a scene graph. We solve for absolute object placement in the scene graph using the proposed backtracking algorithm (Scene Graph Layout), retrieve 3D assets according to the functional and stylistic specifications, and compose the final result in 3D.

Specialized generative models [8,18,44] and data-driven approaches [43,50,61] have demonstrated a remarkable ability to produce diverse and realistic interior layouts. Yet, their performance is governed by the closed-set and limited

datasets used to train them, which inevitably is a biased and incomplete sample of the world. In addition, such methods that accept textual input from the user can only utilize structured text with predefined grammar and rules. Consequently, it is challenging to guide these models toward producing practical designs for real-world, unseen interiors that align with user preferences.

The 3DISS task requires reasoning abilities beyond those of specialized data-driven models. This includes understanding design principles and concepts such as object selection, styles, and spatial arrangement. However, a successful solution must excel in several fundamental steps. Firstly, the underlying generator must proficiently interpret abstract input, identifying the objects to incorporate into the scene to meet user preferences. It should also know about the typical items associated with specific room types, incorporating high-frequency and common-sense objects even if the user does not explicitly mention them. Additionally, it requires an awareness of plausible spatial relationships among objects originating from an unrestricted vocabulary. Addressing these challenges requires a system with a comprehensive understanding of diverse human interior design preferences and an extensive database of 3D objects, encompassing their functionality, dimensions, and stylistic attributes.

Given recent technological advancements, the most viable option to overcome these obstacles is large language models (LLMs) [1,58]. LLMs are trained on internet-scale data and encode a "world model" that can be probed and interacted with using natural language. Still, they cannot use the language as humans do to solve complex tasks without resorting to specialized techniques, such as Chain-of-Thought [49]. Technical aspects (*e.g.* limited context window length, number of parameters) and artifacts (*e.g.* hallucinations) additionally hamper practical applications. Attempts to produce structured output with LLMs without incorporating hard constraints, communication interfaces, and cross-checks are exciting at first glance but leave much room for improvement. LayoutGPT [13], a recent work that utilizes LLMs for 3DISS, directly predicts the absolute positions of objects in the scene. Although this technique may be effective when dealing with small-scale arrangements of a few objects, it proves inadequate in generating realistic scenes containing dozens of objects interlinked in intricate ways. Moreover, the single-step scene generation in LayoutGPT lacks the capability to provide interpretability regarding the resulting objects and their arrangement. One of the most recent works in this field is Holodeck [59], which addresses many of the spatial inconsistencies found in LayoutGPT and is capable of generating personalized 3D scenes. However, the method struggles with complex object configurations and is limited to a single 3D asset dataset.

Recent research [36,57] has also demonstrated that when multiple LLM agents with diverse responsibilities communicate with each other, they can collectively tackle complex tasks with which a single LLM instance may struggle. Hence, we utilize LLM agents to collectively address the challenges posed by 3DISS, leading to the creation of scenes that are spatially more plausible and diverse. Furthermore, drawing inspiration from other works [2,5,46,61], we employ scene graphs as scene representations since they offer a high-level

abstraction by focusing on the objects and their relationships, which can be creatively developed with LLMs, refined through rule-based feedback, and visualized. Additionally, employing the scene graph representation as the interface between LLM agents and algorithms enables interpretable object arrangements.

To this end, we present **I-Design**, a personalized interior designer for 3D Indoor Scene Synthesis. Starting from the user specification of the design preferences in plain text, I-Design queries LLM agents to identify room items, their properties, and relative relationships in the form of a scene graph. It solves for absolute object placement in the scene graph using the proposed backtracking algorithm, retrieves 3D assets according to the functional and stylistic specifications, and composes the final result in 3D. To evaluate the proposed design pipeline, and following in the footsteps of [54], we propose a novel evaluation protocol based on a vision-language model (VLM). Extensive quantitative and qualitative experiments show that I-Design outperforms existing methods in delivering high-quality 3D design solutions that align with abstract concepts in the user input. Our contributions are summarized as follows:

- A novel method that takes an unstructured grammar-free natural language user input and provides 3D design solutions that align with user preferences.
- A new approach to the 3DISS task through the reasoning and conversation of multiple LLM agents.
- A procedural scene graph layout transformation, converting scene graphs with relative node relationships into final absolute 3D representations.
- An interpretable pipeline, providing flexibility and enabling iterative design without needing to redo the entire process.
- A VLM-based evaluation for 3D scenes.

2 Related Work

3DISS via LLMs: Advancements in LLMs have already impacted 3D scene synthesis, although the connections between the two have not been fully explored. Initial scene synthesis methods integrated LLMs to encode user textual input into vector representations that were subsequently used in the object placement process [26,27,43,47]. Feng *et al.* [13] expanded the scope with LayoutGPT by exclusively employing GPT models to generate 3D indoor scene representations. Their work functions like a retrieval system, employing a strategy based on absolute coordinates to position objects in the scene. Wen *et al.* [52] further advance this line of research by developing a dataset-free, open-vocabulary approach for generating 3D home layouts. Such endeavors often encounter challenges stemming from the limitations of existing GPT models in terms of geometric reasoning, resulting in scenes with objects that overlap or are placed outside the scene boundaries. One of the latest advancements in this field is Holodeck [59], which addresses most of the overlap and out-of-bounds issues through a constraint-based placement solution and candidate selection process during object retrieval, while maintaining strong generalization to diverse scenes.

Recent research has demonstrated that when multiple LLM agents with diverse responsibilities communicate with each other, they can collectively tackle complex tasks with which a single LLM instance may struggle. This concept, introduced through projects like AutoGPT [57] and ChatDev [36], has already found diverse applications in data analysis [21], interactive reasoning [25,48], software development [36], and planning [7,40,41]. Our work strives to address the above 3DISS challenges by reasoning with LLMs in a multi-agent setting.

3DISS via Generative Models: One formulation of 3DISS involves generating multi-view consistent image sets or panoramas according to user-defined specifications and converting these 2D representations into 3D scenes. Approaches like Text2Room [18] and LucidDreamer [8] use a virtual camera to navigate the space and iteratively generate images through image inpainting, monocular depth estimation, 3D lifting, and stitching. Alternatively, MVDiffusion [44] utilizes correspondence-aware attention modules to generate multi-view consistent panoramas or image sets in a single pass. Notably, efforts [3,14,39,45,60] focusing on long-term consistent novel view synthesis are also adaptable for 3DISS.

Despite leveraging the capabilities of 2D generative models [11,17,38,63], current methodologies struggle to integrate 3D geometric constraints that are essential for practical interior design applications, such as floor plans and walls. Ctrl-Room [12] attempts to address this challenge but artifacts persist. Besides, monocular image-based depth estimation and 3D lifting introduce uncertainty into the final 3D mesh output [18]. Furthermore, the loose semantic coupling across views poses challenges in indoor scene synthesis, leading to unrealistic scenarios like multi-representations of the same furniture instance throughout the scene [12,44]. The LLM-based approach we employ can also generalize to different settings, such as free scope of room type or object descriptions, while simultaneously ensuring the satisfaction of geometric constraints.

3DISS Through Prior Learning: Another data-driven approach to tackling 3DISS divides the process into two distinct steps: 3D asset selection and layout synthesis, *i.e.* determining the furniture sets for a room and their placements.

For 3D asset selection, two primary approaches emerge: (1) dataset-based retrieval and (2) synthesis via generative models. With the availability of extensive and high-quality asset datasets [6,9,16,55], one can retrieve suitable models, for example, using semantic embedding [28]. Alternatively, generative models can be employed to generate appropriate 3D assets based on text or image inputs [19,29,30,33]. Given the richness of existing large-scale 3D asset repositories within our context, as well as the direct real-world applicability of retrieving objects from product databases, we use the retrieval-based method.

For furniture arrangement, classical approaches apply specific rules to guide the placement of furniture [10,51,56], construct grammars for procedural modeling [22,34,35], or use human interaction for editing [20,31]. However, with the advent of large-scale 3D indoor datasets [16,24,42,64], recent works have shifted towards learning from expert-designed layouts by employing generative models. Pioneering efforts like ATISS [32] and SceneFormer [47] employ transformer-based models to synthesize indoor environments in an autoregressive manner

autonomously, selecting and placing objects sequentially. LEGO-NET [50] refines initial messy room layouts by learning human criteria for regularity through a transformer-based diffusion-like pipeline. Recent works like DiffuScene [43] and Commonscenes [62] utilize diffusion models to generate interior scenes, representing the scene as a scene graph. To liberate layout synthesis from dataset constraints, we choose to employ LLM-based methods. The common-sense knowledge learned by LLMs can assist in creating a wide range of reasonable designs.

3 I-Design

Our method for proposing design solutions from user input in plain text (I-Design) is summarized in Fig. 1. Initially, I-Design examines unstructured textual user input and transforms it into a viable design proposal, represented as a scene graph, through querying LLM agents (Sect. 3.2). Following, the scene graph layout module produces object placement proposals represented in the form of a 2D room floor plan (Sect. 3.3). Finally, I-Design retrieves 3D assets from existing databases (Sect. 3.4) and assembles the final design within a 3D environment, producing functional and stylistic design solutions that reflect user input (Sect. 3.5).

3.1 Problem Formulation

Given a non-structured textual user input T_{user}, the dimensions of a room $(l_{room}, w_{room}, h_{room}) \in \mathbb{R}^3$, and the number of objects to include in the scene $n \in \mathbb{R}$, the objective is to create a 3D scene that aligns with user's requests in the textual input, while ensuring functionality, coherent design, and 3D consistency.

We use a scene graph $\mathcal{G} = (\mathcal{O}, \mathcal{E})$ to represent the spatial relationship between objects, where nodes $\mathcal{O} = [\mathbf{o_1}, ..., \mathbf{o_n}]$ represent object instances in the scene. An additional node type o_r is included for room layout elements such as walls, ceiling, or the floor. Each object $\mathbf{o_i}$ is associated to a set of properties $\mathbf{o_i} = \{\alpha_i, m_i, s_i, r_i, p_i, cs_i\}$, where α_i denotes the object's name and m_i describes its material and architectural style. The associated geometric properties are: $s_i \in \mathbb{R}^3$ the dimensions of its bounding box, $r_i \in \mathbb{R}$ the rotation angle along the z-axis pointing upwards, and $p_i \in \mathbb{R}^3$ the position of the object in the scene. Finally, $cs_i = (x_{neg}, x_{pos}, y_{neg}, y_{pos}) \in \mathbb{R}^4$ is the cluster size, indicating the overall space that an object, along with its children, would occupy in each direction.

Edges in the graph are represented by $\mathcal{E}$ where $e_{ij} = (\text{adj}_{ij}, \text{prep}_{ij})$ is the directed edge from $\mathbf{o}_i$ to $\mathbf{o}_j$ that comprises details regarding the adjacency of the two objects and the prepositional connection between them. $\mathbf{o}_i$ is considered as the parent node, the predecessor node in the graph, to the child node $\mathbf{o}_j$. The root nodes of the scene graph are represented by $\mathbf{o}_r$. Relationships stem from the scene graph's structure as a polytree. These edges describe the spatial locations between objects (e.g. *left*/*right*, *in front*/*behind*, *on*, and *above*/*under*) relative to the canonical orientation of the parent object's bounding box, as well as their connections to the room layout (e.g., *on* and *in the corner*).

3.2 LLM Multi-agents

Generating a 3D indoor scene based on unstructured user input is a challenging task that demands detailed planning. There are two essential steps shared among common methods: based on user input, selecting the objects to populate the scene and arranging them in a meaningful and coherent configuration. To more effectively tackle this complexity, we employ a multi-agent approach, allocating the tasks involved in these steps across multiple LLM agents. As shown in Fig. 2, interpreting and transforming user input into a functional, personalized scene graph involves five distinct agents: Interior Designer, Interior Architect, Engineer, Layout Corrector, and Layout Refiner. The Interior Designer, Architect, and Engineer propose a relative scene graph from the user input. We assume that the agents have appropriately considered contextually bonded objects (e.g., a dining table with the dining chairs) while constructing the relative scene graph, justifying spatial separation and disconnection within the graph. However, the relative scene graph lacks guaranteed spatial consistency, allowing for connections between objects that may defy real 3D scene constraints. Therefore, our pipeline uses a Layout Corrector agent to rectify invalid and implausible connections, while the scene graph contains only relative relations. Finally, to increase the robustness of the pipeline when dealing with multiple objects assigned to the same parent within the scene graph, the Layout Refiner agent proposes relations between such objects, simplifying the subsequent layout stage. Next, we examine each of the agents in more detail.

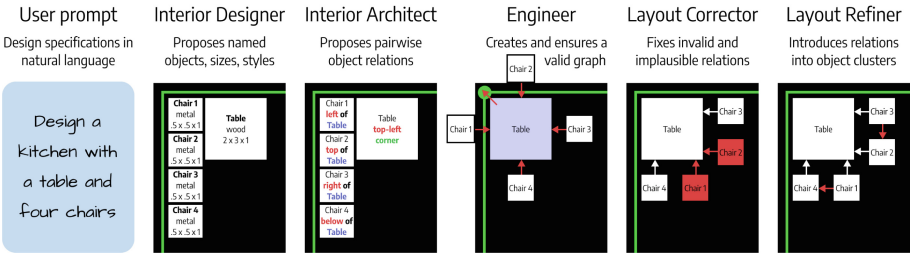

Fig. 2. Scene Graph Generation Pipeline with LLM Agents. Each agent receives the user input and a JSON representation of the previous stages and transforms the output according to the task prompt. Due to the specialized nature of each agent, the generated scene graph reflects user specifications and feasibility constraints, such as topological correctness and semantic plausibility. Red indicates changes introduced by each agent. Complete prompts of every agent are specified in the suppl. material. (Colour figure online)

Interior Designer: The agent receives inputs comprising the free-form user input T_{user}, the room dimensions $(l_{room}, w_{room}, h_{room})$, and the desired number of objects n, and proposes a selection of objects $\{\mathbf{o}_i\}$ tailored to the user's

preferences, while also ensuring their functionality matches the specific room type. Specifically, for each object, the interior designer suggests the properties $\mathbf{o_i} = \{\alpha_i, m_i, s_i\}$, which include its name α_i, material m_i, and 3D size s_i. Note that the agent may propose several identical instances. For example, there might be four instances of a "chair"positioned around a table (see Fig. 2).

Interior Architect: Its primary responsibility is to establish object-to-object connections, as well as object-to-room layout relations. In other words, its role is to establish the edge connections e_{ij} between the object nodes. The agent is not constrained on the number of edges each object can have, allowing for diverse configurations where an object may have edges to room layout elements but none to other objects, and vice versa. Additionally, the Interior Architect creates information on the rotation r_i of each object around the vertical axis.

Engineer: This agent transforms the relative scene graph into a JSON object structured according to a specified JSON schema. Each entry in the JSON file encompasses the details for $\mathbf{o_i} = \{\alpha_i, m_i, s_i, r_i, e_i.\}$ for the corresponding object. Moreover, the agent employs a JSON schema validator that assesses the validity of the generated JSON file based on the provided schema. In cases of non-compliance, a modification from the Engineer is required until the output aligns with the specified schema, ensuring a valid JSON representation.

Layout Corrector: The responsibility of this agent is to fix invalid connections in the graph, which involves removing spatially implausible edges and eliminating ambiguities between nodes, such as objects being placed in a corner with only one wall specified. Before calling the Layout Corrector, an intermediate step algorithmically identifies:

1. *Objects positioned beyond room boundaries (as exemplified in Fig. 2):* This examines children nodes of objects allocated along walls or in room corners. For example, if object **A** is positioned *behind* object **B**, which is placed alongside a wall, this would result in object **A** being out of bounds.
2. *Spatially impossible object-to-object connections:* This involves verifying whether adjacency relations between objects could be violated via conflicting edges in the scene graph. This requires checking if any object has been positioned between two adjacent objects in the scene graph.
3. *Size incompatibility between parent and child objects:* We operate under the assumption that, in a valid scene graph, the bounding box of the parent object is sufficiently spacious to accommodate its children. For example, if a *table* has two *chairs* positioned on its *left* side, it should be wide enough to accommodate them; likewise, a *wall* with multiple objects attached *on* it should have sufficient size to accommodate all of its children objects. This last check ensures this compatibility.

With this information as input, the agent determines the relocation of objects experiencing spatial conflicts, either suggesting alternative edges for them or removing them entirely from the scene.

Layout Refiner: The agent aims to eliminate ambiguities between children nodes that share the same edge to the parent. A notable example of this phenomenon is observed with ornaments placed on a desk. As a desk typically accommodates several objects on its surface, a scene graph designating the *desk* as the parent with the preposition *on* as the edge fails to convey the relative orientation of the objects on the desk to each other. To eliminate this ambiguity, we establish edges between children nodes, ensuring a distinct ordering among them. The agent also verifies that the scene graph maintains its acyclic property. If the graph contains cycles, we must remove edges contributing to these cycles to preserve a distinct hierarchy within the scene graph. This property is then assumed as valid when running the Backtracking Algorithm in Sect. 3.3 for the depth-wise placement of objects.

3.3 Scene Graph Layout

Computing Cluster Sizes: In the final post-processing phase, we compute cluster sizes for each object to facilitate object positioning in the subsequent backtracking algorithm, thereby strategically constraining the search space for object placement within the room. This step is crucial for scenarios like positioning a table with chairs on each side, as it prevents the table from being placed at the room's edge. This approach enables us to determine the clearance – the distance an object must spare in each direction to ensure the successful placement of its children within the scene. Taking such precaution becomes crucial in scenes where the search space for each object is extensive, yet the solution space is relatively constrained. Without this, the randomized nature of the backtracking algorithm may lead to extended processing times, particularly when dealing with many object relationships.

Backtracking Algorithm: The backtracking algorithm converts the relative representation of the scene graph $\mathcal{G}$ into object positions $\mathcal{P}$. The room is initially populated with the root nodes of the scene graph $\mathcal{G}$, representing the fundamental elements of the room layout, such as walls and ceiling. The positions of these root nodes remain fixed while other objects are arranged around them. Conceptually, the algorithm represents the space of plausible positions for each object as a bounding box $\mathcal{B}_i$ and samples a position p_i for each o_i from $\mathcal{B}_i$.

The objects are placed after being topologically sorted, prioritizing nodes higher in the scene graph hierarchy for placement first. To avoid placement issues later in the algorithm, each object o_i is positioned alongside its children according to $\mathcal{G}_i$. The plausible positions for the bounding box of the objects are determined by intersecting the bounding boxes produced by the predefined placement functions $f(o_i, o_j \rightarrow o_i \in \mathcal{G}_j)$.

The nodes are organized into depth groups, where a node's depth $d = |o_i \rightarrow o_r|$ represents its distance to the closest o_r. If an object o_i with depth d cannot be placed into the scene due to either an empty bounding box $\mathcal{B}_i = \varnothing$ or if p_i consistently results in collisions with other objects, we remove p_j for all objects with depth $\geq d$. This step also removes children nodes with $\geq d$ that might have been placed in the scene. Subsequently, we re-sample positions for objects with depth $d-1$. Once all objects with depth $d-1$ are positioned successfully, we increment the depth. An example of scene graph processing is shown in Fig. 3.

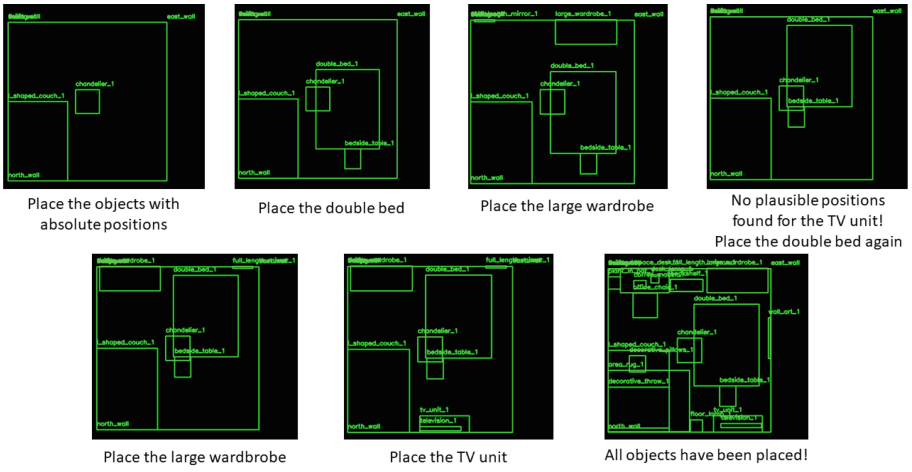

Fig. 3. Backtracking-based Scene Graph Layout. Starting from the scene graph specifying interior items as nodes connected with relative relationships, the backtracking algorithm solves for their absolute placement in the room. The algorithm reverses dead-end configurations and excels at object placement while respecting spatial constraints.

3.4 3D Asset Retrieval

We can generate textual descriptions for each object by utilizing the object name, style, and material information. These textual descriptions are then transformed into text embeddings using the CLIP [37] text encoder. The alignment between OpenShape [28] encodings and CLIP embeddings enables us to measure the distance between our text embeddings and the learned object representations from the database of choice. This facilitates the retrieval of a 3D asset that is closest to our textual description of each object. After the retrieval, the assets can be adjusted to fit the bounding box provided in the scene graph.

3.5 3D Composer

Having produced the final scene graph and retrieved the 3D assets, we employ an off-the-shelf 3D renderer to visualize the room interior in 3D. The output of

I-Design pipeline is a collection of entities, including (a) the scene graph, (b) the floor plan, (c) preconfigured rendered views, and (d) interpretability artifacts, such as the input user prompt and the communication log between the agents.

This output bundle serves multiple purposes. Primarily, it allows for the inspection of every pipeline stage and, if necessary, enables the replay of select stages to introduce variations. For example, the user may want to swap out a piece of furniture or render the scene from a novel viewpoint. Secondarily, the bundle enables diverse quantitative studies to investigate the alignment of the user preference with the produced results. We elaborate on this aspect in Sect. 4.2.

4 Experiments

4.1 Implementation Details

We use Microsoft's AutoGen [53] framework to enable multi-agent conversation, with each agent equipped with GPT-4 model [1]. The `temperature` is configured at 0.7, while `top_p` is set to 1.0. For object retrieval, we rely on OpenShape [28], utilizing text embeddings to retrieve objects from Objaverse [9] and 3D-FUTURE [16]. The scene is then visualized and rendered using Blender [4].

4.2 Quantitative Evaluation

To assess the design quality of the interior rooms generated by our proposed pipeline, we conduct a quantitative comparison with baseline approaches.

Metrics: We conduct quantitative evaluations using the following metrics:

Average Number of Proposed Objects (NObj): The diversity of the generated interior rooms can be partially assessed by considering the number of furniture pieces proposed and placed within the scene. Here, we compute the average number of proposed objects for bedrooms and living rooms separately.

Out-of-Boundary Rates (OOB): The feasibility of a generated layout can be quantified by examining the frequency of objects extending beyond room boundaries [13]. If there are any bounding boxes outside the designated room boundaries within a scene, that scene is deemed invalid. OOB values are calculated as the ratio of invalid scenes to the total number of generated scenes.

Bounding Box Loss (BBL): The viability of a generated layout can also be measured by evaluating the degree of overlap between proposed 3D bounding boxes. This is calculated as the average intersected volume of bounding boxes across generated scenes in cubic decimeters.

GPT-4V Ratings: As highlighted in [54], GPT-4V [1] has been shown to serve as a human-aligned evaluator for 3D content. Inspired by this concept, we employ GPT-4V as an evaluator to assess the visual quality of synthesized rooms based on their renderings. The rating criteria encompass aspects of functionality and activity-based alignment (Func.), layout and furniture (Layout.), color scheme and material choice (Scheme.), as well as overall aesthetic and atmosphere (Atmos.). The integer ratings range from 0, the minimum, to 10, the maximum. The rating process is conducted in a unary manner, where each scene is evaluated individually. Scene renders are given to the evaluator for rating to assess their alignment with the provided user input. Each evaluation is repeated three times in parallel to account for the variability in the LLMs grading. The resulting grades are averaged in each category with standard deviations reported.

Baseline: We compare to LayoutGPT [13] and Holodeck [59]. LayoutGPT [13] utilizes LLM models, including GPT-4 [1], to generate room layouts in a CSS-like format through few-shot learning with examples sourced from the 3D-FRONT [15] dataset. To evaluate our method on more complex prompts, we compare it against Holodeck [59], an LLM-based method for generating diverse 3D environments.

Experimental Settings: As LayoutGPT [13] only supports bedroom and living room layout generation, our comparison is limited to these two room types and variations of their dimensions. We ensure that the prompts given to our pipeline contain only this information for a fair comparison. Furthermore, we let LayoutGPT [13] retrieve objects from Objaverse [9], and use same settings for rendering. We generate ten living room and ten bedroom scenes with the same dimensions for each method. The quantitative comparison results are provided in Table 1. To farely compare with Holodeck, we retrieve assets from the AI2Thor dataset [23] that Holodeck uses for 3D scene generation. We also report our results using a combination of Objaverse and 3D-FUTURE to showcase the full capabilities of I-Design. Since Holodeck can handle multiple room types, we generate ten scenes all of different type. Results are shown in Table 2. Refer to the suppl. material for a complete list of prompts used in both experiments.

Results: Compared to LayoutGPT [13], as illustrated in Table 1, our approach stands out in proposing a higher number of total objects and more physically plausible layouts, exhibiting zero OOB and minimal BBL. For evaluations with GPT-4V, our method outperforms LayoutGPT [13] across all metrics, indicating better visual quality of the synthesized rooms and alignment with user preferences. For the second quantitative analysis in Table 2, our approach proposes richer scenes with higher number of objects, similarly to Holodeck no out-of-bounds objects, while having negligible Bounding Box Loss. According to the GPT-4V-based grading schema, our method outperforms Holodeck in nearly every category. When I-Design uses a combination of Objaverse and

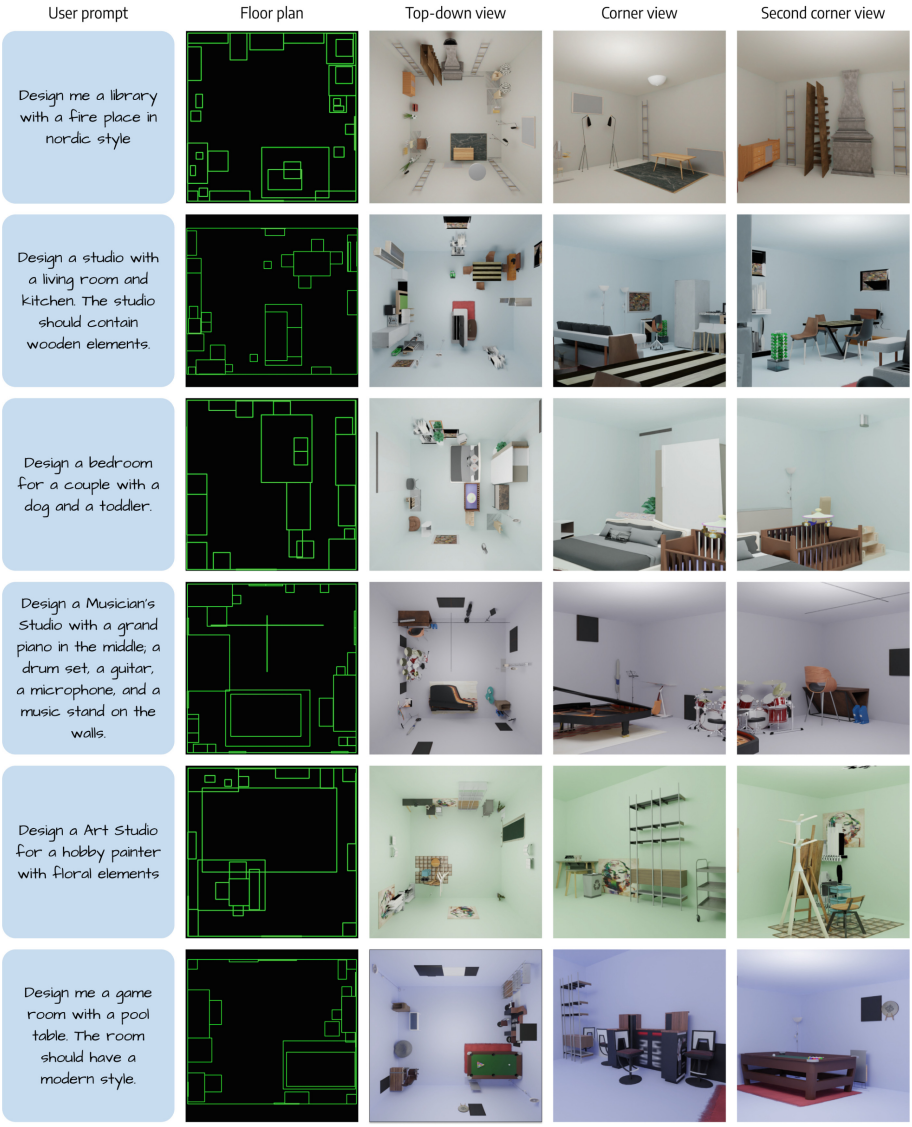

Fig. 4. Gallery of Results Obtained with I-Design. The first column lists user prompts, specifying design preferences, such as functionality, style, and atmosphere. The top-down scene graph layout generated by the LLM Agents is shown in the second column. The third column shows this layout rendered from the top-down view after the Asset Retrieval stage. The last two columns show corner views of the resulting design. Evidently, I-Design is capable of factoring in diverse user specifications and producing vibrant, functional designs.

Table 1. Quantitative comparison with LayoutGPT [13]. Both methods generate 10 "Bedroom" and 10 "Living room" scenes of varying sizes. Objective metrics include average number of objects (NObj), out-of-bounds rate (OOB), and bounding box loss (BBL). Subjective metrics involve using GPT-4V [1] to evaluate functionality (Func.), layout (Layout.), color scheme (Scheme.), and overall aesthetic (Atmos.). The evaluator assigns a grade from 0 to 10 and provides reasoning (Sect. 4.2). Each scene is evaluated 3 times, with mean and standard deviation calculated and averaged across scenes. I-Design produces more realistic scenes, scoring better in all metrics with tight confidence.

Method	Room Type	NObj ↑	OOB ↓	BBL ↓	GPT-4V Criteria ↑				
					Func. ↑	Layout. ↑	Scheme. ↑	Atmos. ↑	Avg. ↑
LayoutGPT [13]	Bedroom	5.5	51.06	14.09	4.8 ± 0.4	4.8 ± 0.8	4.6 ± 0.8	4.9 ± 0.4	4.8
	LivingRoom	6.9	64.15	1.06	4.8 ± 1.3	4.8 ± 0.8	4.8 ± 1.4	4.6 ± 0.8	4.8
	Avg.	6.2	57.6	7.58	4.8	4.8	4.7	4.8	4.8
Ours (I-Design)	Bedroom	12.7	0.0	0.34	5.2 ± 0.4	5.5 ± 0.2	5.6 ± 0.7	5.5 ± 0.2	5.5
	LivingRoom	23.6	0.0	0.31	5.8 ± 2.6	5.6 ± 2.1	5.9 ± 1.1	5.7 ± 1.3	5.8
	Avg.	**18.2**	**0.0**	**0.33**	**5.5**	**5.6**	**5.8**	**5.6**	**5.7**

Table 2. Quantitative comparison with Holodeck [59] Both methods generate 10 different room types with personalized design choices (Functionality Prompts; see Supplementary materials for individual prompts). We compare grades from two setups of our method: (i) using assets from the AI2Thor [23] dataset (same as Holodeck) and (ii) using assets from Objaverse and 3D-FUTURE [16] based on zero-shot category classification. Our method outperforms Holodeck in almost all metrics, while the setup with Objaverse and 3D-FUTURE exceeds Holodeck in all GPT-4V ratings.

Method	NObj ↑	OOB ↓	BBL ↓	GPT-4V Criteria ↑				
				Func. ↑	Layout. ↑	Scheme. ↑	Atmos. ↑	Avg. ↑
Holodeck [59]	16.5	**0.0**	**0.0**	**7.0** ± 0.5	6.5 ± 0.6	6.0 ± 0.5	6.2 ± 0.6	6.4
Ours (AI2THOR)	**22.9**	**0.0**	0.06	6.7 ± 0.8	**6.6** ± 0.5	**6.4** ± 0.6	**6.3** ± 0.6	**6.5**
Ours (Objaverse + 3D-FUTURE)				**7.8** ± 0.3	**7.6** ± 0.5	**7.2** ± 0.4	**7.3** ± 0.3	**7.5**

3D-FUTURE, it results in a significant improvement on the baseline scores. A visual comparison between these methods is provided in the suppl. material.

4.3 Qualitative Study

Figure 4 showcases rooms generated with I-Design given various prompts. Each synthesized room adeptly reflects user specifications. When users explicitly mention furniture preferences and desired positions, as showcased in the fourth example, the resulting room seamlessly aligns with those requirements. Similarly, when specifications are implicit, as seen in the third example depicting a bedroom tailored for a family with a toddler, the synthesized room's functionality satisfies user preferences, with a crib placed next to the king-sized bed.

5 Conclusion

Computing-aided technologies have been widely used to simplify interior design for non-experts. However, current methods for generating interiors often suffer from inconsistencies between views and limitations on number of objects or room types, restricted by the dataset's domain. Our goal is to eliminate these constraints and provide a user-friendly interface for streamlining interior design.

We present I-Design, which leverages an LLM multi-agent architecture to interpret user preferences expressed in text, transforming unstructured input into a structured representation. The framework infers objects that align with the user's requirements and establishes spatial relationships, integrating common sense knowledge. The resulting scene graph visually represents the proposed design. Crucially, I-Design ensures interpretability with a transparent breakdown of each step, making the design process accessible.

However, our framework encounters limitations, particularly termination issues when placing a large number of objects in a small scene. Spatial conflicts may persist despite postprocessing, or there may be insufficient space for placement. Asset retrieval also poses challenges due to varying quality and invalid default orientations in the database. Additionally, discrepancies in object sizes compared to proposed dimensions can lead to artifacts after resizing. A detailed analysis of limitations, as well as insights on future work, can be found in the Discussion section of the suppl. material.

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NimbleD: Enhancing Self-supervised Monocular Depth Estimation with Pseudo-labels and Large-Scale Video Pre-training

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Abstract. We introduce NimbleD, an efficient self-supervised monocular depth estimation learning framework that incorporates supervision from pseudo-labels generated by a large vision model. This framework does not require camera intrinsics, enabling large-scale pre-training on publicly available videos. Our straightforward yet effective learning strategy significantly enhances the performance of fast and lightweight models without introducing any overhead, allowing them to achieve performance comparable to state-of-the-art self-supervised monocular depth estimation models. This advancement is particularly beneficial for virtual and augmented reality applications requiring low latency inference. The source code, model weights, and attributions are available at https://github.com/xapaxca/nimbled.

Keywords: Monocular Depth Estimation · Self-supervised Learning · Knowledge Distillation

1 Introduction

Monocular depth estimation (MDE) is the task of predicting the distances to objects relative to the camera from a single image input. Low latency depth estimation is crucial for metaverse applications as it ensures real-time, accurate spatial awareness and immersive interactions with virtual and real-world objects, enhancing user experience and comfort. Recent advancements in vision transformers (ViT) [6], large vision models [25], and generative models [28] have significantly advanced MDE [1,2,14,18,24,30,35,40]. The two main training paradigms commonly used for MDE are supervised and self-supervised learning (SSL). Supervised MDE relies on ground truth depth labels obtained from LiDAR or RGB-D cameras, whereas SSL MDE leverages geometric constraints from monocular videos or stereo setups. On one hand, SSL MDE models [44] trained exclusively on monocular videos represent the most accessible approach since they require only monocular videos for training, without the need for a

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stereo setup and ground truth depth, potentially allowing for training on large-scale unlabeled data [34]. On the other hand, the recent supervised models employing large-scale training [40] or generative methods [18] have demonstrated superior zero-shot depth estimation capabilities. However, these models are often relatively slow at inference, limiting their use in metaverse applications where low latency is crucial. Another challenge is the feasibility of training these models for companies or individuals with limited resources. In this work, we address these challenges by introducing NimbleD, a straightforward yet effective SSL MDE learning framework enhanced by pseudo-labels generated by a large vision model and large-scale video pre-training. This framework noticeably boosts the depth estimation quality of fast and lightweight models without introducing any overhead, enabling them to achieve the performance of larger state-of-the-art SSL MDE methods.

Our contributions are summarized as follows:

- Develop an effective training strategy that utilizes both self-supervision from monocular videos and pseudo-supervision from disparities generated by a large vision model.
- Introduce a concise loss function that combines SSL and PSL losses.
- Incorporate large-scale video pre-training from publicly available videos without any complex data preparation.
- Boost the depth estimation performance of fast and lightweight models to a level comparable to state-of-the-art SSL MDE methods.

2 Related Work

Recent advancements in zero-shot MDE have yielded significant progress. MiDaS [2,27] has effectively addressed training challenges on diverse datasets by implementing a scale- and shift-invariant loss, ensuring compatibility with various ground truth forms. Similarly, ZoeDepth's [1] innovative two-stage framework integrates relative and metric depth estimation, utilizing large-scale pre-training alongside domain-specific fine-tuning, augmented by a novel bin adjustment mechanism for scale-aware predictions. ZeroDepth's [14] introduction of input-level geometric embeddings and a variational latent representation has facilitated scale-aware depth estimation across diverse domains, leveraging extensive labeled and synthetic datasets. UniDepth [24] introduces a module that learns camera parameters and uses a pseudo-spherical representation to separate the camera and depth components, demonstrating superior metric depth estimation. KBR's [34] self-supervised learning approach capitalizes on a vast dataset derived from YouTube, covering a wide array of environments without necessitating camera intrinsics. Marigold [18] employs diffusion-based image generators alongside a fine-tuned Stable Diffusion [28] model with synthetic data, whereas DMD's [30] diffusion-based method incorporates field-of-view augmentation and log-scale depth representation, further refined through fine-tuning on a varied dataset mix. DepthAnything [40] stands out by proposing a large-scale training

regimen for DPT [26], utilizing an extensive dataset coupled with knowledge distillation, semantic supervision, and CutMix [41] data augmentation, showcasing a wide-ranging and innovative approach within the field.

Self-supervised MDE trained exclusively on monocular videos was first introduced by Zhou *et al*. [44]. Monodepth2 [13] further advanced the field by addressing challenges associated with occluded objects and objects moving at the same speed as the camera. Following these foundational works, significant progress has been made. This includes recent works focusing on depth network architecture enhancements [15,38] and incorporation of additional supervisory signals [23,43].

There are studies aimed at developing lightweight MDE models that employ mobile architectures. LiteMono [42] utilizes a custom hybrid CNN-Transformer encoder, while SwiftDepth [22] uses SwiftFormer [31] as an encoder accompanied by a custom two-stage decoder. Both models exhibit competitive depth estimation quality with fewer parameters and faster inference speeds, making them ideal for augmented and virtual reality applications.

Various works explore knowledge distillation techniques. Wang *et al*. [37] propose a method that distills knowledge from the teacher model at both the feature and output levels, utilizing silog [8], gradient matching [27], and pairwise distillation [19] losses. TIE-KD [3] focuses on output-level knowledge distillation, employing Kullback-Leibler divergence and SSIM [39] for the loss function. MViTDepth [9] combines self-supervised learning with knowledge distillation, using the logarithm of the L1 difference at the output level. MAL [5] is a multi-frame MDE method that leverages additional supervision from a instance segmentation model to identify moving objects and performs knowledge distillation at the output level using the L1 difference.

In this work, we address the progress in zero-shot MDE by leveraging a modern large vision model to generate pseudo-labels, introducing a straightforward yet efficient loss function, and applying large-scale video pre-training. Our approach ensures that lightweight self-supervised MDE methods match the depth estimation quality of larger state-of-the-art models, making them suitable for real-time applications in the metaverse.

3 Method

In this section, we introduce NimbleD learning framework. First, the problem statement of self-supervised MDE enhanced by pseudo-labels is introduced. Next, the combined SLL and PSL loss function is detailed. Finally, the large-scale video pre-training without camera intrinsics is described.

3.1 Self-supervised Monocular Depth Estimation Enhanced by Pseudo-labels

Self-supervised MDE is generally formulated as an image reconstruction problem. Depth estimation serves as a building block of the entire training framework, which is then used independently during inference. The framework consists of

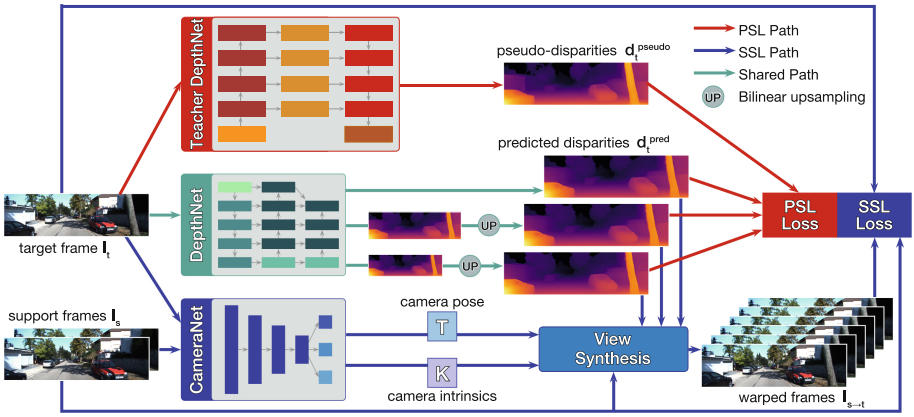

Fig. 1. Overview of the proposed NimbleD learning framework. The framework comprises a depth network that outputs multi-scale disparity predictions (d_t^{pred}), a camera network that outputs the relative camera pose (T) and camera intrinsic parameters (K), and a teacher depth network that generates pseudo-disparities (d_t^{pseudo}). The ultimate objective of the method is to minimize the loss between the predicted and pseudo-disparities (*PSL Loss*) and to minimize the image reconstruction loss (*SSL Loss*).

a depth network that predicts disparities d_t from a target frame I_t and a pose network that predicts the relative camera pose T between the target frame I_t and support frames I_s, where $s \in \{t-1, t+1\}$ indicates the frames temporally adjacent to the target frame. Given depth map D_t, which is the inverse of disparities ($1/d_t$), camera pose T, and camera intrinsics K that must usually be known beforehand, the target frame is reconstructed from support frames I_s as follows:

$$I_{s\to t} = I_s\langle \text{proj}(D_t, T, K)\rangle, \tag{1}$$

where $I_{s\to t}$ represents the reconstructed target image, proj() denotes the projection of depth onto the support frames I_s coordinates, and $\langle\rangle$ signifies a differentiable sampling operator [17].

Following [34], the pose network is assigned to learn camera intrinsics K. The only modification introduced to the network involves adding two additional output heads to the decoder: the first outputs the focal lengths, and the second outputs the principal point coordinates, which together construct the intrinsics matrix K. Since it predicts all camera parameters (both extrinsic and intrinsic), we refer to it as the camera network.

Supervised MDE is generally formulated as the minimization of error between predicted and ground truth depth maps. Considering situations where ground truth information is unavailable, we employ a large vision model to generate pseudo-labels and minimize the error between predictions and these pseudo-labels. This pseudo-supervisory signal enhances our self-supervised image reconstruction problem and can be seen as offline knowledge distillation at the output level.

The proposed NimbleD learning framework is depicted in Fig. 1.

3.2 Combined Self-superivsed and Pseudo-supervised Loss Function

We propose a simple yet effective combination of losses widely used in self-supervised and supervised MDE. Our total loss is a weighted average of SSL and PSL losses, averaged across the upsampled multi-scale outputs of the depth network. It is defined as:

$$L_{\text{total}} = \frac{1}{n}\sum_{i=1}^{n}(\lambda L^i_{\text{SSL}} + (1-\lambda)L^i_{\text{PSL}}), \tag{2}$$

where λ is the loss weighting parameter, and n denotes the number of output scales.

The main building block of the SSL loss is photometric error, defined as:

$$\text{pe}(I_t, I_{s\to t}) = \frac{\alpha}{2}\,(1 - \text{SSIM}(I_t, I_{s\to t})) + (1-\alpha)||I_t - I_{s\to t}||_1, \tag{3}$$

where SSIM denotes the Structural Similarity Index Measure [39], and α is set to 0.85.

The SSL loss is the per-pixel minimum reprojection loss with auto-masking [13], defined as:

$$L_{\text{SSL}} = \text{mean}\left(\mu \min_{s\in\{t-1,t+1\}} \text{pe}(I_t, I_{s\to t})\right), \tag{4}$$

where μ represents the auto-masking function, given by:

$$\mu = [\min_{s\in\{t-1,t+1\}} \text{pe}(I_t, I_{s\to t}) < \min_{s\in\{t-1,t+1\}} \text{pe}(I_t, I_s)], \tag{5}$$

where $[\cdot]$ is the indicator function, which equals 1 if the enclosed condition is true and 0 otherwise.

The PSL loss is defined as a scale-and-shift invariant loss [27], calculated in the disparity space:

$$L_{\text{PSL}} = \text{mean}\left(\left|\hat{d}_{\text{pred}} - \hat{d}_{\text{pseudo}}\right|\right), \tag{6}$$

where $\hat{d}_{\text{pred}}$ and $\hat{d}_{\text{pseudo}}$ are the scaled and shifted predicted and pseudo-disparities, respectively. These are defined as follows:

$$\hat{d} = \frac{d - \text{median}(d)}{\text{mean}\left(|d - \text{median}(d)|\right)}. \tag{7}$$

It is important to note that widely used smoothness regularization loss [12] was not employed due to its negligible impact on the results [33], see ablation studies. This decision is made to save training time during the large-scale video pre-training.

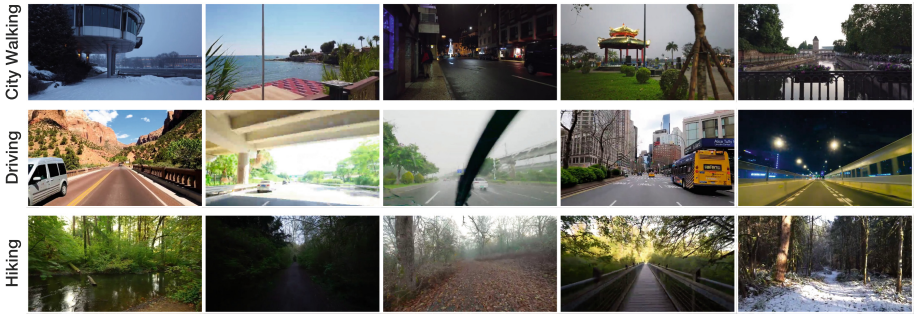

Fig. 2. The dataset for large-scale video pre-training is curated from publicly available videos. It consists of three equally represented classes - city walking, driving, and hiking - offering a diverse range of outdoor environments.

Table 1. Summary of the dataset for large-scale video pre-training. The dataset includes three balanced classes, each characterized by a diverse set of filming conditions, totaling more than 1.1 million frame triplets.

Class	#Videos	Duration (h)	#Frames	#Triplets	
				Large set	Small set
City walking	35	23.1	383,199	363,973	19,156
Driving	35	23.1	400,617	380,518	20,029
Hiking	35	23.1	392,446	372,758	19,618
Total	105	69.7	1,176,262	1,117,249	58,803

3.3 Large-Scale Video Pre-training

NimbleD's camera network, which predicts camera intrinsics, enables the use of any video with moving cameras and objects. Therefore, we select YouTube videos with a CC-BY license for large-scale pre-training.

We categorize the selected videos into three classes to ensure a diverse range of outdoor environments: driving, hiking, and city walking. The driving class comprises videos from the USA and China, featuring a variety of driving conditions. The hiking videos, filmed in the woods of national parks across the USA, provide natural, uneven terrain. The city walking videos, covering urban areas in Europe and Asia, offer a mix of architectural styles and pedestrian scenarios. Each class includes 35 videos, with each totaling approximately 23 h, resulting in a 70-hour dataset. Unlike SlowTV [34], our dataset ensures a more balanced composition by maintaining a similar number of frames in each class, as summarized in Table 1.

The videos are downloaded at a resolution of 480×854 to manage space efficiently, and frames are extracted at a rate of 5 fps to balance motion detail with dataset size. The dataset encompasses a variety of lighting conditions, weather scenarios, and filming challenges. For example, some driving videos are filmed

with a camera installed inside a vehicle, capturing reflections and raindrops on the windshield, as well as the movement of the wipers. City walking videos may display abrupt camera movements, resulting in reduced overlap between frames. Hiking footage often suffers from motion blur and defocusing issues.

Despite these challenges, we decide against pre-processing the videos or the extracted frames, aiming to present the data in its most authentic form, embracing the imperfections and variability inherent in real-world footage. This approach ensures that the dataset reflects genuine outdoor conditions, making it a valuable resource for developing and testing self-supervised MDE models. Some samples from the dataset are presented in Fig. 2.

Our dataset ultimately comprises 1.1M overlapping triplets. Additionally, we create a smaller set of 59K triplets for debugging.

4 Experiments

In this section, we first briefly introduce the datasets used and the evaluation metrics. Then, we describe the training implementation details, followed by the evaluation results. Finally, an extensive ablation study is performed to explore various components of our framework.

4.1 Datasets

In addition to the aforementioned selected publicly available videos, we utilize KITTI [11], NYUv2 [32], and Make3D [29].

For training on KITTI, we use Eigen-Zhou split [44], which consists of 39,810 automotive outdoor frame triplets. For evaluation on KITTI, we employ Eigen split [7], which includes 697 frames with LiDAR-based ground truth depth, adhering to the standard evaluation protocol. This protocol includes Garg crop [10], 80-meter distance cap, and input resolution of 192×640. Additionally, for comparison with the teacher model, we use Eigen-Benchmark split [36], which comprises 652 frames with improved LiDAR-based ground truth depth, and in this context, we forego cropping.

For zero-shot generalization, we turn to NYUv2 and Make3D, which are not seen by the teacher depth network. NYUv2 encompasses 654 indoor frames with ground truth depth sourced from an RGB-D camera. This evaluation leverages Eigen crop [8], 10-meter distance cap, and input resolution of 288×384. Make3D consists of 134 outdoor frames with laser-based ground truth depth. We use 70-meter distance cap, center crop, and input resolution of 192×640.

Across these evaluation scenarios, we consistently employ a set of standard evaluation metrics [8]. For NYUv2 and Make3D, we use AbsRel, SqRel, RMS, and $\text{RMS}_{\log}$. For evaluation on KITTI, we additionally use $\delta_1 < 1.25$, $\delta_2 < 1.25^2$, and $\delta_3 < 1.25^3$.

Table 2. Training details. Training consists of two stages: pre-training on publicly available YouTube videos and fine-tuning on KITTI [11]. *LR* stands for learning rate.

	Pre-training	Fine-tuning
Dataset	YouTube videos	KITTI [11]
Color Jitter	✓	✓
Horizontal flip	✓	✓
Aspect augmentation	✓	
Batch size	16	16
#Epochs	6	20
Training time (h)	32	5
Optimizer	AdamW [21]	AdamW [21]
Weight decay	10^{-3}	10^{-3}
Initial LR	10^{-4}	10^{-4}
LR scheduler	Cosine [20]	Step
Adjusted LR	10^{-5}	10^{-5}
Adjustment epoch	4	10
Loss weighting λ	0.9	0.9 (0.95 for last 5 ep.)

4.2 Implementation Details

For the teacher depth network, we utilize DepthAnything [40], chosen for its high-quality results. Specifically, its ability to generate a large volume of pseudo-labels with limited computational resources stands out, contrasting with diffusion-based models [18]. Notably, DepthAnything is not trained on the KITTI [11], NYUv2 [32], or Make3D [29] datasets. We observe its optimal performance at a resolution of 518×518, rather than at the original input image size. Therefore, we first resize the original images to 518×518 and then interpolate the predictions back to the original image size. To maintain the integrity of the pseudo-labels, we store the disparities without normalization and also preserve the maximum value to facilitate information retrieval.

For the depth network, we utilize several baseline methods: Monodepth2 [13], SwiftDepth [22], and LiteMono [42], each with different configurations. Monodepth2 is the standard baseline model for SSL MDE research, while SwiftDepth and LiteMono are fast and lightweight models. The encoders of all depth networks are pre-trained on ImageNet [4]. To ensure that the predicted disparities are consistent with the teacher model, we replace the output activation function from sigmoid to ReLU, thereby eliminating the need for output rescaling to a specific range.

For the camera network, we employ the widely-used PoseNet [13,44], based on an ImageNet [4] pre-trained ResNet-18 [16] backbone, to ensure compatibility with established methods. We modify the decoder to also output camera intrinsics, in line with KBR [34].

For training, we apply standard data augmentations [13] to the frames fed into the depth network. Additionally, for large-scale video pre-training, we employ aspect ratio augmentations similar to those used in KBR [34], involving center cropping at randomly selected aspect ratios followed by interpolation to sizes that approximate the total pixel counts of a 192×640 input size.

Training is conducted on a single NVIDIA GeForce RTX 4090 GPU. All training details are summarized in Table 2.

Table 3. Results on KITTI [11] Eigen split [7] with median alignment. The input image size is 192×640. *Par* indicates the total number of parameters in millions. The highest scores are highlighted in **bold**, while the second-highest scores underlined.

Method	Venue	Lower is better					Higher is better		
		Par	AbsRel	SqRel	RMS	RMS_{log}	δ_1	δ_2	δ_3
DaCCN [15]	ICCV'23	13.0	0.099	0.661	4.316	0.173	0.897	0.967	0.985
GasMono [43]	ICCV'23	27.9	0.098	–	4.303	0.173	0.903	0.968	0.984
FGTO [23]	CVPR'24	27.9	0.096	0.696	4.327	0.174	0.904	0.968	0.985
SQLdepth [38]	AAAI'24	31.1	**0.091**	0.713	4.204	0.169	**0.914**	0.968	0.984
MD2-R18 [13]	ICCV'19	14.8	0.115	0.903	4.863	0.193	0.877	0.959	0.981
+*NimbleD (ours)*	–	14.8	0.100	0.739	4.440	0.175	0.898	0.967	0.985
MD2-R50 [13]	ICCV'19	34.6	0.110	0.831	4.642	0.187	0.883	0.962	0.982
+*NimbleD (ours)*	–	34.6	0.097	0.721	4.377	0.172	0.904	0.968	0.985
SwiftDepth-S [22]	ISMAR'23	3.6	0.110	0.830	4.700	0.187	0.882	0.962	0.982
+*NimbleD (ours)*	–	3.6	0.098	0.733	4.401	0.174	0.901	0.968	0.985
SwiftDepth [22]	ISMAR'23	6.4	0.107	0.790	4.643	0.182	0.888	0.963	0.983
+*NimbleD (ours)*	–	6.4	0.096	0.697	4.333	0.171	0.905	0.969	**0.986**
LiteMono-S [42]	CVPR'23	**2.5**	0.110	0.802	4.671	0.186	0.879	0.961	0.982
+*NimbleD (ours)*	–	**2.5**	0.099	0.709	4.370	0.172	0.898	0.967	**0.986**
LiteMono [42]	CVPR'23	3.1	0.107	0.765	4.561	0.183	0.886	0.963	0.983
+*NimbleD (ours)*	–	3.1	0.096	0.684	4.304	0.171	0.903	0.969	**0.986**
LiteMono-8M [42]	CVPR'23	8.8	0.101	0.729	4.454	0.178	0.897	0.965	0.983
+*NimbleD (ours)*	–	8.8	0.092	**0.646**	**4.194**	**0.165**	0.910	**0.970**	**0.986**

4.3 Results

Evaluation on KITTI. We evaluate our method on KITTI [11] using Eigen split [7] with median alignment, comparing it with baseline models and leading state-of-the-art single-frame SSL MDE methods, as shown in Table 3. First, it is clear that NimbleD significantly enhances the performance of all the baseline models. The method revitalizes Monodepth2-R18 [13], while SwiftDepth [22] and LiteMono [42] perform on par with FGTO [23]. Meanwhile, LiteMono-S and

Table 4. Results on KITTI [11] KITTI Eigen-Benchmark [36] splits with improved ground truth against the teacher model using least-squares alignment. The input image size is 350 × 350 and 518 × 518 for DepthAnything [40] and 192 × 640 for methods enhanced by NimbleD. *Par* indicates the total number of parameters in millions. The highest scores are highlighted in **bold**, while the second-highest scores underlined.

Method	Lower is better					Higher is better		
	Par	AbsRel	SqRel	RMS	RMS_{log}	δ_1	δ_2	δ_3
DepthAnything [40] 350×350	335.3	0.071	0.432	3.504	0.122	0.947	0.989	0.997
DepthAnything [40] 518×518	335.3	**0.064**	**0.363**	**3.239**	0.111	**0.957**	**0.991**	0.997
MD2-R18 [13] +*NimbleD*	14.8	0.073	0.448	3.610	0.122	0.941	0.988	0.996
MD2-R50 [13] +*NimbleD*	34.6	0.070	0.438	3.571	0.134	0.945	0.989	0.997
SwiftDepth-S [22] +*NimbleD*	3.6	0.072	0.441	3.585	0.120	0.943	0.989	0.997
SwiftDepth [22] +*NimbleD*	6.4	0.069	0.408	3.469	0.117	0.946	0.990	0.997
LiteMono-S [42] +*NimbleD*	**2.5**	0.072	0.437	3.584	0.124	0.940	0.988	0.997
LiteMono [42] +*NimbleD*	3.1	0.070	0.427	3.537	0.126	0.944	0.989	0.997
LiteMono-8M [42] +*NimbleD*	8.8	0.065	0.374	3.338	**0.110**	0.951	**0.991**	**0.998**

SwiftDepth-S, which offer the best speed and accuracy trade-off on the market, are not far behind DaCCN [15] and GasMono [43]. Moreover, LiteMono-8M demonstrates state-of-the-art performance similar to SQLdepth [38].

We also compared our method to the teacher model, DepthAnything [40], on KITTI [11] Eigen-Benchmark [36] split with improved ground truth using least squares alignment, which is the standard for evaluating supervised MDE methods, as shown in Table 4. Since the teacher model uses a higher input image size, we also included its results with a 350×350 input size for a fair comparison, which has a pixel count similar to input size of 192×640. Notably, almost all of our models performed similarly to DepthAnything with an input resolution of 350×350, while LiteMono-8M achieved results close to the teacher model's resolution of 518×518. This demonstrates that NimbleD performs exceptionally well in offline knowledge distillation, considering the number of parameters is reduced by 38 times and the model does not use ground truth data.

The qualitative results compared with DepthAnything, Monodepth2-R50, SwiftDepth, and LiteMono-8M are illustrated in Fig. 3.

Generalization. We evaluate the zero-shot generalization of our method on NYUv2 [32] and Make3D [29] by comparing it to baseline models [22,42]. The results are shown in Table 5. NimbleD significantly enhances the models' generalization ability on NYUv2 (indoor), while the improvement is less notable on Make3D (outdoor). Overall, despite major metric improvements, the results remain unsurprisingly subpar since the models are fine-tuned on KITTI [11]. The qualitative comparison with baseline models on NYUv2 is illustrated in Fig. 4, while the comparison on Make3D is depicted in Fig. 5.

Table 5. Zero-shot results on NYUv2 [32] and Make3D [29] against baseline models with median alignment. The input image size is 288×384 for NYUv2 and 192×640 for Make3D. Lower values are better for all columns.

Method	NYUv2 (indoor) [32]				Make3D (outdoor) [29]			
	AbsRel	SqRel	RMS	$\text{RMS}_{\log}$	AbsRel	SqRel	RMS	$\text{RMS}_{\log}$
MD2-R18	0.362	0.665	1.266	0.390	0.322	3.587	7.417	0.163
+*NimbleD*	0.237	0.279	0.802	0.281	0.287	3.782	7.179	0.146
MD2-R50	0.326	0.556	1.139	0.358	0.322	3.611	7.326	0.160
+*NimbleD*	0.236	0.247	0.764	0.278	0.284	3.477	7.053	0.145
SwiftDepth-S	0.317	0.457	1.070	0.357	0.330	3.909	7.547	0.163
+*NimbleD*	0.223	0.240	0.759	0.267	0.293	3.834	7.197	0.146
SwiftDepth	0.277	0.386	0.969	0.314	0.312	**3.292**	7.125	0.158
+*NimbleD*	**0.204**	**0.200**	**0.689**	**0.244**	0.282	3.708	7.100	0.143
LiteMono-S	0.328	0.537	1.161	0.362	0.332	3.372	7.248	0.167
+*NimbleD*	0.236	0.259	0.782	0.287	0.300	4.074	7.382	0.149
LiteMono	0.317	0.488	1.095	0.350	0.317	3.410	7.253	0.161
+*NimbleD*	0.243	0.310	0.850	0.295	0.278	3.411	6.979	0.143
LiteMono-8M	0.298	0.415	1.026	0.344	0.318	3.451	7.236	0.159
+*NimbleD*	**0.204**	0.206	0.699	0.253	**0.271**	3.408	**6.952**	**0.140**

Table 6. Ablation study of loss components, camera intrinsics, and pre-training on the KITTI [11] Eigen split [7] with median alignment. Input image size: 192×640. *Base*: baseline method which is SwiftDepth [22]. *SSL*: self-supervised learning. *PSL*: pseudo-supervised learning. *ReLU*: output activation function is ReLU instead of sigmoid. *multi*: multi-scale outputs. *photo*: photometric error. *min*: minimum reconstruction. *auto*: automasking. *sm*: smoothness loss. *K*: camera intrinsics required. *LSVP*: large-scale video pre-training. * marks increasing loss weighting λ from 0.9 to 0.95 for the last 5 fine-tuning epochs. The highest scores are highlighted in **bold**, and the second-highest scores are underlined.

Loss	ReLU	multi	photo	min	auto	sm	90K	LSVP	Lower is better				Higher is better		
									AbsRel	SqRel	RMS	$\text{RMS}_{\log}$	δ_1	δ_2	δ_3
Base [22]		✓	✓	✓	✓	✓	✓		0.107	0.790	4.643	0.182	0.888	0.963	0.983
SSL	✓	✓	✓	✓	✓	✓	✓		0.108	0.814	4.624	0.185	0.886	0.963	0.982
SSL	✓	✓	✓	✓	✓		✓		0.107	0.808	4.616	0.184	0.888	0.963	0.982
SSL	✓	✓	✓	✓	✓				0.110	0.878	4.742	0.187	0.887	0.962	0.981
PSL	✓								0.108	0.762	4.735	0.180	0.877	0.963	**0.986**
PSL	✓	✓							0.108	0.764	4.708	0.179	0.878	0.964	**0.986**
SSL+PSL	✓	✓	✓	✓	✓		✓		0.101	0.718	4.415	0.175	0.895	0.967	**0.986**
SSL+PSL	✓	✓	✓	✓	✓				0.101	0.711	4.435	0.174	0.894	0.967	**0.986**
SSL+PSL	✓	✓	✓	✓					0.104	0.770	4.578	0.176	0.890	0.967	0.985
SSL+PSL	✓	✓	✓						0.103	0.731	4.481	0.175	0.890	0.966	0.985
SSL+PSL	✓	✓	✓	✓	✓			✓	0.097	0.711	4.354	**0.171**	0.903	**0.969**	**0.986**
SSL+PSL*	✓	✓	✓	✓	✓			✓	**0.096**	**0.697**	**4.333**	**0.171**	**0.905**	**0.969**	**0.986**

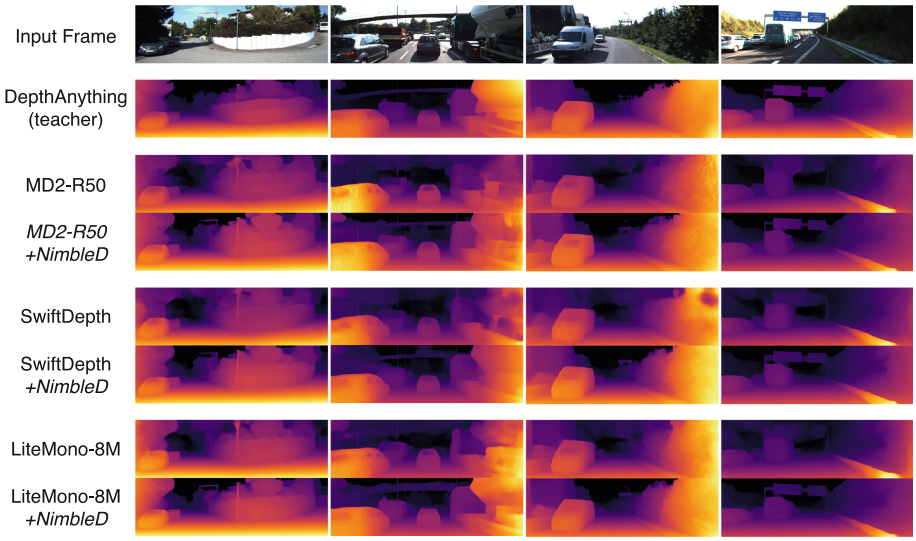

Fig. 3. Qualitative results on KITTI [11] Eigen split [7], compared with the teacher model [40] and baseline models [13,22,42]. NimbleD observably enhances the depth estimation quality of all baseline models. It identifies distant objects not detected by the teacher model and demonstrates improved handling of sky regions compared to the baseline models.

4.4 Ablation Study

We conduct extensive experiments on the components of the loss function, the influence of training without known camera intrinsics, and the impact of large-scale pre-training. SwiftDepth [22] is used as a baseline model. The evaluation results on KITTI [11] Eigen split [7] are presented in Table 6. Initially, we replace the output activation function from sigmoid to ReLU in our baseline model, resulting in a minor deterioration of the results. Subsequently, we eliminate the widely-used smoothness regularization loss [13], which does not affect the outcomes. Training without camera intrinsics notably worsens the SSL results, and we note the model's frequent convergence issues during training. We also evaluate the PSL loss calculation with both single-scale and multi-scale disparity predictions, finding no significant difference between them. To maintain consistency with the SSL loss, we opt for the multi-scale approach. Further experimentation on the combined SSL and PSL loss with known camera intrinsics reveals a significant improvement in results, indicating a synergistic effect of the combination. Interestingly, the results remain unchanged when training without camera intrinsics, highlighting the supportive role of PSL for not only improving metrics but also for making training robust in contrast to KBR [34]. We then explore the necessity of minimum reconstruction and automasking [13]; however, omitting them leads to a deterioration in performance. Lastly, we apply large-scale pre-training to our model using the selected combined loss on YouTube videos,

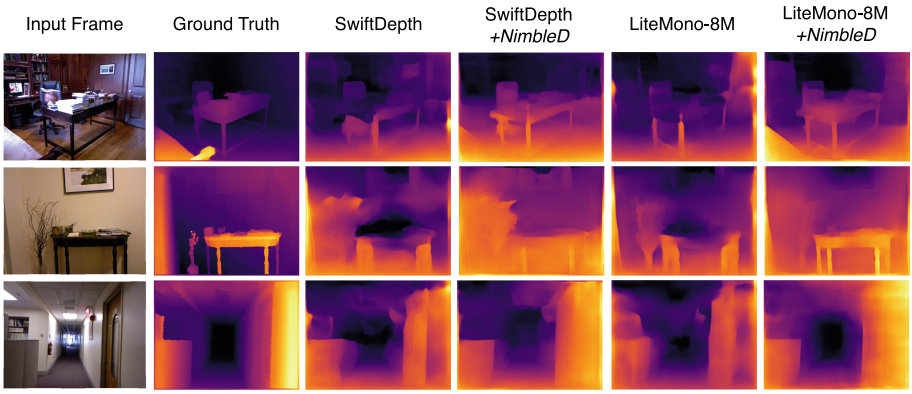

Fig. 4. Zero-shot qualitative results on NYUv2 [32], compared to the baseline models [22,42]. NimbleD noticeably improves the generalization ability of both baseline models.

Fig. 5. Zero-shot qualitative results on Make3D [29], compared to the baseline models [22,42]. NimbleD noticeably improves the generalization ability of both baseline models.

followed by fine-tuning on KITTI, which significantly enhances the evaluation metrics. Combined SSL and PSL experiments use a loss weighting parameter λ set to 0.9, determined by extensive manual search as shown in Table 7. Interestingly, increasing λ to 0.95 for the last 5 fine-tuning epochs slightly improves and stabilizes results.

Table 7. Results on KITTI [11] Eigen split [7] with various values for the loss weighting parameter λ. The table presents averaged results from the last 5 epochs trained solely on KITTI for 20 epochs. The input image size is 192×640. The highest scores are highlighted in **bold**, and the second-highest scores are underlined.

λ	Lower is better				Higher is better		
	AbsRel	SqRel	RMS	$\text{RMS}_{\log}$	δ_1	δ_2	δ_3
0.1	0.107	0.722	4.656	0.179	0.879	0.964	**0.986**
0.3	0.106	0.733	4.651	0.179	0.882	0.964	**0.986**
0.5	0.105	0.719	4.627	0.177	0.882	0.964	**0.986**
0.7	0.104	0.713	4.531	0.176	0.887	0.966	**0.986**
0.8	0.102	0.703	4.458	**0.175**	0.891	**0.967**	**0.986**
0.85	0.102	**0.702**	4.438	0.176	0.892	**0.967**	**0.986**
0.9	**0.101**	0.720	**4.437**	**0.175**	**0.895**	**0.967**	**0.986**
0.95	0.103	0.761	4.518	0.178	**0.895**	0.966	0.985
0.99	0.113	1.013	4.906	0.216	0.883	0.960	0.980

5 Conclusion

In this work, we introduce NimbleD, an efficient MDE learning framework that enhances SSL using pseudo-labels generated by a large vision model. This approach allows for robust training without camera intrinsics and maintains quality, enabling us to leverage large-scale video pre-training. Consequently, NimbleD elevates the depth estimation performance of fast and lightweight models to match state-of-the-art SSL MDE methods without introducing any overhead, which is advantageous for real-time metaverse applications. While some may argue that using pseudo-supervision from larger models is inequitable, we believe that large vision models are a crucial part of the current deep learning landscape and should be fully utilized.

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GeoTransfer: Generalizable Few-Shot Multi-view Reconstruction via Transfer Learning

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Abstract. This paper presents a novel approach for sparse 3D reconstruction by leveraging the expressive power of Neural Radiance Fields (NeRFs) [35] and fast transfer of their features to learn accurate occupancy fields. Existing 3D reconstruction methods from sparse inputs still struggle with capturing intricate geometric details and can suffer from limitations in handling occluded regions. On the other hand, NeRFs [35] excel in modeling complex scenes but do not offer means to extract meaningful geometry. Our proposed method offers the best of both worlds by transferring the information encoded in NeRF [35] features to derive an accurate occupancy field representation. We utilize a pre-trained, generalizable state-of-the-art NeRF network [18] to capture detailed scene radiance information, and rapidly transfer this knowledge to train a generalizable implicit occupancy network. This process helps in leveraging the knowledge of the scene geometry encoded in the generalizable NeRF [18] prior and refining it to learn occupancy fields, facilitating a more precise generalizable representation of 3D space. The transfer learning approach leads to a dramatic reduction in training time, by orders of magnitude (*i.e.* from several days to 3.5 hrs), obviating the need to train generalizable sparse surface reconstruction methods from scratch. Additionally, we introduce a novel loss on volumetric rendering weights that helps in the learning of accurate occupancy fields, along with a normal loss that helps in global smoothing of the occupancy fields. We evaluate our approach on the DTU dataset [1] and demonstrate **state-of-the-art performance** in terms of reconstruction accuracy, especially in challenging scenarios with sparse input data and occluded regions. We furthermore demonstrate the generalization capabilities of our method by showing qualitative results on the Blended MVS [59] dataset without any retraining. Project page: geotransfer.github.io

Keywords: 3D Reconstruction · Volume rendering · Sparse views

1 Introduction

Creating three-dimensional structures from a set of images is a fundamental task in the realm of computer vision, finding broad applications in fields like robotics,

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augmented reality, and virtual reality. The first seminal deep learning approaches tackling this problem used Multi-view stereo techniques (MVS), as demonstrated by MVSNet [58] and its successors [10,14,50,57]. These methods build 3D cost volumes based on the camera frustum, deviating from traditional euclidean space, to achieve accurate depth-map estimation. However, they often necessitate subsequent steps, such as depth-map filtering, fusion, and mesh reconstruction, and exhibit susceptibility to noise, texture-less regions, and gaps.

Unlike seminal work relying on explicit representations (*e.g.* meshes [16, 19,51] and point clouds [2,12,21]), neural implicit reconstruction methods [8,37,38,52,60] constitute another popular class of strategies to address this challenge, creating precise and realistic geometry from multi-view images through the use of volume rendering and neural implicit representations based on the Sign Distance Function (SDF) [44] and its variations. However, despite their effectiveness, these approaches come with inherent limitations, including a lack of cross-scene generalization capabilities and the need for substantial computational resources to train them from scratch for each scene. Moreover, these techniques heavily depend on a large number of input views. However, due to many constrained scenarios (*e.g.* out-of-the-studio, low budget, *etc.*) and in the interest of wider applicability, there is active interest in seeking solutions that can deliver under minimal input.

To solve these issues, recent investigations, in the context of novel-view synthesis [18,36,56] and 3D reconstruction [26,29,47] have sought solutions by relying on learned data priors across many training scenes, by conditioning the implicit representation on spatially local features obtained from the sparse input images through generalizable encoders. This approach has proven effective in achieving remarkable cross-scene generalization capabilities, even with sparse views as input. Most relevant to our approach is GeoNeRF [18] which constructs a cost volume to enable geometry-aware scene reasoning, followed by attention-based view aggregation and volumetric rendering to learn the radiance field of a scene.

In this paper, differently from concurrent works, we explore the idea of using this pre-existing generalizable state-of-the-art NeRF to obtain scene reconstructions through transfer learning. In this context, we show that under the assumption of our scene being composed of solid, non-transparent objects, it is possible to rapidly transform the generalizable sampling-dependent opacity obtained from the density field of GeoNeRF [18] to a generalizable sampling-independent occupancy. This strategy also leads to a drastic reduction of training time from the order of several days to a couple of hours, which removes the need to train generalizable sparse reconstruction methods [26,29,47] from scratch. We introduce a novel volumetric rendering weight loss, in addition to a surface normal based smoothing loss to further refine our occupancy field, leading to the current state-of-the-art results in sparse 3D reconstruction on the DTU dataset [1] without requiring any test-time optimization, and outperforming in this process the current state-of-the-art generalizable SDF based 3D reconstruction networks [26,29,47].

In summary, our contribution can be summarized as:

- We explore a novel strategy of fast adaptation of an existing state-of-the-art generalizable NeRF method to obtain a generalizable occupancy network by transferring and fine-tuning its features. This yields the **state-of-the-art performance on DTU** [1] reconstruction from sparse views 1.
- The decrease in training duration from multiple days to just a few hours, all the while achieving state-of-the-art performance, removes the necessity to train computationally intensive sparse generalizable surface reconstruction techniques from scratch.
- Among the losses we use for our transfer learning framework, we propose a novel volumetric rendering weight loss to impose the properties followed by an ideal occupancy field in a volumetric rendering framework which leads to the learning of a more accurate occupancy functions (5).

2 Related Work

There is a substantial body of work on the subject of 3D reconstruction, and we review here work we deemed most relevant to the context of our contribution.

Neural Surface Reconstruction. In the realm of neural surface reconstruction, the utilization of neural implicit representations enables the depiction of 3D geometries as continuous functions that can be computed at arbitrary spatial locations. Due to their capability to represent complex and detailed shapes in a compact and efficient manner, these representations demonstrate significant potential in tasks such as 3D reconstruction [8,17,20,37,38,52,60–62,64], shape representation [3,13,33,44], and novel view synthesis [27,35,49]. The emergence of NeRF [35] has instigated a significant shift in the paradigm towards employing similar techniques for these tasks. IDR [61] utilizes surface rendering to acquire geometry from multi-view images but necessitates additional object masks. Unisurf [38], which is most relevant to our work, models the local opacity of a NeRF [35] with an occupancy network, which allows them to train on multiple datasets including ones involving forward-facing scenes such as LLFF [34]. Differently, several methods have attempted to rewrite the density function in NeRF [35] using Signed Distance Function (SDF) and its variants, successfully achieving plausible geometry. Significantly, NeuS [52] formulates an unbiased and occlusion aware volumetric weight function equation by employing logistic sigmoid functions. Conversely, Volsdf [60] incorporates a signed distance function into the density formulation and introduces a sampling strategy that meets a determined error bound on the transparency function. HF-NeuS [54] improves upon NeuS [52] by modeling transparency as a transformation of the estimated signed distance field and proposes to decompose the signed distance function into a base function and a displacement function with a coarse-to-fine strategy to gradually increase the high-frequency details. These methods offer a robust approach for multi-view 3D reconstruction from 2D images. However, these methods require prolonged optimization for training each scene independently and also require a substantial number of dense images, making it challenging to generalize to unknown scenes and limiting deployment.

Generalizable NeRFs. Some recent methods [9,15,22,25,36,55,62] synthesize novel views on a single scene with sparse views, albeit facing difficulties in understanding the underlying geometry of the scene, which they attempt to solve using several geometry-based regularization strategies [4,13,40,43]. To attempt to solve this problem, certain methods [6,7,18,24,28,53,63] generate novel views in unknown scenarios through a generalization approach, constructing neural radiance fields on sparse views. These methods can infer on unknown scenarios without fine-tuning after training on multiple known scenarios, which involves incorporating priors derived from a larger model trained on multi-view image datasets. Among such methods, PixelNeRF [63] conditions itself on features extracted by a CNN. MVSNeRF [6] constructs a neural volume from the cost volume obtained by warping image features and conditions itself on this neural volume. IBRNet [53] aggregates features from nearby views to infer geometry and adopts an image-based rendering approach. NeuRay [28] utilizes neural networks to model and handle occlusions, thereby improving the quality and accuracy of image-based rendering. GeoNeRF [18], another recent method and one that we build on, employs a cascaded cost volume and an attention-based technique to aggregate information from different views. However, deriving the scene geometry from the volume density of these NeRF [35] based methods involves meticulous tuning of the density threshold, leading to artifacts due to the inherent ambiguity in the density field, as previously pointed out in Unisurf [38].

Generalizable Neural Surface Reconstruction. Point Cloud input models [5,39,41,42,46] are typically trained with SDF ground truth supervision. To obtain 3D reconstructions from sparse images with an accurate level set, current methods marry generalizable NeRFs with the Signed Distance Function (SDF) based transformation functions that model the volume density, thereby enabling volume rendering. Among these methods, both SparseNeuS [29] and VolRecon [47] achieve generalizable neural surface reconstruction by utilizing information from source images as priors. SparseNeuS [29] does this by encoding geometric information using a regular euclidean volume, while VolRecon [47] introduces multi-view image features through the view transformer to advance this scheme. The most recent work of ReTR [26] utilizes a hybrid extractor to obtain a multi-level euclidean volume and then employs a reconstruction transformer to enhance performance. Contrarily to these methods, we employ the Unisurf [38] based volumetric rendering framework which reconstructs surfaces by predicting occupancy, enabling the seamless transfer of the opacity information from a pretrained GeoNeRF [18] to learn a refined occupancy field.

3 Method

Our goal is to obtain a feed-forward generalizable 3D reconstruction network from a sparse array of images. Namely, this network ($e.g. f_o$) should be able to deliver an implicit shape representation, *i.e.* a binary shape occupancy field for instance, from observed images $\{I_i\}_{i=1}^N$ and their calibrations $\{\pi_i\}_{i=1}^N$ of a test scene (unseen at training), without requiring any optimization on this new scene.

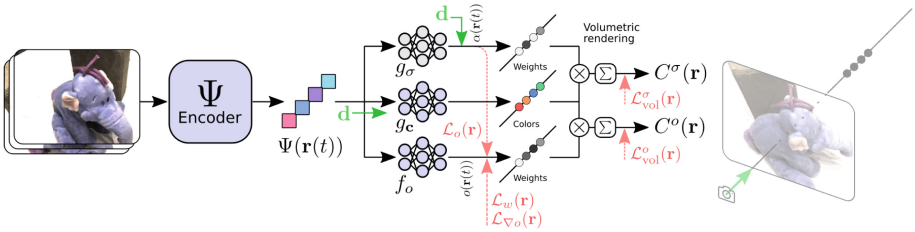

Fig. 1. Overview of our transfer learning: Our final model (in light blue) is comprised of tuned image encoder Ψ and implicit occupancy and color decoders f_o and $g_\mathbf{c}$. The encoder Ψ, the color decoder $g_\mathbf{c}$ and density decoder g_σ are initialized as a pretrained generalizable NeRF. Red dashed lines symbolize our tuning losses. We apply multiple regularizations on our occupancy f_o, while tuning the network with both the density and occupancy guided volumetric renderings. (Color figure online)

The inferred shape $\hat{\mathcal{S}}$ can be obtained as the level set of the learned occupancy f_o:

$$\hat{\mathcal{S}} = \{\mathbf{x} \in \mathbb{R}^3 \mid f_o(\mathbf{x}) = 0.5\}. \tag{1}$$

Practically, an explicit triangle mesh for $\hat{\mathcal{S}}$ can be obtained through the Marching Cubes algorithm [30] while querying the neural network f_o. We propose to adapt an existing generalizable NeRF model for this purpose. We chose the model denoted GeoNeRF [18] without loss of generality as it is one of the best performing models in generalizable novel view synthesis.

Generalizable NeRF model. Let $\mathbf{r}$ be a ray, *i.e.*$\mathbf{r}(t) = \mathbf{o} + t\mathbf{d}$, where $\mathbf{o}$ is the camera origin and $\mathbf{d}$ the ray direction. The color C of the pixel corresponding to a ray $\mathbf{r}$ can be generated by integrating along the ray:

$$C(\mathbf{r}) = \int T(t)\sigma(\mathbf{r}(t))\mathbf{c}(\mathbf{r}(t), \mathbf{d})dt \tag{2}$$

$$= \int w(t)\mathbf{c}(\mathbf{r}(t), \mathbf{d})dt. \tag{3}$$

In the equation above, $\sigma(\mathbf{r}(t))$ denotes volume density, which represents a differential opacity signaling the amount of radiance accumulated by a ray passing through the point $\mathbf{r}(t)$. $T(t)$ denotes transparency, *i.e.* the accumulated transmittance along the ray until t, which can be derived from density accordingly:

$$T(t) = \exp\left(-\int^t \sigma(\mathbf{r}(s))ds\right). \tag{4}$$

Furthermore, $w(t)$ is commonly referred to as the volumetric rendering weighting function.

In practice, the integral in Eq. 3 is approximated using discrete samples $\{t_i\}$ with the quadrature rule [32]. This gives rise to an equation resembling α-compositing, where α represents opacity, which controls the amount of radiance

that is absorbed or transmitted at each point in the scene. This leads to:

$$C_\sigma(\mathbf{r}) = \sum_i T_i \left(1 - e^{-\sigma_i(t_{i+1}-t_i)}\right) \mathbf{c}(\mathbf{r}(t_i), \mathbf{d}) \tag{5}$$

$$= \sum_i \prod_{j=0}^{i-1} (1-\alpha_j)\alpha_i \mathbf{c}(\mathbf{r}(t_i), \mathbf{d}) \tag{6}$$

where $\alpha_i = 1 - \exp(-\sigma_i(t_{i+1} - t_i))$.

Gneralizable NeRFs are typically comprised of an encoder network (*e.g.* Ψ) producing spatially local features. These features are mapped subsequently by a density network (MLP) (*e.g.* g_σ) and a viewing direction dependent color network (MLP) (*e.g.* $g_\mathbf{c}$). (*cf.* Fig. 1). Inference is achieved via volumetric rendering as shown above. The colors and densities necessary for this rendering are thus modelled as follows:

$$\mathbf{c}(\mathbf{r}(t), \mathbf{d}) = g_\sigma(\Psi(\mathbf{r}(t), \{I_i\}), \mathbf{d}) \tag{7}$$

$$\sigma(\mathbf{r}(t)) = g_\mathbf{c}(\Psi(\mathbf{r}(t), \{I_i\})). \tag{8}$$

Adapting a Generalizable NeRF Model. A geometry can be extracted from NeRF [35] models at test time by thresholding the density function, which is view and sampling independent. However, it is not clear how the threshold can be selected, and the obtained geometries tend to be noisy and inaccurate, with high Chamfer errors with respect to the ground truth.

Although the opacity is the closest measure in a NeRF [35] to an occupancy, the opacity as defined in the volumetric rendering framework is view and ray-sampling dependent. Hence it is not clear how a solely spatially dependent geometry can be extracted from it.

Yet, in order for generalizable NeRFs to be able to reason correctly about 3D for accurate novel view synthesis, we hypothesize that they must encompass a good high level representation of geometry, that could be nudged towards an accurate and smooth shape representation per se. Based on this, we propose to tune a Generalizable NeRF (Namely GeoNeRF [18]) into a generalizable occupancy model, that offers view and ray-sampling independent geometry at convergence.

We define a new Sigmoid activated implicit decoder f_o that will represent the occupancy field in feature space:

$$o(\mathbf{r}(t)) = f_o(\Psi(\mathbf{r}(t), \{I_i\}) \tag{9}$$

We propose to tune the encoder Ψ to adapt the feature space to this new prediction task. This can be seen as form of transfer learning. As this tuning is achieved through volumetric rendering, we use the baseline model color network ($\mathbf{c} := g_\mathbf{c}$) to perform volumetric rendering with f_o:

$$C_o(\mathbf{r}) = \sum_i \prod_{j=0}^{i-1} (1-o_j) o_i \mathbf{c}(\mathbf{r}(t_i), \mathbf{d}). \tag{10}$$

Color network $g_{\mathbf{c}}$ has been pretrained using the density based weight functions in volumetric rendering. Hence, it is important to tune it as-well to this new occupancy based rendering. Furthermore, as we also want out initial Generalizable NeRF not to deviate substantially from its original weights and hence lose its original knowledge, we train by backpropagating both the volumetric rendering loss based on occupancy, and the original volumetric rendering loss based on density together:

$$\mathcal{L}^{o}_{vol}(\mathbf{r}) = ||C_o(\mathbf{r}) - C_{\text{GT}}(\mathbf{r})||_2 \tag{11}$$
$$\mathcal{L}^{\sigma}_{vol}(\mathbf{r}) = ||C_\sigma(\mathbf{r}) - C_{\text{GT}}(\mathbf{r})||_2. \tag{12}$$

We also retain the depth supervision used by the geometry reasoner in GeoNeRF [18], with ground truth depths if available, and with their self-supervised depth loss otherwise. Based on the knowledge that assuming solid objects, α becomes a discrete occupancy indicator variable $o \in \{0, 1\}$ which either takes $o = 0$ in free space and $o = 1$ in occupied space, we bootstrap our occupancy with α predictions from the density branch, as a form of warm up or initialization for a few iterations at the beginning of the training:

$$\mathcal{L}_o(\mathbf{r}) = \sum_i ||o_i - \alpha_i||_2. \tag{13}$$

A common problem with learning geometry through volumetric rendering is that the weight function does not peak at the surface [38,52]. This can be observed in Fig. 5, where neither our baseline nor the generalizable NeRF model we build on satisfy this constraint. Hence, we propose a novel loss to remedy this limitation. First, for the current ray $\mathbf{r}$, we perform ray tracing, *i.e.* finding sample pair t_k and t_{k+1} where the occupancy flips from "empty" ($f_o(t_k) < 0.5$) to "occupied" ($f_o(t_{k+1}) \geq 0.5$) for the first time along the ray. Then, we perform secant method based root finding [37,38] between these samples to find the root t^* corresponding to the surface-ray intersection. Ideally, we want our weight function to reach a sharp 1-peak at this location. Hence, we supervise the weight with a Gaussian centered at root t^*, and whose standard deviation we dynamically reduce during training, following the scheduling detailed in Sect. 4:

$$\mathcal{L}_w(\mathbf{r}) = \sum_t || \prod_{j=0}^{i-1} (1 - o_j) o_i - \mathrm{e}^{-((t_i - t^*)/a)^2} ||_2. \tag{14}$$

To reduce the noise in our reconstructions, we follow [38] and implement a smoothing over the normalized spatial gradients of our occupancy function at the surface, *i.e.* using root locations t^*:

$$\mathcal{L}_{\nabla o}(\mathbf{r}) = || \frac{\nabla o(\mathbf{r}(\mathbf{t}^*))}{||\nabla o(\mathbf{r}(\mathbf{t}^*))||_2} - \frac{\nabla o(\mathbf{r}(\mathbf{t}^*) + \epsilon)}{||\nabla o(\mathbf{r}(\mathbf{t}^*) + \epsilon)||_2} ||_2, \tag{15}$$

where $\epsilon \in \mathbb{R}^3$ is a small random perturbation, and gradients can be computed efficiently through automatic differentiation [45].

Table 1. Quantitative comparison on the DTU dataset [1]. Best and second best methods are **emboldened** and underlined respectively. *Ours** refers to the model trained using additional datasets.

Scan	24	37	40	55	63	65	69	83	97	105	106	110	114	118	122	Mean
COLMAP [48]	**0.9**	2.89	1.63	1.08	2.18	1.94	1.61	1.3	2.34	1.28	1.1	1.42	0.76	1.17	1.14	1.52
MVSNet [58]	1.05	2.52	1.71	1.04	1.45	**1.52**	0.88	1.29	1.38	1.05	0.91	0.66	0.61	1.08	1.16	1.22
IDR [61]	4.01	6.4	3.52	1.91	3.96	2.36	4.85	1.62	6.37	5.97	1.23	4.73	0.91	1.72	1.26	3.39
VolSDF [60]	4.03	4.21	6.12	0.91	8.24	1.73	2.74	1.82	5.14	3.09	2.08	4.81	0.6	3.51	2.18	3.41
UNISURF [38]	5.08	7.18	3.96	5.3	4.61	2.24	3.94	3.14	5.63	3.4	5.09	6.38	2.98	4.05	2.81	4.39
NeuS [52]	4.57	4.49	3.97	4.32	4.63	1.95	4.68	3.83	4.15	2.5	1.52	6.47	1.26	5.57	6.11	4.00
IBRNet-ft [53]	1.67	2.97	2.26	1.56	2.52	2.30	1.50	2.05	2.02	1.73	1.66	1.63	1.17	1.84	1.61	1.90
SparseNeuS-ft [29]	1.29	2.27	1.57	**0.88**	1.61	1.86	1.06	1.27	1.42	1.07	0.99	0.87	**0.54**	1.15	1.18	1.27
PixelNerf [63]	5.13	8.07	5.85	4.4	7.11	4.64	5.68	6.76	9.05	6.11	3.95	5.92	6.26	6.89	6.93	6.28
IBRNet [53]	2.29	3.70	2.66	1.83	3.02	2.83	1.77	2.28	2.73	1.96	1.87	2.13	1.58	2.05	2.09	2.32
MVSNeRF [6]	1.96	3.27	2.54	1.93	2.57	2.71	1.82	1.72	2.29	1.75	1.72	1.47	1.29	2.09	2.26	2.09
GeoNeRF [18]	3.40	4.37	3.99	2.94	5.08	4.50	3.42	4.68	4.54	4.05	3.47	3.23	3.34	3.57	3.63	3.88
SparseNeuS [29]	1.68	3.06	2.25	1.1	2.37	2.18	1.28	1.47	1.8	1.23	1.19	1.17	0.75	1.56	1.55	1.64
VolRecon [47]	1.2	2.59	1.56	1.08	1.43	1.92	1.11	1.48	1.42	1.05	1.19	1.38	0.74	1.23	1.27	1.38
ReTR [26]	1.05	2.31	**1.44**	0.98	**1.18**	**1.52**	0.88	1.35	1.3	**0.87**	1.07	0.77	0.59	1.05	1.12	1.17
Ours	1.01	2.24	1.52	**0.88**	1.37	1.82	0.85	1.39	**1.25**	1.0	**0.77**	0.63	0.57	0.96	**1.0**	1.15
*Ours**	0.95	**2.23**	1.45	0.94	1.26	1.67	**0.81**	**1.21**	1.34	1.02	0.84	**0.6**	0.58	**0.94**	1.02	**1.12**

Finally, our full training is done following the combined objective averaged over batches of rays, and we train on the same multi-view data as our original Generalizable NeRF model was initially trained on:

$$\mathcal{L} = \sum_{\mathbf{r}} \mathcal{L}^{o}_{vol}(\mathbf{r}) + \mathcal{L}^{\sigma}_{vol}(\mathbf{r}) + \lambda \mathcal{L}_{\nabla o}(\mathbf{r}) + \mu \mathcal{L}_{w}(\mathbf{r}) + \nu \mathcal{L}_{o}(\mathbf{r}). \tag{16}$$

These loses are analysed separately in Sect. 4, where we show their respective numerical and qualitative contribution to our overall performance.

4 Experiments

In this section, we showcase the efficacy and merits of our proposed approach. Firstly, we offer an intricate overview of our experimental configurations, encompassing implementation specifics, datasets, and baseline methods. Secondly, we present both quantitative and qualitative comparisons on two extensively utilized datasets, namely DTU [1] and BlendedMVS [59]. Finally, we perform thorough ablation studies to scrutinize the impact of distinct components in our proposed methodology.

Datasets. In line with prior research [26,29,47], we employ the DTU dataset [1] for the training phase. Since we perform transfer learning from GeoNeRF [1], which is a NeRF-based framework, to learn generalizable occupancy fields, we are also able to leverage training on non-object centric datasets such as the real forward-facing datasets from LLFF [34] and IBRNet [53] and hence, we also include an additional model to show the effect of training on more data. In both

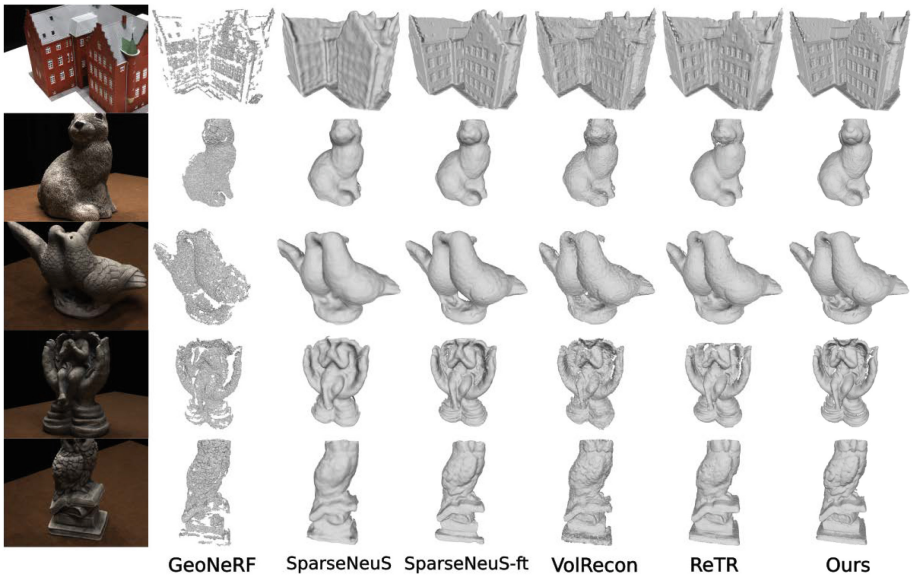

Fig. 2. Qualitative comparison of reconstructions from 3 input views in datatset DTU.

cases (with and without additional training datasets), our backbone GeoNeRF [1] was trained on the same data as our full models to maintain fairness in evaluation performance comparisons. The DTU dataset [1] is characterized by indoor multi-view stereo data, featuring ground truth point clouds from 124 distinct scenes and under 7 different lighting conditions. Throughout our experiments, we utilize the same set of 15 scenes as [26,29,47] for testing purposes, reserving the remaining scenes for training. Concerning the BlendedMVS dataset [59], we opt for 7 scenes in accordance with SparseNeuS [29]. For each scene, we use the same 3 sparse input views following SparseNeuS [29]. To ensure impartial evaluation, we use the foreground masks from IDR [61] to evaluate how well our approach performs on the test set, consistent with prior research [26,29,47]. Additionally, to assess the generalization capability of our proposed framework, we qualitatively compare our method on the BlendedMVS dataset [59] without any fine-tuning. For our novel-view synthesis experiments, we follow the same split of testing images within a scene as in GeoNeRF [1] and the testing scenes are identical to our 3D reconstruction experiments.

Baselines. In order to showcase the efficacy of our method, we conducted comparisons with a) SparseNeus [29], VolRecon [47] and ReTR [26], the leading generalizable neural surface reconstruction approaches and their finetuned (ft) versions on sparse images; b) Generalizable neural rendering methods [6,18,53,63]; c) Neural implicit reconstruction methods [38,52,60,61] that necessitate scene-specific training from the beginning; and finally, d) Well-known multi-view stereo (MVS) [48,58] methods.

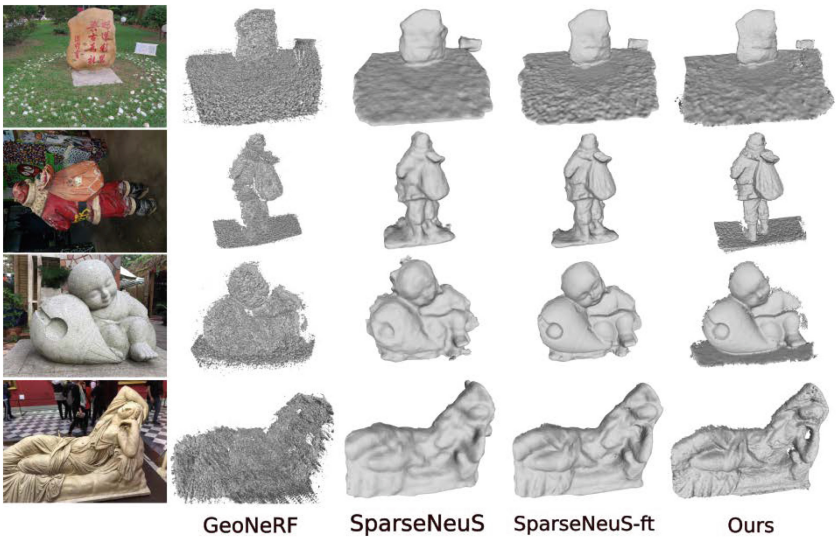

Fig. 3. Qualitative comparison of reconstructions from 3 input views in dataset BMVS. Note that we reconstruct detailed surfaces with our method without any fine-tuning.

Implementation Details. Our model is implemented using PyTorch [45] and PyTorch Lightning [11]. During the training phase, we utilize an image resolution of 800×600, with N (the number of source images) set to 3. Training extends over 5400 steps using the Adam optimizer [23] on a single RTX A6000 GPU, with an initial learning rate of 5×10^{-4}. We apply the occupancy distillation loss for the first 100 warm-up steps. Thereafter, the weight of the distillation loss is reduced to 0. A cosine learning rate scheduler [31] is applied to the optimizer. The ray number sampled per batch and the batch size are configured to 128 and 1, respectively. Employing a hierarchical sampling strategy, we uniformly sample N_{coarse} points on the ray during both training and testing. Subsequently, importance sampling is applied to sample an additional N_{fine} points on top of the coarse probability estimation. In our experiments, we set N_{coarse} to 96 and N_{fine} to 32. During testing, the image resolution is kept the same at 800×600. We also address the scheduling strategy followed for $\mathcal{L}_w$, described in 14. Specifically, we follow:

$$a = \max(a_{max}\mathrm{e}^{-\mathrm{k}\beta}, a_{min}). \tag{17}$$

where k denotes the global iteration number. a_{max}, a_{min} and β are hyperparameters, which we set to 1, 0.04 and 0.001 respectively. Finally, the weights associated with the losses are set as $\lambda = 0.1$, $\mu = 0.2$ and $\nu = 0.2$. We use the Marching Cubes algorithm [30] with a grid resolution of 400 for extracting each surface mesh by thresholding the occupancy field at 0.5.

4.1 Sparse View Reconstruction on DTU

We conduct surface reconstruction with sparse views (only 3 views) on the DTU dataset [1] and assess the predicted surface by comparing it to the ground-truth point clouds using the chamfer distance metric. To facilitate a fair comparison, we followed the evaluation process employed in [26,29,47] and adhered to the same testing split as described in them. As indicated in Table 1, our method (only DTU trained) surpasses SparseNeuS [29] and VolRecon [47] by a considerable margin, *i.e.* by 30% and 17% respectively. When we use additional datasets for training, the gap further increases to 32% and 19% respectively. Furthermore, our approach exhibits superior performance compared to well-known multi-view stereo (MVS) methods like Colmap [48] and MVSNet [58]. Our approach also demonstrates superior performance in comparison to ReTR [26], which is the latest state-of-the-art generalizable neural implicit reconstruction method. Additionally, we present qualitative results of sparse view reconstruction in Fig. 2, showcasing that our reconstructed geometry features more expressive and detailed surfaces which represent the ground truth surfaces more accurately when compared to the current state-of-the-art methods. Another DTU generalization experiment using MVSNeRF [6] as the backbone is included in the supplemetary section to illustrate that the method presented can be extended to other generalizable NeRF baselines.

4.2 Generalization on BlendedMVS

To demonstrate the generalization prowess of our proposed approach, we perform additional evaluations on the BlendedMVS dataset [59] without resorting to any fine-tuning. The high-fidelity reconstructions of large-scale scenes and small objects across diverse domains, as illustrated in Fig. 3, affirms the efficacy of our method in terms of its generalization capabilities. Our method is able to obtain more detailed surfaces in comparison to SparseNeuS [29], even after it is fine-tuned on the sparse testing set.

4.3 Novel-View Synthesis Performance on DTU

Since GeoNeRF [18] is among the state-of-the-art methods for sparse novel-view synthesis, it is reasonable to assume that we also inherit its novel-view synthesis performance since we train our method by transferring its features. To validate this, we evaluate novel-view synthesis results on the DTU [1] dataset. We compare our method against GeoNeRF [18], VolRecon [47] and ReTR [26] on the mean PSNR, SSIM and LPIPS metrics over all scenes. Since we use 3 input source images to build our cost feature volume, for fairness in comparison, we also evaluate the novel-view synthesis results for VolRecon [47] and ReTR [26] with 3 input source images to build the cost feature volume, instead of the 4 input source images used during their respective trainings. Our results can be summarized as below:

Table 2. Novel-view performance on DTU [1].

Method	PSNR ↑	SSIM ↑	LPIPS ↓
VolRecon [47]	19.61	0.81	0.23
ReTR [26]	19.53	0.79	0.24
GeoNeRF [18]	23.48	**0.93**	**0.087**
Ours	**24.08**	**0.93**	0.093

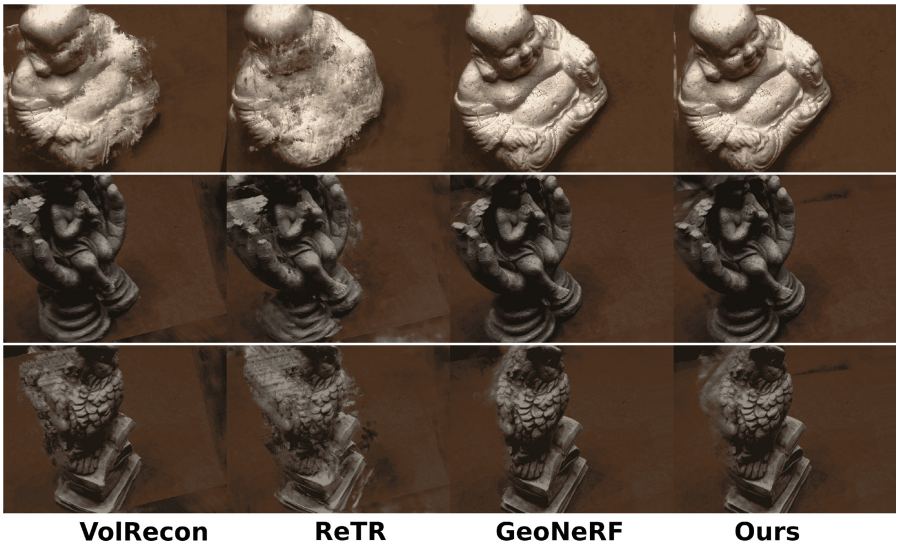

Fig. 4. Novel-View synthesis qualitative evaluation on DTU [1] using 3 source images. We notice that generalizable reconstruction models (ReTR [26], VolRecon [47]) struggle to perform extreme novel-view extrapolation.

As indicated in Table 2, we are very close in performance on novel-view synthesis metrics to GeoNeRF [18]. Note that we use 3 input views, the same setting on which we conduct our novel-view experiments on. The original GeoNeRF [18], however, was trained with 6 input views. The purpose behind this experiment was simply to make sure that our training paradigm does not sacrifice GeoNeRF [18]'s excellent novel-view synthesis performance. Overall, our method offers state-of-the-art reconstruction performance without foregoing the novel-view synthesis performance of one of the state-of-the-art novel-view synthesis methods we transfer features from. This is because our learnt occupancy fields by design and by penalization with the help of occupancy loss $\mathcal{L}_o$ induce volumetric weights associated with the opacity fields of GeoNeRF [18]. We also notice that we far surpass VolRecon [47] and ReTR [26] in terms of novel-view synthesis performance. Hence, we offer a dual advantage, leading to our method offering **state-of-the-art** reconstruction results along with superior novel-view

synthesis results. We also show some qualitative comparisons in Fig. 4 and several other qualitative comparisons on several DTU [1] scenes in the supplementary section.

4.4 Ablation Studies

In this section, we conduct an ablative analysis to justify the choice of our final architecture and choice of losses. We ablate in the full training scenario using all testing scenes of the DTU dataset [1].

Table 3. Effect of fine-tuning encoder/ feature vol. of GeoNeRF [18]

Method	Chamfer ↓
w/o tuning encoder	1.29
w tuning encoder	1.15

Table 4. Normal Consistency

	w/o $\mathcal{L}_{\nabla o}$	w $\mathcal{L}_{\nabla o}$
NC. ↑	0.64	0.68

Effect of Fine-Tuning Strategies. In this study, we showcase the outcomes of training our occupancy model while fine-tuning different components of GeoNeRF [18]. As illustrated in Table 3, a better occupancy field is learnt when the encoder of the pre-trained GeoNeRF [18] model is tuned jointly with the occupancy network. This is because without doing so, the encoder is more suited for learning a density field as is the case for GeoNeRF [18], while we aim to learn an occupancy field.

Effect of Different Loss Components. In this study, we showcase the outcomes of various loss components to illustrate their efficacy. As illustrated in the 1^{st} row of Table 5, the novel volumetric rendering weight loss $\mathcal{L}_w$ is vital in learning a sharp and accurate occupancy field. Without it, the chamfer metric degrades, which demonstrates its importance. The 2^{nd} row illustrates the importance of bootstrapping our occupancy with α predictions. Furthermore, we see in Table 6 that the accuracy chamfer metric expectedly suffers more without $\mathcal{L}_w$ as it is directly responsible for sharper surfaces, and less so for the completeness metric.

Table 5. Effect of discarding different loss components

Method	Chamfer ↓
w/o $\mathcal{L}_w$	1.22
w/o $\mathcal{L}_o$	1.17
Full model	1.15

Table 6. Ablation of accuracy & completeness

	w/o $\mathcal{L}_w$	w $\mathcal{L}_w$
Acc. ↓	0.77	0.67
Comp. ↓	1.67	1.64
Overall. ↓	1.22	1.15

This is also illustrated in Fig. 5, which shows that the loss results in a weight distribution that agrees with one theoretically induced by an occupancy field

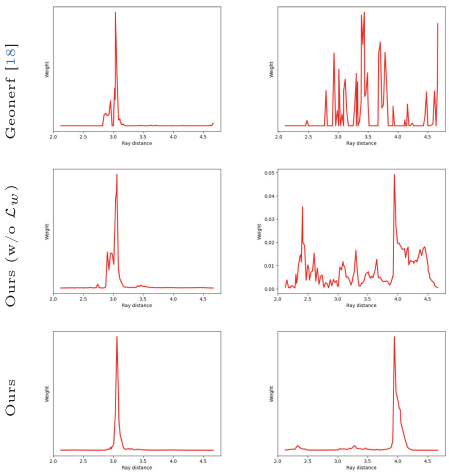

Fig. 5. Comparison between volumetric weight distribution along rays between Geonerf [18] and our method.

Table 7. Comparison of total training duration.

Method	Training duration
SparseNeuS [29]	∼3 d
ReTR [26]	∼3 d
Ours (re-train)	**∼3.5 hrs**

i.e. one that 1-peaks on the surface boundary, as discussed in Unisurf [38]. We also see in the 2^{nd} row that without our bootstrapped occupancy loss $\mathcal{L}_o$, our surface reconstruction performance is impacted negatively. Finally, we investigate the effect of discarding the normal loss $\mathcal{L}_{\nabla o}$. The normal loss is important for reducing noise in our reconstruction as witnessed in our qualitative results. This smoothing, can however, impact chamfer distance and it can be challenging to strike a perfect balance between these conflicting objectives. To illustrate the importance of normal loss, we provide normal consistency results in Table 4 which demonstrates that our normal loss $\mathcal{L}_{\nabla o}$ smoothens the reconstructions and consequently, improves normal consistency between the reconstructed and GT meshes. The whole study demonstrates that all our losses are indispensable to our model's final performance.

Training Time. Here, we address the time of training for our model in Table 7. Given a fully-trained GeoNeRF [18], our model is able to finetune its features rapidly. While training SparseNeuS [29] and ReTR [26] fully on a single RTX A6000 GPU takes almost 3 days, we are able to train to adapt features of a pretrained GeoNeRF [18] in approximately 3.5 hrs to learn occupancy fields.

5 Limitations

Because of the underlying assumption while learning occupancy in a volumetric framework, our method is specifically designed to represent surfaces that are solid and not transparent. Moreover, the accuracy of the reconstructions is diminished in areas that are infrequently visible in the input images.

6 Conclusions

We introduce a novel approach, GeoTransfer, that rapidly transfers the 3D understanding of generalizable NeRFs to obtain accurate occupancy fields for implicit surface reconstruction. Unlike previous methods that introduced improvements to the encoders of generalizable SDF based reconstruction methods, we discovered that it sufficed to learn a occupancy network in feature space that learns to transform sampling-dependent opacities obtained from the state-of-the-art generalizable NeRF to sampling-independent occupancy fields. This approach also proved to be way faster than training a generalizable sparse 3D surface reconstruction method from scratch. Our method achieves state-of-the-art reconstruction quality for sparse inputs, showcasing its efficacy.

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DiVR: Incorporating Context from Diverse VR Scenes for Human Trajectory Prediction

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Abstract. Virtual environments provide a rich and controlled setting for collecting detailed data on human behavior, offering unique opportunities for predicting human trajectories in dynamic scenes. However, most existing approaches have overlooked the potential of these environments, focusing instead on static contexts without considering user-specific factors. Employing the CREATTIVE3D dataset, our work models trajectories recorded in virtual reality (VR) scenes for diverse situations including road-crossing tasks with user interactions and simulated visual impairments. We propose Diverse Context VR Human Motion Prediction (DiVR), a cross-modal transformer based on the Perceiver architecture that integrates both static and dynamic scene context using a heterogeneous graph convolution network. We conduct extensive experiments comparing DiVR against existing architectures including MLP, LSTM, and transformers with gaze and point cloud context. Additionally, we also stress test our model's generalizability across different users, tasks, and scenes. Results show that DiVR achieves higher accuracy and adaptability compared to other models and to static graphs. This work highlights the advantages of using VR datasets for context-aware human trajectory modeling, with potential applications in enhancing user experiences in the metaverse. Our source code is publicly available at https://gitlab.inria.fr/ffrancog/creattive3d-divr-model.

Keywords: Virtual Reality · Scene understanding · Heterogeneous graphs · Cross-modal

1 Introduction

Capturing dynamic interactions between individuals and their environments is crucial for human motion prediction, and including visual scene context enhances the accuracy of these predictions. However, existing approaches primarily rely on frame-by-frame video analysis, and have difficulty to adequately account for these complex interactions [19]. Scene point clouds generated from 3D sensors

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provide spatial data but can't capture temporal changes nor the human intentions [11,30]. These data capture and representation methods, though useful in controlled and simple scenarios, fall short in offering a nuanced understanding of individual intentions and interactions within dynamic environments. This limitation significantly impacts prediction accuracy, particularly in complex navigation scenarios or for individuals with invisible conditions such as visual impairments.

VR provides a secure and controlled media to replicate real-world scenarios to study human behavior. The CREATTIVE3D dataset [23][1], integral to this research, offers ontology-based VR environments, with annotations of 3D scenes, objects, and interactive tasks. Additionally, it includes scenes simulating low-vision conditions with gaze tracking. This allows a deeper investigation into how well models can capture individual behavior, or generalize across behaviors in varied contexts, addressing the limits of existing computer vision approaches and datasets.

Recognizing these limitations, our research introduces DiVR, a novel model that uses heterogeneous graph representations [2] to effectively capture the dynamic nature of human environments, as shown in Fig. 1. We leverage the annotations in the CREATTIVE3D dataset to test and refine DiVR for human motion prediction. By incorporating both static and dynamic variables, DiVR captures interactions and environmental factors, making it capable of generalizing across varying users, task complexities, and scene layouts, a key contribution to this work. Such versatility is crucial to applications like autonomous driving where the accuracy of predicting pedestrian movements can significantly impact safety and operational efficiency. We discuss DiVR's adaptability to diverse scenarios including those involving simulated low-vision conditions – vision loss that cannot be fully corrected – and complex navigation tasks.

Our contributions are thus threefold:

1. Propose DiVR, a novel approach using dynamic heterogeneous graphs that capture both the static and dynamic scene context to improve human motion prediction.
2. Evaluate graph context to improve adaptability to low-vision and task complexity.
3. Explore how scene graph context helps the model generalize better across different user groups, tasks, and scenes.

2 Related Work

Human Trajectory Prediction. Human trajectory prediction is a computer vision task, originally developed using surveillance video data from datasets like ETH and UCY. It is formulated as a sequence-to-sequence prediction task: given past position data to predict future sequences. Up until recently, Recurrent Neural Networks (RNNs) were dominantly used for trajectory prediction, including

[1] https://zenodo.org/doi/10.5281/zenodo.8269108.

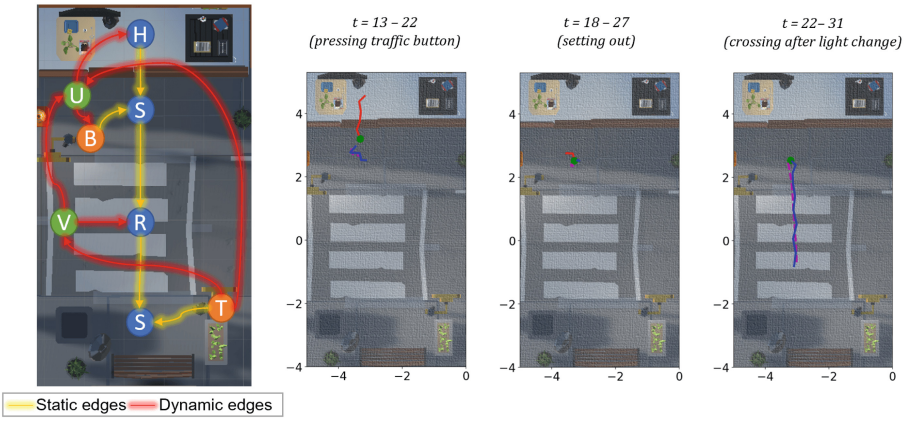

Fig. 1. Left: A heterogeneous graph representing a pedestrian crossing scene, with nodes for different locations (H: Home, S: Sidewalks, R: Road) and objects (U: User, V: Vehicle, T: Traffic Lights, B: Button). Right: Three scenarios of road crossing are depicted. In each scenario, the red line represents the input trajectory ending at the green dot. The blue line indicates the ground truth future motion, while the magenta line shows the predicted future motion using DiVR. (Color figure online)

human-robot interaction scenarios [28]. Corona et al. included context through simple graph representations [4], and Rondón et al. [16] with dedicated Long Short-Term Memory (LSTM) units fed with estimated saliency maps to modulate the importance of context over inertia depending on the prediction horizon. Transformer models [21] marked a significant shift, introducing self-attention that captures temporal relations more effectively than RNNs, and setting new standards for motion prediction [1,27,30]. However, transformers encounter challenges with the quadratic complexity in computational, particularly for high-dimensional inputs like images and videos, leading to significant computational costs during both training and testing phases. The Perceiver architecture [10] was conceived to address this by avoiding costly self-attention operations in the input space, instead restricting them to a low-dimensional latent space, with input and output dimensions handled only through cross-attention operations.

Cross-Modality Learning. Foundational cross-modal learning has been conceived to effectively fuse information from audio-visual [15] and audio-lingual [13] tasks, and are well-adapted to motion prediction tasks. Cross-modal representations enable the inclusion of human behavior and environmental interaction [25] information. GIMO [30] is representative of this and has strongly inspired our work, including motion capture, 3D point clouds, and gaze tracking in a cross-modal transformer architecture for human motion prediction. Zaier et al. [27] introduce a cross-attention model CMAPT to fuse pedestrian trajectory data with kinematic and visual features extracted from camera surveillance videos. They employ a bi-modal transformer to capture spatio-temporal interactions.

Finally, Gallo et al. [6] were the first to benchmark trajectory prediction in fully VR using the CREATTIVE3D dataset including motion, gaze, and scene point cloud data.

Graph-Based Techniques. Graph Neural Networks (GNN) have been recognized early on for their potential to model molecular structures [5,12] and are now applied to video tasks such as action recognition [7]. In pedestrian trajectory prediction, graphs can appropriately capture the interactions between people and other scene elements such as obstacles and pathways to improve accuracy [20,24,32]. Graph models can involve static and dynamic scene elements, which help model continuous changes within interaction networks [14]. Notably, Zhou et al. [31] introduced a hybrid static-dynamic graph approach to represent relations between pedestrians, significantly improving multi-pedestrian trajectory prediction. Sun et al. [20] studied how road graphs improved trajectory prediction on the autonomous driving dataset nuScenes. Spatial Temporal GNN [3] effectively combines RNN mechanisms with GNNs for trajectory prediction in complex real-world scenarios such as urban and parking environments, leveraging temporal dynamics to refine predictions and align with rapidly changing settings. To the best of our knowledge, our approach is the first to leverage heterogeneous graph representations to investigate trajectory prediction of real humans in fully interactive virtual environments, incorporating simulated low-vision conditions. Drawing inspiration from methodologies in autonomous driving, such as those by Carrasco et al. [2] and Zipfl et al. [33], we incorporate temporal heterogeneous graphs to effectively capture and model the dynamic interactions and temporal evolution within urban pedestrian environments.

3 Approach

Our work addresses human trajectory prediction on the CREATTIVE3D dataset, modeled on the 2D head position from past positions and context data. The human model comprises, at a given time t (in frames), the head position $\mathbf{p}_t \in \mathbf{R}^2$ in meters, representing the user's absolute position within a 10 by 4 m tracked space. The problem involves predicting a full motion sequence over a future horizon H, defined as $\mathbf{M}_{t+1:t+H} = \{(\hat{\mathbf{p}}_{t+1}), \ldots, (\hat{\mathbf{p}}_{t+H})\}$ from a given time t. We use 3 s of past motion data sampled at 2 frames per second to predict the trajectory for the subsequent 5 s.

Here, we present the graph representation for scene context, and the DiVR model which decodes this representation and other kinematic information into coordinate prediction.

3.1 Context Representation

While road graph representations are usually obtained from pre-extraction steps for autonomous driving datasets, the CREATTIVE3D dataset offers detailed annotations for entities like pedestrians, vehicles, traffic lights, and movable objects, as well as navigable spaces: house, sidewalks, roads, and crossings. Using

these detailed annotations, we generate scene graphs where nodes represent individuals, entities, and spaces, and the edges represent their relations such as proximity and interactions.

To prepare the dataset, we normalize spatial coordinates and convert categorical attributes into numerical values using one-hot encoding. In our study, we explore two types of graph representations: heterogeneous and homogeneous.

Heterogeneous Graph Representation. We model the scene context as a directed heterogeneous graph, denoted by $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{T}, \mathcal{R})$. Each node $v \in \mathcal{V}$ and edge $e \in \mathcal{E}$ is classified by type, with $\tau(v) : \mathcal{V} \rightarrow \mathcal{T}$ mapping nodes to node types $\mathcal{T} = \{\text{location}, \text{pedestrian}, \text{vehicle}, \text{button}, \text{traffic light}\}$, and $\phi(e) : \mathcal{E} \rightarrow \mathcal{R}$ mapping edges to relation types $\mathcal{R} = \{\text{approach}, \text{adjacent}, \text{interaction}\}$. These types represent typical urban interactions, such as a pedestrian or vehicle approaching a location, adjacency between locations, and interactions between entities like pedestrians and traffic lights.

Node Representation: At a given time t, each node represents urban entity attributes within the graph. Node attributes are detailed in Table 1, and include a unique identifier, interaction capability, mobility, presence status, and spatial positioning. This modeling captures both dynamic and static properties of entities, essential for representing urban interactions in our trajectory prediction task.

Table 1. Node Attributes for the heterogeneous graph

Attribute	Description
id	Unique identifier for the node
interactable $\in \{0, 1\}$	Indicates if the node can engage with other nodes
localization $\in \{0, 1\}$	Indicates if the node represents a fixed location
movable $\in \{0, 1\}$	Indicates if the node is dynamic
color $\in \{-1, 0, 1, 2\}$	Traffic light: red (0), orange (1), green (2), or -1 if N/A
presence $\in \{0, 1\}$	Indicates if the node is present in the scene
$x_{\min}, y_{\min}, x_{\max}, y_{\max}, x_{\text{cent}}, y_{\text{cent}}$	Normalized bounding box and centroid coordinates

Edge Representation: Edges in the heterogeneous graph indicate connectivity and are annotated with features that describe the nature of interactions, such as active or inactive status, type of interaction (e.g., crossing, waiting), and the physical or temporal distance between entities. This attributed edge characterization allows us to distinguish, for example, a pedestrian waiting at a traffic light from one actively crossing the street, providing deeper insights into interactive dynamics. Table 2 outlines the edge attributes.

Graph Embedding (f_c). To extract graph embeddings from the temporal heterogeneous graphs, we employ a Temporal Graph Convolutional Network

Table 2. Edge Attributes for the heterogeneous graph

Attribute	Description
id	Unique identifier for the edge
location_type $\in \{0, 1\}$	Indicates the position of the node
dynamic_type $\in \{0, 1\}$	Indicates if the interaction changes over time
adjacent_type $\in \{0, 1\}$	Indicates spatial adjacency
interaction_type $\in \{0, 1\}$	Indicates an interaction
active $\in \{0, 1\}$	Indicates if the interaction is currently taking place
distance	Normalized spatial distance between the connected nodes

(TemporalGCN) [8,18], as illustrated in Fig. 2. This model uses edge-conditioned convolution operations to transform edge features into dynamic weights for graph convolution layers. This adaptation allows the model to effectively differentiate and represent diverse interaction patterns within the graph. Next, global mean pooling aggregates these features across all nodes, creating a graph-level representation for each timestamp. The temporal embedding obtained from these timestamps are then concatenated, providing a representation of the dynamic graph context. Finally, a linear transformation is applied to the concatenated embedding, resulting in the final output graph embedding that integrates both spatial and temporal information.

Homogeneous Graph Representation. We model the scene context as a homogenous graph denoted by $\mathcal{G}_{\text{h}} = (\mathcal{V}_{\text{h}}, \mathcal{E}_{\text{h}})$, where all nodes $v \in \mathcal{V}_{\text{h}}$ and edges $e \in \mathcal{E}_{\text{h}}$ are treated uniformly without distinct types. In this representation, nodes do not differentiate between specific types of entities, such as pedestrians, vehicles, or traffic infrastructure. Each node is associated with attributes that encompass both the type of entity and its current state.

3.2 The DiVR Model

We present the DiVR model (Fig. 2), a new architecture that leverages cross-modal attention over three modalities of data: (1) gaze-interpolated scene point cloud, (2) past motion data, and (3) the human-scene interaction context through heterogeneous graph representations. Each of these modalities is processed by an individual branch with the PerceiverIO architecture. The first branch employs PointNet++ to extract and encode features from the gaze data interpolated with the scene point cloud into a latent vector f_{gaze}. Concurrently, the second branch transforms raw motion data into a latent representation f_{motion}. The third branch utilizes the TemporalGCN introduced in the previous section to handle temporal heterogeneous graphs, producing a latent graph vector $f_{context}$.

At the core of DiVR's architecture is a cross-modal attention mechanism that fuses f_{motion} and f_{gaze}, enhancing the model's sensitivity to the interplay between gaze direction and movements. This mechanism also integrates

$f_{context}$, combining motion, gaze, and environmental graph data via a predictive cross-modal transformer. As shown in Sect. 4, this fusion of diverse modalities, along with a structured representation of the environment and interactions using heterogeneous graphs, substantially improves the accuracy of future trajectory predictions.

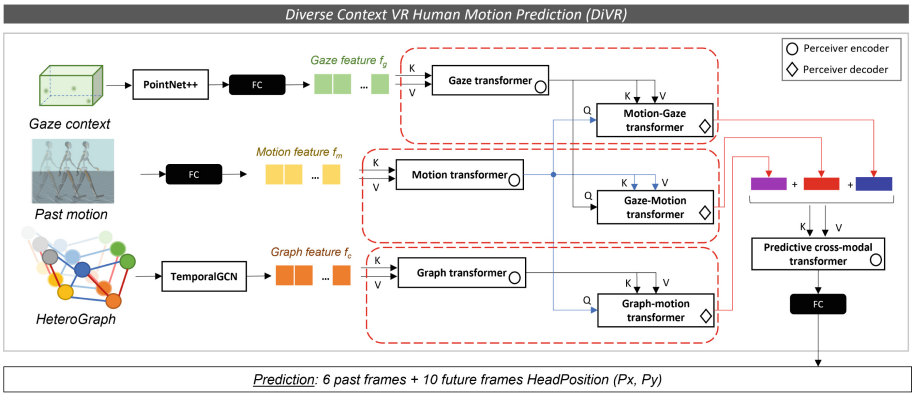

Fig. 2. Overview of the DiVR model.

Training. To effectively train the motion prediction models under various conditions, the loss function L_{total} integrates components to capture key aspects of prediction accuracy: $L_{\text{total}} = L_{\text{trans}} + L_{\text{rec}} + L_{\text{des_trans}}$. The translation loss, L_{trans}, represents the mean of the L1 losses between the predicted and ground-truth trajectories. The reconstruction loss, L_{rec}, penalizes discrepancies between the model's outputs and the inputs during the encoding phase, ensuring temporal consistency. The destination loss, $L_{\text{des_trans}}$, specifically targets the accuracy of the final step predictions.

4 Experimental Evaluation

This section evaluates the DiVR model's performance in predicting human trajectories using the CREATTIVE3D dataset. We assess the model's adaptability comparing it against established benchmarks. The goal is to demonstrate DiVR's robustness and accuracy across different user conditions, scene complexities, and task variations, highlighting its reduced trajectory prediction error compared to conventional models. We aim to show that DiVR integrates diverse contextual data effectively, using advanced graph-based techniques.

4.1 Experimental Setup

Dataset. We use the CREATTIVE3D dataset [22] to analyze human interactions and navigation in VR environments, focusing on road crossing scenarios. This dataset, which records user behaviors in urban settings, is structured around six scenarios, the dataset includes tasks of varying complexity: from simple tasks (ST) requiring navigation to complex tasks (CT) involving interactions, such as activating a traffic light before crossing. These scenarios are tested under two visual conditions: Normal Vision (NV) and Simulated Low Vision (LV). The Low Vision condition specifically simulates a scotoma, a central blind spot. We split the dataset with 70% for training, 15% for validation, and 15% for testing.

Experiments. Multiple experiments are conducted to assess model performance under varying conditions. The **Context Data Evaluation** assesses the impact of different visual conditions and task complexities using the balanced dataset. The **Low Vision and Complex Task Generalization** experiment tests model adaptability through three configurations, beginning with a baseline model trained on NV and ST, and progressively including more complex tasks and varied visual conditions. The **User Generalization** examines the model's ability to generalize across different users by selecting a test group of 10 users at random from a total of 40. The **Scene Generalization** trains models on single-lane road scenarios and tests them on two-lane scenarios to evaluate generalization to different environments. Finally, the **Task Generalization** assesses models on crossing tasks and their adaptability to tasks requiring a return to the starting point.

Evaluation Metrics. Our evaluation adopts two metrics that are endorsed and utilized in existing literature [17,26,27,29]: the Average Displacement Error (ADE) and the Final Displacement Error (FDE). ADE measures the average L2 distance (in meters) between predicted and actual trajectories at each time step from $T_{\text{obs}} + 1$ to T_{pred}, indicating overall prediction consistency. FDE calculates the Euclidean distance (in meters) between the predicted and actual final positions at the last time step, T_{pred}, reflecting the accuracy of endpoint prediction.

Implementation Details. Our model uses the PerceiverIO architecture for motion prediction. It consists of 6 encoding layers, each with a dimension of 256 and 8 attention heads, identical to the original PerceiverIO. The implementation was done using PyTorch on an Ubuntu server, supported by an Nvidia V100 GPU with 32 GB RAM. We trained the DiVR model end-to-end for 100 epochs with a batch size of 16. The Adam optimizer was used with an initial learning rate of 0.0001 and a weight decay of 0.0001. The learning rate was adjusted using an exponential decay method with a gamma value of 0.99.

4.2 Results

Context Data Evaluation

Baseline Models. We compare three prediction models with increasing context integration on the CREATTIVE3D dataset: (1) MLP [9] baseline, using only historical position data, (2) TRACK [16], incorporating real-time gaze data for user attention, and (3) GIMO [30], which includes gaze and 3D point cloud data and inspired our model.

Table 3. Comparison of ADE and FDE on baseline models and DiVR. Lower scores indicate better performance. Each model takes 3 s input to predict the next 5 s.

Architecture	ADE	FDE
MLP [9]	0.854	1.512
TRACK [16]	0.759	1.247
GIMO [30]	0.757	1.240
DiVR-Hom	0.621	0.993
DiVR-Het	**0.588**	**0.842**

DiVR Models. As shown in Table 3, DiVR-Het not only outperforms the MLP baseline, reducing the ADE and FDE by 31.2% and 44.3%, respectively, but also demonstrates the effectiveness of integrating high-level context as compared to GIMO. This context includes elements such as scene layout, and human interactions, which significantly influence trajectory predictions. Similarly, DiVR-Hom shows significant improvement over the baseline with a 27.3% reduction in ADE and 34.3% in FDE. We show in Fig. 1 qualitative results of the DiVR-Het model.

Low Vision and Complex Task Generalization

Baseline. Table 4 presents the baseline performance and generalization capabilities of various models trained under Normal Vision (NV) with Simple Tasks (ST), and their generalizability to complex tasks and low vision conditions. In the NV+ST test, the GIMO model achieves an ADE of 0.509 and an FDE of 0.745, slightly outperforming the TRACK model. Both models initially demonstrate better ADE values than DiVR.

The detailed results show that under complex task test, the DiVR-Het model adapts well, achieving an ADE of 0.910 and an FDE of 1.661. These results represent a 19.9% reduction in ADE and 23.3% in FDE compared to GIMO, and 28.3% and 25.4% reductions, respectively, compared to TRACK. In low vision condition tests, DiVR-Het also outperforms the baseline models, with an ADE of 0.615 and an FDE of 0.860. This reflects a 2.4% improvement in ADE and 9.6% in FDE over GIMO, and a 3.3% improvement in ADE and 8.8% in FDE over TRACK.

Table 4. Assessment of generalization. Comparison of ADE and FDE on baseline models and DiVR across different test conditions: Normal Vision + Simple Task, complex task, and low vision. Trained under normal vision and simple tasks.

Arch	NV + ST		Complex task		Low Vision	
	ADE	FDE	ADE	FDE	ADE	FDE
MLP [9]	0.652	1.139	1.135	2.174	0.648	1.169
TRACK [16]	0.515	0.706	1.270	2.229	0.636	0.943
GIMO [30]	**0.509**	0.745	1.135	2.163	0.630	0.951
DiVR-Hom	0.561	0.830	1.172	2.248	0.669	1.029
DiVR-Het	0.604	**0.696**	**0.910**	**1.661**	**0.615**	**0.860**

Adaptation to Complex Tasks. Differently from the previous setup, where models were trained only on simple tasks, the models are now trained on both simple and complex tasks with normal vision (Diverse Task Training). Table 5(a) shows the performance under these conditions, demonstrating that DiVR models benefit significantly from this training strategy. DiVR-Hom reduces its ADE from 0.592 to 0.470 and its FDE from 0.908 to 0.816, marking a 20.6% improvement in ADE and a 10.1% improvement in FDE, outperforming TRACK and GIMO under the same conditions. Although DiVR-Het's improvement in ADE is less pronounced than that of DiVR-Hom, it still demonstrates strong generalizability and adaptability.

Table 5. Comparison of ADE and FDE between baseline models and DiVR across Different test conditions: Normal Vision + Simple Task, complex task, and low vision. Two train conditions: Diverse Task and Diverse Vision

Arch	(a) Diverse Task Training				(b) Diverse Vision Training			
	NV + ST		Complex Task		NV + ST		Low Vision	
	ADE	FDE	ADE	FDE	ADE	FDE	ADE	FDE
MLP [9]	1.063	2.116	0.668	1.398	1.136	2.273	0.852	1.764
TRACK [16]	0.916	1.526	0.682	1.343	0.866	1.460	0.664	1.126
GIMO [30]	0.617	1.007	0.589	1.109	0.771	1.267	0.662	1.180
DiVR-Hom	0.592	0.908	**0.470**	**0.816**	**0.547**	0.763	**0.560**	0.791
DiVR-Het	**0.484**	**0.694**	0.506	0.852	0.574	**0.750**	**0.560**	**0.767**

Adaptation to Low Vision Conditions. Table 5(b) shows the performance when models are trained on normal and low vision (Diverse Vision Training) conditions, targeting adaptability to low vision scenarios. In the low vision test, both DiVR-Het and DiVR-Hom achieve an ADE of 0.560, with FDEs of 0.791 and 0.767, respectively. These scores are lower than those of the TRACK and GIMO models.

User, Scene, and Task Generalization

This section analyzes the models' adaptability to user variations, scene complexity, and task challenges, as shown in Table 6. For user generalization, 10 randomly selected users formed the test group, while others were used for training and validation. Scene generalization involved training on single-lane roads and testing on two-lane crossings. Task generalization assessed models trained on simple crossing tasks and tested on scenarios requiring a return to the starting point.

Table 6. Comparison of ADE and FDE values on (1) three architectures (GIMO, DiVR-Hom, and DiVR-Het) for User, scene, and task generalizability.

Architecture	User		Scene		Task	
	ADE	FDE	ADE	FDE	ADE	FDE
GIMO [30]	0.621	0.994	**0.997**	1.479	1.296	**2.100**
DiVR-Hom	**0.513**	0.841	1.011	**1.469**	**1.249**	2.183
DiVR-Het	0.559	**0.828**	1.073	**1.469**	1.384	2.605

User Generalization. The models show good adaptability to user variations, with DiVR-Hom achieving the best ADE of 0.513, a 17.4% improvement over the baseline GIMO model. DiVR-Het also excels in FDE, with a 16.7% improvement, indicating both DiVR models effectively generalize across diverse user behaviors. **Scene Generalization.** Adapting to increased scene complexity from single-lane to two-lane crossings, DiVR models show increases in ADE and FDE to around 1 and 1.4, respectively, demonstrating reasonable resilience given the additional 3.5 m complexity in two-lane roads. **Task Generalization.** This presents the most significant challenge, with FDE values around 2 across all models. However, DiVR-Hom achieves the best ADE, suggesting the model can infer some directional or trajectory characteristics despite the increased task difficulty.

Ablation Tests

We conducted additional ablation studies on the DiVR-Het model. Table 7 shows the results, focusing on the impact of graph and gaze data on trajectory prediction accuracy, measured by ADE and FDE. The study evaluates the DiVR-Het model by replacing either graph or gaze inputs with a tensor of zeros.

Graph Ablation Study. The graph ablation study reveals a significant decline in performance when graph data is omitted from the DiVR-Het model as shown in Table 7. Under NV+ST test, the graph-ablated model's ADE and FDE increase to 0.622 and 0.949, respectively, marking increases of 16.3% in ADE

and 34.6% in FDE. The impact is even more pronounced under complex task conditions, with the ADE and FDE increasing to 0.944 and 1.767, jumps of 35.8% and 45.1%, respectively. In low-vision tests, the increases are 13.0% in ADE and 24.2% in FDE, showing again the usefulness of graph data for maintaining lower error rates across vision conditions.

Table 7. Comparison of ADE and FDE between DiVR-Het model with graph and gaze ablation. Lower scores indicate better performance. Each model takes 3 s input to predict the next 5 s.

Architecture	NV + ST		Complex Task		Low Vision	
	ADE	FDE	ADE	FDE	ADE	FDE
DiVR-Het	**0.535**	**0.705**	**0.695**	**1.218**	**0.583**	**0.871**
Graph ablation	0.622	0.949	0.944	1.767	0.659	1.082
Gaze ablation	0.590	0.802	0.793	1.391	0.673	1.008

Gaze Ablation Study. The gaze ablation study shows decreased performance, but with a slightly lesser impact than graph ablation, indicating that while gaze data is important, it is less critical than graph data in the DiVR-Het model. In NV+ST test, removing gaze data results in an increase in ADE and FDE to 0.590 and 0.802, up by 10.3% and 13.8% compared to the non-ablated model. For complex task tests, ADE and FDE rise by 14.1% and 14.2% to 0.793 and 1.391. In low vision conditions, ADE and FDE increase by 15.4% and 15.7%, demonstrating that gaze data significantly contributes to the model's accuracy in predicting human trajectories, especially in challenging scenarios.

Qualitative Results

Context Data Evaluation. We compare GIMO and DiVR-Het models across three urban scenes. In the first scene, "pressing Traffic Button" (left column, Fig. 3), the observed trajectory (red line) shows a pedestrian disposing of trash and then interacting with a traffic button (blue line). GIMO (top row) incorrectly predicts immediate crossing, while DiVR (bottom row) accurately predicts the path toward the button, closely matching the ground truth. In the second scene, "Waiting for Traffic Light" (middle column), GIMO predicts the pedestrian begins crossing, but DiVR captures the stationary behavior, closely aligning with the ground truth. In the final scene, "crossing" (right column), DiVR's predictions better capture the actual crossing path. These comparisons highlight DiVR's superior performance in predicting complex pedestrian behaviors in urban environments.

User Generalization. In the user generalization section, we visually demonstrate DiVR's adaptability across different users in the test set. Figure 4 illustrate scenarios where DiVR has successfully predicted diverse user movements within the CREATTIVE3D dataset, reflecting its robustness in handling variations in

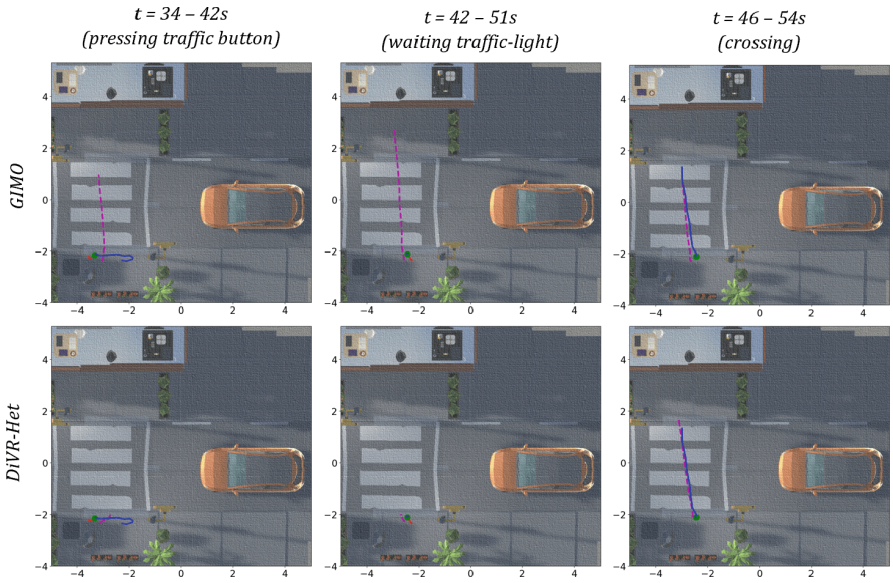

Fig. 3. Context evaluation: pedestrian trajectory prediction comparison between two models, GIMO (scene pointcloud) and DiVR-Het (heterogeneous graphs) in scenes sampled from CREATTIVE3D dataset. In each scene, red lines and green dots show the observed past trajectories and their endpoints, blue lines the ground truth future trajectories, and magenta lines the predicted trajectories. (Color figure online)

pedestrian behavior. Despite varied actions and intentions of users, DiVR consistently aligns closely with the ground truth, showcasing its capacity to accurately generalize across a broad spectrum of user interactions. These qualitative results complement our quantitative findings, where DiVR exhibited robust values on ADE and FDE for user generalization.

Scene Generalization. For scene generalization, the visualizations focus on DiVR's performance in complex urban layouts. In Fig. 5 each plot corresponds to scenarios with increased environmental complexity, from training on single-lane in training to more challenging two-lane crossing tests. While DiVR maintains a reasonable level of accuracy, the predictions reflect the heightened challenge, with slight deviations from the ground truth becoming more apparent. These instances highlight the model's limits and strengths in adapting to spatial variations and complex scene dynamics, aligning with the quantitative results show in Table 6 that indicated a greater challenge in scene generalization compared to user behaviors.

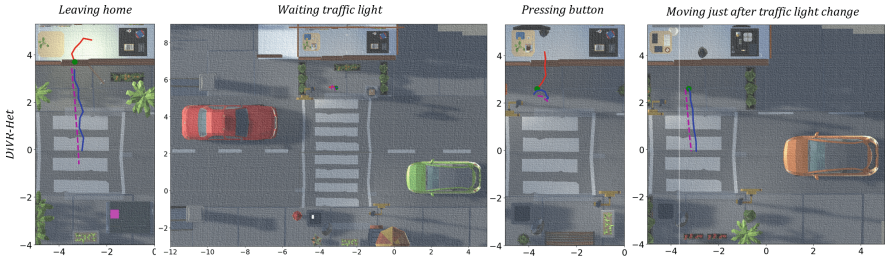

Fig. 4. User Generalization in CREATTIVE3D: Each subplot shows different situations (e.g. leaving home, pressing button, etc.) for users that were not seen on training phase for the DiVT-het model. Predicted trajectories (magenta lines), ground truth (blue lines), observed trajectory (red lines) and green dot marks their endpoints. (Color figure online)

Fig. 5. Scene Generalization: pedestrian trajectory predictions by DiVR trained on single-lane and tested on two-lane crossing. Highlighting its capacity and limitations in adapting to complex urban layouts. Predicted trajectories (magenta lines), ground truth (blue lines), observed trajectory (red lines) and green dot marks their endpoints. (Color figure online)

5 Discussion and Conclusion

We presented DiVR, a multimodal transformer for human trajectory prediction that leverages heterogeneous graphs from rich VR contextual data. To our knowledge, this is the first work to investigate trajectory prediction of real humans in fully interactive virtual environments. Our experiments demonstrated DiVR's robustness across varied conditions, highlighted by extensive generalization tests. A key strength of this study is the detailed evaluation across a wide range of scenarios, showcasing DiVR's effectiveness in handling complex tasks and low-vision conditions through graph-based and temporal modeling techniques.

However, a notable limitation is the model's reliance on high-quality datasets for accurate scene graph creation, which are not widely available. To mitigate this, future work could explore data from smart city infrastructures and autonomous vehicle sensors, offering real-time traffic and pedestrian data. This would enhance the model's applicability and performance in real-world scenarios, aiding in the development of more adaptive urban traffic systems. Additionally,

VR's advantages in incorporating diverse scenes and populations into training highlight its potential for important real-life applications.

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Skeleton-Aware Motion Retargeting Using Masked Pose Modeling

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Abstract. Motion retargeting aims at transferring a given motion from a source character to a target one. The task becomes increasingly challenging as the differences between the body shape and skeletal structure of input and target characters increase. We present a novel approach for motion retargeting between skeletons whose goal is to transfer the motion from a source skeleton to a target one in a different format. Our approach works when the two skeletons differ in scale, bone length, and number of joints. Surpassing previous approaches, our method can also retarget between skeletons that differ in hierarchy and topology, such as retargeting between animals and humans. We train our method as a transformer using a random masking strategy both in time and space, aiming at reconstructing the joints of the masked input skeleton to obtain a deep representation of the motion. At testing time, our proposal can retarget the input motion to different skeletons, reconstructing the disparities between the source and the target. Our method outperforms state-of-the-art results on the Mixamo dataset, which features a high variance between skeleton formats. Moreover, we show how our approach can effectively generalize to different domains by transferring between human motion and quadrupeds, and vice-versa. Our code is available at www.github.com/mmlab-cv/skeleton-aware-motion-retargeting.

1 Introduction

Retargeting the motion from an input subject to a target one can be applied to skeletons [1,15,23], meshes [13,22] or directly on videos [6,26]. Regardless of the chosen medium, the task becomes more difficult as the differences between source and target grow, and as the complexity of the animation to be retargeted increases.

In this work, we tackle the task of motion retargeting between 3D skeletons, which can be employed for multiple applications in motion analysis and computer animation, such as transferring the motion from a real-world character to a

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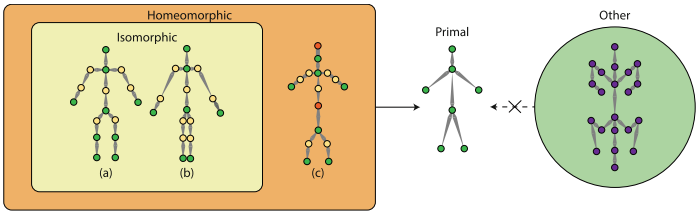

Fig. 1. Examples of isomorphic skeletons are (a) and (b), which have the same joints but varying bone lengths. Both (a) and (b) are homeomorphic to the skeleton (c), which has a different number of joints. They are defined as homeomorphic because they can all be reduced to a common primal skeleton (d). The skeleton of a quadruped (e) is neither homeomorphic nor isomorphic to the others because the topology is changed and thus (e) cannot be reduced to the common primal skeleton.

synthetic one. Such techniques would allow us to accurately animate digital representations in the metaverse, providing a personal set of animation patterns for each character and thus improving the realism and quality of the synthetic environment. In the case of motion retargeting between skeletons, source, and target can differ not only in scale and bone length but also in topology and number of joints.

Skeletons are defined as a graph with vertices and edges represented by bones and joints, with varying topologies. Similarly to graphs, and as shown in Fig. 1, two skeletons can be defined as *isomorphic* when they share the same number of joints (vertices) and bones (edges). If two skeletons are not isomorphic, they can be defined as *homeomorphic*, or topologically equivalent, if they can be reduced to a common primal skeleton [1], by removing the proper number of joints (vertices). While the approaches in the literature tend to confine the retargeting within homeomorphic and isomorphic skeletons [22,23], in our work we deal with non-homeomorphic skeletons, such as retargeting between humans and quadrupeds that differ both in walking pattern and topology (quadrupeds' skeleton also include the tail).

The inability of the existing methods to deal with non-homeomorphic skeletons can be traced back to the use of graph-based algorithms [22,23]. These algorithms usually represent the motion as spatial-temporal graphs and rely on complex encoder-decoder structures [13,26], skeleton pooling and convolution operations [22], or GAN-based approaches [6]. Instead of representing the motion transfer between different skeletons using graphs, we represent each joint as a quaternion that describes the joint's rotation relative to the parent in the hierarchy. Each quaternion is then treated as an individual chunk of data to be processed by our architecture. Since the data chunks are both spatially related (they are joints belonging to the same skeleton) and temporally related (consecutive skeletons in time represent an animation), the task of our method is to model and learn this spatiotemporal relationship.

In computer vision, deep neural networks have exploited the continuous spatial and temporal relationship between pixels of videos and images to improve results on a variety of tasks [9]. However, convolutions and pooling operations do not achieve as good results when dealing with non-continuous data like skeletons, because they do not adequately model the sparsity of the information. Differently, transformers [9] employ a token-based approach and rely on the positional encoder to model the spatial and hierarchical information between image tokens. When working with videos [10,20], the positional encoding can be effectively extended to the temporal dimension. In this work, we show how the transformer architecture is more suited to deal with the 3D spatially distributed and hierarchically organized nature of pose skeleton data. A single simple transformer network can obtain state-of-the-art results without relying on complex encoder-decoder structures, graph convolutional networks, or GANs.

More recently, masked image modeling [12,24] demonstrated how transformers can learn data representations when trained as auto-encoders to reconstruct the masked portion of the input data. When dealing with videos [10,20], the masked portion of the video data can increase up to 90% thanks to both spatial and temporal correlations. Following the same principle, we apply the reconstruction of masked data to skeleton motion modeling. At training time, we randomly mask skeleton joints in space and time, to obtain a rich representation of the input data. At test time, when retargeting a motion from one source topology to a target one, source and target skeletons usually share several common joints (e.g. knees, elbows, wrists). Thus, we selectively mask the missing joints in the target format that need to be retrieved.

Our proposed solution consists of a transformer-based [9,21] architecture, where the encoding models the temporal and spatial relationships between joints, thus allowing for the retargeting of the motion of the skeletons between arbitrary topologies.

The novelties of our work can be summarised as follows:

- we propose a novel self-supervised pipeline to retarget the motion sequence between skeletons;
- we deal with motion retargeting between non-homeomorphic skeletons without paired motion;
- we apply the masking modeling strategy to spatial-temporal skeleton motion, training our architecture to reconstruct random joints of the input image after masking. At test time, the masking is specifically applied depending on the differences between the source and target topology;
- we obtain state-of-the-art results on motion retargeting on the Mixamo dataset [4];
- we retarget the motion between human and quadruped providing results on the quadruped dataset [27].

2 Related Work

Human Motion Retargeting. Skeleton-based motion retargeting aims at transferring the motion from a skeleton topology to a different one [1,15,23]. The skeleton can vary in bone length, scale, and the number of joints.

Skeleton representations have been used to aid both video [6,26] and mesh motion retargeting [13,22].

Video motion retargeting [6,26] aims at transferring a desired input motion to an input video of a person, giving as output a video of the same person performing the alternative input motion. These methods usually rely on separate encoding of video and motion parameters, which are combined to produce the output. To obtain these representations they usually rely on encoding-decoding structures with multiple loss functions.

When dealing with meshes and skeletons, [13] focuses on compact SMPL models, while [22] focuses on avoiding mesh interpenetration while doing retargeting of 3D motion, by detecting contacts between different parts of the mesh, as well as foot-ground contact detection.

Since both video and mesh approaches strongly rely on the skeleton before obtaining good results, we focus on refining the skeleton prediction in challenging scenarios, which can benefit both tasks.

Retargeting Between Isomorphic Skeletons. A basic approach to performing motion retargeting between isomorphic skeletons is the so-called *copy rotation*; starting from a common pose (usually the T-pose), the rotations from one skeleton are copied to another one, without adapting to changes in scale or bone length, as described in [1].

Other works [3,23,29] rely on recurrent neural networks paired with forward kinematics to disentangle the motion from a given character to be transferred to a different one. The authors in [2] use separate encoders to encode motion from videos and feed them to a generator; the generator extracts the relevant motion components, with the same skeleton topology as the input one. On a similar note, GAN-based approaches synthesize novel motion sequences [15] using multi-scale hierarchical generative networks to produce new animations starting from a single motion sequence over multiple scales and time. These methods usually rely on a single input animation, and thus on a single isomorphic skeleton topology. Recent methods [22,28] incorporate mesh data in the retargeting process to address and solve issues like mesh interpenetration. In [28] introduces a skeleton-aware and a shape-aware module. The shape-aware module is trained using an attractive/repulsive field mechanism to resolve collisions while maintaining adherence to the target motion. Despite these advancements and their ability to generalize to various body shapes, our approach achieves better results without relying on a shape-aware module to penalize mesh self-interpenetration.

Retargeting Between Homeomorphic Skeletons. The algorithm presented by Aberman et al. in [1] can retarget the motion between homeomorphic and isomorphic skeletons using an encoder-decoder structure. It relies on a graph

representation for skeletons since this data structure can be easily represented in terms of vertices and edges, achieving state-of-the-art results. However, being graph-based, the solution can not generalize to non-homeomorphic skeletons with different topologies, namely where the number of leaf joints changes. Dealing with non-homeomorphic skeletons is a research area that has been only partially investigated. Several works [19,25] have tried to transfer the motion from humans to non-humanoid characters with non-homeomorphic skeletons. However, the aforementioned methods require either paired motions [25], meaning a sample of the same motion performed by both the input and the target, or an explicit skeleton mapping [19] to be able to perform the motion transfer task.

To the best of our knowledge, we are the first to deal with skeleton-based motion retargeting between non-homeomorphic skeletons without paired motion.

Masked Modeling for Representation Learning. Masked language modeling [8,16] and masked image modeling [12,24] have recently demonstrated how self-supervised approaches can achieve better performances than fully supervised ones, making them scalable representation learners for multiple tasks. In the video domain, where the data is not only spatially but also temporally correlated [10,20], the relationship is strong enough to allow the network to reconstruct the input with up to 90% of the image masked. Transformers architectures [21] have further contributed to improving representation learning when combined with a masking strategy on randomly selected image parts [5,7,9]. Among others, SimMIM [24] has proven to be a simple yet effective strategy, masking image patches by replacing them with random token vectors of the same dimension. In [24], the goal is to train a network to obtain a strong image representation for the prediction task. We apply a similar strategy by randomly masking a subset of skeleton joints in space and time. We demonstrate the effectiveness of the self-supervised learned skeleton representation by using it in the motion retargeting task.

3 Method

Our goal is to train a neural network capable of transferring an input motion A from a source skeleton to a target skeleton with different topologies. The key idea is to use a masking strategy alongside a transformer network to reconstruct the masked portions of the input motion. By masking random parts of skeletons with varying numbers of joints and proportions, the network learns to reconstruct different sections, adapting to their different topologies and developing an awareness of the skeleton structures.

The mathematical notation for the motion representation and the detailed implementation of each step are further explained in the next paragraphs.

Motion Representation. We represent the motion as a sequence of *armatures* (Fig. 2). Each armature is characterized by a dynamic part and a static part. The dynamic representation Q_K consists of N joints (nodes) and each joint j_n is expressed as a 4D quaternion, to describe the relative rotation with respect

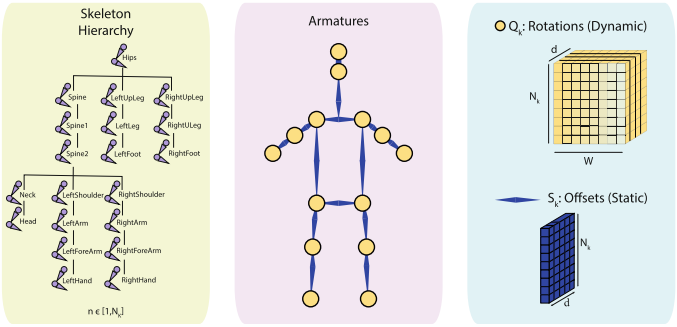

Fig. 2. Motion representation. Each skeleton can be represented as a hierarchical structure, where the parent-child relationship is defined by the kinematic chain. Moreover, each skeleton is described by a static representation S_k containing the offsets (bone lengths), and a dynamic representation Q_k. The length of the animation is a window of size $w \in [1, W]$ frames.

to the parent joint in the skeletal hierarchy. While the static representation S_k does not vary for a character during an animation, the dynamic representation Q_k is subject to changes to obtain the animation.

3.1 Skeleton-Aware Pose Masking Auto-encoder

In the following paragraphs we describe in detail the main components of our transformer-based auto-encoder architecture, illustrated in Fig. 3.

Objective. The motion of a generic character k is defined as $A((S_k, Q_k))$. The skeletal-only motion $A(S_k, Q_k)$ can be divided into the skeletal dynamic motion $A(Q_k)$ and the static representation, which is a learnable token S_k and does not vary across the motion. Given a skeletal motion $A(S_k, Q_k)$, our pose masking auto-encoder performs the retargeting to obtain the same motion $A(S_t, Q_t)$ in a target skeleton format S_T.

Tokenizer. Transformers process chunks of data (tokens) as it happens for words in NLP [21] or image patches in computer vision [9] in their multi-headed architecture. Since we are dealing with human motions, we define individual joint tokens to be embedded. To obtain a descriptor suitable for the motion retargeting task, we represent the joints' position in the 3D space as local rotations. Starting from the root joint (pelvis), each joint is expressed as a 4D quaternion $Q_k(j_n)$ which describes the rotation with respect to the parent joint in the hierarchy. The embedding $E_k(Q_k(j_n))$ is applied to each quaternion using a simple trainable linear layer followed by an activation function φ to obtain a vector of size d for each joint. Thus, each embedded motion A sequence can be expressed as $W \times N \times E_k(Q_k(j_n))$ with dimensions $W \times N \times d$.

Pose Masking Strategy. Similar to the techniques used in image-based masking methods [10,20], our objective is to train an auto-encoder by using masked

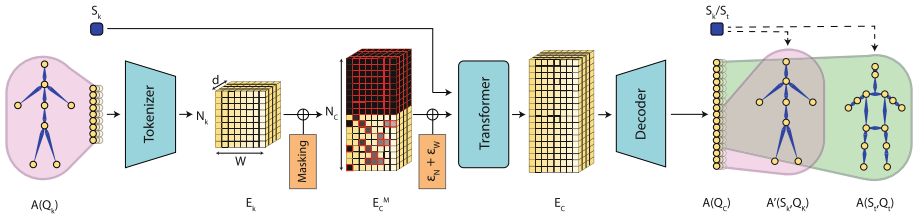

Fig. 3. Skeleton-aware pose masking auto-encoder. Starting with an initial animation $A(Q_k)$, the process begins with a **tokenizer**, which converts each joint of the input into a series of tokens E_k. We then proceed by **masking** (indicated by black squares) a random subset of E_k tokens. The remaining unmasked tokens are concatenated to include every possible topology, forming E_C^M. To capture the relationships between the joints in E_C^M, we add a **spatio-temporal positional embedding**, represented as $\varepsilon_N + \varepsilon_W$. This combined representation is augmented with a learnable token S_k that represents static information, and this forms the input to the **encoding transformer**. The transformer maps the masked input to a latent feature representation E_C, which contains the embedded motion with all masked joints predicted. Subsequently, a **decoder** extracts the super-skeleton motion $A(Q_C)$ from this latent space, using the learned token (S_k during training and S_t during testing). From this, we derive the reconstructed input motion $A'(S_k, Q_k)$ and the retargeted motion $A(S_t, Q_t)$. The auto-encoder is trained to predict the masked joints by minimizing the mean squared error (MSE) loss between the original input $A(S_k, Q_k)$ and the reconstructed output $A'(S_k, Q_k)$.

data as input and reconstructing the absent segments. This approach involves applying a masking strategy to the data, training the auto-encoder to take in incomplete data, and reconstructing the missing portions as output with the information of the unmasked part. Following the tokenization of the input data, we mask a subset of joints M from the embedded animation E_k, which has dimensions $W \times N_k \times d$, resulting in the masked embedded animation. Subsequently, we concatenate $N_C - N_k$ empty, learnable tokens to the masked embedded animation to form E_C^M. This process ensures that the latent representation E_C^M, now of size $W \times N_C \times d$, is sufficiently large to accommodate all possible input skeleton topologies, thereby enhancing the skeleton awareness of our network. Therefore, unlike image-based masking methods, our approach reconstructs not only the missing parts of the input skeleton Q_k but also the empty parts added during the masking process.

Spatial-Temporal Positional Embedding. The purpose of the spatial-temporal positional embedding is to enable the network to learn the inter-dependencies among tokenized joints. Similarly to [10], e adopts separable positional embeddings, thus one for the space and the other for the time. The spatial embedding ε_N with dimensions $N_C \times d$ models the hierarchical graph representation shown in Fig. 2 for all the possible skeleton topologies, and it is repeated for each of the W frames. Its purpose is to learn the parent-child inter-dependencies between all the joints in the kinematic chain. The temporal embedding ε_W with

dimensions $W \times d$ models the time relationship between frames, ensuring the smoothness of the final animation. It is repeated for each of the N_C joints. The total spatial-temporal embeddings are given by $\varepsilon_N + \varepsilon_W$, with dimensions $W \times N_C \times d$. Similarly to [9], the values for both ε_N and ε_W are learnable.

Encoding Transformer. The encoding transformer takes as input the motion sequence combined with the spatial-temporal encoder, represented as $E_C^M + \varepsilon_N + \varepsilon_W$. Additionally, a learnable token vector S_k of size d is appended to model the static components of the skeleton for various skeleton topologies. This transformer, based on a modified ViT architecture [9], utilizes distinct activation functions and multiple attention heads to process the spatiotemporal joint data. The output is a latent space containing the embedded reconstructed motion E_C.

Decoder. In line with the approach from [24], our prediction head is replaced with a straightforward linear layer. The linear layer produces an animation $A(Q_{\mathbf{C}})$, which represents the animation of a super-skeleton $Q_{\mathbf{C}}$ encompassing all possible topologies in $\mathbf{C}$.

During training, we extract the motion $A'(Q_k)$ from $A(Q_{\mathbf{C}})$, corresponding to the input motion $A(Q_k)$ as reconstructed by the auto-encoder. The learned static token S_k serves as a selector for the joints related to a specific skeleton topology. The network is trained to reconstruct all input joints (both masked and unmasked) using Mean Square Error (MSE) loss for each frame of the animation. The training loss is defined as follows:

$$\mathcal{L}_{MSE}(A(Q_k), A'(Q_k)) = \frac{1}{W \times N_k} \sum_{w=1}^{W} \sum_{n=1}^{N_k} (FK(S_k, j_n) - FK(S_k, j'_n))^2 \quad (1)$$

where j_n and j'_n are the n-th joints of a frame w of $A(Q_k)$ and $A'(Q_k)$, respectively, and $FK(-)$ represents a Forward Kinematic layer [1] that allows to express the joints as 3D spatial positions computed from the quaternions.

At test time, given the reconstructed super-skeleton motion $A(Q_{\mathbf{C}})$ we obtain the motion $A(Q_t)$ for each of the possible skeletons Q_t by applying the same Forward Kinematic layer such that $A(S_t, Q_t) = FK(S_t, j'_n)$.

4 Experiments

To evaluate our framework, we perform the retargeting between isomorphic, homeomorphic, and non-homeomorphic skeletons.

Datasets. We evaluate our method on the Mixamo dataset [4], following the experimental setups of [28] and [23] for isomorphic skeletons. This involves collecting 7 characters for training and 11 for testing. For homeomorphic skeletons, we use the same protocol as [1], excluding 3 characters (Liam, Pearl, and Jasper) which are no longer available on Mixamo.

For extra-structural retargeting, we evaluate our method using two datasets: the Ubisoft La Forge Animation Dataset (LAFAN1) [11] and the quadruped

dataset presented in [27]. The LAFAN1 dataset is a human motion dataset containing 5 subjects and 77 different sequences of motions. The quadruped dataset consists of 52 unique dog motion sequences, including idle, walk, run, sit, stand, and a few jumps.

Additionally, we evaluate our method using the Carnegie Mellon University (CMU) Dataset, a real human motion capture dataset comprising 144 actors performing a wide range of motions. Unlike the Mixamo dataset, where retargeting between all characters is achieved through copy rotations, the CMU dataset has been collected using an Optical Motion Capture System. This presents a more challenging scenario since it involves real actors, each exhibiting unique movement patterns and behaviors, characteristics that are not captured in the Mixamo dataset because of the copy rotation technique.

We cannot quantitatively evaluate the retargeting process between the CMU and Mixamo datasets since the motions in these datasets are not paired. Instead, we assess the network's ability to reconstruct parts of the input real motion. This simulates optical capture sessions where occlusions may cause markers to be undetected, resulting in non-smooth animations. By demonstrating our network's ability to reconstruct motion parts from noisy observations, we highlight its potential for enhancing motion capture data post-processing.

In all scenarios, we use an average of 1250 motions for each training character, 100 for validation, and 60 motions for each test character. The test set includes both seen and unseen characters, but only unseen motions. Due to inconsistencies in train/test splitting across different methods, we provide the full list of characters and animations used in the Supplementary Materials to establish a baseline for future research.

Implementation Details. Our network is trained using PyTorch Lightning on a single NVIDIA GeForce RTX 3090. The token vector representing each joint has size $d = 192$, the transformer has 4 heads and the activation function is a Leaky ReLu both in the tokenizer and the transformer. At training time, we randomly mask a subset of $M = 10$ joints. At training time, we randomly mask a subset of $M = 10$ joints for the Mixamo dataset. During training, we set the following hyper-parameters: 25 epochs, learning rate $2e^{-4}$, exponential LR scheduler with $\gamma = 0.95$, batch size 64, and Adam optimizer without weight decay. The optimizer parameters are the same as [17] and $\lambda = 0.7$. As [1], we set the window length to $W = 8$.

Experimental Setup and Evaluation Metrics. As in [1], depending on the variation between the input and target skeletons, the following experiments are defined:

1. retargeting between **isomorphic skeletons**, when the input and target topology have the same number of joints but different bone lengths.
2. retargeting between **homeomorphic skeletons**, when the input and target have different number of joints and bone lengths
3. retargeting between **non-homemorphic skeletons** when input and target have different bone length, number of joints N, and topology as in the human-quadruped retargeting.

To evaluate the performance of our approach we use the Mean Square Error (MSE) between joints for each frame of an animated skeleton. Similarly to [1,28], the MSE is calculated by aligning the root of the retargeted motion to the ground truth (GT), normalizing by each character height.

A crucial aspect of achieving high-quality animation retargeting is addressing mesh interpenetration issues. The Mixamo dataset, while widely used, does not consistently provide clean ground truth animations. In some cases, motions exhibit interpenetration, primarily because the copy-rotation techniques used between characters with significantly different dimensions can lead to these issues.

To enhance the evaluation process comprehensively, we introduce the Face Interpenetration Error (FIE). This metric quantifies the percentage of faces that collide within a given animation. It serves as a direct indicator of the extent of mesh interpenetration present in the retargeted animations:

$$FIE = \frac{1}{W} \sum_{w=1}^{W} \frac{\#\Delta(S_t, Q_t)}{\#F(M_t)}\% \tag{2}$$

where W is the number of frames of the animation, $\#\Delta(S_t, Q_t)$ is the number of colliding faces and $\#F(M_t)$ is the number of total faces of a mesh.

We propose a comprehensive metric called Skeleton and Collisions Error (SCE), which combines both the skeleton-based metric (MSE) and the shape-based metric (FIE) to account for both collision and joint errors. The (SCE) is calculated as $SCE = MSE \times FIE$. This is the first time such a metric has been introduced.

Non-homeomorphic Skeletons. As there is no dataset providing paired motion between two non-homeomorphic characters, it is not possible to compute quantitative results, as no ground truth of the retargeted motion is available.

4.1 Quantitative Results

Isomorphic Skeletons. Table 1 presents the quantitative results for retargeting between isomorphic skeletons, with $N_k = 22$, $N_t = 22$, and $N_C = 25$. For fairness, we evaluated copy-rotation, SAN [1] and R^{2}ET [28] on our data for fairness. We also report the FIE for the ground truth (GT) data to indicate the number of collisions, highlighting the limitations of the Mixamo dataset. When examining the MSE values, both our method and R^{2}ET outperform the baseline copy-rotation approach, demonstrating better retargeting of skeletal animation between input and target characters. However, MSE alone is not a sufficient evaluation metric since low values can still result in numerous mesh collisions. By analyzing the FIE, we see that our approach, despite lacking any mesh collision avoidance implementation, achieves state-of-the-art performance. It delivers a similar MSE to other methods but with fewer collisions, outperforming all approaches except SAN. This indicates that our low MSE corresponds

Table 1. Results for the retargeting between isomorphic skeletons. * indicates values taken from [28].

Isomorphic			
Methods	**MSE**	**FIE↓**	**SCE↓**
GT (Mixamo)	–	4.10	–
Copy	0.045	3.23	0.145
NKN*	0.575	–	–
PMnet*	0.281	–	–
SAN	0.141	**1.53**	0.216
R^2ET	**0.042**	3.96	0.166
Ours	0.043	3.13	**0.134**

Table 2. Results for the retargeting between homeomorphic skeletons.

Homeomorphic			
Methods	**MSE**	**FIE%↓**	**SCE↓**
SAN	0.108	**1.25**	0.135
Ours	**0.025**	3.6	**0.090**

to effectively retargeted motion with fewer collisions. Furthermore, when considering the combined metric SCE, our approach achieves the lowest number of collisions while maintaining poses that closely resemble the original. Moreover, when evaluating the combined metric SCE, our approach achieves the lowest number of collisions while preserving poses that closely resemble the original. This highlights the effectiveness of our method in capturing the static properties of various skeleton types.

Homemorphic Skeletons. Table 2 presents the quantitative results for retargeting between homeomorphic skeletons, with configurations of $N_k = 22$ and $N_t = 25$ or vice-versa $N_k = 25$ and $N_t = 22$, and $N_C = 25$. We compare our method to SAN, the only other approach capable of handling homeomorphic

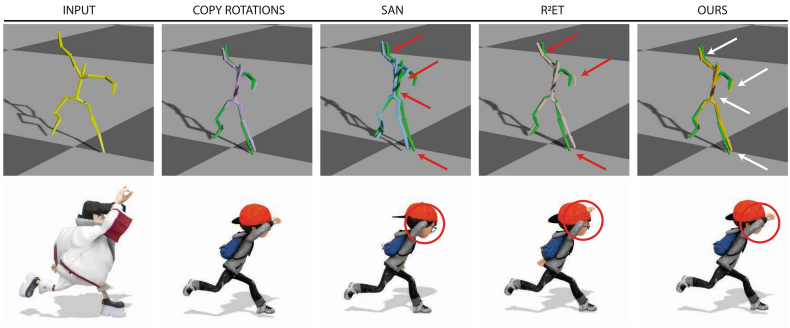

Fig. 4. Qualitative results for the retargeting between **isomorphic skeletons**. The outputs are overlaid with the ground truth, represented by the green skeleton. Our results demonstrate greater stability and alignment with the ground truth. In the second row, we present the mesh visualization. Notably, our method achieves superior qualitative results even without employing any mesh collision penalizer. (Color figure online)

skeletons. Our method outperforms SAN in both MSE and SCE, while SAN is better able to limit collisions as shown by the FIE value. Our approach can achieve comparable results relying only on a simple network approach, that is lightweight and less computationally expensive. At inference time, our method can perform the retargeting at 23 fps compared to the 6 fps of SAN. Moreover, we can generalize to different T-poses, like the quadruped one, which is indicated as a failure case in SAN.

4.2 Qualitative Results

We provide a demo video showcasing the results of motion sequence retargeting across various scenarios and characters in the Supplementary Material.

Isomorphic Skeletons. In Fig. 4, we report the subset of the qualitative results for our approach, compared to copy-rotations, SAN, and R^2ET.

In the first row, we show how other methods such as copy-rotations and SAN cannot fully solve the collisions between the hands. R^2ET solves the collisions, but spreads the hands far apart from each other, losing the original dynamics of the motion. Our method can refine the retargeting using only the skeleton-aware module preserving the original poses and avoiding critical interpenetration between mesh parts.

In the second row, we provide further evidence that our face-based method can better solve the collisions with respect to other methods, being the only one able to avoid interpenetrations between the arm and the head.

Homeomorphic Skeletons. In Fig. 5, we show how our method can avoid interpenetrations compared to the skeleton-aware SAN. It is worth noting that there is a large variation in both the skeleton and the shape, a challenging scenario that causes ambiguities in the position of the arm, also visible in the source taken from Mixamo.

Non-homeomorphic Skeletons. In Fig. 6, we demonstrate the retargeting process from a quadruped to a biped. The two skeletons have distinct joint configurations, as detailed in Fig. 1. The non-homeomorphic skeleton of the quadruped exhibits a significantly different kinematic chain and includes an additional end-effector due to the presence of a tail. As also shown in the Supplementary Material video, we are able to achieve good retargeting between such diverse skeletons and semantically diverse motions, surpassing what was indicated as a failure case in previous approaches [1].

Retargeting from Real-World Character. To demonstrate the robustness and the generalization ability of our method, we show some results from real-world complex motions taken from the Skills from Video (SFV) data [18] in Fig. 7. This unlocks the possibility of transferring to simulated characters virtually any human motion video processed by the common SMPL mesh [14].

Results on CMU Dataset. We evaluate our approach using the CMU dataset, as it consists of real motion capture data from a diverse set of actors performing

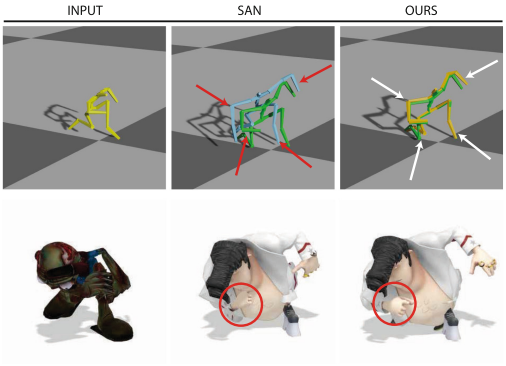

Fig. 5. Qualitative results for the retargeting between **homeomorphic skeletons**. In green the GT.

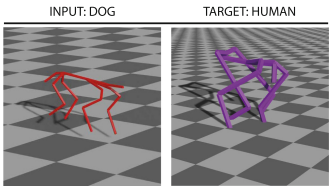

Fig. 6. Qualitative results for the retargeting between **non-homeomorphic skeletons**.

a wide variety of motions. This allows us to validate our proposal on a realistic and varied dataset, unlike the Mixamo dataset, which is generated through copy rotation between characters. Since the motions in the CMU and Mixamo datasets are not paired, we test the network's capability to reconstruct the original data by masking different amounts of joints. This simulates a real capturing session where researchers might encounter issues such as occlusions or prediction errors. As shown in Table 4, our network is capable of reconstructing real data effectively, and the results are consistent with those obtained from the Mixamo dataset. Figure 8 illustrates the corresponding qualitative results.

4.3 Ablation Studies

We conduct the ablation studies using the isomorphic skeletons setup.

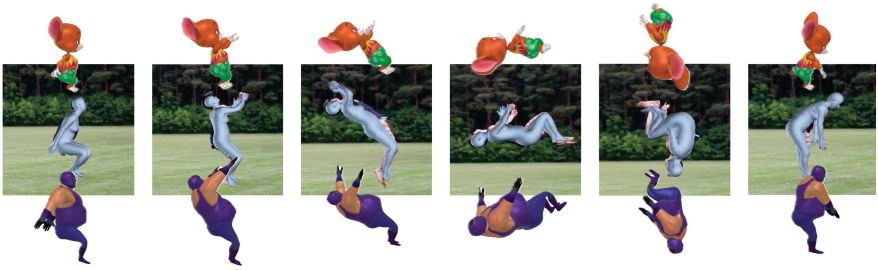

Fig. 7. Our approach can retarget from real-world characters, whose motion can be modeled by a common SMPL mesh, even with challenging and unseen motion such as backflip.

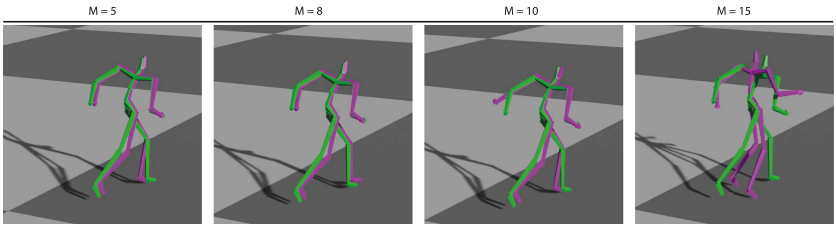

Fig. 8. Qualitative results on the CMU dataset with varying numbers of masked joints. In these visualizations, the ground truth (GT) is shown in green, while the predictions are depicted in purple. The network performs well when up to 10 joints are masked. However, as the number of masked joints exceeds 10, the prediction accuracy decreases significantly. This decline is primarily because masking more than half of the skeleton's joints introduces substantial challenges for the network, leading to increased errors. (Color figure online)

Table 3. Ablation studies on the MSE value for our method. Each column refers to a different experiment: masking of M joints (left), changing the masking strategy (center), ablating the embeddings (right). In bold our best configuration.

M		Masking Strategy		Encoding	
5	0.095	**zero**	**0.043**	No encoding	0.524
10	**0.043**	random	0.095	ε_J	1.025
15	0.094	perturb	1.025	ε_W	1.450
				$\varepsilon_J + \varepsilon_W$	**0.043**

Table 4. Results on CMU dataset with different number of masked joints.

Masked Joints	MSE
5	0.047
8	0.035
10	0.081
15	0.15

In Table 3, we show the ablation studies for our network. Since it focuses only on the skeleton retargeting, we report the MSE values only. Each column reports a different type of experiment. We first evaluate the effect of the number of masked joints M. M plays a crucial role in the learning phase: in fact, if M is low it means that the network does not have many samples in the ground truth to learn from and to reconstruct the $N_C - N_k$ joints. On the other hand, if M is large, it leads the network to reconstruct too many missing tokens and struggles to learn the spatio-temporal relationships between joints, worsening the results. Next, we evaluate the masking strategy, with 0 masking providing a more consistent starting point for the process and leading to better results. Finally, we ablate the spatial and temporal embeddings. Used individually, each models the relationships only through either time or space, penalizing the other dimension. Without both embeddings, the network is neither penalized nor helped to learn the spatial and temporal relationships. We obtain the best results by combining both, meaning that our embedding can model the relationships across both space and time.

5 Conclusions

In this work, we presented a novel approach for skeleton-aware motion retargeting. To our knowledge, our method is the first method that enables fully automatic motion retargeting between isomorphic, homeomorphic, and non-homeomorphic character topologies. We obtain state-of-the-art results on the Mixamo dataset and a visually convincing motion retargeting between the different skeletal topologies. To achieve better results, we aim to develop a shape-aware module to improve collision avoidance between the final meshes. Moreover, we plan to collect a dataset with the paired motion of non-homeomorphic skeletons, to allow for a quantitative comparison.

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LucidDreaming: Controllable Object-Centric 3D Generation

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Abstract. With the recent development of generative models, Text-to-3D generations have also seen significant growth, opening a door for creating video-game 3D assets from a more general public. Nonetheless, people without any professional 3D editing experience would find it hard to achieve precise control over the 3D generation, especially if there are multiple objects in the prompt, as using text to control often leads to missing objects and imprecise locations. In this paper, we present LucidDreaming as an effective pipeline capable of spatial and numerical control over 3D generation from only textual prompt commands or 3D bounding boxes. Specifically, our research demonstrates that Large Language Models (LLMs) possess 3D spatial awareness and can effectively translate textual 3D information into precise 3D bounding boxes. We leverage LLMs to get individual object information and their 3D bounding boxes as the initial step of our process. Then with the bounding boxes, We further propose clipped ray sampling and object-centric density blob bias to generate 3D objects aligning with the bounding boxes. We show that our method exhibits remarkable adaptability across a spectrum of mainstream Score Distillation Sampling-based 3D generation frameworks and our pipeline can even used to insert objects into an existing NeRF scene. Moreover, we also provide a dataset of prompts with 3D bounding boxes, benchmarking 3D spatial controllability. With extensive qualitative and quantitative experiments, we demonstrate that LucidDreaming achieves superior results in object placement precision and generation fidelity compared to current approaches, while maintaining flexibility and ease of use for non-expert users. **Videos are available at:** https://www.zhaoningwang.com/LucidDreaming/.

1 Introduction

As gaming and media industries continue to expand and evolve, 3D content creation has become a crucial element in their development pipelines, ensuring immersive and visually captivating experiences for users. Amidst the notable progress in text-to-image diffusion models [14,37], 3D content generation has

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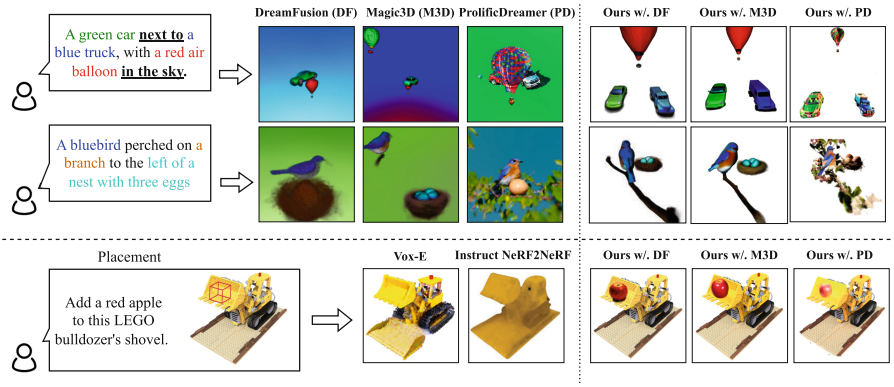

Fig. 1. Our pipeline enables numerical and spatial controls with just textual information, while using baseline methods [21,33,48] only would often fail to fully capture the controlling logic (top). At the bottom, we show an application of our pipeline generating and placing specified objects with a NeRF with user-provided bounding boxes and prompts. This is typically difficult due to NeRF's implicit representation.

seen significant enhancements [5,33,48] with the introduction of Score Distillation Sampling (SDS) [33], which distills knowledge from 2D diffusion models into a Nerual Radiance Field (NeRF) [27], a popular 3D representation. Such progression empowers individuals to instantiate intricate 3D models from mere textual descriptions.

Although it is a great way to create 3D content in spectacular detail, controlling the generation is not a trivial task. Generating and positioning multiple objects often require iterative processes and specialized 3D software skills, such as model editing, limiting complex asset creation to professionals rather than the general public. This limitation creates a barrier to broader adoption and creative exploration in 3D content creation, particularly in gaming and interactive. One straightforward way of exerting control would be using the text prompt directly in the 3D generation, such as *"6 apples arranged in a 3 by 2 grid"*. Nonetheless, diffusion models are deficient in comprehending such underlying logic in the prompts, especially with spatial relations or numbers [15,20]. Furthermore, the distillation of 2D models handles occlusions poorly. For example, the SDS will always try to adhere to "six apples" in every single view in the former prompt, contributing to missing/extra objects and wrong positions. As shown by the generative baselines in Fig. 1 (top-left), objects are either omitted or fused into composite entities when using naive control from the prompts. Consequently, relying solely on textual input to steer the 3D generative process proves to be quite impractical under the current context.

To overcome these constraints, recent methods have implemented controllability from a compositional standpoint, utilizing conditional diffusion models to constraint generation through the projection of 3D boxes within 2D images [32,43]. However, such approaches can not adapt to enhanced diffusion models in a plug-and-play manner without finetuning. We also observe that the 3D bounding boxes used in the current work [32] have to be connected, i.e.,

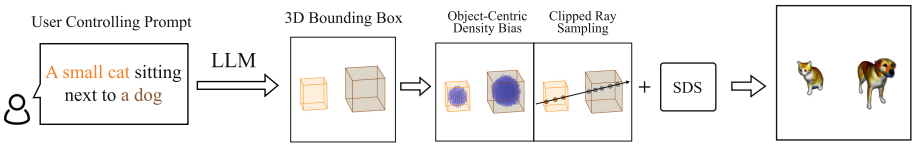

Fig. 2. A high-level overview of our pipeline, controlling prompts are decomposed into 3D bounding boxes with LLMs, such as GPT4. Then in LucidDreaming, object-centric density bias and clipped ray sampling are used with Score Distillation Sampling (SDS) loss to align the generation with the user's control.

all boxes have to share at least one common face with the others. This would prevent such methods from generating objects with distinctness. In this paper, we aim to create a pipeline that unlocks the possibility for people with no professional experience to create more complex multi-objects while also adaptive to different popular SDS-based 3D asset creation methods. This capability is particularly relevant for video game development, where designers often need to rapidly prototype and iterate on complex scenes with multiple, precisely placed objects.

To this end, we propose **LucidDreaming** as a straightforward yet effective method to solve the problem. To begin with, previous work fits the NeRF scenes globally, rendering each view as separate images and only enforcing conditions on the 2D images individually. Manipulating only in 2D image space would lead to inaccuracies in text-to-3D frameworks or the need for conditional diffusion models to maintain control. To resolve such limitations, we propose clipped ray sampling, enabling isolated object rendering in 3D space, and thereby enhancing controllability significantly.

However, the naive integration of clipped ray sampling does not solve the omission or clustering of objects in the baseline methods. Such issues arise due to the common use of uni-sphere density blob initialization in 3D generation [21,33], predisposing towards single-entity creation. In this light, we propose object-centric density initialization, further promoting distinctiveness between generated objects. Moreover, to achieve an end-to-end methodology, we integrate a Large Language Model, enabling control over generation through a simple prompt that translates into bounding boxes.

Apart from existing generation methods which only generate from scratch, our approach is also applicable to 3D content generation within a pre-trained NeRF scene, as shown in Fig. 1 (bottom). Alike changes to an existing scene would normally be achievable through previous 3D editing methods [13,40,56], but only limited to slight semantic modifications to existing objects, like adding accessories or altering object styles. Conversely, our method can generate entirely new objects within existing spaces, independent of the scene's current objects.

Figure 2 illustrates a high-level overview of our framework. Due to the modular design of the method, our pipeline is compatible with a majority of SDS-based 3D generation methods, exhibiting its wide adaptability, exemplified in Fig. 1 (right). We further provide a dataset comprising prompts alongside corresponding 3D bounding boxes and object descriptions, aimed at facilitating

benchmarking in 3D generation controllability. In summary, our contributions are listed as follows:

- We propose LucidDreaming, a plug-and-play pipeline for generating multiple objects with spatial and numerical relationships simply from text prompts, allowing asset creation without any profession 3D software knowledge.
- We present clipped ray sampling and object-centric density bias initialization, for individual object rendering and optimization. This enables controllable 3D content generation both from scratch and in pre-trained scenes.
- We show our method precisely adheres to the controlling condition and is adaptive to multiple SDS-based methods. We illustrate that Large Language Models have 3D awareness and Large Multi-Models such as GPT4-V can be used to evaluate controllability.
- We also provide a dataset of scene prompts with individual 3D bounding boxes with object descriptions for benchmarking controllability.

2 Related Work

2D Diffusion Models. With the introduction of the diffusion models [14,44], there has been significant progress in text-guided image generation, showing remarkable capabilities in creating high-fidelity image synthesis [36,37,39]. Due to the enriched semantics and elevated controllability of pre-trained text-to-image diffusion models, numerous research efforts [20,23,49,54] have proposed controlling image generation with conditioning priors, such as bounding boxes, to meticulously dictate the locations where objects should be generated. Nonetheless, extending 2D controlling methods to a 3D context is not a trivial task due to the increased complexity and dimension in 3D representation.

3D Generation with Diffusion. Recently, implicit 3D representations, notably Neural Radiance Fields (NeRF) [2,3,27,30] have garnered substantial popularity. The fast-growing of 2D diffusion models has facilitated multiple 3D synthesis methods with such representation. In particular, the Score Distillation Sampling (SDS) is introduced in DreamFusion [33] and concurrently in Score Jacobian Chaining [47] to generate 3D contents from the text. The SDS applies 2D Text-to-Image diffusion models to optimize the NeRF representations. A series of subsequent studies [1,4,5,16,21,22,25,34,42,45,48,52,55] based on SDS has improved the quality of outcomes significantly with revised 3D representations and sampling. However, imposing vocabulary constraints, like spatial relationship or numeracy, often results in generation failures, as diffusion models struggle to grasp the inherent reasoning [9,15].

3D Controllability in Generation. Prior to the advent of proficient generative models, the majority of 3D manupulation research has primarily focused on object deformation [17,50,53], animation [51], removal [29], and translation [11,51]. Recently, with the emergence of diffusion models and SDS-based techniques,

controlling 3D representations with generative capabilities has surged, enabling semantic control [41], and compositional manipulation [6,32,43]. However, existing methods are either limited to demanding custom diffusion intermediate [32, 43], or requiring explicit 3D geometry priors [4,6,19]. In contrast, our method enables precise 3D generation control using simple constraints, such as bounding boxes, instead of elaborated geometry priors. We also eliminate the need for specialized mediums like DreamBooth [38,43] or conditional diffusion models [32], allowing flexible adaptation to established SDS-based 3D generation frameworks [33,48] with minimal effort.

3D Controlability in Editing. With the prevalence of NeRF representations, a wide range of research [7,10,13,28,40,46,56] are conducted on controlled edit with a pre-trained NeRF Scene. However, they primarily concentrate on editing a pre-existing object in the editing location, mostly focusing on adding accessories [10,19,40,56], removing objects [46], or altering object styles [7,13]. Our method complements these by focusing on the creation of new objects the void space of the NeRF scene, thereby extending the possibilities of 3D scene manipulation.

3 Preliminary

NeRF and Instant-NGP. Neural Radiance Fields (NeRF) [27] is a concise 3D scene representation through a function f_θ, utilizing a parameterized MLP (multi-layer perceptron) with adjustable parameters (θ). Essentially, NeRF transforms a 3D coordinate ($\mathbf{x}$) and viewing angle (d) to density (σ) and RGB color ($\mathbf{c}$) via $f_\theta : (\mathbf{x}, d) \rightarrow (\sigma, \mathbf{c})$. Given a posed camera, each of its rendered image $\mathbf{I}$'s pixels determines a ray represented by an origin $\mathbf{o}$ and direction $\mathbf{d}$. Then the 3D points can be represented with the ray function: $\mathbf{r}(t) = \mathbf{o} + t\mathbf{d}$, where t is the depth of the ray. For each ray r we sample m points from $\{t_1, t_2, ..., t_m\} \in M$ and we query the MLP and accumulate the outcome to ray color $\hat{\mathbf{C}}(\mathbf{r})$:

$$\hat{\mathbf{C}}(\mathbf{r}) = \sum_{i=1}^{M} T_i \left(1 - \exp\left(-\sigma_i \delta_i\right)\right) \mathbf{c}_i, \tag{1}$$

where the distance of adjacent samples $\delta_i = t_{i+1} - t_i$ and the accumulated transmittance $T_i = \exp\left(-\sum_{j=1}^{i} \sigma_j \delta_j\right)$.

Instant-NGP [30] is an optimized NeRF representation with hash encoding, enabling better performance. It is widely used by numerous 3D generation frameworks. Notably, along with the encoding, it also introduces a binary occupancy grid $G(x)$ to speed up the ray sampling with c and σ in Eq. 1:

$$(\mathbf{c}, \sigma) = \begin{cases} f_\theta(\mathbf{x}, d) & \text{if } G(\mathbf{x}) = 1, \\ (0, 0) & \text{otherwise.} \end{cases} \tag{2}$$

In this way, the empty positions are skipped for faster sampling. The G is updated every 16 steps such that $G(\mathbf{x})$ is set to 1 for all $\mathbf{x}$ where $\sigma_{\mathbf{x}} > threshold$ [30].

Score Distillation Sampling. DreamFusion [33] proposes the SDS loss that distills the knowledge in 2D diffusion models for 3D generation. With the rendered image $\mathbf{I}$ from NeRF θ, it adds Gaussian noise ϵ to $\mathbf{I}$ and applies the loss with the predicted noised ϵ_ϕ from diffusion model ϕ:

$$\nabla_\theta \mathcal{L}_{\text{SDS}}(\phi, \mathbf{I}) = \mathbb{E}_{\epsilon,t}\left[w(t)\left(\epsilon_\phi\left(\mathbf{I}_t; y, t\right) - \epsilon\right)\frac{\partial \mathbf{I}}{\partial \theta}\right], \tag{3}$$

where y would be the input text and t is the level of noise.

Particularly, to guarantee centered position and efficiency, DreamFusion [33] applies a spatial density bias at NeRF initialization with a Gaussian probability density function:

$$\sigma_{\text{init}}\left(\mathbf{x}\right) = \lambda_\sigma \cdot \exp\left(-\frac{\|\mathbf{x}\|^2}{2s_\sigma^2}\right), \tag{4}$$

where s_σ represents the standard deviation regarding density σ and $\mathbf{x}$ is the 3D position. Although such initialization keeps generated 3D objects centered, it adversely affects controllability. As shown in Fig. 4 (a), this initialization creates a sphere-shape density, which leads to poor performance if objects are not exactly at the center. Moreover, combined with the occupancy grid, this would further induce gradient vanishing and no object would be generated.

4 Method

We propose LucidDreaming, an effective way to control 3D generation with bounding boxes. Specifically, we introduce clipped ray sampling in Sect. 4.1 to ensure the individual rendering within the controlling boxes, and object-centric density bias initialization in Sect. 4.2 to place the objects strictly centered within their boxes. Eventually, to make our pipeline end-to-end, we utilize a Large Language Model to decompose a complex prompt into corresponding bounding boxes and descriptions, as detailed in Sect. 4.3.

4.1 Clipped Ray Sampling

The volume rendering in NeRF processes the scene in its entirety, complicating discrete object control. To this end, we propose clipped ray sampling to achieve controllable and individual 3D object generation. Formally, a ray is represented as $\mathbf{r}(t) = \mathbf{o} + t\mathbf{d}$, with origin $\mathbf{o}$, direction $\mathbf{d}$, and depth t. For an object with a bounding box, we can calculate the intersections of the ray with its bounding box, t_{entry} and t_{exit}, as the intersect points the ray gets in and out of the box with methods covered in [26]. Then during the volume rendering, we only keep the points with depth t greater than t_{entry} and less the t_{exit}. Along one ray, for all $\mathbf{x} = \mathbf{o} + t_{\mathbf{x}}\mathbf{d}$, we denote color $\mathbf{c}$ and density σ as:

$$(\mathbf{c}, \sigma) = \begin{cases} f_\theta(\mathbf{x}, d) & \text{if } (t_{\mathbf{x}} > t_{entry} \text{ and } t_{\mathbf{x}} < t_{exit}), \\ (0, 0) & \text{otherwise.} \end{cases} \tag{5}$$

In this way, we can perform rendering separately for each object with Eq. 1 to get the final color $\hat{\mathbf{C}}(\mathbf{r})$. This enables us to easily aggregate the colors into rendered image $\mathbf{I}$ and apply SDS loss $\mathcal{L}^i_{\text{SDS}}$ separately with Eq. 3, where i is the index of objects. By this means, we guarantee each object is rendered within its corresponding bounding box and will not go beyond the range. As exemplified in Fig. 3, the 3D objects of *"a red apple"* and *"a Lego figure"* can be optimized and generated separately with individual SDS losses.

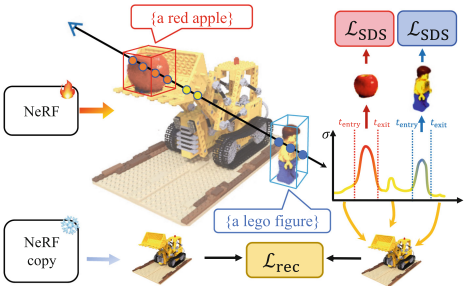

Fig. 3. Clipped Ray Sampling. Points within object boxes are sampled between t_{entry} and t_{exit} for $\mathcal{L}$SDS. Outside points use $\mathcal{L}$rec against frozen NeRF.

Scene Preservation. While the clipped ray sampling ensures that the generation is confined to the specified bounding boxes, it simultaneously discards anything existing outside of these bounds during the sampling. This results in an unpredictable outcome from NeRF in these positions, frequently leading to floaters and noises like those shown in Fig. 8. Additionally, our focus extends to generative processes not solely from a blank canvas but also within the context of a pre-trained NeRF scene. To address both these aspects effectively, we employ inverse clipped ray sampling in the pipeline as a comprehensive solution.

To begin with, we keep a copy of the initialized NeRF model, either from an empty canvas or from a pre-trained scene, and freeze it throughout the generation. In the case of 3D generation, the frozen model would be a network with totally empty occupancy grid, rendering nothing but void. For the pre-trained scenes, it would be the same as the existing scene. Then during the optimization, we render all the points outside the bounding boxes with an inverse clipped ray sampling. Formally, given n bounding boxes, we define inverse clipped ray sampling as:

$$(\mathbf{c}, \sigma) = \begin{cases} f_\theta(\mathbf{x}, d) & \text{if } \left[\bigwedge_{i=1}^{n}(t_{\mathbf{x}} < t^i_{entry} \text{ or } t_{\mathbf{x}} > t^i_{exit})\right], \\ (0, 0) & \text{otherwise.} \end{cases} \tag{6}$$

Then we aggregate the pixels into image $\mathbf{I}_{inv}$, as the rendered image from all 3D points outside the bounding boxes. We also normally render an image from the frozen copied model, with the same camera positions, to get the reference image $\hat{\mathbf{I}}$. After that, we can calculate the reconstruction loss $\mathcal{L}_{\text{rec}}$, which defined as:

$$\mathcal{L}_{\text{rec}} = |\mathbf{I}_{inv} - \hat{\mathbf{I}}|. \tag{7}$$

Finally, the overall loss to update the NeRF model for n objects is formulated as:

$$\mathcal{L} = \sum_{i=1}^{n} \mathcal{L}^i_{\text{SDS}} + \alpha \mathcal{L}_{\text{rec}}, \tag{8}$$

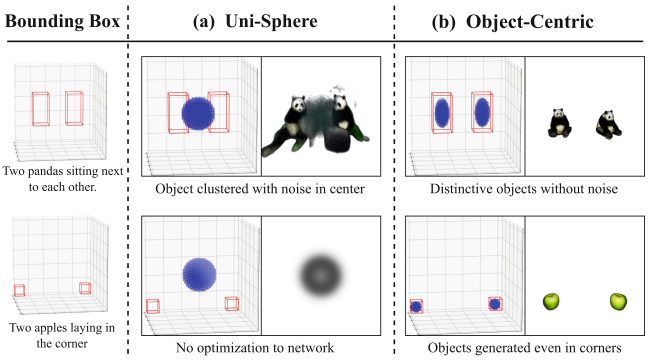

Fig. 4. We show two toy examples in illustration of the occupancy grids with clipped ray sampling. With default uni-sphere density bias (a), the objects are either clustered to the center (top), or totally missing due to gradient vanishing (bottom), while our object-centric bias (b) aligns the object's initial density with the given bounding boxes.

where α is a hyperparameter used to weigh two losses. The overall illustration of clipped ray sampling is shown in Fig. 3. We find using such scene preservation with reconstruction loss greatly helps in preventing the noise in generation and persevering the scene in placement. The effects are demonstrated in Figs. 8 and 7, along with ablation studies in Sect. 5.4.

Camera Pose Sampling. In scenarios with small target bounding boxes, the SDS loss with clipped ray sampling tends to be disproportionately influenced by larger objects, which contain more sampled points under constant camera distances. Such sample point discrepancy would lead to suboptimal representation or even the vanishment of smaller ones. To address this, we update the object-centric camera sampling [32] and keep the camera distance proportional to the longest side of the bounding boxes, pulling the camera closer to smaller objects. More comprehensive technical details can be found in the Supplementary Material.

4.2 Object-Centric Density Bias Initialization

While clipped ray-sampling is a great way to separate the volume rendering of different objects, directly applying such a method to 3D generation frameworks, such as DreamFusion [33], would normally lead to poor outcomes due to the initial uni-sphere density bias in Eq. 4. As shown in the top figure of Fig. 4 (a), the objects would cluster into the center, and any density outside the bounding box would remain as noise in the final outcome. Even worse, if an object's bounding box is far from the center and has no intersection with the initial density uni-sphere, it leads to gradient vanishing and results in no optimization to objects. This occurrence can be attributed to the combination of Eq. 6 with the occupancy grid in Eq. 2, wherein the condition of feed-forward becomes ($t_{\mathbf{x}} > t_{entry}$ and $t_{\mathbf{x}} < t_{exit}$ and $G(\mathbf{x}) = 1$). With the occupancy grid being created from initialized density, this would entail no 3D point $\mathbf{x}$ being feed-forwarded into the MLP, yielding no optimization to the network. As shown in

Fig. 4 (a) bottom, with uni-sphere density initialization, no objects are rendered and the network stays in its initial state.

To this end, we introduce object-centric density bias to accurately position the initial density within the designated bounding boxes. We adapt and modify the initialization method described in [21] to incorporate an object-centric approach. More formally, for the n objects with bounding boxes, the initial density bias is formulated as:

$$\sigma_i(\mathbf{x}) = \lambda_\sigma \cdot \left(1 - \frac{\| \frac{\mathbf{x}-c_i}{l_i} \|_2}{s_\sigma}\right), \qquad (9)$$
$$\sigma_{\text{init}}\ (\mathbf{x}) = max(\sigma_1(\mathbf{x}), \sigma_2(\mathbf{x}), ..., \sigma_n(\mathbf{x})),$$

where c_i and l_i are the center and length of sides of the i-th bounding box, and $i \in \{1, 2, \ldots, n\}$. The overall density bias would be the max of all bias from the bounding boxes. In this way, the initialized density would be within the bounding box, with enough points satisfying the condition in both Eqs. 2 and 6 while leaving minimum density occupancy outside the boundary to prevent noise. The density will also be centered in the boxes, avoiding clustering towards the scene center. As shown in Fig. 4 (b), the initialized density is aligned with the bounding boxes, without center-clustering (top) or gradient vanishing (bottom). Moreover, the initialized density blobs are no longer strict spheres but are ellipsoids in regards to the shape of the bounding box, facilitating shape alignment.

4.3 Generating 3D Bounding Boxes with Prompts

The bounding boxes offer a manageable way to direct the generation process. However, configuring these boxes still necessitates some manual labor, belittling the end-to-end intent of our pipeline. As the Large Language Models (LLM) excel in understanding complex logics [20], we adopt a similar approach in LLM-guided diffusion [20] to use the LLM as a layout generator, creating 3D bounding boxes for a given text prompt. We follow LLM-guided diffusion in the prompting and in-context learning of the LLM as a layout generator. We make the following adjustments in the context of 3D space:

3D Coordinate System. We instruct the LLM (*e.g.* GPT-3.5/4) to generate in accordance with [12] in space of $[512, 512, 512]$, and provide layout examples for in-context learning.

Containment Relationship. Our findings suggest that conceptualizing the containment relationship (the inside/outside dynamic) as a single object could lead to improved results. Consequently, we prompt the Language Model to combine such relationships.

An example of responses can be shown below:
Caption: A pair of brown shoes placed neatly next to a black briefcase with a blue tie draped over it.
Objects: [('a pair of brown shoes', [0, 0, 0, 256, 256, 200]), ('a black briefcase with a blue tie draped over it', [256, 0, 0, 256, 256, 300])]

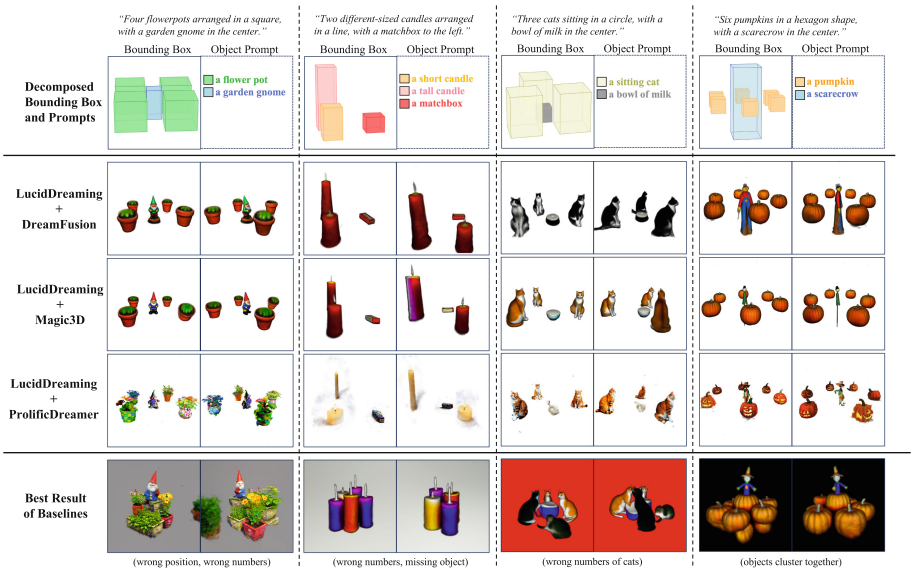

Fig. 5. Examples of controlled 3D generation. The bounding boxes and prompts are decomposed from the scene prompt with an LLM. We show our method is adaptable to multiple SDS-based 3D generation methods to generate Bounding Box-controlled 3D content. In the last row, we show the best of three baseline methods with the scene prompt. Clustered objects, missing items, and wrong spatial are the most common issues in the baseline methods. Please refer to the supplementary for more results and frameworks adapted to ours.

The objects are represented as tuples of a description utilized in SDS loss calculation and the bounding box specified by *[x, y, z, depth, width, height]*, inspired by [20]. This output can then directly serve as bounding boxes to guide the controlled 3D generation with methods discussed in Sects. 4.1 and 4.2. We show the complete prompt and more textual and visualizations of the response in the Supplementary Material.

5 Experimental Results

This section describes the experimental results on controllable 3D generation with LucidDreaming. First, we describe the implementation detail and setup in Sect. 5.1. Then, we show the quantitative results of our method compared to baselines in Sect. 5.2. In Sect. 5.3, we present qualitative results of controlled 3D generation and placement, and Sect. 5.4 offers ablation studies on the employment of scene preservation and effects of individual components. More details are provided in the Supplementary Material.

Table 1. Quantitative evaluation results with three baselines. The reported metrics are the average of individual-rated scores.

Model	BLIP-VQA	GPT4-V	Human
DreamFusion	20.89	31.08	41.67
+ **Ours**	**30.96**	**62.96**	**62.37**
Magic3D	27.08	42.04	43.70
+ **Ours**	**35.87**	**69.23**	**69.81**
ProlificDreamer	35.09	41.14	42.60
+ **Ours**	**38.73**	**65.65**	**64.80**

Table 2. Ablation of each component of our LucidDreaming.

CRS	OCDB	SP	BLIP-VQA	GPT4-V
✗	✗	✗	27.08	28.13
✓	✗	✗	13.95	0.46
✗	✓	✗	28.86	43.47
✓	✓	✗	34.63	66.08
✓	✓	✓	35.87	69.23

5.1 Experiment Setup

Implementation. In our experiments, we adopt the implementation from Three-Studio [12] for Text-to-3D generation frameworks and build our methods upon their codebase. We kept most of the default settings the same for generative frameworks. Notably, We adjusted all sampling resolutions for SDS to 128 and the training steps to 10,000 for fair comparisons. We also use Instant-NGP as NeRF representation for all methods in accordance with [12].

Baseline Models. Our experiments mainly focus on implementing and comparing with three SDS-based Text-to-3D generative frameworks: DreamFusion [33], Magic3D [21], and Prolific Dreamer [48]. Following [12], we use Deep Floyd IF [8,39] as the Distillation Model for DreamFusion and Magic3d, and Stable Diffusion 2.1 [37] for ProlificDreamer.

Dataset. To facilitate controllable 3D generation and establish a standard benchmark for future work, we provide a dataset that consists of 150 scene prompts, each accompanied by corresponding bounding boxes and object descriptions. The prompts predominantly feature scenes with multiple objects and emphasize spatial relationships, numeracy, and various sizes of 3D objects. Specifically, we gather 3D bounding boxes generated from GPT-4 [31] and subsequently validate them with human annotators. We will release the dataset for future research.

5.2 Quantitative Results

Evaluation Metrics. In this paper, we include three metrics to quantify the controllability of 3D object generation. First, we follow the T2I benchmark [15] for object-wise evaluation with BLIP-VQA [18]. The BLIP-VQA independently verifies each object in the prompt and aggregates the probabilities to a final score. Second, we also adopt the Chain-of-Thought evaluation prompt from T2I benchmark [15], which requires the model to explain the image first and then relate to the given prompt and give a rating from 0 to 100. We use such prompting with the GPT4-Vision (GPT4-V) [31][1] to evaluate the results. Eventually, we

[1] About 1% of evaluated images inexplicably trigger the OpenAI safety protocol, so we omit those in the calculation of the average ratings.

gather feedback from human annotators for the alignment between text and image. We ask the annotator to rate from 1 to 5 and normalize the options to 0-100. To avoid human biases, we gather at least three annotations per image and report the average.

We show the quantitative results in Table 1. It can be clearly seen that LucidDreaming achieves significant improvements in controllability than the baseline methods. For example, we achieve 8.8% and 26.9% improvements compared to Magic3D results with BLIP-VQA and GPT4-V, which demonstrates our method outperforms the baseline results with significant improvements in terms of both 3D quality and controllability. In addition, our method shows notable enhancements in human evaluations and preferences, emphasizing its superiority in controlled 3D generation with improved quality and precision.

Discussion. While CLIP scores are commonly used for evaluating 3D generation for quality measurement, their suitability for assessing controllability is limited [15]. We provide detailed comparative results and discussions in the Supplementary Material.

5.3 Qualitative Results

Controlled 3D Generation. We provide qualitative results of controlled 3D generation in Figs. 1 (top) and 5. Results demonstrate that our method can effectively exert control over 3D generation given merely a scene prompt. More specifically, we show the scene prompt and decomposed 3D bounding boxes with object descriptions in the first row of Fig. 5, which are used as conditions for controllable object-centric 3D generation with LucidDreaming. The results in the middle rows of Fig. 5 and the top-right of Fig. 1 highlight that our method is capable of generating objects closely adhered to the controlling bounding boxes and the scene prompts. Moreover, these results serve as examples demonstrating the integration of LucidDreaming with various 3D generation methodologies, confirming its adaptability and compatibility across a wide range of 3D generation frameworks.

In comparison, we display the best (in terms of GPT4-V rating) among the three baseline methods in the bottom row of Fig. 5. It is observed that missing items, wrong positions, and object clustering to the center are not uncommon in baseline generations. This is largely attributed to the diffusion model's deficiency in understanding complex prompts, such as those with numeracy and spatial relationships, and the limitations of constraining only from 2D rendering, lacking explicit 3D controls. This confirms that relying exclusively on text prompts for conditioning and controlling 3D model generations leads to unreliable outcomes.

Controlled Object Placement to Scene. In Figs. 1 (bottom) and 6, we show qualitative results of content generation and placement within a pre-trained NeRF representation on the blender-synthetic dataset [27]. In Fig. 6 (right), we demonstrate our capacity to generate objects given defined bounding boxes while maintaining the overall integrity and composition of the original scene. In our experiments, we choose Vox-E [40] as the representative editing method for the

Fig. 6. Unlike existing methods focused on global scene editing (*e.g.*, Vox-E), our approach enables the controllable generation of high-quality 3D content while preserving the original scene.

placement comparison, and generate content in blender-synthetic objects in an editing approach. As depicted in Fig. 6 (middle), when conditioned solely on text prompts and lacking precise control mechanisms, Vox-E primarily alters the semantic properties of existing objects rather than creating new elements in the areas designated by the prompts. Instead, LucidDreaming is able to create full 3D objects with more fine-grained control.

5.4 Ablation Studies

We find the scene preservation discussed in Sect. 4.1 plays an important role in the generation quality. As shown in Fig. 8(Appendix), this approach effectively eliminates noise in the area outside the bounding boxes. In addition, it preserves

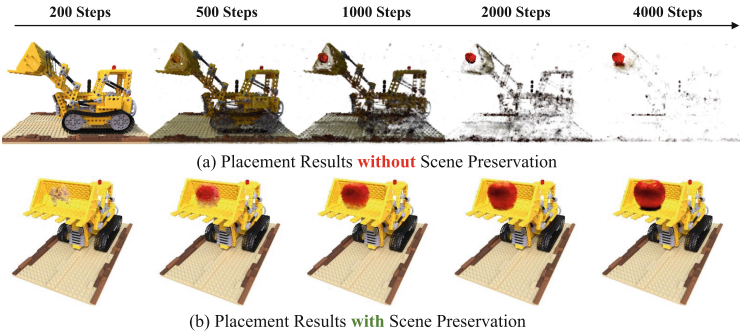

Fig. 7. Ablation on scene preservation with reconstruction loss in pre-trained scenes, validating its effectiveness in preserving the integrity of the original objects.

the original object intact when placing new content in a pre-trained NeRF scene. In contrast, naively resuming from the pre-trained NeRF and generating object would lead to catastrophic forgetting and destruction of the integrity of the original object. We present a progressive comparison in Fig. 7, demonstrating that scene preservation effectively prevents the bulldozer from diminishing.

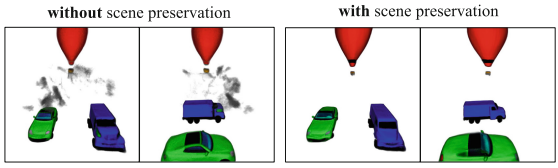

Fig. 8. Ablation experiments on scene preservation in generation from scratch. It helps to prevent noises during the generation.

Ablations on LucidDreaming Components In Table 2, we provide ablations studies on components in LucidDreaming with Magic3D as the backbone generation framework, including the proposed Clipped Ray Sampling (CRS), Object-Centric Density Bias (OCDB), and Scene Preservation (SP). Specifically, with only CRS, LucidDreaming fails hard with unsatisfying BLIP-VQA and GPT4-V rating, primarily due to gradient vanishing discussed in Fig. 4. Using OCDB slightly improves the controlled generation, but it achieves a much higher rating when combined with CRS and constrained by bounding boxes. The SP further prevents floaters and noises and enhances the final quality.

6 Conclusion

In this paper, we demonstrate that the existing Text-to-3D generation methods lack fine-grained controllability, while controllable counterparts are limited due to their reliance on customized diffusion models and their inability to easily adapt to other pipelines. To overcome these challenges, we introduce LucidDreaming to achieve controllable object-centric 3D generation from scratch or

pre-trained scenes. We propose clipped ray sampling and object-centric density bias initialization to generate discrete 3D objects that conform to given specifications. Our method can enhance the controllability of various 3D generation techniques in a plug-and-play manner, achieving fine-grained control over the created contents. We include more discussions on limitations and future works in the Supplementary Material.

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BehAVE: Behaviour Alignment of Video Game Encodings

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Abstract. Domain randomisation enhances the transferability of vision models across visually distinct domains with similar content. However, current methods heavily depend on intricate simulation engines, hampering feasibility and scalability. This paper introduces BehAVE (https://sites.google.com/view/behavefw/home), a video understanding framework that utilises existing commercial video games for domain randomisation without accessing their simulation engines. BehAVE taps into the visual diversity of video games for randomisation and uses textual descriptions of player actions to *align* videos with similar content. We evaluate BehAVE across 25 first-person shooter (FPS) games using various video and text foundation models, demonstrating its robustness in domain randomisation. BehAVE effectively aligns player behavioural patterns and achieves zero-shot transfer to multiple unseen FPS games when trained on just one game. In a more challenging scenario, BehAVE enhances the zero-shot transferability of foundation models to unseen FPS games, even when trained on a game of a different genre, with improvements of up to 22%. BehAVE is available online (https://github.com/nrasajski/BehAVE).

1 Introduction

Video game engines uphold an internal representation of the game environment [12,17], encompassing essential variables such as player position and map layout. Upon undergoing processing by the game graphics *renderer*, this data becomes intricately entwined with the game's visual style, resulting in the images presented to the player on screen. Securing access to game engine data, however, proves challenging, if not impossible in practice, particularly for commercial

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video games. Consequently, this circumstance directs the trajectory of game artificial intelligence (AI) research towards the utilisation of more accessible game representations such as pixels [14,39]. Unfortunately even with state of the art pre-trained computer vision (CV) models the resulting game pixel encodings do not generalise well, even between games of the same genre, and suffer from what is known as the *domain gap* problem [35] (see Fig. 1a).

(a) Foundation Encodings **(b)** BehAVE Encodings

Fig. 1. High level overview of the **BehAVE** framework. The t-SNE plots show encodings of short video sequences from 5 distinct FPS games: (a) indicates the *domain gap* between encodings of different games from a video foundation model, while (b) shows encodings aligned by BehAVE. The framework positions similar player behaviour encodings (*e.g.*, aim gun) closely across visually diverse games like *PUBG* (left) and *Apex Legends* (right).

Improving model generalisation stands as a pivotal problem within machine learning (ML) research, encompassing various research topics such as domain adaptation, meta-learning, and transfer learning. Domain generalisation [33] has emerged as a focal point of interest aiming to develop models capable of effectively generalising across unseen testing domains by leveraging training data from diverse but related domains. One highly promising technique for enhancing the transferability of CV models in games is **domain randomisation** [28], a simple technique that improves the robustness of a CV model by training it on visuals derived from randomising rendering parameters of a simulator engine. Building accurate large-scale simulators, however, is a formidable challenge which requires significant time, expertise, and effort [6]. Identifying and randomising relevant simulation parameters adds further to the complexity, making the entire process a resource-intensive endeavour. Additionally the randomisation process can occasionally lead to inaccurate and infeasible results.

To address the above mentioned issues, in this paper we introduce a novel approach to domain randomisation that leverages the rich visual diversity inherent in **video games**; see Fig. 1. Specifically our framework distinguishes itself by not relying on game engine access for the randomisation process. This unlocks the potential of CV to be trained and tested on videos from commercial-grade video game titles, a capability that has, until now, remained elusive. Our Behaviour Alignment of Video Game Encodings (**BehAVE**) framework harnesses the unique characteristic of gameplay videos opposed to any other videos available. Gameplay video footage is generated by sequential player actions

(*i.e.,* controller inputs) that control on-screen animations characterised as *player behaviour*. Using player actions, BehAVE is able to align video encodings of similar player behaviours across visually diverse games (*i.e.*, different *domains*). Crucially, it employs *semantic action encoding*, a method that infuses semantic information about behaviour through textual descriptions of actions, which are encoded using pre-trained text encoders [15]. As a result, the text encodings of player actions guide the behavioural alignment of video encodings.

We train BehAVE's alignment module, implemented on top of foundation video encoders [30], across a diverse array of games from the first person shooter (FPS) game genre, namely our introduced *SMG-25* dataset. Our findings suggest that BehAVE is able to uncover similar behavioural patterns—despite visual distinctions, such as variations in game style or aesthetics—across unseen games of the SMG-25 dataset (see Fig. 1 for an illustrative example). The alignment training proves efficient even with small datasets comprising only a few games, and it demonstrates robustness across various tested video and text encoders.

To assess the *transferability* of BehAVE we test it on a video understanding task across various FPS games, while solely being trained on the FPS game *Counter Strike: GO* (Valve, 2012). Further, we test a more challenging scenario evaluating the transfer performance to the FPS genre from *Minecraft* (Mojang, 2011) a first person game from a different genre (non-FPS). Our findings indicate higher transferability when learning to classify behaviour from our aligned representation space as compared to without, showcasing up to 22% higher classification accuracies across the different behaviour categories tested. We view this as a potential avenue for annotating extensive datasets of online gameplay videos with behaviour labels, thereby serving as a stepping stone towards learning generalised representations of behaviour in videos. Our contributions can be summarised as follows: (1) We introduce the **BehAVE** framework for domain randomisation via commercial video games; (2) We propose **Semantic Action Encoding** for representing player actions as textual descriptions processed through a pretrained text encoder; (3) We introduce the **SMG-25** dataset of synchronised gameplay and actions.

2 Background

Video Understanding in CV. Video understanding methods seek to interpret visual information embedded within temporal image-sequences. Recent strides in deep learning have led to attaining remarkable performance in diverse video understanding tasks, including but not limited to *video classification* [4], *video summarisation* [1], *short and long-form video understanding* [38], and *object tracking* [44]. Current endeavours focus on training strategies that are independent of any specific downstream task. The resulting *video foundation models* [34,36] yield powerful video representations, readily applicable across a diverse range of tasks. We use such foundation models in our study courtesy of their out-of-the-box performance and employ them as is (*i.e.* frozen) bounded by limited computational resources [29]. This underscores the computational efficiency of our video understanding framework, ultimately enhancing its accessibility.

Transferable CV and Domain Randomisation. Despite their impressive out-of-the-box performance, foundation models showcase limited capacities on transferring knowledge from one domain to another *visually distinct* domain, primarily due to the "domain gap" challenge [31,32]. Tobin et al. [28] introduced the technique of *domain randomisation* to train transferable vision models by injecting variability during learning. This is achieved by randomising the rendering of a simulator that generates training data. Leiprecht [11] showcases the efficacy of domain randomisation in CARLA [6], a large-scale driving simulator. Mishra et al. [13], however, bring to light the numerous complexities associated with identifying and tweaking relevant parameters of such simulators. Furthermore, Kim et al., [8] emphasise the *limited* variability that can be attained from a single simulator, impacting the transfer capacity [42]. Hence, in this work, we adopt a simulator-free approach for visual domain randomisation.

Video Games for CV. Inspired by insights from [20,23] suggesting that procedurally generated sets of diverse games enhance generality in machine learning, we explore the use of existing video games in CV. Several recent studies investigate the use of commercial-standard games as an alternative to dedicated in-lab simulators or procedural game level generation approaches, in an attempt to circumvent limitations related to inaccessible game engines. Notably, *Grand Theft Auto 5* (Rockstar, 2013) serves as a popular video game for collecting annotated data, achieved by intercepting rendering communication between graphics hardware and the screen buffer [10,22,27] or employing a game modification such as "infrared vision mod" [7]. Alternatively, Pearce and Zhu [16] gather internal game state information from *CS:GO* by probing the machine's memory. In contrast to such prior works involving the "reverse-engineering" of game engines, our approach captures high-level game information such as player actions using raw inputs from the machine's I/O devices, thereby simplifying the collection of annotated gameplay and boosting the scalability of our method across numerous commercial video games.

Multimodal Alignment of CV Models. Given that BehAVE considers different modalities of input such as videos of gameplay and corresponding player actions, we draw inspiration from contemporary work in *video action recognition* [43]. In their work with a paired video-text caption dataset, Song et al. [26] extract *verbs* from captions and use them as *action labels*. Our framework builds on similar principles utilising language models [15], but instead, encodes player actions; BehAVE then uses these action encodings for alignment with another modality, namely gameplay videos. To achieve this, we rely on *multimodal alignment frameworks* that operate with and align vision and language such as CLIP [18] and VideoCLIP [9,40]. Drawing upon insights from the aforementioned studies, we propose a novel method for performing visual domain randomisation with commercial games by aligning gameplay videos with semantically represented player actions.

3 The BehAVE Framework

As introduced earlier, we present *BehAVE*, a framework operating on paired visuals-and-actions datasets derived from commercial games, with the aim of aligning video encodings based on similar player behaviour. The BehAVE method is presented in Algorithm 1 and visually depicted in Fig. 2a. In Sect. 3.1, we explain the special structured dataset of games imperative for our framework, followed by Sect. 3.2 covering the encoder models used for both modalities. Finally, Sect. 3.3 details the training method employed.

Algorithm 1. Behaviour-Alignment Training with BehAVE

1: **Inputs:** Games Dataset $\mathbb{D}$, semantic action mapper m, pre-trained text encoder h, pre-trained video encoder f and trainable alignment projector p.
2: **for** (video V, actions A) in $\mathbb{D}$ **do**
3: Compute video encoding $z^{\text{video}} = f(V)$
4: Compute action encoding $z^{\text{caption}} = h(m(A))$
5: Project to aligned encoding $z^{\text{align}} = p(z^{\text{video}})$
6: Calculate loss $\mathcal{L}_{\cos} = 1 - cosine(z^{\text{align}}, z^{\text{caption}})$
7: Update projector network parameters p_θ
8: **end for**
9: **Output:** Trained alignment projector p.

3.1 Games Dataset for Training BehAVE

A crucial component of our domain randomisation framework involves the meticulous preparation of a dataset adhering to a specific *structure* that accommodates semantically similar visual content represented across diverse visual styles. We enforce this structure via the selection criteria of the various commercial games in the training dataset. Note that since BehAVE is trained upon player-game interaction data, we do not require access to the game engines, making it a viable strategy to use commercial games.

Game Selection for Domain Randomisation. Let $\mathcal{G}$ represent a game, with a frame-renderer g, and $\mathcal{G} \in \mathcal{C}$, where $\mathcal{C}$ denotes the family of games of a certain game genre category. Given that domain randomisation with customisable simulators involves the adjustment of simulator render parameters, we proceed to identify and formally define comparable parameters ξ within the context of video games. Each game's renderer encompasses certain game-specific parameters denoted as "game style parameters" ($\xi^{\mathcal{G}}$), associated with either the visual aesthetics of the game, such as *textures and colours of objects*, or the underlying rules governing the game, such as *game physics*. These parameters are considered invariant throughout the game, reflecting game design choices made during

development, and are less likely to be shared across all games of this genre category. In the context of our analysis involving multiple commercial games, we observe $\xi^{\mathcal{G}} \in \Xi$ where Ξ represents the diverse *global game-design space*, introducing implicit "randomisation" into our framework. This unique characteristic of games makes them ideal for the purpose of visual domain randomisation.

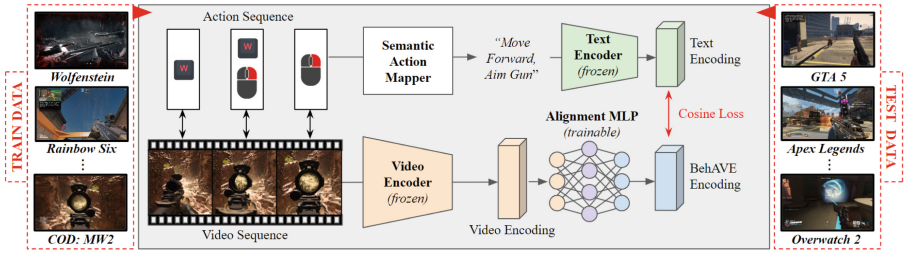
Additionally, we also characterise all game-state-specific parameters of the renderer, including the *player's spatial coordinates*, *health or ammunition status*, and *camera perspective*, as "game content parameters" ($\xi_t^{\mathcal{C}}$), which dynamically evolve at each timestep t in response to player interactions with the game environment. Note that these parameters remain largely consistent across different games that are categorised under the same genre.

Synchronised Gameplay Recording. At each timestep t, we obtain two synchronised information streams-visuals and actions. Player inputs or actions are selected from the shared action space of the game genre $\mathcal{C}$ and are represented by a set of $N^{\mathcal{C}}$ unique keypresses as $A_t = \{a_n\}_{n=1}^{N^{\mathcal{C}}}$, where $a_n \in \{0, 1\}$. Visuals are recorded in the form of RGB frames $F_t \subset \mathbb{R}^{h \times w \times 3}$ where h is height and w is width. The visuals of a game can be regarded as dynamic sequences of frames that arise from the interactions between the player and the game, as follows: $F_{t+1} = g(F_t, A_t, \xi_t^{\mathcal{C}} \mid \xi^{\mathcal{G}})$. Thus, the video frames are generated sequentially by the game renderer g processing the game content and player action information at every timestep for the given predefined game style. This inherent characteristic of gameplay visuals, derived from player interactions, enables us to employ actions for effectively discerning visual content.

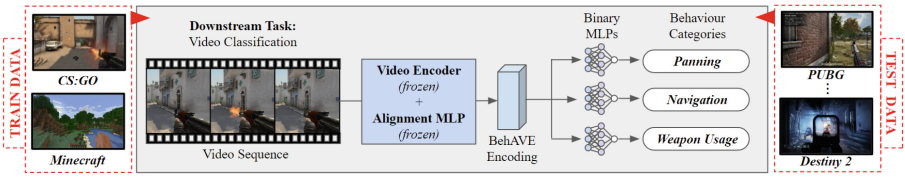
Data Pre-processing. Although we collect data at the timestep level, our framework operates on videos for identifying behaviour. To this end, we aggregate data over consecutive timesteps, forming a window of length T to obtain video sequences $V = (F_1, F_2, ..., F_T)$ and action sequences $A = (A_1, A_2, ..., A_T)$. Consequently, for each game, the synchronised gameplay-actions dataset is denoted as $D^{\mathcal{G}} = \{(V_i, A_i)\}_{i=1}^{I}$ where $|D^{\mathcal{G}}| = I$. We construct the *overall* dataset $\mathbb{D} = \bigcup_{\mathcal{G} \in \mathcal{C}} D^{\mathcal{G}}$ comprising of $k = |\mathbb{D}|$ "distinct" games from the game genre $\mathcal{C}$, adhering to the previously outlined selection strategy. This dataset forms the basis for behaviour alignment training.

3.2 Encoding the Modalities

Video Encoding and Alignment. As previously stated, we harness the capabilities of a pre-trained video foundation model in our study for video understanding. Let f denote the backbone model of a *video encoder*; thus, the latent representation of the backbone's video encoding can be given by $z^{\text{video}} = f(V)$ (Algorithm 1, line 3), where V denotes a 1 s video consisting of 16 consecutive frames. Note that we employ f within our training in a frozen state, a decision influenced by our evaluation of foundational models as well as other computational constraints. To facilitate the alignment of representation spaces of different



(a) Behaviour Alignment



(b) Behaviour Classification

Fig. 2. Overview of experiments and datasets used: **(a) Behaviour Alignment:** BehAVE is trained on synchronised gameplay video and player actions from the SMG-25 train dataset, and evaluated on unseen games from the SMG-25 test dataset. **(b) Behavior Classification:** We test the transferability of a *video classification* task. BehAVE is trained independently on CS:GO and Minecraft, and transferred to the SMG-25 test dataset.

modalities within our framework, we employ a trainable MLP model p that acts as an *alignment projector* and operates on top of the video encoder, yielding the aligned projection encoding $z^{\text{align}} = p(z^{\text{video}})$ (Algorithm 1, line 5).

Semantic Action Encoding. As previously explained, each video is associated with a sequence of binary actions A, indicating the presence or absence of a key-press at every timestep. Binary labels for actions, however, offer limited insights into the inter-relationships among various sub-actions. For instance, in FPS games, the binary encodings of the four actions-*move left*, *move right*, *shoot gun*, and *aim gun*-are equidistant from one another. This encoding type, however, fails to capture the underlying semantic similarity between the first two actions (*i.e.*, related to movement), the last two actions (*i.e.*, related to weapon use) and also the semantic difference between these two behavioural categories.

To address the above limitation, we propose to equip BehAVE with a hand-crafted *semantic action mapper* function m which injects semantic information into the action encodings via text. It maps the binary sequence of keypresses to a behaviour text caption. Then, we use a pre-trained text foundation model in the form of a *text encoder* h that gives the caption's text encoding $z^{\text{caption}} = h(m(A))$ (Algorithm 1, line 4). We argue that such pre-trained encoders will be able to better capture the inter-relationships among the joint distribution of actions that are otherwise difficult to represent with binary action encodings.

3.3 Alignment Training

Upon obtaining the encodings of videos and actions, we initiate the training phase of the framework using the specified dataset of games. We observe different video sequences exhibiting similar behaviour across different games. Consequently, to align the representation space of the video encoder to match that of the text encoding of behaviour, we choose to train our projector head as attached to the video encoder. Subsequently, we use a loss $\mathcal{L}_{\cos}$ (Algorithm 1, line 6) based on the cosine similarity between the video projector encoding and the behaviour text encoding so that the former aligns with the latter on the same (shared) representation space. The loss is defined as follows:

$$\mathcal{L}_{\cos}(z^{\text{align}}, z^{\text{caption}}) = 1 - \frac{z^{\text{align}} \cdot z^{\text{caption}}}{\|z^{\text{align}}\|_2 \, \|z^{\text{caption}}\|_2} \tag{1}$$

Upon completion of the alignment training, the video encoder equipped with the trained alignment projector can be utilised on any other visual content for video understanding, without requiring access to any other modalities—such as player actions—that only pertain to games.

In summary, the introduced BehAVE framework operates as follows. The structurally enriched dataset of the framework facilitates domain randomisation, the pre-trained video encoder enables video understanding, the semantic action encoding introduces the semantic notion of behaviour, and the alignment training module ensures enhanced transferability in video understanding.

4 Experiments

Figure 2 outlines the two primary experiments conducted in our study: 2a *Behaviour Alignment*, where we perform alignment training (Sect. 4.2), and 2b *Behaviour Classification*, where we assess transferability of the aligned models in a downstream classification task (Sect. 4.3). Before delving into the experiments, in Sect. 4.1 we introduce the datasets and evaluation metrics employed. Note that all experiments have been carried out on a single GTX 1070 (8GB) GPU, highlighting the cost-effective nature of our method.

4.1 Dataset and Metrics

We test BehAVE on three datasets, namely *SMG-25*, *CS:GO*, and *Minecraft*, across the two experiments of our study. The **SMG-25** (Synchronised Multi-Game FPS Dataset) is our newly introduced dataset illustrated in Fig. 3 that encompasses synchronised gameplay visuals and player action data from multiple commercial First Person Shooter (FPS) games, gathered following the structure outlined in Sect. 3.1. It comprises over ∼250K data points spanning 25 visually diverse FPS games, encompassing actions related to player behaviour categories such as *panning* (player looking around), *navigation* (player moving in the environment) and *weapon usage* (player engaging the equipped weapon). We partition it into an SMG-25 *train set* for use in Experiment (a) and an SMG-25 *test*

set used for evaluations in both Experiments (a) and (b). The train-test splits consist of disjoint sets of games, enabling evaluation of zero-shot performance on unseen games. Further details about this dataset are available in supplementary material. Additionally, we source similar gameplay-actions data from other commercial games, namely **CS:GO** (180K data points of *cs-dust* level from [16]) and **Minecraft** (50K data points of *contractor demonstrations* from [3]), which serve as training datasets for Experiment (b).

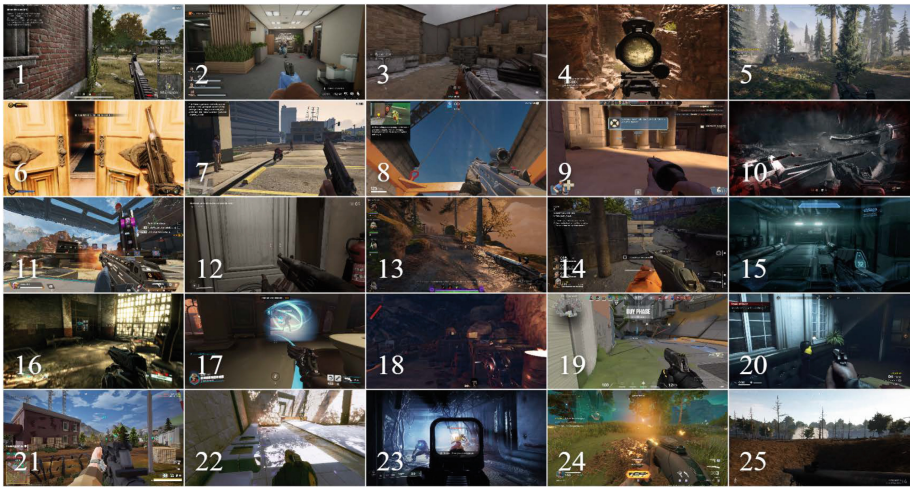


Fig. 3. Screenshots from all games of the SMG-25 dataset: **1)** PUBG: Battlegrounds (*PUBG Studios, 2017*); **2)** Payday 3 (*Starbreeze Studios, 2023*); **3)** Insurgency: Sandstorm (*New World Interactive, 2021*); **4)** Call of Duty: MW2 (*Infinity Ward, 2022*); **5)** Far Cry 5 (*Ubisoft, 2018*); **6)** Bioshock Infinite (*Irrational Games, 2013*); **7)** Grand Theft Auto 5 (*Rockstar, 2013*); **8)** Rainbow Six: Siege (*Ubisoft, 2015*); **9)** Team Fortress 2 (*Valve, 2007*); **10)** Wolfenstein (*Machine Games, 2014*); **11)** Apex Legends (*Respawn Entertainment, 2019*); **12)** Atomic Heart (*Mundfish, 2023*); **13)** Warhammer: Vermintide 2 (*Fatshark, 2018*); **14)** Back 4 Blood (*Turtle Rock Studios, 2021*); **15)** Halo 4 (*343 Industries, 2012*); **16)** Crysis 2 (*Crytek, 2011*); **17)** Overwatch 2 (*Blizzard Entertainment, 2022*); **18)** Deathloop (*Arkane Lyon, 2021*); **19)** Valorant (*Riot Games, 2020*); **20)** Generation Zero (*Systemic Reaction, 2019*); **21)** Polygon (*Readaster Studio, 2020*); **22)** Titanfall 2 (*Respawn Entertainment, 2016*); **23)** Destiny 2 (*Bungie, 2017*); **24)** Shatterline (*Frag Lab, 2022*); **25)** Operation Harsh Doorstep (*Drakeling Labs, 2023*).

Evaluation Metrics. To comprehensively assess the alignment quality in Experiment (a) and the transferability of aligned models in Experiment (b), we employ several metrics. The *Silhouette Score* [24], ranging from -1 to 1, quantifies the cluster quality of embeddings. In Experiment (a), it is used to gauge the

effectiveness of the alignment projection based on behaviour categories as cluster labels, with a higher score indicating better-defined clusters. Additionally, the *Transferability Score* evaluates results in Experiment (b) by considering the percentage difference in classification test accuracy between models trained on BehAVE encodings and those trained on foundation video encodings. A positive difference signifies better transferability for the alignment method relative to the corresponding foundation method, and vice versa.

4.2 Behaviour Alignment

In this experiment, depicted in Fig. 2b, we test the BehAVE framework on the SMG-25 dataset. Table 1 provides details on various pre-trained video and text encoders analysed, while a comprehensive report on the configurations tested is available in supplementary material. For the alignment projector, we opt for a 4-layer MLP with ReLU activations and 50% dropout. The size of the final layer of the MLP is adjusted to match the encoding size of the selected text encoder. As elaborated next, we perform a thorough analysis of various design choices incorporated into our framework.

Table 1. The video and text foundation models used in our experiments. (†) For video and text models, respectively, input size indicates timesteps of RGB frames and maximum token length. The number of parameters and the encoding size of the models are also listed.

Input	PreTrain Method	Model	Input Size†	#Params	Encoding Size
Video	I3D [5]	3D-ConvNet	$16 \times 3 \times 224 \times 224$	79M	512
	VideoMAEv2 [34]	ViT-Base	$16 \times 3 \times 224 \times 224$	87M	768
	MVD [36]				
Text	GPT-2 [19]	Transformer	512 tokens	110M	768
	CLIP [18]		77 tokens	63M	512
	BERT [21]		256 tokens	33M	384

Action Encoding Schemes. We conduct a comparative study on our semantic action encodings, focusing on actions alone, without videos, in our dataset. We compare the cluster quality of actions encoded traditionally in binary form (*i.e., keypress labels*) to text encodings (*i.e., behaviour captions*), with the aim of highlighting benefits of infusing semantics into our BehAVE framework. Results using only the unique set of actions from SMG-25 are presented in Sect. 5.1.

Impact of Alignment Training. Subsequently, we train BehAVE with the previously mentioned pre-trained video and text encoders and analyse the benefits of aligning the representation space of the video encoder with that of the

text encoder. The alignment projector is trained for 10 epochs using the *adam* optimiser with a learning rate of $1e^{-3}$. The training data comprises a set of 15 games from SMG-25, processed in batch sizes of 128. The resulting representation space post-alignment is evaluated using the *silhouette score* metric on 10 games from the SMG-25 test set, allowing us to assess the "zero-shot" performance of our methods on unseen games. In Sect. 5.1 we present results from 5 independent runs, with randomised train-test splits for each run.

Sensitivity Analysis of k. For practical applications it is important to analyse how the number of games (k) used during training affects alignment performance. Thus, we explore the impact of varying the number of games included in the training set, ranging from 1 to 15 and compare them on a fixed test set of 10 games. We randomise the selection of games in the train and test sets across the 5 runs reported in Sect. 5.1.

Table 2. Impact of Alignment Training. Average silhouette scores (with standard deviations) over 5 runs for behaviour and game labels on the SMG-25 test set. Higher scores indicate better performance for behaviour categories, while lower scores indicate better performance for game labels. Best performing models are highlighted in bold.

Alignment (Approach)	Encoder Methods (Video - Action)	Alignment Dimension	Behaviour Labels			Game
			Panning ↑	Navigation ↑	Weapon ↑	Label ↓
Foundation (Baseline)	I3D - None	512	$0.03_{\pm 0.04}$	$0.04_{\pm 0.04}$	$0.00_{\pm 0.01}$	$0.07_{\pm 0.07}$
	VideoMAEv2 - None	768	$0.08_{\pm 0.00}$	$0.13_{\pm 0.00}$	$0.03_{\pm 0.00}$	$0.10_{\pm 0.00}$
	MVD - None	768	$0.14_{\pm 0.02}$	$0.09_{\pm 0.04}$	$0.01_{\pm 0.01}$	$-0.05_{\pm 0.02}$
Keypress (Naive)	I3D - Binary	16	$0.36_{\pm 0.01}$	$0.45_{\pm 0.01}$	$0.30_{\pm 0.02}$	$-0.21_{\pm 0.01}$
	VideoMAEv2 - Binary	16	$0.35_{\pm 0.00}$	$0.48_{\pm 0.01}$	$0.32_{\pm 0.01}$	$-0.16_{\pm 0.00}$
	MVD - Binary	16	$0.43_{\pm 0.02}$	$0.44_{\pm 0.03}$	$0.09_{\pm 0.01}$	$-0.24_{\pm 0.02}$
BehAVE (Ours)	VideoMAEv2 - GPT-2	768	$\mathbf{0.51_{\pm 0.02}}$	$\mathbf{0.58_{\pm 0.05}}$	$0.20_{\pm 0.05}$	$\mathbf{-0.29_{\pm 0.06}}$
	VideoMAEv2 - CLIP	512	$0.40_{\pm 0.01}$	$0.49_{\pm 0.01}$	$\mathbf{0.35_{\pm 0.00}}$	$-0.20_{\pm 0.01}$
	VideoMAEv2 - BERT	384	$0.41_{\pm 0.03}$	$0.48_{\pm 0.04}$	$\mathbf{0.35_{\pm 0.02}}$	$-0.17_{\pm 0.03}$

4.3 Behaviour Classification

Following alignment training, we emphasise the benefits of BehAVE encodings for transferring a downstream task across visually distinct domains. For this purpose, we select video classification as a representative task in video understanding, with behaviour categories serving as the class labels. As depicted in Fig. 2b, for each behaviour category, we train a classifier (3-layer MLP with binary output) on video encodings of a single game (source domain) not included in our SMG-25 dataset, and evaluate the performance of this classifier on multiple FPS games from the SMG-25 test set (target domains). We first use foundation video encodings as input to this classifier, and then BehAVE encodings are employed

in the same fashion. We ultimately compare the classifiers' performance on FPS games between the two methods using the *transferability score* metric. We first perform this experiment with our source domain being CS:GO, a game of the *same* genre, followed by a more challenging experiment with our source domain being Minecraft, a *similar* first-person perspective game but from non-shooter genre. In both cases, we evaluate classification performance on the SMG-25 test set of games unseen in alignment training to gauge the zero-shot transfer capacity of our classifiers; results are reported in Sect. 5.2.

5 Results

5.1 Behaviour Alignment

Comparing Action Encoding Schemes. Figure 4a presents the comparison between binary action encoding and semantic text encoding from the pre-trained CLIP text encoder [18]. The silhouette score for binary actions (0.11) is significantly lower than that for text encodings (0.41), showing that text encodings produce better cluster quality. Findings are further supported by t-SNE projections, where actions in the same behaviour categories appear to form more compact and distinct clusters. These results highlight the advantage of encoding actions as text based on semantics, rather than just as binary keypresses.

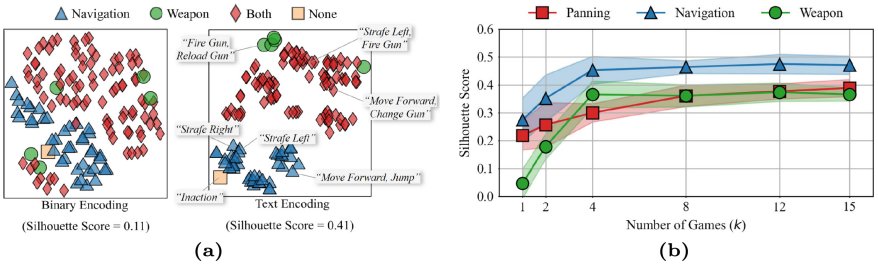
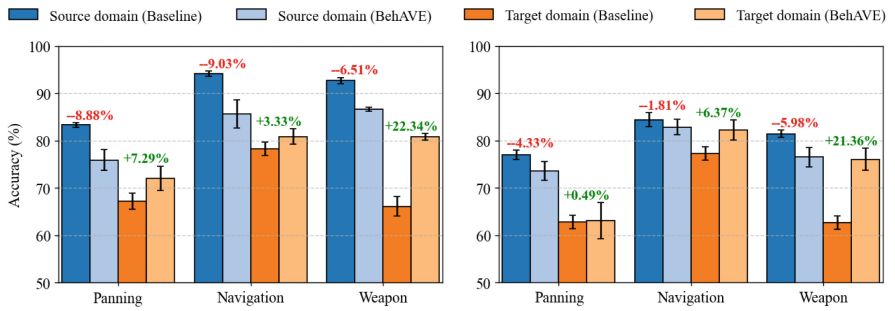


Fig. 4. Behaviour alignment experiments: (a) t-SNE embeddings and corresponding silhouette scores of actions encoded as binary labels (left) compared to pretrained text encoders (right). (b) Effect of varying the number of games in alignment training on behaviour category clustering across 10 test games.

Impact of Alignment Training. Table 2 presents the analysis of the representation spaces before and after running BehAVE's alignment training. First, we observe that for all three behaviour categories, even alignment based on a naive approach of binary encoding of actions (see middle block of table) improves clustering quality across all video foundation models tested (upper block). Second, more interestingly, we observe even bigger improvements when using BehAVE's text encoding scheme (lower block). As a consequence of this alignment, we also

observe that the domain gap, indicated by clustering on game labels, reduces for all alignment approaches compared to the foundation approach. The benefits of BehAVE are apparent across all tested configurations; a comprehensive list of all video and text encoders are provided in supplementary material.

Sensitivity Analysis of k. Figure 4b For practical applications it is important domain randomisation process to the number of games (k) used in training. Performance varies across different behaviour categories. For *weapon usage* and *navigation*, we observe convergence between 4 to 6 games, indicating that a relatively small number of games is sufficient for identifying these categories. Surprisingly, for *panning*, alignment continues to improve beyond 10 games. We argue that this phenomenon is attributed to the substantial variability in panning actions across different games, particularly due to different mouse sensitivity presets in SMG-25 that were approximated through visual inspection rather than precise extraction from the game engine (additional details in supplementary material). As a result, in our experiments with SMG-25, we chose 15 games for training and 10 for testing to balance the effectiveness of domain randomisation while maintaining a sufficiently large test set for zero-shot performance evaluation.



(a) CS:GO (Source) Unseen games (Target) **(b)** Minecraft (Source) Unseen games (Target)

Fig. 5. Behaviour classification accuracy across 3 behaviour categories when transferring from (a) CS:GO (FPS game) and (b) Minecraft (non-FPS game) to unseen FPS games. Although BehAVE (aligned) encodings perform slightly worse on source domain test sets than foundation (unaligned) encodings, they show significant improvements in generalisation to target domains, highlighting BehAVE's enhanced transfer capacities.

5.2 Behaviour Classification

Transfer from CS:GO (*Same* Genre). Figure 5a shows transferability scores across all three behaviour categories, with CS:GO as the source domain and multiple unseen FPS games from the SMG-25 test set as target domains. Notably, we notice the poor transfer capacity of classifiers based on foundation encodings. In the case of BehAVE, while the absolute performance of the classifiers

on source domains declines (highlighted in red above the respective bars), we observe a 3% to 22% improvement in transfer to target domains (highlighted in green). These findings are in line with contemporary studies, such as those in [2,37], suggesting that minimising training error leads machines to absorb all correlations, including spurious ones from data biases. As shown in Fig. 1a, foundation models retain game-specific style information, leading to the exploitation of spurious correlations and poor generalisation. Conversely, BehAVE reduces data bias impact, enabling generalisation to new test distributions by discarding game-specific information, as seen in Fig. 1b.

Transfer from Minecraft (*Similar* Genre). Figure 5b showcases similar benefits when using Minecraft as the source domain, despite its non-shooter nature. While *panning* category obtains minimal improvement, we see a 6% to 22% improvement in classifying *navigation* and *weapon* categories. Although the weapon equipped by the Minecraft player is a pick-axe, the classifier is still able to transfer this knowledge to the gun-related behaviours of SMG-25 games. This indicates that our model, trained on a diverse dataset covering multiple game genres with aligned semantic behaviour patterns, can generalise to new genres with overlapping characteristics. However, while the results demonstrate BehAVE's transferability across genres, this exploration is preliminary; further investigation into cross-genre transferability is left for future work.

6 Discussion

Downstream Applications. BehAVE's zero-shot performance shows promising applications in imitation learning across both first-person shooter and non-shooter game genres. While our focus has been on transferable video understanding, a complementary study could explore representation learning optimised for specific downstream tasks. Supplementary material includes initial experiments on BehAVE's potential in learning inverse dynamics models.

Future Work. Future experiments could employ BehAVE to various game genres, such as fighting, platformer, and driving games, with some, like driving games, requiring additional action preprocessing like discretising steering [41]. Experiments could also assess training across multiple genres and evaluate transferability to unseen ones. Another direction is to keep pace with advancements in foundation models like large language and vision-language Models, deepening action understanding. With more computational resources and a larger team [29], fine-tuning these models could enhance performance and transferability, making BehAVE more adaptable to specific applications.

Scalability. As outlined in Sect. 4, we operate within a limited compute and with a small team of annotators, hindering data collection expansion and comparisons with end-to-end trained models. BehAVE could potentially perform better with increased data and computational resources; however, our focus is

on introducing an accessible method that proves effective even at a smaller scale [29].

Ethical Impact. This paper introduces a dataset of fully anonymised annotated FPS gameplay videos. Both the gameplay videos and the annotations were collected in-house from participants in a laboratory setting under a data collection protocol approved by the University Research Ethics Committee of the University of Malta. The protocol was inspired by previously peer-reviewed work cited in Sect. 2. We acknowledge the emerging trends in machine learning for autonomous weapons research [25] and wish to clarify that this project has no military applications and was not funded by military sources. Additionally, the data collection protocol is exclusively applicable to video games, and the dataset aims to support research in video games and artificial intelligence.

Dataset. The dataset utilised in this research was derived from various commercial games. All game data consisting of screen captured game graphics, is the intellectual property of their respective game developers and publishers (see Fig. 3). To the best of our knowledge the inclusion of the data in this study falls under the provisions of fair use for the purpose of academic research, analysis, and non-commercial study under US law, and text and data mining (TDM) for research purposes under EU law. The dataset will be publicly available for future studies under a non-commercial license to ensure scientific reproducibility.

7 Conclusion

In this paper, we introduced BehAVE, a video understanding framework that uses a simulator-free domain randomization method, leveraging the inherent variations in graphics and animations in commercial video games. Tested on first-person shooter games, BehAVE effectively aligns video encodings of similar player behaviours across different games. It outperforms foundation models in zero-shot transferability to unseen FPS games during a behaviour classification task, even when trained on games from different genres like Minecraft. BehAVE offers a strong and efficient method for generalising perception across visually diverse environments.

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Collaborative Control for Geometry-Conditioned PBR Image Generation

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Abstract. Graphics pipelines require physically-based rendering (PBR) materials, yet current 3D content generation approaches are built on RGB models. We propose to model the PBR image distribution directly, avoiding photometric inaccuracies in RGB generation and the inherent ambiguity in extracting PBR from RGB. As existing paradigms for cross-modal fine-tuning are not suited for PBR generation due to both a lack of data and the high dimensionality of the output modalities, we propose to train a new PBR model that is tightly linked to a frozen RGB model using a novel cross-network communication paradigm. As the base RGB model is fully frozen, the proposed method retains its general performance and remains compatible with *e.g.* IPAdapters for that base model.

Keywords: Image Generation · Material Properties · Multi-Modal Generation · Physically-Based Rendering

1 Introduction

The recent meteoric rise of diffusion models has made at-scale generation of high-quality RGB image content more accessible than ever and Text-to-Texture and Text-to-3D approaches successfully lift this to 3D [34]. But to maximize the usefulness of the generated textures in downstream 3D workflows, generated content must be compatible with physically-based rendering (PBR) pipelines for proper shading and relighting. Current approaches rely on generated RGB images and subsequent PBR extraction through inverse rendering, suffering from the physically **in**accurate lighting in the generated RGB images as well as from significant ambiguities in the inverse rendering. We propose a solution for geometry-conditioned generation of PBR images by modeling the joint distribution directly, avoiding the issues around photometric consistency and inverse rendering (Fig. 1).

M. Boss—Work done while at Unity Technologies, Core Technical Contributions.
S. Vainer and S. Donné—Equal Contributions

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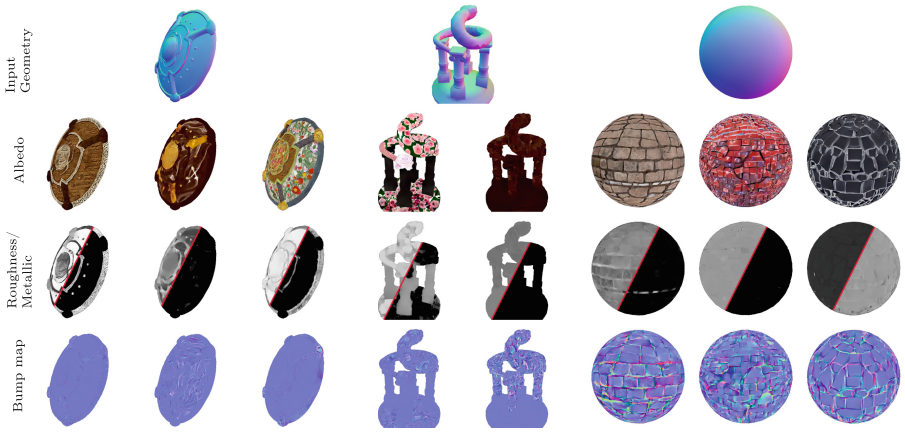


Fig. 1. Generated PBR materials. By tightly linking the PBR diffusion model with a frozen RGB model, we produce high-quality PBR images conditioned on geometry and prompts. Visit the project page at https://unity-research.github.io/holo-gen.

To model the distribution of non-RGB modalities, existing approaches typically fine-tune the weights of a base RGB model [11,29,33,40,41,58]. Applied to PBR images, this means either directly predicting the entire PBR image stack or sequentially predicting them conditioned on one another. Neither is sufficient for our use-case: jointly predicting the entire PBR image stack is problematic as the higher-dimensional modality does not compress well into the established latent spaces (as we show in Sect. 5), and sequentially predicting the elements of the PBR image stack is significantly more expensive and risks compounding errors in the sequential generation. Furthermore, while state-of-the-art RGB diffusion models are trained on billions of images [52], there is unfortunately no dataset of such size at our disposal for PBR content generation. Instead, the largest available dataset of PBR content is Objaverse [9], containing around 800,000 objects with associated PBR textures, limited to "everyday" appearances of the objects. In light of the restricted training data available, fine-tuning the base model results in catastrophic forgetting, forfeiting generalizability, as we illustrate in Sect. 5.

Instead, we keep a pre-trained RGB image model frozen and train a parallel model to generate PBR images, as shown in Fig. 2. We tightly link the PBR model to the frozen RGB model using our proposed cross-network control paradigm, in order to leverage its expressivity and rich internal state. As a result we are able to generate qualitative and diverse PBR content, even for unlikely appearances of objects (far out-of-distribution for the Objaverse dataset). Crucially, the frozen RGB model safeguards against catastrophic forgetting *and* remains compatible with techniques such as IPAdapter [73]. In summary, we:

1. Propose the novel *Collaborative Control* paradigm to tightly link the PBR generator to a fully frozen pre-trained RGB model, modeling the joint distribution of RGB and PBR images directly (see Sect. 4.1),

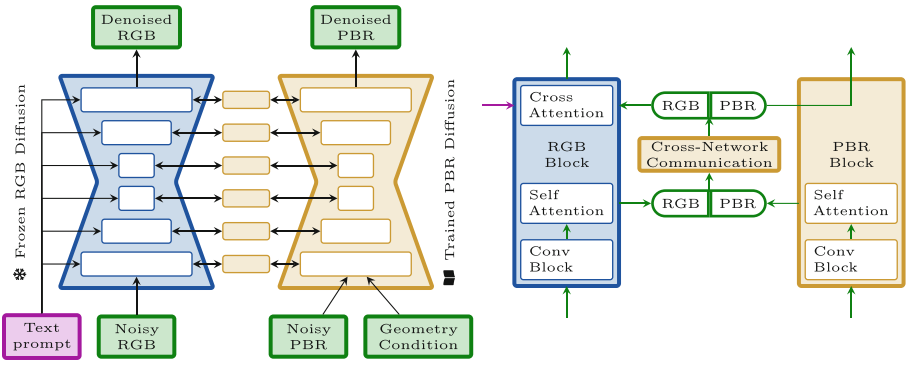


Fig. 2. Collaborative Control. Two parallel models collaborate to generate pixel-aligned outputs of different modalities. We freeze the left pre-trained RGB model and train the right PBR model with its cross-network communication layers. The cross-communication concatenates the states of both models, processes them with a small MLP, and residually distributes the result back to the respective models. As discussed in Sect. 5, prompt cross-attention in the PBR model is counter-productive.

2. Illustrate that the proposed control mechanism is data-efficient, and generates high-quality images even from a very restricted training set,
3. Demonstrate the compatibility with IPAdapter [73] specifically, and
4. Ablate our design choices to show the improvement over existing paradigms in literature and the issues with existing paradigms.

2 Related Work

Generating Natural Images from Text Prompts. Natural image generation has a long history: from GANs [14,25–28] and VAEs [30], to autoregressive models [45,64]. More recently, the introduction of diffusion models [18,56,57] was a breakthrough in the generative field—far more stable than typical GAN training, albeit slow and computationally expensive, and easier to control and condition. Unfortunately, these approaches require billions of images to train from scratch [52], and for PBR image generation the largest commonly available dataset is Objaverse [9]. With 800,000+ objects it is still several orders of magnitude smaller than LAION-5B [52] and proves insufficient to train generative models that can generalize to unlikely semantics (as illustrated in Sect. 5). While pre-trained RGB models encode rich prior knowledge around structure, semantics, and materials [11,53,59], Sarkar *et al.* warn that the models are often still geometrically inaccurate [50]: we have found that this extends to material properties, as diffusion models prefer idealized and artistic appearances over photometric accuracy.

Generating Non-RGB Modalities. Existing works fine-tune pre-trained RGB models to predict *e.g.* Depth [29,58], semantics [33] or intrinsic properties [11,40,41], either directly or through LoRA's [19]. Sadly, this is not plausible for PBR image generation: compressing PBR images into the existing

low-dimensional latent space overloads it, and the alternative of sequentially predicting channel triplets is too costly and slow. Wonder3D [41] and UniDream [40] perform joint RGB and normal diffusion using a cross-domain self-attention aligning the two parallel branches, yet this scales poorly an increasing number of output modalities. Our proposed approach instead uses a frozen RGB model, negating the risk of catastrophic forgetting, and trains a parallel branch for all additional modalities jointly (in the latent space of a PBR VAE), reducing cost.

Image-Based Conditioning. Existing pixel-accurate control techniques come in two flavors: re-training of the base model with modified input spaces [12,29], and training of a parallel model that affects the base model's state [10,20,79]. We find in Sect. 5 that the former risks losing the base model's expressiveness and quality. In ControlNet [79] and ControlNet-XS [10], the controlling model only influences the base RGB model's output while in AnimateAnyone [20] the parallel model is only tasked with generating its own output. Instead, we leave the RGB base model's weights fully frozen and residually edit its internal states from a parallel model that is itself tasked with generating PBR images: our PBR model both *controls* the base model (to guide it towards the domain of rendered images), and *generates* its own PBR output (based on the RGB model's internal state); therefore, our proposed approach requires full bidirectional connections between both branches as shown in Fig. 5. To condition on input geometry we concatenate it to the input of the PBR branch, as Ke *et al.* [29].

Text-to-3D. describes the task of generating 3D objects from text prompts, often with the aim to support downstream graphics pipelines such as game engines. Earlier methods leverage Score Distillation Sampling [46] (SDS) to iteratively optimize a 3D representation by backpropagating the diffusion model's noise predictions [15,22,35,37,42,43,55,61,62,66–68,71,80,81] through the RGB model, or building on viewpoint-aware image models [21,38,39,41,47,54,76] for direct fusion. Such RGB methods ignore that object appearance varies with viewing angle, often resulting in artifacts around highlights, and their RGB output is not useful in graphics pipelines. More recent work generates PBR properties so using inverse rendering with a differentiable renderer [7,40,70,72,74]: a major concern is lighting being baked into the material channels (*e.g.* HyperDreamer [70] uses an ad-hoc regularization to reduce these artifacts). **Text-to-Texture** methods restrict the Text-to-3D problem to objects with known structure by conditioning the diffusion model on the object geometry [4,6,31,32,75,77,78], but face similar issues by operating in the RGB domain. Paint3D [77] also discusses the lighting artifacts typical with inverse rendering and introduces a custom post-processing diffusion model to alleviate these. By directly generating PBR content, our proposed technique promises to resolve issues related to inverse rendering in the latter methods, all the while retaining the simplicity of the former methods.

Evaluation metrics for generative methods compare the output distributions with known ground-truth distributions, typically with the Inception

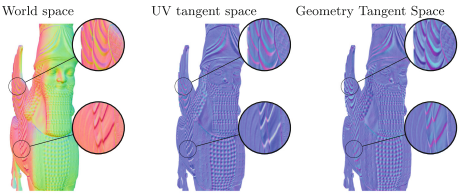


Fig. 3. Bump map. Similar surface bumps in world space (left) are dissimilar in the UV tangent space (middle) because of the arbitrary UV mapping. Representing the bump map in a tangent space solely dependent on the geometry (right) resolves this issue.

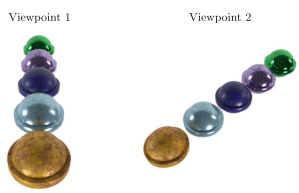


Fig. 4. Rendering function. The dataset is constructed so that the lighting remains constant with respect to the camera, simplifying the rendering function f_{RGB}: notice the similar highlight location.

Score [49] (IS) or the Fréchet Inception Distance [17] (FID). CMMD [23] argues that neither is well suited to modern generative models, and compare the distributions of CLIP embeddings rather than Inceptionv3 [60] internal states.

Aside from comparing modelled distributions with the ground truth, we also wish to evaluate the alignment of the generated images to their text prompts. CLIPScore [16] compares the image's CLIP embedding with that of the prompt: whether all the relevant elements are represented and whether any extraneous elements were introduced. We also report the OneAlign *aesthetics* and *quality* metrics of the generated images [69], which have been shown to align well with human perception, to provide a more quantitative indication of quality.

3 Preliminaries

PBR materials are a compact representation of the bidirectional reflectance distribution function (BRDF), which describes how light is reflected from the surface of an object. We use the popular Cook-Torrance analytical BRDF model [8], using specifically the Disney BRDF Basecolor-Metallic parametrization [3] as it inherently promotes physical correctness. In this parametrization, the BRDF comprises *Albedo* ($\boldsymbol{b}_a \in \mathbb{R}^3$), *Metallic* ($\boldsymbol{b}_m \in \mathbb{R}$), and *Roughness* ($\boldsymbol{b}_r \in \mathbb{R}$) components. To increase realism during rendering beyond the resolution of the underlying geometry (often a mesh), graphics pipelines add small details such as wood grain or grout between tiles by encoding them as offsets to the surface normals in an additional *bump map* ($\boldsymbol{b}_n \in \mathbb{R}^3$). As this bump map is typically defined in a tangent space based on an arbitrary UV-unwrapping, it entangles the surface property with this arbitrary UV mapping. Instead, we propose to predict the bump map defined in a tangent space based solely on the object geometry, disentangling the texture from the UV mapping as shown in Fig. 3. To construct this geometry tangent space for a point $\boldsymbol{p} = [p_x, p_y, p_z]^T$ with geometry normal $\boldsymbol{n}$, we construct the local tangent vector as $\boldsymbol{t} = \boldsymbol{n} \times ([-p_y, p_x, 0]^T \times \boldsymbol{n})$, corresponding to Blender's *Radial Z* geometry tangent. The geometry tangent space is then constructed as $(\boldsymbol{t}/\|\boldsymbol{t}\|, \boldsymbol{n} \times \boldsymbol{t}/\|\boldsymbol{t}\|, \boldsymbol{n})^T$.

Diffusion models [18,56] iteratively invert a forward degradation process to generate high-quality images from pure noise (typically white Gaussian noise). Formally, the forward process iteratively degrades images from the data distribution $\boldsymbol{z}_0 \sim p(\boldsymbol{z})$ to standard-normal samples $\boldsymbol{z}_T \sim \mathcal{N}(0, I)$ over the course of T degradation steps as $\boldsymbol{z}_t \sim \mathcal{N}(\alpha_t \boldsymbol{z}_{t-1}, (1-\alpha_t)I)$, where α_t denotes the noise schedule for timestep t. Practically, the forward process can be condensed into the direct distribution $\boldsymbol{z}_t \sim \mathcal{N}(\sqrt{\bar{\alpha}_t}\boldsymbol{z}_0, (1-\bar{\alpha}_t)I)$ with the appropriate choice of $\bar{\alpha}_t$. The diffusion model $\mathcal{D}$ is trained to sample the stochastic reverse process $\mathcal{D}_t(\boldsymbol{z}_t) \sim p(\boldsymbol{z}_{t-1}|\boldsymbol{z}_t)$ to iteratively generate $\boldsymbol{z}_0$ from $\boldsymbol{z}_T$.

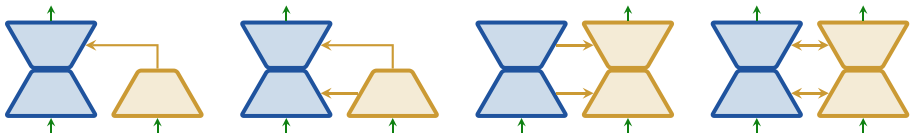
4 Method

We wish to train a PBR diffusion model $\mathcal{D}_{pbr}$ that models the reverse denoising process for PBR images as represented in the latent space of a VAE [48], representing the data distribution $p(\boldsymbol{z}_{pbr})$. We find that we lack the data required to train this model directly, and instead propose to model $p(\boldsymbol{z}'_{rgb} := f_{rgb}(\boldsymbol{z}_{pbr}), \boldsymbol{z}_{pbr})$ based on an RGB diffusion model $\mathcal{D}_{rgb}$ for the RGB data distribution $p(\boldsymbol{z}_{rgb})$; f_{rgb} is a rendering function that projects the PBR images onto the RGB domain. To motivate this, we split the joint reverse process into two separate processes using Bayes' rule:

$$\begin{aligned} &p(\boldsymbol{z}'_{rgb,t-1}, \boldsymbol{z}_{pbr,t-1} | \boldsymbol{z}'_{rgb,t}, \boldsymbol{z}_{pbr,t}) \\ &\quad \sim p(\boldsymbol{z}'_{rgb,t-1} | \boldsymbol{z}'_{rgb,t}, \boldsymbol{z}_{pbr,t}) p(\boldsymbol{z}_{pbr,t-1} | \boldsymbol{z}'_{rgb,t-1}, \boldsymbol{z}'_{rgb,t}, \boldsymbol{z}_{pbr,t}) \end{aligned} \tag{1}$$

The RGB model is implemented based on $\mathcal{D}_{rgb}(\boldsymbol{z}_{rgb,t-1}) \sim p(\boldsymbol{z}_{rgb,t-1}|\boldsymbol{z}_{rgb,t})$: we align the current RGB sample with the PBR sample and restrict it to $\mathrm{Im}(f)$ (the domain of rendered images with the fixed environment map) so that its internal states are more easily interpreted by the PBR branch[1]. To simplify this alignment problem, the rendering function f_{rgb} uses fixed camera settings and a fixed environment map as shown in Fig. 4. The PBR model now no longer models $p(\boldsymbol{z}_{pbr,t-1}|\boldsymbol{z}_{pbr,t})$: it additionally has access to the RGB context $(\boldsymbol{z}'_{rgb,t-1}, \boldsymbol{z}'_{rgb,t})$ which simplifies the problem. The RGB and PBR models are in practice much more intertwined than Eq. (1) implies: this derivation serves mostly as an intuitive indication for why the joint problem is more tractable. Note that $\boldsymbol{z}'_{rgb,t}$ is a degraded version of $\boldsymbol{z}'_{rgb,0}$, and *not* a rendered version of $\boldsymbol{z}_{pbr,t}$: the PBR model does not learn to do inverse rendering in degraded image space but rather learns to denoise PBR images given additional RGB context.

[1] Our intuition as to why a fixed environment map is beneficial is that it makes the RGB model's internal states more consistent to interpret, and makes the control problem of projecting to $\mathrm{Im}(f)$ simpler. Early in training, generated sample quality can be boosted significantly by applying the foreground mask to the RGB estimate for the first few timesteps; a rough projection to bring the estimate much closer to $\mathrm{Im}(f)$. After longer training, this is no longer necessary, as the PBR branch is capable enough to restrict the RGB branch to $\mathrm{Im}(f)$.



(a) ControlNet (b) ControlNet-Xs (c) AnimateAnyone (d) Proposed

Fig. 5. High-level overview of communication in (a) ControlNet [79], (b) ControlNet-XS [10], (c) AnimateAnyone [20] and (d) our proposed Collaborative Control approach. Blue represents frozen blocks, while orange elements are optimized during training.

4.1 Collaborative Control

In summary, our proposed approach comprises two models working in tandem: a pre-trained RGB image model and a new PBR model (see Fig. 5 for a high-level overview of our proposed control scheme). The previous section identifies two tasks for this cross-network communication: aligning the RGB model's output with both the PBR model's output and the map of the rendering function $\mathrm{Im}(f)$, and communicating knowledge in the RGB model to the PBR model. ControlNet [79] and ControlNet-XS [10] discuss solutions to the former control problem—the authors conclude that communication from the base model's encoder to the controlling model's encoder, and from the controlling model's decoder to the base model's decoder, is sufficient. AnimateAnyone [20] addresses the latter problem and concludes that, there, uni-directional communication from the left model to the right model is sufficient. We have found that full bidirectional communication is crucial for our approach: the PBR branch needs to extract relevant information from the RGB model's hidden state, while simultaneously guiding the RGB output towards render-like images (*i.e.* with a black background and compatible lighting) to ensure those hidden states are consistent with its own expectations. We dub this *Collaborative Control*.

We implement the cross-network communication as a connecting layer between the two models after every self-attention module; its inputs are the concatenation of the model states and its outputs are residually distributed to both models again. During training, we only optimize the weights of the PBR model and the cross-network communication links against both models' outputs, while the RGB model remains fully frozen. By adopting this approach, we safeguard the base RGB model's weights, and do not risk catastrophic forgetting for that base model. As we discuss in Sect. 5, we have found that a single per-pixel linear layer is sufficient, although we also evaluate the other control schemes from Fig. 5 as well as an attention-based communication layer. Notably, we have also found that disabling the text cross-attention in the PBR model is crucial to out-of-distribution performance; we attribute this to overfitting on the restricted dataset, as this problem worsens with reduced training data. Only allowing prompt attention through the frozen RGB model prevents such overfitting.

4.2 Implementation

Compressing PBR Images Into Latent Space. RGB diffusion models benefit immensely from a dedicated VAE to encode the images into a lower-dimensional latent space [48]. Existing solutions that generate an alternate modality typically encode that modality with the RGB VAE, but PBR images cannot be compressed into the same latent space due to the higher dimensionality. Instead, we could select channel triplets $\boldsymbol{b}_a$, $[\boldsymbol{b}_m, \boldsymbol{b}_r, \mathbf{0}]$, and $\boldsymbol{b}_n$ and process those with the RGB VAE, but we instead choose to train a dedicated PBR VAE—our ablation studies indicate that the distribution mismatch between the PBR channels and the RGB space is too large, and performance would otherwise suffer. We adopt the VAE architecture and training code from StableDiffusion v1.5 [48], although following Vecchio et al. [65] we set the latent space channel count to 14 for the optimal balance between quality and compression when processing PBR images.

Conditioning on Existing Geometry. We concatenate the screen-space geometry normals to the PBR model's inputs to condition the joint output. Referring to Fig. 5, Collaborative Control encapsulates the ControlNet scheme that would typically be used for this conditioning [10]: as we jointly train from scratch, this does not introduce additional cost.

Generating Training Data. Our dataset for training both the PBR VAE and the Collaborative Control scheme is based on Objaverse [9]: a dataset containing 800,000+ 3D models with annotations for what the models represent (describing both shape and texture). After sanitizing and filtering the dataset we retain roughly 300,000 objects. Each of the objects is rendered with Blender from 16 viewpoints encircling the object using a fixed pinhole camera model and a fixed (camera colocated) environment map[2] as in Fig. 4. For the evaluations in Sect. 5, we leave out 2% randomly selected elements.

Training Collaborative Control. For most of the experiments in Sect. 5, ZeroDiffusion [36,63] is the base RGB model, a zero-terminal-SNR version fine-tuned from StableDiffusion v1.5 [48]. As Collaborative Control is agnostic to the base model, we also illustrate StableDiffusion v1.5 and v2.1 as base models in Sect. 5. We optimize the PBR model's weights as well as the cross-network communication layers to minimize the training loss for the RGB and PBR denoising jointly, while keeping the RGB model fully frozen. Unless otherwise stated, we directly train on 512×512 resolution for a total of 200,000 update steps with a batch size of 12 and a learning rate of 3×10^{-5} (on one 80 GB VRAM A100, taking roughly two days). We evaluate the effect of a larger training budget by training on 8 A100's for the same number of steps, increasing the batch size by a factor of 8 without affecting training time—for environmental and cost purposes, the training budget is kept low for the main ablation study.

[2] https://polyhaven.com/a/studio_small_08.

5 Results

Distribution Match Metrics. As an evaluation of how well the data distribution is modeled, distribution match is considered a proxy to both quality and diversity. The Inception Score (IS [49]), which checks the distribution match against ImageNet, is not relevant in a PBR context as it applies only to RGB images. The Fréchet Inception Distance (FID [17]), which compares the distributions of the last hidden state of the Inceptionv3 [60] network on both real and generated images, has been found to better align to perceptual quality. Finally, the recently introduced CLIP Maximum-Mean Discrepancy (CMMD [23]) compares the distribution of the CLIP embeddings of generated images to that of a reference dataset. It offers significantly improved sample efficiency, and was shown by the authors to be a better indicator of low-level image quality than FID. However, as these metrics are intended for three-channel color images, we evaluate them on PBR images following Chambon et al. [5], by averaging the relevant scores of multiple triplets. We report as PBR distribution match the average of the scores over each of the PBR channels independently, as well as over three additional triplets, as the full set of triplets is prohibitively expensive to compute (the supplementary contains all the constituting scores). The additional triplets are (grayscale albedo, roughness, metallic), (roughness, metallic, normal XY norm) and (grayscale albedo, normal X, normal Y) for a balance between the full cartesian product (which is costly) and mixing channels that are normally relatively independent.

Out-of-distribution (OOD) performance metrics indicate the level to which our generator can align to conditioning that it was not trained on. Recent work has introduced the CLIP alignment score [13,16], which estimates the average distance between the text prompt CLIP embedding and the generated image's CLIP embedding, indicating how faithfully the prompt was followed. Additionally, OneAlign [69] is a neural model that estimates aesthetics and quality scores for images, shown to align well with human opinions, summarized in a QAlign score for both aesthetics and general quality. In order to evaluate the OOD performance, we randomly select 50 objects from Objaverse, and generate unlikely appearance prompts for them using ChatGPT4 [1]. These results, as well as a t-SNE plot of the embeddings of the original and OOD prompts, are integrally shown in the supplementary material.

5.1 Comparisons and Ablations

To the best of our knowledge, there are no published PBR generation models that generate PBR images for entire objects or scenes (only for generation of single materials [51,65]). Therefore, we perform an extensive ablation study on our design choices, taking care to include typical approaches from techniques that generate other modalities than PBR. Please refer to Fig. 6 for the qualitative comparisons, while Table 1 contains quantitative results.

Comparison Between Control Paradigms. We compare the performance of the proposed bidirectional cross-network communication layer against two other paradigms: one inspired by ControlNet-XS [10], and one inspired by AnimateAnyone [20]. In the former, dubbed *one-way* communication, the communication layers receive as input only the RGB model's internal state, and they only affect the PBR model's internal state. The latter, dubbed *clockwise* communication, functions in the same way for the encoder part of the architecture, but reverses the information flow to go from the PBR model to the RGB model for the decoder half of the architecture. We see that the *one-way* attention does not perform well, with lower distribution match scores as well as OOD performance scores; the frozen RGB model cannot realign to the conditional distribution required from it in Eq. (1), and we see that the positions it generates for the objects does not align at all with the mask from the normal image. The *clockwise* attention performs significantly better, but is likely still hampered by $\boldsymbol{z}'_{rgb,t-1}$ not being easily available to the PBR model—a similar reasoning as to why the authors of ControlNet-XS included the direct communication link between the base and controlling models' encoders.

Comparison Between Communication Types. In terms of the type of communication, we compare the proposed single-layer per-pixel communication against a per-pixel MLP-based communication layer, and a global attention layer. The latter performs surprisingly well considering that it lacks pixel correspondences; it is hard to enforce pixel-wise alignment through a global attention layer, which we hypothesize to be the reason for the lower quantitative performance. As Jin et al. [24] discuss, an attention-based architecture is also less robust to resolution changes. The per-pixel MLP, containing four hidden per-pixel linear layers with normalization layers [2] in-between, does not qualitatively perform notably better than the single-layer communication layer, so that we settle for the simpler and more computationally efficient choice.

Comparison Against Fine-Tuning. We also compare Collaborative Control against the alternative where we edit the first and last layers of the pre-trained network to match the dimensionality of the PBR images (optionally with the rendered image), and then fine-tune the entire network end-to-end. Although the distribution match scores for these fine-tuning variants are similar to Collaborative Control, the fine-tuning methods strongly overfit to the training data and perform very poorly in a qualitative OOD comparison.

PBR-Specific VAE Vs RGB VAE. We compare the performance of Collaborative Control with a PBR-specific VAE against a version that uses the triplets-based RGB VAE mentioned in Sect. 4 to encode the PBR channels (encoding albedo, roughness+metallic, and bump maps in separate triplets and concatenating their latent representations). The mismatch within the PBR domain is clear, both quantitatively through the worse distribution matching scores, and qualitatively in the produced images.

Fig. 6. Generated albedo, roughness/metallic and bump map images from the ablation studies. While significant quality differences are visible, only the fine-tuning approach and the data-sparse regime *with* PBR prompt cross-attention fail completely. The version that was trained on a smaller resolution does not break but does not result in maximum detail either. Best viewed digitally.

Table 1. Quantative results for all evaluated variants. The ablation baseline is highlighted in bold, duplicated for easier comparisons within the individual ablations.

		2% held-out evaluation data								OOD					
		CMMD ↓		FID ↓		QAlign ↑ Ae		QAlign ↑ Q		QAlign ↑ Ae		QAlign ↑ Q		CLIPScore ↑	
		PBR	Relit	PBR	Relit	Albedo	Relit	Albedo	Relit	Albedo	Relit	Albedo	Relit	Albedo	Relit
Communication	one-way	16.44	13.38	20.90	16.39	1.95	1.97	2.37	2.48	1.91	1.59	2.35	1.69	23.08	23.40
	clockwise	6.78	2.76	12.21	11.53	2.04	2.02	2.63	2.64	**2.14**	1.70	2.77	**1.74**	26.45	24.53
	bi-directional	**6.30**	**1.79**	**11.65**	**10.64**	**2.11**	2.03	**2.75**	**2.66**	2.12	1.76	**2.78**	1.73	**26.76**	**25.41**
	Pixel-wise zero-conv	6.30	**1.79**	11.65	**10.64**	**2.11**	**2.03**	**2.75**	**2.66**	2.12	1.76	2.78	1.73	26.76	25.41
	Pixel-wise MLP	**5.43**	1.87	**11.43**	10.67	2.10	2.02	2.74	**2.66**	**2.26**	1.75	**2.96**	**1.81**	**27.15**	**25.95**
	Global Attention	7.60	5.22	13.61	11.93	1.94	1.98	2.51	2.60	1.99	1.72	2.71	1.80	24.50	24.01
	Collaborative Control	6.30	**1.79**	11.65	**10.64**	**2.11**	2.03	**2.75**	**2.66**	2.12	1.76	2.78	1.73	**26.76**	**25.41**
	Fine-tuning (with RGB output)	13.40	2.78	14.42	10.79	2.05	2.02	2.60	2.61	2.10	1.76	2.62	**1.86**	25.04	22.69
	Fine-tuning (without RGB output)	**5.25**	2.88	**11.41**	11.37	2.03	1.99	2.58	2.58	**2.26**	1.71	**2.97**	1.81	25.66	23.31
	PBR VAE	**6.30**	**1.79**	**11.65**	**10.64**	2.11	**2.03**	**2.75**	**2.66**	2.12	1.76	2.78	1.73	**26.76**	**25.41**
	RGB VAE on triplets	84.66	5.99	25.81	11.63	**2.16**	1.99	2.67	2.55	**2.30**	1.71	**2.95**	**1.80**	25.27	23.98
Training budget	**1 A100, two days**	6.30	1.79	11.65	10.64	**2.11**	2.03	**2.75**	2.66	**2.12**	1.76	2.78	1.73	26.76	**25.41**
	8 A100s, two days	**2.96**	**1.12**	**9.55**	**9.76**	2.08	2.04	2.68	2.67	2.01	1.73	**2.82**	**1.82**	**26.78**	25.22
Training resolution	256×256	**2.23**	**1.44**	**9.82**	**10.20**	2.10	2.04	2.73	2.68	2.20	1.78	**3.13**	**1.80**	26.71	25.21
	512×512	6.30	1.79	11.65	10.64	**2.11**	2.03	**2.75**	2.66	**2.12**	1.76	2.78	1.73	**26.76**	**25.41**
Training data	**No PBR prompt attention** 1%	6.25	**1.43**	11.87	10.79	2.18	2.04	**2.86**	2.69	**2.35**	1.76	**3.35**	1.89	26.58	25.11
	No PBR prompt attention 5%	**5.77**	1.45	**11.49**	**10.54**	2.13	2.04	2.78	2.69	2.09	1.73	3.01	1.84	**27.28**	25.04
	No PBR prompt attention 20%	5.97	1.68	11.50	10.61	2.12	2.03	2.78	2.67	2.23	1.76	3.23	1.88	25.72	24.99
	No PBR prompt attention **98%**	6.30	1.79	11.65	10.64	2.11	2.03	2.75	2.66	2.12	1.76	2.78	1.73	26.76	25.41
	PBR prompt attention 1%	20.61	4.25	18.35	12.16	**2.19**	2.03	2.76	2.62	2.25	1.61	2.83	1.80	24.75	23.25
	PBR prompt attention 5%	12.17	2.58	14.95	10.97	2.13	2.04	2.71	2.65	2.27	1.80	2.97	**1.98**	26.89	**25.53**
	PBR prompt attention 20%	11.35	2.33	14.78	10.78	2.13	2.03	2.74	2.65	2.29	1.76	3.07	1.77	26.79	25.39
	PBR prompt attention 98%	9.18	2.57	13.25	11.02	2.10	2.03	2.68	2.64	2.17	1.80	2.84	1.87	25.80	24.83

Impact of the Training Budget. Comparing the version training on a single A100 with the version trained with 8 A100s (for eight times the batch size), we see that the latter performs significantly better quantitatively in terms of distribution match, but not quality. Visually, the differences are less clear, although the higher-budget model appears to follow complex prompts slightly better.

Impact of the Training Resolution. We compare the performance of Collaborative Control with two training resolutions: 256×256 and 512×512, both evaluated on 512×512 (ZeroDiffusion's native resolution). While the low-resolution model quantitatively performs better, visually it is clear that it does not capture the same level of detail as the high-resolution model—we explain this through the metrics focusing on high-level encoding of the images, and the lower resolution enables smoother training through a larger batch size (42).

Impact of Training Dataset Size. Now, we evaluate the performance of Collaborative Control under data sparsity by evaluating models trained on 98%, 20%, 5% and 1% of the full 6M training images. The proposed approach proves very data-efficient and performs well even when trained on only a few thousand images (1%). We observe that it is necessary to disable prompt cross-attention in the PBR model, and that this effect gets more pronounced with fewer data: we hypothesize that the model overfits to the training data and that forcing prompt attention to occur through the frozen RGB base model prevents this overfitting.

Fig. 7. A re-lit generated texture in Peppermint Powerplant and Pine Attic and lightfield slice under environment rotation.

Fig. 8. Interpolation on text embeddings and initial noise shows the stability of the proposed approach in both of these spaces.

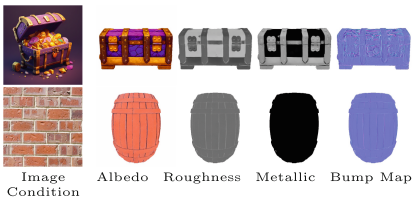


Fig. 9. Our PBR diffusion remains compatible with control techniques trained for the base frozen RGB model. We illustrate this using StableDiffusion 1.5 as the base model using a public IP-Adapter [73].

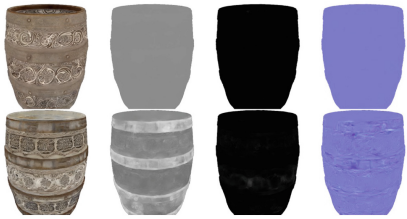


Fig. 10. Our most common failure case is the constant roughness, metallic or bump maps. Prompting a *porcelain barrel with intricate designs* for two different random seeds illustrates this behaviour.

Compatibility with Other Control Techniques. As a closing experiment, we illustrate that Collaborative Control is compatible with other control techniques [10,44,73], which drastically expands the practical applications of our proposed method. We demonstrate this specifically with IP-Adapter [73], which allows us to condition the final output on a style image by introducing additional style cross-attention layers within the base model. We can apply an IP-Adapter to the base model and still generate PBR content, as illustrated in Fig. 9.

Relighting. For a small qualitative indication that the PBR materials we generate also look natural under different environment lighting, Fig. 7 shows a generated texture when relit in two novel environment maps. Furthermore, a slice of the lightfield under rotation of the second environment map shows that the highlights and shadows move smoothly (hinting that the bump map is meaningful).

Interpolation. To illustrate the stability of our proposed approach in terms of both initial noise and the text prompt, we (separately) interpolate between two prompts and between two initial noises in Fig. 8. We perform prompt interpolation in CLIP space, and for noise interpolation we rescale the final image to have unit standard deviation and zero mean after blending. The resulting images appear natural yet meaningfully interpolate between both extremes.

5.2 Limitations and Failure Cases

We identify two major failure cases: lack of detail in the roughness, metallic, and bump maps, and a failure to follow OOD prompts. In the former, we see (*e.g.* Fig. 10) that the model outputs a constant (though varying per instance) roughness and metallic value, and a flat bump map. We attribute this to the training data: Objaverse contains many objects with constant roughness and metallic properties and a flat bump map—likely biasing the model towards such outputs. Anecdotally, we have found that selecting a different random seed will often succeed where the first generation disappointed—practically, the model produces very diverse results even for the same prompt and the same conditioning geometry, so that we argue that this is either not a significant issue or can be resolved with better training data.

A failure to follow out-of-distribution prompts happens mostly when structural features in the prompt are incompatible with the conditioning geometry, such as for example *a gilded lion* for a table mesh. We hypothesize that the control signal from the PBR model conflicts with the text cross-attention in the frozen RGB model, resulting in lackluster outputs. Different random seeds occasionally resolve this issue, albeit more rarely.

Finally, we note that the model does come at the cost of executing two parallel diffusion models. We note that the motivation for this was mainly to retain a frozen copy of the base RGB model: in applications where these requirements are prohibitively expensive, distilling our approach into a direct PBR model is likely to bring relief.

6 Conclusion

In this work, we have proposed Collaborative Control, a new paradigm for leveraging a pre-trained image-based RGB diffusion model for generating high-quality PBR image content conditioned on object geometry. We have shown that this bi-directional control paradigm is extremely data-efficient while retaining the high quality and expressiveness of the base RGB model, even when faced with text queries completely out of distribution for the PBR training data. The plug-and-play nature of our proposed approach is compatible with existing adaptations of the base RGB model, which we have illustrated with IP-Adapter for style guidance of the PBR content. The availability of high-quality PBR content generation as offered by our proposed approach opens up new avenues for graphics applications, specifically in Text-to-Texture.

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Real-Time Neural Cloth Deformation Using a Compact Latent Space and a Latent Vector Predictor

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Abstract. We propose a method for real-time cloth deformation using neural networks. The computational overhead of most of the existing learning methods for cloth simulation often limits their use in interactive applications. Employing a two-stage training process, our method predicts garment deformations in real-time. In the first stage, a graph neural network extracts cloth vertex features which are compressed into a latent vector with a mesh convolution network. We then decode the latent vector to blend shape weights, which are fed to a trainable blend shape module. In the second stage, we freeze the latent extraction and train a temporal latent predictor network. The temporal latent predictor uses a subset of the inputs from the first stage, ensuring that inputs are restricted to those which are readily available in a typical game engine. Then, during inference, the latent predictor predicts the compacted latent which is processed by the decoder and blend shape networks from the first stage. The latent predictor is the crucial component to speed up our inference time by replacing the resource-intensive graph neural network from the first stage. Our experiments demonstrate that our method effectively balances computational efficiency and realistic cloth deformation, making it suitable for real-time use in applications such as games.

Keywords: Cloth synthesis · Deep neural network · Real time

1 Introduction

Modeling, synthesizing and rendering clothing on virtual characters is a critical task in many applications such as games, special effects, telepresence, and

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VR environments. Physical simulation has been typically used with excellent results [1,18], but with significant limitations related especially to performance, stability, and controllability [18].

Modern video games contain many complex components related to graphics and animation working together with a limited computational budget and with a high number of assets (i.e. characters, garment types, accessories, motion types). Therefore, garment synthesis methods aimed at games require, in addition to the high quality of the results, high performance as well as a high degree of generalization to adapt to the large number of assets that typically appear in a game.

Most used methods in games predict displacement on a template garment in the canonical pose and subsequently drive cloth deformations using a combination of blend shapes and Linear Blend Skinning (LBS) [13,16]. However, as noted by Grigorev et al. [8], this pose-driven deformation has difficulties both in correctly representing loose garments and with the dynamic behaviour of clothing. This is because pose-driven methods are not supervised by any real physics simulation. The HOOD method [8] tries to address these issues by learning complex cloth deformations using direct physics supervision at a vertex level and, as such, is agnostic to the underlying body. One major concern with HOOD though is that it is far too slow to be used in video games.

In this work, we propose a garment synthesis method specifically designed with game requirements in mind. Our method uses the blend shapes and LBS methods while producing good realistic cloth deformation in real time. Our method uses two key ideas. First, we distil the per-vertex feature vector obtained from a powerful but slower method such as HOOD [8] that encodes the complex information about the garment deformation to a per-garment compact latent space that can be efficiently decoded. Then we compute the posed garment from the compact latent space. This addresses only half of the problem: even if the decoding is efficient, the encoding based on the graph embedding and message passing in HOOD is quite slow. To address this issue, our second key idea is to create a fast latent predictor that computes the latent code of the next frame of the deformation. We demonstrate that this approach produces high-quality results at very fast speeds. Moreover, it generalizes over different body shapes thus allowing only one latent space for a variety of character shapes and sizes. We further demonstrate its suitability for game-like applications by showing a real-time demo where the body poses are generated ad-hoc and in real-time using motion matching [6,9], a common method used in games for character pose synthesis. Our main contributions can be summarized as: (i) a network architecture for effective distillation of per-vertex features to a compact latent space, (ii) an efficient latent predictor for real-time purposes relying on a compact set of inputs readily available in game engines, and (iii) a two-stage training strategy to achieve quality and speed in computing cloth dynamics.

The rest of the paper is organized as follows: Sect. 2 presents in more detail the related work, Sect. 3 presents in detail our method, Sect. 4 presents our

results, comparisons and analysis and, finally, Sect. 5 presents our conclusions, limitations and future work.

2 Related Work

In the last decade, an enormous amount of work has been levied toward applying deep learning-based methods to various academic fields including computer graphics and computer animation. Character garment animation has recently received increasing attention for said research. Garment synthesis using neural networks can be largely classified into pose-driven methods where the prediction is conditioned on the body poses and the outputs of the neural networks are displacement for a template garment [5,19,21,23] and/or deformations encoded as blend shapes [2,12,14]. These approaches are appealing as the computation of linear blend skinning and blend shapes can be done extremely efficiently on GPU.

An early example of displacement based methods is TailorNet [19], which is a supervised method for predicting character clothing using pose, body shape, and garment style as input. The key to the methodology involves explicitly separating low-frequency and high-frequency garment deformations in-order to avoid the common problem of overly smooth output that neural networks suffer from. Low and high-frequency displacement are generated as functions of body shape, pose, and style using a Multi-Layered Perceptron (MLP). The high-frequency displacement is further refined using the mixture weights predicted by the garment style and the body shape. Self-Supervised Neural Dynamic Garments (SNUG) [21] formulates garment physical-based constraints as loss terms which are minimized during training, allowing for the model to learn displacement for dynamic garment deformations in a self-supervised manner removing the need of simulated data to train. Swish [12] is a quasi-static garment deformer, taking as input the character pose and outputting PCA weights which are used to reconstruct displacement with PCA vectors, similar to blend shapes. This is one of the only two methods that are suited for deployment in games and in fact was used in Electronic Arts Madden NFL 21 to deform football player jerseys. However, this method is limited to tight garments and it does not take into account the dynamics of the cloth. The largest limitation of the LBS-based methods is their failure on loose garments. Zhang et al. [23] address this by learning a generative space of plausible garment geometries. Then, their method learns a mapping to this space to capture the motion-dependent dynamic deformations, conditioned on the previous state of the garment as well as its relative position with respect to the underlying body. SMPLicit [7] is another generative model capable of representing body pose, shape, and clothing geometry. It can represent multiple garment topologies with the same model. This is achieved via a learned implicit function.

Another way to address the failure cases for loose garments is to model the deformed garment using blend shapes [2,3,11,14]. In the training stage, a set of blend shapes is created to span the deformation space of the garment thus

improving over the LBS-only methods. Physically-Based Neural Solver (PBNS) [2] uses a self-supervised learning approach for learning garment deformations similar to SNUG [21]. Neural cloth simulation (NCS) [3] utilizes an encoder-decoder neural network architecture which explicitly disentangles static and dynamic cloth deformations.

HOOD [8] departs from the pose-based generation framework to get the initially posed garment and uses physics supervision at the vertex level to drive garment deformation. While the results are impressive, due to its usage of Graph Neural Networks coupled with the necessity of computing per-vertex features at every time step, this approach is ill suited to real-time in-game cloth deformation.

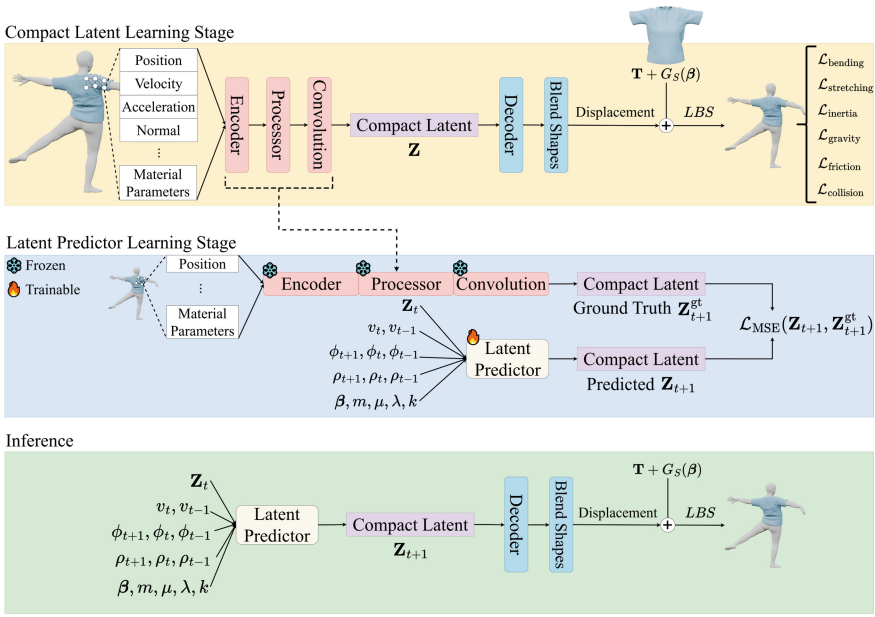


Fig. 1. Overview of our method. We develop a compact latent space for cloth deformation in the first stage (a), and then in the second stage (b) we train a latent predictor to efficiently predict a vector in the compact latent space. During inference (c), we utilize lightweight MLPs (the latent predictor and the decoder) and blend shapes for fast inference.

3 Method

Our method, depicted in Fig. 1, predicts displacement on a template garment in the canonical space and uses LBS to perform garment synthesis for computational efficiency and ease of application. Designed for real-time applications, such as games, our method utilizes a two-stage training process to model dynamic

cloth behavior. Our key idea is to obtain a compact latent representation for cloth deformation that can be predicted given sparse inputs and converted into displacement on a template garment.

While a method like HOOD [8] can simulate high-quality garment deformation with dynamics using time integration by predicting acceleration, it results in a high-dimensional latent representation and demands significant computational resources at inference time, making it unsuitable for game engines and real-time applications. Specifically, its encoder generates high-dimensional latent features, its processor updates these features with graph neural networks and message passing, and then its decoder predicts acceleration with the processed features. Utilizing the architecture of the encoder and the processor, which are effective at capturing garment deformation and dynamics, our compact latent learning stage aims to train a network that predicts displacement by compressing the high-dimensional latent features from the encoder and the processor into a compact latent vector using Mesh Convolution [25]. Leveraging the compact latent vector, we predict the displacement with our decoder and learnable blend shapes. The decoder predicts blend shape weights from the compact latent vector, and then the displacement is computed as a weighted sum of the learnable blend shapes with the blend shape weights. Finally, we perform LBS on the deformed garment with the displacement. Training these networks in an end-to-end manner, as illustrated in Fig. 1(a), enables us to learn a compact latent space that effectively represents garment deformation and dynamics.

Our latent predictor learning stage trains a lightweight MLP to get rid of the encoder, the processor, and the convolutional neural network of the compact latent learning stage, ensuring fast inference, as shown in Fig. 1(b). This block is trained to predict the latent vector at the next time step $\mathbf{Z}_{t+1}$ using the current latent vector $\mathbf{Z}_t$, root joint velocities of two frames $\{v_t, v_{t-1}\}$, joint rotations of three frames $\{\phi_{t+1}, \phi_t, \phi_{t-1}\}$, joint positions of three frames $\{\rho_{t+1}, \rho_t, \rho_{t-1}\}$, the body shape parameter $\boldsymbol{\beta}$, and material parameters $\{m, \mu, \lambda, k\}$, which are detailed in Sect. 3.3. These inputs are necessary and sufficient to predict the next step latent vector within interactive frame rates. During inference time, using this simple MLP with the decoder and the blend shapes enables our method to be applied in real-time applications, as presented in Fig. 1(c).

3.1 Garment Model

Our garment model $G(\boldsymbol{\beta}, \boldsymbol{\theta}, \mathbf{X})$ is defined as:

$$G(\boldsymbol{\beta}, \boldsymbol{\theta}, \mathbf{X}) = LBS(T(\boldsymbol{\beta}, \boldsymbol{\theta}, \mathbf{X}), J(\boldsymbol{\beta}), \boldsymbol{\theta}, \tilde{\mathcal{W}}) \tag{1}$$

where $\boldsymbol{\beta}$ and $\boldsymbol{\theta}$ are shape and pose parameters used by the SMPL body model [15], $\mathbf{X}$ is a feature vector defined in Sects. 3.2 and 3.3, and LBS is the linear blend skinning function. This function transforms the deformed garment $T(\boldsymbol{\beta}, \boldsymbol{\theta}, \mathbf{X})$ from the canonical space to the posed space with joint locations $J(\boldsymbol{\beta})$ of the body in shape $\boldsymbol{\beta}$, pose $\boldsymbol{\theta}$, and the diffused skinning weights $\tilde{\mathcal{W}}$.

The core of our garment model is the deformed garment $T(\boldsymbol{\beta}, \boldsymbol{\theta}, \mathbf{X})$. This involves deforming the template garment $\mathbf{T}$ with diffused shape blend-shapes

$G_S(\boldsymbol{\beta})$ based on the body shape $\boldsymbol{\beta}$ and further deforming the shaped garment $(\mathbf{T} + G_S(\boldsymbol{\beta}))$ using displacements predicted by network inputs $\mathbf{X}$. The diffusion method for skinning weights $\tilde{\mathcal{W}}$ and shape blend shapes $G_S(\boldsymbol{\beta})$, introduced by Grigorev et al. [8], is necessary due to the challenge posed by the diverse SMPL body shapes. This method allows garments, which are initially designed to fit the template SMPL body, to be aligned with different shapes of the template body, ensuring appropriate deformation.

3.2 Compact Latent Learning Stage

The goal of this learning stage is to train our network Φ and obtain a low-dimensional latent representation for garment deformation and dynamics. Our network Φ follows an Encode-Process-Convolve-Decode architecture, which integrates the Encode-Process architecture [20] of HOOD with a mesh convolution network and our decoder. The network Φ is trained in an unsupervised manner to predict displacement on the shaped garment for the next time step. This is different from HOOD which predicts acceleration for the next time step. Refer to supplementary material for implementation details. With no ground-truth simulation data requirement, we optimize physics-based losses on the final garment vertices $G(\boldsymbol{\beta}, \boldsymbol{\theta}, \mathbf{X}_{H+})$ using the deformed garment $T(\boldsymbol{\beta}, \boldsymbol{\theta}, \mathbf{X}_{H+})$. The deformed garment in this stage is defined as:

$$T(\boldsymbol{\beta}, \boldsymbol{\theta}, \mathbf{X}_{\mathrm{H}+}) = \mathbf{T} + G_S(\boldsymbol{\beta}) + \Phi(\mathbf{X}_{\mathrm{H}+}), \tag{2}$$

where $\Phi(\mathbf{X}_{\mathrm{H}+})$ is the displacement predicted by our network for the given inputs $\mathbf{X}_{\mathrm{H}+}$.

Inputs $\mathbf{X}_{\mathrm{H}+}$: The network Φ takes inputs $\mathbf{X}_{\mathrm{H}+}$, similar to those in Grigorev et al. [8]. The inputs $\mathbf{X}_{\mathrm{H}+}$ consist of per-vertex and per-edge feature vectors from the garment mesh and the body mesh at time t. The feature vectors for each vertex include velocity, normals, material parameters, vertex type, and vertex level. Material parameters for each vertex consist of mass m, Lamé parameters μ and λ, and the bending coefficient k. The vertex type indicates whether a vertex is pinned to prevent a garment from falling down from a body, and the vertex level represents the coarse level of a vertex (HOOD uses coarsened graphs for its multi-level message passing). Additionally, we incorporate positions and accelerations for garment and body vertices to enhance garment behavior modelling. The feature vectors for each edge of the garment include the relative position of connected vertices at the current time t and in the rest pose, with norms of these relative positions, along with material parameters and delta time. The same per-edge feature vectors are included from coarsened graphs of the garment. For edges connecting body and garment vertices by their proximity, the edge feature vectors include relative positions at the current time t and the next time $t + 1$, with norms and delta time.

Encode and **Process:** The encoder comprises MLPs in the same way as HOOD. It transforms the input feature vectors $\mathbf{X}_{\mathrm{H}+}$ into latent vertex and edge features,

which are then processed by a series of message passing blocks in the processor. This processor updates the features to capture garment deformation and dynamics, following the same approach and inputs of latent vertex and edge features as HOOD. Since HOOD passes only the processed vertex features to its decoder after the processor, indicating that the vertex features contain the necessary information for garment deformation and dynamics, we also pass only the processed vertex features to the mesh convolution network without the edge features.

Convolve: We compress the processed vertex features into a compact latent vector $\mathbf{Z}$ with the mesh convolution network by Zhou et al. [25]. This network excels at constructing localized latent features, making it ideal for our compression task. We utilize its convolution and residual layers to reduce the dimensionality of the processed vertex features and flatten the compressed vertex features into a latent vector. Since the compressed latent vector $\mathbf{Z} \in \mathbb{R}^L$ with the dimension L is used at inference time, the dimension L of the latent vector has to be small enough to be evaluated in interactive frame rates, and large enough to encode garment deformation and dynamics. This latent vector is then passed to the decoder.

Decode: When predicting garment deformations, learnable blend shapes have proven effective in modeling non-linear garment behavior [2–4]. Following this approach, we build our decoder $f_{\text{dec}} : \mathbb{R}^L \rightarrow \mathbb{R}^D$ with an MLP, which takes the latent vector $\mathbf{Z} \in \mathbb{R}^L$ and predicts D blend shape weights for our learnable blend shapes. The learnable blend shapes consist of D blend shape matrices. Thus, the final displacement for each vertex is defined as:

$$\sum_{j}^{D} \mathbf{D}_{j,i} f_{\text{dec}}(\mathbf{Z})_j, \tag{3}$$

where $\mathbf{D}$ is the array of the learnable blend shape matrices and $\mathbf{D}_{j,i}$ indicates the blend shape basis for the ith garment vertex of the jth blend shape matrix in the array. A single blend shape matrix $\mathbf{D}_j \in \mathbb{R}^{N\times 3}$ consists of blend shape bases for N garment vertices. The combination of the MLP with the blend shapes is fast enough to enable real-time inference. Details about the implementation are elaborated in the supplementary material.

Loss Functions: We train our network with the same losses as described by Grigorev et al. [8]. We employ the bending loss $\mathcal{L}_{\text{bending}}$ that introduces smoothness by penalizing sharp bends, measured through the dihedral angles between adjacent triangles [1]. The stretching loss $\mathcal{L}_{\text{stretching}}$, based on the Saint Venant-Kirchhoff (StVK) model, enforces hyperelastic material behavior. The inertia loss $\mathcal{L}_{\text{inertia}}$ is used to generate realistic dynamic motion by resisting drastic changes in velocity. The gravity loss $\mathcal{L}_{\text{gravity}}$ applies a constant downward force on the vertices of the garment mesh, creating realistic drapes and falls. The friction loss $\mathcal{L}_{\text{friction}}$ prevents the sliding motion of the garment, enhancing its stability and realism. Since all the above losses can cause interpenetration between the garment and the body, the collision loss $\mathcal{L}_{\text{collision}}$ is used to move the garment

vertices away from the body vertices. Therefore, the total loss $\mathcal{L}$ is defined as a weighted sum of these individual losses:

$$\begin{aligned}\mathcal{L} = & w_b \mathcal{L}_{\text{bending}} + w_s \mathcal{L}_{\text{stretching}} \\ & + w_i \mathcal{L}_{\text{inertia}} + w_g \mathcal{L}_{\text{gravity}} \\ & + w_f \mathcal{L}_{\text{friction}} + w_c \mathcal{L}_{\text{collision}},\end{aligned} \tag{4}$$

where w_b, w_s, w_i, w_g, w_f, and w_c are scalar weights that control the contributions of each loss term. By adjusting these weights, we fine-tune the balance between different aspects of cloth behavior. Following unsupervised training methods with physics-based losses [2,3,8,21], optimizing our network using these losses allows for realistic cloth deformation, similar to solving equations of motion for cloth simulation through energy optimization.

3.3 Latent Predictor Learning Stage

The goal of this stage is to train a latent predictor f_{LP}, which will replace the computationally intensive Encode-Process-Convolve components of the network Φ in the compact latent learning stage.

To ensure that the latent predictor is lightweight and suitable for real-time use, it is constructed solely with an MLP. Refer to the supplementary material for implementation details. The latent predictor takes a set of simpler inputs $\mathbf{X}_c$ compared to the inputs $\mathbf{X}_{\text{H+}}$. The inputs $\mathbf{X}_c$ are optimized for instantaneous preparation in each frame while being robust enough to accurately predict the compact latent vector $\mathbf{Z}$. These inputs are defined as $\mathbf{X}_c = \{\mathbf{Z}_t, \boldsymbol{v}, \boldsymbol{\phi}, \boldsymbol{\rho}, \boldsymbol{\beta}, m, \mu, \lambda, k\}$. The latent vector at the current time step $\mathbf{Z}_t$ provides temporal context, aiding in the prediction of the next latent vector. This input latent vector is set to zero in the first frame. The inputs include joint velocities $\boldsymbol{v}$, local joint rotations $\boldsymbol{\phi}$, and global joint positions $\boldsymbol{\rho}$. Specifically, the joint velocities comprise root joint velocities over two frames $\{v_t, v_{t-1}\}$, and the pose consists of local joint rotations over three frames $\{\phi_{t+1}, \phi_t, \phi_{t-1}\}$ in 6D representation [24] and global joint positions over three frames $\{\rho_{t+1}, \rho_t, \rho_{t-1}\}$. We include the global joint positions which are evaluated by forward kinematics with animation sequences, in order to enable the latent predictor to find a better mapping function from the inputs to a latent vector. To prevent the latent predictor from overfitting to the global joint positions of animation sequences, we subtract global joint positions with the root joint translation and remove the root joint translation from the inputs. We also exclude joints such as wrists and hands from the inputs, due to their minimal impact on garment deformation. We additionally incorporate the body shape $\boldsymbol{\beta}$ and garment material parameters. The material parameters are mass m, Lamé parameters μ and λ, and the bending coefficient k, which are also used for the compact latent learning stage. Through these inputs, we can reduce the number of variables in $\mathbf{X}_{\text{H+}}$ from hundreds of thousands of parameters down to thousands in $\mathbf{X}_c$. For example, we reduce 886,165 floating-point variables in $\mathbf{X}_{\text{H+}}$ to 2,591 floating-point variables in $\mathbf{X}_c$ for a t-shirt garment in our training dataset.

Utilizing these inputs, the latent predictor predicts the next latent vector $\mathbf{Z}_{t+1} = f_{\mathrm{LP}}(\mathbf{X}_c)$. To optimize the latent predictor, we compute the ground truth latent vector $\mathbf{Z}^{\mathrm{gt}}_{t+1}$ using the Encode-Process-Convolve components with the inputs $\mathbf{X}_{\mathrm{H}+}$, while keeping the weights of the components frozen. The optimization is performed by minimizing the mean squared error loss between the predicted latent vector and the ground truth:

$$\mathcal{L}_{\mathrm{MSE}} = \frac{1}{L}\sum_{i}^{L}\left(\mathbf{Z}^{\mathrm{gt}}_{t+1,i} - \mathbf{Z}_{t+1,i}\right)^2 \tag{5}$$

During inference, our garment model $G(\boldsymbol{\beta}, \boldsymbol{\theta}, \mathbf{X}_c)$ gets the final transformed vertices using the deformed garment $T(\boldsymbol{\beta}, \boldsymbol{\theta}, \mathbf{X}_c)$. The deformed garment at inference time is defined as:

$$T(\boldsymbol{\beta}, \boldsymbol{\theta}, \mathbf{X}_c) = \mathbf{T} + G_S(\boldsymbol{\beta}) + \Psi(\mathbf{X}_c), \tag{6}$$

where the function Ψ predicts the displacement on the shaped garment using the latent predictor, the decoder, and the learnable blend shapes with the given inputs $\mathbf{X}_c$. The displacement for each vertex is computed as $\sum_j^D \mathbf{D}_{j,i} f_{\mathrm{dec}}(f_{\mathrm{LP}}(\mathbf{X}_c))_j$.

4 Results and Discussion

In this section, we evaluate the effectiveness and performance of our networks. We first detail our training setup. Next, we qualitatively compare our networks with a state-of-the-art method, NCS [3]. Additionally, we measure and compare the computational performance of our method against NCS. Furthermore, we validate the effectiveness of our two-stage training process by an ablation study, and we demonstrate the efficiency and applicability of our method by implementing it in a real-time application demo.

4.1 Training Setup

Training Dataset: For our experiments, we use pose sequences from the AMASS dataset [17]. We adopt the sequence list from the VTO dataset [22] for training purposes. The VTO dataset takes sequences from the AMASS dataset, but we replace some unavailable sequences of the list with others in similar pose categories of the AMASS dataset (The list of sequences is found in the supplementary material). This sequence list contains 56 sequences with a total of 19,145 frames, comprising 13 walking, 12 running, 10 jumping, 10 arm movements, 6 torso movements, 4 dancing, and 1 avoidance movement sequence. For testing, we use other sequences from the AMASS dataset, excluding those used for training. The garments for our experiments include a dress, a long-sleeve top, pants, a tank top, and a t-shirt, provided by Grigorev et al. [8] for training their HOOD network. These garments are aligned with the template female body of SMPL. Accordingly, we used the SMPL female body for our experiments.

Training Details: We trained five networks for compact latent learning and five latent predictors with the five garments in the training set. We implemented our method and trained the networks on a PC, equipped with an Intel Xeon W-2135 CPU, an NVIDIA RTX A4000 GPU, and 64 GB of RAM. Training for the compact latent learning stage took approximately 40 h per garment, with 120,000 iterations each. We used the Adam optimizer for training the networks in the compact latent learning stage, with a learning rate 5×10^{-5}, and applied gradient clipping with a max norm of 1.0. Since the training method from HOOD does not support batch training, we used a batch size of one. For each training pose, We randomly sampled the shape $\boldsymbol{\beta}$ from the uniform distribution $\mathcal{U}(-3, 3)$. We also randomly sampled material parameters for the inputs by sampling a value from the uniform distribution $\mathcal{U}(0, 1)$ and scaling it with its minimum and maximum values. The mass for each vertex of a garment m ranged from 4.34×10^{-2} to 7×10^{-1}, the range of Lamé's second parameter μ from 15909 to 63636, the range of Lamé's first parameter λ from 3535.41 to 93333.73, and the range of the bending coefficient k from 6.37×10^{-8} to 1.31×10^{-3}. To advance our garment deformation over time, we apply the autoregressive training from HOOD, predicting one next step at the beginning and increasing the number of prediction steps every 5000 iterations to 5. We further set the inertia loss weight w_i from 3.0 to 5.0 and the gravity loss weight w_g from 2.0 to 3.0 to make our garment deformation more dynamic, with the other weights w_b, w_s, w_f set to 1.0. The collision loss w_c starts at 5×10^3 at the beginning of training and

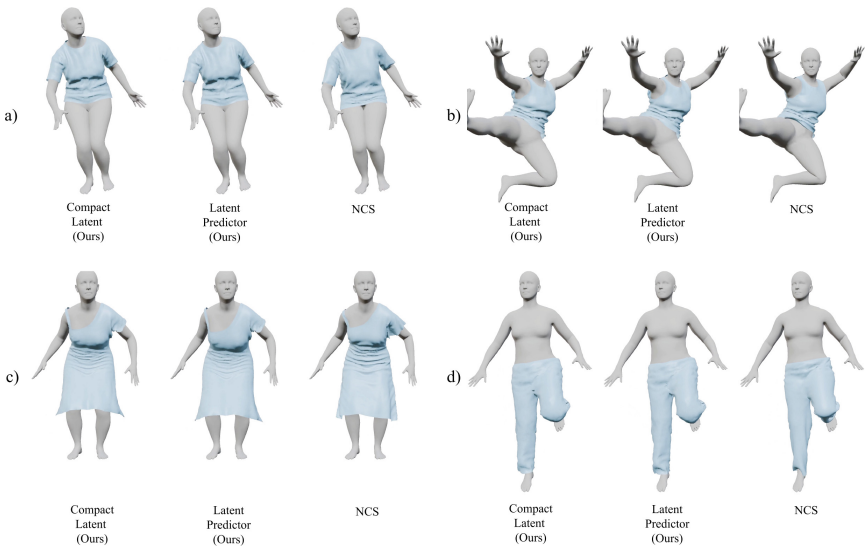


Fig. 2. Qualitative comparison with NCS [3]. For each garment, the first column shows the results from networks Φ in the compact latent learning stage, the second column shows results from networks Ψ using latent predictors f_{LP}, and the third column shows results from NCS networks.

increases linearly to 8×10^6 from 50,000 iterations to 100,000 iterations. Each latent predictor was trained for 5 million iterations, which took approximately 12 h. We trained latent predictors for each garment using an Adam optimizer with a learning rate of 1×10^{-4} and a batch size of 512.

Normalization: When training a network in the compact latent learning stage, we adopted the inputs and outputs normalization from HOOD. Unlike the normalization method for the outputs from HOOD, which gathers statistics for acceleration from linearly-skinned garments, we collect statistics for displacement from our prediction and evaluate the final displacement by denormalizing the predicted outputs. When training a latent predictor, we also normalize the inputs $\mathbf{X}_c$ and the outputs $\mathbf{Z}$ by mean and standard deviations for each input and each output except for the material parameters. The material parameters in the inputs are normalized to fall between 0.0 and 1.0 based on their value ranges.

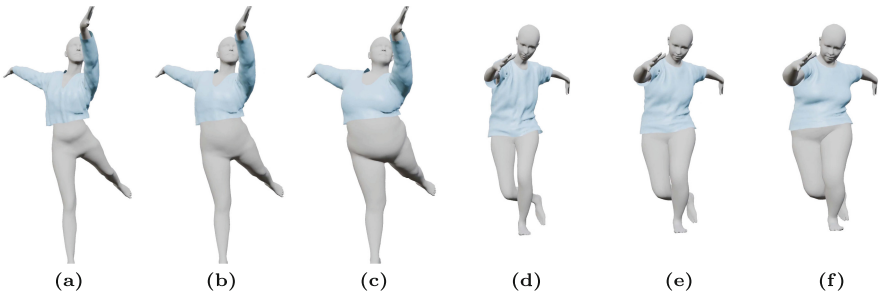


Fig. 3. The generalization capacity to different body shapes from networks Ψ using the latent predictor, the decoder, and the blend shapes. **(a)** and **(d)** are the thin bodies, **(b)** and **(e)** are the template bodies, and **(c)** and **(f)** are the heavy bodies.

4.2 Comparison with State-of-the-Art

For this comparison, we trained five NCS networks for each garment in the training set with the template SMPL female body, as NCS supports only a fixed-shaped body with a garment. We used the public code released by Bertiche et al. [3] and a batch size of 512. We chose the cloth model of Baraff et al. [1], since it is more optimal than the StVK model for convergence according to Bertiche et al. [3]. We further applied the heuristic method of Li et al. [14], which gradually increases the inertia loss weight from 0.1 to 1.0. This heuristic allows us to get the best cloth dynamics.

Unsupervised methods lack ground-truth data for quantitative analysis. Furthermore, the variety of network architectures, loss function implementations, and parameter ranges complicates a fair quantitative comparison. Moreover,

the sole comparison of loss values does not work. For instance, a lower inertia loss value does not guarantee superior quality or more dynamic deformation, as demonstrated by Bertiche et al. [3]. Consequently, our comparison is solely qualitative.

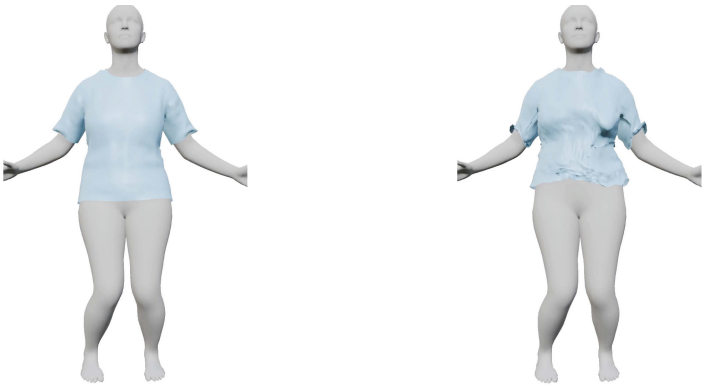
We present our qualitative results with NCS in Fig. 2 and the supplementary video. The first column shows the inference from the compact latent learning stage network, the second column shows the inference using the latent predictor with the decoder and blend shapes, and the third column shows the inference from NCS. As seen in Fig. 2 and the supplementary video, our method produces wrinkles and dynamics of similar quality to NCS. Unlike NCS, our method supports generalization to different body shapes with the same trained network. This generalization is shown in Fig. 3 and the supplementary video, which includes a thin body, the template body, and a heavy body. Our latent predictor can handle different body shape parameters $\boldsymbol{\beta}$ and predicts a compact latent vector accordingly.

For computational performance comparisons, a wide range of neural models for cloth simulation exist. Graph neural network models such as HOOD [8] and SwinGar [14] do not claim to achieve engine-ready real-time performance, as their architecture limits their performance to orders of magnitude slower. Only pose-driven models such as PBNS [2], SNUG [21], and NCS [3] achieve the desired performance criteria. Out of these models, PBNS is purely a static model, not being capable of achieving garment dynamics, and SNUG has dynamics limited to three frames, as thoroughly discussed in Bertiche et al. [3]. As such, we limit our comparison to the closest competitor, NCS [3].

Performance Evaluation: We evaluated the per-frame inference times on GPU for the "full model" (including both model inference and blend shape resolution). We also separately evaluated the runtimes for solely the neural network inference without the blend shape resolution step. Our method takes 523 microseconds per frame for the full model and 238 microseconds for the neural network inference. In contrast, NCS takes 1266 microseconds per frame for the full model and 1030 microseconds for the neural network inference. This result shows that our method outperforms NCS on GPU in the computational performance. We believe this behavior is explained by architecture differences between our model and NCS: our model utilizes more neurons but a simpler structure than NCS, as we do not utilize a GRU layer. As such, when computing on GPU, our model benefits more from GPU parallelism than NCS, while when computing on CPU, the lack of parallelism hurts our performance. The evaluation was done using models exported to ONNX Runtime on a t-shirt garment with 4424 vertices. The hardware utilized was an Intel Xeon W-2255 CPU @ 3.7GHz CPU, and an NVIDIA RTX 2070 Super GPU.

4.3 Ablation Study

To validate the effectiveness of our two-stage training process in learning garment deformation and dynamics, we performed an ablation study. We directly



(a) Full method **(b)** Training only the inference block (Fig. 1 (c))

Fig. 4. (a) Our method shows accurate garment deformation and realistic dynamics (results from network Ψ using the latent predictor, the decoder, and the blend shapes). **(b)** The ablation study, trained with only the inference block (Fig. 1(c)), results in poor and invalid deformations, highlighting the necessity of the compact latent learning stage.

trained only the components used during inference (Fig. 1(c)), specifically the latent predictor, the decoder, and the learnable blend shapes. We used the same training setup with the t-shirt garment as in the compact latent learning stage, utilized the same losses (Eq. (4)) from the compact latent learning stage, and additionally supported batch training with a batch size of 32. The results, presented in Fig. 4 and the supplementary video, demonstrate that training with only the MLPs and blend shapes does not achieve proper garment deformation and dynamics. The results from the ablation study exhibit less realistic dynamics and invalid deformations in parts of the garment. This outcome validates that our compact latent learning stage is essential for effectively learning garment deformation and dynamics.

4.4 Application on Real-Time Demo

We implemented our inference model with the trained latent predictor, decoder, and blend shapes using C++ and OpenGL. In the supplementary video, we show a SMPL body controlled using motion matching [6,9]. This shows the ease with which our method can be applied in real time, thanks to our inference model consisting of only two MLPs with blend shapes.

5 Conclusion, Limitations and Future Work

Realistic cloth deformation in game applications is complex due to varying character body shapes, poses, and movements, and many garments with different geometry, topology and materials. On top of these requirements, real-time cloth

deformation is essential for an enjoyable game experience. Procedural physics-based techniques are being replaced by neural network approaches. The latter seems more robust and yields high-quality results. However, the representation models required to capture realistic cloth deformation behavior are of very high dimension and correspondingly difficult to incorporate in game engines with limited resources and also far too slow for use in games. The two-stage training strategy presented in this work balances computational resource constraints and realistic cloth deformation effectively for application in games. The first stage learns a compact representation from the high-dimensional feature representation of a complex trained network, and the second stage learns a latent predictor with inputs in a format best suited for input to game engines. At inference time, the latent predictor predicts the compacted latent which is processed by the decoder and blend shape networks from the first stage, enabling framewise inference in real-time. This two-stage strategy is general and could be used in other neural network methods, wherein high-quality results mandate very high dimensional feature representations, and correspondingly high computational resources, limiting their application in real-time environments. Our experiments demonstrate the effectiveness and applicability of our method compared to NCS.

Our method has some limitations. Similar to other learning-based methods for garment synthesis, there are instances in test sequences where interpenetration between the garment and the body is not fully resolved. Since we utilize the previous compact latent in the latent predictor, these failures can lead to invalid garment deformation. This issue is especially problematic on loose garments. Future work could focus on enhancing collision handling by implementing an edge-based collision loss, which may provide more robust results compared to the vertex-based collision loss currently used [10]. The issue of drift is likewise a potential problem, notably for applications such as games where inference is run for long periods of time. In practice, our model proved to be robust to this: all of our experiments were exempt from drift. Our explanation for this robustness against drift is that our method first predicts learned latents before outputting absolute canonical space displacements. However, there are no explicit guarantees on the robustness of the model output with respect to drift. Another area of improvement is the generalization to garments with different topologies. The mesh convolution and the learnable blend shapes in our method are not agnostic to garment topology. In the future, we will explore other ways to compress the high-dimensional per-vertex features and compute the displacement from them, regardless of the garment topology.

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Level Up Your Tutorials: VLMs for Game Tutorials Quality Assessment

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Abstract. Designing effective game tutorials is crucial for a smooth learning curve for new players, especially in games with many rules and complex core mechanics. Evaluating the effectiveness of these tutorials usually requires multiple iterations with testers who have no prior knowledge of the game. Recent Vision-Language Models (VLMs) have demonstrated significant capabilities in understanding and interpreting visual content. VLMs can analyze images, provide detailed insights, and answer questions about their content. They can recognize objects, actions, and contexts in visual data, making them valuable tools for various applications, including automated game testing. In this work, we propose an automated game-testing solution to evaluate the quality of game tutorials. Our approach leverages VLMs to analyze frames from video game tutorials, answer relevant questions to simulate human perception, and provide feedback. This feedback is compared with expected results to identify confusing or problematic scenes and highlight potential errors for developers. In addition, we publish complete tutorial videos and annotated frames from different game versions used in our tests. This solution reduces the need for extensive manual testing, especially by speeding up and simplifying the initial development stages of the tutorial to improve the final game experience.

Keywords: Video Games · Vision-Language Model · Testing · Tutorial · Quality Assessment

1 Introduction

A crucial component of this initial experience is the game tutorial. Effective tutorials are essential for ensuring that players can quickly learn the fundamental mechanics and objectives of the game, thereby enhancing their overall enjoyment and engagement. This is particularly important for games with unique or complex core mechanics [6]. However, designing and evaluating effective game tutorials traditionally involves multiple iterations with human testers who have no prior knowledge of the game. This process is time-consuming and resource-intensive and prone to inconsistencies and delays in identifying critical issues [5].

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Fig. 1. Proposed framework. VLM is asked to answer questions about tutorial frames. Actual answers by the VLM are then compared with the expected answers provided by the developers to provide a quality score. The score informs about possible areas of improvement that can be used to improve the final User Interaction (UI).

Recent Vision-Language Models (VLMs) demonstrated remarkable capabilities in understanding and interpreting visual content [9,20]. They can analyze images, provide detailed insights, and answer questions about their content, recognizing objects, actions, and contexts within visual data. These capabilities make VLMs valuable tools for various applications, including automated game testing.

In this work, we propose an innovative automated game-testing solution that leverages VLMs to evaluate the quality of game tutorials. Our approach involves analyzing frames from video game tutorials and using VLMs to answer relevant questions, simulating human perception. This method provides immediate feedback, which is then compared to expected outcomes defined by the developers. By identifying confusing or problematic scenes and highlighting potential bugs, our solution helps developers refine and improve the tutorials, ensuring a smoother learning curve for players.

Our solution involves extracting significant frames from game tutorials and labeling them with questions that a developer would ask a human tester. These questions, along with their expected answers, serve as a ground truth to evaluate the VLMs' answers. By comparing the VLMs' responses to the expected answers, we can determine the clarity and comprehensibility of the tutorials as shown in Fig. 1. This process is analogous to writing automated software tests, aiming to identify and correct issues early in the development cycle.

Automating the testing process enables developers to focus on refining game mechanics and delivering a better player experience. Compared to traditional game testing, which relies on iterative human feedback, this solution speeds up the evaluation process, allowing for more frequent and thorough assessments without the need for extensive human resources. Moreover, unlike human testers, VLMs provide consistent feedback through each iteration, and they can be easily scaled to different games and tutorial formats.

Our contributions can be summarized as follows:

1. We propose a new solution to automatically evaluate the clarity of tutorials in games. Such solution can be applied to the development process of any part of a videogame;
2. We release the frame tutorials annotated by developers in two different versions of the game and the associated videos and the code at https://github.com/DarthReca/level-up-your-tutorials;

3. We benchmarked various open-source and closed-source state-of-the-art models for the proposed task

The rest of this article is structured as follows. In Sect. 2 the related works are presented. Section 3 describes the game, the tutorials and the dataset collection process adopted in the experimental section. Instead, Sect. 4 introduces the proposed solution, whose experimental results are presented in Sect. 5. Section 6 draws conclusions and provides suggestions for future work.

2 Related Works

This section presents contributions from two relevant areas: Visual Language Models and Machine Learning for game testing.

2.1 Vision Language Models

In recent years, Vision-Language Models (VLMs) have gained significant attention within the computer vision and natural language processing communities, particularly for their ability to integrate visual and textual information. Key advancements in this domain have been driven by models such as CLIP [24] and ALIGN [12], which leverage large-scale datasets to learn joint representations of images and texts. Recent advancements have seen the creation of Flamingo [3], which shows remarkable few-shot performance for visual question answering. Subsequently, GPT-4 [20], and LLaVA [17] have brought in visual instruction tuning to improve the instruction-following ability of VLMs. Concurrently, models such as KOSMOS-2 [22] have improved VLMs with visual grounding capabilities. The latest open-source models such as Intern-VL [8,9] and DragonFly [7] have reached comparable performance to closed-source solutions like GPT-4 [20] in a few amount of parameters offering more scalable and customizable solutions for a large scale adoption.

2.2 Machine Learning for Game Testing

Software product testing generally falls into two categories: manual testing and automated testing. Manual testing, particularly end-to-end play testing, remains the primary technique used by game developers. However, this method is often considered time-consuming and costly within the industry [23]. While manual testing provides valuable feedback that enhances software quality, there has been a significant shift towards applying existing automated testing techniques or developing new ones specifically for gaming. This shift aims to provide developers with tools to evaluate software quality continuously and on time to provide a better experience for end users. By creating surrogates of human perception, it is possible to integrate testing into the entire software lifecycle instead of relying solely on predetermined testing phases with human testers.

Machine learning and deep learning techniques have been pivotal in this evolution. Gameplay was proposed as a source to investigate bugs in games [16],

although the investigation was limited to video metadata and simple machine learning and deep learning solutions. Reinforcement learning and inverse reinforcement learning were also employed to create agents for defect detection learning from human tester [4]. Frameworks such as RiverGame are built to automate various aspects of video game testing [21], analyzing visual and sound aspects and using a behavior-driven development methodology for test specifications.

The latest advancements, such as Large Language Models (LLMs), have been explored for detecting bugs in video games by interpreting textual descriptions of game events [27]. The effectiveness of Vision Language Models has been proven in Graphic User Interface (GUI) testing, as shown in VisionDroid [19] and CogAgent [11]. In the game testing field, only GlitchBench evaluates their effectiveness [26] to identify game glitches from Reddit images.

Although many solutions have investigated game testing from different perspectives, to our knowledge, no solution has been proposed to investigate the effectiveness of game tutorials, which are critical to the success of a game in the early stages of gameplay. Furthermore, the capabilities of vision language models in this domain have not yet been sufficiently explored. This paper proposes to fill these gaps and pay more attention to these two underestimated aspects.

3 Dataset

Publicly available games lack annotations about the developers' intention for specific choices in the tutorials, so we selected a game for which we can obtain such information from the developing team to provide an effective way to measure if the visual and textual components in the game effectively convey the intentions. The chosen game is *But They Are Cats*[1].

3.1 Game and Tutorials

The game is a tower defense with some extra mechanics explained in the four tutorials. This section summarizes the game's rules, describing what each tutorial wants to explain to players.

Tutorial 1. The first tutorial introduces the player to basic mechanics, which can be summarized as follows: (1) The player has to place cats-in-boxes (the turrets) to defend the cheese from rats (enemies); (2) There are waves of enemies interleaved by some breaks; (3) If a rat reaches the cheese, the player can still save the cheese if he/she manages to destroy the rat before it can reach the base; (4) If a rat holding cheese is destroyed, the dropped cheese will slowly move toward its original place before being considered safe; (5) The level reward depends on the number of pieces of cheese rescued plus a base reward.

[1] https://thefellowshipofthebox.itch.io/but-they-are-cats.

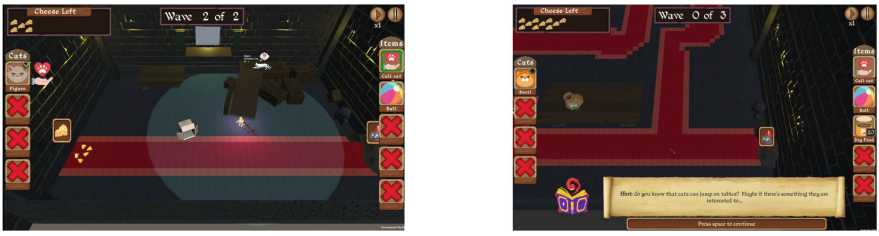
Tutorial 2. The second tutorial introduces the player to the behavior of the cats and can be summarized as follows: (1) Events can distract cats during the break between waves. As a result, cats will exit their boxes; (2) Player can attract cats to desired positions using different tools available; (3) Each cat has a favorite tool that speeds up the entire process.

Tutorial 3. The third tutorial introduces the player to the differences between types of cats and can be summarized as follows: (1) Different types of cats are available. The second type of cat is presented. It is faster and does less damage than the previous one; (2) There exist different types of rats; (3) Each type of cat reacts differently depending on distractions and tools used by the player.

Tutorial 4. The fourth tutorial introduces the player to the interactions with the environment and can be summarized as follows: (1) It is possible to move the camera around in the level; (2) The third type of cat is presented. It is slower than the others, but it can attack more rats at once; (3) Cats can jump on furniture; (4) Cats can break certain items placed around the level and alter the rats' path; (5) Each broken item around the level negatively affects the final reward; (6) It is possible to find special items in the levels.

3.2 Data and Annotation

Every video game tutorial alternates between two modes: a descriptive mode, which introduces the game mechanics to the player, and a playable mode, which requires player interaction as shown in Fig. 2. In contrast to automated testing methods that use game agents, this work focuses on analyzing the presentation part of each tutorial. We aim to determine whether visual language models (VLMs) can effectively substitute human perception to assess the comprehensibility of the visual and textual information presented to the user.



(a) Playable mode (b) Descriptive mode

Fig. 2. Example of the two modes of the video game tutorial.

To this end, our technique involves extracting significant frames from each recorded video of the tutorial. Each frame is labeled with questions that a

developer would ask a human tester to determine if the purpose of that part of the tutorial is clear, along with the expected answers (ground truth) that demonstrate understanding. Figure 3 shows one labeled frame. This process was repeated for frames from two different versions of the same game: one from an older development iteration (namely version P), and the latest released version (namely version L).

This approach allows us to test whether the improvements introduced in the latest version correspond to the quality assessment results obtained through our analysis. It is important to note that the game is a prototype with room for improvement, so even the scores obtained for the latest version are not for a final, polished game.

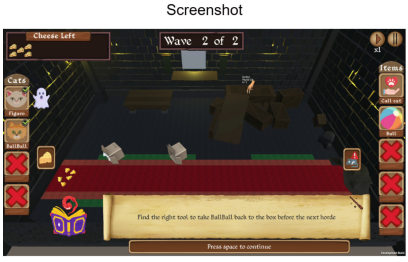


Fig. 3. Example annotation. For each frame, we provide a list of questions and answers related to the frame.

The answers are direct and the outputs generated by the VLM must be concise. This is enforced to avoid lower evaluation scores against the original developer's expected answer due to a strong mismatch of the lexicon or the semantic meaning of the phrases, since the purpose is to understand if specific concepts and rules about the game are sufficiently clear and understandable.

Quality Assessment Technique. Our objective is to create a tool that helps developers evaluate the quality of a game, one section at a time, following a methodology conceptually similar to writing automated tests for their codebase. The choice to label each frame with one or more question-answer pairs stems from the basic method of writing functional tests in the field of software engineering. The classical framework involves defining a test case and constructing the test following the AAA pattern (Arrange-Act-Assert) or the equivalent Given-When-Then in Behavior-Driven Development (BDD) [13].

In our context, the question represents the system-under-test, the frame serves as the test context, and the answer is the expected value. In our case, the Arrange step involves defining the input and the objective: selecting the scene to test, extracting the frame, and identifying the detail to be assessed for comprehension by a human player, which translates to the question. The Act step involves submitting the question and the frame to the VLM and collecting

the answers, which in the testing framework is often referred to as the *actual value*.

The assertion step evaluates the correspondence between the actual and expected values (the ground truth provided by the developer). This process determines whether the test passes or fails. As detailed in Sect. 4.3, we have established thresholds based on chosen metrics to provide developers with an indication of the test outcome. In our quality assessment approach, the developer receives feedback on whether the frame is acceptable, needs revision, or is rejected. The matching between AAA and our proposed method is illustrated in Fig. 4.

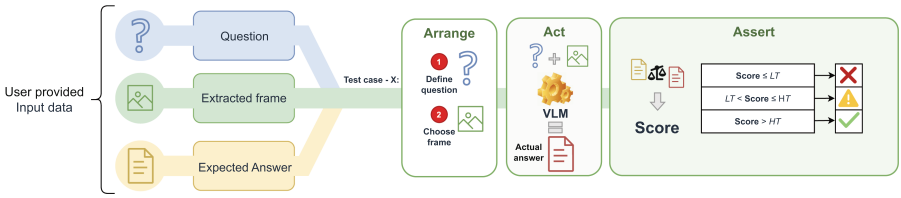


Fig. 4. Matching of the proposed method with AAA pattern. The Arrange defines the question-frame couple to make a test on. The Act produces the Actual Answer given the text-image pair using a VLM. The Assert computes a score evaluating the Expected against the Actual Answer. This score is then compared to Lower and High Thresholds (LT, HT) to indicate whether the test is passed, requires revision, or failed.

4 Methodology

In this section, we describe the models employed for the study, the metrics adopted in the experimental evaluation, and how they can be interpreted to evaluate the quality of the videogame's tutorial.

4.1 Models

In this section, we describe the VLMs employed for our study. GPT-4 [20] is the closed-source model that achieves state-of-the-art performance on many benchmarks. More in detail, we evaluated the GPT-4o model in the tutorial evaluation task. Instead, regarding open models, we considered the InternVL family and DragonFly model. The first one [8,9] is a family of open models that proved good performance on many benchmarks (comparable to GPT-4). Different sizes of InternVL models are available, ranging from 2B to 108B parameters. Instead, DragonFly [7] is a recent small open-source model that supports multiple image resolutions.

GPT-4. GPT-4o [20] is a large-scale, multimodal model developed by OpenAI, capable of processing both text and image inputs to generate text outputs. Based on the Transformer architecture, it was pre-trained on a vast corpus of data and fine-tuned using Reinforcement Learning from Human Feedback (RLHF). GPT-4 demonstrates human-level performance on several professional and academic benchmarks, including scoring in the top 10% on a simulated bar exam. It excels in various NLP tasks, surpassing previous models and state-of-the-art systems in multiple languages.

InternVL. InternVL [9] is a large-scale vision-language model designed to integrate a robust vision encoder with a large language model, utilizing a custom vision transformer, InternViT-6B, and a language middleware, QLLaMA. Its training strategy is composed of vision-language contrastive training, generative training, and supervised fine-tuning. InternVL 1.5 [8] enhances the original with continuous pre-training and dynamic resolution achieving state-of-the-art results across benchmarks and available in different sizes. The latest, InternVL 2.0, introduces progressive alignment training, supports multiple input modalities and multitask outputs, and achieves superior performance in multimodal tasks, available in sizes from 2B to 108B parameters.

DragonFly. DragonFly [7] is designed to enhance fine-grained visual understanding and reasoning, particularly for high-resolution images. It employs a multi-resolution visual encoding strategy, processing images at low, medium, and high resolutions and a zoom-in patch selection method that focuses on the most informative high-resolution sub-images. Visual tokens from these images are projected into the language space and combined with text tokens for input into a large language model. This architecture allows Dragonfly to efficiently handle high-resolution images, making it especially effective for tasks requiring detailed image analysis. Dragonfly-Med, the biomedical version, demonstrates state-of-the-art performance on various medical image benchmarks, validating its design.

All models' characteristics and their respective scores in multimodal understanding benchmarks are summarized in Table 1.

4.2 Tutorial Evaluation

As introduced in Sect. 3.2, we extracted from the four tutorials different frames depicting and introducing to the players several game mechanisms. Data acquisition was conducted on two different versions of the videogame tutorial: its *latest* release, namely L, and the *previous* deployment, which we identify as P. Each tutorial in our dataset is characterized by several screenshot acquisitions, and each capture is associated with one or more questions and their corresponding answers. Each pair question/answer was formulated by the developers and entails details about the game's mechanics, which are described in the tutorial phase.

Table 1. Models' comparison on MMMU [28] and MMBench [18] benchmarks according to Open VLM Leaderboard [10].

Model	Open	Language Model	Vision Model	MMMU	MMBench
GPT-4o [20]	✗	-	-	**69.2**	**82.2**
InternVL2-26B [9]	✓	InternLM2-20B	InternViT-6B	55.2	81.2
InternVL2-8B [9]	✓	InternLM2-7B	InternViT-300M	51.2	79.4
InternVL2-4B [9]	✓	Phi-3-mini [1]	InternViT-300M	48.3	73.6
InternVL1.5-26B [8]	✓	InternLM2-20B	InternViT-6B	46.8	79.7
InternVL1.5-4B [8]	✓	Phi-3-mini [1]	InternViT-300M	45.1	69.7
DragonFly [7]	✓	Llama3-8B [2]	CLIP [24]	36.2	-

The changes that were made after revising the video game led to differences in tutorial details, so the number of acquisitions between P and L is different.

All the considered VLMs in this paper were evaluated in two different experimental configurations:

- with history: at every prompt, the model received as input the entirety of chat history for the selected tutorial, in conjunction with the new frame to be evaluated as well as all its related questions, each being appropriately numbered
- without history: at every prompt, the VLM received as input the new frame and all its related questions without any previous information

The former configurations were proposed to evaluate the performance difference in scene understanding, also considering game mechanics that were presented in previous images, in contrast with the sole information contained within the new frame.

Additionally, as an initial prompt, we provided the following phrase to the system: *"You are a gamer. You are playing a game tutorial. I will provide you some screenshots of the tutorial. Answer the questions related to the screenshot. Be concise and direct."*. Such prompt was provided at the beginning of every chat session, i.e., the "with history" configuration provided such phrase as input only for the first question, whereas the "without history" configuration appended it at the beginning of every question.

The evaluation procedure split all the numbered answers to match the corresponding question. Metric computation was performed on a per-question basis, i.e., we evaluated each response separately with the expected answer. This solution permits the identification of the unclear aspects of each frame.

4.3 Metrics

To understand if the visual explanations are sufficiently clear and to provide automatic evaluation of the output quality, we need a quantitative measure of the results. We rely on established metrics for summary evaluation: ROUGE [15] and BERT-Score [29].

ROUGE-N [15] evaluates the syntactical overlap between N-grams (contiguous sequences of N items from a given text) of the prediction and of the ground truth. In this study, we employed ROUGE-1 (the overlap of unigrams), ROUGE-2 (the overlap of bigrams), and ROUGE-L (that measures the longest common subsequence) as ROUGE-N metrics.

Instead, BERT-Score [29] relies on encoder-only language models to measure the semantical overlap. For each token in the generated text, BERT-Score computes the cosine similarity with every token in the reference text. This results in a similarity matrix where each element represents the semantic similarity between a token in the generated text and a token in the reference text. The precision and recall are calculated to measure how many tokens are in common. The F1-Score aggregates the two metrics.

5 Experimental Results

This section presents experimental settings together with qualitative and quantitative performances.

5.1 Experimental Settings

All experiments were conducted on a workstation with Intel Core i9-10980XE, 128GB of RAM and an NVIDIA RTX A6000 GPU with 48GB of VRAM. All open-source models were employed with zero temperature for determinism. The images were resized at 1920×1080 resolution and preprocessed according to each model requirements. BERT-Score was calculated employing all-mpnet-base-v2 of Sentence Transformers [25] trained for the sentence embedding task. The GPT-4 version employed is GPT-4o accessed on July 2024.

5.2 Comparative Results Between Versions

In this section, we compare the scores achieved by all the models on common questions between the two tutorial versions (*latest* and *previous*). The results are shown in Table 2 for both the "with history" and "without history" configurations. It can be observed that, for all the considered models, metrics, and configurations, the latest version demonstrates higher scores and improvements compared to the previous version. This supports the fact that our proposed approach is able to capture improvements over the revised version of the game, ranging from +1.61% to +46.75% with history information and ranging from +2.10% and +45.21% without any previous interaction with the model. In both cases, the most significant increase was observed in ROUGE-2 metric, independently from the model. These results also suggest the historical information is not necessarily relevant for evaluating the frames since we can see differences in any case.

Table 2. Results obtained for the previous version (P) and the last one (L) by tested models for common questions

	Model	ROUGE-1		ROUGE-2		ROUGE-L		BERT-SCORE	
		P	L	P	L	P	L	P	L
with hystory	GPT-4o	0.4125	**0.4998**	0.1844	**0.2706**	0.3664	**0.4361**	0.6505	**0.6973**
	InternVL2-26B	0.4136	**0.4540**	0.1775	**0.2488**	0.3563	**0.4080**	0.6480	**0.6679**
	InternVL2-8B	0.4338	**0.4466**	0.1883	**0.2479**	0.3646	**0.3872**	0.6641	**0.6748**
	InternVL2-4B	0.4352	**0.4485**	0.1957	**0.2383**	0.383	**0.4025**	0.6504	**0.6771**
	InternVL1.5-26B	0.3811	**0.4214**	0.1659	**0.2156**	0.3168	**0.3676**	0.6390	**0.6563**
	InternVL1.5-4B	0.4555	**0.4780**	0.2119	**0.2459**	0.3841	**0.4173**	0.6544	**0.6848**
	DragonFly	0.2764	**0.3073**	0.1261	**0.1565**	0.2459	**0.2788**	0.5897	**0.6369**
without hystory	InternVL2-26B	0.402	**0.4187**	0.1862	**0.2311**	0.3525	**0.3751**	0.6424	**0.6578**
	InternVL2-8B	0.3862	**0.4273**	0.1617	**0.2348**	0.3274	**0.3744**	0.6442	**0.6633**
	InternVL2-4B	0.4381	**0.4819**	0.2016	**0.2527**	0.3783	**0.4169**	0.6611	**0.6848**
	InternVL1.5-26B	0.3566	**0.3706**	0.1563	**0.1889**	0.3036	**0.3236**	0.6367	**0.6625**
	InternVL1.5-4B	0.4398	**0.4620**	0.204	**0.2410**	0.3748	**0.3886**	0.6482	**0.6618**
	DragonFly	0.3189	**0.3646**	0.1394	**0.1954**	0.2845	**0.3275**	0.5985	**0.6146**

5.3 Models Consistency

Given that we have tested different models, it might be beneficial to consider whether it would be possible to employ one model in place of another, depending on whether they are able to identify good and bad samples consistently. We compute the Spearman Rank correlation in Fig. 5 to understand the model agreement. The correlations are all medium-high and the lowest are achieved by the worst model in performance: DragonFly. The highest agreement is shown by InternVL V2 8B and 26B. This fact proves that it is not necessary to use the larger version because similar results can be obtained using the smaller version, which is more portable. They both show correlations higher than 0.7 with GPT-4, representing good open-source alternatives.

5.4 Analysis of Tutorials Clarity

In this section, we evaluate each tutorial from the *latest* version to understand if there could be any differences between their clarity level and which tutorial requires higher attention. We compute the mean metrics over tutorials and models. In Table 3, we can see the differences between tutorials and metrics are consistent, i.e. the ROUGEs ranks coincide with the BERT-Score rank. As a consequence, independently from the considered metric, Tutorial 3 can be identified as the one that should prioritized and have larger room for improvement Instead, Tutorial 1 showed better clarity. From this perspective, we can conclude the third tutorial needs attention due to possible mistakes or unclear passages in the teaching and learning process.

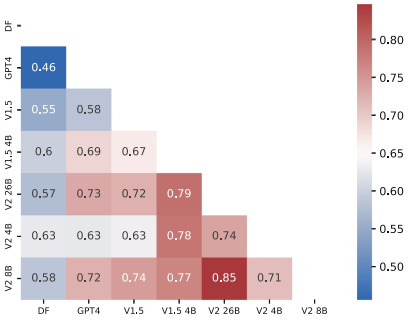


Fig. 5. Mean Spearman Correlation across metrics by model. *DF* is DragonFly, and *VX* indicates the InternVL version and eventually the size.

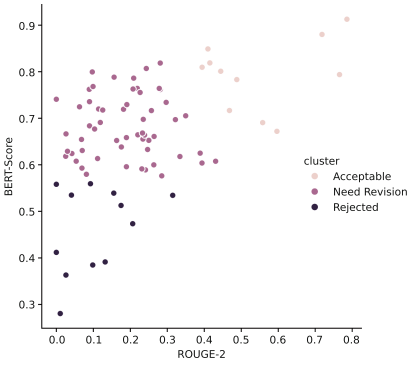


Fig. 6. Clusters of quality level according to ROUGE-2 and BERT-Score.

Table 3. The mean results by tutorial and models. The best one is in bold, and the worst one is underlined.

	ROUGE-1	ROUGE-2	ROUGE-L	BERT-SCORE
Tutorial 1	**0.4535**	**0.26**	**0.4004**	**0.6893**
Tutorial 2	0.4126	0.1882	0.3371	0.6546
Tutorial 3	0.3261	0.1226	0.2779	0.6345
Tutorial 4	0.4182	0.237	0.3834	0.6398

5.5 Automated Testing and Qualitative Analysis

It is possible to adopt many strategies to select the frames that require more attention due to clarity and should be revised by developers to improve the quality of the tutorial. To provide an automatic evaluation, we could consider three levels of attention for each answer: correct, requires attention, and wrong. Consequentially the related frame of the game could be considered acceptable, needing revision, or rejected, since its clarity is not sufficient.

Since we do not know apriori a good threshold to partition each frame in the three classes, we apply K-Means clustering using ROUGE-2 and BERT-Score of each model as features to create three clusters of similar samples. ROUGE metrics are highly correlated with each other, so we take only ROUGE-2 which is commonly employed in benchmarking. After such step, we obtain three clusters for the three different aforementioned quality levels as shown in Fig. 6. One simple solution to classify a frame based on its score is to apply K-NN (since they cannot be easily represented by linear equations ensuring a certain robustness).

An alternative approach, which provides simpler boundaries, is applying automatic thresholding algorithms like the multi-otsu one, which provides thresholds based on histogram analysis [14]. In this case, we get $BS \leq 0.5149$,

Question	What is the Treasure?
Expectation	The cheese to protect is the Treasure.
Answer	The Treasure is an item located on the left side of the screen, indicated by the label 'Treasure'.
Score	BS: 0.3871, R2: 0.087

(a) Example 1

Question	How can you know what is distracting a cat?
Expectation	The indicator near the cat in the Cats panel indicates what distracts the cat.
Answer	Distractions may be indicated by visual or audio cues in the game environment, but the specific indicator is not shown in the screenshot.
Score	BS: 0.4898, R2: 0.0571

(b) Example 2

Fig. 7. Example of bad samples using the thresholds from clustering. Each example shows the reference frame, the question, the expected answer (Expectation), the predicted answer (Answer), and the relative ROUGE-2 (R2) and BERT-Score (BS).

$R2 \leq 0.1785$ for low quality, and $BS > 0.7215$, $R2 > 0.4496$ for high quality ones. It's important to note that this method can only consider a single feature at a time, so it may not consider the relationship between R2 and BS in order to select the most suitable threshold.

In Fig. 7, we show two samples that are below the provided thresholds with the answers provided by GPT-4. In the first case, the model does not understand the visual cue, probably due to a missing explanation. In the second case, the other visual cue is not clearly explained and could be unnoticed by both players and models for the same reason as before. In this way, we can understand easily were we should pay attention and which section could be improved.

6 Conclusion

This study presents an innovative approach to evaluating the quality of video game tutorials by leveraging Vision-Language Models (VLMs). Our research demonstrates that VLMs can effectively simulate human perception to assess the clarity and comprehensibility of game tutorials. By analyzing frames from tutorial videos and comparing VLM-generated responses to expected answers, we can identify confusing or problematic scenes, providing valuable feedback to developers. We evaluated the proposed methodology on a single game for which we had access to two versions of the tutorial, comparing the scores achieved by two versions of the tutorial. Without any loss of generalization, it can be applied to other videogames.

Our experimental results show that the latest version of the game tutorials, which had undergone refinements, outperformed the previous version in terms of clarity and comprehensibility. This indicates that our automated approach can reliably detect improvements and issues within game tutorials, making it a useful tool for developers.

Additionally, the study reveals that historical information is not always necessary for evaluating frames, as both configurations with and without history showed improvements. This flexibility can help the evaluation process, reducing the need for a large context and using this approach for different lengths of tutorials. The clustering technique and Otsu's method used to classify frames into different quality levels proved practical and effective in offering a method for developers to focus on specific areas needing improvement. However, one limitation is ensuring that the model's perception aligns with that of actual players, which could vary based on experience and familiarity with the game.

Future work will aim to compare VLM assessments with human evaluations more comprehensively and extend the methodology to detect other issues such as verbal harassment, bugs, and non-inclusive content. This will enhance the robustness of our approach, providing a more thorough assessment of game tutorial quality and overall player experience.

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Across-Game Engagement Modelling via Few-Shot Learning

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Abstract. Domain generalisation involves learning artificial intelligence (AI) models that can maintain high performance across diverse domains within a specific task. In video games, for instance, such AI models can supposedly learn to detect player actions across different games. Despite recent advancements in AI, domain generalisation for modelling the users' experience remains largely unexplored. While video games present unique challenges and opportunities for the analysis of user experience—due to their dynamic and rich contextual nature—modelling such experiences is limited by generally small datasets. As a result, conventional modelling methods often struggle to bridge the domain gap between users and games due to their reliance on large labelled training data and assumptions of common distributions of user experience. In this paper, we tackle this challenge by introducing a framework that decomposes the *general* domain-agnostic modelling of user experience into several domain-*specific* and game-dependent tasks that can be solved via few-shot learning. We test our framework on a variation of the publicly available *GameVibe* corpus, designed specifically to test a model's ability to predict user engagement across different first-person shooter games. Our findings demonstrate the superior performance of few-shot learners over traditional modelling methods and thus showcase the potential of few-shot learning for robust experience modelling in video games and beyond.

Keywords: Few-Shot Learning · Video Games · Engagement · Affective Computing

1 Introduction

Domain generalisation involves the creation of models that can perform well across various contextual factors (domains) within the same task (e.g., object recognition) [42]. Building generalisable models becomes even more challenging when one has only limited samples per domain. Conventional methods may struggle to bridge the differences in data distribution between domains as they

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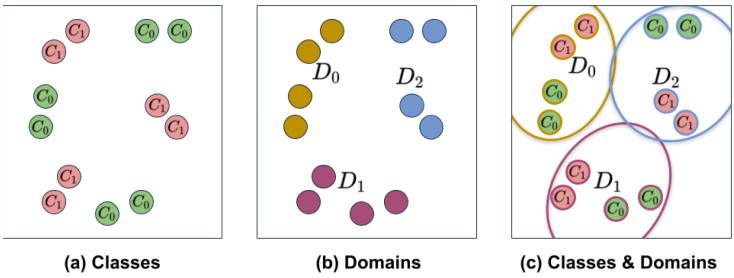

Fig. 1. Illustration of a classification problem—viewed as 2D plots of projected embeddings—containing three domains D (yellow, magenta, blue) and two classes C (green, pink). Such a problem is the prediction of user engagement classes (high vs. low) across several games (domains). Plot (a) showcases that the two classes cannot be easily separated across the entire dataset of domains. Plot (b) shows that points within the same domain are clustered together independently of their class. The rightmost plot (c) illustrates the method introduced in this paper combining both class and domain information: the method results in isolated domains allowing for classification within these more homogeneous groups. (Color figure online)

depend on large labelled training data and similar test data distributions. Various approaches have been developed over the years to address these challenges in computer vision subareas like object recognition [12]. Few-Shot Learning (FSL) [36] is a widely used method, enabling models to generalise to new data with minimal labelled samples by overcoming the inductive bias from the source domain distribution, thus adapting better to the target domain distribution.

Despite the rapid progress towards this direction, the area of domain generalisation within video games and user experience modelling remains underexplored. FSL is particularly well-suited for experience modelling because user experience annotations are inherently subjective, context-dependent and cannot be reliably collected in vast amounts [39]. Moreover, video games, with their complex narratives, mechanics, levels and rich audiovisual stimuli, present unique challenges that FSL can address by using minimal yet diverse training samples to build robust and generalisable models [40]. Additionally, video games present a distinct opportunity for studying the dynamics of user experience, whether for players or viewers [40]. Unlike traditional media, video games feature numerous multimodal contextual factors varying widely across games—from visuals and audio, to text and navigation—that impact user experience directly. This creates, in turn, a vast and complex multidomain problem (as defined by different contextual factors, such as different games, game genres, and players) for any artificial intelligence (AI) method that wishes to generalise within.

Motivated by the lack of few-shot domain generalisation studies in video games, in this paper, we propose a formulation for multidomain few-shot user experience modelling. Assuming consistent experience labelling behaviour, our framework treats each game as a separate domain, decomposing the multidomain problem into several non-overlapping domain-specific tasks (see Fig. 1) by incor-

porating domain-specific information into the labels. These tasks can then be solved simultaneously using few-shot learning approaches. To test our approach, we introduce the *GameVibe* Few-Shot (GVFS) dataset, a few-shot variation of the *GameVibe* [1] corpus. *GameVibe* contains data from 30 First-Person Shooter (FPS) games annotated for viewer engagement, aimed at testing models' ability to generalise across contextual factors for the same task. Our initial study of FSL-based modeling focuses on *viewer engagement*, which encompasses cognitive and affective user states like attention, interest, and enjoyment, offering a comprehensive understanding of user experience in games [39].

The paper presents several notable contributions. First, we propose an effective approach for turning multidomain classification tasks into few-shot learning tasks. Second, following the above-mentioned formulation, we devise GVFS, a few-shot learning variation of a publicly available game dataset specifically collected for learning models of viewer engagement that can generalise across various contextual factors. Third, we compare several few-shot learning approaches (e.g., metric learning and contrastive learning) against the conventional end-to-end engagement modelling approach, which is the standard practice in the field of affective computing and experience modelling. Finally, we employ various pretrained backbones and conduct experiments across 4 different scenarios, including 5- and 10-way, and 1- and 5-shot classification tasks. The results demonstrate that the proposed formulation is better suited for such problems since the few-shot learners achieve significantly higher accuracy values than the conventional domain-agnostic baseline in the vast majority of experiments performed. It should be noted that the proposed formulation is versatile and thus can be extended to any multidomain problem, beyond games, serving as a valuable tool for broader applications in machine learning and AI.

2 Background

Few-Shot Classification. Few-shot learning is a paradigm focused on training models to classify new, unseen samples using only a few labelled examples. FSL methods are categorised into optimisation-based, metric learning, and hybrid approaches. Optimisation-based methods enable models to adapt quickly to new tasks with minimal data. Examples include MAML [5], which optimises model parameters for better adaptation; Reptile [24], a MAML variant that reduces computational load by removing task-specific reinitialisation; and Meta-SGD [13], which optimises both parameters and learning rates. Metric learning approaches involve projecting inputs into a shared space where a metric is used to distinguish between classes. For instance, Prototypical Networks [30] learn a prototype for each class within a known metric space, ensuring that samples from the same class are closer to their prototype than to prototypes of other classes. Relation Networks [31] capture the relationship between data points by learning a deep distance metric that compares a small number of samples per iteration. Matching Networks [33] employ an attention mechanism to create a weighted nearest neighbour classifier using the support set of each learning

iteration. Recent work focuses at the intersection of FSL and contrastive learning (CL). Notably, Liu et al. [14], used CL with noise contrastive estimation for few-shot image classification. ContrastNet [4], a CL framework, addresses representation and overfitting in text classification. Zhen et al. [41] proposed mixed-supervised hierarchical CL for aligning temporal clips. Jian et al. [9] combines supervised CL with masked language modelling in few-shot learning across various language tasks. In this work, we focus on domain generalisation within video games, an underexplored area in FSL research. We use FSL to model viewer engagement across multiple gaming contexts, and test our models' generalisation capacity by introducing the GVFS dataset. We treat each game as a distinct domain paving the way for future research in domain generalisation within this field.

Modelling Experience in Games. When it comes to games, experience modelling refers to the development of models that predict how a person behaves and feels while interacting with a game [40]. The majority of the early studies employed models focusing on hand-crafted features. Frommel et al. [6] utilised the input from a graphics tablet and gameplay performance to predict the emotional state of players. Melhart et al. [21] employed hand-crafted features that describe the gameplay context to learn general models of player arousal. Assuming that raw gameplay footage can be an effective elicitor to affect Makantasis et al. [16] used CNN architectures to predict player arousal from gameplay footage. Lastly, Pinitas et al. [26] evolved the parameters of a preference learner to predict arousal in gameplay videos. Apart from the core dimensions of emotion, significant progress has been made in modelling more complex constructs of experience in games, such as engagement. Xue et al. [37] proposed a Dynamic Difficulty Adjustment framework to maximise player engagement. Huang et al. [8] introduced a two-stage player engagement modelling method via Hidden Markov Models. Melhart et al. [20] used chat logs as a proxy for engagement and predicted moment-to-moment gameplay engagement. Pinitas et al. [28] employed pretrained CNN models and time-conditioning to predict long-term engagement in *Tom Clancy's The Division 2* (Ubisoft, 2020). Recently, Pan et al. [25] proposed a CNN model for estimating streamers' engagement using gameplay footage, audio, and facial expressions. Unlike the above-mentioned studies, this paper explores the potential of FSL for robust experience modelling in video games with only few labelled samples. We evaluate several methods, including metric and CL ones, against end-to-end domain-agnostic engagement modelling methods. Our models use four different backbone architectures covering various FSL scenarios. The results demonstrate the superior performance of FSL against the conventional end-to-end approaches.

3 Method

3.1 Problem Setting

A primary contribution of this work is introducing a framework that improves learning from limited data with a high domain gap. Conventional end-to-end modelling methods struggle to generalise amid varying contextual factors within the same task, even in binary classification, especially when labels are noisy due to the subjective nature of engagement [38]. Our framework exploits domain knowledge to decompose the classification problem into non-overlapping domain-specific sub-problems, with different classes falling under different domains. This ensures each domain has unique classes, which simplifies the learning process by reducing the variability and complexity within each sub-problem. Hence, Few-Shot Learning techniques can more effectively learn from the limited data available within each domain, as the model can focus on the specific features and patterns relevant to each unique domain-specific class.

Modified Classification Objective: Let f be a function that projects data from multiple domains D_n with $n \in \{0...N\}$ into a common space of lower dimensions. We can decompose $f(\cdot)$ in the following manner:

$$f(x) = \sum_{n=1}^{N} f_n(x)\mathbf{1}_{D_n}(x) \tag{1}$$

were $f_n(x)$ corresponds to the function within n-th domain D_n and $\mathbf{1}_{D_n}(x)$ is the indicator function. It should be noted that $f_n(x)$ retains the core properties of $f(x)$ such as its invariance to specific transformations of x.

Our objective is to learn a function $g(\cdot)$ that maps $f(\cdot)$ to the probability of discrete categories $y \in Y$, where Y is the set of all possible classes. Consequently, the probability of class y given $f(x)$ can be defined as follows:

$$g(f(x)) = p(y|f(x)) = \sum_{n=1}^{N} p(y|f_n(x))\mathbf{1}_{D_n}(x) \tag{2}$$

It is evident that the predicted probability distribution depends on the domain D_n. Thus we define y_n to be the event $y|f_n(x)$ and consequently $p(y|f_n(x)) = p(y_n)$ is the probability of class y given the domain D_n. It is important to note that y_n cannot occur outside of D_n and due to that $p(y_n) = 0$ for all $D_i \neq D_n$. As a result, the new classification objective is to learn a function $g(\cdot)$ that maps $f(\cdot)$ to discrete categories $y_n \in Y_{D_n}$, for $n = 1, \cdots, N$, where Y_{D_n} is the set of all possible y_n classes within domain D_n.

In practice, we essentially need to define a set of distinct domain-specific classes with each class to exist only within its corresponding domain. To achieve this we define the relabelling function R_Y as follows:

$$R_Y(y, n) = |Y|n + y \tag{3}$$

with Y being the set of all possible labels of the domain-agnostic classification labels (e.g., $\{0, 1\}$ for binary classification as illustrated in Fig. 1a), $y \in Y$, n a unique identifier of each domain and $|Y|$ is the cardinality of Y. For a binary classification problem with 3 domains (Fig. 1c), $|Y| = 2$ and $n \in \{0, 1, 2\}$. Consequently, the relabelled classes for these domains would be $\{0, 1\}$, $\{2, 3\}$, and $\{4, 5\}$ respectively. This ensures that the same class label in different domains is treated as a different label, thus creating non-overlapping domain-specific tasks.

It is worth noting that the domain identifier n can be derived either by problem-specific knowledge (e.g., each game constitutes a different domain) or by applying a clustering algorithm on top of the initial projection function f.

Learning From Limited Data: An obvious drawback of the method described above is that it is not always possible to collect large amounts of data for every domain present within a dataset making the training of end-to-end classifiers infeasible. Additionally, even if we manage to train a conventional classifier, such a model will not be able to generalise in unseen categories of new domains. Few-shot classification allows models to learn and adapt to new, unseen categories using a few labelled examples. We define D^{train}, D^{val}, and D^{test} as distinct and non-overlapping datasets for training, validation, and testing purposes. During each iteration, called an episode, data is sampled from D^{train}, D^{val}, or D^{test}. Each episode consists of N classes (referred to as N-way) and K samples per class (referred to as K-shot). Within an episode, a support set (S) and a query set (Q) containing labelled samples are defined from D^{train}, D^{val} and D^{test} during training, validation and testing, respectively. The model is trained to classify the samples in the query set using the information from the support set. The purpose of D^{train}, D^{val}, and D^{test} (consisting of different domains) in FSL is to ensure the model can generalise well from a small number of examples, adapt to new tasks, and be evaluated fairly on its ability to handle unseen data with minimal supervision. Formally, the i-th sample in the support and query sets is represented, respectively, as (x_i^s, y_i^s) and (x_i^q, y_i^q). Note that N (number of classes) and K (number of samples per class) are hyperparameters that influence the difficulty of the few-shot classification setting. In particular higher values of N increase the number of comparisons and lower values of K reduce the amount of information available in the support set per episode.

3.2 Representation Components

The overall methodology employed in this paper is illustrated in Fig. 2. In representation learning, an encoder is a neural network model that, after training, efficiently encodes input into low-dimensional, high-level representations. This work tests the capacity of learned representations to predict engagement from raw gameplay footage within FSL. We use four different backbones: one CNN-based and three Transformer-based, which are then finetuned for FSL tasks. The first encoder is I3D [3], using inflated 3D convolutions with an embedding size of 512. The Transformer encoders include the base versions of MVD [35], VideoMAE [32], and VideoMAEv2 [34], each with a patch size of 16 and an embedding

size of 768. All models are pretrained on Kinetics [10] and accept a tensor of 16 RGB frames as input. Additionally, a ReLU-activated layer, optimised by the FSL objectives described in Sect. 3.3, is added on top of the frozen backbones. This trainable layer has the same dimension as the output of the corresponding backbone, and $L2$-normalisation projects the embedding on the unit sphere.

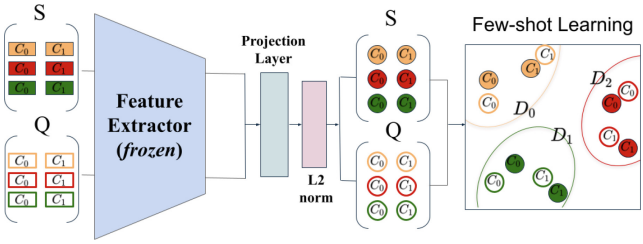

Fig. 2. A problem with 3 domains (yellow, red, green) and 2 classes (C_0 and C_1) per domain. S and Q represent the support and query sets, respectively. We first extract embeddings using a pre-trained frozen feature extractor. Following this step, we pass the extracted embeddings through a trainable projection layer and perform $L2$ normalisation. Finally, we optimise the few-shot losses using the resulting S and Q. (Color figure online)

3.3 Learning Objectives

Prototypical Network Loss: The Prototypical Network (PN) loss [30] is being used to learn a metric space in which classification can be performed by computing the distances between the query samples and the prototypes derived for each of the N classes within the support set. The PN loss is defined as

$$L_{PN} = -\frac{1}{|Q|} \sum_{(x_i^q, y_i^q) \in Q} \sum_{n=1}^{N_q} \mathbb{I}(y_i^q, n) \log(p_\theta(y_i^q = n | x_i^q)), \tag{4}$$

where $p_\theta(y_i^q = n|x_i^q) = softmax(-d(f_\theta(x_i^q), c_n^s))$ is the probability of a query sample x_i^q to fall into the class n, $c_n^s = \frac{1}{|S_n|} \sum_{(x_i^s, y_i^s) \in S} (y_i^s = n) f_\theta(x_i^s)$ is the prototype of class n, $d(\cdot)$ corresponds to the Euclidean distance, $f_\theta(\cdot)$ represents the learnable embedding functions, $|S_n|$ is the number of samples of class n in the support set, and $|Q|$ is the cardinality of the query set.

Matching Network Loss: Similarly to PN, the Matching Network (MN) Loss [33] aims to map an unlabelled example to a latent space defined by a small labelled set, enabling adaptation to new classes without fine-tuning. The optimisation objective uses a simple attention mechanism to weight sample distances between the support and query sets. The MN loss is defined as

$$L_{MN} = -\frac{1}{|Q|}\sum_{i=1}^{|Q|}\log p_\theta(y_i|x_i, S) \tag{5}$$

where $|Q|$ is the number of query samples, x_i is the i-th query sample, y_i is the true label of the i-th query sample, and S is the support set. Additionally $p_\theta(y_i|x_i, S) = \sum_{(x_j,y_j)\in S} a(x_i, x_j) \cdot \mathbb{I}(y_j = y_i)$ where $a(x_i, x_j)$ is the attention mechanism defined as $a(x_i, x_j) = softmax(f_\theta(x_i) \cdot f_\theta(x_j))$ which is the probability of a query sample x_i to be similar with the support sample x_j when projected on the space $f_\theta(\cdot)$ defined by the support set S.

Supervised Contrastive Loss: The objective of supervised contrastive (SC) learning [11] is to derive representations that make samples with the same label (positive pairs) more similar and samples with different labels (negative pairs) more dissimilar. The minimisation of this loss yields distinct and separable representations for each class. Inspired by previous work in few-shot representation learning for image classification [14] we formulate SC as follows:

$$L_{SC} = \frac{1}{|S|}\sum_{s\in S}\frac{-1}{|P_s^q|}\sum_{p\in P_s^q}\log\frac{\exp(r_s \cdot r_p/\tau)}{\sum_{q\in Q}\exp(r_s \cdot r_q/\tau)}, \tag{6}$$

where S is a set of all samples in the support set and P_s^q is the set of only the query set samples that are assigned to the same class as s while $q \in Q$ denotes any element in the query set. With r_s, r_p and r_q we denote the latent representations produced from a function f_θ that samples x_s, x_p and x_q, respectively. τ is a non-negative temperature hyperparameter that transforms the representation similarity distribution. Finally, s, p and q correspond to the index of the current support set sample, a query set sample positive to the current support sample and a sample in the query set, respectively.

4 Experiments

4.1 The GameVibe Few-Shot Dataset

The dataset is derived from the *GameVibe* corpus [1], a publicly available dataset of gameplay footage from 30 dissimilar commercial FPS games annotated for viewer engagement. The videos include audiovisual stimuli for engagement annotation, featuring diverse graphical styles and gameplay modes. Additionally, the videos contain only in-game sounds. All videos are limited to a maximum of 15 s of non-gameplay content, such as cut scenes or transition animations. Figure 3 illustrates the games of the corpus. The GameVibe corpus consists of four subcorpora (ids 1–4), each containing different gameplay clips from 30 FPS games and it is annotated by five randomly assigned participants, ensuring a variety of gameplay experiences and perspectives.

Corpus Modalities: The corpus includes two modalities: video frames and in-game audio. The video modality comprises high-resolution and low-resolution

Fig. 3. Screenshots from the 30 different FPS games annotated for engagement. List of game titles: (1) Apex Legends; (2) Battlefield 1942; (3) Blitz Brigade; (4) Borderlands 3; (5) Corridor 7; (6) Counter-Strike 2016; (7) Counter-Strike 2018; (8) Counter-Strike 2019; (9) Counter-Strike: Global Offensive; (10) Doom; (11) Dusk; (12) Far Cry 1; (13) Fortnite; (14) Heretic; (15) Hrot; (16) Insurgency; (17) Modern Combat: Sandstorm; (18) Medal of Honor 2010; (19) Medal of Honor 1999; (20) Medal of Honor: Pacific Assault; (21) Operation Bodycount; (22) Outlaws; (23) Overwatch 2; (24) PUBG; (25) Superhot; (26) Team Fortress 2; (27) Void Bastards; (28) Wolfenstein 3D; (29) Wolfenstein New Order; (30) Wolfram Wolfenstein.

gameplay videos captured at 30 Hz. Recent games have a resolution of 1280×720 pixels, while older games have 541×650 pixels. Each clip is 60 s long. The audio is stereo sound recorded at 44 kHz, extracted from the video. However, this study focuses solely on modelling engagement from raw gameplay footage.

Engagement Annotation: Participants annotated 30 one-minute gameplay videos (one per game) based on the following engagement definition: *"A high level of engagement is associated with a feeling of tension, excitement, and readiness. A low level of engagement is associated with boredom, low interest, and disassociation with the game"*. This definition of engagement is used due to its relevance and applicability within the specific context of FPS games [1,28], where continuous attention and responsiveness are critical. The games were presented randomly to avoid habituation effects. Due to high cognitive load, annotating engagement for multiple games simultaneously is impractical [23]. Short videos were chosen to balance annotation reliability and engagement stimuli richness. The 20 annotators (5 per subcorpus) were research staff and graduate students from the University of Malta. All annotation tasks were conducted in the same room with consistent conditions and equipment. Engagement annotations were collected with the RankTrace [15] tool and were tested for reliability via annotator QA methods [2]. Data was collected and analysed respecting GDPR and national ethics guidelines [22] (see **ethical considerations** section below).

Data Processing: In this work, we focus on vision-based experience modelling in games following best practices from games research and affect modelling [16,17,25]. Each session video is divided into non-overlapping 1-second

time windows. To account for reaction time between gameplay and annotation, the input is shifted by 1 s relative to the annotation time window [19]. Video segments are converted into RGB frames (30 per second). To reduce computational load, 16 RGB frames ($224 \times 224 \times 3$ pixels) are sampled at regular intervals within each window. Engagement traces (one per annotator, five per gameplay video) undergo min-max normalisation, scaling values to [0, 1] for each trace, and the median trace is derived to mitigate inter-annotator disagreement [7]. The resulting engagement trace is segmented into 1-second windows, and the average engagement value is calculated for each window.

Defining Labels per Game: Predicting user experience labels is challenging due to the subjective nature of such labels and the systematic reporting errors they might contain. Thus, following best practices from recent studies in the literature [18,27,29] we model engagement as categorical classification and we denote e as the engagement value within a time window. Assuming that the annotators exhibit consistent behaviour across games, we initially define game-agnostic classification labels by binarising the e values based on the median ground truth value of the entire set of affect annotation traces ($\bar{e}$) within a subcorpus (same participants within an annotation session). Consequently, the i_{th} time window falls under the *high engagement* and the *low engagement* class, respectively, when $e_i > \bar{e} + \epsilon$ and $e_i < \bar{e} - \epsilon$. Notably, the threshold $\epsilon = 0.1$ is used to eliminate ambiguous windows with annotation values close to $\bar{e}$ which may deteriorate the stability of the learning process (e.g., Fig. 1a). Since each game constitutes a domain of its own, we apply the relabelling method discussed in Sect. 3.1 to construct game-specific labels of engagement.

By utilising the unique game identifier $g_{ID} \in \{0, \ldots, 29\}$ (domain identifier n of Eq. 3), the set of (binary) labels derived in the previous step and the relabelling function of Eq. 3 we define low and high engagement class labels for each game. For the game with $g_{ID} = 0$ low and high engagement classes are represented by 0 and 1, respectively. For the game with $g_{ID} = 1$ the same classes are represented by 2 and 3. In general, for a game with $g_{ID} = n$, low and high engagement classes are represented by $2n$ and $2n + 1$, respectively. Since those classes are categories without a natural order, the game id value does not affect the output of our models. Finally, based on those labels we discard games that yield less than 10 samples per class since they don't allow for sampling Q and S sets from both classes and can lead to overinflated performance within the few-shot learning setting. The resulting dataset—*GameVibe* Few-Shot—is the first dataset for few-shot experience modelling within video games. We refer to the four subcorpora of the dataset as GVFS_1, GVFS_2, GVFS_3, and GVFS_4, the key properties of which are summarised in Table 1.

4.2 Experiment Protocol

In this study, we compare models optimised by the losses mentioned in Sect. 3.3, using the N-way K-shot FSL evaluation setting. For each episode, we randomly sample N classes and K samples per class for both training and validation. Following best practices from earlier studies [4,14], we report results on 5-way (5w)

Table 1. High-level statistics of the GVFS dataset. Each subcorpus includes 5 unique annotators. The train/valid/test columns refer to the number of games in the train validation and test set, respectively. Values within parentheses correspond to the number of distinct classes (2 per game).

subcorpus	#samples	#games	#train/valid/test	Binary Majority
$GVFS_1$	1054	23	8 (16)/8 (16)/7 (14)	52.47%
$GVFS_2$	1026	22	8 (16)/7 (14)/7 (14)	52.14%
$GVFS_3$	797	19	7 (14)/6 (12)/6 (12)	54.20%
$GVFS_4$	1186	27	9 (18)/9 (18)/9 (18)	50.34%

using both 1-shot (1 s) and 5-shot (5 s) settings, and we extend this evaluation protocol to the more challenging 10-way (10w) setting.

The models are trained using early-stopping after 10 epochs (40–200 episodes, depending on the experiment) without validation accuracy improvement, returning the best model based on validation accuracy. All models are optimised via SGD with a scheduler that halves the learning rate, α, every 5 epochs (20–100 episodes). Preliminary experiments indicate optimal learning rate values within $\alpha \in (10^{-3}, 10^{-2})$. Hyperparameters are selected via greedy search on the validation set. Model performance is evaluated in terms of accuracy following the evaluation protocol of prototypical networks for fair comparison across FSL methodologies. We repeat the model training experiments 5 times and sample 200 test episodes per run. All reported significance tests are at a 95% confidence interval (CI) with $p < 0.05$. Finally, we test a conventional end-to-end baseline model that employs the same architecture as discussed in Sect. 3.2, specifically leveraging pretrained backbones with identical layers and hyperparameters. The end-to-end model is trained on binary game-agnostic labels by optimising the cross-entropy loss. This baseline predicts high and low engagement on unseen games, serving as the lower bound of performance.

4.3 Results

The proposed methodology is tested extensively across the four different subcorpora of the GVFS dataset and four few-shot settings. Table 2 showcases the average accuracy of the models for the 5-way few-shot experimental setting (e.g., 5 classes are compared per episode) and the backbone architectures employed. It is evident that FSL significantly outperforms the end-to-end baseline (B) across both 5-shot and 1-shot experiments demonstrating the robustness of the proposed approach and the ability of the resulting FSL models to generalise across different domains within the same subcorpus. In particular, the SC method yields the highest accuracy in 20 out of 32 5-way experiments showcasing the reliability of the contrastive optimisation objective. However, all few-shot learners perform statistically on par in 22 out of 32 experiments indicating that the main source of performance improvement is the training framework (see Sect. 3.1) and not the few-shot learning objectives themselves.

Table 2. 5-way few-shot experiments (1-shot and 5-shot) across the GVFS sub-corpora and on average. Mean accuracy on the End-to-end baseline (B), Matching Network Loss (MN) Prototypical Network Loss (PN) and Supervised Contrastive Loss (SC). Bold values indicate the highest accuracy obtained for each sub-corpus and backbone used. Underlined values denote methods whose accuracy is statistically equivalent to the highest accuracy obtained as determined by the 95% CI.

Backbone	Method	$GVFS_1$		$GVFS_2$		$GVFS_3$		$GVFS_4$		Average	
		5 s	1 s	5 s	1 s	5 s	1 s	5 s	1 s	5 s	1 s
I3D	B	55.25	55.25	54.10	54.10	47.50	47.50	48.63	48.63	51.37	51.37
	MN	**91.21**	84.71	**91.34**	83.30	**89.02**	79.87	92.88	**85.42**	91.11	83.33
	PN	**91.21**	84.04	**91.34**	83.09	88.83	80.18	**93.12**	84.80	**91.13**	83.03
	SC	91.19	**84.80**	89.75	**83.76**	89.02	**82.53**	91.83	85.02	90.45	**84.03**
MVD	B	49.27	49.27	54.26	54.26	52.30	52.30	46.64	46.64	50.62	50.62
	MN	**81.73**	74.62	**84.95**	75.00	80.06	**74.36**	**84.53**	75.16	**82.82**	74.79
	PN	**81.73**	74.13	**84.95**	74.69	**80.07**	**74.36**	84.52	74.31	**82.82**	74.37
	SC	77.77	**75.96**	82.36	**75.71**	78.25	72.89	81.36	**77.6**	79.94	**75.54**
VideoMAE	B	53.68	53.68	49.45	49.45	52.92	52.92	49.25	49.25	51.33	51.33
	MN	78.86	66.31	75.83	64.14	73.54	61.69	79.96	64.00	77.05	64.04
	PN	77.95	64.80	75.83	62.96	73.35	61.51	79.96	63.64	76.77	63.23
	SC	**81.73**	**74.27**	**79.77**	**69.00**	**76.44**	**62.93**	**83.10**	**74.67**	**80.26**	**70.22**
VideoMAEv2	B	50.06	50.06	54.73	54.73	50.45	50.45	45.93	45.93	50.29	50.29
	MN	91.20	**83.82**	90.99	81.78	89.40	78.98	92.88	82.62	91.12	81.80
	PN	91.89	83.2	90.93	82.11	89.40	**80.27**	92.95	83.73	91.29	82.33
	SC	**94.08**	83.33	**92.36**	**82.56**	**90.45**	79.11	**93.83**	**84.49**	**92.68**	**82.37**

Table 3 illustrates the average accuracy of the models for the 10-way few-shot experimental setting (e.g., 10 classes are compared per episode) and the backbone architectures employed. In the same vein, the few-shot learners significantly outperform the baseline models (B) across all 5-shot, and in the vast majority of the 1-shot experimental settings (e.g., 13 out of 16) further highlighting the capacity of the proposed approach to build models that are able to generalise well across domains within the same corpus. The few-shot learners and the end-to-end model performed statistically on par in the remaining 3 experiments, where VideoMAE was used as the backbone. Once again, the SC method marks the highest accuracy in 17 out of 32 10-way experiments. Moreover, all few-shot learners performed statistically on par in 27 out of 32 experiments further strengthening the efficacy of the proposed framework to generalise across domains with limited data regardless of the choice of the backbone and few-shot learning objective.

Importantly, the average performance of the models across the GVFS sub-corpora showcases that accuracy improves significantly from 1-shot to 5-shot settings across all methods and backbones, reflecting the advantage of having more

Table 3. 10-way few-shot experiments (1-shot and 5-shot) across the GVFS sub-corpora and on average. Mean accuracy on the End-to-end baseline (B), Matching Network Loss (MN) Prototypical Network Loss (PN) and Supervised Contrastive Loss (SC). Bold values indicate the highest accuracy obtained for each sub-corpus and backbone used. Underlined values denote methods whose accuracy is statistically equivalent to the highest accuracy obtained as determined by the 95% CI.

Backbone	Method	$GVFS_1$		$GVFS_2$		$GVFS_3$		$GVFS_4$		Average	
		5 s	1 s	5 s	1 s	5 s	1 s	5 s	1 s	5 s	1 s
I3D	B	55.25	55.25	54.10	54.10	47.50	47.50	48.63	48.63	51.37	51.37
	MN	85.18	**78.22**	**84.53**	**77.53**	**82.65**	**75.47**	**87.16**	78.29	**84.88**	**77.38**
	PN	**84.95**	77.49	83.79	77.38	**82.65**	75.20	**87.16**	78.47	84.64	77.14
	SC	83.46	77.02	82.69	77.25	82.61	75.11	86.08	**78.56**	83.71	76.99
MVD	B	49.27	49.27	54.26	54.26	52.30	52.30	46.64	46.64	50.62	50.62
	MN	**73.37**	67.2	**74.89**	67.29	**70.23**	65.93	**75.31**	68.11	**73.45**	67.13
	PN	73.13	**68.09**	74.88	**67.53**	**70.23**	65.93	**75.31**	**68.18**	73.39	**67.43**
	SC	72.51	66.44	74.34	66.89	69.37	**66.00**	73.92	68.04	72.54	66.84
VideoMAE	B	53.68	53.68	49.45	49.45	52.92	52.92	49.25	49.25	51.33	51.33
	MN	69.08	58.31	66.36	57.17	63.92	54.62	70.25	58.02	67.40	57.03
	PN	68.36	**59.73**	66.36	56.95	63.92	54.93	70.25	57.11	67.22	57.18
	SC	**70.23**	58.87	**69.73**	**57.19**	**68.55**	**55.76**	**74.56**	**58.29**	**70.77**	**57.53**
VideoMAEv2	B	50.06	50.06	54.73	54.73	50.45	50.45	45.93	45.93	50.29	50.29
	MN	84.56	75.8	84.08	74.88	82.38	72.71	86.34	76.40	84.34	74.95
	PN	84.27	76.67	83.72	74.88	82.16	72.13	86.33	76.22	84.12	74.98
	SC	**86.2**	**78.27**	**85.40**	**76.15**	**83.89**	**73.22**	**87.64**	**77.62**	**85.78**	**76.32**

examples in few-shot learning. Furthermore, as the number of classes increases from 5-way to 10-way, a notable drop in performance is observed across all methods and backbones. This is due to the increased complexity and higher intra-class variability, making it harder for the models to distinguish between more classes with limited samples. Additional adversity comes from the subjective nature of experience annotation that inherently leads to noisy mappings between embeddings and labels. The baseline models (B) struggle to learn engagement patterns likely due to the significant domain gaps existent between different video games, thereby yielding poor generalisation. Unlike our approach, which uses domain-specific information to address these challenges, the baseline models assume a single distribution across games and thus fail to capture their unique features.

5 Discussion

This work introduced a framework for learning from limited data across multiple domains, focusing on engagement modelling in FPS games, where different games

represent different domains. The framework decomposed the general problem (e.g., predicting engagement in unseen games) into domain-specific problems by incorporating domain-specific knowledge into engagement labels. This approach was tested on the *GameVibe Few-Shot* dataset for few-shot experience modelling. We compared three FSL approaches with the conventional end-to-end method across the dataset's subcorpora. Results suggest that FSL models can effectively distinguish viewer engagement in unseen games, even with only one labelled sample per engagement level.

While the results demonstrate the method's efficacy in predicting viewer engagement in FPS games, this study does not consider individual player characteristics like personal interests, skill level or playing behaviour. Developing models that consider and adapt to individual player characteristics could yield a more personalised and accurate measure of engagement, thereby, enhancing the applicability and robustness of our approach across different gaming contexts. Future research should also attempt to extend this approach to other game genres, such as arcade, sport, puzzle or strategy games, which may exhibit more complex engagement patterns. The dissimilar FSL objectives we explored offered us valuable insights into the stability and generalisation of engagement models since the baseline model we compared them against follows the conventional classification training process in affective computing [16,20,27]. Despite notable improvements in accuracy, there might be trade-offs in other aspects of model performance, such as increased training time and computational complexity, that we do not consider here. While focusing solely on accuracy as the primary evaluation metric is standard practice in the literature it limits our assessment's scope. Future work should provide a more detailed analysis of both the baseline and FSL performance while exploring additional metrics, such as robustness to noisy data and the impact of data quality. Addressing these aspects will further enhance the model's applicability and reliability in real-world gaming environments.

Additional directions for future research include the thorough testing of the framework's efficacy across other modalities, such as sound and graph-based structures. Another direction involves tackling more challenging problems by relaxing annotation consistency, requiring the model to handle visual diversity and label shifts. Extending the framework to address domain-incremental scenarios, where models detect novel domains and assign labels during training, is an obvious next step. We also plan to evaluate our method's effectiveness in generating representations for tasks like few-shot segmentation and preference learning. Despite these open directions, the framework remains versatile and applicable to any few-shot learning task in video games, including gameplaying agents and content generation [40].

Ethical Considerations: The experiments were conducted on a publicly available dataset of gameplay footage and engagement annotations. Any personally identifiable information was replaced by untraceable ids. The Research Ethics Committee of the University of Malta approved the protocol. The dataset used contains no potentially offensive data, and the data processing has been described

in detail to assist scientific reproducibility. Finally, to the best of our knowledge, our work does not contribute to the development of deceptive applications or the escalation of existing privacy or discriminatory issues.

6 Conclusion

In this paper, we addressed the problem of few-shot domain generalisation in video games. We particularly introduced a new approach for multidomain few-shot experience modelling, utilising the *GameVibe* Few-Shot dataset, a variant of the *GameVibe* corpus, to test our models' generalisation capabilities across different contexts. We compared our method against existing modelling approaches and we thus tested several few-shot learning methods, such as metric learning and contrastive learning, and traditional end-to-end domain-agnostic approaches. Our experiments, across 4 few-shot classification settings and pretrained backbones, showed that few-shot learning methods outperform conventional techniques. These results highlight the potential of few-shot learning to effectively model noisy data that come from multiple domains. Thus our findings suggest that few-shot learning can advance AI and computer vision research within games and beyond, by reducing reliance on large annotated datasets, thereby making AI development more accessible and scalable even in highly subjective downstream tasks such as user engagement modelling.

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Hand2Any: Hand-to-Any Motion Mapping with Few-Shot User Adaptation for Avatar Manipulation

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Abstract. We explore user-agnostic and user-specific mapping techniques for manipulating avatars using hand gestures in virtual reality (VR). User-agnostic mapping allows users to control various avatars based on common user agreements, while user-specific mapping adapts to individual preferences using few-shot adaptation. Both approaches use supervised learning with paired datasets of motion data. Our evaluation, including quantitative assessments and user studies, shows that both techniques offer finer control, require less physical effort, and provide higher user satisfaction compared to existing methods. However, only a few users preferred user-specific mapping, indicating that the benefits of personalized mapping may vary. Finally, we demonstrate our methods' ability to manipulate avatars with variable joint structures, surpassing current methods.

Keywords: avatar manipulation · hand avatar · motion mapping

1 Introduction

Virtual avatars are essential for interacting within virtual environments, allowing users to express themselves and engage with others. A common method for avatar manipulation involves synchronizing the avatar's 3D motion with the user's movements, making the user's movements the controller for the avatar. However, this synchronization can be challenging, especially with morphological differences between the user and the avatar. While humanoid avatars have straightforward mappings due to their similarity to the human body, non-humanoid avatars, such as animals, require unique approaches to map the controller's actions to the avatar's movements. Therefore, it is crucial to develop methods that seamlessly map controller movements to avatar actions, regardless of their morphological differences. Additionally, diverse mapping methods are needed because manipulating avatars with distinct morphologies may require variations tailored to individual users' preferences and capabilities. Some users may prefer specific gestures for controlling certain avatars, highlighting the need for user-specific mapping strategies to ensure comfortable and intuitive avatar manipulation.

In this study, we explore both user-agnostic and user-specific mapping techniques for manipulating avatars using hand gestures in virtual reality (VR), as illustrated

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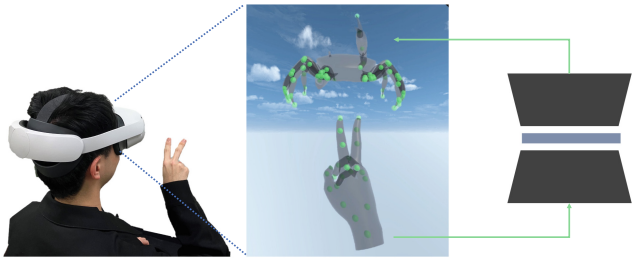

Fig. 1. Our proposed avatar manipulation. We design a neural network to create a mapping function that connects input hand motion to any avatar motion through supervised learning.

in Fig. 1. User-agnostic mapping allows users to control various avatars based on common user agreements, while user-specific mapping adapts to individual preferences using few-shot adaptation. Both approaches use supervised learning with paired datasets of motion data. Our evaluation, including quantitative assessments and user studies, shows that both techniques offer finer control, require less physical effort, and provide higher user satisfaction compared to existing methods. However, only a few users preferred user-specific mapping, indicating that the benefits of personalized mapping may vary. Finally, we demonstrate our methods' ability to manipulate avatars with variable joint structures, surpassing current methods.

2 Related Works

2.1 Motion Retargeting for Similar Morphology

Motion retargeting translates motions from a controller to an avatar. Various techniques have been extensively studied, particularly for morphologically similar objects like humanoid avatars [6,9,20]. These methods address height and joint configuration discrepancies using spatial constraints and human kinematics but are limited to repetitive motions (e.g., walking and running) and require time-consuming expert adjustments. Hence, there is a need for general mapping techniques that are easier to create and can handle a wider range of motions.

Deep learning-based motion retargeting addresses these manual adjustment issues and can handle various types of motions [3,23]. It leverages motion capture datasets, using markers attached to the body, to automatically optimize parameters in neural networks. This approach efficiently manages a wide range of motions without laborious adjustments. Holden et al. demonstrated robust avatar control using Autoencoder [12–14], while Harvey et al. proposed future keyframe interpolation using Long Short Term Memory [11]. Aberman et al. achieved mapping from avatars with many joints to those with fewer joints using convolutional networks [2]. However, these studies assume morphological similarity between the controller and avatar, limiting their effectiveness for different morphologies. Motion retargeting techniques that do not rely on morphological constraints can significantly expand avatar manipulation possibilities, allowing for more flexible user expression. Therefore, we propose a method for mapping motions between any controller and avatars with different morphologies.

2.2 Mapping Techniques for Different-Morphology Avatars

We classify motion retargeting methods into two groups: direct mapping and semantic mapping. Direct mapping correlates the controller's control points directly to the avatar using joints, centers of gravity, or meshes. For example, in VRChat[1], avatar manipulation is achieved by matching the positions of devices on the user's head and hands with corresponding avatar parts. Previous research has proposed methods exploiting partial similarities between the controller and avatar [5,15,28,30], often based on preliminary experiments [1,4,19,21]. However, these methods assume morphological similarity, limiting their effectiveness when the avatar's joint structure changes and requiring expert parameter adjustments.

Semantic mapping allows avatar manipulations without relying on morphological similarities by encoding motion to extract latent information and decoding it to reproduce the target motion. Parameters are optimized automatically using motion datasets, eliminating the need for expert adjustments. Previous studies have used both linear mixed models [5,29,32] and Neural Networks (NNs) [7,22]. Linear mixed models struggle with larger datasets, whereas NNs can efficiently optimize parameters even with extensive datasets. Notably, ACE [22] developed a framework for manipulating non-humanoid avatars using distinct motion datasets. However, it assumed identical distributions of the datasets, which is rarely the case, leading to unintended manipulations. Therefore, pairing the motions between the controller and the avatar is crucial.

Calibration is typically expensive and many users prefer to avoid it. User adaptation requires a user dataset and involves calibration efforts. To address these challenges, we propose optimizing mappings for each user using a compact dataset, streamlining the calibration process and enabling users to achieve their preferred avatar manipulation with minimal effort.

3 Method

3.1 Problem Definition

We formulate the problem of avatar manipulation as follows. Let C and A be variables that represent a controller pose and an avatar pose, defined as

$$C = \{\boldsymbol{c}^i \in \mathbb{R}^7 | 0 < i \leq I\}$$
$$A = \{\boldsymbol{a}^j \in \mathbb{R}^7 | 0 < j \leq J\},$$

where $\boldsymbol{c}^i$ and $\boldsymbol{a}^j$ represent the poses of the i-th and j-th joints, respectively. Each joint comprises a concatenation of a 3-dimensional position vector in local coordinates and a 4-dimensional rotation vector in quaternion form..

The goal of this problem is to estimate a mapping function $f : \mathbb{R}^{I\times 7} \rightarrow \mathbb{R}^{J\times 7}$ that captures the true distribution of $P(A|C)$. Generally, such a true distribution is inaccessible. Therefore, in a general machine learning framework, we predict the optimal parameter $\hat{\theta}$ based on a dataset $\mathfrak{D} = \{(C_n, A_n) | 0 < n \leq N\}$ as follows.

[1] https://hello.vrchat.com/.

$$\hat{\theta} = \arg\min_{\theta} \mathbb{E}_{C,A\sim\mathfrak{D}} L(f(C_n;\theta), A_n), \tag{1}$$

where N is the number of samples, $\mathbb{E}$ denotes expectation, and L is a loss function.

Let us assume $\mathfrak{D}$ contains data obtained from multiple users. Under this assumption, Eq. (1) performs a user-agnostic optimization. In addition, we design a user-specific optimization. It is achieved by adding $\boldsymbol{u}$, a $d_{\boldsymbol{u}}$-dimensional vector, to the input as a user representation, thus $f^u : (\mathbb{R}^{I\times 7}, \mathbb{R}^{d_u}) \rightarrow \mathbb{R}^{J\times 7}$. Extending the dataset to $\mathfrak{D}^u = \{(C_n, \boldsymbol{u}_n, A_n) | 0 < n \leq N\}$, a user-specific optimal mapping is obtained by

$$\hat{\theta}^u = \arg\min_{\theta} \mathbb{E}_{C,A\sim\mathfrak{D}^u} L(f^u(C_n, \boldsymbol{u}_n;\theta), A_n). \tag{2}$$

Here, in a test phase, users are not in $\mathfrak{D}^u$, and thus their $\boldsymbol{u}$ is unknown. In this case, we additionally need to estimate $\hat{\boldsymbol{u}}$ for each test user in user-specific optimization. We perform a few-shot fine-tuning for this additional optimization task.

3.2 Temporal Alignment of Controller-Avatar Poses

Previous study [19] obtained a dataset by restricting avatar motion to "walking" and "running." These actions do not require temporal alignment since they are cyclic, and humans can easily synchronize them. However, in more general cases of avatar motion, there may be an inevitable time gap in controller poses due to delays, followed by unpredictable dynamics. Thus, simply mapping the time series $C_{0:T} = \{C_t | 0 < t \leq T\}$ and $A_{0:T} = \{A_t | 0 < t \leq T\}$ according to t does not necessarily achieve the correspondence that the user intends.

To address this issue, we employ dynamic time-warping (DTW) [25] for synchronization between two time series datasets. The goal is to obtain a time-synchronized sequence $\mathfrak{D} = \{(C_t, A_t) | 0 < t \leq T\}$ from $\mathfrak{D}' = \{(C_t, A_t) | 0 < t \leq T\}$. Here, T represents the number of samples measured. Since the delay and following avatar dynamics result in a non-linear gap between the two time series $H_{0:T}$ and $A_{0:T}$, we use DTW to align them , which we give as

$$d(C_{t_1}, A_{t_2}) = |c'^{\max}_{t_1} - a'^{\max}_{t_2}| \tag{3}$$

$$c'^{\max}_{t_1} = \max_{0<i\leq I} \mathrm{sqrt}(||\boldsymbol{c}^i_t - \boldsymbol{c}^i_{t-1}||) \tag{4}$$

$$a'^{\max}_{t_2} = \max_{0<j\leq J} \mathrm{sqrt}(||\boldsymbol{a}^i_t - \boldsymbol{a}^i_{t-1}||). \tag{5}$$

Here, we use the maximum velocity among I joints of C_t and J joints of A_t (i.e., $c'^{\max}_t$ and $a'^{\max}_t$) because users typically move their controller while focusing on the most active part of the avatar.

3.3 User-Agnostic Mapping Optimization

The previous study [24] employed Bayesian linear regression for avatar manipulation, encountering challenges in setting up prior distributions and dealing with computational costs. In contrast, deep learning approaches offer a straightforward means of implicitly

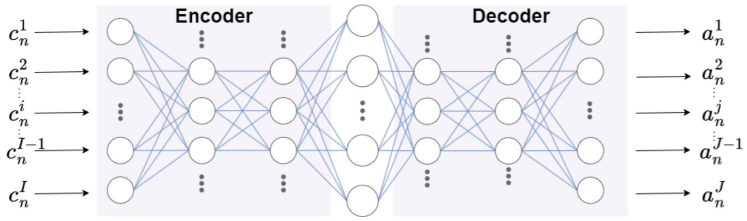

Fig. 2. Our mapping architecture translates the controller's position and rotation (C_t) to the avatar's position and rotation (A_t). I and J denote the number of joints in the controller and avatar, respectively.

expressing nonlinear relationships between inputs and outputs, as well as efficiently extracting features from large datasets.

Therefore, we leverage the encoder-decoder model in a multilayer perceptron considering C and A are vectors of length $7 \times I/J$. Let $g : \mathbb{R}^{I\times 7} \rightarrow \mathbb{R}^{d_z}$ be the encoder and $h : \mathbb{R}^{d_z} \rightarrow \mathbb{R}^{J\times 7}$ be the decoder, where $d_{\boldsymbol{z}}$ is the channels of a latent vector $\boldsymbol{z}$. Our proposed architecture is illustrated in Fig. 2. The mean squared error between avatar pose prediction and ground truth is used for the loss function L as

$$L = ||A_n - h(g(C_n))|| \tag{6}$$

3.4 User-Specific Mapping Optimization

We refer to the model described in the previous section as "user-agnostic mapping" because the dataset $\mathfrak{D}$ typically encompasses a broad spectrum of user data as common agreements. In this section, we extend it to "user-specific mapping" as illustrated in Fig. 3. We introduce a user embedding module into the user-specific mapping to capture user attributes. A simple linear layer takes userID as a one-hot-vector input and calculates a five-dimensional user vector $\boldsymbol{u}$ as an output. Each node in this vector corresponds to a userID. The user-specific mapping allows each user to manipulate avatars according to their intentions, facilitated by the user vector $\boldsymbol{u}$ inserted into the intermediate layer. Based on $\boldsymbol{u}$, the user-specific mapping f^u is defined as follows:

$$f^u(H, \boldsymbol{u}) = h(\mathrm{cat}(g(H), \boldsymbol{u})), \tag{7}$$

where $\mathrm{cat}(,)$ is the operation of concatenation.

After training f^u with $\mathfrak{D}^u$ and minimizing the loss L, we can adapt f^u to a specific user by determining the optimal $\boldsymbol{u}$ that best represents the user's data. To achieve this, we freeze the trained model parameters θ, leaving $\boldsymbol{u}$ as the only trainable variable. Subsequently, we remove the user embedding module from the model architecture and optimize $\boldsymbol{u}$ to minimize the loss L for the user-specific data. Given that $\boldsymbol{u}$ is only a 5-dimensional vector ($d_{\boldsymbol{u}} = 5$), we anticipate efficient optimization using few-shot data.

During the few-shot learning, the initial value of $\boldsymbol{u}$ significantly impacts accuracy [17]. Hence, we initialize $\boldsymbol{u}$ with the average value observed in the training set, facilitating efficient expression by the user vector $\boldsymbol{u}$ optimized with a few-shot dataset.

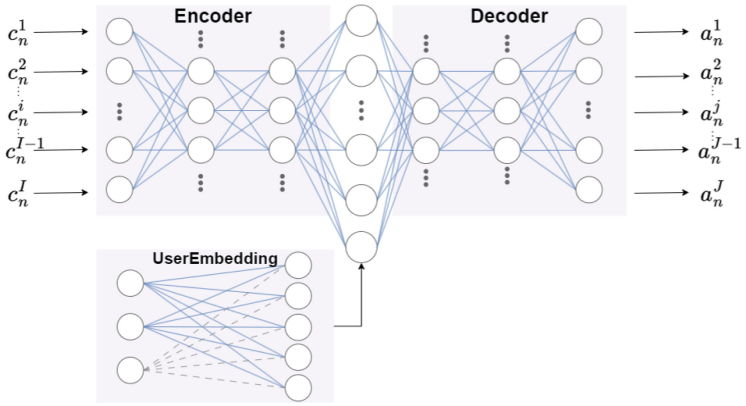

Fig. 3. User-adaptable architecture. The user embedding module takes a userID and computes a user vector ($\boldsymbol{u}$), which is then inserted into the intermediate layer to enable user-specific avatar manipulations.

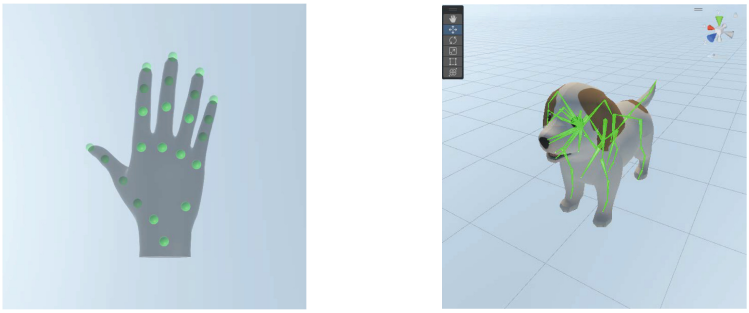

Fig. 4. Recording of motion. The left figure shows the user's right hand model with 23 joints. The right figure shows a pre-rigged avatar with joints assigned to limbs, tail, and other parts.

4 Dataset Acquisition Flow

4.1 Recording System

We developed the recording system and used to generate dataset $\mathfrak{D}$. We regard C as a right hand. First, the avatar and avatar animation buttons appear on the screen. Pressing the avatar animation button triggers the avatar's animation. Recording begins at a rate of 60 Hz after a three-second countdown, which is initiated by pressing the record button. The joints of the right hand C and the avatar A are to be measured as illustrated in Fig. 4. The experimental platform was developed in Unity 2021 for the Meta Quest 2 headset powered by an Intel Core i7 CPU and NVIDIA GeForce RTX 2080 GPU.

4.2 Procedure

A set of avatars was used and presented to the users, one at a time. After selecting an avatar, the scene was displayed. The user pressed the designated animation button to

Fig. 5. Frog avatar.

Fig. 6. Scorpion Avatar.

observe the avatar's motion and practiced with the displayed right hand while observing. When ready, the user started the measurement by pressing a button. The dataset $\mathfrak{D}$ was obtained in this manner.

5 Evaluation of User-Specific Mapping

We conducted cross-validation [27] to evaluate the effectiveness of user-specific mapping optimization in the deep learning model shown in Fig. 3. The user dataset was used as test data, while the remaining dataset served as training data.

Avatars of frogs and scorpions from the Unity Asset Store were selected, each with three measured motions: jumping, rolling, and flipping for the frog; walking, attacking, and falling for the scorpion (see Fig. 5 and Fig. 6). Seven participants provided 3,780 frames for the frog, and five participants provided 2,790 frames for the scorpion. Few-shot adaptation used five frames per motion, totaling 15 frames.

The user vector, expressing user attributes, was evaluated using mean squared error (MSE) under three conditions:

random: the user vector assigned a random value.
average: the average value of the user vector obtained in the pre-training phase
fine_tuning: the user vector obtained in the fine-tuning phase

The encoder and decoder hidden layers were set to 32 dimensions. The middle layer to 16 ($d_z = 16$). Adam optimizer [18] was used for optimization. MSE loss function calculated the loss for position and rotation separately, with the total error as a linear weighted sum. Hyper-parameter tuning was performed empirically for each avatar.

6 User Study

6.1 Hypothesis

We formulated the following two hypotheses to investigate the effectiveness of our avatar manipulations.

H1: Users prefer implicit mapping over explicit mapping for aligning the joints of the right hand with the avatar's joints during manipulation.
H2: Users prefer user-specific mapping over user-agnostic mapping for avatar manipulation.

Fig. 7. Dog avatar.

Fig. 8. Shark avatar.

Fig. 9. Crab avatar.

Fig. 10. Crow avatar.

6.2 Participants

Sixteen participants (including three females; mean age = 23.88; standard deviation = 0.86) participated in the study. They were recruited via a social networking service and a university email service. The experiment approximately lasted 90 min, and each participant was compensated with 1,500 yen (roughly 10 USD). This study was approved by our University's IRB.

6.3 Experimental Conditions

The experiment was a within-participant design and conducted individually for each participant. Each participant manipulated avatars using the following three controllers:

- Baseline mapping (BS): The mapping where the joints of the right hand correspond to the joints of the avatar.
- User-agnostic mapping (UA): The user-agnostic mapping model in Fig. 2.
- User-specific mapping (US): The user-specific mapping model in Fig. 3.

Each participant manipulated the following four avatars:

- Dog: The dog (see Fig. 7) has four types of animations such as rolling and turning its head around.
- Shark: The shark (see Fig. 8) has three types of animations such as waving its left hand and swimming.
- Crab: The crab (see Fig. 9) has three types of animations such as walking and raising its right hand.
- Crow: The crow (see Fig. 10) has three types of animations such as flying and attacking with its legs.

6.4 Baseline Controller

As the baseline controller, we selected one with right-hand joints corresponding to the avatar joints, based on the HandAvatar experiment procedure [16]. The mapping was defined through a mathematical optimization problem based on hand biomechanics. Since our method requires only the right hand, a preliminary experiment with five participants was conducted to create the mapping for right-hand avatar manipulation.

6.5 Experimental Procedure

First, participants underwent user dataset measurement to create the user-specific controller using tools explained in Sect. 4.1. A user optimization controller was created from the obtained dataset. Then, participants manipulated the avatar using three controllers. They selected the controller and avatar as instructed, practiced imitating the avatar's motions for about three minutes, and then recorded the avatar animation. Afterward, they answered a questionnaire regarding avatar manipulation. This process was repeated for all three controllers and four avatars. After completing all experiments, participants were interviewed about their experiences and comments. To reduce bias, the order of controller manipulation was randomized.

6.6 Evaluation Metrics

The pose deviation between the imitated and target animation was used for objective metrics. This deviation was calculated as the MSE of the position and rotation in two time-synchronized animations.

Additionally, participants were asked to answer questions about embodiment, precision, comfort, physical load, and satisfaction during a series of experiences on a seven-point Likert scale. The questions are adapted from previous studies by Jiang et al. [16] (Q1, Q3, Q5, Q7, and Q9) and newly added based on the study by Gonzalez-Franco et al. [10] (Q2, Q4, Q6, Q8, and Q10).

Q1 (Embodiment1): You feel very immersive and think the avatar is yourself.
Q2 (Embodiment2): The avatar's motions were caused by your motions.
Q3 (Precision1): You can control the motions of the avatar precisely.
Q4 (Precision2): You can closely match the avatar's posture to the task animation.
Q5 (Comfort1): You are comfortable using this gesture to control the avatar.
Q6 (Comfort2): There is no difference between your imagined gesture and the actual gesture.
Q7 (Physical Load1): It's physically easy to complete the task.
Q8 (Physical Load2): There's no strain on my hands or fingers while performing the task.
Q9 (Satisfaction1): You are generally satisfied with the experience.
Q10 (Satisfaction2): You can smoothly perform the task (anyone can easily imitate the avatar's posture).

7 Result

7.1 User-Specific Mapping Optimization Optimization

The results for the frog and scorpion avatars are shown in Fig. 11. For the frog avatar (see Fig. 11 right), the error under the conditions of user-specific vectors was either equivalent to or lower than that of the average user vector. For the scorpion avatar (see Fig. 11 left), the error under user-specific vectors was the lowest among all users. When the average user vector was compared to the fine-tuning user vector, the latter was found to be 4% and 10% lower in the frog and scorpion avatar results, respectively.

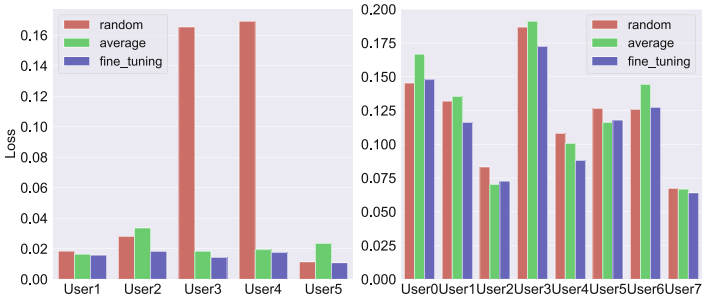

Fig. 11. Evaluation of user-specific mapping optimization. The left and right figures show the results for the scorpion and the frog avatars, respectively. The red, green and purple graphs show the random user vector, the average user vector, and the user optimization vector, respectively. (Color figure online)

7.2 Objective Evaluation

The objective evaluation was based on the deviation between the user-controlled and target avatar motion. The deviation was measured using the mean squared error of the differences in posture between each corresponding joint of the target avatar and the controlled avatar, with their root positions aligned. The results are shown in Fig. 12.

Two-way ANOVA was performed for Avatar factor v.s. Controller factor and adjustments to the degrees of freedom were made using the Greenhouse-Geisser correction in case of violation of the sphericity assumption (Mauchly's sphericity test). Table 1 shows the Two-way ANOVA results and F-values, p-values, and Effect size η^2. Subsequently, Shaffer's modified sequentially rejective Bonferroni post-hoc tests [26] were conducted for multiple comparisons. The avatar factor was fixed, and paired t-tests were conducted between controller conditions. Post-hoc test showed that user-agnostic mapping has a lower MSE value than baseline in Crab ($p < 0.001$), Crow ($p < 0.001$), Dog ($p < 0.001$), and Shark ($p < 0.001$) conditions. Similarly, user-specific mapping is also lower than baseline mapping as follows.

Crab
BS-UA condition: $(t(15) = 21.52, p < 0.001, r = 0.99)$
BS-US condition: $(t(15) = 18.25, p < 0.001, r = 0.98)$
Crow
BS-UA condition: $(t(15) = 10.80, p < 0.001, r = 0.94)$
BS-US condition: $(t(15) = 6.18, p < 0.001, r = 0.85)$
Dog
BS-UA condition: $(t(15) = 26.79, p < 0.001, r = 0.99)$
BS-US condition: $(t(15) = 23.45, p < 0.001, r = 0.99)$
Shark
BS-UA condition: $(t(15) = 10.24, p < 0.001, r = 0.94)$
BS-US condition: $(t(15) = 10.13, p < 0.001, r = 0.94)$

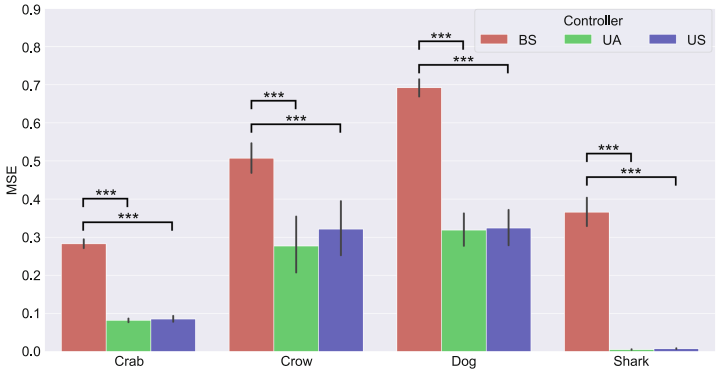

Fig. 12. Avatar pose deviation score for each avatar and controller (n = 16). Representative values indicate the mean, and error bars indicate the SE. *: $p < 0.05$, **: $p < 0.01$, ***: $p < 0.001$. The baseline, user-agnostic and user-specific are abbreviated as "BS," "UA," and "US," respectively.

7.3 Subjective Evaluation

The results are shown in Fig. 13. Two-way ANOVA (Avatar v.s. Controller) was conducted. Table 1 shows the Two-way ANOVA results and F-values F, p-values p, and Effect size η^2. Aligned Rank Transform (ART) [8,31] was performed for the data since the normality assumption was violated. When the assumption of sphericity was violated according to Mauchly's sphericity test, adjustments to the degrees of freedom were made using the Greenhouse-Geisser correction. Then, post-hoc tests were conducted using Shaffer's modified sequentially rejective Bonferroni post-hoc tests.

Post-hoc test showed that user-agnostic mapping has a higher score than baseline mapping precision1 ($p < 0.001$), precision2 ($p < 0.001$), physical load1 ($p < 0.001$), physical load2 ($p < 0.001$), satisfaction1 ($p < 0.001$), satisfaction2 ($p < 0.001$). Furthermore, it showed that user-specific mapping has a higher score than baseline mapping in precision1 ($p < 0.036$), precision2 ($p < 0.035$), physical load1 ($p < 0.001$), physical load2 ($p < 0.001$), satisfaction1 ($p = 0.007$), satisfaction2 ($p = 0.003$).

8 Discussion

8.1 User-Specific Mapping Optimization

From Fig. 11, our model shows that user-specific optimization effectively reduces MSE values in avatar manipulation. The green graph, representing the user-specific vector, demonstrates this improvement.

However, there is a notable difference between the scorpion and frog avatars. Users tend to make similar hand motions for familiar and simple avatar actions. For instance, frog motions, such as jumping and rolling, are easy to imagine and mimic, leading to less impact from user-specific optimization. Conversely, scorpion avatars, being less familiar and more complex, benefit significantly from user-specific optimization due to the diverse hand motion data in the dataset.

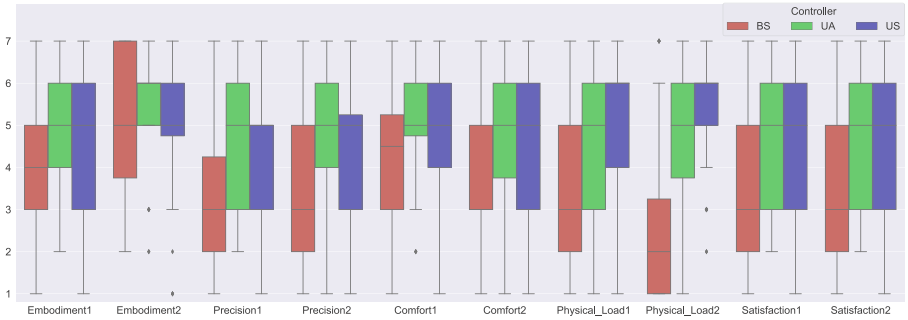

Fig. 13. Subjective ratings comparing the dimensions of embodiment, precision, comfort, physical load, and satisfaction towards the avatar manipulation (n = 16). The red, green, and purple boxplots are the baseline, user-agnostic, and user-specific mappings, respectively. (Color figure online)

8.2 Subjective Evaluation

In precision, physical load, and satisfaction metrics, significant main effects were found between baseline mapping and both user-agnostic and user-specific mappings, showing user preference for our methods. Users achieved higher precision, less physical load, and greater satisfaction, supporting [H1].

In embodiment metrics, no significant effects were found, likely due to differences in participant mindsets (controller as a tool vs. avatar embodiment). These results do not critically relate to [H1] and [H2] as users did not focus on embodiment during manipulations.

Similarly, no significant effects were observed in comfort metrics. The three controllers developed ensured unbiased and generally applicable mappings.

No significant main effect was found between user-agnostic and user-specific mappings, thus not supporting [H2]. Most users were satisfied with user-agnostic mapping, similar to default key settings in games, which are often sufficient without modification.

Despite this, some users preferred advanced settings for user-specific manipulation, with five participants rating user-specific mapping higher than user-agnostic mapping, indicating its usefulness for certain individuals.

8.3 Objective Evaluation

In avatar pose deviation results, there were significant main effects between baseline and user-agnostic, baseline and user-specific conditions. These results support [H1]. Furthermore, the fact that there was no significant difference between user-agnostic and user-specific conditions does not imply a lack of support for [H2]. Avatar pose deviation metrics indicate how users manipulate avatars precisely. So, the results ensure the precision of our methods.

8.4 Manipulating Avatars with Variable Joint Structures

Our method offers unique advantages that cannot be demonstrated through previous studies. For instance, consider an avatar with a variable joint structure, such as the robot avatar shown in Fig. 14. In this robot avatar, the position of the connection point in the foot segment changes, and the base segment also rotates. Manipulating such an avatar is challenging with a direct joint-to-joint mapping approach. In contrast, our method,

Table 1. The results of the F and p values and effect sizes for the Two-way ANOVA. The avatar and controller conditions are abbreviated as "Ava" and "Con," respectively. Items with a significant difference are in bold.

Questions	Condition	F		p	η^2
Embodiment1[Q1]	Ava	F(1.47, 19.08)	1.5340	0.2390	0.0390
	Con	F(3, 39)	0.0895	0.9654	0.0012
	Ava:Con	F(6, 78)	0.7172	0.6369	0.0143
Embodiment2[Q2]	Ava	F(1.28, 16.58)	0.5130	0.5273	0.0130
	Con	F(1.65, 21.49)	0.1882	0.7889	0.0016
	Ava:Con	F(6, 78)	0.1431	0.9899	0.0024
Precision1[Q3]	**Ava**	**F(2, 26)**	**14.009**	**0.0001**	**0.2240**
	Con	F(3, 39)	0.3680	0.7765	0.0057
	Ava:Con	F(6, 78)	0.3920	0.8821	0.0079
Precision2[Q4]	**Ava**	**F(2, 26)**	**10.935**	**0.0004**	**0.2001**
	Con	F(3, 39)	0.4185	0.7407	0.0041
	Ava:Con	F(6, 78)	0.7097	0.6427	0.0134
Comfort1[Q5]	Ava	F(1.3, 16.85)	3.7867	0.0597	0.0803
	Con	F(3, 39)	0.1070	0.9555	0.0010
	Ava:Con	F(6, 78)	0.2403	0.9618	0.0038
Comfort2[Q6]	Ava	F(1.48, 19.23)	2.1205	0.1558	0.0585
	Con	F(3, 39)	0.2318	0.8737	0.0033
	Ava:Con	F(6, 78)	0.1507	0.9884	0.0033
Physical Load1[Q7]	**Ava**	**F(2, 26)**	**16.094**	**0.0001**	**0.2215**
	Con	F(3, 39)	0.0576	0.9816	0.0004
	Ava:Con	F(6, 78)	0.1162	0.9942	0.0013
Physical Load2[Q8]	**Ava**	**F(1.43, 18.6)**	**22.5580**	**0.0001**	**0.3357**
	Con	F(1.95, 25.38)	0.2003	0.8146	0.0009
	Ava:Con	F(6, 78)	0.0541	0.9993	0.0010
Satisfaction1[Q9]	**Ava**	**F(1.35, 17.57)**	**5.4941**	**0.0229**	**0.1033**
	Con	F(3, 39)	0.1994	0.8961	0.0024
	Ava:Con	F(6, 78)	0.1756	0.9827	0.0042
Satisfaction2[Q10]	**Ava**	**F(1.41, 18.29)**	**7.8898**	**0.0066**	**0.1666**
	Con	F(2.05, 26.68)	0.0491	0.9551	0.0006
	Ava:Con	F(6, 78)	0.2378	0.9627	0.0050
Pose Deviation	**Ava**	**F(2.06, 30.91)**	**277.50**	**0.0001**	**0.4500**
	Con	**F(1.25, 18.74)**	**587.70**	**0.0001**	**0.4300**
	Ava:Con	**F(1.93, 28.93)**	**10.94**	**0.0001**	**0.0036**

Fig. 14. Robot avatar with a variable joint structure. Left) Running pose, where the base of the foot is positioned on the circumference of the lower part of the red eye. Right) Flying pose, where the base of the foot is positioned on the circumference of the upper part of the red eye. (Color figure online)

which employs a mapping technique based on semantic interpretation, proves to be effective in this scenario.

8.5 Limitations

In the user study, we chose HandAvatar [16] as the baseline, considering its high degrees of freedom for hand-based avatar manipulation. However, our baseline was designed through a preliminary experiment, not optimized computationally like HandAvatar, due to its requirement for both hands and lack of publicly available code, making perfect reproduction difficult. Future work should implement HandAvatar for both hands to allow a comprehensive and fair comparison.

We evaluated four avatars, each with three animations, selected from HandAvatar categories. While this selection aimed to vary the range of motion, it was not based on specific criteria, potentially introducing bias. To ensure validity in future experiments, we should formalize the classification method for avatars and movements to reduce potential bias.

9 Conclusion

In this study, we proposed user-agnostic and user-specific mapping methods and investigated their effects on avatar manipulations. Our method creates a mapping between a hand and an arbitrary avatar from paired datasets. Experimental results demonstrated that our method significantly improved accuracy, reduced physical burden, and increased user satisfaction compared to the baseline method that directly maps joints to each other. However, no significant differences were found between the user-agnostic and user-specific methods. The user-specific method was found to be more effective for some users, suggesting that personalized mapping approaches may benefit certain individuals.

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SkelFormer: Markerless 3D Pose and Shape Estimation Using Skeletal Transformers

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Abstract. We introduce SkelFormer, a novel markerless motion capture pipeline for multi-view human pose and shape estimation. Our method first uses off-the-shelf 2D keypoint estimators, pre-trained on large-scale in-the-wild data, to obtain 3D joint positions. Next, we design a regression-based inverse-kinematic skeletal transformer that maps the joint positions to pose and shape representations from heavily noisy observations. This module integrates prior knowledge about pose space and infers the full pose state at runtime. Separating the 3D keypoint detection and inverse-kinematic problems, along with the expressive representations learned by our skeletal transformer, enhance the generalization of our method to unseen noisy data. We evaluate our method on three public datasets in both in-distribution and out-of-distribution settings and observe better performance than prior works. Moreover, ablation experiments demonstrate the impact of each of the modules of our architecture. Finally, we study the performance of our method in dealing with noise and heavy occlusions and find considerable robustness with respect to other solutions. Supplementary materials and demonstration video available at https://vdavoodnia.github.io/projects/skelformer/.

Keywords: Markerless Motion Capture · Multi-view Human Pose Estimation · Inverse-kinematics · Skeletal Transformers

1 Introduction

Motion capture is an active field of research with applications in sports, entertainment, health, and human-computer interaction. Currently, optical motion capture technology offers the most reliable and accurate solution by using a large number of cameras that detect markers attached to the actor's body. As a result, optical motion capture is costly and has a time-consuming setup process, preventing its practical application in low-budget or outdoor settings. In contrast, markerless optical motion capture offers a more convenient and portable

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solution for capturing the pose, generally at the cost of accuracy. Therefore, a significant amount of research has been dedicated to improving markerless motion capture in recent years, delivering high-quality animations by using only a few RGB cameras [1,2]. Yet, markerless approaches either take a long time to process, *e.g.*, 6 min for a 30-second video [3], or they struggle to perform well in in-the-wild environments [4]. Optimization-based solutions that fit a parametric model to the detected keypoints often exhibit long run-times, while regression models trained on controlled and in-studio datasets lack generalization due to low-diversity backgrounds, appearance, and lighting conditions [5].

In this paper, we propose a novel pipeline for markerless motion capture, which we name SkelFormer. At a high level, SkelFormer consists of two main modules: a 3D keypoint estimator and a skeletal transformer. First, to simplify 3D keypoint detection while maintaining generalizability to a wider distribution of scenarios, our method uses a Direct-Linear-Transformation (DLT) [6] triangulation method on the output of off-the-shelf 2D keypoint estimators trained on in-the-wild data. Next, we propose a skeletal transformer motivated by Inverse-Kinematics (IK) approaches to generate body pose and shape parameters rather than relying on the commonly used optimization methods. This module significantly reduces computational overhead while exhibiting more accurate performance. However, the misalignment between the estimated 3D keypoints and the body joint configuration of motion capture data makes the integration of keypoint estimators and our IK module challenging. To address this, we propose a simple joint regressor, trained on a small set of synthetic, and use it to generate synthetic keypoints from motion capture data that are aligned with 2D keypoint estimators. Lastly, we apply several augmentations on the acquired keypoints and train our IK component using the noisy data.

To rigorously test the performance of our method, we evaluate SkelFormer in both In-Distribution (InD) and Out-of-Distribution (OoD) settings against prior works while noting that most previous works have been tested in InD settings. Next, detailed ablation experiments demonstrate the impact of each component. Finally, we study the performance of our method and examine its robustness to highly noisy and occluded data.

In summary, we propose SkelFormer, a regression-based IK solution that converts 3D body keypoint positions to a full-body pose and shape. Our model bridges the gap between the most accurate 3D keypoint detection algorithms and human pose and body mesh estimation with negligible performance degradation. SkelFormer achieves robust results and outperforms others in InD scenarios. Next, we find strong cross-dataset generalizability through OoD evaluation on two unseen datasets, achieving competitive performance to InD multi-view solutions. Additionally, our method exhibits high robustness (less than half of the error of optimization-based solutions) in severely noisy and occluded scenarios.

2 Related Work

2.1 Keypoint Detection

The 2D keypoint estimation field has seen substantial progress in recent years. Generally, 2D keypoint estimation models are categorized into top-down and

bottom-up approaches, each with their trade-offs in speed and accuracy [7–9]. Due to the availability of large-scale datasets, such as COCO WholeBody [10,11] and Halpe [12], 2D estimators have expanded into whole-body keypoints, potentially impacting 3D human pose and shape estimation. Previous works have proposed several strategies to infer the 3D keypoints of a subject, including semi-supervised learning [13,14], temporal [15,16], and multi-view [17,18] modelling. PoseBert [15] and volumetric Learnable Triangulation (LT) [17] are notable examples of temporal and multi-view methods, respectively, reporting 3D keypoint estimation with an error of below pixel-level accuracy. Our method leverages the advances in 3D keypoint estimation by using off-the-shelf models.

2.2 Pose and Shape Estimation

Regression-based methods generally predict the parameters of a body model, *e.g.*, SMPL [19,20] represented by body shape and pose components. The research on body pose and shape regression can be categorized into single-view and multi-view problems. Single-view approaches generally suffer from the inherent 2D image to 3D pose ambiguities, resulting in worse performance compared to multi-view methods. For instance, Pose2Mesh [21] uses a GraphCNN, consisting of a mesh coarsening encoder-decoder architecture, to regress the human body and shape from a single image. Similarly, GTRS [22] proposes a lightweight graph-based transformer network to uplift 2D keypoints to 3D pose and shape parameters. PyMAF [23] also explores visual encoder feature maps for pose regression. We design a multi-view markerless motion capture pipeline by taking inspiration from the advances in single-view research [1,22].

In the context of markerless motion capture via multiple views, the majority of methods are supervised, utilizing strategies such as collaborative learning [24], volumetric feature aggregation [25], multi-view feature fusion via attention [2], and pixel-aligned feedback fusion [4]. Since regression models typically rely heavily on the availability of annotated data and the diversity of postures in the training data, they tend to be limited to in-studio quality and do not perform well on OoD evaluations with background and appearance shifts [4]. Although a common solution to this problem is to pre-train the network on in-the-wild datasets, it does not guarantee better generalizability as the neural networks are susceptible to over-fitting and catastrophic forgetting. This is also evident by the best estimation error of 93 *mm* on the in-the-wild Ski-Pose [26] dataset reported in a recent work [2], which is three times bigger than their 33 *mm* error on the Human3.6 m [5] dataset. We address the OoD generalization limitation in multi-view setups by incorporating prior knowledge of human pose into the solution in an IK solver by training on a large set of motion capture data.

Optimization-based approaches fit the parameter of the SMPL model to features extracted from an image, such as 2D/3D keypoints and silhouettes [3]. Simplify-x [27] introduced VPoser, a human variational pose prior trained on a large collection of motion capture data, to reduce the complexity of the optimization space. Subsequently, the majority of recent works rely on VPoser to fit the SMPL model to 3D predictions, which is a time-consuming process that can

take up to a day to process a one-hour-long video [3,28]. Furthermore, variational Gaussian models are prone to mean collapse due to the prior distribution assumption. Additionally, as the optimization process is sensitive to initialization, their potential to fit noisy data accurately is hindered [29]. To address this challenge, several works like ProHMR [30] have proposed using regression-based models to initialize the optimization variables. Although this approach can speed up and improve accuracy, optimization algorithms remain susceptible to reaching undesirable local minima, especially on noisy, exotic, or unseen poses as discussed in [29]. In summary, though capable of modelling complex movements and interactions, optimization-based models often require careful hyper-parameter tuning for each recording sequence, making them difficult to use in the face of multiple constraints and impractical for fast-paced production. We show that compared to optimization solutions, our skeletal transformer performs more accurately with better noise and occlusion robustness.

2.3 Inverse-Kinematics Models

IK is the task of obtaining body joint rotations given several pose constraints, with applications in robotics [31] and animation [32]. In the context of human pose and shape estimation, HybrIK [1] proposed a hybrid analytical-neural IK solution that obtains the SMPL body rotations given the 3D keypoints estimated from monocular images. They designed their model to disambiguate the 2D image to 3D pose estimation by considering the shape of the human body and breaking the joint rotations to their swing and twist components. However, to our knowledge, IK applications of neural networks have not been explored for multi-view pose and shape estimation. This may be due to the superior performance of optimization methods, such as VPoser [27], yet at a high computational cost.

2.4 Skeletal Neural Networks

Previous research has reported superior performance for human motion modelling [33], 3D keypoint refinement [34], and 2D to 3D uplifting [15] using skeletal neural networks. In these approaches, the human body structure is modelled as a skeletal graph, and the neural networks exploit the graph structure by learning local and global pose features. Similar techniques have been proposed for hand pose estimation, where the margins for error are much lower than human body pose estimation [35]. Motivated by the recent success of transformers in several fields, such as natural language processing [36] and computer vision [37], they have been used for motion inbetweening and completion [38,39] of human poses, achieving high-quality results. We design a transformer model to capture the full pose state via contextualized latent representations from 3D joint positions.

3 Methodology

Overview. As illustrated in Fig. 1, our pipeline starts with a 3D keypoint estimator consisting of different sub-modules for tracking, 2D keypoint estimation,

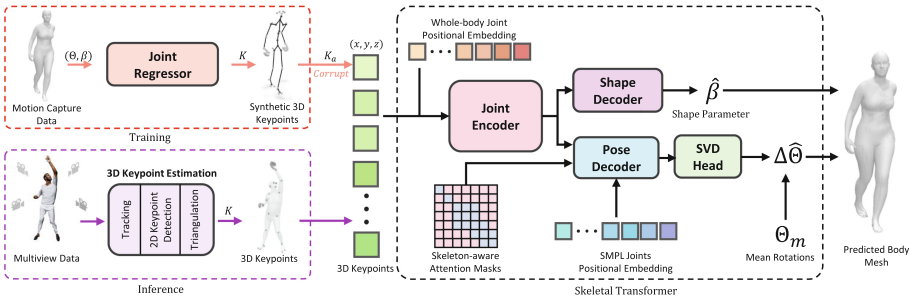

Fig. 1. An overview of the proposed skeletal transformer pipeline is demonstrated. During training, noisy 3D keypoints are generated using our joint regressor, while during inference, 3D keypoint are provided by off-the-shelf models. Then, our proposed skeletal transformer maps the keypoints onto the SMPL pose and shape parameters.

and triangulation, for which we use off-the-shelf models. Our proposed skeletal transformer then maps the estimated 3D keypoints onto the SMPL pose and shape parameters. The details of each part are given below.

3.1 3D Keypoint Estimation

Human Tracking. We employ Faster R-CNN [40], a well-established object detection model, to track the subjects in the input frames. Although more advanced methods, such as 3D skeleton tracking modules for crowded scenes [41], can be used, we did not observe any misidentification during single-person experiments.

2D Pose Estimation. To estimate the 2D joints, we employ HRNet-W48+Dark [7,9] trained on COCO WholeBody dataset [11]. Since this model is trained on in-the-wild datasets, it helps with the generalizability of our pipeline.

Triangulation. We choose a simple triangulation method by employing DLT [6] on 2D keypoints given the extrinsic camera parameters. For this purpose, we consider 2D detection scores for assigning point occlusions.

3.2 Skeletal Transformer

Traditional IK solvers often assume noise-free constraints, which is not always the case for observations, *e.g.*, 3D keypoints in markerless motion capture. As a result, iterative IK solvers generally perform better than regression models in reaching the local minima given the ground-truth joints, at the cost of additional computations. However, noise and occlusions can cause iterative IK solvers to reach sub-optimal solutions. To address this issue and speed up the process, we introduce an end-to-end learnable pose reconstruction model as illustrated in Fig. 1. Following, we present the detailed components of our skeletal transformer.

Joint Encoder. As depicted in Fig. 1, our network consists of a 3D joint encoder followed by pose and shape decoders. To account for the order of the joints in

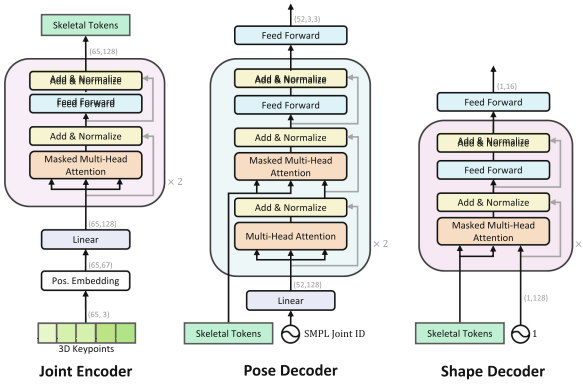

Fig. 2. Detailed architectures of our modules are presented.

a skeleton, we first inject information about their ordering by concatenating a positional embedding (*i.e.*, joint ID embedding) vector to each joint input [42]. The results are then passed through an embedding layer to match the dimensionality of the transformer's hidden layers. Next, we pass the embedded data through a series of transformer encoder blocks consisting of self-attention and feed-forward layers, followed by layer normalization, to obtain the skeletal tokens (see Fig. 2). To model joint occlusions, we mask out corresponding connections, encouraging the network to use contextual information embedded within the rest of the joints. The result is a latent representation of the joint positions, which is passed to the following decoders.

Pose Decoder. Our pose decoder estimates the body pose given the latent representations of joint positions. A positional embedding is fed into the decoder to relay information about all 52 SMPL+H skeleton joints. Then, the skeletal tokens and occlusion masks are passed through the multi-head attention layers in the decoder. Specifically, the decoder consists of several blocks of self-attention, multi-head attention, and feed-forward layers followed by layer normalization (see Fig. 2). Additionally, to block the unwanted correlations between far-away joints (*e.g.*, left and right hands) that existed in the training set, we set the attention weights such that each joint only attends to the other joints that are in a distance of less than 4 nodes in the kinematic tree.

Symmetric Orthogonalization. The next step is to obtain the joint rotations from the decoder's output. Although previous works [27,30,43] have suggested using 6-DoF representations of rotation matrices, we find that SVD symmetric orthogonalization proposed in a recent work [44] yields more accurate results and converges faster. Therefore, the output of the pose decoder is passed through a fully connected residual layer that outputs a square matrix $M_{3\times3}$ with SVD of $U\Sigma V^T$. Then, its symmetric orthogonalization $\Theta \in SO(3)$ is obtained by:

$$\Theta = U\Sigma_o V^T, where\ \Sigma_o = diag(1, 1, det(UV^T)). \tag{1}$$

Shape Decoder. To obtain the shape parameters, we design a shape decoder with a similar architecture to our pose decoder. Since the order of joints does not affect the shape decoder, we remove the self-attention layers and set the sequence length of the shape decoder to one, effectively reducing it to a feed-forward network with multi-head attention (see Fig. 2).

3.3 Joint Regressor

Our goal is to train our skeletal transformer on a large collection of motion capture data consisting of samples represented by SMPL shape and pose parameters. However, the skeleton configuration extracted from whole-body keypoint detectors is not aligned with the SMPL joints. To solve the alignment issue, we use a joint regressor to convert SMPL representations to our desired skeletal configurations. Currently available joint regressors have been reported to be either inaccurate [45], or simply obtained by selecting vertices on the *surface* of the body, which is not biomechanically correct nor in accordance with existing body models. Therefore, we propose a novel joint regressor and training scheme using a small amount of synthetic data to align the SMPL with the 3D keypoints.

The joint regressor, defined as a linear layer $K = \mathcal{J}V$, is trained to map the body mesh vertices $V \in \mathbb{R}^{6890\times 3}$ to 3D keypoints $K \in \mathbb{R}^{J\times 3}$, where J is the number of joints in the 3D keypoint configuration. In order to train the joint regressor, we randomly take 10,000 SMPL body samples from the AMASS dataset [46] and render them from four orthogonal views after removing the root translation and adding random root rotations augmentation. Next, following the process of Sect. 3, we use HRNet-W48+Dark [7,9] to estimate whole-body keypoints followed by DLT triangulation for obtaining 3D keypoints.

To encourage sparsity and avoid out-of-body predictions, previous works [45] have suggested using an L_2 regularization on the joint regressor with the goal of (a) enforcing a sum of 1 for vertex weights of each 3D joint; and (b) encouraging all of the weights of the joint regressor to lie between 0 and 1. However, doing so creates a trade-off between regularization and accuracy. To solve this issue, we apply a temperature-scaled Softmax function over the trainable parameters of the joint regressor ϕ, thus automatically satisfying both constraints. Doing so also gives us control over the vertex sparsity for each of the 3D joints. The joint regressor's weights for the i_{th} keypoint are computed as:

$$\mathcal{J}_i(\phi) = \frac{e^{\phi_i/T}}{\sum_{j=1}^{J} e^{\phi_j/T}}, \tag{2}$$

where T controls the sharpness of the distribution of the vertex weights. We use an L-BFGS optimizer [47] to increase the training efficiency. We empirically set the temperature to $T = 10$ so that 3 to 10 vertices contribute to each joint.

3.4 Data Preparation

We extract pairs of 3D keypoints K (using our joint regressor), body joint rotation matrices $\Theta \in \mathbb{R}^{52\times 3\times 3}$, and shape parameters $\beta \in \mathbb{R}^{16}$ from human motion

capture data. Next, we measure the average joint rotations across the dataset to normalize Θ. As some layers in the skeletal transformer share weights across all the joints, the symmetry of the left and right rotations is important. Yet, the mean pose Θ_m provided and used in previous works [30,43,48] is not symmetrical. To address this, we add the mediolateral mirrored samples to the dataset and use quaternion averaging as described in [49] to measure Θ_m, subsequently normalizing the pose by $\Delta\Theta = \Theta_m^{-1}\Theta$. During training, we apply a range of online augmentations to diversify the data and enhance our model's robustness. The augmentations include joint occlusions, body rotation, keypoint noise addition, shape augmentation, outlier addition, and mediolateral mirroring.

3.5 Training

To train our skeletal transformer, we feed the model with augmented data K_a to obtain pose and shape parameters, which are then passed through the SMPL Forward Kinematics (FK) layer. We then use a combination of rotational, positional, and shape losses to leverage different scopes and granularities to help with training [50]. Our final loss is calculated as the sum of the following losses.

Rotation Loss. Our rotational loss is the sum of the global and local geodesic distances between the predicted $\hat{\Theta}$ and ground-truth rotations Θ of every joint:

$$L_R(\Theta, \hat{\Theta}) = \arccos\left(\frac{\mathrm{tr}(\Theta\hat{\Theta}^\intercal) - 1}{2}\right). \tag{3}$$

Position Loss. Our model uses the SMPL FK layer to obtain the keypoints $\hat{K}$ and vertices $\hat{V}$. We then define a positional loss using $L_P = L_{1;s}(\hat{K}, K) + L_{1;s}(\hat{V}, V)$, where the $L_{1;s}$ represents a smoothed L_1 loss [51]. This loss can be seen as a combination of L_1 and L_2 distances, which is less susceptible to outliers than L_2, and it has less near-zero penalty than the L_1 distance.

Shape Loss. Finally, we minimize the $L_S = L_2(\hat{\beta}, \beta)$ distance between the estimated $\hat{\beta}$ and ground-truth shape parameters β to train our shape decoder.

4 Experiments

4.1 Datasets

AMASS. The Archive of Motion Capture as Surface Shapes (AMASS) [46] is a collection of 3D human pose and shape information collected from multiple motion capture databases. It contains over 40 h of motion capture data from more than 300 subjects and spans over 11,000 actions. We follow the standard train, test, and validation splits used in prior works [3,27]. This dataset is used solely to train the skeletal transformer IK solver.

Human3.6 m. Human3.6 m [5] is the standard benchmark for evaluating 3D human pose, shape, and body estimation in multi-view and single-view approaches. Following previous works [52], we perform our evaluations using Protocol-I, where the root-centred MPJPE and PA-MPJPE on subjects 9 and 11 are measured. We use this dataset for ***InD*** evaluation of our method.

RICH. Real scenes, Interaction, Contact and Humans (RICH) dataset [53] is a recently published dataset of multi-view videos with accurate markerless motion-captured bodies and scenes. The test set contains one withheld scene and 7 unseen subjects in 52 scenarios, captured using four cameras. We use this dataset for ***OoD*** evaluation of our method in outdoor settings.

MPI-INF-3DHP. The Max Planck Institute for Informatics 3D Human Pose dataset (MPI-INF-3DHP) [54] is a collection of over 1.5 million frames captured from eight angles, featuring eight actors performing various actions like sitting, dancing, and exercising. We follow the previous works by evaluating our model on subject 8 of the training set [25,55]. We use this dataset for ***OoD*** evaluations.

4.2 Evaluation Metrics

To evaluate the performance of our method, we employ standard evaluation metrics in 3D pose estimation literature. Mean-Per-Joint-Position-Error (**MPJPE**) measures the Euclidean distance between the estimated joint positions and the ground-truth joint positions, averaged over all joints in the skeleton. **PA-MPJPE** is an extension of MPJPE, where a rigid alignment between the estimated and ground-truth keypoints is applied prior to error measurement. This metric shows how well the skeleton is estimated, regardless of scaling and rotation. Additionally, we report **MPVPE** and **PA-MPVPE** to show the error between ground-truth and predicted vertices of body mesh. Additionally, we report **AUC** and **PCK** at a threshold of 150 *mm* according to MPI-INF-3DHP [54] evaluation criteria. Finally, **Rotation Error** is measured by the geodesic distance between the ground-truth and predicted poses.

4.3 Implementation Details

Hyper-parameters. We select 65 joints from the 133 joints of the whole-body skeleton configuration, excluding most facial landmarks while keeping the eyes, ears, and nose. We choose two transformer blocks for the encoder and each of the decoders. As illustrated in Fig. 2, We use a positional embedding of size 64 in the encoder and decoder while setting hidden layer dimensions to 128 within all layers. The shape and pose decoder heads use 1024-dimensional residual layers.

Optimization. We train our model using AdamW optimizer [56] with a batch size of 1024 on an NVIDIA A4000 GPU. We chose a learning rate of 1e-3, which is warmed up with a factor of 1e-4 for the first 2000 iterations and gradually reduced with a cosine annealing scheduler over 50000 iterations until it reaches 1e-7. The training of the network takes less than 18 h to complete.

Mirror Test. Similar to the flipping test performed in 2D keypoint estimation methods [7,37], we perform a mirroring test during inference to reduce the model's biases towards left and right body parts.

Computation Cost. Our model contains 6.631 M parameters with 159.482 M Floating Point Operations (FLOPs) for a single input. As a result, our skeletal transformer can solve the IK problem in 66 *ms* for a batch size of 512 using approximately 4G of GPU memory. Consequently, our SkelFormer pipeline takes 274 *ms* to predict body pose and shape parameters from a four-camera frame.

Table 1. The comparison of our method in InD settings against prior multi-view works on the full test set of the Human3.6 m dataset. * denotes the results from using ground-truth 3D keypoints as input

Method	MPJPE↓	PA-MPJPE↓	Output
CPN+DLT [57]	32.1	27.8	Joint Pos. Only
LT [17]	20.7	17.0	Joint Pos. Only
Pose2Mesh [21]*	29.0	23.0	Mesh Only
Huang *et al.* [58]	58.2	47.1	Joint Rot.+Mesh
Shin and Halilaj (SPIN^{4,cal}) [25]	49.8	35.4	Joint Rot.+Mesh
Shin and Halilaj (main) [25]	46.9	32.5	Joint Rot.+Mesh
Gong *et al.* [59]	53.8	42.4	Joint Rot.+Mesh
Jiang *et al.* [60]	50.2	37.3	Joint Rot.+Mesh
Jia *et al.* [4]	33.0	26.9	Joint Rot.+Mesh
SMPLify-X (LT) [27]	26.3	21.2	Joint Rot.+Mesh
SkelFormer (CPN)	33.5	27.8	Joint Rot.+Mesh
SkelFormer (LT)	**25.2**	**20.6**	Joint Rot.+Mesh

Table 2. Comparison of our method in OoD settings against prior works.

Method	MPVPE↓	MPJPE↓	PA-MPJPE↓	PCK↑	AUC↑	OoD
RICH Dataset						
METRO [61]	134.5	129.6	–	–	–	✓
METRO [61]	107.9	98.8	–	–	–	✗
SA-HMR [62]	103.0	93.9	–	–	–	✗
IPMAN-R [63]	89.9	79.0	47.6	–	–	✗
SPIN [43]	129.5	112.2	71.5	–	–	✓
PARE [64]	125.0	107.0	73.1	–	–	✓
CLIFF [65]	122.3	107.0	67.2	–	–	✓
SkelFormer (HRNet)	**39.9**	**44.2**	**35.6**	–	–	✓
MPI-INF-3DHP Dataset						
Liang and Lin [55]	–	–	59.0	95.0	65.0	✗
Shin and Halilaj [25]	–	–	50.2	97.4	65.5	✗
Jia *et al.* [4]	–	–	**48.4**	**98.6**	67.3	✗
SkelFormer (HRNet)	–	–	54.8	97.5	**67.4**	✓

4.4 Results

InD Testing. To evaluate SkelFormer, we first compare its performance to prior works on the Human3.6 m [5] dataset. In this experiment, most benchmarks pre-train their models on multiple datasets and fine-tune them on the Human3.6 m training set. Accordingly, we report the performance of our model using keypoint estimators, *i.e.*, CPN [57] and LT [17], trained on Human3.6 m dataset (InD). To this end, we use 3D predictions from CPN (followed by DLT triangulation [6]) and LT, both of which are trained following the standard evaluation protocol [5]. Moreover, we train our skeletal transformer on the AMASS dataset [46] using a 17-joint configuration as per Human3.6M. Delving deep into the results in Table 1, we observe that our method outperforms the best regression model [4]. Additionally, SkelFormer outperforms optimization solutions from multi-view

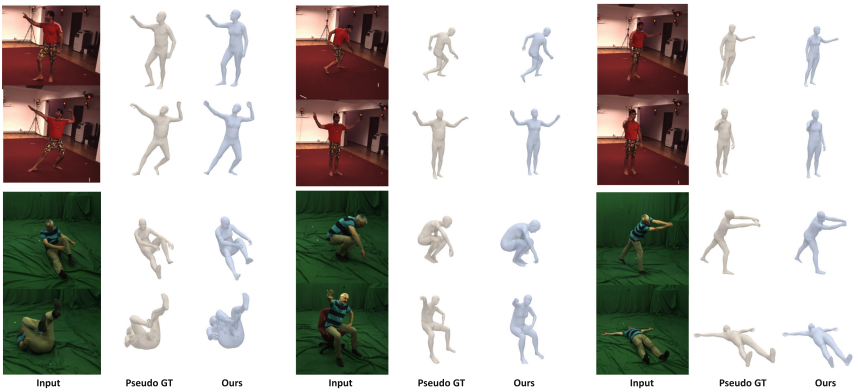

Fig. 3. A visual comparison with the pseudo-ground-truth from [52] is provided, presenting the realism and accuracy of our SkelFormer.

2D keypoints, 3D keypoint [58], and SMPLify-X (LT) [27]. A notable solution is Pose2Mesh [21], another regression-based approach that exploits the 3D ground truth as input. However, SkelFormer outperform Pose2Mesh despite not using the ground-truth information. Finally, although not directly comparable, our method achieves the closest performance to solutions that only predict joint positions, namely, LT [17] and CPN+DLT [57].

OoD Testing. Next, to test our method in OoD settings, we use HRNet-W48+Dark [7,9] as the keypoint extractor, which has not been trained on the RICH [53] or the MPI-INF-3DHP [54] datasets. Moreover, we keep our entire pipeline frozen and do not fine-tune any of its components on any portion of these datasets. The results are presented in Table 2. It should be noted that while prior works on the RICH dataset are monocular pose estimation approaches, all prior works on MPI-INF-3DHP are multi-view solutions and use all available views. On the RICH dataset, Table 2 shows that other solutions, such as SPIN [43] and CLIFF [65], suffer greatly in OoD setups (obtaining 172% and 127% additional error w.r.t. their InD performance on Human3.6 m dataset). In contrast, our method shows competitive results compared to InD solutions by outperforming prior works on RICH while showing relatively minor degradation on the MPI-INF-3DHP dataset. Finally, we demonstrate the fitting quality of our model compared to the pseudo ground truth [52] on the Human3.6 m and the MPI-INF-3DHP datasets in Fig. 3, showing improvements in the feet and hands.

Ablation Study. We test the importance of different network components and report the results in Table 3. Given our goal of increasing generalizability in the presence of noise and occlusions, we conduct experiments on motion capture data from the AMASS [46] testing set at the presence of 20% occlusion and additive Gaussian noise with $\sigma = 20$ *mm*. First, we demonstrate the effectiveness of symmetric orthogonalization by replacing our SVD operation with the commonly used 6-DoF representation, showing a drop of 0.8 *mm* and 0.19° of MPVPE and rotational error when SVD is removed. Next, we experiment with different

Table 3. Ablation Study results in the presence of 20% occlusion and a Gaussian noise of $\sigma = 20$ *mm*.

Experiments	MPVPE↓	PA-MPVPE↓	Rot. Error↓
SVD Symmetric Orthogonalization	**18.6**	**12.1**	**2.94**
6-DoF Symmetric Orthogonalization	19.2	12.6	3.13
Local Rot. Loss	19.9	13.3	3.01
Global Rot. Loss	19.3	13.2	3.63
Global+Local Rot. Loss	**18.6**	**12.1**	**2.94**
Part-based Decoder Att. Weights	18.8	12.1	2.96
$d = 1$ Decoder Att. Weights	19.6	12.8	2.97
$d = 2$ Decoder Att. Weights	19.1	12.4	2.93
$d = 3$ Decoder Att. Weights	19.3	12.6	2.96
$d = 4$ **Decoder Att. Weights**	**18.6**	**12.1**	**2.94**
$d = 5$ Decoder Att. Weights	18.9	12.6	2.97
w/o Decoder Att. Weights	19.7	12.8	3.04
w/o Rotation Normalization	19.5	13.1	3.01
w/o Mirror Test	27.7	19.7	3.56
w/o Shape Aug.	19.8	12.8	2.92

combinations of local and global (after FK) rotational loss functions to train our model. We observe that combining global and local rotation losses results in a better performance. Next, we showcase the effectiveness of our decoder masking strategy using the attention weights in three experiments: *i*) part-based experiment, where we restrict the attention within upper right, upper left, lower right, lower left, and center regions of the body; *ii*) node distance experiments, where we restrict the attention based on the node distance d in the skeleton kinematic tree with values between 1 and 5; and *iii*) without skeleton-aware attention masks, where we allow each joint to attend to any other joints in the decoder. We observe that the best results are obtained for $d = 4$, improving over the vanilla transformer decoder by 1.1 *mm* MPVPE and 0.2° rotation error. Lastly, we show the impact of rotation normalization, mirror testing, and shape augmentation, where a significant drop in performance is seen in the individual absence of each of these components, highlighting their significance. Finally, we perform a qualitative experiment on our joint regressor. To this end, we use VPoser [27] to fit onto 3D keypoints using our joint regressor and the one provided in prior works [27] on Human3.6 m dataset [5]. In Fig. 4, we show the visual fidelity of the effect of our joint regressor in comparison to [52], specifically, in better fitting to the feet regions.

Robustness to Noise and Occlusions. To evaluate the performance of our model on motion capture data, we perform experiments by simulating noise and occlusion. For these experiments, we compare our skeletal transformer with VPoser [27] and its temporal version, VPoser-t [3], which tries to maximize tem-

Fig. 4. The fitting performance of VPoser is demonstrated while using (a) our proposed joint regressor; and (b) the joint regressor from [52].

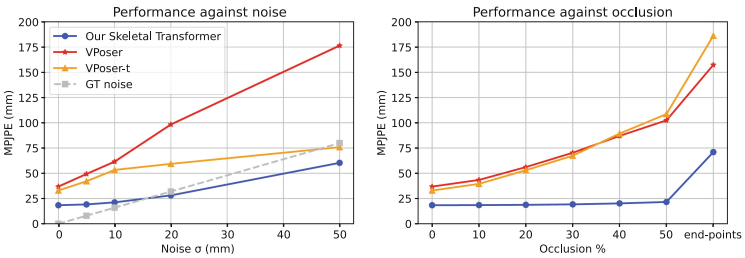

Fig. 5. Robustness of our skeletal transformer is highlighted in the presence of different levels of noise and occlusion by comparing it against VPoser and VPoser-t.

poral consistency during optimization. Figure 5 demonstrates the performance of different models on the AMASS [46] testing set. In the first experiment, we introduce varying noise levels to the input data and evaluate the robustness of our method. More specifically, Gaussian noise with varying standard deviations up to 50 *mm* is added to the input, effectively increasing the MPJPE of ground truth up to 80 *mm* (referred to as GT noise). However, our skeletal transformer predicts body pose and shape parameters, which result in lower error after $\sigma = 15$ *mm*. Additionally, its performance only degrades by 19.7 *mm* at maximum noise level, while VPoser fails to predict less noisy poses. Lastly, by comparing our method to a temporal model (VPoser-t), we demonstrate the model's robustness to noisy scenarios. Next, we report the performance of our model in the presence of occlusions, where each joint is randomly masked with varying occlusion amounts of up to 50%. Figure 5 shows that the performance of our skeletal transformer barely changes between 0 and 50% occlusion, thus showing the model's capability to exploit local and global joint information. In contrast, VPoser's performance drops by more than 20 *mm* in the presence of only 20% occlusion. We finally report the performance in an extreme occlusion scenario, where only 7 end-point keypoints are provided. Our model outperforms other solutions while maintaining a reasonable accuracy.

Fig. 6. Sample results on the RICH dataset are presented.

Qualitative Assessments. Figure 6 demonstrates the visual quality of SkelFormer on RICH [53] dataset in equally-sampled frames from the testing set. Our method yields the correct pose and shape with high overlaps with the subject.

5 Conclusion

In this paper, we presented SkelFormer, a novel multi-stage pipeline consisting of keypoint estimators and a skeletal transformer for markerless human motion capture. Our method leverages large amounts of motion capture data to address the poor generalization of multi-view human shape and pose estimation approaches while outperforming optimization approaches in accuracy. Through extensive experiments, we demonstrated the effectiveness of SkelFormer in several challenging conditions, including InD and OoD settings. Specifically, We achieve the best results in InD experiments among prior multi-view approaches and show competitive OoD performance, demonstrating the generalization of our pipeline.

Furthermore, we show the effectiveness of the proposed elements in our skeletal transformer through our ablation study. Finally, our single-frame skeletal transformer exhibits higher noise and occlusion robustness than optimization approaches that rely on temporal data.

Future Work. The major limitation of this work is the accumulation of errors in the multi-stage pipeline. Although our network mitigates jitters and occlusions better than other approaches, it is still prone to artifacts caused by mediolateral mix-ups and incorrect tracking. More accurate human trackers [40,41], 2D keypoint estimators, and triangulation techniques may be explored to remedy such issues. Another promising direction is to use SkelFormer to initialize generative models such as DMMR [28] and HuMoR [3] to further refine the motions.

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PlaMo: Plan and Move in Rich 3D Physical Environments

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Abstract. Controlling humanoids in complex physically simulated worlds is a long-standing challenge with numerous applications in gaming, simulation, and visual content creation. In our setup, given a rich and complex 3D scene, the user provides a list of instructions composed of target locations and locomotion types. To solve this task we present PlaMo, a scene-aware path planner and a robust physics-based controller. The path planner produces a sequence of motion paths, considering the various limitations the scene imposes on the motion, such as location, height, and speed. Complementing the planner, our control policy generates rich and realistic physical motion adhering to the plan. We demonstrate how the combination of both modules enables traversing complex landscapes in diverse forms while responding to real-time changes in the environment. **Video:** https://youtu.be/wWlqSQlRZ9M

1 Introduction

Physics-consistent animation of human characters is a challenging task with numerous applications, from automating game design through simulating factories and urban areas to training autonomous vehicles in pedestrian scenarios. In some cases, one may want to control the animation using text instructions that capture high-level goals, like "Run to the road once the car passes by" or "Hide behind the large tree". These tasks involve two types of control problems: First, planning a sequence of motions for reaching the goal with a natural behavior; Then, controlling the actual motion of the character to generate a natural-looking movement.

Previous studies in physics-based animation have usually focused on one of these two challenges, often addressing a simplified setup. For planning, [28] developed a method for long-range planning of a bipedal character, and [33] generates a distribution of trajectories based on user-defined characteristics. Other studies focused on movement control: mimicking [27], interleaving motion types [37], interactions with objects [41] or people [49] and crowd simulation [33] (see related method section).

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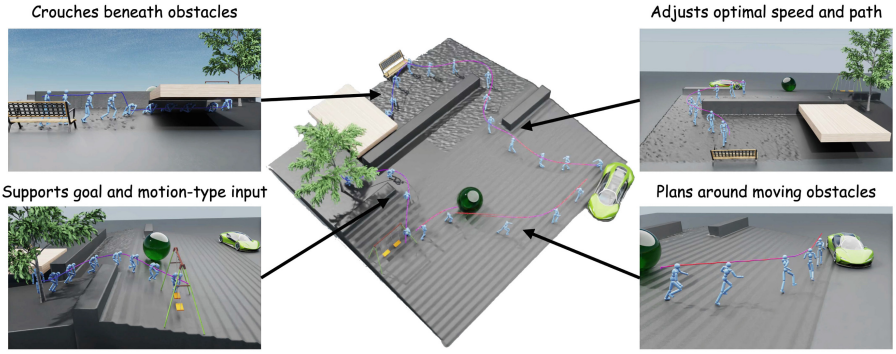


Fig. 1. Path planning and motion control *PlaMo* animates a humanoid in a complex 3D scene. The input consists of a physically simulated 3D scene and a series of text instructions describing high-level navigation landmarks and locomotion type ("Crouch-walk from the tree to the swing"). The scene may contain diverse terrain (gravel, stairs), 3D obstacles, and dynamic obstacles (here, a green ball). The output is a sequence of motor actuations controlling a humanoid character. (Color figure online)

However, the two tasks are interdependent. Planning which motions to take depends on the physical characteristics of each motion and the character. Then, executing the motion depends on the sequence and longer-horizon goal. Importantly, this interdependence becomes stronger when environments become more complex. Handling 3D obstacles, uneven terrain, and dynamic objects may require the planner to override user instruction and revert to motions that are feasible in the specific environment. In this work, we address the problem of training physics-based motion models for humanoids that can handle long-horizon planning and motion control in intricate environments.

Our approach *PlaMo*, for *Plan and Move*, combines long-horizon path planning with a novel matching motion controller. This fusion creates an agent that responds to semantic instructions from users but can adapt the plan and motion for navigating naturally through various terrains and obstacles.

When designing a path planner for this task, we need to consider several aspects. First, the planner must only produce plans that can be executed by the character's physics-based motion controller. More specifically, the character typically has a range of possible motions, such as walking, running, or crouching under an obstacle. The planner should devise plans that are within the feasibility envelope of the motion controller. Second, the capabilities of the humanoid character depend on the environment. For example, it is harder to run uphill than downhill. The planner therefore must be aware of the slope and obstacles. A similar problem occurs when turning around obstacles. Running too fast makes it harder to control a character, so the planner must match the character's velocity to the curvature of the path.

To address these challenges, we develop a planner inspired by state-of-the-art planners in the fields of robotics and autonomous driving [1,9,40]. These techniques have proved very useful in hard real-world problems and we show

here how they can be applied in physics-based character animation. Our planner operates in three stages. First, it constructs a low-resolution path using the A^* algorithm. Second, it creates a high-resolution height trajectory that determines the position of a character's head along the path, taking into account 3D obstacles. Third, it sets the locomotion velocity along the path, based on the slope and curvature of the path and on the feasible movement envelope of the character, which we learn offline. This is achieved by solving a quadratic program that finds the optimal speed along the path [1]. At the end of planning, it produces a sequence of head positions and velocity along the path and sends that to the motion controller. As far as we know, this is the first long-horizon planner that is aware of a complex 3D environment and character motion envelope.

Guided by the planned path, the motion controller actuates the character's joints at every time step. We condition the controller on the pose and velocity of the humanoid, surrounding terrain, and near-term path waypoints. To train the motion controller, we designed a reward function with two main components: path-following and style. The path-following component rewards the agent for successfully reaching target 3D waypoints at designated times. Our reward design considers 3D path tracking, coupled with additional terms for improving orientation and posture. For style, we use a discriminative objective [30, AMP] encouraging natural movement. This controller is trained on random paths that take into account the correlation between height and speed, for example, crawling is typically performed slowly.

In summary, this paper makes the following novel contributions: (1) We introduce a methodology for combining path planning and motion control within a detailed physical simulation, which we call *PlaMo*. (2) We present an efficient method for planning long-horizon paths for animated humanoid characters, producing a feasible and safe path across uneven terrain around static, dynamic, and three-dimensional obstacles. (3) We develop a robust motion controller trained through reinforcement learning. Training to imitate human motions, the controller produces motion guided by a sequence of 3D head positions and velocities.

2 Related Work

Physics-Based Animation. Prior work shows that physics simulation can be utilized for learning scene-aware controllers [18,20,42,46,47]. These controllers are robust to changes in the scene like irregular terrain, obstacles, and adversarial perturbations [14,19,39]. Typically, prior work focused either on complex motions in simple scenes [10,29,37], or simple motions within complex scenes [7,43]. Closest to our work is TRACE and PACE [33], which focuses on pedestrian locomotion with a diffusion-based planner. To generate paths, the planning module TRACE is trained on pedestrian data, typically upright walking motions. In contrast, we focus on long-term 3D-aware navigation and locomotion. *PlaMo* combines a short-sighted terrain-aware locomotion controller with a long-term and adaptive scene-aware path generator. The path generator plans a 3D trajectory. This plan takes into account the scene and the capabilities of the motion controller. It determines the speed and height to avoid both static and dynamic

obstacles. This plan is provided to the controller. The controller then tracks the requested path along rich and complex terrains.

Scene-Aware Animation. Most prior work has used kinematic animated characters to tackle a wide range of problems such as object interaction [15,35], long-term motion generation [16] and storytelling [32]. However, as physics-based animation is less forgiving, prior works that use simulation have typically focused on solving specific object-interaction tasks [7], or simple locomotion (walking or running) with short-term planning across complex terrains [33]. We tackle the problem of long-term motion generation across complex scenes using physics-based control.

Path Planning. There are numerous works in the world of planning motion in different levels of hierarchy. For high-level planning, the A* algorithm and its variations [3,5,44] use dynamic programming to produce a discrete path that considers the cost and connectivity of moving across the grid. Alternatively, navigation meshes [11,12,38] use the geometric structure of the obstacle to find the shortest path. Finer control models such as Model Predictive Control (MPC; [8,31]) construct an optimization problem by considering the dynamics of all objects and the physical constraints of the agent while using a lookahead of a receding horizon, these models usually require an explicit knowledge on the dynamics of the agent itself. In cases where the dynamics in the scene are unknown or complicated, learning methods can be employed instead such as RL for a given simulator [4,24,48] or diffusion models when there is access to relevant data [2,13]. In this context, our works are most related to recent approaches for planning in autonomous driving, extending A* with additional smoothing and obstacle avoidance modules to form feasible trajectories [23,45,50].

3 Preliminaries

We begin with a formal introduction of the reinforcement learning (RL) setup for training our motion controller, together with the mathematical representation of our simulated scene.

3.1 Reinforcement Learning

We train a goal-conditioned motion controller which we model as an RL [36] problem. In RL, an agent interacts with an environment according to a policy π. At each step t, the agent observes a state s_t and samples an action a_t from the policy $a_t \sim \pi(a_t|s_t)$. The environment then transitions to the next state s_{t+1} based on the transition probability $p(s_{t+1}|s_t, a_t)$. The goal is to maximize the discounted cumulative reward, defined as

$$J = \mathbb{E}_{p(\sigma|\pi)}\left[\sum_{t=0}^{T} \gamma^t r_t \middle| s_0 = s\right],$$

where $p(\sigma|\pi) = p(s_0)\Pi_{t=0}^{T-1} p(s_{t+1}|s_t, a_t)\pi(a_t|s_t)$ is the likelihood of a trajectory $\sigma = (s_0, a_0, r_0, \ldots, s_{T-1}, a_{T-1}, r_{T-1}, s_T)$, and $\gamma \in [0, 1)$ is a discount factor that determines the effective horizon of the policy.

To find the optimal policy π^* that maximizes the discounted cumulative reward, the popular policy gradient paradigm seeks to directly optimize the policy using the utility of complete trajectories. Perhaps the most widely used algorithm in this context is PPO [34], which gradually improves the policy while maintaining stability without drifting too far from it.

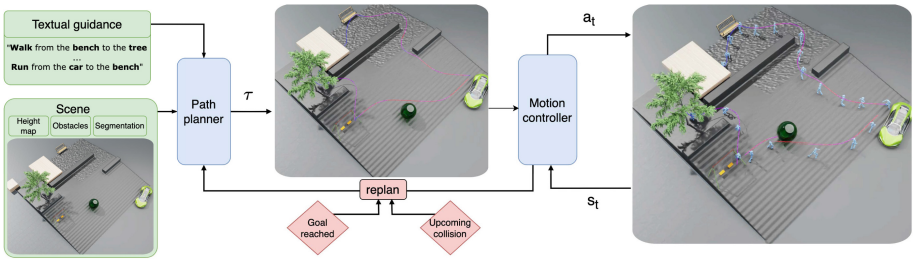


Fig. 2. Method overview. A simulated scene is provided to the path planner, together with a series of textual instructions, requesting a humanoid to reach landmarks in the scene using various locomotion types. The high-level planner computes a path that is fed to a reinforcement-learning-based low-level motion controller, which controls a humanoid to follow the path using the request locomotion type.

3.2 3D Scenes

In this work our characters navigate across rich 3D scenes. We describe the simulated scene as:

$$Scene = \{h, \mathcal{O}_{\text{static}}, \mathcal{O}_{\text{dynamic}}, \mathcal{O}_{\text{top}}, \mathcal{L}\}\,. \tag{1}$$

The 5 components of the scene are as follows:

1. A terrain given as a height map $h(x, y) \in \mathbb{R}$, specifying the height for each location in the grid.
2. A set of non-passable static obstacles $\mathcal{O}_{\text{static}}$.
3. A set of dynamic obstacles and their properties $\mathcal{O}_{\text{dynamic}}$.
4. A set of height-limiting ("top") obstacles requiring the agent to crouch or crawl when traversing through them $\mathcal{O}_{\text{top}}$.
5. A set of landmarks $\mathcal{L} = \{L_i\}_{i=0}^{l}$.

The locations and speeds of the dynamic objects are updated in each simulation step, based on their provided properties. The goal in these scenes is to plan a path between a series of landmarks $L_1, \ldots, L_k$ and successfully execute the plan.

4 Method

Our method, *PlaMo*, consists of two main components (Fig. 2):

High-Level Path Planner. The planner receives two inputs: a physically simulated 3D scene and a templated text instruction describing a sequence of (locomotion type, start point, target point). The planner then outputs a path τ consisting of velocity and 3D head positions at each time-step. The path is optimized to lead the agent through a smooth, physically feasible path to reach the goal while avoiding obstacles. It is then sent to the motion controller.

Low-Level Locomotion Controller. This controller receives as input the height map h of the local terrain and a desired path τ. It outputs a sequence of motor actions for the humanoid joints, resulting in the humanoid following the desired path. The locomotion controller is trained on real-world motion captures using RL to generate realistic human-like behavior.

Together, at each time-step of the simulation, the path planner translates the current humanoid location and target landmark to a local path that the controller aims to follow. The controller then outputs actions on the joints to the simulator, which then advances to the next simulation step.

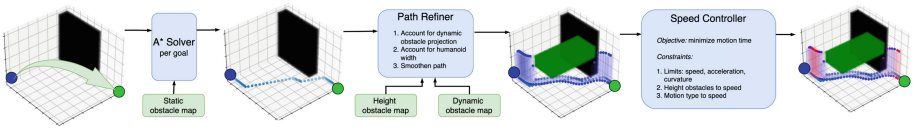


Fig. 3. The three stages of our dynamic path planner: (i) A^* solver, (ii) path refiner, and (iii) speed controller.

4.1 Dynamic Path Planner

We present a comprehensive module for path planning, structured into distinct phases. The full planning flow is visualized in Fig. 3.

Textual Guidance Parser. The input to the planner is a 3D simulated scene *Scene* and a series of k text instructions in the form "Run from the tree to the lake". These instructions refer to known landmarks in the scene where we assign each landmark to its corresponding set of (x, y) coordinates. We then translate each instruction to a triplet of (source landmark, target landmark, locomotion type). Together, they form a sequence of instructions that go through the series of landmarks $L_1, \ldots, L_k$ connected by paths. We observe that two main characteristics that differentiate between desired forms of locomotion are speed

and height. As such, each path is represented as a series of 3D positions. We determine velocities by controlling the distance between the points on the path. These paths are fed to the motion controller.

Our planner is inspired by state-of-the-art robotics and autonomous driving techniques [1,9,40]. It operates in three stages: low-resolution path construction using the A* algorithm, high-resolution height trajectory creation considering 3D obstacles, and locomotion velocity setting based on path slope and curvature. It ultimately produces a sequence of head positions and velocities for the motion controller, being the first long-horizon planner aware of complex 3D environments and character motion envelopes.

A* with Awareness to Slope. We form a 2D grid on the terrain level, connecting every point to its eight neighbors. We then remove connections based on walls, top obstacles too low to bypass, and infeasible slopes. The remaining connections are assigned a weight based on the distance and height difference, representing how difficult and time-consuming it is for the agent to cross it. Then, for every landmark L_i, we mark all its assigned (x, y) coordinates as goals for A*. Finally, the A* algorithm [3] constructs a shortest-path solution to the nearest coordinate of the landmark. To support long paths across large areas, we implement A* efficiently using KD-trees [51] and perform it once.

Path Refiner. The path created at the previous A* phase is coarse and impervious to dynamic obstacles and height constraints. We refine it to achieve smoother paths that avoid all obstacles. Thus, these paths are easier for the low-level controller to follow, increasing its success rate. For *dynamic obstacles*, we detect potential collisions by estimating the movement pattern of the obstacle and comparing it with the last designated path τ. If a collision is likely to occur within a $1.5\,[s]$ time frame, we replan the path, with the dynamic obstacle set at its future location as an added constraint. Finally, the path refiner adjusts the trajectory for smoothness and obstacle avoidance. The path refiner then sets the desired head height (z-axis) values that guide the character beneath top obstacles.

Motion-Aware Speed Controller. The path created in the previous phases contains only spatial information and lacks consideration of the humanoid character's performance envelope. The third step assigns a velocity vector to each frame in the path, consistent with the requested motion type (walk, run, crouch, crawl) and the terrain. Specifically, a quadratic program (QP) is solved to minimize travel time along the refined path while avoiding infeasible areas. The QP is constrained by valid speed ranges for each locomotion type and by bounds on longitudinal and lateral accelerations that the character can achieve. The speed controller accounts for path curvature, adapting longitudinal speed based on lateral acceleration and adjusting the speed according to elevation and slope. This results in a refined speed profile that ensures the character slows down before turns and around obstacles.

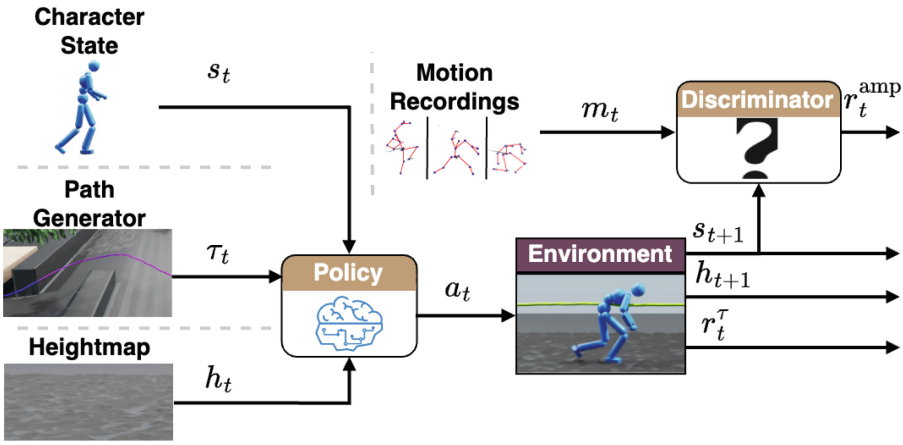


Fig. 4. Locomotion controller module. The locomotion policy observes the character state s_t, the height map of the terrain h, and the requested path τ. During training, the trajectories are randomly sampled. In inference, the trajectories consider the terrain characteristics, such as obstacles. After simulating the predicted action the resulting state is used for providing the style reward r^{amp} and for the path following reward r^{τ}.

Dependence Between Control and Planning: The effectiveness of our method hinges on the strong interdependence between the control and planning phases. The A* algorithm incorporates height differences into its cost function, considering the difficulty and time required for the character to traverse different elevations. This notion can be easily extended to different agents with different capabilities; for example, an accomplished humanoid climber will choose a steeper, shorter path than an agent having difficulties with high slopes. The tight coupling continues through the path refinement and speed control phases, where adjustments for slope, curvature, and terrain are directly informed by the character's performance limits. This ensures that the final path is not only theoretically optimal, but also practically executable, maintaining the balance and stability of the character in real-world conditions.

4.2 Locomotion Controller

Plans produced by the high-level path planner are provided to a low-level locomotion controller. This controller, illustrated in Fig. 4, is tasked with executing the provided plan using human-like physical motion. The provided plans control the height, location, and speed at which the character should move. This is performed across a wide range of irregular terrains. To achieve this we train our locomotion controller using reinforcement learning. We provide a reward with two components: one for following the path and the other for producing realistic motion. In this paper, we accommodate four forms of locomotion: running, walking, crouch walking, and crawling. It is interesting to note that the planner does

not explicitly guide a specific locomotion type, like running or crouching; but since the motion controller is tasked with creating a natural-looking movement, the right type of locomotion naturally emerges for every speed and head height.

State. At each step the controller observes the current pose of the character s_t, the target trajectory τ_t, and the surrounding heightmap h_t. The trajectory is provided as 5 s into the future with ten samples, one every 0.5 s. The height map allows the controller to be aware of its surroundings, ensuring motions are robust to varying scenes. We represent the height map as a rectangular grid rotated to the character's root orientation. It consists of 64 equidistant samples spanning $0.8\,[m]$ on each axis.

Reward. The goal of the controller is to track a target path τ, generated by the planner (Sect. 4.1). We identify that speed and height are the two main characteristics for controlling our desired forms of locomotion (run, walk, crouch-walk, crawl). As such, the path is constructed in 3D. At each time step t, the point τ_t represents the 3D coordinates for the head. By controlling the distance between the points, we determine the requested speed, and changing the Z-axis provides control over the desired character's height.

Due to the importance of height control, we split the reward into four components. The first considers the x-y displacement of the character. This guides the path the character should follow, in addition to speed. The second component considers the height of the head. The third considers the head's direction (pitch and yaw), aligning it with the path. Since all path rewards target the head, to produce more realistic movement when following lower heights we add the reward term: $r_t^{\text{body}} = c_{\text{body}}(z_{\text{head}} - \max_{\text{body}} z)$ with $c_{\text{body}} = 0.1$. This term encourages the head to be the highest point to prevent a hunchback posture when crouching and crawling.

$$r_t^{\tau} = e^{-c_{\text{pos}}||\tau_t^{x,y} - s_t^{x,y}||^2} + e^{-c_{\text{height}}||\tau_t^{z} - s_t^{z}||^2} + e^{-c_{\text{dir}}||\tau_t^{y,p} - s_t^{y,p}||^2} + r_t^{\text{body}}\,,$$

where $c_{\text{pos}} = 2$, $c_{\text{height}} = 10$ and $c_{\text{dir}} = 20$ are fixed.

Complementing the task reward, we leverage AMP [30] for motion stylization. AMP trains a discriminator to differentiate between reference recorded motions m_t and those simulated by the motion controller s_t. The resulting reward combines the tracking objective with the discriminative reward: $r_t = r_t^{\text{amp}} + r_t^{\tau}$.

Reference Motions. The adversarial reward in AMP is maximized when the simulated pose distribution matches the data. Naively providing a diverse dataset of human motions may result in unwanted behaviors. For example, back-flips and break-dancing are distinctly different from locomotion. To ensure that the motion distribution matches the motions needed to solve the task, we hand-picked 75 distinct motions from the AMASS dataset [21], covering the four locomotion forms performed in various styles. An appropriate ablation study is provided in Sect. 5.

Random Path Generation. During training, the path generator samples random paths for the character to follow. If the task demands unlikely motion given the data, we observe a degradation in both motion quality and tracking performance. For example, crawling at $v = 5\,m/s$ is unrealistically fast and will not appear in our motion capture data. Thus, the task and discriminative rewards will contradict each other, impairing the training process. To provide a realistic path, the generator first samples the requested heights along the path, ranging from $0.4\,[m]$ (crawling) to $1.47\,[m]$ (the simulated humanoid's height), with smooth transitions in between. Then, the speed is sampled uniformly between 0 and $v_t^{\max}$, where:

$$v_t^{\max} = \min\left(1 + 4 \cdot (z_t - z^{\min})/(1.2 - z^{\min}),\, V_{\max}\right),$$

for a minimal height $z^{\min} = 0.4\,[m]$ and maximal speed $V_{\max} = 5\,[m/s]$. This results in a linearly increasing max-speed, ranging from $v_t^{\max} = 1\,[m/s]$ for crawling motions, and up to $v_t^{\max} = 5\,[m/s]$ for any height above $1.2\,[m]$.

Actions. Similar to prior work [27,33], we opt for proportional derivative (PD) control. We represent the action distribution using a multi-dimensional Gaussian with a fixed diagonal covariance matrix.

5 Experiments

We test *PlaMo* both qualitatively and quantitatively. First, we evaluate the motion controller with several common forms of motion across a range of randomized terrains. In addition, we evaluate our design decision regarding the workflow for training the controller, using ablation experiments regarding both data curation and height-speed coupled path generation.

Then, we analyze the quality of the path planner. We focus on evaluating the paths produced by the QP solver in terms of how well they fit the capability envelope of the locomotion controller. Finally, we provide qualitative examples both for the locomotion controller and the whole system in a complex scene with multiple landmarks. We stress that our test scenes were not observed during training. Our agent can navigate during inference in every scene as defined in (1), without requiring any further training. We provide video demonstrations and comparisons in the supplementary material.

5.1 Experiment Details

We focus here on the SMPL humanoid [17] with a neutral body structure. The humanoid is simulated in IsaacGym [22], operating at $120\,[Hz]$. The controller is trained using PPO [34] and takes decisions at a rate of $30\,[Hz]$. The motion controller is trained for one week on a single NVIDIA V100 GPU.

Training Data. We utilize motion capture data from the AMASS dataset [21] that contains more than 10 h of diverse motion recordings across thousands of distinct motions. Complementing AMASS, HumanML3D [6] provides textual labels for each motion.

6 Results

We now discuss the resulting paths and motions produced by *PlaMo*.

6.1 Locomotion

We test our locomotion module on various terrains for every motion type; **example videos are given in the supplementary material**, and screen captures are shown in Fig. 7. Table 2, reports the XY-displacement along the path, which shows how well the controller tracks the requested path, and the Z-displacement, which measures how well it maintains the requested height. The results show a hierarchy of complexity between tasks such that stairs are harder than rough gravel-like terrain, which is harder than locomotion across a flat and smooth sloped surface. Although the data only contained motions traversing flat, smooth floors, the controller learned to generalize to a wide range of uneven terrains and produce robust motions resembling those in the data. Also, as expected, moving at lower speeds enables the character to follow the path more precisely, thus maintaining a lower error.

Table 1. Locomotion ablation. We compare various design choices for data purification and path generation during training. The analysis focuses on crawling on rough terrain, with additional results provided in the supplementary material. The results were averaged over 10^6 runs.

Data purification	Path generator	XY-err	Z-err
Ours (Curated + Coupled)		0.25	0.10
Curated	Naive	0.50	0.13
Naive	Naive	0.78	0.09
Naive	Coupled	1.16	0.17
None	Coupled	0.66	0.09
None	Naive	1.39	0.19

Locomotion Ablation. We consider the crawling on rough terrain. We focus on this task as it was shown in Table 2 as the hardest type of motion to generate. For our ablation study, we consider two design decisions – data curation, and random path generation. **Data curation:**[1] We evaluate the importance of data curation

[1] We provide additional information in the supplementary material.

by comparing training on the full dataset (*none*), automated curation (*naive*, based on the HumanML3D [6] textual labels), and manual curation (*curated*). **Random path generation:** To showcase the importance of coupling height and speed in the randomly sampled paths, we also compare our *coupled* motion-aware path sampler that samples the requested speed along a random path according to the requested height, with a *naive* baseline of uncoupled sampling of both speed and height.

The results, shown in Table 1, emphasize the importance of these design decisions. We find that the best results are obtained by combining the manually curated dataset, alongside the coupled random path generator. By ensuring a tight alignment between the requested motions, the data representing them, and the randomly generated paths, we find an improvement of 50% in the XY-err, compared to the next best combination.

Table 2. Mean path displacement for various locomotion types. We test four motion types over four types of terrains. For each, we sample a path with constant speed and height, and compute the mean error between the position of the humanoid and the desired path. The results were averaged over 10^6 runs.

	Crawl		Crouch-walk		Walk		Run	
Terrain	XY-err $[m]$	Z-err $[m]$	XY-err $[m]$	Z-err $[m]$	XY-err $[m]$	Z-err $[m]$	XY-err $[m]$	Z-err $[m]$
Flat	0.20	0.07	0.20	0.03	0.24	0.21	0.52	0.26
Slope	0.21	0.09	0.22	0.03	0.25	0.21	0.60	0.27
Rough	0.25	0.10	0.25	0.04	0.29	0.24	0.79	0.31
Stairs	0.45	0.13	0.41	0.07	0.39	0.27	1.01	0.35

6.2 Terrain-Dependent Path Planning

When animating a character we often want it to exhibit behaviors that align with the character's story. For example, while a young athlete may be capable of running up a hill, an old person may prefer to take a longer and less physically challenging path. Here, we show that these preferences can be encoded into our path planner. To do so, we re-weight the connectivity graph based on the slope. The distance between each two points is increased by $e^{c \cdot \text{slope}} - 1$. As illustrated in Fig. 5, as a result of increasing the weight, our planner produces a path around the hill instead of directly across it.

6.3 Speed Control

The third phase of our path planner uses a QP solver to optimize the speed along the path curves. On the one hand, if the speed is too high for the curvature, the character will not be able to maintain the desired route, resulting in large displacement errors and possible collisions with obstacles. On the other hand, if

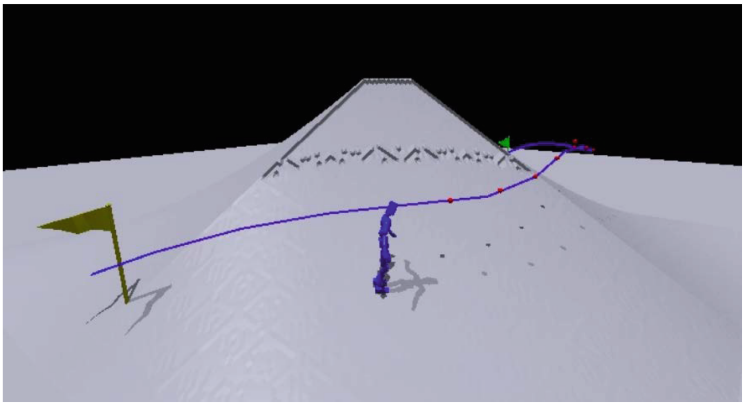


Fig. 5. Terrain dependent path planning. The humanoid does not go through the shortest path between the two landmarks but prefers a smoother, flatter surface due to the height differences.

the speed is too low, it takes the character an unnecessarily long time to complete the route. To avoid jerks, the optimization program also considers the maximum acceleration ($a_{\max} = 0.5\,[m/s^2]$) and deceleration ($a_{\min} = -0.1\,[m/s^2]$).

We demonstrate this trade-off in the "Slalom" scene (Fig. 8). This scene requires the character to take steep turns. Here, we compare constant speed versus adaptive speed. We generate both routes using our A^* solver and path refiner. However, we only refine the "adaptive speed" run using our speed controller module. Without adapting the acceleration profile to the curvature of the path, the humanoid fails to follow the path accurately and often collides with the walls.

For a quantitative evaluation of the Slalom test, we compare various constant speeds, as well as various configurations of our QP solver. The latter includes different combinations of maximal lateral and longitudinal acceleration. We summarize the results in Fig. 10a and 10b. Each choice of constant speed and QP configuration corresponds to a single point in the plots and is the average of 100 runs. The constant speeds were chosen between $1\,[m/s]$ and $3.5\,[m/s]$. In Fig. 10a, we report the success rate versus the average completion time. Success is defined as successfully reaching the target landmark. In Fig. 10b, we show the same, but with the average 3D disposition error on the y-axis. As seen, for every QP configuration we tested, our planner enables the humanoid to traverse its path faster and without compromising on accuracy thus achieving the Pareto front.

6.4 Randomized Routes in a Complex Randomized Scenes

Finally, we showcase *PlaMo*'s ability to solve previously unseen scenes in varying forms (Fig. 6). We automatically construct random scenes consisting of: **(1)** multiple forms of terrains (smooth, gravel-like, stairs and sloped), with **(2)** several

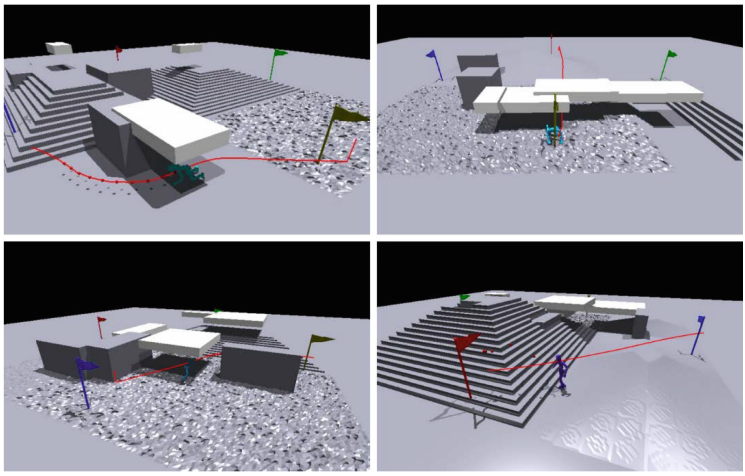


Fig. 6. Randomized complex scenes. We randomize several terrain types, walls and top obstacles to generate a variety of scenes.

types of obstacles (walls and low-ceiling obstacles), and four landmarks in each scene. *PlaMo* is given a random scene with navigation instructions to a random sequence of landmarks. Figure 6 illustrates some of these environments, showing

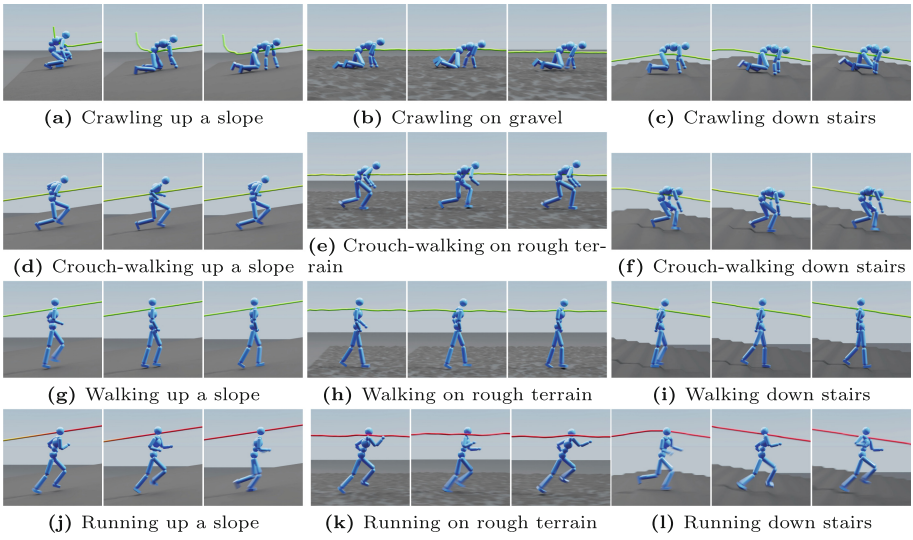


Fig. 7. Locomotion: Screen captures of our trained controller when conditioned on various forms of locomotion across varying types of terrains. Crawling is achieved by requesting a head-height of 0.4 $[m]$, crouching at 0.8 $[m]$, walking and running at upright 1.47 $[m]$. The corresponding speeds are 1 $[m/s]$, 2 $[m/s]$, 2 $[m/s]$, and 4 $[m/s]$.

that *PlaMo* successfully plans realistic routes. In our experiments, our agent has an 81% chance to fully solve a randomized 4-landmarks scene.

As shown in Figs. 1 and 9, *PlaMo* solves this task in varying forms while adhering to the scene constraints. It slows down and starts to crawl before passing under the low ceiling (top left), speeds up during straight lines while carefully slowing during a tight curve, and finally, it dynamically adapts the path when a collision with a dynamically moving object is deemed likely[2].

7 Discussion, Limitations and Future Work

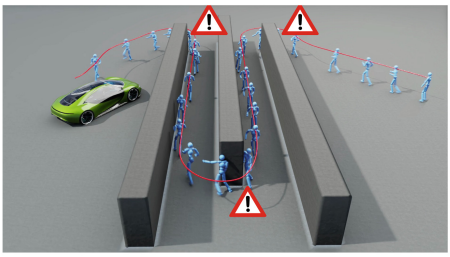
We described *PlaMo*, a framework for planning long paths and controlling the motion of humanoid characters in simulated environments for physics-based animation. Given a complex simulated scene and textual instructions, the system efficiently plans a path that a humanoid character can traverse using natural locomotion. The planner takes into account the character's performance envelope in uneven terrain. In addition, it takes into account both static and dynamic obstacles. Then, the trained motion controller successfully tracks these paths, producing complex, scene-aware motions, for example, crawling under a low-hanging ceiling or quickly avoiding an incoming projectile.

PlaMo can handle various scene complexities, but has some limitations which we outline as future work. **Interaction with objects and multiple agents.** Many interesting scenarios for gaming and simulation would require physical interactions with dynamic objects or other agents. Learning such interactions not only requires specific data but may require extending the human model and potentially the controller training, for example with hands [26]. **Dynamic objects.** *PlaMo* handles objects whose motion can be easily predicted using rules. Modeling more complex dynamic objects may require learning-based approaches to model the dynamic scene. **Rich language and scene understanding.** *PlaMo* focuses on planning and control. It opens opportunities for combining it with modern language models and 3D scene understanding. With these expansions in mind, we see *PlaMo* as a stepping stone in a much greater system where non-playable characters (NPCs) are given roles to play, forming a rich simulated virtual world.

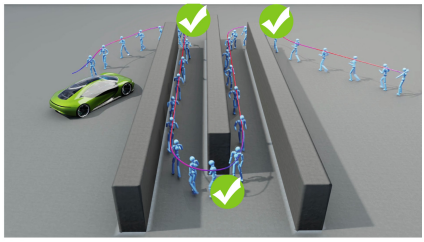
A Data Curation

To train our motion controller, we consider three datasets. First, the entire AMASS dataset contains various motions, such as dancing, cartwheels, and sitting. Second, we consider an automatic curation method. This uses a large language model (LLM, ChatGPT4 [25]). We the LLM with the textual descriptions of each movement and ask it to categorize the motions into five classes: ['walking', 'running', 'crouching', 'crawling', and others]. These textual labels are extracted from the HumanML3D dataset [6]. This filtering reduced the size

[2] The dynamic re-planning is best seen in the supplementary video.

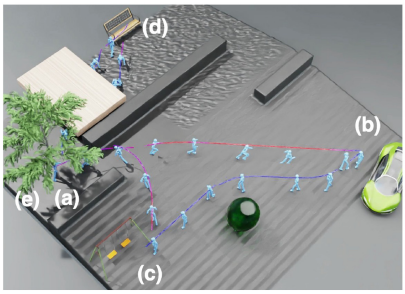


(a) Slalom test with constant speed (baseline). The humanoid follows the path poorly, hitting the walls on every turn.

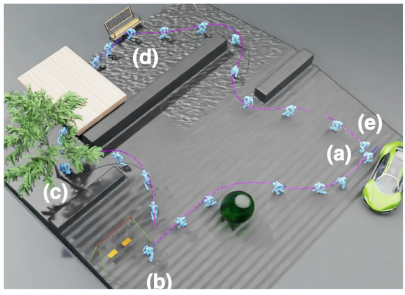


(b) Slalom test with adaptive speed (ours). The humanoid follows the path well, never hitting the walls.

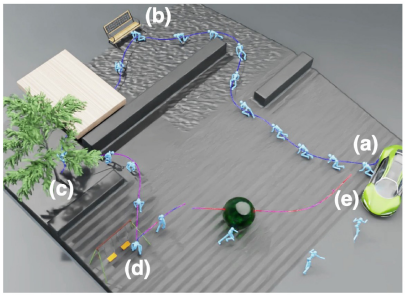
Fig. 8. Comparison of consistent and adaptive speed profiles on the slalom test. The colors represent the speed, with red indicating speeds above 3.5 m/s and blue below 1 m/s. (Color figure online)



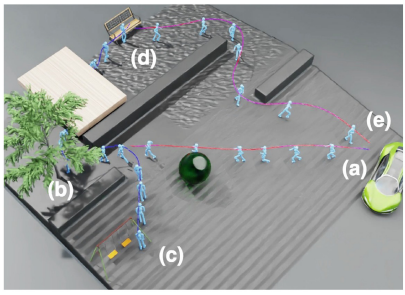
(a) Starting at the tree: "Run to the car" then "walk to the swing" then "run to the bench" then "walk to the tree".



(b) Starting at the car: "Walk crouching to the bench" then "walk crouching to the tree" then "walk crouching to the bench" then "walk crouching to the car".

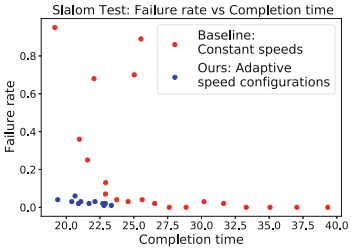


(c) Starting at the car: "Crawl to the bench" then "crawl to the tree" then "walk crouching to the bench" then "sprint to the car"

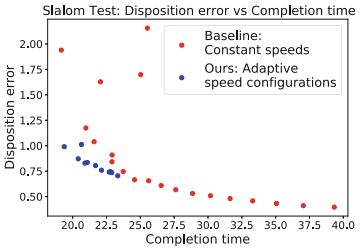


(d) Starting at the car: "Run to the tree" then "walk to the swing" then "walk to the bench" then "run to the car".

Fig. 9. Diverse solutions: Our system can solve unseen tasks in diverse forms. Here, we show how the character traverses a random sequence of segments within the scene. Each example showcases a randomly sampled sequence of waypoints to be performed with a randomly sampled motion.



(a) Failure rate and Completion Time: Pareto curve in the Slalom test. Testing our speed controller vs. constant speed. Our planner obtains lower failure rate while retaining faster completion.



(b) Disposition error and Completion Time: Pareto curve in the Slalom test. Testing our our speed controller vs. constant speed. Our planner obtains lower disposition error while retaining faster completion.

Fig. 10. .

of the dataset down to 600 motions. Finally, we also tested a manual curation procedure. For manual curation, we handpick 75 motion captures, roughly 20 from each desired motion type, and create a small, tailor-made dataset.

B Additional Results

In Tables 3 – 7 we present the full results for the ablation study given in Table 1. We show the XY and Z errors for every choice of data filtering and path generator speed samples, for every terrain and every motion type.

Table 3. No data filtering. Naive path generator. Locomotion controller errors.

	Crawl		Crouch-walk		Walk		Run	
Terrain	XY-err $[m]$	Z-err$[m]$	XY-err $[m]$	Z-err $[m]$	XY-err$[m]$	Z-err $[m]$	XY-err$[m]$	Z-err$[m]$
Flat	1.11	0.20	0.43	0.11	0.35	0.31	0.76	0.36
Slope	1.12	0.20	0.46	0.11	0.38	0.32	0.82	0.36
Rough	1.39	0.19	0.53	0.13	0.44	0.34	1.06	0.40
Stairs	1.44	0.21	0.78	0.16	0.53	0.36	1.19	0.42

C Additional Technical Details for Section 4.1

As a final step, the output format of the path planner (x,y,z,v) is translated into a format suitable for the motion controller. This controller requires (x, y, z) waypoints spaced at fixed time intervals, where a larger waypoint spacing corresponds to a higher speed. The conversion from the spatially defined path to this temporal trajectory format is accomplished through an iterative algorithm supplemented by linear interpolation.

Table 4. No data filtering. Coupled path generator. Locomotion controller errors.

	Crawl		Crouch-walk		Walk		Run	
Terrain	XY-err [m]	Z-err[m]	XY-err [m]	Z-err [m]	XY-err[m]	Z-err [m]	XY-err[m]	Z-err[m]
Flat	0.43	0.07	0.39	0.07	0.37	0.31	0.74	0.36
Slope	0.49	0.08	0.42	0.07	0.40	0.32	0.81	0.38
Rough	0.66	0.09	0.49	0.08	0.45	0.34	1.03	0.42
Stairs	0.94	0.13	0.67	0.10	0.53	0.35	1.13	0.43

Table 5. Naive data filtering. Naive path generator. Locomotion controller errors.

	Crawl		Crouch-walk		Walk		Run	
Terrain	XY-err [m]	Z-err[m]	XY-err [m]	Z-err [m]	XY-err[m]	Z-err [m]	XY-err[m]	Z-err[m]
Flat	0.62	0.15	0.26	0.05	0.29	0.27	0.60	0.30
Rough	1.16	0.17	0.37	0.07	0.35	0.28	0.87	0.36
Stairs	1.24	0.19	0.52	0.11	0.43	0.30	1.02	0.37

Table 6. Naive data filtering. Coupled path generator. Locomotion controller errors.

	Crawl		Crouch-walk		Walk		Run	
Terrain	XY-err [m]	Z-err[m]	XY-err [m]	Z-err [m]	XY-err[m]	Z-err [m]	XY-err[m]	Z-err[m]
Flat	0.67	0.06	0.25	0.06	0.25	0.23	0.57	0.29
Slope	0.55	0.07	0.26	0.07	0.30	0.23	0.64	0.30
Rough	0.78	0.09	0.32	0.07	0.34	0.26	0.85	0.35
Stairs	0.80	0.13	0.48	0.10	0.42	0.28	1.01	0.36

D Additional Experiments for the Speed Controller

Here, we provide additional results on the slalom test from Fig. 8. We record the paths the humanoid traversed with a constant speed, as well as our speed controller output, and compare them with the desired reference path. The constant speed was selected so the two runs had the same completion time. The result is shown in Fig. 11. Despite having the same average speeds, in the constant-speed run, the humanoid has a difficult time matching the speed during the steep turns and hits the wall at its corners. This does not happen with the adaptive speed run, which follows the desired path better.

E Supplementary Files

We provide as addition to the paper supplementary videos to demonstrate PlaMo:

Table 7. Curated data filtering. Naive path generator. Locomotion controller errors.

	Crawl		Crouch-walk		Walk		Run	
Terrain	XY-err $[m]$	Z-err$[m]$	XY-err $[m]$	Z-err $[m]$	XY-err$[m]$	Z-err $[m]$	XY-err$[m]$	Z-err$[m]$
Flat	0.29	0.09	0.25	0.04	0.21	0.21	0.49	0.27
Slope	0.34	0.12	0.26	0.04	0.24	0.22	0.57	0.28
Rough	0.50	0.13	0.29	0.05	0.27	0.23	0.78	0.32
Stairs	0.81	0.17	0.43	0.07	0.36	0.26	0.97	0.35

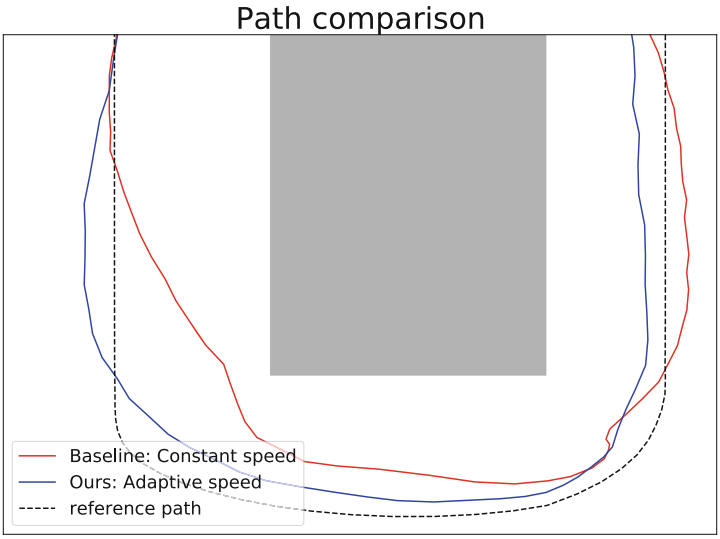


Fig. 11. Comparison of planned paths (blue) Our path with adaptive speed. (red) A path with a constant speed that has the same average speed as the adaptive-speed path. (dashed black) the desired reference path. When using a constant speed, the humanoid hits the wall at its corners. Using the adaptive speed generates a fast but smoother path, that is far from the obstacle but still follows well the desired path better. (Color figure online)

1. Videos of the locomotion controller in terrain X motion type that correspond to Table 2 and Fig. 7.
2. Videos of the humanoid doing the Slalom scene presented in Fig. 8.
3. Videos of randomized route and motion types in a diverse scene, corresponding to Fig. 9

We also provide yaml files that correspond to the filtered dataset considered for the ablation study given in Table 1.

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